full_study

March 16, 2026

1 US Power Outage Analysis

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Repository Link: https://github.com/k-phantastic/US-Power-Outage-Analysis
Website Link: https://khanhphan.com/US-Power-Outage-Analysis

[98]:
```python
from pathlib import Path
import time

from utils import *

pd.options.plotting.backend = 'plotly'
pio.renderers.default = "vscode"
# For widescreen display, overrides utils.py settings
pd.set_option('display.max_columns', None)
pd.set_option("display.max_rows", None)

# Machine Learning Stack
from sklearn.model_selection import train_test_split, cross_val_score, RandomizedSearchCV
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler, QuantileTransformer
from sklearn.linear_model import LinearRegression, Ridge
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score, root_mean_squared_error
from sklearn.base import BaseEstimator, RegressorMixin
from sklearn.ensemble import RandomForestRegressor, RandomForestClassifier, HistGradientBoostingRegressor

from xgboost import XGBRegressor
import optuna
from scipy.stats import randint, uniform
```

1.1 Step 1: Introduction

1.1.1 U.S. Power Outage Dataset from Purdue University

Source: https://engineering.purdue.edu/LASCI/research-data/outages/outage.xlsx
Data Dictionary: https://www.sciencedirect.com/science/article/pii/S2352340918307182?via%3Dihub#t0005

> This dataset includes the major outages witnessed by different states in the continental U.S. Besides major outages, this data contains information on geographical location of the outages, regional climatic information, land-use characteristics, electricity consumption patterns and economic characteristics of the states affected by the outages.
>
> Column information is located in Table 1 of Data Dictionary, Variable descriptions of the article

[146]:
```python
# Load the raw dataset
file_path = 'data/outage.xlsx'

raw_df = pd.read_excel(file_path)
raw_df.head(10)
```

[146]:
```
   Major power outage events in the continental U.S. Unnamed: 1 Unnamed: 2  \
0              Time period: January 2000 - July 2016        NaN        NaN
1  Regions affected: Outages reported in this dat…        NaN        NaN
2                                                NaN        NaN        NaN
3                                                NaN        NaN        NaN
4                                          variables        OBS       YEAR
5                                              Units        NaN        NaN
6                                                NaN          1       2011
7                                                NaN          2       2014
8                                                NaN          3       2010
9                                                NaN          4       2012

  Unnamed: 3 Unnamed: 4   Unnamed: 5   Unnamed: 6         Unnamed: 7  \
0        NaN        NaN          NaN          NaN                NaN
1        NaN        NaN          NaN          NaN                NaN
2        NaN        NaN          NaN          NaN                NaN
3        NaN        NaN          NaN          NaN                NaN
4      MONTH U.S._STATE  POSTAL.CODE  NERC.REGION     CLIMATE.REGION
5        NaN        NaN          NaN          NaN                NaN
6          7  Minnesota           MN          MRO  East North Central
7          5  Minnesota           MN          MRO  East North Central
8         10  Minnesota           MN          MRO  East North Central
9          6  Minnesota           MN          MRO  East North Central

      Unnamed: 8       Unnamed: 9                          Unnamed: 10  \
0            NaN              NaN                                  NaN
1            NaN              NaN                                  NaN
2            NaN              NaN                                  NaN
```

```
3            NaN              NaN                                NaN
4  ANOMALY.LEVEL  CLIMATE.CATEGORY                 OUTAGE.START.DATE
5        numeric              NaN  Day of the week, Month Day, Year
6           -0.3           normal               2011-07-01 00:00:00
7           -0.1           normal               2014-05-11 00:00:00
8           -1.5             cold               2010-10-26 00:00:00
9           -0.1           normal               2012-06-19 00:00:00

                 Unnamed: 11                       Unnamed: 12  \
0                        NaN                               NaN
1                        NaN                               NaN
2                        NaN                               NaN
3                        NaN                               NaN
4          OUTAGE.START.TIME          OUTAGE.RESTORATION.DATE
5  Hour:Minute:Second (AM / PM)  Day of the week, Month Day, Year
6                   17:00:00               2011-07-03 00:00:00
7                   18:38:00               2014-05-11 00:00:00
8                   20:00:00               2010-10-28 00:00:00
9                   04:30:00               2012-06-20 00:00:00

                 Unnamed: 13         Unnamed: 14            Unnamed: 15  \
0                        NaN                 NaN                    NaN
1                        NaN                 NaN                    NaN
2                        NaN                 NaN                    NaN
3                        NaN                 NaN                    NaN
4      OUTAGE.RESTORATION.TIME      CAUSE.CATEGORY  CAUSE.CATEGORY.DETAIL
5  Hour:Minute:Second (AM / PM)                NaN                    NaN
6                   20:00:00      severe weather                    NaN
7                   18:39:00  intentional attack              vandalism
8                   22:00:00      severe weather             heavy wind
9                   23:00:00      severe weather           thunderstorm

       Unnamed: 16      Unnamed: 17     Unnamed: 18         Unnamed: 19  \
0              NaN              NaN             NaN                 NaN
1              NaN              NaN             NaN                 NaN
2              NaN              NaN             NaN                 NaN
3              NaN              NaN             NaN                 NaN
4  HURRICANE.NAMES  OUTAGE.DURATION  DEMAND.LOSS.MW  CUSTOMERS.AFFECTED
5              NaN             mins        Megawatt                 NaN
6              NaN             3060             NaN               70000
7              NaN                1             NaN                 NaN
8              NaN             3000             NaN               70000
9              NaN             2550             NaN               68200

            Unnamed: 20           Unnamed: 21            Unnamed: 22  \
0                   NaN                   NaN                    NaN
1                   NaN                   NaN                    NaN
```

```
2                    NaN                    NaN                    NaN
3                    NaN                    NaN                    NaN
4              RES.PRICE              COM.PRICE              IND.PRICE
5  cents / kilowatt-hour  cents / kilowatt-hour  cents / kilowatt-hour
6                   11.6                   9.18                   6.81
7                  12.12                   9.71                   6.49
8                  10.87                   8.19                   6.07
9                  11.79                   9.25                   6.71

             Unnamed: 23    Unnamed: 24    Unnamed: 25    Unnamed: 26  \
0                    NaN            NaN            NaN            NaN
1                    NaN            NaN            NaN            NaN
2                    NaN            NaN            NaN            NaN
3                    NaN            NaN            NaN            NaN
4            TOTAL.PRICE      RES.SALES      COM.SALES      IND.SALES
5  cents / kilowatt-hour  Megawatt-hour  Megawatt-hour  Megawatt-hour
6                   9.28        2332915        2114774        2113291
7                   9.28        1586986        1807756        1887927
8                   8.15        1467293        1801683        1951295
9                   9.19        1851519        1941174        1993026

     Unnamed: 27 Unnamed: 28 Unnamed: 29 Unnamed: 30    Unnamed: 31  \
0            NaN         NaN         NaN         NaN            NaN
1            NaN         NaN         NaN         NaN            NaN
2            NaN         NaN         NaN         NaN            NaN
3            NaN         NaN         NaN         NaN            NaN
4    TOTAL.SALES  RES.PERCEN  COM.PERCEN  IND.PERCEN  RES.CUSTOMERS
5  Megawatt-hour           %           %           %            NaN
6        6562520       35.55       32.23        32.2        2308736
7        5284231       30.03       34.21       35.73        2345860
8        5222116        28.1        34.5       37.37        2300291
9        5787064       31.99       33.54       34.44        2317336

     Unnamed: 32    Unnamed: 33      Unnamed: 34   Unnamed: 35   Unnamed: 36  \
0            NaN            NaN              NaN           NaN           NaN
1            NaN            NaN              NaN           NaN           NaN
2            NaN            NaN              NaN           NaN           NaN
3            NaN            NaN              NaN           NaN           NaN
4  COM.CUSTOMERS  IND.CUSTOMERS  TOTAL.CUSTOMERS  RES.CUST.PCT  COM.CUST.PCT
5            NaN            NaN              NaN             %             %
6         276286          10673          2595696         88.94         10.64
7         284978           9898          2640737         88.83         10.79
8         276463          10150          2586905         88.92         10.69
9         278466          11010          2606813          88.9         10.68

    Unnamed: 37       Unnamed: 38      Unnamed: 39      Unnamed: 40  \
0           NaN               NaN              NaN              NaN
```

```
1          NaN              NaN             NaN             NaN
2          NaN              NaN             NaN             NaN
3          NaN              NaN             NaN             NaN
4  IND.CUST.PCT  PC.REALGSP.STATE  PC.REALGSP.USA  PC.REALGSP.REL
5            %              USD             USD        fraction
6         0.41            51268           47586            1.08
7         0.37            53499           49091            1.09
8         0.39            50447           47287            1.07
9         0.42            51598           48156            1.07

        Unnamed: 41   Unnamed: 42    Unnamed: 43  Unnamed: 44    Unnamed: 45  \
0                NaN           NaN            NaN          NaN            NaN
1                NaN           NaN            NaN          NaN            NaN
2                NaN           NaN            NaN          NaN            NaN
3                NaN           NaN            NaN          NaN            NaN
4  PC.REALGSP.CHANGE  UTIL.REALGSP  TOTAL.REALGSP  UTIL.CONTRI  PI.UTIL.OFUSA
5                  %           USD            USD            %              %
6                1.6          4802         274182         1.75            2.2
7                1.9          5226         291955         1.79            2.2
8                2.7          4571         267895         1.71            2.1
9                0.6          5364         277627         1.93            2.2

  Unnamed: 46   Unnamed: 47 Unnamed: 48             Unnamed: 49  \
0         NaN           NaN         NaN                     NaN
1         NaN           NaN         NaN                     NaN
2         NaN           NaN         NaN                     NaN
3         NaN           NaN         NaN                     NaN
4  POPULATION  POPPCT_URBAN   POPPCT_UC            POPDEN_URBAN
5         NaN             %           %  persons per square mile
6     5348119         73.27       15.28                    2279
7     5457125         73.27       15.28                    2279
8     5310903         73.27       15.28                    2279
9     5380443         73.27       15.28                    2279

               Unnamed: 50              Unnamed: 51    Unnamed: 52  \
0                      NaN                      NaN            NaN
1                      NaN                      NaN            NaN
2                      NaN                      NaN            NaN
3                      NaN                      NaN            NaN
4                POPDEN_UC             POPDEN_RURAL  AREAPCT_URBAN
5  persons per square mile  persons per square mile              %
6                   1700.5                     18.2           2.14
7                   1700.5                     18.2           2.14
8                   1700.5                     18.2           2.14
9                   1700.5                     18.2           2.14

  Unnamed: 53 Unnamed: 54    Unnamed: 55       Unnamed: 56
```

```
0          NaN          NaN            NaN                NaN
1          NaN          NaN            NaN                NaN
2          NaN          NaN            NaN                NaN
3          NaN          NaN            NaN                NaN
4   AREAPCT_UC     PCT_LAND  PCT_WATER_TOT  PCT_WATER_INLAND
5            %            %              %                  %
6          0.6        91.59           8.41               5.48
7          0.6        91.59           8.41               5.48
8          0.6        91.59           8.41               5.48
9          0.6        91.59           8.41               5.48
```

[100]:

```
# Initial file has the header in row 5, with first column being blank and␣
 ↪second column being index
df = pd.read_excel(file_path, header=5, usecols=range(2, 57), )

# Skip the units row
df = df.drop(index=0)
df.head()
```

[100]:

```
     YEAR  MONTH U.S._STATE POSTAL.CODE NERC.REGION      CLIMATE.REGION  \
1  2011.0    7.0  Minnesota          MN         MRO  East North Central
2  2014.0    5.0  Minnesota          MN         MRO  East North Central
3  2010.0   10.0  Minnesota          MN         MRO  East North Central
4  2012.0    6.0  Minnesota          MN         MRO  East North Central
5  2015.0    7.0  Minnesota          MN         MRO  East North Central

  ANOMALY.LEVEL CLIMATE.CATEGORY    OUTAGE.START.DATE OUTAGE.START.TIME  \
1          -0.3           normal  2011-07-01 00:00:00          17:00:00
2          -0.1           normal  2014-05-11 00:00:00          18:38:00
3          -1.5             cold  2010-10-26 00:00:00          20:00:00
4          -0.1           normal  2012-06-19 00:00:00          04:30:00
5           1.2             warm  2015-07-18 00:00:00          02:00:00

  OUTAGE.RESTORATION.DATE OUTAGE.RESTORATION.TIME      CAUSE.CATEGORY  \
1     2011-07-03 00:00:00                20:00:00      severe weather
2     2014-05-11 00:00:00                18:39:00  intentional attack
3     2010-10-28 00:00:00                22:00:00      severe weather
4     2012-06-20 00:00:00                23:00:00      severe weather
5     2015-07-19 00:00:00                07:00:00      severe weather

  CAUSE.CATEGORY.DETAIL HURRICANE.NAMES OUTAGE.DURATION DEMAND.LOSS.MW  \
1                   NaN             NaN            3060            NaN
2             vandalism             NaN               1            NaN
3            heavy wind             NaN            3000            NaN
4          thunderstorm             NaN            2550            NaN
5                   NaN             NaN            1740            250
```

	CUSTOMERS.AFFECTED	RES.PRICE	COM.PRICE	IND.PRICE	TOTAL.PRICE	RES.SALES	\
1	70000.0	11.6	9.18	6.81	9.28	2332915	
2	NaN	12.12	9.71	6.49	9.28	1586986	
3	70000.0	10.87	8.19	6.07	8.15	1467293	
4	68200.0	11.79	9.25	6.71	9.19	1851519	
5	250000.0	13.07	10.16	7.74	10.43	2028875	

	COM.SALES	IND.SALES	TOTAL.SALES	RES.PERCEN	COM.PERCEN	IND.PERCEN	\
1	2114774	2113291	6562520	35.55	32.23	32.2	
2	1807756	1887927	5284231	30.03	34.21	35.73	
3	1801683	1951295	5222116	28.1	34.5	37.37	
4	1941174	1993026	5787064	31.99	33.54	34.44	
5	2161612	1777937	5970339	33.98	36.21	29.78	

	RES.CUSTOMERS	COM.CUSTOMERS	IND.CUSTOMERS	TOTAL.CUSTOMERS	RES.CUST.PCT	\
1	2.31e+06	276286.0	10673.0	2.60e+06	88.94	
2	2.35e+06	284978.0	9898.0	2.64e+06	88.83	
3	2.30e+06	276463.0	10150.0	2.59e+06	88.92	
4	2.32e+06	278466.0	11010.0	2.61e+06	88.9	
5	2.37e+06	289044.0	9812.0	2.67e+06	88.82	

	COM.CUST.PCT	IND.CUST.PCT	PC.REALGSP.STATE	PC.REALGSP.USA	PC.REALGSP.REL	\
1	10.64	0.41	51268	47586	1.08	
2	10.79	0.37	53499	49091	1.09	
3	10.69	0.39	50447	47287	1.07	
4	10.68	0.42	51598	48156	1.07	
5	10.81	0.37	54431	49844	1.09	

	PC.REALGSP.CHANGE	UTIL.REALGSP	TOTAL.REALGSP	UTIL.CONTRI	PI.UTIL.OFUSA	\
1	1.6	4802	274182	1.75	2.2	
2	1.9	5226	291955	1.79	2.2	
3	2.7	4571	267895	1.71	2.1	
4	0.6	5364	277627	1.93	2.2	
5	1.7	4873	292023	1.67	2.2	

	POPULATION	POPPCT_URBAN	POPPCT_UC	POPDEN_URBAN	POPDEN_UC	POPDEN_RURAL	\
1	5.35e+06	73.27	15.28	2279	1700.5	18.2	
2	5.46e+06	73.27	15.28	2279	1700.5	18.2	
3	5.31e+06	73.27	15.28	2279	1700.5	18.2	
4	5.38e+06	73.27	15.28	2279	1700.5	18.2	
5	5.49e+06	73.27	15.28	2279	1700.5	18.2	

	AREAPCT_URBAN	AREAPCT_UC	PCT_LAND	PCT_WATER_TOT	PCT_WATER_INLAND
1	2.14	0.6	91.59	8.41	5.48
2	2.14	0.6	91.59	8.41	5.48
3	2.14	0.6	91.59	8.41	5.48
4	2.14	0.6	91.59	8.41	5.48

```
5          2.14        0.6    91.59          8.41              5.48
```

[101]:
```python
# DataFrame info
print(df.info())
print(f"\nColumns: {df.columns.tolist()}")
print(f"\nShape: {df.shape}")
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1534 entries, 1 to 1534
Data columns (total 55 columns):
 #   Column                   Non-Null Count  Dtype
---  ------                   --------------  -----
 0   YEAR                     1534 non-null   float64
 1   MONTH                    1525 non-null   float64
 2   U.S._STATE               1534 non-null   object
 3   POSTAL.CODE              1534 non-null   object
 4   NERC.REGION              1534 non-null   object
 5   CLIMATE.REGION           1528 non-null   object
 6   ANOMALY.LEVEL            1525 non-null   object
 7   CLIMATE.CATEGORY         1525 non-null   object
 8   OUTAGE.START.DATE        1525 non-null   object
 9   OUTAGE.START.TIME        1525 non-null   object
 10  OUTAGE.RESTORATION.DATE  1476 non-null   object
 11  OUTAGE.RESTORATION.TIME  1476 non-null   object
 12  CAUSE.CATEGORY           1534 non-null   object
 13  CAUSE.CATEGORY.DETAIL    1063 non-null   object
 14  HURRICANE.NAMES          72 non-null     object
 15  OUTAGE.DURATION          1476 non-null   object
 16  DEMAND.LOSS.MW           829 non-null    object
 17  CUSTOMERS.AFFECTED       1091 non-null   float64
 18  RES.PRICE                1512 non-null   object
 19  COM.PRICE                1512 non-null   object
 20  IND.PRICE                1512 non-null   object
 21  TOTAL.PRICE              1512 non-null   object
 22  RES.SALES                1512 non-null   object
 23  COM.SALES                1512 non-null   object
 24  IND.SALES                1512 non-null   object
 25  TOTAL.SALES              1512 non-null   object
 26  RES.PERCEN               1512 non-null   object
 27  COM.PERCEN               1512 non-null   object
 28  IND.PERCEN               1512 non-null   object
 29  RES.CUSTOMERS            1534 non-null   float64
 30  COM.CUSTOMERS            1534 non-null   float64
 31  IND.CUSTOMERS            1534 non-null   float64
 32  TOTAL.CUSTOMERS          1534 non-null   float64
 33  RES.CUST.PCT             1534 non-null   object
 34  COM.CUST.PCT             1534 non-null   object
 35  IND.CUST.PCT             1534 non-null   object
```

```
 36   PC.REALGSP.STATE         1534 non-null   object
 37   PC.REALGSP.USA           1534 non-null   object
 38   PC.REALGSP.REL           1534 non-null   object
 39   PC.REALGSP.CHANGE        1534 non-null   object
 40   UTIL.REALGSP             1534 non-null   object
 41   TOTAL.REALGSP            1534 non-null   object
 42   UTIL.CONTRI              1534 non-null   object
 43   PI.UTIL.OFUSA            1534 non-null   object
 44   POPULATION               1534 non-null   float64
 45   POPPCT_URBAN             1534 non-null   object
 46   POPPCT_UC                1534 non-null   object
 47   POPDEN_URBAN             1534 non-null   object
 48   POPDEN_UC                1524 non-null   object
 49   POPDEN_RURAL             1524 non-null   object
 50   AREAPCT_URBAN            1534 non-null   object
 51   AREAPCT_UC               1534 non-null   object
 52   PCT_LAND                 1534 non-null   object
 53   PCT_WATER_TOT            1534 non-null   object
 54   PCT_WATER_INLAND         1534 non-null   object
dtypes: float64(8), object(47)
memory usage: 659.3+ KB
None

Columns: ['YEAR', 'MONTH', 'U.S._STATE', 'POSTAL.CODE', 'NERC.REGION',
'CLIMATE.REGION', 'ANOMALY.LEVEL', 'CLIMATE.CATEGORY', 'OUTAGE.START.DATE',
'OUTAGE.START.TIME', 'OUTAGE.RESTORATION.DATE', 'OUTAGE.RESTORATION.TIME',
'CAUSE.CATEGORY', 'CAUSE.CATEGORY.DETAIL', 'HURRICANE.NAMES', 'OUTAGE.DURATION',
'DEMAND.LOSS.MW', 'CUSTOMERS.AFFECTED', 'RES.PRICE', 'COM.PRICE', 'IND.PRICE',
'TOTAL.PRICE', 'RES.SALES', 'COM.SALES', 'IND.SALES', 'TOTAL.SALES',
'RES.PERCEN', 'COM.PERCEN', 'IND.PERCEN', 'RES.CUSTOMERS', 'COM.CUSTOMERS',
'IND.CUSTOMERS', 'TOTAL.CUSTOMERS', 'RES.CUST.PCT', 'COM.CUST.PCT',
'IND.CUST.PCT', 'PC.REALGSP.STATE', 'PC.REALGSP.USA', 'PC.REALGSP.REL',
'PC.REALGSP.CHANGE', 'UTIL.REALGSP', 'TOTAL.REALGSP', 'UTIL.CONTRI',
'PI.UTIL.OFUSA', 'POPULATION', 'POPPCT_URBAN', 'POPPCT_UC', 'POPDEN_URBAN',
'POPDEN_UC', 'POPDEN_RURAL', 'AREAPCT_URBAN', 'AREAPCT_UC', 'PCT_LAND',
'PCT_WATER_TOT', 'PCT_WATER_INLAND']

Shape: (1534, 55)
```

1.2 Step 2: Data Cleaning and Exploratory Data Analysis

[102]:
```python
def fix_data_types(df):
    '''
    Fixes data types of columns in the DataFrame based on expected types.
    '''
    datetime_cols = [
        'OUTAGE.START.DATE',
```

```
        #'OUTAGE.START.TIME',       # datetime.time object
        'OUTAGE.RESTORATION.DATE',
        #'OUTAGE.RESTORATION.TIME'  # datetime.time object
        ]
    int_cols = [
        'YEAR', 'MONTH', 'OUTAGE.DURATION', 'DEMAND.LOSS.MW', 'CUSTOMERS.
↪AFFECTED', 'POPULATION'
        ]
    float_cols = [
        'ANOMALY.LEVEL', 'RES.PRICE', 'COM.PRICE', 'IND.PRICE', 'TOTAL.PRICE',␣
↪'RES.SALES', 'COM.SALES', 'IND.SALES',
        'TOTAL.SALES', 'RES.PERCEN', 'COM.PERCEN', 'IND.PERCEN', 'RES.
↪CUSTOMERS', 'COM.CUSTOMERS', 'IND.CUSTOMERS',
        'TOTAL.CUSTOMERS', 'RES.CUST.PCT', 'COM.CUST.PCT', 'IND.CUST.PCT', 'PC.
↪REALGSP.STATE', 'PC.REALGSP.USA',
        'PC.REALGSP.REL', 'PC.REALGSP.CHANGE', 'UTIL.REALGSP', 'TOTAL.REALGSP',␣
↪'UTIL.CONTRI', 'PI.UTIL.OFUSA',
        'POPPCT_URBAN', 'POPPCT_UC', 'POPDEN_URBAN', 'POPDEN_UC',␣
↪'POPDEN_RURAL', 'AREAPCT_URBAN', 'AREAPCT_UC', 'PCT_LAND',
        'PCT_WATER_TOT', 'PCT_WATER_INLAND'
        ]
    # cat_cols = [
    #     'U.S._STATE', 'POSTAL.CODE', 'NERC.REGION', 'CLIMATE.REGION',␣
↪'CLIMATE.CATEGORY', 'CAUSE.CATEGORY',
    #     'CAUSE.CATEGORY.DETAIL'
    # ]
    # HURRICANE.NAMES will be treated as object automatically

    for col in datetime_cols:
        df[col] = pd.to_datetime(df[col], errors='coerce')
    for col in int_cols:
        df[col] = pd.to_numeric(df[col], errors='coerce').astype('Int64')
    for col in float_cols:
        df[col] = pd.to_numeric(df[col], errors='coerce').astype('float64')
    # for col in cat_cols:
    #     df[col] = df[col].astype('category')
    return df
```

[103]:
```
# Columns with Missing Values
missing_values = df.isnull().sum()
print("\nMissing Values per Column:")
print(missing_values[missing_values > 0].sort_values(ascending=False))
```

```
Missing Values per Column:
HURRICANE.NAMES            1462
DEMAND.LOSS.MW              705
```

```
CAUSE.CATEGORY.DETAIL       471
CUSTOMERS.AFFECTED          443
OUTAGE.RESTORATION.DATE      58
OUTAGE.RESTORATION.TIME      58
OUTAGE.DURATION              58
RES.SALES                    22
TOTAL.PRICE                  22
IND.PRICE                    22
COM.PRICE                    22
TOTAL.SALES                  22
RES.PRICE                    22
IND.SALES                    22
RES.PERCEN                   22
COM.PERCEN                   22
IND.PERCEN                   22
COM.SALES                    22
POPDEN_UC                    10
POPDEN_RURAL                 10
OUTAGE.START.TIME             9
OUTAGE.START.DATE             9
CLIMATE.CATEGORY              9
ANOMALY.LEVEL                 9
MONTH                         9
CLIMATE.REGION                6
dtype: int64
```

[104]:
```
# Missing CLIMATE.REGION values --> Hawaii and Alaska, non-continental states
missing_climate_region = df[df['CLIMATE.REGION'].isnull()]
missing_climate_region
```

[104]:
```
        YEAR  MONTH U.S._STATE POSTAL.CODE NERC.REGION CLIMATE.REGION  \
1516  2008.0   12.0     Hawaii          HI          HI            NaN
1517  2011.0    5.0     Hawaii          HI          PR            NaN
1518  2006.0   10.0     Hawaii          HI        HECO            NaN
1519  2006.0    6.0     Hawaii          HI        HECO            NaN
1520  2006.0   10.0     Hawaii          HI        HECO            NaN
1534  2000.0    NaN     Alaska          AK        ASCC            NaN

     ANOMALY.LEVEL CLIMATE.CATEGORY    OUTAGE.START.DATE OUTAGE.START.TIME  \
1516          -0.7             cold  2008-12-26 00:00:00          18:13:00
1517          -0.4           normal  2011-05-02 00:00:00          17:06:00
1518           0.7             warm  2006-10-15 00:00:00          07:09:00
1519             0           normal  2006-06-01 00:00:00          14:12:00
1520           0.7             warm  2006-10-15 00:00:00          07:09:00
1534           NaN              NaN                  NaN               NaN

     OUTAGE.RESTORATION.DATE OUTAGE.RESTORATION.TIME  \
```

1516	2008-12-27 00:00:00	17:00:00
1517	2011-05-02 00:00:00	20:00:00
1518	2006-10-15 00:00:00	16:12:00
1519	2006-06-01 00:00:00	18:09:00
1520	2006-10-16 00:00:00	14:55:00
1534	NaN	NaN

	CAUSE.CATEGORY	CAUSE.CATEGORY.DETAIL	HURRICANE.NAMES \
1516	severe weather	thunderstorm	NaN
1517	severe weather	NaN	NaN
1518	severe weather	earthquake	NaN
1519	system operability disruption	NaN	NaN
1520	severe weather	earthquake	NaN
1534	equipment failure	failure	NaN

	OUTAGE.DURATION	DEMAND.LOSS.MW	CUSTOMERS.AFFECTED	RES.PRICE	COM.PRICE \
1516	1367	1060	294000.0	29.28	26.28
1517	174	220	62000.0	34.58	32.14
1518	543	110	59886.0	23.25	21.26
1519	237	120	29300.0	24.09	22.06
1520	1906	1170	291000.0	23.25	21.26
1534	NaN	35	14273.0	NaN	NaN

	IND.PRICE	TOTAL.PRICE	RES.SALES	COM.SALES	IND.SALES	TOTAL.SALES \
1516	22.43	25.78	253625	275383	306397	835405
1517	27.85	31.29	249369	292304	310682	852355
1518	17.71	20.54	274609	307570	340735	922915
1519	18.74	21.45	265474	298487	327618	891580
1520	17.71	20.54	274609	307570	340735	922915
1534	NaN	NaN	NaN	NaN	NaN	NaN

	RES.PERCEN	COM.PERCEN	IND.PERCEN	RES.CUSTOMERS	COM.CUSTOMERS \
1516	30.36	32.96	36.68	409668.0	61684.0
1517	29.26	34.29	36.45	417531.0	60043.0
1518	29.75	33.33	36.92	401592.0	61334.0
1519	29.78	33.48	36.75	401592.0	61334.0
1520	29.75	33.33	36.92	401592.0	61334.0
1534	NaN	NaN	NaN	230534.0	38074.0

	IND.CUSTOMERS	TOTAL.CUSTOMERS	RES.CUST.PCT	COM.CUST.PCT	IND.CUST.PCT \
1516	673.0	472025.0	86.79	13.07	0.14
1517	698.0	478272.0	87.3	12.55	0.15
1518	689.0	463615.0	86.62	13.23	0.15
1519	689.0	463615.0	86.62	13.23	0.15
1520	689.0	463615.0	86.62	13.23	0.15
1534	854.0	273530.0	84.28	13.92	0.31

```
     PC.REALGSP.STATE PC.REALGSP.USA PC.REALGSP.REL PC.REALGSP.CHANGE  \
1516            50565          48401           1.04              -0.7
1517            49110          47586           1.03               0.4
1518            50358          48909           1.03               1.1
1519            50358          48909           1.03               1.1
1520            50358          48909           1.03               1.1
1534            57401          44745           1.28              -2.2

     UTIL.REALGSP TOTAL.REALGSP UTIL.CONTRI PI.UTIL.OFUSA  POPULATION  \
1516         1646         67364        2.44           0.5    1.33e+06
1517         1935         67686        2.86           0.6    1.38e+06
1518         1436         65956        2.18           0.5    1.31e+06
1519         1436         65956        2.18           0.5    1.31e+06
1520         1436         65956        2.18           0.5    1.31e+06
1534          724         36046        2.01           0.2    6.28e+05

     POPPCT_URBAN POPPCT_UC POPDEN_URBAN POPDEN_UC POPDEN_RURAL AREAPCT_URBAN  \
1516        91.93     20.47       3180.8    1664.7         18.2          6.12
1517        91.93     20.47       3180.8    1664.7         18.2          6.12
1518        91.93     20.47       3180.8    1664.7         18.2          6.12
1519        91.93     20.47       3180.8    1664.7         18.2          6.12
1520        91.93     20.47       3180.8    1664.7         18.2          6.12
1534        66.02     21.56       1802.6      1276          0.4          0.05

     AREAPCT_UC PCT_LAND PCT_WATER_TOT PCT_WATER_INLAND
1516        2.6    58.75         41.25             0.38
1517        2.6    58.75         41.25             0.38
1518        2.6    58.75         41.25             0.38
1519        2.6    58.75         41.25             0.38
1520        2.6    58.75         41.25             0.38
1534       0.02    85.76         14.24              2.9
```

[105]:
```python
# Missing POPDEN_UC and POPDEN_RURAL values --> District of Columbia
missing_popden_uc_rural = df[df['POPDEN_UC'].isnull() | df['POPDEN_RURAL'].
 ↪isnull()]
missing_popden_uc_rural
```

[105]:
```
       YEAR  MONTH            U.S._STATE POSTAL.CODE NERC.REGION  \
841  2010.0    2.0  District of Columbia          DC         RFC
842  2011.0    8.0  District of Columbia          DC         RFC
843  2010.0    8.0  District of Columbia          DC         RFC
844  2003.0    9.0  District of Columbia          DC         RFC
845  2011.0    1.0  District of Columbia          DC         RFC
846  2013.0    2.0  District of Columbia          DC         RFC
847  2014.0    1.0  District of Columbia          DC         RFC
848  2013.0    6.0  District of Columbia          DC         RFC
849  2012.0    6.0  District of Columbia          DC         RFC
```

850	2010.0	8.0	District of Columbia	DC	RFC

	CLIMATE.REGION	ANOMALY.LEVEL	CLIMATE.CATEGORY	OUTAGE.START.DATE	\
841	Northeast	1.2	warm	2010-02-05 00:00:00	
842	Northeast	-0.6	cold	2011-08-27 00:00:00	
843	Northeast	-1.2	cold	2010-08-05 00:00:00	
844	Northeast	0.2	normal	2003-09-18 00:00:00	
845	Northeast	-1.3	cold	2011-01-26 00:00:00	
846	Northeast	-0.4	normal	2013-02-08 00:00:00	
847	Northeast	-0.5	cold	2014-01-06 00:00:00	
848	Northeast	-0.2	normal	2013-06-13 00:00:00	
849	Northeast	-0.1	normal	2012-06-29 00:00:00	
850	Northeast	-1.2	cold	2010-08-12 00:00:00	

	OUTAGE.START.TIME	OUTAGE.RESTORATION.DATE	OUTAGE.RESTORATION.TIME	\
841	19:00:00	2010-02-12 00:00:00	15:46:00	
842	23:05:00	2011-08-29 00:00:00	15:30:00	
843	15:30:00	2010-08-05 00:00:00	22:00:00	
844	16:20:00	2003-09-28 00:00:00	18:00:00	
845	17:00:00	2011-01-31 00:00:00	08:00:00	
846	11:38:00	2013-02-08 00:00:00	14:17:00	
847	19:50:00	2014-01-06 00:00:00	20:44:00	
848	15:30:00	2013-06-13 00:00:00	16:00:00	
849	22:15:00	2012-07-05 00:00:00	12:52:00	
850	06:45:00	2010-08-12 00:00:00	21:00:00	

	CAUSE.CATEGORY	CAUSE.CATEGORY.DETAIL	HURRICANE.NAMES	OUTAGE.DURATION	\
841	severe weather	winter storm	NaN	9886	
842	severe weather	NaN	NaN	2425	
843	severe weather	thunderstorm	NaN	390	
844	severe weather	hurricanes	Isabel	14500	
845	severe weather	winter storm	NaN	6660	
846	equipment failure	generator trip	NaN	159	
847	severe weather	winter	NaN	54	
848	severe weather	uncontrolled loss	NaN	30	
849	severe weather	thunderstorm	NaN	8077	
850	severe weather	NaN	NaN	855	

	DEMAND.LOSS.MW	CUSTOMERS.AFFECTED	RES.PRICE	COM.PRICE	IND.PRICE	\
841	NaN	97651.0	13.41	13.8	7.11	
842	NaN	220000.0	13.06	12.77	8.29	
843	NaN	76729.0	14.76	13.31	8.13	
844	NaN	530000.0	8.36	8.61	5.98	
845	NaN	210000.0	13.62	13.17	7.48	
846	140	52000.0	11.97	11.88	5.4	
847	NaN	NaN	12.59	12.67	8.66	
848	700	40000.0	12.65	11.77	5.33	

849	3000	425000.0	13.18	12.03	5.71
850	NaN	101003.0	14.76	13.31	8.13

	TOTAL.PRICE	RES.SALES	COM.SALES	IND.SALES	TOTAL.SALES	RES.PERCEN	\
841	13.55	181006	682695	18213	902386	20.06	
842	12.69	239882	801888	16927	1084951	22.11	
843	13.47	236540	883096	18171	1167163	20.27	
844	8.46	152325	721287	21491	920981	16.54	
845	13.13	229985	744163	19395	1018592	22.58	
846	11.73	180750	627931	15521	846772	21.35	
847	12.49	218682	736898	20478	1004907	21.76	
848	11.72	175324	789494	20375	1015794	17.26	
849	12.03	179048	772663	18518	999646	17.91	
850	13.47	236540	883096	18171	1167163	20.27	

	COM.PERCEN	IND.PERCEN	RES.CUSTOMERS	COM.CUSTOMERS	IND.CUSTOMERS	\
841	75.65	2.02	227550.0	26449.0	1.0	
842	73.91	1.56	229450.0	26496.0	1.0	
843	75.66	1.56	227550.0	26449.0	1.0	
844	78.32	2.33	199215.0	26283.0	1.0	
845	73.06	1.9	229450.0	26496.0	1.0	
846	74.16	1.83	235322.0	26530.0	1.0	
847	73.33	2.04	239355.0	26463.0	1.0	
848	77.72	2.01	235322.0	26530.0	1.0	
849	77.29	1.85	231550.0	26547.0	1.0	
850	75.66	1.56	227550.0	26449.0	1.0	

	TOTAL.CUSTOMERS	RES.CUST.PCT	COM.CUST.PCT	IND.CUST.PCT	PC.REALGSP.STATE	\
841	254001.0	89.59	10.41	0.0	168377	
842	255948.0	89.65	10.35	0.0	167337	
843	254001.0	89.59	10.41	0.0	168377	
844	225500.0	88.34	11.66	0.0	155044	
845	255948.0	89.65	10.35	0.0	167337	
846	261854.0	89.87	10.13	0.0	159340	
847	265820.0	90.04	9.96	0.0	159831	
848	261854.0	89.87	10.13	0.0	159340	
849	258099.0	89.71	10.29	0.0	163461	
850	254001.0	89.59	10.41	0.0	168377	

	PC.REALGSP.USA	PC.REALGSP.REL	PC.REALGSP.CHANGE	UTIL.REALGSP	\
841	47287	3.56	0.8	1416	
842	47586	3.52	-0.6	1300	
843	47287	3.56	0.8	1416	
844	45858	3.38	3.3	1115	
845	47586	3.52	-0.6	1300	
846	48396	3.29	-2.5	1165	
847	49091	3.26	0.3	1107	

```
848         48396         3.29            -2.5        1165
849         48156         3.39            -2.3        1071
850         47287         3.56             0.8        1416

    TOTAL.REALGSP UTIL.CONTRI PI.UTIL.OFUSA  POPULATION POPPCT_URBAN  \
841        101904        1.39           0.3    605126.0          100
842        103820        1.25           0.3    620472.0          100
843        101904        1.39           0.3    605126.0          100
844         88143        1.26           0.4    568502.0          100
845        103820        1.25           0.3    620472.0          100
846        103430        1.13           0.3    649540.0          100
847        105312        1.05           0.3    659836.0          100
848        103430        1.13           0.3    649540.0          100
849        103804        1.03           0.3    635342.0          100
850        101904        1.39           0.3    605126.0          100

    POPPCT_UC POPDEN_URBAN POPDEN_UC POPDEN_RURAL AREAPCT_URBAN AREAPCT_UC  \
841         0       9856.5       NaN          NaN           100          0
842         0       9856.5       NaN          NaN           100          0
843         0       9856.5       NaN          NaN           100          0
844         0       9856.5       NaN          NaN           100          0
845         0       9856.5       NaN          NaN           100          0
846         0       9856.5       NaN          NaN           100          0
847         0       9856.5       NaN          NaN           100          0
848         0       9856.5       NaN          NaN           100          0
849         0       9856.5       NaN          NaN           100          0
850         0       9856.5       NaN          NaN           100          0

    PCT_LAND PCT_WATER_TOT PCT_WATER_INLAND
841    89.71         10.29            10.29
842    89.71         10.29            10.29
843    89.71         10.29            10.29
844    89.71         10.29            10.29
845    89.71         10.29            10.29
846    89.71         10.29            10.29
847    89.71         10.29            10.29
848    89.71         10.29            10.29
849    89.71         10.29            10.29
850    89.71         10.29            10.29
```

[106]:
```
# Show duplicates
df_duplicates = df[df.duplicated(keep=False)]
df_duplicates
```

[106]:
```
        YEAR  MONTH  U.S._STATE POSTAL.CODE NERC.REGION CLIMATE.REGION  \
25    2014.0    1.0   Tennessee          TN        SERC        Central
44    2014.0    1.0   Tennessee          TN        SERC        Central
```

```
220   2013.0   12.0       Texas         TX       TRE        South
274   2013.0   12.0       Texas         TX       TRE        South
500   2004.0    7.0     Arizona         AZ      WECC    Southwest
514   2004.0    7.0     Arizona         AZ      WECC    Southwest
994   2010.0    8.0   Louisiana         LA      SERC        South
998   2010.0    8.0   Louisiana         LA      SERC        South
1000  2010.0    8.0   Louisiana         LA      SERC        South
1042  2004.0    5.0     Florida         FL      FRCC    Southeast
1043  2004.0    5.0     Florida         FL      FRCC    Southeast
1049  2004.0    5.0     Florida         FL      FRCC    Southeast
1194  2010.0    7.0  California         CA      WECC         West
1240  2010.0    7.0  California         CA      WECC         West

     ANOMALY.LEVEL CLIMATE.CATEGORY    OUTAGE.START.DATE OUTAGE.START.TIME  \
25            -0.5             cold  2014-01-07 00:00:00          06:00:00
44            -0.5             cold  2014-01-07 00:00:00          06:00:00
220           -0.3           normal  2013-12-13 00:00:00          11:00:00
274           -0.3           normal  2013-12-13 00:00:00          11:00:00
500            0.5             warm  2004-07-06 00:00:00          06:00:00
514            0.5             warm  2004-07-06 00:00:00          06:00:00
994           -1.2             cold  2010-08-02 00:00:00          12:45:00
998           -1.2             cold  2010-08-02 00:00:00          12:45:00
1000          -1.2             cold  2010-08-02 00:00:00          12:45:00
1042           0.2           normal  2004-05-28 00:00:00          12:00:00
1043           0.2           normal  2004-05-28 00:00:00          12:00:00
1049           0.2           normal  2004-05-28 00:00:00          12:00:00
1194          -0.9             cold  2010-07-29 00:00:00          18:39:00
1240          -0.9             cold  2010-07-29 00:00:00          18:39:00

     OUTAGE.RESTORATION.DATE OUTAGE.RESTORATION.TIME         CAUSE.CATEGORY  \
25       2014-01-07 00:00:00                08:30:00         severe weather
44       2014-01-07 00:00:00                08:30:00         severe weather
220      2013-12-27 00:00:00                11:00:00  fuel supply emergency
274      2013-12-27 00:00:00                11:00:00  fuel supply emergency
500      2004-08-09 00:00:00                12:00:00         severe weather
514      2004-08-09 00:00:00                12:00:00         severe weather
994      2010-08-04 00:00:00                11:00:00          public appeal
998      2010-08-04 00:00:00                11:00:00          public appeal
1000     2010-08-04 00:00:00                11:00:00          public appeal
1042     2004-05-31 00:00:00                12:00:00          public appeal
1043     2004-05-31 00:00:00                12:00:00          public appeal
1049     2004-05-31 00:00:00                12:00:00          public appeal
1194     2010-07-29 00:00:00                19:26:00         severe weather
1240     2010-07-29 00:00:00                19:26:00         severe weather

     CAUSE.CATEGORY.DETAIL HURRICANE.NAMES OUTAGE.DURATION DEMAND.LOSS.MW  \
25                  winter             NaN             150            NaN
```

```
44                winter            NaN            150            NaN
220                 Coal            NaN          20160            NaN
274                 Coal            NaN          20160            NaN
500             wildfire            NaN          49320            NaN
514             wildfire            NaN          49320            NaN
994                  NaN            NaN           2775            NaN
998                  NaN            NaN           2775            NaN
1000                 NaN            NaN           2775            NaN
1042                 NaN            NaN           4320              0
1043                 NaN            NaN           4320              0
1049                 NaN            NaN           4320              0
1194            wildfire            NaN             47            522
1240            wildfire            NaN             47            522

      CUSTOMERS.AFFECTED RES.PRICE COM.PRICE IND.PRICE TOTAL.PRICE RES.SALES  \
25                   NaN      9.73     10.02      6.27        9.13   4795011
44                   NaN      9.73     10.02      6.27        9.13   4795011
220                  NaN     11.27         8      5.79        8.67  11901122
274                  NaN     11.27         8      5.79        8.67  11901122
500                  NaN      8.88      7.57      5.55        7.91   3466457
514                  NaN      8.88      7.57      5.55        7.91   3466457
994                  NaN      9.29       8.5      5.99        8.13   3695516
998                  NaN      9.29       8.5      5.99        8.13   3695516
1000                 NaN      9.29       8.5      5.99        8.13   3695516
1042                 0.0      8.95      7.59      5.84        8.07   8238817
1043                 0.0      8.95      7.59      5.84        8.07   8238817
1049                 0.0      8.95      7.59      5.84        8.07   8238817
1194                 NaN     15.02     15.27     11.11       14.39   8105267
1240                 NaN     15.02     15.27     11.11       14.39   8105267

     COM.SALES IND.SALES TOTAL.SALES RES.PERCEN COM.PERCEN IND.PERCEN  \
25     2841462   1912538     9549176      50.21      29.76      20.03
44     2841462   1912538     9549176      50.21      29.76      20.03
220   10736864   8269474    30908046       38.5      34.74      26.76
274   10736864   8269474    30908046       38.5      34.74      26.76
500    2563737   1048662     7078856      48.97      36.22      14.81
514    2563737   1048662     7078856      48.97      36.22      14.81
994    2407508   2401355     8505369      43.45      28.31      28.23
998    2407508   2401355     8505369      43.45      28.31      28.23
1000   2407508   2401355     8505369      43.45      28.31      28.23
1042   7132397   1728226    17107025      48.16      41.69       10.1
1043   7132397   1728226    17107025      48.16      41.69       10.1
1049   7132397   1728226    17107025      48.16      41.69       10.1
1194  11041978   4398786    23616311      34.32      46.76      18.63
1240  11041978   4398786    23616311      34.32      46.76      18.63

      RES.CUSTOMERS  COM.CUSTOMERS  IND.CUSTOMERS  TOTAL.CUSTOMERS  \
```

25	2.76e+06	4.72e+05	1271.0	3.23e+06
44	2.76e+06	4.72e+05	1271.0	3.23e+06
220	9.95e+06	1.37e+06	95881.0	1.14e+07
274	9.95e+06	1.37e+06	95881.0	1.14e+07
500	2.26e+06	2.59e+05	7419.0	2.52e+06
514	2.26e+06	2.59e+05	7419.0	2.52e+06
994	1.97e+06	2.74e+05	18128.0	2.27e+06
998	1.97e+06	2.74e+05	18128.0	2.27e+06
1000	1.97e+06	2.74e+05	18128.0	2.27e+06
1042	7.89e+06	1.05e+06	27417.0	8.96e+06
1043	7.89e+06	1.05e+06	27417.0	8.96e+06
1049	7.89e+06	1.05e+06	27417.0	8.96e+06
1194	1.29e+07	1.81e+06	75160.0	1.48e+07
1240	1.29e+07	1.81e+06	75160.0	1.48e+07

	RES.CUST.PCT	COM.CUST.PCT	IND.CUST.PCT	PC.REALGSP.STATE	PC.REALGSP.USA \
25	85.35	14.61	0.04	41955	49091
44	85.35	14.61	0.04	41955	49091
220	87.19	11.97	0.84	51695	48396
274	87.19	11.97	0.84	51695	48396
500	89.44	10.26	0.29	41169	47037
514	89.44	10.26	0.29	41169	47037
994	87.1	12.1	0.8	48305	47287
998	87.1	12.1	0.8	48305	47287
1000	87.1	12.1	0.8	48305	47287
1042	88.0	11.69	0.31	41618	47037
1043	88.0	11.69	0.31	41618	47037
1049	88.0	11.69	0.31	41618	47037
1194	87.26	12.23	0.51	51733	47287
1240	87.26	12.23	0.51	51733	47287

	PC.REALGSP.REL	PC.REALGSP.CHANGE	UTIL.REALGSP	TOTAL.REALGSP	UTIL.CONTRI \
25	0.85	0.8	1739	274775	0.63
44	0.85	0.8	1739	274775	0.63
220	1.07	2.7	28880	1370216	2.11
274	1.07	2.7	28880	1370216	2.11
500	0.88	1	5113	232702	2.2
514	0.88	1	5113	232702	2.2
994	1.02	3.6	4509	219576	2.05
998	1.02	3.6	4509	219576	2.05
1000	1.02	3.6	4509	219576	2.05
1042	0.88	3.1	15193	724792	2.1
1043	0.88	3.1	15193	724792	2.1
1049	0.88	3.1	15193	724792	2.1
1194	1.09	-0.1	28360	1931514	1.47
1240	1.09	-0.1	28360	1931514	1.47

	PI.UTIL.OFUSA	POPULATION	POPPCT_URBAN	POPPCT_UC	POPDEN_URBAN	POPDEN_UC	\
25	0.4	6.55e+06	66.39	12.02	1450.3	1076.2	
44	0.4	6.55e+06	66.39	12.02	1450.3	1076.2	
220	10.5	2.65e+07	84.7	9.35	2435.3	1539.9	
274	10.5	2.65e+07	84.7	9.35	2435.3	1539.9	
500	1.7	5.65e+06	89.81	9.74	2625.4	1669	
514	1.7	5.65e+06	89.81	9.74	2625.4	1669	
994	1.4	4.54e+06	73.19	11.85	1685.8	1280.1	
998	1.4	4.54e+06	73.19	11.85	1685.8	1280.1	
1000	1.4	4.54e+06	73.19	11.85	1685.8	1280.1	
1042	3.9	1.74e+07	91.16	3.72	2315.2	1230.3	
1043	3.9	1.74e+07	91.16	3.72	2315.2	1230.3	
1049	3.9	1.74e+07	91.16	3.72	2315.2	1230.3	
1194	11.5	3.73e+07	94.95	5.22	4303.7	2124.1	
1240	11.5	3.73e+07	94.95	5.22	4303.7	2124.1	

	POPDEN_RURAL	AREAPCT_URBAN	AREAPCT_UC	PCT_LAND	PCT_WATER_TOT	\
25	55.6	7.05	1.72	97.84	2.16	
44	55.6	7.05	1.72	97.84	2.16	
220	15.2	3.35	0.58	97.26	2.74	
274	15.2	3.35	0.58	97.26	2.74	
500	5.8	1.92	0.33	99.65	0.35	
514	5.8	1.92	0.33	99.65	0.35	
994	29.5	4.56	0.97	82.49	17.51	
998	29.5	4.56	0.97	82.49	17.51	
1000	29.5	4.56	0.97	82.49	17.51	
1042	35.9	13.81	1.06	81.55	18.45	
1043	35.9	13.81	1.06	81.55	18.45	
1049	35.9	13.81	1.06	81.55	18.45	
1194	12.7	5.28	0.59	95.16	4.84	
1240	12.7	5.28	0.59	95.16	4.84	

	PCT_WATER_INLAND
25	2.16
44	2.16
220	2.09
274	2.09
500	0.35
514	0.35
994	8.71
998	8.71
1000	8.71
1042	7.64
1043	7.64
1049	7.64
1194	1.73
1240	1.73

[107]:
```
df[df['OUTAGE.DURATION'] > 30000]
```

[107]:
```
        YEAR  MONTH  U.S._STATE POSTAL.CODE NERC.REGION      CLIMATE.REGION  \
54    2014.0    1.0   Wisconsin          WI         RFC  East North Central
118   2003.0    5.0    Michigan          MI         RFC  East North Central
121   2009.0    3.0    Michigan          MI         RFC  East North Central
500   2004.0    7.0     Arizona          AZ        WECC           Southwest
514   2004.0    7.0     Arizona          AZ        WECC           Southwest
990   2014.0    2.0    New York          NY        NPCC           Northeast
1182  2003.0   10.0  California          CA        WECC                West
1208  2016.0    3.0  California          CA        WECC                West

     ANOMALY.LEVEL CLIMATE.CATEGORY    OUTAGE.START.DATE OUTAGE.START.TIME  \
54            -0.5             cold  2014-01-24 00:00:00          00:00:00
118           -0.2           normal  2003-05-15 00:00:00          14:00:00
121           -0.4           normal  2009-03-03 00:00:00          06:48:00
500            0.5             warm  2004-07-06 00:00:00          06:00:00
514            0.5             warm  2004-07-06 00:00:00          06:00:00
990           -0.5             cold  2014-02-07 00:00:00          07:00:00
1182           0.3           normal  2003-10-26 00:00:00          01:44:00
1208           1.6             warm  2016-03-03 00:00:00          11:00:00

     OUTAGE.RESTORATION.DATE OUTAGE.RESTORATION.TIME        CAUSE.CATEGORY  \
54       2014-04-09 00:00:00                11:53:00  fuel supply emergency
118      2003-06-16 00:00:00                14:00:00        severe weather
121      2009-04-26 00:00:00                18:05:00     equipment failure
500      2004-08-09 00:00:00                12:00:00        severe weather
514      2004-08-09 00:00:00                12:00:00        severe weather
990      2014-03-21 00:00:00                08:00:00  fuel supply emergency
1182     2003-11-18 00:00:00                22:54:00        severe weather
1208     2016-04-06 00:00:00                19:47:00  fuel supply emergency

     CAUSE.CATEGORY.DETAIL HURRICANE.NAMES OUTAGE.DURATION DEMAND.LOSS.MW  \
54                    Coal             NaN          108653            NaN
118               flooding             NaN           46080            240
121     transformer outage             NaN           78377            378
500               wildfire             NaN           49320            NaN
514               wildfire             NaN           49320            NaN
990                   Coal             NaN           60480            675
1182              wildfire             NaN           34390            NaN
1208                   NaN             NaN           49427              0

      CUSTOMERS.AFFECTED RES.PRICE COM.PRICE IND.PRICE TOTAL.PRICE RES.SALES  \
54                  NaN     12.86     10.24      7.16       10.28   2393637
118                 2.0      8.23      7.45      4.95        6.67   2244807
121                 NaN     10.66      8.78      6.63        8.84   2825441
500                 NaN      8.88      7.57      5.55        7.91   3466457
```

514	NaN	8.88	7.57	5.55	7.91	3466457
990	NaN	21.69	17.48	8.19	17.81	4550076
1182	108000.0	10.47	11.64	9.17	10.72	7096893
1208	0.0	17.66	13.83	10.39	14.34	6448036

	COM.SALES	IND.SALES	TOTAL.SALES	RES.PERCEN	COM.PERCEN	IND.PERCEN	\
54	2089309	1959985	6442931	37.15	32.43	30.42	
118	2789455	3273564	8308074	27.02	33.58	39.4	
121	3019167	2251830	8096999	34.89	37.29	27.81	
500	2563737	1048662	7078856	48.97	36.22	14.81	
514	2563737	1048662	7078856	48.97	36.22	14.81	
990	6390265	1497619	12699456	35.83	50.32	11.79	
1182	10217148	4590049	21974546	32.3	46.5	20.89	
1208	9366373	4149987	20033300	32.19	46.75	20.72	

	RES.CUSTOMERS	COM.CUSTOMERS	IND.CUSTOMERS	TOTAL.CUSTOMERS	\
54	2.63e+06	3.46e+05	5465.0	2.98e+06	
118	4.22e+06	4.83e+05	14224.0	4.71e+06	
121	4.25e+06	5.21e+05	13065.0	4.79e+06	
500	2.26e+06	2.59e+05	7419.0	2.52e+06	
514	2.26e+06	2.59e+05	7419.0	2.52e+06	
990	7.05e+06	1.05e+06	7862.0	8.10e+06	
1182	1.21e+07	1.77e+06	82579.0	1.40e+07	
1208	1.34e+07	1.69e+06	148549.0	1.53e+07	

	RES.CUST.PCT	COM.CUST.PCT	IND.CUST.PCT	PC.REALGSP.STATE	PC.REALGSP.USA	\
54	88.22	11.6	0.18	46676	49091	
118	89.45	10.25	0.3	42768	45858	
121	88.85	10.87	0.27	37004	46680	
500	89.44	10.26	0.29	41169	47037	
514	89.44	10.26	0.29	41169	47037	
990	87.0	12.91	0.1	63217	49091	
1182	86.74	12.66	0.59	49815	45858	
1208	87.96	11.07	0.97	58974	50660	

	PC.REALGSP.REL	PC.REALGSP.CHANGE	UTIL.REALGSP	TOTAL.REALGSP	UTIL.CONTRI	\
54	0.95	1.9	4680	268742	1.74	
118	0.93	1.7	8288	429436	1.93	
121	0.79	-7.7	7769	366401	2.12	
500	0.88	1	5113	232702	2.2	
514	0.88	1	5113	232702	2.2	
990	1.29	1	18048	1248299	1.45	
1182	1.09	2.9	28280	1756127	1.61	
1208	1.16	4.6	27735	2317466	1.2	

	PI.UTIL.OFUSA	POPULATION	POPPCT_URBAN	POPPCT_UC	POPDEN_URBAN	POPDEN_UC	\
54	1.9	5.76e+06	70.15	14.35	2123.3	1671.5	

```
118             3.8    1.00e+07        74.57      8.19       2034.1    1390.4
121             3.7    9.90e+06        74.57      8.19       2034.1    1390.4
500             1.7    5.65e+06        89.81      9.74       2625.4      1669
514             1.7    5.65e+06        89.81      9.74       2625.4      1669
990             7.3    1.97e+07        87.87      5.21       4161.4      1700
1182           10.8    3.53e+07        94.95      5.22       4303.7    2124.1
1208             12    3.93e+07        94.95      5.22       4303.7    2124.1

     POPDEN_RURAL AREAPCT_URBAN AREAPCT_UC PCT_LAND PCT_WATER_TOT  \
54           32.5          3.47        0.9    82.69         17.31
118          47.5          6.41       1.03    58.46         41.54
121          47.5          6.41       1.03    58.46         41.54
500           5.8          1.92       0.33    99.65          0.35
514           5.8          1.92       0.33    99.65          0.35
990          54.6          8.68       1.26    86.38         13.62
1182         12.7          5.28       0.59    95.16          4.84
1208         12.7          5.28       0.59    95.16          4.84

     PCT_WATER_INLAND
54               3.05
118              2.07
121              2.07
500              0.35
514              0.35
990              3.65
1182             1.73
1208             1.73
```

[108]:

```python
# Data Cleaning Functions
def remove_duplicate_rows(df):
    # Remove duplicate rows
    df = df.drop_duplicates()
    return df

def combine_outage_start(df):
    #Combine OUTAGE.START.DATE and OUTAGE.START.TIME into a single column␣
 ↪OUTAGE.START.
    time_as_td = pd.to_timedelta(df['OUTAGE.START.TIME'].astype(str),␣
 ↪errors='coerce')
    df['OUTAGE.START'] = df['OUTAGE.START.DATE'] + time_as_td
    return df

def combine_outage_restoration(df):
    # Combine OUTAGE.RESTORATION.DATE and OUTAGE.RESTORATION.TIME into a single␣
 ↪column OUTAGE.RESTORATION.
    time_as_td = pd.to_timedelta(df['OUTAGE.RESTORATION.TIME'].astype(str),␣
 ↪errors='coerce')
```

```python
    df['OUTAGE.RESTORATION'] = df['OUTAGE.RESTORATION.DATE'] + time_as_td
    return df

def add_month_names(df):
    # Map numeric MONTH to MONTH.NAME.
    MONTH_NAMES = {1:'Jan',2:'Feb',3:'Mar',4:'Apr',5:'May',6:'Jun',
                   7:'Jul',8:'Aug',9:'Sep',10:'Oct',11:'Nov',12:'Dec'}
    df['MONTH.NAME'] = df['MONTH'].map(MONTH_NAMES)
    return df
```

[109]:
```python
# Apply all cleaning functions in a pipeline

df_cleaned = (
    df.pipe(remove_duplicate_rows)
      .pipe(fix_data_types)
      .pipe(combine_outage_start)
      .pipe(combine_outage_restoration)
      .pipe(add_month_names)
)
```

[110]:
```python
df_cleaned.head()
```

[110]:
```
   YEAR  MONTH U.S._STATE POSTAL.CODE NERC.REGION      CLIMATE.REGION  \
1  2011      7  Minnesota          MN         MRO  East North Central
2  2014      5  Minnesota          MN         MRO  East North Central
3  2010     10  Minnesota          MN         MRO  East North Central
4  2012      6  Minnesota          MN         MRO  East North Central
5  2015      7  Minnesota          MN         MRO  East North Central

   ANOMALY.LEVEL CLIMATE.CATEGORY OUTAGE.START.DATE OUTAGE.START.TIME  \
1           -0.3           normal        2011-07-01          17:00:00
2           -0.1           normal        2014-05-11          18:38:00
3           -1.5             cold        2010-10-26          20:00:00
4           -0.1           normal        2012-06-19          04:30:00
5            1.2             warm        2015-07-18          02:00:00

  OUTAGE.RESTORATION.DATE OUTAGE.RESTORATION.TIME      CAUSE.CATEGORY  \
1              2011-07-03                20:00:00      severe weather
2              2014-05-11                18:39:00  intentional attack
3              2010-10-28                22:00:00      severe weather
4              2012-06-20                23:00:00      severe weather
5              2015-07-19                07:00:00      severe weather

  CAUSE.CATEGORY.DETAIL HURRICANE.NAMES  OUTAGE.DURATION  DEMAND.LOSS.MW  \
1                   NaN             NaN             3060            <NA>
2             vandalism             NaN                1            <NA>
3            heavy wind             NaN             3000            <NA>
```

```
4         thunderstorm              NaN            2550            <NA>
5                  NaN              NaN            1740             250

   CUSTOMERS.AFFECTED  RES.PRICE  COM.PRICE  IND.PRICE  TOTAL.PRICE  \
1               70000      11.60       9.18       6.81         9.28
2                <NA>      12.12       9.71       6.49         9.28
3               70000      10.87       8.19       6.07         8.15
4               68200      11.79       9.25       6.71         9.19
5              250000      13.07      10.16       7.74        10.43

   RES.SALES  COM.SALES  IND.SALES  TOTAL.SALES  RES.PERCEN  COM.PERCEN  \
1   2.33e+06   2.11e+06   2.11e+06     6.56e+06       35.55       32.23
2   1.59e+06   1.81e+06   1.89e+06     5.28e+06       30.03       34.21
3   1.47e+06   1.80e+06   1.95e+06     5.22e+06       28.10       34.50
4   1.85e+06   1.94e+06   1.99e+06     5.79e+06       31.99       33.54
5   2.03e+06   2.16e+06   1.78e+06     5.97e+06       33.98       36.21

   IND.PERCEN  RES.CUSTOMERS  COM.CUSTOMERS  IND.CUSTOMERS  TOTAL.CUSTOMERS  \
1       32.20       2.31e+06       276286.0        10673.0         2.60e+06
2       35.73       2.35e+06       284978.0         9898.0         2.64e+06
3       37.37       2.30e+06       276463.0        10150.0         2.59e+06
4       34.44       2.32e+06       278466.0        11010.0         2.61e+06
5       29.78       2.37e+06       289044.0         9812.0         2.67e+06

   RES.CUST.PCT  COM.CUST.PCT  IND.CUST.PCT  PC.REALGSP.STATE  PC.REALGSP.USA  \
1         88.94         10.64          0.41           51268.0         47586.0
2         88.83         10.79          0.37           53499.0         49091.0
3         88.92         10.69          0.39           50447.0         47287.0
4         88.90         10.68          0.42           51598.0         48156.0
5         88.82         10.81          0.37           54431.0         49844.0

   PC.REALGSP.REL  PC.REALGSP.CHANGE  UTIL.REALGSP  TOTAL.REALGSP  \
1            1.08                1.6        4802.0       274182.0
2            1.09                1.9        5226.0       291955.0
3            1.07                2.7        4571.0       267895.0
4            1.07                0.6        5364.0       277627.0
5            1.09                1.7        4873.0       292023.0

   UTIL.CONTRI  PI.UTIL.OFUSA  POPULATION  POPPCT_URBAN  POPPCT_UC  \
1         1.75            2.2     5348119         73.27      15.28
2         1.79            2.2     5457125         73.27      15.28
3         1.71            2.1     5310903         73.27      15.28
4         1.93            2.2     5380443         73.27      15.28
5         1.67            2.2     5489594         73.27      15.28

   POPDEN_URBAN  POPDEN_UC  POPDEN_RURAL  AREAPCT_URBAN  AREAPCT_UC  PCT_LAND  \
1        2279.0     1700.5          18.2           2.14         0.6     91.59
```

```
2        2279.0     1700.5          18.2          2.14        0.6      91.59
3        2279.0     1700.5          18.2          2.14        0.6      91.59
4        2279.0     1700.5          18.2          2.14        0.6      91.59
5        2279.0     1700.5          18.2          2.14        0.6      91.59

   PCT_WATER_TOT  PCT_WATER_INLAND        OUTAGE.START  OUTAGE.RESTORATION  \
1           8.41              5.48 2011-07-01 17:00:00 2011-07-03 20:00:00
2           8.41              5.48 2014-05-11 18:38:00 2014-05-11 18:39:00
3           8.41              5.48 2010-10-26 20:00:00 2010-10-28 22:00:00
4           8.41              5.48 2012-06-19 04:30:00 2012-06-20 23:00:00
5           8.41              5.48 2015-07-18 02:00:00 2015-07-19 07:00:00

  MONTH.NAME
1        Jul
2        May
3        Oct
4        Jun
5        Jul
```

[111]:
```
# Export cleaned dataset to CSV
output_path = 'data/cleaned_outage_dataset.csv'
df_cleaned.to_csv(output_path, index=False)
print(f"Cleaned dataset exported to {output_path}")
```

```
Cleaned dataset exported to data/cleaned_outage_dataset.csv
```

2 Exploratory Data Analysis

[112]:
```
# Outage Frequency over the Years
yearly = df_cleaned.groupby('YEAR', observed=True).size().
 ↪reset_index(name='Number of Outages')
fig = px.scatter(
    yearly,
    x='YEAR',
    y='Number of Outages',
    title='<b>Number of Outages per Year</b>',
    labels={'Number of Outages': 'Number of Outages', 'YEAR': 'Year'},
    trendline='ols'
)

fig.add_trace(go.Scatter(
    x=yearly['YEAR'],
    y=yearly['Number of Outages'],
    mode='lines',
    name='Trend',
    line=dict(color='teal', width=1),
    showlegend=False
```

```
))
fig.data[1].line.color = 'red'
fig.data[1].line.dash = 'dot'
fig.update_layout(
    template="plotly_white" # Match style of other plots
)
fig.show()

# Export plot for website
fig.write_html('docs/assets/plots/outages_per_year_all.html',␣
 ↪include_plotlyjs='cdn')
```

[113]:
```
yearly_by_region = df_cleaned.groupby(['YEAR', 'CLIMATE.REGION'],␣
 ↪observed=True).size().unstack(fill_value=0)
fig = px.line(
    yearly_by_region.reset_index(),
    x='YEAR',
    y=yearly_by_region.columns.tolist(),
    title='<b>Number of Outages by Year and Region</b>',
    labels={'value': 'Number of Outages', 'variable': 'Region', 'YEAR': 'Year'},
)
fig.update_layout(legend_title_text='NCEI Climate Region')
fig.update_layout(
    template="plotly_white" # Match style of other plots
)
fig.show()

# Export plot for website
fig.write_html('docs/assets/plots/outages_per_year_by_region.html',␣
 ↪include_plotlyjs='cdn')
```

[114]:
```
# Outage Frequency by Month
month_order =␣
 ↪['Jan','Feb','Mar','Apr','May','Jun','Jul','Aug','Sep','Oct','Nov','Dec']

by_month = df_cleaned.groupby('MONTH.NAME', observed=True).size().
 ↪reindex(month_order)
fig = px.bar(
    by_month.reset_index(name='Number of Outages'),
    x='MONTH.NAME',
    y='Number of Outages',
    title='<b>Number of Outages by Month (all years combined)</b>',
    labels={'MONTH.NAME': 'Month', 'Number of Outages': 'Number of Outages'},
    color='Number of Outages',
    color_continuous_scale='teal',
)
fig.update_layout(legend_title_text='Count')
```

```python
fig.update_layout(
    template="plotly_white" # Match style of other plots
)
fig.show()

# Export plot for website
fig.write_html('docs/assets/plots/outages_by_month.html',␣
 ↪include_plotlyjs='cdn')
```

[115]:
```python
by_region = df_cleaned.groupby('CLIMATE.REGION', observed=True).size().
 ↪sort_values(ascending=False)
fig = px.bar(
    by_region.reset_index(name='Number of Outages'),
    x='CLIMATE.REGION',
    y='Number of Outages',
    title='<b>Total Number of Outages by Climate Region</b>',
    labels={'CLIMATE.REGION': 'NCEI Climate Region', 'Number of Outages':␣
 ↪'Number of Outages'},
    color='Number of Outages',
    color_continuous_scale='teal',
)
fig.update_layout(
    template="plotly_white" # Match style of other plots
)
fig.show()

# Export plot for website
fig.write_html('docs/assets/plots/outages_by_climate_region.html',␣
 ↪include_plotlyjs='cdn')
```

[116]:
```python
# Regional Analysis
top_15_states = df_cleaned['U.S._STATE'].value_counts().nlargest(15).
 ↪sort_values(ascending=True)
fig = px.bar(
    top_15_states.reset_index(name='Number of Outages'),
    x='Number of Outages',
    y='U.S._STATE',
    title='<b>Top 15 States by Number of Outages</b>',
    labels={'U.S._STATE': 'State', 'Number of Outages': 'Number of Outages'},
    color='Number of Outages',
    color_continuous_scale='teal',
    orientation='h'
)
fig.update_layout(
    template="plotly_white" # Match style of other plots
)
fig.show()
```

```
# Export plot for website
fig.write_html('docs/assets/plots/top_15_states_by_outages.html',␣
 ↪include_plotlyjs='cdn')
```

[157]:
```
# Plotly chloropleth using POSTAL.CODE for state-level outage counts, with a␣
 ↪selector to switch to average outage duration per state
state_counts = df_cleaned.groupby('POSTAL.CODE', observed=True).size().
 ↪reset_index(name='Number of Outages')
state_duration = df_cleaned.groupby('POSTAL.CODE', observed=True)['OUTAGE.
 ↪DURATION'].mean().reset_index(name='Average Outage Duration (min)')
fig = px.choropleth(
    state_counts,
    locations='POSTAL.CODE',
    locationmode='USA-states',
    color='Number of Outages',
    scope='usa',
    title='<b>State-Level Outage Statistics</b>',
    color_continuous_scale='teal',
)
fig.update_layout(
    template="plotly_white",
    updatemenus=[
        dict(
            type='buttons',
            direction='down',
            buttons=list([
                dict(
                    label='Outage Counts',
                    method='update',
                    args=[{'z': [state_counts['Number of Outages']]},
                          {'title': '<b>State-Level Outage Counts</b>',
                           'coloraxis.colorbar.title.text': 'Number␣
 ↪of<br>Outages'}]
                ),
                dict(
                    label='Average Outage Duration',
                    method='update',
                    args=[{'z': [state_duration['Average Outage Duration␣
 ↪(min)']]},
                          {'title': '<b>State-Level Average Outage Duration</
 ↪b>',
                           'coloraxis.colorbar.title.text':␣
 ↪'Average<br>Outage<br>Duration (min)'}]
                )
            ]),
```

```
            showactive=True,
        )
    ]
)
fig.show()

# Export plot for website
fig.write_html('docs/assets/plots/state_level_outage_stats.html',␣
 ↪include_plotlyjs='cdn')
```

[117]:
```
by_cause = (
    df_cleaned
    .groupby('CAUSE.CATEGORY', observed=True)
    .agg(
        num_outages=('OUTAGE.DURATION', 'count'),
        avg_duration=('OUTAGE.DURATION', 'mean'),
    )
    .sort_values('num_outages')
    .reset_index()
)

fig = px.bar(
    by_cause,
    x='num_outages',
    y='CAUSE.CATEGORY',
    orientation='h',
    color='avg_duration',
    color_continuous_scale='teal',
    title='<b>Number of Outages by Cause Category</b><br>Colored by Average␣
 ↪Outage Duration (in minutes)',
    labels={
        'num_outages':     'Number of Outages',
        'CAUSE.CATEGORY':  'Cause Category',
        'avg_duration':    'Avg Outage Duration (min)',
    },
    hover_data={'avg_duration': ':.0f'},
)
fig.update_layout(coloraxis_colorbar_title='Avg Duration<br>(min)')
fig.update_layout(
    template="plotly_white" # Match style of other plots
)

fig.show()

# Export plot for website
fig.write_html('docs/assets/plots/outages_by_cause_category.html',␣
 ↪include_plotlyjs='cdn')
```

[118]:

```
# Relevant climate region map: https://www.ncei.noaa.gov/access/monitoring/
↪reference-maps/us-climate-regions
pivot_df = (
    df_cleaned
    .groupby(['CAUSE.CATEGORY', 'CLIMATE.REGION'], observed=True)
    .agg(avg_outage_duration = ('OUTAGE.DURATION', 'mean'))
    .reset_index()
)

pivot_table = pivot_df.pivot_table(
    index='CAUSE.CATEGORY',
    columns='CLIMATE.REGION',
    values='avg_outage_duration',
)

region_abbrev = {
    'East North Central': 'E. N. Central',
    'West North Central': 'W. N. Central',
}

cause_abbrev = {
    'equipment failure': 'Equipment Failure',
    'fuel supply emergency': 'Fuel Supply',
    'intentional attack': 'Intentional Attack',
    'islanding': 'Islanding',
    'public appeal': 'Public Appeal',
    'severe weather': 'Severe Weather',
    'system operability disruption': 'System Disruption',
}

pivot_table = pivot_table.rename(columns=region_abbrev, index=cause_abbrev)
pivot_table
```

[118]:

```
CLIMATE.REGION       Central  E. N. Central  Northeast  Northwest    South  \
CAUSE.CATEGORY
Equipment Failure      322.0       26435.33      215.8      702.0   295.78
Fuel Supply         10035.25       33971.25   14629.57        1.0  16590.0
Intentional Attack    346.06        2376.05     195.98     373.81   325.61
Islanding             125.33            1.0      881.0      73.33    493.5
Public Appeal         1410.0          733.0     2655.0      898.0  1083.42
Severe Weather       3273.49        4434.82     4429.9     4838.0  4391.35
System Disruption     2695.2         2610.0      773.5      141.0   866.07

CLIMATE.REGION      Southeast  Southwest     West  W. N. Central
CAUSE.CATEGORY
Equipment Failure       554.5      113.8   524.81           61.0
Fuel Supply              <NA>       76.0   6154.6           <NA>
```

```
Intentional Attack      504.67    265.67   857.68          23.5
Islanding                 <NA>       2.0   214.86          68.2
Public Appeal          1895.67    2275.0  2028.11         439.5
Severe Weather         2662.56   7378.78  2972.03        2442.5
System Disruption       169.31    329.22   363.67          <NA>
```

[119]:
```python
# Used LLM to fix table styling
styled_pivot_table = (
    pivot_table
    .style
    .background_gradient(cmap='Blues', axis=None)
    .format('{:,.0f}', na_rep='N/A')
    .set_properties(**{
        'text-align': 'center',
        'padding': '4px 8px',
        'font-size': '0.75rem',    # back to readable size, scale handles the␣
 ↪rest
        'white-space': 'nowrap',
    })
    .set_table_styles([
        {'selector': 'th', 'props': [
            ('padding', '4px 8px'),
            ('font-size', '0.75rem'),
            ('white-space', 'nowrap'),
        ]},
        {'selector': 'table', 'props': [
            ('width', '100%'),
        ]},
    ])
)
# Export to HTML string
pivot_table_html = styled_pivot_table.to_html()

# Save to assets folder
with open('docs/assets/plots/pivot_table.html', 'w') as f:
    f.write(pivot_table_html)
```

2.1 Step 3: Assessment of Missingness

[121]:
```python
df_cleaned['duration_missing'] = df_cleaned['OUTAGE.DURATION'].isna()
df_cleaned[df_cleaned['duration_missing']].to_csv('data/
 ↪outages_with_missing_duration.csv', index=False)
```

[23]:
```python
def calculate_tvd(df, col, target):
    pivoted = df.pivot_table(index=col, columns=target, aggfunc='size',␣
 ↪fill_value=0, observed=True)
    dist = pivoted / pivoted.sum()
```

```
    return dist.diff(axis=1).iloc[:, -1].abs().sum() / 2

obs_tvd = calculate_tvd(df_cleaned, 'CLIMATE.REGION', 'duration_missing')

tvds = []
for _ in range(1000):
    shuffled = df_cleaned['CLIMATE.REGION'].sample(frac=1).
 ↪reset_index(drop=True)
    temp_df = df_cleaned.assign(shuffled_region=shuffled)
    tvds.append(calculate_tvd(temp_df, 'shuffled_region', 'duration_missing'))

p_val_dep = np.mean(np.array(tvds) >= obs_tvd)
print(f"CLIMATE.REGION Observed TVD: {obs_tvd}, P-value: {p_val_dep}")
```

```
Observed TVD: 0.25449988607883345, P-value: 0.008
```

[24]:
```
month_df = df_cleaned.dropna(subset=['MONTH']).copy()

obs_tvd_month = calculate_tvd(month_df, 'MONTH', 'duration_missing')

month_tvds = []
for _ in range(1000):
    shuffled_missing = month_df['duration_missing'].sample(frac=1).values

    temp_df = month_df.assign(shuffled_m = shuffled_missing)

    tvd = calculate_tvd(temp_df, 'MONTH', 'shuffled_m')
    month_tvds.append(tvd)

p_val_month = np.mean(np.array(month_tvds) >= obs_tvd_month)
print(f"MONTH Observed TVD: {obs_tvd_month:.4f}, P-value: {p_val_month:.4f}")
```

```
MONTH Observed TVD: 0.2219, P-value: 0.2470
```

[122]:
```
clean_df = df_cleaned.dropna(subset=['ANOMALY.LEVEL']).copy()
m_obs = clean_df.loc[clean_df['duration_missing'] == True, 'ANOMALY.LEVEL'].
 ↪mean()
nm_obs = clean_df.loc[clean_df['duration_missing'] == False, 'ANOMALY.LEVEL'].
 ↪mean()
observed_diff = abs(m_obs - nm_obs)

diffs = []
for _ in range(1000):
    shuffled_labels = clean_df['duration_missing'].sample(frac=1,␣
 ↪replace=False).values
    m = clean_df.loc[shuffled_labels, 'ANOMALY.LEVEL'].mean()
    nm = clean_df.loc[~shuffled_labels, 'ANOMALY.LEVEL'].mean()
    diffs.append(abs(m - nm))
```

```
p_val_anomaly = np.mean(np.array(diffs) >= observed_diff)
print(f"ANOMALY.LEVEL Observed Diff: {observed_diff:.4f}, P-value:␣
 ↪{p_val_anomaly:.4f}")

fig_dist = ff.create_distplot(
    [diffs],
    ['Shuffled Mean Differences'],
    show_hist=True,
    show_rug=False,
    bin_size=0.005
)

fig_dist.add_vline(
    x=observed_diff,
    line_dash="dash",
    line_color="red",
    annotation_text=f"Observed Diff: {observed_diff:.4f}",
    annotation_position="top right"
)

fig_dist.update_layout(
    title="<b>Duration Missingness vs Anomaly Level</b><br>Empirical␣
 ↪Distribution of the Test Statistic",
    xaxis_title="Absolute Difference in Mean Anomaly Level",
    yaxis_title="Density",
    template="plotly_white",
    bargap=0.02,
    legend=dict(
        orientation='h',
        x=0.8,
        xanchor='center',
        y=1.05,
        yanchor='bottom',
    ),
)

fig_dist.show()

# Export plot for website
fig_dist.write_html('docs/assets/plots/duration_missingness_vs_anomaly_level.
 ↪html', include_plotlyjs='cdn')
```

```
ANOMALY.LEVEL Observed Diff: 0.7100, P-value: 0.0000
```

[26]:
```
fig_box = px.box(
    df_cleaned,
```

```
    x='duration_missing',
    y='ANOMALY.LEVEL',
    color='duration_missing',
    notched=True,
    title='<b>Missingness Analysis: Anomaly Level Distribution</b><br>Duration␣
↪Missing (True) vs. Not Missing (False)',
    labels={'duration_missing': 'Is Outage Duration Missing?', 'ANOMALY.LEVEL':␣
↪'Anomaly Level (ONI)'},
    template='plotly_white'
)

fig_box.show()

# Export plot for website
fig_box.write_html('docs/assets/plots/anomaly_level_by_duration_missingness.
↪html', include_plotlyjs='cdn')
```

[27]:
```
fig_month = px.box(
    df_cleaned,
    x='duration_missing',
    y='MONTH',
    color='duration_missing',
    notched=True,
    title='<b>Missingness Analysis: Month Distribution</b><br>Outage Duration␣
↪Missing (True) vs. Not Missing (False)',
    labels={'duration_missing': 'Is Outage Duration Missing?', 'MONTH': 'Month␣
↪(1-12)'},
    template='plotly_white',
    category_orders={'duration_missing': [False, True]}
)

# If the boxes are at the same level, it proves the missingness is MCAR with␣
↪respect to time
fig_month.show()

# Export plot for website
fig_month.write_html('docs/assets/plots/month_by_duration_missingness.html',␣
↪include_plotlyjs='cdn')
```

[28]:
```
fig_dist_month = ff.create_distplot(
    [month_tvds],
    ['Null Distribution (Shuffled Months)'],
    show_hist=True,
    show_rug=False,
    bin_size=0.005
)
```

```
fig_dist_month.add_vline(
    x=obs_tvd_month,
    line_dash="dash",
    line_color="red",
    annotation_text=f"Observed TVD: {obs_tvd_month:.4f}"
)

fig_dist_month.update_layout(
    title="<b>Permutation Test Results: Month vs. Missingness</b>",
    xaxis_title="Total Variation Distance (TVD)",
    template="plotly_white",
    bargap=0.02,
    legend=dict(
        orientation='h',
        x=0.5,
        xanchor='center',
        y=1.05,
        yanchor='bottom',
    ),
)

fig_dist_month.show()

# Export plot for website
fig_dist_month.write_html('docs/assets/plots/month_missingness_permutation_test.
 ↪html', include_plotlyjs='cdn')
```

2.2 Step 4: Hypothesis Testing

3 Three options for hypothesis testing:

3.1 Comparing Outage Durations by Cause (Permutation Test)

Our EDA shows that different causes lead to different outage times. We can test if this difference is statistically significant.

Null Hypothesis (H0): The distribution of outage durations is the same for outages caused by "severe weather" and "equipment failure." Any observed difference in sample means is due to random chance.

Alternative Hypothesis (H1): Outages caused by "severe weather" have a longer average duration than outages caused by "equipment failure."

Test Statistic: Difference in group means (Mean Duration of Severe Weather - Mean Duration of Equipment Failure).

3.2 Climate Categories and Outage Causes (Chi-Square or TVD)

We have a CLIMATE.CATEGORY column (warm, cold, normal) and a CAUSE.CATEGORY column. We could see if the types of outages that occur depend on the climate conditions.

Null Hypothesis (H0): The distribution of outage causes is independent of the climate category (warm vs. cold).

Alternative Hypothesis (H1): The distribution of outage causes depends on the climate category (e.g., certain causes are more likely in cold climates vs. warm climates).

Test Statistic: Total Variation Distance (TVD) if you are comparing exactly two distributions, or the Chi-Square statistic if comparing across multiple categories.

3.3 Customers Affected by Region (Permutation Test)

We can compare the impact of outages across different NERC or Climate regions.

Null Hypothesis (H0): Outages in the "Northeast" climate region affect the same number of customers, on average, as outages in the "South" climate region.

Alternative Hypothesis (H1): Outages in the "Northeast" climate region affect a different number of customers, on average, compared to the "South" climate region.

Test Statistic: Absolute difference in sample means Mean Customers Northeast −Mean Customers South . (Using the absolute difference makes this a two-sided test).

Test Statistic: Absolute difference in sample means Mean Customers Northeast −Mean Customers South . (Using the absolute difference makes this a two-sided test).

[29]:
```
# Option 1: Severe Weather vs. Equipment Failure
mask = df_cleaned['CAUSE.CATEGORY'].isin(['severe weather', 'equipment␣
↪failure'])
df_hyp = df_cleaned[mask].dropna(subset=['OUTAGE.DURATION']).copy()

means = df_hyp.groupby('CAUSE.CATEGORY', observed=True)['OUTAGE.DURATION'].
↪mean()
obs_diff = means['severe weather'] - means['equipment failure']

diffs = []
for _ in range(1000):
    shuffled_causes = df_hyp['CAUSE.CATEGORY'].sample(frac=1, replace=False).
↪values

    temp_df = df_hyp.assign(shuffled=shuffled_causes)
    shuffled_means = temp_df.groupby('shuffled', observed=True)['OUTAGE.
↪DURATION'].mean()

    diffs.append(shuffled_means['severe weather'] - shuffled_means['equipment␣
↪failure'])

p_val = np.mean(np.array(diffs) >= obs_diff)

print(f"Observed Difference: {obs_diff:.2f} minutes")
print(f"P-value: {p_val:.4f}")
```

```
Observed Difference: 2015.98 minutes
P-value: 0.0000
```

[30]:
```python
# Climate Category vs. Cause Category
climate_mask = df_cleaned['CLIMATE.CATEGORY'].isin(['warm', 'cold'])
df_hyp_b = df_cleaned[climate_mask].copy()

obs_tvd = calculate_tvd(df_hyp_b, 'CAUSE.CATEGORY', 'CLIMATE.CATEGORY')

tvd_diffs = []
for _ in range(1000):
    shuffled_climate = df_hyp_b['CLIMATE.CATEGORY'].sample(frac=1,␣
↪replace=False).values

    temp_df = df_hyp_b.assign(shuffled_climate=shuffled_climate)

    tvd = calculate_tvd(temp_df, 'CAUSE.CATEGORY', 'shuffled_climate')
    tvd_diffs.append(tvd)

p_val_b = np.mean(np.array(tvd_diffs) >= obs_tvd)

print(f"Observed TVD: {obs_tvd:.4f}")
print(f"P-value: {p_val_b:.4f}")
```

```
Observed TVD: 0.0646
P-value: 0.3260
```

While our permutation test suggested that outage causes are independent of climate categories (p=0.3270), this may be due to the broad nature of the 'Climate Category' variable. Real-world events, like the Texas Power Crisis, suggest that localized extreme weather has a profound impact that might be masked when aggregating data at a national, multi-year level.

3.4 Step 5: Framing a Prediction Problem

We would like to predict the duration of an outage given its early statistics when the incident occurs. As such, we will look to train a regression model and look at key statistics being RMSE, MAE, and R^2.

There is also heavy skew in our target variable, with most outages lasting $<$ 2 days. We will look to use the log distribution.

[31]:
```python
# Outage Duration Distribution
df_cleaned['OUTAGE.DURATION'].describe()
```

[31]:
```
count      1468.0
mean      2582.58
std       5813.37
min          0.0
25%        100.0
```

```
50%         691.0
75%        2880.0
max      108653.0
Name: OUTAGE.DURATION, dtype: Float64
```

[32]:
```
# Plot outage duration and log outage duration distributions side by side
fig = make_subplots(rows=1, cols=2)
fig.add_trace(
    go.Histogram(x=df_cleaned['OUTAGE.DURATION'], nbinsx=100, name='Outage␣
 ↪Duration'),
    row=1, col=1
)
fig.add_trace(
    go.Histogram(x=np.log1p(df_cleaned['OUTAGE.DURATION']), nbinsx=30,␣
 ↪name='Log Outage Duration'),
    row=1, col=2,
)
fig.update_layout(
    title='<b>Outage Duration Distribution</b><br>Original vs. Log-Transformed',
    xaxis_title='Outage Duration (Minutes)',
    xaxis2_title='Log Outage Duration',
    yaxis_title='Count',
    template='plotly_white',
    showlegend=False,
    bargap=0.02
)
fig.show()

# Export plot for website
fig.write_html('docs/assets/plots/outage_duration_distribution.html',␣
 ↪include_plotlyjs='cdn')
```

3.4.1 Further investigation on missing rows

Previous dataframe was cleaned for preliminary analysis and EDA, we will look to further investigate missing data and perform imputation accordingly. End result will be our engineered dataframe with key features for model training and analysis.

[33]:
```
# All missing rows in "cleaned" dataset:
missing_count = df_cleaned.isna().sum()[df_cleaned.isna().sum() > 0].
 ↪sort_values(ascending=False)
missing_count
```

[33]:
```
HURRICANE.NAMES           1454
DEMAND.LOSS.MW             700
CAUSE.CATEGORY.DETAIL      467
CUSTOMERS.AFFECTED         437
```

```
OUTAGE.RESTORATION          58
OUTAGE.RESTORATION.DATE     58
OUTAGE.RESTORATION.TIME     58
OUTAGE.DURATION             58
IND.PRICE                   22
TOTAL.SALES                 22
IND.SALES                   22
COM.SALES                   22
RES.SALES                   22
TOTAL.PRICE                 22
COM.PRICE                   22
IND.PERCEN                  22
RES.PRICE                   22
COM.PERCEN                  22
RES.PERCEN                  22
POPDEN_UC                   10
POPDEN_RURAL                10
OUTAGE.START                 9
MONTH                        9
OUTAGE.START.TIME            9
OUTAGE.START.DATE            9
CLIMATE.CATEGORY             9
ANOMALY.LEVEL                9
MONTH.NAME                   9
CLIMATE.REGION               6
dtype: int64
```

[34]:

```python
# CAUSE.CATEGORY.DETAIL has 467 missing values, can investigate CAUSE.CATEGORY␣
↪to impute missing values
print(df_cleaned['CAUSE.CATEGORY'].value_counts())

# CAUSE.CATEGORY of missing CAUSE.CATEGORY.DETAIL values
print(f"\n---> Filtered by missing CAUSE.CATEGORY.
↪DETAIL\n{df_cleaned[df_cleaned['CAUSE.CATEGORY.DETAIL'].isna()]['CAUSE.
↪CATEGORY'].value_counts()}")
```

```
CAUSE.CATEGORY
severe weather                   760
intentional attack               418
system operability disruption    127
public appeal                     65
equipment failure                 60
fuel supply emergency             50
islanding                         46
Name: count, dtype: int64

---> Filtered by missing CAUSE.CATEGORY.DETAIL
CAUSE.CATEGORY
```

```
severe weather                 187
system operability disruption   90
public appeal                   65
intentional attack              48
islanding                       46
fuel supply emergency           19
equipment failure               12
Name: count, dtype: int64
```

[35]:
```python
# Investigate missing clinmate regions --> Hawaii and Alaska only
df_cleaned[df_cleaned['CLIMATE.REGION'].isna()]
```

[35]:
```
      YEAR  MONTH U.S._STATE POSTAL.CODE NERC.REGION CLIMATE.REGION  \
1516  2008     12     Hawaii          HI          HI            NaN
1517  2011      5     Hawaii          HI          PR            NaN
1518  2006     10     Hawaii          HI        HECO            NaN
1519  2006      6     Hawaii          HI        HECO            NaN
1520  2006     10     Hawaii          HI        HECO            NaN
1534  2000   <NA>     Alaska          AK        ASCC            NaN

      ANOMALY.LEVEL CLIMATE.CATEGORY OUTAGE.START.DATE OUTAGE.START.TIME  \
1516           -0.7             cold        2008-12-26          18:13:00
1517           -0.4           normal        2011-05-02          17:06:00
1518            0.7             warm        2006-10-15          07:09:00
1519            0.0           normal        2006-06-01          14:12:00
1520            0.7             warm        2006-10-15          07:09:00
1534            NaN              NaN               NaT               NaN

     OUTAGE.RESTORATION.DATE OUTAGE.RESTORATION.TIME  \
1516              2008-12-27                17:00:00
1517              2011-05-02                20:00:00
1518              2006-10-15                16:12:00
1519              2006-06-01                18:09:00
1520              2006-10-16                14:55:00
1534                     NaT                     NaN

                     CAUSE.CATEGORY CAUSE.CATEGORY.DETAIL HURRICANE.NAMES  \
1516                 severe weather          thunderstorm             NaN
1517                 severe weather                   NaN             NaN
1518                 severe weather            earthquake             NaN
1519  system operability disruption                   NaN             NaN
1520                 severe weather            earthquake             NaN
1534              equipment failure               failure             NaN

      OUTAGE.DURATION  DEMAND.LOSS.MW  CUSTOMERS.AFFECTED  RES.PRICE  \
1516             1367            1060              294000      29.28
1517              174             220               62000      34.58
```

```
1518            543            110                  59886      23.25
1519            237            120                  29300      24.09
1520           1906           1170                 291000      23.25
1534           <NA>             35                  14273        NaN

      COM.PRICE  IND.PRICE  TOTAL.PRICE  RES.SALES  COM.SALES  IND.SALES  \
1516      26.28      22.43        25.78   253625.0   275383.0   306397.0
1517      32.14      27.85        31.29   249369.0   292304.0   310682.0
1518      21.26      17.71        20.54   274609.0   307570.0   340735.0
1519      22.06      18.74        21.45   265474.0   298487.0   327618.0
1520      21.26      17.71        20.54   274609.0   307570.0   340735.0
1534        NaN        NaN          NaN        NaN        NaN        NaN

      TOTAL.SALES  RES.PERCEN  COM.PERCEN  IND.PERCEN  RES.CUSTOMERS  \
1516     835405.0       30.36       32.96       36.68       409668.0
1517     852355.0       29.26       34.29       36.45       417531.0
1518     922915.0       29.75       33.33       36.92       401592.0
1519     891580.0       29.78       33.48       36.75       401592.0
1520     922915.0       29.75       33.33       36.92       401592.0
1534          NaN         NaN         NaN         NaN       230534.0

      COM.CUSTOMERS  IND.CUSTOMERS  TOTAL.CUSTOMERS  RES.CUST.PCT  \
1516        61684.0          673.0         472025.0         86.79
1517        60043.0          698.0         478272.0         87.30
1518        61334.0          689.0         463615.0         86.62
1519        61334.0          689.0         463615.0         86.62
1520        61334.0          689.0         463615.0         86.62
1534        38074.0          854.0         273530.0         84.28

      COM.CUST.PCT  IND.CUST.PCT  PC.REALGSP.STATE  PC.REALGSP.USA  \
1516         13.07          0.14           50565.0         48401.0
1517         12.55          0.15           49110.0         47586.0
1518         13.23          0.15           50358.0         48909.0
1519         13.23          0.15           50358.0         48909.0
1520         13.23          0.15           50358.0         48909.0
1534         13.92          0.31           57401.0         44745.0

      PC.REALGSP.REL  PC.REALGSP.CHANGE  UTIL.REALGSP  TOTAL.REALGSP  \
1516            1.04               -0.7        1646.0        67364.0
1517            1.03                0.4        1935.0        67686.0
1518            1.03                1.1        1436.0        65956.0
1519            1.03                1.1        1436.0        65956.0
1520            1.03                1.1        1436.0        65956.0
1534            1.28               -2.2         724.0        36046.0

      UTIL.CONTRI  PI.UTIL.OFUSA  POPULATION  POPPCT_URBAN  POPPCT_UC  \
1516         2.44            0.5     1332213         91.93      20.47
```

```
1517          2.86             0.6     1378227          91.93      20.47
1518          2.18             0.5     1309731          91.93      20.47
1519          2.18             0.5     1309731          91.93      20.47
1520          2.18             0.5     1309731          91.93      20.47
1534          2.01             0.2      627963          66.02      21.56

      POPDEN_URBAN  POPDEN_UC  POPDEN_RURAL  AREAPCT_URBAN  AREAPCT_UC  \
1516        3180.8     1664.7          18.2           6.12        2.60
1517        3180.8     1664.7          18.2           6.12        2.60
1518        3180.8     1664.7          18.2           6.12        2.60
1519        3180.8     1664.7          18.2           6.12        2.60
1520        3180.8     1664.7          18.2           6.12        2.60
1534        1802.6     1276.0           0.4           0.05        0.02

      PCT_LAND  PCT_WATER_TOT  PCT_WATER_INLAND        OUTAGE.START  \
1516     58.75          41.25              0.38 2008-12-26 18:13:00
1517     58.75          41.25              0.38 2011-05-02 17:06:00
1518     58.75          41.25              0.38 2006-10-15 07:09:00
1519     58.75          41.25              0.38 2006-06-01 14:12:00
1520     58.75          41.25              0.38 2006-10-15 07:09:00
1534     85.76          14.24              2.90                 NaT

      OUTAGE.RESTORATION MONTH.NAME  duration_missing
1516 2008-12-27 17:00:00        Dec             False
1517 2011-05-02 20:00:00        May             False
1518 2006-10-15 16:12:00        Oct             False
1519 2006-06-01 18:09:00        Jun             False
1520 2006-10-16 14:55:00        Oct             False
1534                 NaT        NaN              True
```

[36]:
```python
# Can just use CAUSE.CATEGORY.DETAIL since it includes hurricane information
hurricanes_detail = (df_cleaned['CAUSE.CATEGORY.DETAIL'] == 'hurricanes')
no_names = (df_cleaned['HURRICANE.NAMES'].isna())
len(df_cleaned[~no_names]), len(df_cleaned[hurricanes_detail]), len(df_cleaned[hurricanes_detail & no_names])
```

[36]:
```
(72, 74, 2)
```

[37]:
```python
# Imputations and Feature engineering functions

def impute_climate_region(df):
    # Hawaii and Alaska don't have climate regions in dataset.
    df.loc[df['U.S._STATE'] == 'Hawaii', 'CLIMATE.REGION'] = 'Hawaii'
    df.loc[df['U.S._STATE'] == 'Alaska', 'CLIMATE.REGION'] = 'Alaska'
    return df

def impute_cause_category_detail(df):
```

```
    # Impute based on CAUSE.CATEGORY
    fallback = {
        'severe weather':                   'Other Weather',
        'intentional attack':               'Other Intentional Attack',
        'system operability disruption':    'Other Equipment Failure',
        'equipment failure':                'Other Equipment Failure',
        'fuel supply emergency':            'Other Fuel Supply Emergency',
        'public appeal':                    'Other Public Appeal',
        'islanding':                        'Other Islanding',
    }

    df['CAUSE.CATEGORY.DETAIL'] = df['CAUSE.CATEGORY.DETAIL'].fillna(
        df['CAUSE.CATEGORY'].map(fallback)
    )
    return df

def drop_missing_months(df):
    df = df.dropna(subset=['OUTAGE.START'])
    return df

def drop_missing_price(df):
    df = df.dropna(subset=['RES.PRICE'])
    return df

def median_impute_customers_affected(df):
    # Median imputation for CUSTOMERS.AFFECTED, grouped by CAUSE.CATEGORY
    df['CUSTOMERS.AFFECTED'] = df.groupby('CAUSE.CATEGORY')['CUSTOMERS.AFFECTED'].transform(
        lambda x: x.fillna(x.median().round())
    )
    return df

def drop_missing_durations(df):
    df = df.dropna(subset=['OUTAGE.DURATION'])
    return df

def identify_weekend(df):
    df['OUTAGE.START'] = pd.to_datetime(df['OUTAGE.START'])
    df['START_ON_WEEKEND'] = df['OUTAGE.START'].dt.weekday.apply(lambda x: 1 if x >= 5 else 0) # 1 for Weekends, 0 for weekdays
    return df

def identify_season(df):
    outage_month = pd.to_datetime(df['OUTAGE.START']).dt.month
    df['SEASON'] = outage_month.map({
            12: 'Winter', 1: 'Winter',  2: 'Winter',
            3:  'Spring', 4: 'Spring',  5: 'Spring',
```

```
            6:  'Summer', 7: 'Summer',  8: 'Summer',
            9:  'Fall',   10: 'Fall',   11: 'Fall',
        })
    return df

def cyclical_encode_month(df):
    # Cyclical encoding for MONTH
    df['MONTH_SIN'] = np.sin(2 * np.pi * df['MONTH'] / 12)
    df['MONTH_COS'] = np.cos(2 * np.pi * df['MONTH'] / 12)
    return df

def remove_outage_outliers(df):
    # Remove extreme outliers in OUTAGE.DURATION (e.g., above 30,000 minutes or␣
 ↪~20 days)
    df = df[df['OUTAGE.DURATION'] <= 30000]
    return df
```

[38]:
```
df_engineered = (
    df_cleaned.pipe(impute_climate_region)
      .pipe(impute_cause_category_detail)
      .pipe(drop_missing_months)
      .pipe(drop_missing_price)
      .pipe(drop_missing_durations)
      .pipe(median_impute_customers_affected)
      .pipe(identify_weekend)
      .pipe(identify_season)
      .pipe(cyclical_encode_month)
      .pipe(remove_outage_outliers)
)

print(f"Rows dropped: {len(df_cleaned) - len(df_engineered)} ({(len(df_cleaned)␣
 ↪- len(df_engineered)) / len(df_cleaned) * 100:.1f}% of original)")
```

```
Rows dropped: 77 (5.0% of original)
```

[39]:
```
df_engineered.head()
```

[39]:
```
   YEAR  MONTH U.S._STATE POSTAL.CODE NERC.REGION       CLIMATE.REGION  \
1  2011      7  Minnesota          MN         MRO  East North Central
2  2014      5  Minnesota          MN         MRO  East North Central
3  2010     10  Minnesota          MN         MRO  East North Central
4  2012      6  Minnesota          MN         MRO  East North Central
5  2015      7  Minnesota          MN         MRO  East North Central

   ANOMALY.LEVEL CLIMATE.CATEGORY OUTAGE.START.DATE OUTAGE.START.TIME  \
1           -0.3           normal        2011-07-01          17:00:00
2           -0.1           normal        2014-05-11          18:38:00
3           -1.5             cold        2010-10-26          20:00:00
```

```
4          -0.1          normal       2012-06-19          04:30:00
5           1.2            warm       2015-07-18          02:00:00

  OUTAGE.RESTORATION.DATE OUTAGE.RESTORATION.TIME      CAUSE.CATEGORY  \
1              2011-07-03                20:00:00      severe weather
2              2014-05-11                18:39:00  intentional attack
3              2010-10-28                22:00:00      severe weather
4              2012-06-20                23:00:00      severe weather
5              2015-07-19                07:00:00      severe weather

  CAUSE.CATEGORY.DETAIL HURRICANE.NAMES  OUTAGE.DURATION  DEMAND.LOSS.MW  \
1         Other Weather             NaN             3060            <NA>
2             vandalism             NaN                1            <NA>
3            heavy wind             NaN             3000            <NA>
4          thunderstorm             NaN             2550            <NA>
5         Other Weather             NaN             1740             250

   CUSTOMERS.AFFECTED  RES.PRICE  COM.PRICE  IND.PRICE  TOTAL.PRICE  \
1               70000      11.60       9.18       6.81         9.28
2                   0      12.12       9.71       6.49         9.28
3               70000      10.87       8.19       6.07         8.15
4               68200      11.79       9.25       6.71         9.19
5              250000      13.07      10.16       7.74        10.43

   RES.SALES  COM.SALES  IND.SALES  TOTAL.SALES  RES.PERCEN  COM.PERCEN  \
1   2.33e+06   2.11e+06   2.11e+06     6.56e+06       35.55       32.23
2   1.59e+06   1.81e+06   1.89e+06     5.28e+06       30.03       34.21
3   1.47e+06   1.80e+06   1.95e+06     5.22e+06       28.10       34.50
4   1.85e+06   1.94e+06   1.99e+06     5.79e+06       31.99       33.54
5   2.03e+06   2.16e+06   1.78e+06     5.97e+06       33.98       36.21

   IND.PERCEN  RES.CUSTOMERS  COM.CUSTOMERS  IND.CUSTOMERS  TOTAL.CUSTOMERS  \
1       32.20       2.31e+06       276286.0        10673.0         2.60e+06
2       35.73       2.35e+06       284978.0         9898.0         2.64e+06
3       37.37       2.30e+06       276463.0        10150.0         2.59e+06
4       34.44       2.32e+06       278466.0        11010.0         2.61e+06
5       29.78       2.37e+06       289044.0         9812.0         2.67e+06

   RES.CUST.PCT  COM.CUST.PCT  IND.CUST.PCT  PC.REALGSP.STATE  PC.REALGSP.USA  \
1         88.94         10.64          0.41           51268.0         47586.0
2         88.83         10.79          0.37           53499.0         49091.0
3         88.92         10.69          0.39           50447.0         47287.0
4         88.90         10.68          0.42           51598.0         48156.0
5         88.82         10.81          0.37           54431.0         49844.0

   PC.REALGSP.REL  PC.REALGSP.CHANGE  UTIL.REALGSP  TOTAL.REALGSP  \
1            1.08                1.6        4802.0       274182.0
```

```
2          1.09              1.9      5226.0     291955.0
3          1.07              2.7      4571.0     267895.0
4          1.07              0.6      5364.0     277627.0
5          1.09              1.7      4873.0     292023.0

   UTIL.CONTRI  PI.UTIL.OFUSA  POPULATION  POPPCT_URBAN  POPPCT_UC  \
1         1.75            2.2     5348119         73.27      15.28
2         1.79            2.2     5457125         73.27      15.28
3         1.71            2.1     5310903         73.27      15.28
4         1.93            2.2     5380443         73.27      15.28
5         1.67            2.2     5489594         73.27      15.28

   POPDEN_URBAN  POPDEN_UC  POPDEN_RURAL  AREAPCT_URBAN  AREAPCT_UC  PCT_LAND  \
1        2279.0     1700.5          18.2           2.14         0.6     91.59
2        2279.0     1700.5          18.2           2.14         0.6     91.59
3        2279.0     1700.5          18.2           2.14         0.6     91.59
4        2279.0     1700.5          18.2           2.14         0.6     91.59
5        2279.0     1700.5          18.2           2.14         0.6     91.59

   PCT_WATER_TOT  PCT_WATER_INLAND        OUTAGE.START  OUTAGE.RESTORATION  \
1           8.41              5.48 2011-07-01 17:00:00 2011-07-03 20:00:00
2           8.41              5.48 2014-05-11 18:38:00 2014-05-11 18:39:00
3           8.41              5.48 2010-10-26 20:00:00 2010-10-28 22:00:00
4           8.41              5.48 2012-06-19 04:30:00 2012-06-20 23:00:00
5           8.41              5.48 2015-07-18 02:00:00 2015-07-19 07:00:00

  MONTH.NAME  duration_missing  START_ON_WEEKEND  SEASON  MONTH_SIN  MONTH_COS
1        Jul             False                 0  Summer       -0.5      -0.87
2        May             False                 1  Spring        0.5      -0.87
3        Oct             False                 0    Fall      -0.87        0.5
4        Jun             False                 0  Summer        0.0       -1.0
5        Jul             False                 1  Summer       -0.5      -0.87
```

[145]:
```python
df.shape[0], df_cleaned.shape[0], df_engineered.shape[0]
```

[145]:
```
(1534, 1526, 1449)
```

[40]:
```python
# Some columns will still have missing values, but our selected features will␣
↪only be using a subset
df_engineered.isna().sum()[df_engineered.isna().sum() > 0]
```

[40]:
```
HURRICANE.NAMES    1378
DEMAND.LOSS.MW      660
POPDEN_UC            10
POPDEN_RURAL         10
dtype: int64
```

3.5 Feature setup and Training functions

[41]:
```
# Feature selection

cat_features = [
    'SEASON',
    'U.S._STATE',
    'NERC.REGION',
    'CLIMATE.REGION',
    'CLIMATE.CATEGORY',
    'CAUSE.CATEGORY',
    'CAUSE.CATEGORY.DETAIL',
]
num_features = [
    'MONTH_SIN',
    'MONTH_COS',
    'START_ON_WEEKEND',
    'ANOMALY.LEVEL',
    'CUSTOMERS.AFFECTED',
    'RES.PRICE',
    'COM.PRICE',
    'IND.PRICE',
    'UTIL.CONTRI',
    'PI.UTIL.OFUSA',
]

# Preprocessing Pipeline
preprocessor = ColumnTransformer(
    transformers=[
        ('num', StandardScaler(), num_features),
        ('cat', OneHotEncoder(handle_unknown='ignore'), cat_features)
    ]
)

preprocessor_dense = ColumnTransformer(
    transformers=[
        ('num', StandardScaler(), num_features),
        ('cat', OneHotEncoder(handle_unknown='ignore', sparse_output=False),␣
↪cat_features)
    ]
)

features = cat_features + num_features
target = 'OUTAGE.DURATION'
```

[42]:
```
print(f"Initial cleaned dataset shape:  {df.shape}")
print(f"Engineered dataset shape:       {df_engineered.shape}")
```

```
X = df_engineered[features]
y = np.log1p(df_engineered[target]) # Log transform target to reduce skew
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,␣
↪random_state=42)

print(f"Training set shape:             {X_train.shape}, {y_train.shape}")
print(f"Test set shape:                 {X_test.shape}, {y_test.shape}")
```

```
Initial cleaned dataset shape:  (1534, 55)
Engineered dataset shape:       (1449, 63)
Training set shape:             (1159, 17), (1159,)
Test set shape:                 (290, 17), (290,)
```

[43]:
```
# General Setup for testing different models

results_log = [] # For final dataframe creation

def plot_log_and_reg_residuals(residual_df, model_name):
    fig = make_subplots(rows=1, cols=2, subplot_titles=[
        'Residuals vs Predicted (minutes)',
        'Residuals vs Predicted (log scale)',
    ])
    fig.add_trace(
        go.Scatter(
            x=residual_df['Predicted Duration (min)'],
            y=residual_df['Residuals (min)'],
            mode='markers', opacity=0.5, marker=dict(size=4),
            name='minutes',
        ),
        row=1, col=1,
    )
    fig.add_trace(
        go.Scatter(
            x=residual_df['Predicted Log Duration'],
            y=residual_df['Residuals (log)'],
            mode='markers', opacity=0.5, marker=dict(size=4),
            name='log scale',
        ),
        row=1, col=2,
    )


    # Zero lines for reference
    fig.add_hline(y=0, line_dash='dash', line_color='black', row=1, col=1)
    fig.add_hline(y=0, line_dash='dash', line_color='black', row=1, col=2)
```

```python
    fig.update_layout(
        title=f'<b>Residuals - {model_name}</b>',
        title_pad=dict(t=20),
        margin=dict(t=80),
        autosize=True,
        height=450, # seems best for visibility
        showlegend=False,
        template='plotly_white'
    )
    fig.update_xaxes(title_text='Predicted (log min)', row=1, col=2)
    fig.update_xaxes(title_text='Predicted (min)', row=1, col=1)

    # Set y-axis limits to be symmetric around zero and slightly larger than max absolute residual
    abs_max_min = residual_df['Residuals (min)'].abs().max() * 1.1
    abs_max_log = residual_df['Residuals (log)'].abs().max() * 1.1

    fig.update_yaxes(title_text='Residuals', range=[-abs_max_min, abs_max_min], row=1, col=1)
    fig.update_yaxes(title_text='Residuals', range=[-abs_max_log, abs_max_log], row=1, col=2)
    fig.show()
    return fig

def train_and_evaluate_model(model, X_train, y_train, X_test, y_test, label=None):
    start_time = time.time()
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)

    y_pred_minutes = np.expm1(y_pred) # Convert back to minutes for residuals plot
    y_test_minutes = np.expm1(y_test)

    metrics = {
        'Model': label or model.__class__.__name__,
        'MAE (log)': mean_absolute_error(y_test, y_pred),
        'RMSE (log)': root_mean_squared_error(y_test, y_pred),
        'MAE (minutes)': mean_absolute_error(y_test_minutes, y_pred_minutes),
        'RMSE (minutes)': root_mean_squared_error(y_test_minutes, y_pred_minutes),
        'R^2': r2_score(y_test, y_pred),
    }

    print(f"Model:                      {metrics['Model']}")
    print(f"MAE (log):                  {metrics['MAE (log)']:.4f}")
```

```
    print(f"RMSE (log):              {metrics['RMSE (log)']:.4f}")
    print(f"MAE (minutes):           {metrics['MAE (minutes)']:.2f} minutes")
    print(f"RMSE (minutes):          {metrics['RMSE (minutes)']:.2f} minutes")
    print(f"R^2:                     {metrics['R^2']:.4f}")
    print(f"Training Time:           {time.time() - start_time:.2f} seconds")
    # Plot residuals
    residual_df = pd.DataFrame({
        'Predicted Log Duration': y_pred,
        'Residuals (log)': y_test - y_pred,
        'Predicted Duration (min)': y_pred_minutes,
        'Residuals (min)': y_test_minutes - y_pred_minutes
    })
    fig = plot_log_and_reg_residuals(residual_df, metrics['Model'])

    results_log.append(metrics)
    return model, y_pred, fig

def train_tune_evaluate(model, param_distributions, X_train, y_train, X_test,␣
 ↪y_test, n_iter=50, label=None):
    random_search = RandomizedSearchCV(
        estimator=model,
        param_distributions=param_distributions,
        n_iter=n_iter,
        scoring='neg_mean_absolute_error',
        cv=3,
        verbose=1,
        random_state=42,
        n_jobs=-1
    )
    print(f"Starting hyperparameter tuning for {label or model.__class__.
 ↪__name__}...")
    start_time = time.time()
    random_search.fit(X_train, y_train)
    print(f"Completed in {time.time() - start_time:.2f} seconds")

    print(f"Best Hyperparameters: {random_search.best_params_}")
    best_model = random_search.best_estimator_

    tuned_label = f"{label or model.__class__.__name__} (tuned)"
    return train_and_evaluate_model(best_model, X_train, y_train, X_test,␣
 ↪y_test, label=tuned_label)
```

3.6 Step 6: Baseline Model

Use naive linear regression, without any tuning

[44]:
```python
# Linear Regression - Baseline
linear_pipeline = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('regressor', LinearRegression())
])

linear_reg, linear_pred, linear_fig = train_and_evaluate_model(linear_pipeline,␣
↪X_train, y_train, X_test, y_test, label="Linear Regression (Baseline)")

# Export plot for website
linear_fig.write_html('docs/assets/plots/linear_regression_residuals.html',␣
↪include_plotlyjs='cdn')
```

```
Model:                    Linear Regression (Baseline)
MAE (log):                1.4645
RMSE (log):               1.9254
MAE (minutes):            1812.06 minutes
RMSE (minutes):           4378.00 minutes
R^2:                      0.5185
Training Time:            0.02 seconds
```

3.7 Step 7: Final Model

[45]:
```python
# Ridge Regression - Hyperparameter Tuning
ridge_pipeline = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('regressor', Ridge())
])
ridge_param_dist = {
    'regressor__alpha': np.logspace(-3, 3, 50)
}
ridge_reg, ridge_pred, ridge_fig = train_tune_evaluate(ridge_pipeline,␣
↪ridge_param_dist, X_train, y_train, X_test, y_test, n_iter=1000,label="Ridge␣
↪Regression")

# Export plot for website
ridge_fig.write_html('docs/assets/plots/ridge_regression_residuals.html',␣
↪include_plotlyjs='cdn')
```

```
Starting hyperparameter tuning for Ridge Regression…
Fitting 3 folds for each of 50 candidates, totalling 150 fits
```

```
c:\ProgramData\anaconda3\Lib\site-
packages\sklearn\model_selection\_search.py:320: UserWarning:

The total space of parameters 50 is smaller than n_iter=1000. Running 50
iterations. For exhaustive searches, use GridSearchCV.
```

```
Completed in 2.39 seconds
Best Hyperparameters: {'regressor__alpha': 8.286427728546842}
Model:                   Ridge Regression (tuned)
MAE (log):               1.4789
RMSE (log):              1.9142
MAE (minutes):           1802.69 minutes
RMSE (minutes):          3925.18 minutes
R^2:                     0.5240
Training Time:           0.01 seconds
```

[46]:

```python
# Random Forest Regressor
rf_pipeline = Pipeline(steps=[
    ('preprocessor', preprocessor_dense),
    ('regressor', RandomForestRegressor(random_state=42))
])
rf_param_dist = {
    'regressor__n_estimators': randint(100, 1000),
    'regressor__max_depth': [None, 1, 2, 3, 4, 5, 10, 20, 30, 40, 50],
    'regressor__min_samples_split': randint(2, 30),
    'regressor__min_samples_leaf': randint(1, 20),
    'regressor__max_features': ['sqrt', 'log2', 0.3, 0.5, 0.7, 1.0],
    'regressor__bootstrap': [True, False],
    'regressor__oob_score': [True, False],
    'regressor__warm_start': [True, False],
    'regressor__ccp_alpha': uniform(0.0, 0.01),
    'regressor__max_samples': [0.6, 0.7, 0.8, 0.9, None],
}
rf_reg, rf_pred, rf_fig = train_tune_evaluate(rf_pipeline, rf_param_dist,␣
 ↪X_train, y_train, X_test, y_test, n_iter=500, label="Random Forest␣
 ↪Regressor")

# Export plot for website
rf_fig.write_html('docs/assets/plots/random_forest_residuals.html',␣
 ↪include_plotlyjs='cdn')
```

```
Starting hyperparameter tuning for Random Forest Regressor…
Fitting 3 folds for each of 500 candidates, totalling 1500 fits
```

```
c:\ProgramData\anaconda3\Lib\site-
packages\sklearn\model_selection\_validation.py:540: FitFailedWarning:


711 fits failed out of a total of 1500.
The score on these train-test partitions for these parameters will be set to
nan.
If these failures are not expected, you can try to debug them by setting
error_score='raise'.
```

```
Below are more details about the failures:
--------------------------------------------------------------------------------
624 fits failed with the following error:
Traceback (most recent call last):
  File "c:\ProgramData\anaconda3\Lib\site-
packages\sklearn\model_selection\_validation.py", line 888, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "c:\ProgramData\anaconda3\Lib\site-packages\sklearn\base.py", line 1473,
in wrapper
    return fit_method(estimator, *args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "c:\ProgramData\anaconda3\Lib\site-packages\sklearn\pipeline.py", line
473, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
  File "c:\ProgramData\anaconda3\Lib\site-packages\sklearn\base.py", line 1473,
in wrapper
    return fit_method(estimator, *args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "c:\ProgramData\anaconda3\Lib\site-packages\sklearn\ensemble\_forest.py",
line 433, in fit
    raise ValueError(
ValueError: `max_sample` cannot be set if `bootstrap=False`. Either switch to
`bootstrap=True` or set `max_sample=None`.

--------------------------------------------------------------------------------
87 fits failed with the following error:
Traceback (most recent call last):
  File "c:\ProgramData\anaconda3\Lib\site-
packages\sklearn\model_selection\_validation.py", line 888, in _fit_and_score
    estimator.fit(X_train, y_train, **fit_params)
  File "c:\ProgramData\anaconda3\Lib\site-packages\sklearn\base.py", line 1473,
in wrapper
    return fit_method(estimator, *args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "c:\ProgramData\anaconda3\Lib\site-packages\sklearn\pipeline.py", line
473, in fit
    self._final_estimator.fit(Xt, y, **last_step_params["fit"])
  File "c:\ProgramData\anaconda3\Lib\site-packages\sklearn\base.py", line 1473,
in wrapper
    return fit_method(estimator, *args, **kwargs)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "c:\ProgramData\anaconda3\Lib\site-packages\sklearn\ensemble\_forest.py",
line 450, in fit
    raise ValueError("Out of bag estimation only available if bootstrap=True")
ValueError: Out of bag estimation only available if bootstrap=True


c:\ProgramData\anaconda3\Lib\site-packages\numpy\ma\core.py:2820:
```

```
RuntimeWarning:

invalid value encountered in cast

c:\ProgramData\anaconda3\Lib\site-
packages\sklearn\model_selection\_search.py:1102: UserWarning:

One or more of the test scores are non-finite: [-1.41 -1.54   nan … -1.77
-1.44   nan]


Completed in 52.61 seconds
Best Hyperparameters: {'regressor__bootstrap': False, 'regressor__ccp_alpha':
0.005464303160778473, 'regressor__max_depth': None, 'regressor__max_features':
0.3, 'regressor__max_samples': None, 'regressor__min_samples_leaf': 1,
'regressor__min_samples_split': 6, 'regressor__n_estimators': 382,
'regressor__oob_score': False, 'regressor__warm_start': False}
Model:                    Random Forest Regressor (tuned)
MAE (log):                1.3368
RMSE (log):               1.8286
MAE (minutes):            1739.02 minutes
RMSE (minutes):           3719.70 minutes
R^2:                      0.5657
Training Time:            1.32 seconds
```

[47]:

```python
# XGBoost Regressor
xgb_pipeline = Pipeline(steps=[
    ('preprocessor', preprocessor_dense),
    ('regressor', XGBRegressor(random_state=42, objective='reg:squarederror'))
])

xgb_param_dist = {
    'regressor__n_estimators': randint(100, 1000),
    'regressor__learning_rate': uniform(0.01, 0.2),
    'regressor__max_depth': randint(3, 15),
    'regressor__subsample': uniform(0.6, 0.4),
    'regressor__colsample_bytree': uniform(0.4, 0.6),
    'regressor__min_child_weight': randint(1, 10),
    'regressor__gamma': uniform(0, 0.5),
    'regressor__reg_alpha': uniform(0, 1),
    'regressor__reg_lambda': uniform(0.5, 2),
}

xgb_reg, xgb_pred, xgb_fig = train_tune_evaluate(xgb_pipeline, xgb_param_dist,
 ↪X_train, y_train, X_test, y_test, n_iter=1000, label="XGBoost Regressor")

# Export plot for website
```

```
xgb_fig.write_html('docs/assets/plots/xgboost_residuals.html',
    include_plotlyjs='cdn')
```

```
Starting hyperparameter tuning for XGBoost Regressor…
Fitting 3 folds for each of 1000 candidates, totalling 3000 fits
Completed in 61.82 seconds
Best Hyperparameters: {'regressor__colsample_bytree': 0.44895923200290305,
'regressor__gamma': 0.129295795567392, 'regressor__learning_rate':
0.015570448164214882, 'regressor__max_depth': 14, 'regressor__min_child_weight':
2, 'regressor__n_estimators': 420, 'regressor__reg_alpha': 0.6728471334885447,
'regressor__reg_lambda': 1.9485090549982664, 'regressor__subsample':
0.7970396020887256}
Model:                    XGBoost Regressor (tuned)
MAE (log):                1.3745
RMSE (log):               1.8994
MAE (minutes):            1755.67 minutes
RMSE (minutes):           3729.29 minutes
R^2:                      0.5314
Training Time:            0.43 seconds
```

[48]:

```
# HistGradientBoostingRegressor, typically better for larger datasets, but can
    still be competitive here
hgb_pipeline = Pipeline(steps=[
    ('preprocessor', preprocessor_dense),
    ('regressor', HistGradientBoostingRegressor(random_state=42))
])

hgb_param_dist = {
    'regressor__learning_rate':    uniform(0.01, 0.2),
    'regressor__max_iter':         randint(100, 500),
    'regressor__max_depth':        randint(3, 10),
    'regressor__min_samples_leaf': randint(10, 50),
    'regressor__l2_regularization': uniform(0, 1),
    'regressor__max_leaf_nodes':   randint(20, 100),
}

hgb_reg, hgb_pred, hgb_fig = train_tune_evaluate(hgb_pipeline, hgb_param_dist,
    X_train, y_train, X_test, y_test, n_iter=1000,
    label="HistGradientBoostingRegressor")

# Export plot for website
hgb_fig.write_html('docs/assets/plots/hist_gradient_boosting_residuals.html',
    include_plotlyjs='cdn')
```

```
Starting hyperparameter tuning for HistGradientBoostingRegressor…
Fitting 3 folds for each of 1000 candidates, totalling 3000 fits
Completed in 86.51 seconds
Best Hyperparameters: {'regressor__l2_regularization': 0.4395116886329101,
```

```
'regressor__learning_rate': 0.031132936852348782, 'regressor__max_depth': 8,
'regressor__max_iter': 259, 'regressor__max_leaf_nodes': 62,
'regressor__min_samples_leaf': 11}
Model:                   HistGradientBoostingRegressor (tuned)
MAE (log):               1.3421
RMSE (log):              1.8624
MAE (minutes):           1691.86 minutes
RMSE (minutes):          3557.68 minutes
R^2:                     0.5495
Training Time:           0.49 seconds
```

[53]:

```python
# Additional investment into XGBoost with Optuna

from optuna.samplers import TPESampler
from optuna.pruners import MedianPruner

def objective(trial):
    param = {
        'n_estimators': trial.suggest_int('n_estimators', 100, 1000),
        'learning_rate': trial.suggest_float('learning_rate', 0.01, 0.3,␣
 ↪log=True),
        'max_depth': trial.suggest_int('max_depth', 2, 12),
        'subsample': trial.suggest_float('subsample', 0.5, 1.0),
        'colsample_bytree': trial.suggest_float('colsample_bytree', 0.4, 1.0),
        'min_child_weight': trial.suggest_int('min_child_weight', 1, 10),
        'gamma': trial.suggest_float('gamma', 1e-8, 1.0, log=True),
        'reg_alpha': trial.suggest_float('reg_alpha', 1e-8, 1.0, log=True),
        'reg_lambda': trial.suggest_float('reg_lambda', 1e-8, 1.0, log=True),
        'tree_method': 'hist',
    }

    pipeline = Pipeline(steps=[
        ('preprocessor', preprocessor_dense),
        ('regressor', XGBRegressor(
            random_state=42,
            objective='reg:squarederror',
            **param
        ))
    ])

    scores = cross_val_score(
        pipeline, X_train, y_train,
        cv=3,
        scoring='neg_mean_absolute_error',
        n_jobs=-1
    )
    return scores.mean()
```

```python
# Run the Optuna study
print("Starting Optuna study for XGBoost")
start_time = time.time()

study = optuna.create_study(
    direction='maximize',
    sampler=TPESampler(seed=42, n_startup_trials=50),
    pruner=MedianPruner(n_startup_trials=10, n_warmup_steps=1),
)
study.optimize(objective, n_trials=1000)

print(f"Best Optuna parameters: {study.best_params}")

# Train the final model with best params
best_xgb_pipeline = Pipeline(steps=[
    ('preprocessor', preprocessor_dense),
    ('regressor', XGBRegressor(
        random_state=42,
        objective='reg:squarederror',
        **study.best_params
    ))
])

print("\n\nTuning completed in {:.2f} seconds".format(time.time() - start_time))
optuna_xgb, optuna_xgb_pred, optuna_xgb_fig =␣
 ↪train_and_evaluate_model(best_xgb_pipeline, X_train, y_train, X_test,␣
 ↪y_test, label="XGBoost (Optuna tuned)")

# Export plot for website
optuna_xgb_fig.write_html('docs/assets/plots/xgboost_optuna_residuals.html',␣
 ↪include_plotlyjs='cdn')
```

```
[I 2026-03-13 19:09:24,577] A new study created in memory with name:
no-name-b877beff-c977-45f5-b878-b78415417996
```

```
Starting Optuna study for XGBoost
```

```
[I 2026-03-13 19:09:26,369] Trial 0 finished with value:
-1.45118023753422 and parameters: {'n_estimators': 437, 'learning_rate':
0.2536999076681772, 'max_depth': 10, 'subsample': 0.7993292420985183,
'colsample_bytree': 0.4936111842654619, 'min_child_weight': 2, 'gamma':
2.9152036385288193e-08, 'reg_alpha': 0.08499808989182997, 'reg_lambda':
0.0006440507553993703}. Best is trial 0 with value: -1.45118023753422.
[I 2026-03-13 19:09:28,613] Trial 1 finished with value:
-1.3262023998385273 and parameters: {'n_estimators': 737, 'learning_rate':
0.010725209743171996, 'max_depth': 12, 'subsample': 0.9162213204002109,
'colsample_bytree': 0.5274034664069657, 'min_child_weight': 2, 'gamma':
2.9324868872723725e-07, 'reg_alpha': 2.716051144654844e-06, 'reg_lambda':
```

```
0.00015777981883364995}. Best is trial 1 with value: -1.3262023998385273.
[I 2026-03-13 19:09:29,888] Trial 2 finished with value:
-1.3304785651812543 and parameters: {'n_estimators': 489, 'learning_rate':
0.02692655251486473, 'max_depth': 8, 'subsample': 0.569746930326021,
'colsample_bytree': 0.5752867891211308, 'min_child_weight': 4, 'gamma':
4.452048365748842e-05, 'reg_alpha': 0.019116469627784252, 'reg_lambda':
3.9572205641009174e-07}. Best is trial 1 with value: -1.3262023998385273.
[I 2026-03-13 19:09:30,933] Trial 3 finished with value:
-1.4117987734941895 and parameters: {'n_estimators': 563, 'learning_rate':
0.07500118950416987, 'max_depth': 2, 'subsample': 0.8037724259507192,
'colsample_bytree': 0.502314474212375, 'min_child_weight': 1, 'gamma':
0.39001768308022033, 'reg_alpha': 0.530953226900921, 'reg_lambda':
0.02932100047183291}. Best is trial 1 with value: -1.3262023998385273.
[I 2026-03-13 19:09:32,225] Trial 4 finished with value:
-1.313665675840023 and parameters: {'n_estimators': 374, 'learning_rate':
0.013940346079873234, 'max_depth': 9, 'subsample': 0.7200762468698007,
'colsample_bytree': 0.47322294090686734, 'min_child_weight': 5, 'gamma':
1.8841183049085085e-08, 'reg_alpha': 0.1881755597772026, 'reg_lambda':
1.1755466083160747e-06}. Best is trial 4 with value: -1.313665675840023.
[I 2026-03-13 19:09:33,418] Trial 5 finished with value:
-1.3349671482102698 and parameters: {'n_estimators': 696, 'learning_rate':
0.028869220380495747, 'max_depth': 7, 'subsample': 0.7733551396716398,
'colsample_bytree': 0.5109126733153162, 'min_child_weight': 10, 'gamma':
0.01588775693167255, 'reg_alpha': 0.32808889626606236, 'reg_lambda':
0.14408501080722544}. Best is trial 4 with value: -1.313665675840023.
[I 2026-03-13 19:09:33,526] Trial 6 finished with value:
-1.4639322798125125 and parameters: {'n_estimators': 638, 'learning_rate':
0.22999586428143728, 'max_depth': 2, 'subsample': 0.5979914312095727,
'colsample_bytree': 0.4271363733463229, 'min_child_weight': 4, 'gamma':
1.2865252594826764e-05, 'reg_alpha': 1.481809088646707e-06, 'reg_lambda':
0.04264813784432918}. Best is trial 4 with value: -1.313665675840023.
[I 2026-03-13 19:09:33,813] Trial 7 finished with value:
-1.3277821992204915 and parameters: {'n_estimators': 421, 'learning_rate':
0.026000059117302653, 'max_depth': 7, 'subsample': 0.5704621124873813,
'colsample_bytree': 0.8813181884524238, 'min_child_weight': 1, 'gamma':
0.7854083114461319, 'reg_alpha': 0.015064619068942013, 'reg_lambda':
3.8879928024075543e-07}. Best is trial 4 with value: -1.313665675840023.
[I 2026-03-13 19:09:33,975] Trial 8 finished with value:
-1.3878529673047189 and parameters: {'n_estimators': 104, 'learning_rate':
0.1601531217136121, 'max_depth': 9, 'subsample': 0.8645035840204937,
'colsample_bytree': 0.8627622080115674, 'min_child_weight': 1, 'gamma':
7.374385355858303e-06, 'reg_alpha': 8.451863533931625e-08, 'reg_lambda':
0.08032068562667222}. Best is trial 4 with value: -1.313665675840023.
[I 2026-03-13 19:09:34,094] Trial 9 finished with value:
-1.4145578995682413 and parameters: {'n_estimators': 661, 'learning_rate':
0.030816017044468066, 'max_depth': 2, 'subsample': 0.6554911608578311,
'colsample_bytree': 0.5951099932160482, 'min_child_weight': 8, 'gamma':
0.0012602588933700108, 'reg_alpha': 0.12522814303053625, 'reg_lambda':
```

```
5.994036749692399e-05}. Best is trial 4 with value: -1.313665675840023.
[I 2026-03-13 19:09:34,286] Trial 10 finished with value:
-1.3801874370167566 and parameters: {'n_estimators': 207, 'learning_rate':
0.1131225105716033, 'max_depth': 10, 'subsample': 0.7806385987847482,
'colsample_bytree': 0.8625803079727365, 'min_child_weight': 5, 'gamma':
0.00015200666650051572, 'reg_alpha': 2.632256136809142e-05, 'reg_lambda':
1.5971768764426203e-08}. Best is trial 4 with value: -1.313665675840023.
[I 2026-03-13 19:09:34,447] Trial 11 finished with value:
-1.4024530753849602 and parameters: {'n_estimators': 197, 'learning_rate':
0.011128194768838964, 'max_depth': 9, 'subsample': 0.6571779905381634,
'colsample_bytree': 0.7051424146988217, 'min_child_weight': 10, 'gamma':
9.8704700073667e-07, 'reg_alpha': 1.918948786414487e-05, 'reg_lambda':
0.011076667222301087}. Best is trial 4 with value: -1.313665675840023.
[I 2026-03-13 19:09:34,586] Trial 12 finished with value:
-1.3822059079707396 and parameters: {'n_estimators': 306, 'learning_rate':
0.012992976740520353, 'max_depth': 5, 'subsample': 0.5806106436270022,
'colsample_bytree': 0.9578185914055438, 'min_child_weight': 9, 'gamma':
0.001167427856762126, 'reg_alpha': 0.0936881625557718, 'reg_lambda':
0.026876741555610546}. Best is trial 4 with value: -1.313665675840023.
[I 2026-03-13 19:09:34,757] Trial 13 finished with value:
-1.4409088972732798 and parameters: {'n_estimators': 268, 'learning_rate':
0.20816986844858934, 'max_depth': 7, 'subsample': 0.9037200775820313,
'colsample_bytree': 0.937654779954096, 'min_child_weight': 4, 'gamma':
7.593034903208851e-08, 'reg_alpha': 6.660108438734075e-07, 'reg_lambda':
2.6113333123958975e-05}. Best is trial 4 with value: -1.313665675840023.
[I 2026-03-13 19:09:34,897] Trial 14 finished with value:
-1.4589217997385677 and parameters: {'n_estimators': 837, 'learning_rate':
0.18681142751959703, 'max_depth': 2, 'subsample': 0.7553736512887829,
'colsample_bytree': 0.6504466018892674, 'min_child_weight': 3, 'gamma':
9.09751831698106e-08, 'reg_alpha': 5.022516489125307e-06, 'reg_lambda':
0.3493635816569516}. Best is trial 4 with value: -1.313665675840023.
[I 2026-03-13 19:09:35,174] Trial 15 finished with value:
-1.3744776200243594 and parameters: {'n_estimators': 391, 'learning_rate':
0.05838706626832953, 'max_depth': 9, 'subsample': 0.681814801189647,
'colsample_bytree': 0.9830692496325764, 'min_child_weight': 10, 'gamma':
1.033375989766645e-06, 'reg_alpha': 9.505786310154948e-05, 'reg_lambda':
2.5528569432507475e-06}. Best is trial 4 with value: -1.313665675840023.
[I 2026-03-13 19:09:35,429] Trial 16 finished with value:
-1.329456322791981 and parameters: {'n_estimators': 356, 'learning_rate':
0.011336695817840537, 'max_depth': 8, 'subsample': 0.7513395116144308,
'colsample_bytree': 0.4308872507499936, 'min_child_weight': 3, 'gamma':
0.18455554056364123, 'reg_alpha': 8.25078224477252e-07, 'reg_lambda':
1.4426433452224793e-07}. Best is trial 4 with value: -1.313665675840023.
[I 2026-03-13 19:09:35,590] Trial 17 finished with value:
-1.5024512317253258 and parameters: {'n_estimators': 540, 'learning_rate':
0.28570986468059545, 'max_depth': 4, 'subsample': 0.8360677737029393,
'colsample_bytree': 0.8569717691972305, 'min_child_weight': 3, 'gamma':
0.0066946948903108105, 'reg_alpha': 8.755179646617154e-06, 'reg_lambda':
```

```
0.001144054318382189}. Best is trial 4 with value: -1.313665675840023.
[I 2026-03-13 19:09:35,708] Trial 18 finished with value:
-1.4036960483085723 and parameters: {'n_estimators': 670, 'learning_rate':
0.06185918211556436, 'max_depth': 2, 'subsample': 0.917651247794619,
'colsample_bytree': 0.5924680389830415, 'min_child_weight': 2, 'gamma':
2.1193424275165945e-08, 'reg_alpha': 0.0005335112055356552, 'reg_lambda':
0.002633388257869831}. Best is trial 4 with value: -1.313665675840023.
[I 2026-03-13 19:09:35,764] Trial 19 finished with value:
-1.3901989865313027 and parameters: {'n_estimators': 114, 'learning_rate':
0.05707205898008386, 'max_depth': 4, 'subsample': 0.822586395204725,
'colsample_bytree': 0.5046198574029949, 'min_child_weight': 7, 'gamma':
1.2413171922260375e-05, 'reg_alpha': 0.31177401074836186, 'reg_lambda':
1.2594112057549154e-07}. Best is trial 4 with value: -1.313665675840023.
[I 2026-03-13 19:09:36,142] Trial 20 finished with value:
-1.3129350180233963 and parameters: {'n_estimators': 407, 'learning_rate':
0.01471005003927122, 'max_depth': 12, 'subsample': 0.9386696766904905,
'colsample_bytree': 0.5547649766290934, 'min_child_weight': 7, 'gamma':
0.03449670395050202, 'reg_alpha': 0.00027644356998183205, 'reg_lambda':
0.0001726651247889248}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:36,639] Trial 21 finished with value:
-1.328243263849597 and parameters: {'n_estimators': 317, 'learning_rate':
0.01372537052078299, 'max_depth': 11, 'subsample': 0.9502090285816652,
'colsample_bytree': 0.7798608743639608, 'min_child_weight': 4, 'gamma':
6.218370535731491e-06, 'reg_alpha': 0.006421632248858025, 'reg_lambda':
0.15027338858202638}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:37,073] Trial 22 finished with value:
-1.4638285335390162 and parameters: {'n_estimators': 899, 'learning_rate':
0.1418961933595484, 'max_depth': 9, 'subsample': 0.5420699824975244,
'colsample_bytree': 0.4969772284567683, 'min_child_weight': 9, 'gamma':
0.0007102847286095368, 'reg_alpha': 1.1846127780587211e-08, 'reg_lambda':
6.482945086231278e-08}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:37,233] Trial 23 finished with value:
-1.4099499911890623 and parameters: {'n_estimators': 697, 'learning_rate':
0.01017364485544501, 'max_depth': 3, 'subsample': 0.7743668946832931,
'colsample_bytree': 0.815137118615616, 'min_child_weight': 7, 'gamma':
6.225216726492816e-07, 'reg_alpha': 0.004982344702102993, 'reg_lambda':
7.906653376497142e-07}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:37,498] Trial 24 finished with value:
-1.3753774384980986 and parameters: {'n_estimators': 393, 'learning_rate':
0.12666549392400608, 'max_depth': 9, 'subsample': 0.924611705247089,
'colsample_bytree': 0.7945677353802061, 'min_child_weight': 6, 'gamma':
5.61562556681518e-08, 'reg_alpha': 8.74432769454909e-06, 'reg_lambda':
1.323180041438319e-06}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:37,638] Trial 25 finished with value:
-1.4458483939223064 and parameters: {'n_estimators': 319, 'learning_rate':
0.2736872535680783, 'max_depth': 6, 'subsample': 0.9460232775885566,
'colsample_bytree': 0.7786831755983578, 'min_child_weight': 8, 'gamma':
0.0001049776254156589, 'reg_alpha': 0.00041231684401416913, 'reg_lambda':
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8.712475112920381e-05}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:37,771] Trial 26 finished with value:
-1.4028927265925117 and parameters: {'n_estimators': 275, 'learning_rate':
0.1167211008595864, 'max_depth': 5, 'subsample': 0.5121579832157269,
'colsample_bytree': 0.7872833775443007, 'min_child_weight': 2, 'gamma':
0.333940178385249, 'reg_alpha': 0.4279850668891566, 'reg_lambda':
0.2084083531336826}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:38,163] Trial 27 finished with value:
-1.3264623985811126 and parameters: {'n_estimators': 433, 'learning_rate':
0.010539773956821628, 'max_depth': 12, 'subsample': 0.7140920741586572,
'colsample_bytree': 0.9799928914262017, 'min_child_weight': 10, 'gamma':
0.06669229211616488, 'reg_alpha': 2.2677289093956233e-06, 'reg_lambda':
1.2044307332953856e-05}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:38,355] Trial 28 finished with value:
-1.388179066961964 and parameters: {'n_estimators': 866, 'learning_rate':
0.029385442161128397, 'max_depth': 3, 'subsample': 0.7784006312291751,
'colsample_bytree': 0.9616928644964686, 'min_child_weight': 7, 'gamma':
0.0003634873997738577, 'reg_alpha': 5.989794878601609e-08, 'reg_lambda':
0.0008318745034635068}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:38,862] Trial 29 finished with value:
-1.328342203090962 and parameters: {'n_estimators': 992, 'learning_rate':
0.01610352961694889, 'max_depth': 7, 'subsample': 0.9386865359639778,
'colsample_bytree': 0.8444611706525227, 'min_child_weight': 7, 'gamma':
0.0041674718203826185, 'reg_alpha': 7.515003890531393e-06, 'reg_lambda':
2.2322084823690722e-06}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:39,127] Trial 30 finished with value:
-1.370774992970681 and parameters: {'n_estimators': 829, 'learning_rate':
0.1572663162015696, 'max_depth': 11, 'subsample': 0.9566202762782356,
'colsample_bytree': 0.7068054393165627, 'min_child_weight': 6, 'gamma':
0.02434228998214791, 'reg_alpha': 0.0015838405080674095, 'reg_lambda':
0.0041279556062565675}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:39,361] Trial 31 finished with value:
-1.4521844233071917 and parameters: {'n_estimators': 817, 'learning_rate':
0.2063696403063293, 'max_depth': 5, 'subsample': 0.687791476319972,
'colsample_bytree': 0.45638916390452144, 'min_child_weight': 6, 'gamma':
1.9388231219417792e-08, 'reg_alpha': 5.3062064728788974e-05, 'reg_lambda':
0.0002193598182745354}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:39,438] Trial 32 finished with value:
-1.4200858107272556 and parameters: {'n_estimators': 358, 'learning_rate':
0.07459889030569367, 'max_depth': 2, 'subsample': 0.5186740943746072,
'colsample_bytree': 0.8935603363957949, 'min_child_weight': 4, 'gamma':
1.0386855772080306e-07, 'reg_alpha': 0.00015064200259336973, 'reg_lambda':
0.014452681258412942}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:39,505] Trial 33 finished with value:
-1.4211498454059386 and parameters: {'n_estimators': 294, 'learning_rate':
0.08319261221667239, 'max_depth': 2, 'subsample': 0.5258408605843039,
'colsample_bytree': 0.7188127789408888, 'min_child_weight': 6, 'gamma':
0.001257300858183282, 'reg_alpha': 0.006437699040892684, 'reg_lambda':
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0.6409389074134212}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:40,243] Trial 34 finished with value:
-1.326923299828324 and parameters: {'n_estimators': 565, 'learning_rate':
0.029994793096631372, 'max_depth': 10, 'subsample': 0.6354161256310371,
'colsample_bytree': 0.6633828524233817, 'min_child_weight': 1, 'gamma':
1.595166231333429e-08, 'reg_alpha': 0.502559303800778, 'reg_lambda':
0.04873499929269755}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:40,393] Trial 35 finished with value:
-1.3751390421761254 and parameters: {'n_estimators': 727, 'learning_rate':
0.04018584329485903, 'max_depth': 3, 'subsample': 0.5782185213355431,
'colsample_bytree': 0.5501457388987572, 'min_child_weight': 6, 'gamma':
0.005209156216369749, 'reg_alpha': 0.001912401227325431, 'reg_lambda':
1.7356860603349111e-06}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:40,959] Trial 36 finished with value:
-1.3913939189363835 and parameters: {'n_estimators': 960, 'learning_rate':
0.12301645833176054, 'max_depth': 8, 'subsample': 0.8058603731171761,
'colsample_bytree': 0.6517600374566739, 'min_child_weight': 3, 'gamma':
7.043385017900836e-06, 'reg_alpha': 0.011554971634978302, 'reg_lambda':
1.3036106776477418e-08}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:41,015] Trial 37 finished with value:
-1.5494056522560264 and parameters: {'n_estimators': 204, 'learning_rate':
0.011693687409928539, 'max_depth': 2, 'subsample': 0.9277302920055036,
'colsample_bytree': 0.8221947156280143, 'min_child_weight': 5, 'gamma':
6.06280060403243e-08, 'reg_alpha': 8.568938007937746e-05, 'reg_lambda':
6.134429310056972e-05}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:41,174] Trial 38 finished with value:
-1.3302952503970606 and parameters: {'n_estimators': 256, 'learning_rate':
0.043737251953017454, 'max_depth': 6, 'subsample': 0.8079250490261083,
'colsample_bytree': 0.7810561905205863, 'min_child_weight': 1, 'gamma':
9.928894978066514e-06, 'reg_alpha': 0.0010159663543783963, 'reg_lambda':
0.00010594734273401198}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:41,367] Trial 39 finished with value:
-1.4115992773274915 and parameters: {'n_estimators': 871, 'learning_rate':
0.09396591857488666, 'max_depth': 3, 'subsample': 0.5352843737002149,
'colsample_bytree': 0.7854515669237894, 'min_child_weight': 1, 'gamma':
0.00048551723386735715, 'reg_alpha': 0.33253849842431976, 'reg_lambda':
0.00040159974003412744}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:41,558] Trial 40 finished with value:
-1.3705073670774157 and parameters: {'n_estimators': 449, 'learning_rate':
0.08916916511121768, 'max_depth': 7, 'subsample': 0.7728083946579675,
'colsample_bytree': 0.9648788852659151, 'min_child_weight': 4, 'gamma':
0.48924287610878064, 'reg_alpha': 0.17490617229671115, 'reg_lambda':
3.684080220204264e-07}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:41,614] Trial 41 finished with value:
-1.5486747463288033 and parameters: {'n_estimators': 162, 'learning_rate':
0.014088388780829922, 'max_depth': 2, 'subsample': 0.5472214803779643,
'colsample_bytree': 0.8098040640498141, 'min_child_weight': 1, 'gamma':
3.5629115607768247e-06, 'reg_alpha': 0.05741197509103047, 'reg_lambda':
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1.5352372026024416e-08}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:41,795] Trial 42 finished with value:
-1.381671723202144 and parameters: {'n_estimators': 833, 'learning_rate':
0.026081566878799544, 'max_depth': 3, 'subsample': 0.8483685826820753,
'colsample_bytree': 0.7773657080679304, 'min_child_weight': 9, 'gamma':
0.007595709565769595, 'reg_alpha': 0.026782273644595064, 'reg_lambda':
1.8041663589561627e-06}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:41,945] Trial 43 finished with value:
-1.372570565771585 and parameters: {'n_estimators': 259, 'learning_rate':
0.12845440557093132, 'max_depth': 10, 'subsample': 0.9952525710003366,
'colsample_bytree': 0.6475706061468559, 'min_child_weight': 4, 'gamma':
0.016266858388908328, 'reg_alpha': 5.326333087961553e-06, 'reg_lambda':
0.27929206998030887}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:42,423] Trial 44 finished with value:
-1.3683107057230532 and parameters: {'n_estimators': 873, 'learning_rate':
0.04302057353833569, 'max_depth': 10, 'subsample': 0.8772714370423411,
'colsample_bytree': 0.4618743213015596, 'min_child_weight': 10, 'gamma':
0.00011015874501794107, 'reg_alpha': 0.04089401282581006, 'reg_lambda':
3.634099452501576e-06}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:42,562] Trial 45 finished with value:
-1.4100250249560602 and parameters: {'n_estimators': 906, 'learning_rate':
0.03757492064184006, 'max_depth': 2, 'subsample': 0.9526909882096318,
'colsample_bytree': 0.45477200607168017, 'min_child_weight': 4, 'gamma':
0.398561859498043, 'reg_alpha': 0.40258461633605375, 'reg_lambda':
0.0003868148201538438}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:42,796] Trial 46 finished with value:
-1.3569401587389687 and parameters: {'n_estimators': 669, 'learning_rate':
0.04596300474168577, 'max_depth': 5, 'subsample': 0.664332272684958,
'colsample_bytree': 0.803511073646223, 'min_child_weight': 8, 'gamma':
0.021509600774385743, 'reg_alpha': 0.020746514425671027, 'reg_lambda':
5.3659772168521995e-08}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:43,251] Trial 47 finished with value:
-1.31545051205981 and parameters: {'n_estimators': 545, 'learning_rate':
0.012162455598956208, 'max_depth': 8, 'subsample': 0.7207652506866885,
'colsample_bytree': 0.9326225096549798, 'min_child_weight': 4, 'gamma':
8.640445424597341e-08, 'reg_alpha': 1.3929433566793024e-07, 'reg_lambda':
0.01236189509386435}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:43,360] Trial 48 finished with value:
-1.4468537446290757 and parameters: {'n_estimators': 657, 'learning_rate':
0.014104914380026676, 'max_depth': 2, 'subsample': 0.85048456572956,
'colsample_bytree': 0.44365780381851616, 'min_child_weight': 9, 'gamma':
0.004466196400953584, 'reg_alpha': 4.474971205201382e-08, 'reg_lambda':
4.772014039122303e-08}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:43,762] Trial 49 finished with value:
-1.3615266467584315 and parameters: {'n_estimators': 988, 'learning_rate':
0.035714399098231424, 'max_depth': 6, 'subsample': 0.9063997836287513,
'colsample_bytree': 0.9683491464303152, 'min_child_weight': 10, 'gamma':
0.010642054582625494, 'reg_alpha': 1.0234736931287434e-05, 'reg_lambda':
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4.6559224026716126e-08}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:44,171] Trial 50 finished with value:
-1.3248255782091396 and parameters: {'n_estimators': 479, 'learning_rate':
0.02010955519009952, 'max_depth': 12, 'subsample': 0.7241136310388194,
'colsample_bytree': 0.4058758016349803, 'min_child_weight': 5, 'gamma':
0.07554522855973493, 'reg_alpha': 2.234594608119641e-07, 'reg_lambda':
8.28602811973198e-06}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:44,633] Trial 51 finished with value:
-1.3159968576818188 and parameters: {'n_estimators': 506, 'learning_rate':
0.020034225290250236, 'max_depth': 12, 'subsample': 0.721316338398581,
'colsample_bytree': 0.533005459947811, 'min_child_weight': 5, 'gamma':
0.06439773020095037, 'reg_alpha': 2.2123702369864033e-07, 'reg_lambda':
1.2234085522731741e-05}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:45,105] Trial 52 finished with value:
-1.3162595101626138 and parameters: {'n_estimators': 505, 'learning_rate':
0.019786504340349172, 'max_depth': 12, 'subsample': 0.7258681378060184,
'colsample_bytree': 0.5338955304265807, 'min_child_weight': 5, 'gamma':
2.1657654267029158e-07, 'reg_alpha': 2.487090366313271e-07, 'reg_lambda':
1.0964910539853257e-05}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:45,523] Trial 53 finished with value:
-1.3289876550398405 and parameters: {'n_estimators': 500, 'learning_rate':
0.020674700972030908, 'max_depth': 12, 'subsample': 0.7190650837455671,
'colsample_bytree': 0.401824894514932, 'min_child_weight': 5, 'gamma':
2.082863834829511e-07, 'reg_alpha': 2.8637822957167554e-07, 'reg_lambda':
8.723241769969185e-06}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:46,006] Trial 54 finished with value:
-1.327634492648012 and parameters: {'n_estimators': 601, 'learning_rate':
0.02069372349435501, 'max_depth': 11, 'subsample': 0.7298517107118269,
'colsample_bytree': 0.530918580724198, 'min_child_weight': 5, 'gamma':
0.06651927757295294, 'reg_alpha': 2.1108153857037968e-07, 'reg_lambda':
1.5530023903525703e-05}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:46,425] Trial 55 finished with value:
-1.3250606296518495 and parameters: {'n_estimators': 484, 'learning_rate':
0.019293369292093035, 'max_depth': 12, 'subsample': 0.6239915229880224,
'colsample_bytree': 0.47784370595888703, 'min_child_weight': 5, 'gamma':
0.08006055101948535, 'reg_alpha': 1.4850675363063481e-08, 'reg_lambda':
6.215695430953203e-06}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:46,897] Trial 56 finished with value:
-1.3162814531596754 and parameters: {'n_estimators': 525, 'learning_rate':
0.018203125201386502, 'max_depth': 11, 'subsample': 0.7280932764613514,
'colsample_bytree': 0.5530342415283377, 'min_child_weight': 5, 'gamma':
1.0109227306287384e-08, 'reg_alpha': 1.6538167467988516e-07, 'reg_lambda':
3.0794032586027305e-05}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:47,377] Trial 57 finished with value:
-1.316613076904268 and parameters: {'n_estimators': 542, 'learning_rate':
0.01630141562090787, 'max_depth': 11, 'subsample': 0.6927311881149363,
'colsample_bytree': 0.5625320713185895, 'min_child_weight': 5, 'gamma':
1.2501539671681375e-08, 'reg_alpha': 8.243447731829919e-07, 'reg_lambda':
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3.1206262141740314e-05}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:47,962] Trial 58 finished with value:
-1.313287462417452 and parameters: {'n_estimators': 602, 'learning_rate':
0.01718764658860914, 'max_depth': 11, 'subsample': 0.7025927495469761,
'colsample_bytree': 0.5921153520474509, 'min_child_weight': 4, 'gamma':
2.784279686578764e-07, 'reg_alpha': 1.1503667535860077e-07, 'reg_lambda':
6.977765423911515e-07}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:48,683] Trial 59 finished with value:
-1.3198342751408696 and parameters: {'n_estimators': 594, 'learning_rate':
0.01637808002299408, 'max_depth': 12, 'subsample': 0.7485072479744433,
'colsample_bytree': 0.5969910499950863, 'min_child_weight': 3, 'gamma':
3.5661585126827086e-07, 'reg_alpha': 2.073826870497199e-08, 'reg_lambda':
1.5966730960326164e-07}. Best is trial 20 with value: -1.3129350180233963.
[I 2026-03-13 19:09:49,277] Trial 60 finished with value:
-1.311806050304501 and parameters: {'n_estimators': 600, 'learning_rate':
0.012963173115853209, 'max_depth': 10, 'subsample': 0.7085722164808883,
'colsample_bytree': 0.6022191553025031, 'min_child_weight': 3, 'gamma':
1.4835544397950506e-06, 'reg_alpha': 4.2962938559257e-07, 'reg_lambda':
8.179592868336501e-07}. Best is trial 60 with value: -1.311806050304501.
[I 2026-03-13 19:09:49,842] Trial 61 finished with value:
-1.3178699926754451 and parameters: {'n_estimators': 455, 'learning_rate':
0.012720477170240446, 'max_depth': 10, 'subsample': 0.7013368595060636,
'colsample_bytree': 0.6106145921298084, 'min_child_weight': 2, 'gamma':
3.216851337836046e-05, 'reg_alpha': 4.7493570631238006e-07, 'reg_lambda':
5.029104880681705e-07}. Best is trial 60 with value: -1.311806050304501.
[I 2026-03-13 19:09:50,426] Trial 62 finished with value:
-1.3367850718126395 and parameters: {'n_estimators': 614, 'learning_rate':
0.022671938487097355, 'max_depth': 11, 'subsample': 0.6729647480881621,
'colsample_bytree': 0.5299130660052394, 'min_child_weight': 3, 'gamma':
2.315444489636024e-06, 'reg_alpha': 8.827857823315936e-08, 'reg_lambda':
3.776482713923282e-06}. Best is trial 60 with value: -1.311806050304501.
[I 2026-03-13 19:09:50,740] Trial 63 finished with value:
-1.317608490121244 and parameters: {'n_estimators': 399, 'learning_rate':
0.02263478962331882, 'max_depth': 9, 'subsample': 0.7434270605538412,
'colsample_bytree': 0.5817869424064224, 'min_child_weight': 4, 'gamma':
1.1916116408548623e-06, 'reg_alpha': 2.270308564674768e-06, 'reg_lambda':
8.665795014777379e-07}. Best is trial 60 with value: -1.311806050304501.
[I 2026-03-13 19:09:51,117] Trial 64 finished with value:
-1.3116770028468556 and parameters: {'n_estimators': 569, 'learning_rate':
0.0156375991565408, 'max_depth': 8, 'subsample': 0.7041589570330636,
'colsample_bytree': 0.6212425904679308, 'min_child_weight': 4, 'gamma':
1.574013485660075e-07, 'reg_alpha': 1.1603412402383825e-07, 'reg_lambda':
4.578239311431802e-06}. Best is trial 64 with value: -1.3116770028468556.
[I 2026-03-13 19:09:51,539] Trial 65 finished with value:
-1.308516462605467 and parameters: {'n_estimators': 584, 'learning_rate':
0.012094268972097493, 'max_depth': 8, 'subsample': 0.6473046911793393,
'colsample_bytree': 0.615740459512827, 'min_child_weight': 3, 'gamma':
4.560203237812357e-08, 'reg_alpha': 2.848399752714048e-08, 'reg_lambda':
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2.4064293942942266e-07}. Best is trial 65 with value: -1.308516462605467.
[I 2026-03-13 19:09:52,000] Trial 66 finished with value:
-1.3137879396713499 and parameters: {'n_estimators': 562, 'learning_rate':
0.011926166963179292, 'max_depth': 8, 'subsample': 0.6459201461297663,
'colsample_bytree': 0.6188174393177989, 'min_child_weight': 2, 'gamma':
1.4346390707834547e-07, 'reg_alpha': 3.365727753256263e-08, 'reg_lambda':
2.0567020149858406e-07}. Best is trial 65 with value: -1.308516462605467.
[I 2026-03-13 19:09:52,482] Trial 67 finished with value:
-1.3188668077501287 and parameters: {'n_estimators': 633, 'learning_rate':
0.014899292894139627, 'max_depth': 8, 'subsample': 0.6371456159072038,
'colsample_bytree': 0.620425789403036, 'min_child_weight': 2, 'gamma':
4.401507024746326e-08, 'reg_alpha': 2.6444542423274264e-08, 'reg_lambda':
1.90851875383529e-07}. Best is trial 65 with value: -1.308516462605467.
[I 2026-03-13 19:09:52,996] Trial 68 finished with value:
-1.3151160155194745 and parameters: {'n_estimators': 716, 'learning_rate':
0.01008290617319699, 'max_depth': 8, 'subsample': 0.6194108511432502,
'colsample_bytree': 0.674265480424767, 'min_child_weight': 3, 'gamma':
2.7977094638741255e-08, 'reg_alpha': 3.431385048527318e-08, 'reg_lambda':
3.624814042921952e-07}. Best is trial 65 with value: -1.308516462605467.
[I 2026-03-13 19:09:53,519] Trial 69 finished with value:
-1.3202186550112982 and parameters: {'n_estimators': 574, 'learning_rate':
0.01736428101089938, 'max_depth': 9, 'subsample': 0.593855342324098,
'colsample_bytree': 0.6251198713306719, 'min_child_weight': 2, 'gamma':
4.854095877172464e-07, 'reg_alpha': 7.561208389955407e-08, 'reg_lambda':
9.691082456543286e-07}. Best is trial 65 with value: -1.308516462605467.
[I 2026-03-13 19:09:54,189] Trial 70 finished with value:
-1.305420030126178 and parameters: {'n_estimators': 762, 'learning_rate':
0.011307704969362937, 'max_depth': 9, 'subsample': 0.6481767408141583,
'colsample_bytree': 0.737780177933441, 'min_child_weight': 3, 'gamma':
2.2586633778326592e-07, 'reg_alpha': 1.1602245572832635e-07, 'reg_lambda':
2.5849389604804524e-08}. Best is trial 70 with value: -1.305420030126178.
[I 2026-03-13 19:09:54,846] Trial 71 finished with value:
-1.3125216234432695 and parameters: {'n_estimators': 777, 'learning_rate':
0.012627319524452446, 'max_depth': 9, 'subsample': 0.6521862955940185,
'colsample_bytree': 0.7233009724853503, 'min_child_weight': 3, 'gamma':
1.4587828416179727e-07, 'reg_alpha': 1.0092213414272698e-08, 'reg_lambda':
2.876050133704177e-08}. Best is trial 70 with value: -1.305420030126178.
[I 2026-03-13 19:09:55,525] Trial 72 finished with value:
-1.3169924698059328 and parameters: {'n_estimators': 806, 'learning_rate':
0.01514549272445513, 'max_depth': 9, 'subsample': 0.7037306022786002,
'colsample_bytree': 0.7392190818697733, 'min_child_weight': 3, 'gamma':
4.2681497505175777e-08, 'reg_alpha': 1.1057796749420819e-08, 'reg_lambda':
2.5459342669791657e-08}. Best is trial 70 with value: -1.305420030126178.
[I 2026-03-13 19:09:56,309] Trial 73 finished with value:
-1.3157867258574287 and parameters: {'n_estimators': 767, 'learning_rate':
0.011103901343409375, 'max_depth': 10, 'subsample': 0.6751001526930811,
'colsample_bytree': 0.7261428234218371, 'min_child_weight': 3, 'gamma':
7.968230154721994e-07, 'reg_alpha': 4.738086800840227e-07, 'reg_lambda':
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2.302576669329599e-08}. Best is trial 70 with value: -1.305420030126178.
[I 2026-03-13 19:09:56,938] Trial 74 finished with value:
-1.3129854853583902 and parameters: {'n_estimators': 764, 'learning_rate':
0.013029948975736486, 'max_depth': 9, 'subsample': 0.6552790972022726,
'colsample_bytree': 0.6849785535896667, 'min_child_weight': 3, 'gamma':
1.429080839068681e-07, 'reg_alpha': 1.1676605857575204e-07, 'reg_lambda':
1.0487177444469863e-07}. Best is trial 70 with value: -1.305420030126178.
[I 2026-03-13 19:09:57,557] Trial 75 finished with value:
-1.321934492569004 and parameters: {'n_estimators': 765, 'learning_rate':
0.013176755997120601, 'max_depth': 9, 'subsample': 0.6093069337507918,
'colsample_bytree': 0.6815362247902117, 'min_child_weight': 3, 'gamma':
1.562216740231921e-06, 'reg_alpha': 1.4273291781810705e-06, 'reg_lambda':
9.99501425722932e-08}. Best is trial 70 with value: -1.305420030126178.
[I 2026-03-13 19:09:58,323] Trial 76 finished with value:
-1.3200689174431013 and parameters: {'n_estimators': 778, 'learning_rate':
0.01271081056012699, 'max_depth': 10, 'subsample': 0.6511743504639513,
'colsample_bytree': 0.7531470287564548, 'min_child_weight': 3, 'gamma':
1.397060417299155e-07, 'reg_alpha': 8.868509968210481e-08, 'reg_lambda':
8.351846258233333e-08}. Best is trial 70 with value: -1.305420030126178.
[I 2026-03-13 19:09:58,733] Trial 77 finished with value:
-1.3097707207662126 and parameters: {'n_estimators': 747, 'learning_rate':
0.015337319916676072, 'max_depth': 7, 'subsample': 0.655382572003552,
'colsample_bytree': 0.6946623506738189, 'min_child_weight': 4, 'gamma':
3.3026030506127726e-07, 'reg_alpha': 0.0002227098755309329, 'reg_lambda':
2.771990046553235e-08}. Best is trial 70 with value: -1.305420030126178.
[I 2026-03-13 19:09:59,130] Trial 78 finished with value:
-1.317292181554034 and parameters: {'n_estimators': 698, 'learning_rate':
0.014975373361687652, 'max_depth': 7, 'subsample': 0.6611130988535905,
'colsample_bytree': 0.6989780100826285, 'min_child_weight': 3, 'gamma':
4.914649602860212e-07, 'reg_alpha': 0.0004523571981798758, 'reg_lambda':
1.0047035957436515e-08}. Best is trial 70 with value: -1.305420030126178.
[I 2026-03-13 19:09:59,610] Trial 79 finished with value:
-1.3199067136434008 and parameters: {'n_estimators': 746, 'learning_rate':
0.010944017778477694, 'max_depth': 7, 'subsample': 0.5948834502385237,
'colsample_bytree': 0.7573145875246478, 'min_child_weight': 2, 'gamma':
1.6023256393801358e-07, 'reg_alpha': 0.0002746665626752249, 'reg_lambda':
2.8624642598378873e-08}. Best is trial 70 with value: -1.305420030126178.
[I 2026-03-13 19:10:00,123] Trial 80 finished with value:
-1.3183081326650083 and parameters: {'n_estimators': 806, 'learning_rate':
0.013740127024088764, 'max_depth': 8, 'subsample': 0.682021515111558,
'colsample_bytree': 0.635935541226027, 'min_child_weight': 4, 'gamma':
3.941011330532732e-06, 'reg_alpha': 0.00020354846614850574, 'reg_lambda':
2.665579753706631e-07}. Best is trial 70 with value: -1.305420030126178.
[I 2026-03-13 19:10:00,614] Trial 81 finished with value:
-1.3132324891966076 and parameters: {'n_estimators': 646, 'learning_rate':
0.017729725480977163, 'max_depth': 9, 'subsample': 0.6363774488410026,
'colsample_bytree': 0.6843027432843674, 'min_child_weight': 4, 'gamma':
2.457501706606142e-07, 'reg_alpha': 4.854963593004428e-08, 'reg_lambda':
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4.899273633902032e-07}. Best is trial 70 with value: -1.305420030126178.
[I 2026-03-13 19:10:01,085] Trial 82 finished with value:
-1.3128325612330436 and parameters: {'n_estimators': 638, 'learning_rate':
0.011710775599459555, 'max_depth': 8, 'subsample': 0.6360751542011542,
'colsample_bytree': 0.6963021387511413, 'min_child_weight': 3, 'gamma':
3.753166873953469e-07, 'reg_alpha': 4.5403810105727215e-08, 'reg_lambda':
3.867003740849314e-08}. Best is trial 70 with value: -1.305420030126178.
[I 2026-03-13 19:10:01,596] Trial 83 finished with value:
-1.3058719977070636 and parameters: {'n_estimators': 692, 'learning_rate':
0.01166250794789554, 'max_depth': 8, 'subsample': 0.6466383378649535,
'colsample_bytree': 0.7273950784275942, 'min_child_weight': 3, 'gamma':
2.7508862671297476e-05, 'reg_alpha': 1.719562925381163e-08, 'reg_lambda':
3.545839440018068e-08}. Best is trial 70 with value: -1.305420030126178.
[I 2026-03-13 19:10:02,047] Trial 84 finished with value:
-1.3197453634670622 and parameters: {'n_estimators': 692, 'learning_rate':
0.010105005142967638, 'max_depth': 7, 'subsample': 0.622350800937441,
'colsample_bytree': 0.716795725954746, 'min_child_weight': 2, 'gamma':
2.212111384579829e-05, 'reg_alpha': 1.8692475700100642e-08, 'reg_lambda':
2.6856241192316653e-08}. Best is trial 70 with value: -1.305420030126178.
[I 2026-03-13 19:10:02,538] Trial 85 finished with value:
-1.308840275598642 and parameters: {'n_estimators': 691, 'learning_rate':
0.0116552321449637, 'max_depth': 8, 'subsample': 0.670333126056827,
'colsample_bytree': 0.6633150276630054, 'min_child_weight': 3, 'gamma':
6.666315322139942e-05, 'reg_alpha': 3.3397325876026175e-05, 'reg_lambda':
3.3095323180288025e-08}. Best is trial 70 with value: -1.305420030126178.
[I 2026-03-13 19:10:02,987] Trial 86 finished with value:
-1.3050789204668365 and parameters: {'n_estimators': 625, 'learning_rate':
0.011846328490064239, 'max_depth': 8, 'subsample': 0.6693346900339943,
'colsample_bytree': 0.6579656796771788, 'min_child_weight': 3, 'gamma':
4.5785254759642116e-05, 'reg_alpha': 5.5265528051005855e-05, 'reg_lambda':
4.295275706848416e-08}. Best is trial 86 with value: -1.3050789204668365.
[I 2026-03-13 19:10:03,385] Trial 87 finished with value:
-1.3165631680715664 and parameters: {'n_estimators': 679, 'learning_rate':
0.011173873573147173, 'max_depth': 7, 'subsample': 0.6666309386520657,
'colsample_bytree': 0.6656467225895544, 'min_child_weight': 3, 'gamma':
6.740865983252215e-05, 'reg_alpha': 2.834124191381479e-05, 'reg_lambda':
7.558074214957483e-08}. Best is trial 86 with value: -1.3050789204668365.
[I 2026-03-13 19:10:03,857] Trial 88 finished with value:
-1.3072420135731992 and parameters: {'n_estimators': 729, 'learning_rate':
0.012248586290158894, 'max_depth': 8, 'subsample': 0.6108627927127818,
'colsample_bytree': 0.6420294554376753, 'min_child_weight': 4, 'gamma':
0.0002235108703559542, 'reg_alpha': 4.2088621365836735e-05, 'reg_lambda':
1.7794881647049256e-08}. Best is trial 86 with value: -1.3050789204668365.
[I 2026-03-13 19:10:04,319] Trial 89 finished with value:
-1.3144003097194688 and parameters: {'n_estimators': 735, 'learning_rate':
0.015659987095679253, 'max_depth': 8, 'subsample': 0.6124288718966509,
'colsample_bytree': 0.6386107269905704, 'min_child_weight': 4, 'gamma':
0.0003259060150157194, 'reg_alpha': 3.3403876987363906e-05, 'reg_lambda':
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1.610338977834909e-08}. Best is trial 86 with value: -1.3050789204668365.
[I 2026-03-13 19:10:04,898] Trial 90 finished with value:
-1.308890054348874 and parameters: {'n_estimators': 715, 'learning_rate':
0.011027595690284955, 'max_depth': 8, 'subsample': 0.6887273146821905,
'colsample_bytree': 0.6587234732958218, 'min_child_weight': 2, 'gamma':
0.00016368262894235562, 'reg_alpha': 1.6807564331034628e-05, 'reg_lambda':
1.729881342571979e-08}. Best is trial 86 with value: -1.3050789204668365.
[I 2026-03-13 19:10:05,474] Trial 91 finished with value:
-1.3170315266692627 and parameters: {'n_estimators': 713, 'learning_rate':
0.010578082601871333, 'max_depth': 8, 'subsample': 0.6929909776521799,
'colsample_bytree': 0.6567130119972518, 'min_child_weight': 2, 'gamma':
0.0001757749196671735, 'reg_alpha': 7.356852941077462e-05, 'reg_lambda':
1.9080089777384837e-08}. Best is trial 86 with value: -1.3050789204668365.
[I 2026-03-13 19:10:05,883] Trial 92 finished with value:
-1.3133868719210333 and parameters: {'n_estimators': 622, 'learning_rate':
0.01377563130231946, 'max_depth': 8, 'subsample': 0.6786929742718293,
'colsample_bytree': 0.6403261328028623, 'min_child_weight': 4, 'gamma':
5.9135606283504435e-05, 'reg_alpha': 1.3461392995487278e-05, 'reg_lambda':
6.401710843545481e-08}. Best is trial 86 with value: -1.3050789204668365.
[I 2026-03-13 19:10:06,323] Trial 93 finished with value:
-1.3227909892326677 and parameters: {'n_estimators': 746, 'learning_rate':
0.012024210225171654, 'max_depth': 7, 'subsample': 0.5677232224546118,
'colsample_bytree': 0.6032669744847978, 'min_child_weight': 2, 'gamma':
0.0024465415541832175, 'reg_alpha': 6.113200059265304e-05, 'reg_lambda':
1.2373366292646295e-08}. Best is trial 86 with value: -1.3050789204668365.
[I 2026-03-13 19:10:06,627] Trial 94 finished with value:
-1.3261575602577809 and parameters: {'n_estimators': 678, 'learning_rate':
0.011659542265512997, 'max_depth': 6, 'subsample': 0.7072778923232627,
'colsample_bytree': 0.5754032899149746, 'min_child_weight': 3, 'gamma':
0.00023309662575540378, 'reg_alpha': 3.756387209157232e-05, 'reg_lambda':
4.012384248679804e-08}. Best is trial 86 with value: -1.3050789204668365.
[I 2026-03-13 19:10:07,046] Trial 95 finished with value:
-1.308299370893655 and parameters: {'n_estimators': 583, 'learning_rate':
0.010050343478570745, 'max_depth': 8, 'subsample': 0.6881509637242503,
'colsample_bytree': 0.6698492572326913, 'min_child_weight': 4, 'gamma':
1.855403370268866e-05, 'reg_alpha': 0.0001322630447247313, 'reg_lambda':
1.2110689540063757e-07}. Best is trial 86 with value: -1.3050789204668365.
[I 2026-03-13 19:10:07,455] Trial 96 finished with value:
-1.3082181973197509 and parameters: {'n_estimators': 576, 'learning_rate':
0.010783689234512256, 'max_depth': 8, 'subsample': 0.6923175560659803,
'colsample_bytree': 0.6690987324203029, 'min_child_weight': 4, 'gamma':
4.527688352507385e-05, 'reg_alpha': 0.0001395001888793167, 'reg_lambda':
1.2613856522614583e-07}. Best is trial 86 with value: -1.3050789204668365.
[I 2026-03-13 19:10:07,852] Trial 97 finished with value:
-1.3186615933349888 and parameters: {'n_estimators': 702, 'learning_rate':
0.010139440676189678, 'max_depth': 7, 'subsample': 0.6073820423299494,
'colsample_bytree': 0.7397052631544357, 'min_child_weight': 4, 'gamma':
1.9910302918285176e-05, 'reg_alpha': 1.9833558025627278e-05, 'reg_lambda':
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1.2464802416165343e-07}. Best is trial 86 with value: -1.3050789204668365.
[I 2026-03-13 19:10:08,304] Trial 98 finished with value:
-1.307459304057131 and parameters: {'n_estimators': 652, 'learning_rate':
0.010885155334922238, 'max_depth': 8, 'subsample': 0.6688055934351378,
'colsample_bytree': 0.6676220978510534, 'min_child_weight': 4, 'gamma':
3.6885821416934176e-05, 'reg_alpha': 0.00014661128739851332, 'reg_lambda':
3.8563221235647134e-08}. Best is trial 86 with value: -1.3050789204668365.
[I 2026-03-13 19:10:08,786] Trial 99 finished with value:
-1.3082879894675055 and parameters: {'n_estimators': 649, 'learning_rate':
0.011049573841840415, 'max_depth': 8, 'subsample': 0.6884626149844215,
'colsample_bytree': 0.6691137199198619, 'min_child_weight': 3, 'gamma':
4.004971304134273e-05, 'reg_alpha': 0.0008073122084446744, 'reg_lambda':
5.6631365204029e-08}. Best is trial 86 with value: -1.3050789204668365.
[I 2026-03-13 19:10:09,226] Trial 100 finished with value:
-1.3089427487906438 and parameters: {'n_estimators': 582, 'learning_rate':
0.0109321126107217, 'max_depth': 8, 'subsample': 0.6709689705284396,
'colsample_bytree': 0.6684179226904906, 'min_child_weight': 3, 'gamma':
4.2425656700217054e-05, 'reg_alpha': 0.00012068827284818496, 'reg_lambda':
6.255189719127394e-08}. Best is trial 86 with value: -1.3050789204668365.
[I 2026-03-13 19:10:09,699] Trial 101 finished with value:
-1.3085728862025705 and parameters: {'n_estimators': 654, 'learning_rate':
0.011142616597532172, 'max_depth': 8, 'subsample': 0.6918041972365001,
'colsample_bytree': 0.6555797195367005, 'min_child_weight': 3, 'gamma':
8.691920169447006e-05, 'reg_alpha': 0.0007671062603537313, 'reg_lambda':
3.819437453190723e-08}. Best is trial 86 with value: -1.3050789204668365.
[I 2026-03-13 19:10:10,184] Trial 102 finished with value:
-1.3090203903304563 and parameters: {'n_estimators': 660, 'learning_rate':
0.01211906359768678, 'max_depth': 8, 'subsample': 0.6461004055318655,
'colsample_bytree': 0.7073839263698201, 'min_child_weight': 3, 'gamma':
8.012408493304282e-05, 'reg_alpha': 0.0007981228812783192, 'reg_lambda':
2.627857855678755e-07}. Best is trial 86 with value: -1.3050789204668365.
[I 2026-03-13 19:10:10,678] Trial 103 finished with value:
-1.3083811842060638 and parameters: {'n_estimators': 624, 'learning_rate':
0.010575961451163612, 'max_depth': 9, 'subsample': 0.6282685260107216,
'colsample_bytree': 0.6462939826827291, 'min_child_weight': 4, 'gamma':
1.9510544028375857e-05, 'reg_alpha': 0.002941943348405677, 'reg_lambda':
3.82007809214537e-08}. Best is trial 86 with value: -1.3050789204668365.
[I 2026-03-13 19:10:11,089] Trial 104 finished with value:
-1.3036923296755922 and parameters: {'n_estimators': 524, 'learning_rate':
0.014177180658964828, 'max_depth': 9, 'subsample': 0.6260492850306295,
'colsample_bytree': 0.6459086282880708, 'min_child_weight': 4, 'gamma':
1.767133806392745e-05, 'reg_alpha': 0.0023760159119510637, 'reg_lambda':
5.146451104172687e-08}. Best is trial 104 with value: -1.3036923296755922.
[I 2026-03-13 19:10:11,575] Trial 105 finished with value:
-1.3076583634547476 and parameters: {'n_estimators': 621, 'learning_rate':
0.013385170581066107, 'max_depth': 9, 'subsample': 0.631619364055115,
'colsample_bytree': 0.6295231408152168, 'min_child_weight': 4, 'gamma':
1.713169168693549e-05, 'reg_alpha': 0.001836876262799637, 'reg_lambda':
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1.02931084359729e-07}. Best is trial 104 with value: -1.3036923296755922.
[I 2026-03-13 19:10:11,975] Trial 106 finished with value:
-1.3107581242554958 and parameters: {'n_estimators': 522, 'learning_rate':
0.013857993114428152, 'max_depth': 9, 'subsample': 0.5862523532781042,
'colsample_bytree': 0.635422937568741, 'min_child_weight': 4, 'gamma':
2.8182404187413335e-05, 'reg_alpha': 0.005009372083804252, 'reg_lambda':
1.1841948553693703e-07}. Best is trial 104 with value: -1.3036923296755922.
[I 2026-03-13 19:10:12,487] Trial 107 finished with value:
-1.3085439641650425 and parameters: {'n_estimators': 634, 'learning_rate':
0.010054947807441914, 'max_depth': 9, 'subsample': 0.629053651302495,
'colsample_bytree': 0.676647775945966, 'min_child_weight': 4, 'gamma':
1.1493335655415116e-05, 'reg_alpha': 0.0030213864755869375, 'reg_lambda':
6.231076193883922e-08}. Best is trial 104 with value: -1.3036923296755922.
[I 2026-03-13 19:10:12,980] Trial 108 finished with value:
-1.3169978676078642 and parameters: {'n_estimators': 617, 'learning_rate':
0.014258504813690943, 'max_depth': 9, 'subsample': 0.6053277113363532,
'colsample_bytree': 0.7629131937183558, 'min_child_weight': 4, 'gamma':
1.668257221127503e-05, 'reg_alpha': 0.0016970750399360052, 'reg_lambda':
1.0048338014399165e-08}. Best is trial 104 with value: -1.3036923296755922.
[I 2026-03-13 19:10:13,442] Trial 109 finished with value:
-1.3140265828531443 and parameters: {'n_estimators': 616, 'learning_rate':
0.010631540160299643, 'max_depth': 9, 'subsample': 0.5678729358332743,
'colsample_bytree': 0.6463765208195468, 'min_child_weight': 4, 'gamma':
4.058932678855267e-05, 'reg_alpha': 0.002374952749488232, 'reg_lambda':
1.4154226869704822e-07}. Best is trial 104 with value: -1.3036923296755922.
[I 2026-03-13 19:10:13,841] Trial 110 finished with value:
-1.313132222270909 and parameters: {'n_estimators': 538, 'learning_rate':
0.012939623024491266, 'max_depth': 9, 'subsample': 0.6280904299479746,
'colsample_bytree': 0.6297709419143152, 'min_child_weight': 5, 'gamma':
8.129894028404083e-06, 'reg_alpha': 0.010855290023102584, 'reg_lambda':
8.487763055248584e-08}. Best is trial 104 with value: -1.3036923296755922.
[I 2026-03-13 19:10:14,229] Trial 111 finished with value:
-1.3096014300418246 and parameters: {'n_estimators': 590, 'learning_rate':
0.01229275915299607, 'max_depth': 8, 'subsample': 0.6438857126722967,
'colsample_bytree': 0.6141553437657515, 'min_child_weight': 4, 'gamma':
4.223738836103102e-06, 'reg_alpha': 0.001251298128633134, 'reg_lambda':
2.1276486884335165e-07}. Best is trial 104 with value: -1.3036923296755922.
[I 2026-03-13 19:10:14,681] Trial 112 finished with value:
-1.3084813528578314 and parameters: {'n_estimators': 561, 'learning_rate':
0.013407275786931995, 'max_depth': 9, 'subsample': 0.6164735585933276,
'colsample_bytree': 0.6888540616112481, 'min_child_weight': 4, 'gamma':
1.6481754412373602e-05, 'reg_alpha': 0.00014301180012967854, 'reg_lambda':
1.9421383521410793e-08}. Best is trial 104 with value: -1.3036923296755922.
[I 2026-03-13 19:10:15,123] Trial 113 finished with value:
-1.3087215730483903 and parameters: {'n_estimators': 553, 'learning_rate':
0.01323940452443047, 'max_depth': 9, 'subsample': 0.6008792126198013,
'colsample_bytree': 0.6896061030130598, 'min_child_weight': 4, 'gamma':
1.2922135445450355e-05, 'reg_alpha': 0.0003346649840432319, 'reg_lambda':
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4.566396563916317e-08}. Best is trial 104 with value: -1.3036923296755922.
[I 2026-03-13 19:10:15,562] Trial 114 finished with value:
-1.3048982083094751 and parameters: {'n_estimators': 530, 'learning_rate':
0.010675497346150318, 'max_depth': 9, 'subsample': 0.6170311583003166,
'colsample_bytree': 0.7093273828425491, 'min_child_weight': 4, 'gamma':
3.20055780674823e-05, 'reg_alpha': 0.0006279100121629379, 'reg_lambda':
1.9147416363110292e-08}. Best is trial 104 with value: -1.3036923296755922.
[I 2026-03-13 19:10:16,128] Trial 115 finished with value:
-1.3062361681537622 and parameters: {'n_estimators': 676, 'learning_rate':
0.010615473267422761, 'max_depth': 10, 'subsample': 0.6275889423797004,
'colsample_bytree': 0.7104413289172945, 'min_child_weight': 5, 'gamma':
2.627610816770887e-05, 'reg_alpha': 0.0005339246828602433, 'reg_lambda':
5.2941508985812e-08}. Best is trial 104 with value: -1.3036923296755922.
[I 2026-03-13 19:10:16,557] Trial 116 finished with value:
-1.314060910977096 and parameters: {'n_estimators': 468, 'learning_rate':
0.011530196469898502, 'max_depth': 10, 'subsample': 0.6637112956641327,
'colsample_bytree': 0.7341891834308202, 'min_child_weight': 5, 'gamma':
3.2279694214022544e-05, 'reg_alpha': 0.0006403383120322698, 'reg_lambda':
6.152175868789963e-08}. Best is trial 104 with value: -1.3036923296755922.
[I 2026-03-13 19:10:16,904] Trial 117 finished with value:
-1.3180092158143406 and parameters: {'n_estimators': 527, 'learning_rate':
0.014525340222424661, 'max_depth': 8, 'subsample': 0.6408928658929673,
'colsample_bytree': 0.7117567079390232, 'min_child_weight': 5, 'gamma':
0.0007043069733817256, 'reg_alpha': 0.0005138234773573555, 'reg_lambda':
1.4003009904060833e-08}. Best is trial 104 with value: -1.3036923296755922.
[I 2026-03-13 19:10:17,460] Trial 118 finished with value:
-1.3156655056298023 and parameters: {'n_estimators': 659, 'learning_rate':
0.01005587692200544, 'max_depth': 10, 'subsample': 0.6599144451675129,
'colsample_bytree': 0.7099001449648443, 'min_child_weight': 5, 'gamma':
0.00013505976879790558, 'reg_alpha': 0.0010860918157268443, 'reg_lambda':
8.887889728972354e-08}. Best is trial 104 with value: -1.3036923296755922.
[I 2026-03-13 19:10:17,984] Trial 119 finished with value:
-1.31501217322757 and parameters: {'n_estimators': 675, 'learning_rate':
0.012523731680872315, 'max_depth': 10, 'subsample': 0.6791957158960409,
'colsample_bytree': 0.6770890514464523, 'min_child_weight': 6, 'gamma':
2.716207190694462e-05, 'reg_alpha': 0.00010456627814999734, 'reg_lambda':
1.5172327845979518e-07}. Best is trial 104 with value: -1.3036923296755922.
[I 2026-03-13 19:10:18,414] Trial 120 finished with value:
-1.3186106187827578 and parameters: {'n_estimators': 641, 'learning_rate':
0.01641536127285815, 'max_depth': 8, 'subsample': 0.581761479519366,
'colsample_bytree': 0.7440913194140274, 'min_child_weight': 4, 'gamma':
5.067949068907541e-05, 'reg_alpha': 0.00016175837229984595, 'reg_lambda':
1.9545722050145256e-08}. Best is trial 104 with value: -1.3036923296755922.
[I 2026-03-13 19:10:18,926] Trial 121 finished with value:
-1.302869249991674 and parameters: {'n_estimators': 622, 'learning_rate':
0.010694463434274677, 'max_depth': 9, 'subsample': 0.6261209787377542,
'colsample_bytree': 0.7271965899188532, 'min_child_weight': 4, 'gamma':
5.466843694145246e-06, 'reg_alpha': 0.00036025862525885106, 'reg_lambda':
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4.101955359053519e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:19,430] Trial 122 finished with value:
-1.309243381283214 and parameters: {'n_estimators': 601, 'learning_rate':
0.010902199429469463, 'max_depth': 9, 'subsample': 0.6162555632664102,
'colsample_bytree': 0.7696367860367679, 'min_child_weight': 4, 'gamma':
5.04103866260417e-06, 'reg_alpha': 0.0003562672069344903, 'reg_lambda':
4.4658025312328e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:19,922] Trial 123 finished with value:
-1.314269328036556 and parameters: {'n_estimators': 725, 'learning_rate':
0.011543184913614779, 'max_depth': 8, 'subsample': 0.5564359069886154,
'colsample_bytree': 0.723530201740752, 'min_child_weight': 4, 'gamma':
7.609910980994749e-06, 'reg_alpha': 5.0407569188968284e-05, 'reg_lambda':
2.145920370102458e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:20,393] Trial 124 finished with value:
-1.306668449467272 and parameters: {'n_estimators': 584, 'learning_rate':
0.012405935424256845, 'max_depth': 9, 'subsample': 0.6278930205459092,
'colsample_bytree': 0.7044743509550216, 'min_child_weight': 4, 'gamma':
4.166836872336902e-05, 'reg_alpha': 0.0012897934791088245, 'reg_lambda':
5.391038172524053e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:20,835] Trial 125 finished with value:
-1.3089255467622392 and parameters: {'n_estimators': 514, 'learning_rate':
0.012398761292275151, 'max_depth': 10, 'subsample': 0.5964732537044138,
'colsample_bytree': 0.731124830586932, 'min_child_weight': 5, 'gamma':
9.99010178678871e-05, 'reg_alpha': 0.0011547912574578676, 'reg_lambda':
4.7890893474071844e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:21,485] Trial 126 finished with value:
-1.322993415181341 and parameters: {'n_estimators': 680, 'learning_rate':
0.014202446150714032, 'max_depth': 9, 'subsample': 0.6285547224382557,
'colsample_bytree': 0.7017806239294555, 'min_child_weight': 3, 'gamma':
4.497589145922731e-05, 'reg_alpha': 0.8447536641536932, 'reg_lambda':
2.9980059362424365e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:21,925] Trial 127 finished with value:
-1.3521526616713562 and parameters: {'n_estimators': 650, 'learning_rate':
0.06892973332408786, 'max_depth': 9, 'subsample': 0.6374344003481283,
'colsample_bytree': 0.74389902421328, 'min_child_weight': 4, 'gamma':
1.20120200915065e-05, 'reg_alpha': 0.0036847384371875274, 'reg_lambda':
1.3384928118894942e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:22,419] Trial 128 finished with value:
-1.3086736933555885 and parameters: {'n_estimators': 606, 'learning_rate':
0.011404218674620074, 'max_depth': 9, 'subsample': 0.6132057773143045,
'colsample_bytree': 0.7073786062477021, 'min_child_weight': 4, 'gamma':
0.00023785061056947612, 'reg_alpha': 0.0006313669592072216, 'reg_lambda':
7.821200591228435e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:22,892] Trial 129 finished with value:
-1.3082248751502386 and parameters: {'n_estimators': 552, 'learning_rate':
0.01334688324442052, 'max_depth': 10, 'subsample': 0.6513449392180916,
'colsample_bytree': 0.7180115540876265, 'min_child_weight': 5, 'gamma':
3.14246846732064e-05, 'reg_alpha': 0.0003196154604826234, 'reg_lambda':
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5.537287288613054e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:23,333] Trial 130 finished with value:
-1.3157953201240267 and parameters: {'n_estimators': 486, 'learning_rate':
0.01329261069994949, 'max_depth': 10, 'subsample': 0.6543699924791587,
'colsample_bytree': 0.7506329998106586, 'min_child_weight': 5, 'gamma':
2.536716911040176e-06, 'reg_alpha': 0.00022141296529436117, 'reg_lambda':
2.23563056420656e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:23,795] Trial 131 finished with value:
-1.3123761647852763 and parameters: {'n_estimators': 547, 'learning_rate':
0.010805518454608198, 'max_depth': 10, 'subsample': 0.5889587937505136,
'colsample_bytree': 0.6951063057889251, 'min_child_weight': 5, 'gamma':
3.3986612624200646e-05, 'reg_alpha': 0.0019491591799264294, 'reg_lambda':
5.626638122951145e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:24,266] Trial 132 finished with value:
-1.3031618578184452 and parameters: {'n_estimators': 570, 'learning_rate':
0.012262022621957178, 'max_depth': 9, 'subsample': 0.6280419616388685,
'colsample_bytree': 0.7216529688184621, 'min_child_weight': 4, 'gamma':
6.105887263602787e-05, 'reg_alpha': 0.00042417705881981667, 'reg_lambda':
2.824900382612784e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:24,623] Trial 133 finished with value:
-1.3547314475408043 and parameters: {'n_estimators': 560, 'learning_rate':
0.050501490787241206, 'max_depth': 9, 'subsample': 0.6206381738125031,
'colsample_bytree': 0.7196237018856692, 'min_child_weight': 6, 'gamma':
6.227906045639388e-05, 'reg_alpha': 0.00032703262595378105, 'reg_lambda':
2.610967622931256e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:25,087] Trial 134 finished with value:
-1.3043365239845377 and parameters: {'n_estimators': 539, 'learning_rate':
0.01228915803972928, 'max_depth': 10, 'subsample': 0.6030392387013574,
'colsample_bytree': 0.7345090488545393, 'min_child_weight': 5, 'gamma':
2.6309887242043176e-05, 'reg_alpha': 0.00023985108638082284, 'reg_lambda':
3.525434440811095e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:25,537] Trial 135 finished with value:
-1.3116846559932618 and parameters: {'n_estimators': 530, 'learning_rate':
0.012236547876717067, 'max_depth': 9, 'subsample': 0.608004638550961,
'colsample_bytree': 0.7631565928566033, 'min_child_weight': 4, 'gamma':
0.00013449430512267266, 'reg_alpha': 0.0001871959021398589, 'reg_lambda':
3.4944440809912377e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:25,937] Trial 136 finished with value:
-1.3129888639197362 and parameters: {'n_estimators': 497, 'learning_rate':
0.014856767538887096, 'max_depth': 9, 'subsample': 0.5786147709829683,
'colsample_bytree': 0.7314089006418629, 'min_child_weight': 4, 'gamma':
9.509652844170257e-06, 'reg_alpha': 8.657672416772689e-05, 'reg_lambda':
9.117060210404386e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:26,470] Trial 137 finished with value:
-1.3091254972918351 and parameters: {'n_estimators': 578, 'learning_rate':
0.012265168449140046, 'max_depth': 10, 'subsample': 0.599986502609751,
'colsample_bytree': 0.7005511844386525, 'min_child_weight': 4, 'gamma':
9.07517965023322e-05, 'reg_alpha': 0.00043721847022054925, 'reg_lambda':
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1.1363507700901513e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:26,956] Trial 138 finished with value:
-1.3108684825824242 and parameters: {'n_estimators': 599, 'learning_rate':
0.011619887474347142, 'max_depth': 9, 'subsample': 0.6311021668402884,
'colsample_bytree': 0.7972044385604032, 'min_child_weight': 5, 'gamma':
1.4734730398697561e-05, 'reg_alpha': 0.0012828940031227591, 'reg_lambda':
1.5486164775509832e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:27,430] Trial 139 finished with value:
-1.3377237894984273 and parameters: {'n_estimators': 628, 'learning_rate':
0.033511949555078475, 'max_depth': 9, 'subsample': 0.6219522265909044,
'colsample_bytree': 0.7503367446973626, 'min_child_weight': 4, 'gamma':
2.383450034220334e-05, 'reg_alpha': 0.006312281518488809, 'reg_lambda':
3.725778876058315e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:27,964] Trial 140 finished with value:
-1.309661044886731 and parameters: {'n_estimators': 568, 'learning_rate':
0.010570140569004379, 'max_depth': 10, 'subsample': 0.6399533458028439,
'colsample_bytree': 0.6843904076776965, 'min_child_weight': 4, 'gamma':
6.812402726005554e-06, 'reg_alpha': 0.0005079171318436428, 'reg_lambda':
2.1934873022532496e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:28,435] Trial 141 finished with value:
-1.3101581851486837 and parameters: {'n_estimators': 545, 'learning_rate':
0.013533175203281667, 'max_depth': 10, 'subsample': 0.6476945145940304,
'colsample_bytree': 0.718552737598733, 'min_child_weight': 5, 'gamma':
3.461745507310631e-05, 'reg_alpha': 0.0002367265382911403, 'reg_lambda':
7.077390977322491e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:28,947] Trial 142 finished with value:
-1.309574859116526 and parameters: {'n_estimators': 515, 'learning_rate':
0.012602832725249143, 'max_depth': 11, 'subsample': 0.6591439089199504,
'colsample_bytree': 0.7278135963430621, 'min_child_weight': 5, 'gamma':
6.603165586154363e-05, 'reg_alpha': 0.00031769327946490475, 'reg_lambda':
0.00225061382435394}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:29,440] Trial 143 finished with value:
-1.3164762395051022 and parameters: {'n_estimators': 579, 'learning_rate':
0.014407193997068373, 'max_depth': 10, 'subsample': 0.6354971272583065,
'colsample_bytree': 0.7143580798835768, 'min_child_weight': 5, 'gamma':
5.129111783421473e-05, 'reg_alpha': 0.0016283215696124092, 'reg_lambda':
3.3451736089871303e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:29,799] Trial 144 finished with value:
-1.3205444401094875 and parameters: {'n_estimators': 464, 'learning_rate':
0.016292897017813456, 'max_depth': 10, 'subsample': 0.6195728264688859,
'colsample_bytree': 0.6538319047368638, 'min_child_weight': 6, 'gamma':
2.6480231110322995e-05, 'reg_alpha': 0.0009393743568160603, 'reg_lambda':
4.9953692444943244e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:30,199] Trial 145 finished with value:
-1.310112915746473 and parameters: {'n_estimators': 440, 'learning_rate':
0.013250102642316861, 'max_depth': 9, 'subsample': 0.6469798119626522,
'colsample_bytree': 0.7702597859685576, 'min_child_weight': 4, 'gamma':
0.00011619895356858736, 'reg_alpha': 5.332502760548962e-05, 'reg_lambda':
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1.660958014159565e-07}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:30,724] Trial 146 finished with value:
-1.3113207610953237 and parameters: {'n_estimators': 704, 'learning_rate':
0.011685481086166618, 'max_depth': 9, 'subsample': 0.6023011523218182,
'colsample_bytree': 0.7348396692141581, 'min_child_weight': 5, 'gamma':
2.6042704529767634e-05, 'reg_alpha': 0.0005535035438281652, 'reg_lambda':
1.1284557209560941e-07}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:31,332] Trial 147 finished with value:
-1.3108941204243443 and parameters: {'n_estimators': 796, 'learning_rate':
0.011099925199358346, 'max_depth': 9, 'subsample': 0.6650972709521149,
'colsample_bytree': 0.6770032260316264, 'min_child_weight': 4, 'gamma':
0.0002377068752713315, 'reg_alpha': 9.336833969643951e-05, 'reg_lambda':
2.995003577553587e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:31,793] Trial 148 finished with value:
-1.308828637604651 and parameters: {'n_estimators': 608, 'learning_rate':
0.010619490531543796, 'max_depth': 8, 'subsample': 0.628175823793345,
'colsample_bytree': 0.6887721925687642, 'min_child_weight': 3, 'gamma':
0.0004854367024888822, 'reg_alpha': 0.0001654878672655065, 'reg_lambda':
1.6079627168881524e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:32,351] Trial 149 finished with value:
-1.3174200844453703 and parameters: {'n_estimators': 552, 'learning_rate':
0.015364115548313305, 'max_depth': 11, 'subsample': 0.6097240185953277,
'colsample_bytree': 0.7059980605774238, 'min_child_weight': 4, 'gamma':
1.2288501374394282e-05, 'reg_alpha': 5.0237926102286804e-06, 'reg_lambda':
8.764249516263532e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:32,801] Trial 150 finished with value:
-1.3118312410676507 and parameters: {'n_estimators': 625, 'learning_rate':
0.012544172776313608, 'max_depth': 9, 'subsample': 0.5892939765151234,
'colsample_bytree': 0.6614436368777592, 'min_child_weight': 5, 'gamma':
4.982042835706074e-05, 'reg_alpha': 0.000276501847617077, 'reg_lambda':
1.0070695301765887e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:33,274] Trial 151 finished with value:
-1.3146674164943368 and parameters: {'n_estimators': 661, 'learning_rate':
0.010020720745533866, 'max_depth': 8, 'subsample': 0.6732244705059766,
'colsample_bytree': 0.6320100744741299, 'min_child_weight': 3, 'gamma':
4.057075249342082e-05, 'reg_alpha': 0.0007949186323930993, 'reg_lambda':
4.948342305887971e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:33,766] Trial 152 finished with value:
-1.3049835944487491 and parameters: {'n_estimators': 686, 'learning_rate':
0.01148896166003639, 'max_depth': 8, 'subsample': 0.6531438748719002,
'colsample_bytree': 0.6675943317322581, 'min_child_weight': 3, 'gamma':
1.9025407031547906e-05, 'reg_alpha': 0.0008934852985523501, 'reg_lambda':
6.014613363080604e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:34,155] Trial 153 finished with value:
-1.3124129009459327 and parameters: {'n_estimators': 675, 'learning_rate':
0.011723209015061103, 'max_depth': 7, 'subsample': 0.6532237113131415,
'colsample_bytree': 0.6505466665339291, 'min_child_weight': 3, 'gamma':
2.1467311236939924e-05, 'reg_alpha': 0.002116705460079831, 'reg_lambda':
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2.8286424663644612e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:34,555] Trial 154 finished with value:
-1.3054643766148013 and parameters: {'n_estimators': 588, 'learning_rate':
0.013675466514476l6, 'max_depth': 8, 'subsample': 0.6380100691189007,
'colsample_bytree': 0.6984002738124225, 'min_child_weight': 4, 'gamma':
1.6323715836852094e-05, 'reg_alpha': 0.0004261352048428586, 'reg_lambda':
6.912401248169205e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:35,028] Trial 155 finished with value:
-1.3092588938974121 and parameters: {'n_estimators': 729, 'learning_rate':
0.014191629080673025, 'max_depth': 8, 'subsample': 0.6398133330117238,
'colsample_bytree': 0.6732889723885159, 'min_child_weight': 4, 'gamma':
1.570154106239512e-05, 'reg_alpha': 0.0004059685698649202, 'reg_lambda':
1.9702393020704463e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:35,428] Trial 156 finished with value:
-1.3095693304863678 and parameters: {'n_estimators': 590, 'learning_rate':
0.011406608775093112, 'max_depth': 8, 'subsample': 0.6196623499378069,
'colsample_bytree': 0.6440580242120187, 'min_child_weight': 4, 'gamma':
9.152349199239115e-06, 'reg_alpha': 0.001330561763177828, 'reg_lambda':
7.095454494037828e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:35,901] Trial 157 finished with value:
-1.3146648616359933 and parameters: {'n_estimators': 637, 'learning_rate':
0.01225875025806495, 'max_depth': 8, 'subsample': 0.6326149538896662,
'colsample_bytree': 0.693373448943871, 'min_child_weight': 3, 'gamma':
8.049102982824813e-05, 'reg_alpha': 0.0037994020893200114, 'reg_lambda':
2.771679528566749e-07}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:36,362] Trial 158 finished with value:
-1.3079865234497585 and parameters: {'n_estimators': 702, 'learning_rate':
0.010667245258966722, 'max_depth': 8, 'subsample': 0.6118718745993199,
'colsample_bytree': 0.663173318197509, 'min_child_weight': 4, 'gamma':
6.066152610397198e-06, 'reg_alpha': 0.0007710310880763141, 'reg_lambda':
1.144303809879426e-07}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:36,761] Trial 159 finished with value:
-1.3053804951739043 and parameters: {'n_estimators': 691, 'learning_rate':
0.012978599240173982, 'max_depth': 7, 'subsample': 0.6112749284244979,
'colsample_bytree': 0.6272724705102507, 'min_child_weight': 3, 'gamma':
5.891224446033121e-06, 'reg_alpha': 0.0007297905108932699, 'reg_lambda':
3.61322971504023e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:37,044] Trial 160 finished with value:
-1.5670639060052463 and parameters: {'n_estimators': 687, 'learning_rate':
0.2410339817859637, 'max_depth': 6, 'subsample': 0.5948587859848778,
'colsample_bytree': 0.6092045682911984, 'min_child_weight': 3, 'gamma':
2.2857472289553754e-06, 'reg_alpha': 0.00855510630908187, 'reg_lambda':
3.7464827876182924e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:37,443] Trial 161 finished with value:
-1.3159787302297816 and parameters: {'n_estimators': 709, 'learning_rate':
0.012809532227734372, 'max_depth': 7, 'subsample': 0.6095383623770669,
'colsample_bytree': 0.624653316199229, 'min_child_weight': 3, 'gamma':
5.143564710508432e-06, 'reg_alpha': 0.0006023525274776058, 'reg_lambda':
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2.329925280614733e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:37,937] Trial 162 finished with value:
-1.3175048106306009 and parameters: {'n_estimators': 747, 'learning_rate':
0.011817128597986518, 'max_depth': 8, 'subsample': 0.6237569237022772,
'colsample_bytree': 0.5858798556559788, 'min_child_weight': 3, 'gamma':
7.608964568316316e-06, 'reg_alpha': 0.0010963540119484812, 'reg_lambda':
4.869041147461659e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:38,305] Trial 163 finished with value:
-1.3171164790431626 and parameters: {'n_estimators': 665, 'learning_rate':
0.010001677523596073, 'max_depth': 7, 'subsample': 0.6124572221494853,
'colsample_bytree': 0.6371136114359865, 'min_child_weight': 4, 'gamma':
1.9542665496916638e-05, 'reg_alpha': 0.0007331260745063308, 'reg_lambda':
9.261459957666061e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:38,809] Trial 164 finished with value:
-1.3152961941783838 and parameters: {'n_estimators': 701, 'learning_rate':
0.013635501886901208, 'max_depth': 8, 'subsample': 0.6322054982849276,
'colsample_bytree': 0.7025664587416623, 'min_child_weight': 3, 'gamma':
5.639046944808029e-06, 'reg_alpha': 0.002092332089346082, 'reg_lambda':
1.6005299692066836e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:39,356] Trial 165 finished with value:
-1.3105747402193424 and parameters: {'n_estimators': 717, 'learning_rate':
0.010754645028216124, 'max_depth': 9, 'subsample': 0.6042170342193235,
'colsample_bytree': 0.6581412110761258, 'min_child_weight': 4, 'gamma':
2.9550654016496493e-06, 'reg_alpha': 0.0004687222233972067, 'reg_lambda':
3.201759154093497e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:39,769] Trial 166 finished with value:
-1.3097974831796897 and parameters: {'n_estimators': 739, 'learning_rate':
0.011344019383879021, 'max_depth': 7, 'subsample': 0.6235810440649643,
'colsample_bytree': 0.6836604436075903, 'min_child_weight': 4, 'gamma':
1.2646407385499982e-05, 'reg_alpha': 0.0027433732026151724, 'reg_lambda':
7.022195615747122e-08}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:40,202] Trial 167 finished with value:
-1.3274260669845808 and parameters: {'n_estimators': 648, 'learning_rate':
0.018466896113566805, 'max_depth': 8, 'subsample': 0.5740805064088259,
'colsample_bytree': 0.7452528146356886, 'min_child_weight': 4, 'gamma':
1.0411805243600766e-05, 'reg_alpha': 0.0015782522705352134, 'reg_lambda':
1.678842903112643e-07}. Best is trial 121 with value: -1.302869249991674.
[I 2026-03-13 19:10:40,799] Trial 168 finished with value:
-1.3028225035370153 and parameters: {'n_estimators': 695, 'learning_rate':
0.012868850873011217, 'max_depth': 9, 'subsample': 0.6453536201905438,
'colsample_bytree': 0.7298899193411906, 'min_child_weight': 3, 'gamma':
1.6784663741063516e-05, 'reg_alpha': 0.000883607414349492, 'reg_lambda':
4.3906870457998645e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:41,458] Trial 169 finished with value:
-1.317035035921398 and parameters: {'n_estimators': 781, 'learning_rate':
0.015143262710546357, 'max_depth': 9, 'subsample': 0.6385943282485351,
'colsample_bytree': 0.731399462974937, 'min_child_weight': 3, 'gamma':
2.083684284385134e-05, 'reg_alpha': 0.0009592352076242207, 'reg_lambda':
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4.252220294890514e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:42,065] Trial 170 finished with value:
-1.314798752682522 and parameters: {'n_estimators': 685, 'learning_rate':
0.013069997642027544, 'max_depth': 9, 'subsample': 0.6424627224608389,
'colsample_bytree': 0.7566877050375689, 'min_child_weight': 3, 'gamma':
2.9835263420803516e-05, 'reg_alpha': 0.0004192630508796568, 'reg_lambda':
2.2734273024475505e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:42,590] Trial 171 finished with value:
-1.3111618721045268 and parameters: {'n_estimators': 727, 'learning_rate':
0.012473769102419027, 'max_depth': 8, 'subsample': 0.6149797479747758,
'colsample_bytree': 0.7225686281447434, 'min_child_weight': 3, 'gamma':
1.6437787542858325e-05, 'reg_alpha': 0.0007554815908742008, 'reg_lambda':
5.291496402830748e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:43,200] Trial 172 finished with value:
-1.3151167881444812 and parameters: {'n_estimators': 754, 'learning_rate':
0.011706507828835748, 'max_depth': 9, 'subsample': 0.6594234274666195,
'colsample_bytree': 0.7407371032442522, 'min_child_weight': 4, 'gamma':
5.913582543527151e-06, 'reg_alpha': 0.00022304534768508987, 'reg_lambda':
1.0315456293283602e-07}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:43,702] Trial 173 finished with value:
-1.3100748314764645 and parameters: {'n_estimators': 701, 'learning_rate':
0.014085877827194886, 'max_depth': 8, 'subsample': 0.648355651796784,
'colsample_bytree': 0.7101290188321026, 'min_child_weight': 3, 'gamma':
9.323887241488838e-06, 'reg_alpha': 1.4424506973034374e-06, 'reg_lambda':
3.437788166073847e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:44,290] Trial 174 finished with value:
-1.3058454041764347 and parameters: {'n_estimators': 681, 'learning_rate':
0.010717309046866106, 'max_depth': 9, 'subsample': 0.6284852499975967,
'colsample_bytree': 0.6961822971551587, 'min_child_weight': 3, 'gamma':
6.385539675954548e-05, 'reg_alpha': 0.001317097971091468, 'reg_lambda':
6.773924149551698e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:44,867] Trial 175 finished with value:
-1.311705891527984 and parameters: {'n_estimators': 661, 'learning_rate':
0.01205415687178052, 'max_depth': 9, 'subsample': 0.6297990242640983,
'colsample_bytree': 0.696869582057317, 'min_child_weight': 3, 'gamma':
6.518566628726369e-05, 'reg_alpha': 0.00422185040131753, 'reg_lambda':
1.4887148570216947e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:45,426] Trial 176 finished with value:
-1.3131195262807374 and parameters: {'n_estimators': 618, 'learning_rate':
0.012835621030275484, 'max_depth': 9, 'subsample': 0.6669093007835589,
'colsample_bytree': 0.7173586497197646, 'min_child_weight': 3, 'gamma':
3.783644912459923e-05, 'reg_alpha': 0.0011803535336357286, 'reg_lambda':
6.592868120222161e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:45,974] Trial 177 finished with value:
-1.397923921465592 and parameters: {'n_estimators': 686, 'learning_rate':
0.0993272145723612, 'max_depth': 9, 'subsample': 0.6242842546666807,
'colsample_bytree': 0.7282747548967389, 'min_child_weight': 2, 'gamma':
0.0001086424539425178, 'reg_alpha': 0.0015366060214934887, 'reg_lambda':
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2.4409200515438658e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:46,521] Trial 178 finished with value:
-1.3106115306601536 and parameters: {'n_estimators': 640, 'learning_rate':
0.011200870648948072, 'max_depth': 9, 'subsample': 0.5981932803026512,
'colsample_bytree': 0.6831154768345571, 'min_child_weight': 3, 'gamma':
2.5166014245651316e-05, 'reg_alpha': 0.0005762653773178827, 'reg_lambda':
3.97695313754889e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:47,088] Trial 179 finished with value:
-1.3063412106155232 and parameters: {'n_estimators': 671, 'learning_rate':
0.013745531812217673, 'max_depth': 9, 'subsample': 0.6423845764510896,
'colsample_bytree': 0.6983340138050195, 'min_child_weight': 3, 'gamma':
0.00019009397194260584, 'reg_alpha': 0.0025374697636156578, 'reg_lambda':
7.012897421737421e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:47,394] Trial 180 finished with value:
-1.329782428725501 and parameters: {'n_estimators': 672, 'learning_rate':
0.014594995628986753, 'max_depth': 6, 'subsample': 0.5000151597844791,
'colsample_bytree': 0.7028213898916971, 'min_child_weight': 3, 'gamma':
0.0001912414941264226, 'reg_alpha': 0.00026492302802395704, 'reg_lambda':
6.278390571867275e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:47,960] Trial 181 finished with value:
-1.3083592856270287 and parameters: {'n_estimators': 668, 'learning_rate':
0.013677895331820757, 'max_depth': 9, 'subsample': 0.6433241914499321,
'colsample_bytree': 0.6894416365021584, 'min_child_weight': 3, 'gamma':
0.00030431737688792416, 'reg_alpha': 0.0027339316444725086, 'reg_lambda':
3.216319739543944e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:48,506] Trial 182 finished with value:
-1.318571457971636 and parameters: {'n_estimators': 630, 'learning_rate':
0.01683998183071197, 'max_depth': 9, 'subsample': 0.6545299219624285,
'colsample_bytree': 0.7088325847952036, 'min_child_weight': 3, 'gamma':
6.212511628048769e-05, 'reg_alpha': 0.00205378220898817, 'reg_lambda':
1.9649708897731294e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:49,084] Trial 183 finished with value:
-1.3116239857363865 and parameters: {'n_estimators': 649, 'learning_rate':
0.012337671353857273, 'max_depth': 9, 'subsample': 0.6348093204886566,
'colsample_bytree': 0.7413306355650314, 'min_child_weight': 3, 'gamma':
4.362920566365368e-05, 'reg_alpha': 0.000960898961665684, 'reg_lambda':
4.375128584218629e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:49,704] Trial 184 finished with value:
-1.310520513597823 and parameters: {'n_estimators': 609, 'learning_rate':
0.011115580235337937, 'max_depth': 10, 'subsample': 0.6473260586025937,
'colsample_bytree': 0.6776724256494967, 'min_child_weight': 3, 'gamma':
1.5942276871515715e-05, 'reg_alpha': 0.00043983319553259504, 'reg_lambda':
8.239351052178291e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:50,251] Trial 185 finished with value:
-1.4417758408350478 and parameters: {'n_estimators': 719, 'learning_rate':
0.17658344080210536, 'max_depth': 9, 'subsample': 0.6189012498886377,
'colsample_bytree': 0.6496185275358649, 'min_child_weight': 2, 'gamma':
8.236001207499686e-05, 'reg_alpha': 0.0060617593922954426, 'reg_lambda':
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2.695444564504357e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:50,694] Trial 186 finished with value:
-1.3053108824421351 and parameters: {'n_estimators': 533, 'learning_rate':
0.010575478751937099, 'max_depth': 9, 'subsample': 0.6316871018855439,
'colsample_bytree': 0.6288689487685142, 'min_child_weight': 4, 'gamma':
2.9280983693242013e-05, 'reg_alpha': 0.0014409973816979682, 'reg_lambda':
5.411739429462039e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:51,148] Trial 187 finished with value:
-1.3090849600291103 and parameters: {'n_estimators': 503, 'learning_rate':
0.010178876884402078, 'max_depth': 9, 'subsample': 0.6625343924074045,
'colsample_bytree': 0.7254935289151381, 'min_child_weight': 4, 'gamma':
0.00015554883507464757, 'reg_alpha': 3.3751043624868204e-06, 'reg_lambda':
5.481625777720638e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:51,598] Trial 188 finished with value:
-1.312629548984028 and parameters: {'n_estimators': 533, 'learning_rate':
0.011694033903276382, 'max_depth': 8, 'subsample': 0.6392649547596831,
'colsample_bytree': 0.6974494789481008, 'min_child_weight': 2, 'gamma':
3.0231554733053227e-05, 'reg_alpha': 0.0013562377836749226, 'reg_lambda':
1.4742149667998225e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:52,099] Trial 189 finished with value:
-1.3075907084011196 and parameters: {'n_estimators': 531, 'learning_rate':
0.010568659319359854, 'max_depth': 10, 'subsample': 0.6030378402209104,
'colsample_bytree': 0.7165807448112667, 'min_child_weight': 4, 'gamma':
4.990224191289151e-05, 'reg_alpha': 0.0006266407208742843, 'reg_lambda':
3.332996998031588e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:52,604] Trial 190 finished with value:
-1.3093506850661485 and parameters: {'n_estimators': 571, 'learning_rate':
0.011036268315952905, 'max_depth': 9, 'subsample': 0.6244794271320234,
'colsample_bytree': 0.6737062040682891, 'min_child_weight': 3, 'gamma':
0.00010910384736045615, 'reg_alpha': 0.00018329411778478626, 'reg_lambda':
4.713221634940857e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:53,151] Trial 191 finished with value:
-1.3099412132605133 and parameters: {'n_estimators': 511, 'learning_rate':
0.010596642130973036, 'max_depth': 10, 'subsample': 0.7910966327976291,
'colsample_bytree': 0.7219522705668233, 'min_child_weight': 4, 'gamma':
5.1525552441493266e-05, 'reg_alpha': 0.0005653122386889133, 'reg_lambda':
2.6391803911043282e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:53,655] Trial 192 finished with value:
-1.3055361976438293 and parameters: {'n_estimators': 541, 'learning_rate':
0.01056569737593728, 'max_depth': 10, 'subsample': 0.6034229636011875,
'colsample_bytree': 0.7116971333456996, 'min_child_weight': 4, 'gamma':
2.4192829285993874e-05, 'reg_alpha': 0.00034395243627813294, 'reg_lambda':
3.858833725463598e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:54,189] Trial 193 finished with value:
-1.3144557361041806 and parameters: {'n_estimators': 554, 'learning_rate':
0.010010484969309317, 'max_depth': 10, 'subsample': 0.586761506639664,
'colsample_bytree': 0.7558359121577085, 'min_child_weight': 4, 'gamma':
2.5392273762760684e-05, 'reg_alpha': 0.00036301545130882815, 'reg_lambda':
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6.596475308389923e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:54,660] Trial 194 finished with value:
-1.3120724965823036 and parameters: {'n_estimators': 479, 'learning_rate':
0.012191065409715167, 'max_depth': 10, 'subsample': 0.6136362431821967,
'colsample_bytree': 0.7351857682160137, 'min_child_weight': 4, 'gamma':
2.231035581555789e-05, 'reg_alpha': 0.00026202061318562127, 'reg_lambda':
2.0298810170948495e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:55,194] Trial 195 finished with value:
-1.30697572126062 and parameters: {'n_estimators': 593, 'learning_rate':
0.01165187573040522, 'max_depth': 9, 'subsample': 0.6328422445484684,
'colsample_bytree': 0.7107770269071486, 'min_child_weight': 3, 'gamma':
3.463841622995684e-05, 'reg_alpha': 0.0033362469392600306, 'reg_lambda':
4.1441119183117156e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:55,717] Trial 196 finished with value:
-1.3087588494912488 and parameters: {'n_estimators': 586, 'learning_rate':
0.01180984594863881, 'max_depth': 9, 'subsample': 0.6268774479872399,
'colsample_bytree': 0.7047931122300646, 'min_child_weight': 3, 'gamma':
0.000455454027393822, 'reg_alpha': 0.0029948905524515195, 'reg_lambda':
7.65416064369952e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:56,210] Trial 197 finished with value:
-1.310216612182824 and parameters: {'n_estimators': 539, 'learning_rate':
0.012870430237357232, 'max_depth': 9, 'subsample': 0.6372131277267189,
'colsample_bytree': 0.7134454767333047, 'min_child_weight': 3, 'gamma':
3.3865642489375004e-05, 'reg_alpha': 0.0009766593234216074, 'reg_lambda':
1.4212131335187173e-07}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:56,748] Trial 198 finished with value:
-1.3113592410723314 and parameters: {'n_estimators': 518, 'learning_rate':
0.011599492430635287, 'max_depth': 10, 'subsample': 0.5957972805194166,
'colsample_bytree': 0.6933331933799469, 'min_child_weight': 3, 'gamma':
1.481654529497039e-05, 'reg_alpha': 0.004318962389092204, 'reg_lambda':
4.77313926093859e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:57,252] Trial 199 finished with value:
-1.3135821513708865 and parameters: {'n_estimators': 569, 'learning_rate':
0.012723902663255942, 'max_depth': 9, 'subsample': 0.6065663094410751,
'colsample_bytree': 0.7341436725877075, 'min_child_weight': 3, 'gamma':
7.357464255070856e-05, 'reg_alpha': 0.0013570793871603324, 'reg_lambda':
3.220354915148718e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:57,775] Trial 200 finished with value:
-1.3109711805930868 and parameters: {'n_estimators': 594, 'learning_rate':
0.01391443402813394, 'max_depth': 9, 'subsample': 0.6161610889767278,
'colsample_bytree': 0.7438078919913513, 'min_child_weight': 3, 'gamma':
2.5430927225887656e-05, 'reg_alpha': 0.0024616798066978535, 'reg_lambda':
1.2589177739856468e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:58,217] Trial 201 finished with value:
-1.308267353348009 and parameters: {'n_estimators': 675, 'learning_rate':
0.010950392998278696, 'max_depth': 8, 'subsample': 0.650667283959537,
'colsample_bytree': 0.6391013682132248, 'min_child_weight': 4, 'gamma':
3.827871980794682e-05, 'reg_alpha': 0.00013144863285237965, 'reg_lambda':
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3.945944614236304e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:58,678] Trial 202 finished with value:
-1.3075341304571229 and parameters: {'n_estimators': 558, 'learning_rate':
0.011127421777764542, 'max_depth': 9, 'subsample': 0.6281545560155208,
'colsample_bytree': 0.7108459552383222, 'min_child_weight': 4, 'gamma':
5.921400518625745e-05, 'reg_alpha': 0.0003761527513721596, 'reg_lambda':
5.8595566707130313e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:59,161] Trial 203 finished with value:
-1.308715100468061 and parameters: {'n_estimators': 689, 'learning_rate':
0.01206214853789608, 'max_depth': 8, 'subsample': 0.6429205287653642,
'colsample_bytree': 0.6652457688895295, 'min_child_weight': 4, 'gamma':
0.0007801064574403913, 'reg_alpha': 7.121782671956618e-05, 'reg_lambda':
0.9478792555266731}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:10:59,821] Trial 204 finished with value:
-1.315063328386748 and parameters: {'n_estimators': 732, 'learning_rate':
0.010493483243986032, 'max_depth': 9, 'subsample': 0.7634913820673326,
'colsample_bytree': 0.6886084357657174, 'min_child_weight': 3, 'gamma':
1.1619703290898288e-05, 'reg_alpha': 0.0008838060255216849, 'reg_lambda':
2.131779484794002e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:00,357] Trial 205 finished with value:
-1.3055463551520774 and parameters: {'n_estimators': 693, 'learning_rate':
0.010005412792842554, 'max_depth': 9, 'subsample': 0.6735868367296336,
'colsample_bytree': 0.6196178303971234, 'min_child_weight': 4, 'gamma':
1.93361150049005e-05, 'reg_alpha': 0.001867165127727538, 'reg_lambda':
3.6409307187045184e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:00,883] Trial 206 finished with value:
-1.3078132400307305 and parameters: {'n_estimators': 704, 'learning_rate':
0.011594307738659356, 'max_depth': 9, 'subsample': 0.6549042445326534,
'colsample_bytree': 0.598328462385857, 'min_child_weight': 4, 'gamma':
1.8791097816474715e-05, 'reg_alpha': 0.0017987598659103487, 'reg_lambda':
8.720576002142419e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:01,471] Trial 207 finished with value:
-1.3083171576168744 and parameters: {'n_estimators': 684, 'learning_rate':
0.010004595437851697, 'max_depth': 9, 'subsample': 0.6340202310195652,
'colsample_bytree': 0.6144849025593787, 'min_child_weight': 3, 'gamma':
1.252922030711433e-05, 'reg_alpha': 0.0032043864332992004, 'reg_lambda':
2.8965627083897667e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:02,069] Trial 208 finished with value:
-1.3101580516359959 and parameters: {'n_estimators': 719, 'learning_rate':
0.012870283791611559, 'max_depth': 10, 'subsample': 0.6190055698867756,
'colsample_bytree': 0.6277912676838493, 'min_child_weight': 4, 'gamma':
2.2335813489082807e-05, 'reg_alpha': 0.0006791853328117957, 'reg_lambda':
4.6173985442232427e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:02,738] Trial 209 finished with value:
-1.3135237953761594 and parameters: {'n_estimators': 757, 'learning_rate':
0.010797964324763805, 'max_depth': 9, 'subsample': 0.677618119745418,
'colsample_bytree': 0.7211680024697295, 'min_child_weight': 3, 'gamma':
4.192925302679042e-05, 'reg_alpha': 0.0013777519805155069, 'reg_lambda':
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1.877226771738527e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:03,171] Trial 210 finished with value:
-1.3068698845450974 and parameters: {'n_estimators': 492, 'learning_rate':
0.012012707240929196, 'max_depth': 9, 'subsample': 0.6056452043296926,
'colsample_bytree': 0.6455720868610275, 'min_child_weight': 3, 'gamma':
3.0147128571265486e-05, 'reg_alpha': 0.0022277758735192205, 'reg_lambda':
6.800686273535338e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:03,633] Trial 211 finished with value:
-1.3058537093218359 and parameters: {'n_estimators': 528, 'learning_rate':
0.011987223356740857, 'max_depth': 9, 'subsample': 0.6028781366933708,
'colsample_bytree': 0.6519243317305672, 'min_child_weight': 3, 'gamma':
3.116445040233749e-05, 'reg_alpha': 0.008282150081932262, 'reg_lambda':
6.826960200465854e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:04,066] Trial 212 finished with value:
-1.3077139743498443 and parameters: {'n_estimators': 503, 'learning_rate':
0.01146374946024456, 'max_depth': 9, 'subsample': 0.6022267687025176,
'colsample_bytree': 0.6192501722724371, 'min_child_weight': 3, 'gamma':
3.206130258450743e-05, 'reg_alpha': 0.011487322204427986, 'reg_lambda':
1.0945006599096062e-07}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:04,507] Trial 213 finished with value:
-1.3150188827750318 and parameters: {'n_estimators': 487, 'learning_rate':
0.012123542333871676, 'max_depth': 9, 'subsample': 0.6220383474930258,
'colsample_bytree': 0.6548446988614648, 'min_child_weight': 3, 'gamma':
1.9482228095357835e-05, 'reg_alpha': 0.006134671253851384, 'reg_lambda':
7.290456256073481e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:05,023] Trial 214 finished with value:
-1.3155068393503775 and parameters: {'n_estimators': 531, 'learning_rate':
0.013084435202437725, 'max_depth': 9, 'subsample': 0.5858900460126847,
'colsample_bytree': 0.635770551101088, 'min_child_weight': 2, 'gamma':
2.965664430247579e-05, 'reg_alpha': 0.007652812712049612, 'reg_lambda':
5.654066739848879e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:05,517] Trial 215 finished with value:
-1.3085425670651434 and parameters: {'n_estimators': 516, 'learning_rate':
0.01062507439969513, 'max_depth': 9, 'subsample': 0.6338216308280614,
'colsample_bytree': 0.7299825768010564, 'min_child_weight': 3, 'gamma':
4.7626803644939686e-05, 'reg_alpha': 0.017120973599677885, 'reg_lambda':
1.6394699891240285e-07}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:06,094] Trial 216 finished with value:
-1.3103535849199996 and parameters: {'n_estimators': 548, 'learning_rate':
0.011309086157271817, 'max_depth': 10, 'subsample': 0.6445830840239951,
'colsample_bytree': 0.7057219094534359, 'min_child_weight': 3, 'gamma':
1.4938901602702395e-05, 'reg_alpha': 0.0036810181808307023, 'reg_lambda':
3.805470885957407e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:06,473] Trial 217 finished with value:
-1.3244229472994917 and parameters: {'n_estimators': 473, 'learning_rate':
0.027226620283031865, 'max_depth': 9, 'subsample': 0.608352408702928,
'colsample_bytree': 0.6465532916732819, 'min_child_weight': 3, 'gamma':
2.677258185504239e-05, 'reg_alpha': 0.002159618566892243, 'reg_lambda':
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8.37903854725827e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:06,895] Trial 218 finished with value: -1.308315625030425 and parameters: {'n_estimators': 497, 'learning_rate': 0.013752454955227457, 'max_depth': 9, 'subsample': 0.5917552666067156, 'colsample_bytree': 0.6100561020497154, 'min_child_weight': 3, 'gamma': 9.369190317471038e-06, 'reg_alpha': 0.03151894454628283, 'reg_lambda': 5.450508652221252e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:07,599] Trial 219 finished with value: -1.3148180058302472 and parameters: {'n_estimators': 849, 'learning_rate': 0.01211921026482714, 'max_depth': 9, 'subsample': 0.6592127562882228, 'colsample_bytree': 0.6979374892331442, 'min_child_weight': 3, 'gamma': 6.68892905079513e-05, 'reg_alpha': 0.004631997564897016, 'reg_lambda': 1.1632523522407225e-07}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:08,126] Trial 220 finished with value: -1.306625773361038 and parameters: {'n_estimators': 568, 'learning_rate': 0.01063477659057646, 'max_depth': 9, 'subsample': 0.6282608858303885, 'colsample_bytree': 0.7210841054341073, 'min_child_weight': 3, 'gamma': 1.6573635970805805e-05, 'reg_alpha': 0.001095591590753064, 'reg_lambda': 3.201250270903575e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:08,664] Trial 221 finished with value: -1.307773563124462 and parameters: {'n_estimators': 567, 'learning_rate': 0.010053418648025792, 'max_depth': 9, 'subsample': 0.6268402658717974, 'colsample_bytree': 0.721495695862999, 'min_child_weight': 3, 'gamma': 1.8547839283662656e-05, 'reg_alpha': 0.001038238292835354, 'reg_lambda': 3.303167130179961e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:09,191] Trial 222 finished with value: -1.3085556133833818 and parameters: {'n_estimators': 589, 'learning_rate': 0.011213798259891907, 'max_depth': 9, 'subsample': 0.613750024387902, 'colsample_bytree': 0.7140025359591593, 'min_child_weight': 3, 'gamma': 3.4038802901340785e-05, 'reg_alpha': 0.0019245889680889617, 'reg_lambda': 4.335128908848362e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:09,705] Trial 223 finished with value: -1.3120537608364111 and parameters: {'n_estimators': 535, 'learning_rate': 0.010561392318832932, 'max_depth': 9, 'subsample': 0.641209013646593, 'colsample_bytree': 0.7518384766002187, 'min_child_weight': 3, 'gamma': 1.4217397564481449e-05, 'reg_alpha': 0.001368252470106476, 'reg_lambda': 2.4993016873757974e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:10,224] Trial 224 finished with value: -1.3050606016973194 and parameters: {'n_estimators': 572, 'learning_rate': 0.01247277847939886, 'max_depth': 9, 'subsample': 0.6321263336535269, 'colsample_bytree': 0.7379298810079018, 'min_child_weight': 3, 'gamma': 2.142858718867712e-05, 'reg_alpha': 0.0007066298939272569, 'reg_lambda': 6.921764507130232e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:10,804] Trial 225 finished with value: -1.30508084074535 and parameters: {'n_estimators': 561, 'learning_rate': 0.012909388920131827, 'max_depth': 10, 'subsample': 0.6194179680161371, 'colsample_bytree': 0.7389852368401539, 'min_child_weight': 3, 'gamma': 2.2174365681861997e-05, 'reg_alpha': 0.0005650190238894832, 'reg_lambda':

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6.634273096668609e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:11,394] Trial 226 finished with value:
-1.3057176242571742 and parameters: {'n_estimators': 559, 'learning_rate':
0.013183414321146402, 'max_depth': 10, 'subsample': 0.619157146391701,
'colsample_bytree': 0.7719742120396446, 'min_child_weight': 3, 'gamma':
1.0888680246959484e-05, 'reg_alpha': 0.0005750054691359149, 'reg_lambda':
7.560959806483558e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:11,995] Trial 227 finished with value:
-1.3121128494487841 and parameters: {'n_estimators': 564, 'learning_rate':
0.014992958926896552, 'max_depth': 10, 'subsample': 0.6186639680142237,
'colsample_bytree': 0.7589452391233088, 'min_child_weight': 3, 'gamma':
8.792570405189077e-06, 'reg_alpha': 0.0004805445702757365, 'reg_lambda':
9.041768596998271e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:12,595] Trial 228 finished with value:
-1.310879142131083 and parameters: {'n_estimators': 549, 'learning_rate':
0.013333495282684458, 'max_depth': 10, 'subsample': 0.6486156456813041,
'colsample_bytree': 0.7643096899323766, 'min_child_weight': 3, 'gamma':
1.1853222371853743e-05, 'reg_alpha': 0.0006427639663135658, 'reg_lambda':
6.686742819308459e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:13,155] Trial 229 finished with value:
-1.3124827034274233 and parameters: {'n_estimators': 532, 'learning_rate':
0.014414763055490682, 'max_depth': 10, 'subsample': 0.6216328455697211,
'colsample_bytree': 0.7432453039894995, 'min_child_weight': 3, 'gamma':
7.610279140924158e-06, 'reg_alpha': 0.0003566581740484673, 'reg_lambda':
2.965828433345607e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:13,870] Trial 230 finished with value:
-1.3125265409944251 and parameters: {'n_estimators': 572, 'learning_rate':
0.013283681837965368, 'max_depth': 11, 'subsample': 0.6402563685333827,
'colsample_bytree': 0.7922963952487696, 'min_child_weight': 3, 'gamma':
4.158196409033993e-06, 'reg_alpha': 0.0008045976368346358, 'reg_lambda':
1.2433232222962164e-07}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:14,293] Trial 231 finished with value:
-1.3146710606354963 and parameters: {'n_estimators': 553, 'learning_rate':
0.012628053778189832, 'max_depth': 10, 'subsample': 0.6289299457776131,
'colsample_bytree': 0.7340910352588712, 'min_child_weight': 7, 'gamma':
1.8750555671848356e-05, 'reg_alpha': 0.0005465937913935522, 'reg_lambda':
5.394201890108665e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:14,663] Trial 232 finished with value:
-1.3118344966122668 and parameters: {'n_estimators': 575, 'learning_rate':
0.011243601172048475, 'max_depth': 7, 'subsample': 0.6312788467484476,
'colsample_bytree': 0.7287566543379689, 'min_child_weight': 3, 'gamma':
2.1460325617191154e-05, 'reg_alpha': 0.0009762627970467659, 'reg_lambda':
0.07659623091150014}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:15,243] Trial 233 finished with value:
-1.3133615228094488 and parameters: {'n_estimators': 603, 'learning_rate':
0.012553741872906917, 'max_depth': 10, 'subsample': 0.6148818147393694,
'colsample_bytree': 0.7424280151813291, 'min_child_weight': 4, 'gamma':
1.2335978137337607e-05, 'reg_alpha': 0.10100667138085966, 'reg_lambda':
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4.312784378325444e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:15,969] Trial 234 finished with value:
-1.335161000113308 and parameters: {'n_estimators': 521, 'learning_rate':
0.010705312046556302, 'max_depth': 10, 'subsample': 0.9697285171126336,
'colsample_bytree': 0.725502728958582, 'min_child_weight': 3, 'gamma':
2.3864411338790297e-05, 'reg_alpha': 0.0004263271355271761, 'reg_lambda':
6.960128110006604e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:16,580] Trial 235 finished with value:
-1.307741205176896 and parameters: {'n_estimators': 542, 'learning_rate':
0.012324336874422772, 'max_depth': 9, 'subsample': 0.6519007998392241,
'colsample_bytree': 0.7686284380589655, 'min_child_weight': 3, 'gamma':
1.5959344887482306e-05, 'reg_alpha': 0.0007161615375469671, 'reg_lambda':
3.181519549435169e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:17,190] Trial 236 finished with value:
-1.3170847725550605 and parameters: {'n_estimators': 580, 'learning_rate':
0.013886004994968275, 'max_depth': 10, 'subsample': 0.5989404104909504,
'colsample_bytree': 0.778475905714463, 'min_child_weight': 4, 'gamma':
4.823479879419198e-05, 'reg_alpha': 0.00027439914915513945, 'reg_lambda':
9.029690913668254e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:17,768] Trial 237 finished with value:
-1.3105743157715881 and parameters: {'n_estimators': 559, 'learning_rate':
0.011513645359004192, 'max_depth': 9, 'subsample': 0.6221621202154831,
'colsample_bytree': 0.8186632799532068, 'min_child_weight': 4, 'gamma':
1.0279847606899015e-05, 'reg_alpha': 0.0010287721989475946, 'reg_lambda':
5.294108501815535e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:18,496] Trial 238 finished with value:
-1.3118816102454793 and parameters: {'n_estimators': 523, 'learning_rate':
0.015733410690416238, 'max_depth': 10, 'subsample': 0.6402060969601172,
'colsample_bytree': 0.6995982483239086, 'min_child_weight': 3, 'gamma':
1.6976122009941787e-05, 'reg_alpha': 0.00045364435709402347, 'reg_lambda':
2.3392427675391884e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:19,748] Trial 239 finished with value:
-1.3208238878421381 and parameters: {'n_estimators': 609, 'learning_rate':
0.010082296530191964, 'max_depth': 11, 'subsample': 0.6103764646891309,
'colsample_bytree': 0.738484639897774, 'min_child_weight': 2, 'gamma':
2.5080540327053208e-05, 'reg_alpha': 0.057957448092659646, 'reg_lambda':
3.7652303869320104e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:20,235] Trial 240 finished with value:
-1.3116194999180077 and parameters: {'n_estimators': 585, 'learning_rate':
0.013018768210104009, 'max_depth': 7, 'subsample': 0.6299369190419418,
'colsample_bytree': 0.7192690697443508, 'min_child_weight': 4, 'gamma':
4.197948630233309e-05, 'reg_alpha': 0.0006515833149899142, 'reg_lambda':
1.7673952386021516e-07}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:20,867] Trial 241 finished with value:
-1.3084473612132024 and parameters: {'n_estimators': 510, 'learning_rate':
0.012028553173336315, 'max_depth': 9, 'subsample': 0.6051379249810445,
'colsample_bytree': 0.6251806269304333, 'min_child_weight': 3, 'gamma':
2.9062023372316365e-05, 'reg_alpha': 0.0012722406251505316, 'reg_lambda':
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5.913817935247446e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:21,553] Trial 242 finished with value:
-1.3121680524784631 and parameters: {'n_estimators': 546, 'learning_rate':
0.011377926279914137, 'max_depth': 9, 'subsample': 0.6065392861714273,
'colsample_bytree': 0.7517231365375732, 'min_child_weight': 3, 'gamma':
2.446811459607021e-05, 'reg_alpha': 0.0015191844996518334, 'reg_lambda':
7.146398908593861e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:22,457] Trial 243 finished with value:
-1.3205673486185843 and parameters: {'n_estimators': 695, 'learning_rate':
0.01204471335691723, 'max_depth': 9, 'subsample': 0.6193272752465575,
'colsample_bytree': 0.8382230243407627, 'min_child_weight': 3, 'gamma':
1.4303058355966887e-05, 'reg_alpha': 0.000882440656242605, 'reg_lambda':
8.944601238952737e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:23,236] Trial 244 finished with value:
-1.3124243051504154 and parameters: {'n_estimators': 672, 'learning_rate':
0.010603614063678848, 'max_depth': 9, 'subsample': 0.5968087674625948,
'colsample_bytree': 0.6502596893321327, 'min_child_weight': 3, 'gamma':
3.439020451804e-05, 'reg_alpha': 0.00046765006504576983, 'reg_lambda':
4.56163548630036e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:23,809] Trial 245 finished with value:
-1.317964426892585 and parameters: {'n_estimators': 565, 'learning_rate':
0.012660726143729017, 'max_depth': 9, 'subsample': 0.6368503271570599,
'colsample_bytree': 0.6813992507216392, 'min_child_weight': 6, 'gamma':
8.701889379705667e-05, 'reg_alpha': 0.0002919960583082252, 'reg_lambda':
6.340596252276355e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:24,515] Trial 246 finished with value:
-1.3082185122637775 and parameters: {'n_estimators': 543, 'learning_rate':
0.010975008166382477, 'max_depth': 9, 'subsample': 0.625262218562889,
'colsample_bytree': 0.7306417223404644, 'min_child_weight': 3, 'gamma':
5.5582532174430063e-05, 'reg_alpha': 0.001963558753046814, 'reg_lambda':
2.991564093890918e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:25,169] Trial 247 finished with value:
-1.311060019009678 and parameters: {'n_estimators': 528, 'learning_rate':
0.01394707741079941, 'max_depth': 10, 'subsample': 0.648027929669472,
'colsample_bytree': 0.7054896560159282, 'min_child_weight': 3, 'gamma':
2.5324474693101663e-05, 'reg_alpha': 0.0007497972019449636, 'reg_lambda':
1.1118649756017288e-07}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:25,714] Trial 248 finished with value:
-1.3107596857044206 and parameters: {'n_estimators': 654, 'learning_rate':
0.011818207087808702, 'max_depth': 9, 'subsample': 0.6134744845945392,
'colsample_bytree': 0.7184794910562271, 'min_child_weight': 4, 'gamma':
1.9168911372291284e-05, 'reg_alpha': 1.9171268618501578e-08, 'reg_lambda':
4.403266590818328e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:26,252] Trial 249 finished with value:
-1.3184423154608826 and parameters: {'n_estimators': 623, 'learning_rate':
0.01314971144119692, 'max_depth': 9, 'subsample': 0.6588059708004027,
'colsample_bytree': 0.6353143766356673, 'min_child_weight': 3, 'gamma':
7.681811270678365e-06, 'reg_alpha': 0.0011037666445006702, 'reg_lambda':
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2.6221179394357594e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:26,807] Trial 250 finished with value:
-1.3096153192168591 and parameters: {'n_estimators': 506, 'learning_rate':
0.010629414161055112, 'max_depth': 10, 'subsample': 0.6339990761707929,
'colsample_bytree': 0.6606004390709613, 'min_child_weight': 3, 'gamma':
3.9755760033458364e-05, 'reg_alpha': 0.0003464581719594625, 'reg_lambda':
6.777446157291481e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:27,249] Trial 251 finished with value:
-1.3123549070792926 and parameters: {'n_estimators': 494, 'learning_rate':
0.010017032256978075, 'max_depth': 9, 'subsample': 0.6028612134549679,
'colsample_bytree': 0.6929091473656813, 'min_child_weight': 4, 'gamma':
1.1489880477172176e-05, 'reg_alpha': 0.0006205927944129343, 'reg_lambda':
1.6858438807763483e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:27,764] Trial 252 finished with value:
-1.307787915517128 and parameters: {'n_estimators': 587, 'learning_rate':
0.011301837936803907, 'max_depth': 9, 'subsample': 0.6697912841742039,
'colsample_bytree': 0.7479491418090625, 'min_child_weight': 4, 'gamma':
1.9319678250715353e-05, 'reg_alpha': 0.0022581509357911217, 'reg_lambda':
3.840400933962526e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:28,258] Trial 253 finished with value:
-1.3256787767435174 and parameters: {'n_estimators': 698, 'learning_rate':
0.012289575610801523, 'max_depth': 10, 'subsample': 0.6239807243249442,
'colsample_bytree': 0.7115847068360152, 'min_child_weight': 8, 'gamma':
5.9233863710387716e-05, 'reg_alpha': 0.0014841021520301085, 'reg_lambda':
5.516469444306563e-08}. Best is trial 168 with value: -1.3028225035370153.
[I 2026-03-13 19:11:28,804] Trial 254 finished with value:
-1.3023358402942897 and parameters: {'n_estimators': 561, 'learning_rate':
0.01170093236964921, 'max_depth': 9, 'subsample': 0.6447552755495245,
'colsample_bytree': 0.7361122256381499, 'min_child_weight': 3, 'gamma':
3.168190711790191e-05, 'reg_alpha': 3.480040398751083e-07, 'reg_lambda':
8.729287181189514e-08}. Best is trial 254 with value: -1.3023358402942897.
[I 2026-03-13 19:11:29,330] Trial 255 finished with value:
-1.3056975977745153 and parameters: {'n_estimators': 566, 'learning_rate':
0.014552923159208591, 'max_depth': 9, 'subsample': 0.6459558406229036,
'colsample_bytree': 0.7350292293336734, 'min_child_weight': 3, 'gamma':
1.4384793964957195e-05, 'reg_alpha': 6.467784214648959e-07, 'reg_lambda':
1.283426256729406e-07}. Best is trial 254 with value: -1.3023358402942897.
[I 2026-03-13 19:11:29,553] Trial 256 finished with value:
-1.3413991985907823 and parameters: {'n_estimators': 557, 'learning_rate':
0.014322387452949533, 'max_depth': 5, 'subsample': 0.6512473697777997,
'colsample_bytree': 0.7367264748072552, 'min_child_weight': 3, 'gamma':
1.0312707501626443e-05, 'reg_alpha': 7.030970232883953e-07, 'reg_lambda':
1.1647650611078444e-07}. Best is trial 254 with value: -1.3023358402942897.
[I 2026-03-13 19:11:30,006] Trial 257 finished with value:
-1.3141642989501057 and parameters: {'n_estimators': 562, 'learning_rate':
0.01497463262509982, 'max_depth': 8, 'subsample': 0.6438842846220186,
'colsample_bytree': 0.755691107679797, 'min_child_weight': 3, 'gamma':
1.5769506278031984e-05, 'reg_alpha': 7.7060139136438e-08, 'reg_lambda':
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2.2189214266319906e-07}. Best is trial 254 with value: -1.3023358402942897.
[I 2026-03-13 19:11:30,544] Trial 258 finished with value:
-1.304429345058878 and parameters: {'n_estimators': 543, 'learning_rate':
0.01091480771892869, 'max_depth': 9, 'subsample': 0.6434907693840083,
'colsample_bytree': 0.7266993829648128, 'min_child_weight': 3, 'gamma':
7.040068093010404e-06, 'reg_alpha': 1.5063579906329002e-07, 'reg_lambda':
8.579677913519152e-08}. Best is trial 254 with value: -1.3023358402942897.
[I 2026-03-13 19:11:31,049] Trial 259 finished with value:
-1.311797059616812 and parameters: {'n_estimators': 535, 'learning_rate':
0.013573973978495569, 'max_depth': 9, 'subsample': 0.66355577225907,
'colsample_bytree': 0.7356747109163234, 'min_child_weight': 3, 'gamma':
6.136509590756116e-06, 'reg_alpha': 3.4941117761741527e-07, 'reg_lambda':
1.5339598049936935e-07}. Best is trial 254 with value: -1.3023358402942897.
[I 2026-03-13 19:11:31,491] Trial 260 finished with value:
-1.3162941078010764 and parameters: {'n_estimators': 541, 'learning_rate':
0.011441200952819574, 'max_depth': 8, 'subsample': 0.657569571901996,
'colsample_bytree': 0.7500569355009649, 'min_child_weight': 3, 'gamma':
6.088105217485701e-06, 'reg_alpha': 2.9581181025492113e-07, 'reg_lambda':
1.260019386158948e-07}. Best is trial 254 with value: -1.3023358402942897.
[I 2026-03-13 19:11:32,172] Trial 261 finished with value:
-1.3260105097613184 and parameters: {'n_estimators': 678, 'learning_rate':
0.012997686821671066, 'max_depth': 10, 'subsample': 0.6453363960466781,
'colsample_bytree': 0.9287630043963916, 'min_child_weight': 5, 'gamma':
1.946717569281418e-06, 'reg_alpha': 1.124060852208634e-07, 'reg_lambda':
0.00865793382808464}. Best is trial 254 with value: -1.3023358402942897.
[I 2026-03-13 19:11:32,697] Trial 262 finished with value:
-1.3038822533389316 and parameters: {'n_estimators': 520, 'learning_rate':
0.011217749091027192, 'max_depth': 9, 'subsample': 0.6422354114251015,
'colsample_bytree': 0.7256484881689387, 'min_child_weight': 3, 'gamma':
4.327502642692222e-06, 'reg_alpha': 4.5387551478253875e-08, 'reg_lambda':
8.079821925656267e-08}. Best is trial 254 with value: -1.3023358402942897.
[I 2026-03-13 19:11:33,210] Trial 263 finished with value:
-1.3083258189421827 and parameters: {'n_estimators': 523, 'learning_rate':
0.01131331559626449, 'max_depth': 9, 'subsample': 0.6726171669716041,
'colsample_bytree': 0.7299382440969795, 'min_child_weight': 3, 'gamma':
3.303093282432725e-06, 'reg_alpha': 3.8826669286876425e-08, 'reg_lambda':
1.003120219134565e-07}. Best is trial 254 with value: -1.3023358402942897.
[I 2026-03-13 19:11:33,382] Trial 264 finished with value:
-1.383558993558527 and parameters: {'n_estimators': 550, 'learning_rate':
0.01102880144636436, 'max_depth': 4, 'subsample': 0.6602328255380351,
'colsample_bytree': 0.739898228043649, 'min_child_weight': 3, 'gamma':
3.910010658974532e-06, 'reg_alpha': 1.814992782518468e-07, 'reg_lambda':
8.52734842550977e-08}. Best is trial 254 with value: -1.3023358402942897.
[I 2026-03-13 19:11:33,688] Trial 265 finished with value:
-1.3367560572667079 and parameters: {'n_estimators': 515, 'learning_rate':
0.010547337888438963, 'max_depth': 6, 'subsample': 0.6363818343468306,
'colsample_bytree': 0.7654163785966073, 'min_child_weight': 2, 'gamma':
8.153342428532362e-06, 'reg_alpha': 2.6719028352928273e-08, 'reg_lambda':
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1.826245121293896e-07}. Best is trial 254 with value: -1.3023358402942897.
[I 2026-03-13 19:11:34,320] Trial 266 finished with value:
-1.3224201853493582 and parameters: {'n_estimators': 534, 'learning_rate':
0.02278823717767853, 'max_depth': 10, 'subsample': 0.6517112466888308,
'colsample_bytree': 0.7310750234383281, 'min_child_weight': 2, 'gamma':
9.465104991254594e-06, 'reg_alpha': 5.844602876279178e-07, 'reg_lambda':
9.033460426836679e-08}. Best is trial 254 with value: -1.3023358402942897.
[I 2026-03-13 19:11:34,940] Trial 267 finished with value:
-1.3151143973795263 and parameters: {'n_estimators': 577, 'learning_rate':
0.010018782565283227, 'max_depth': 9, 'subsample': 0.8888211434914295,
'colsample_bytree': 0.7469780612818311, 'min_child_weight': 3, 'gamma':
4.292968700638914e-06, 'reg_alpha': 5.8622996215268044e-08, 'reg_lambda':
4.5689445755716746e-08}. Best is trial 254 with value: -1.3023358402942897.
[I 2026-03-13 19:11:35,695] Trial 268 finished with value:
-1.3162028363662024 and parameters: {'n_estimators': 900, 'learning_rate':
0.011830947672284167, 'max_depth': 9, 'subsample': 0.6384184539331208,
'colsample_bytree': 0.7247124034415495, 'min_child_weight': 3, 'gamma':
7.790460404647424e-07, 'reg_alpha': 9.555273421393976e-07, 'reg_lambda':
0.00037153397024324274}. Best is trial 254 with value: -1.3023358402942897.
[I 2026-03-13 19:11:36,119] Trial 269 finished with value:
-1.3164140748890805 and parameters: {'n_estimators': 605, 'learning_rate':
0.012297666236686608, 'max_depth': 8, 'subsample': 0.652468603234962,
'colsample_bytree': 0.772943453750078, 'min_child_weight': 5, 'gamma':
1.2724825502389824e-06, 'reg_alpha': 2.108193914073404e-07, 'reg_lambda':
2.2281528179648458e-08}. Best is trial 254 with value: -1.3023358402942897.
[I 2026-03-13 19:11:36,657] Trial 270 finished with value:
-1.3014346401217736 and parameters: {'n_estimators': 554, 'learning_rate':
0.011172725753833365, 'max_depth': 10, 'subsample': 0.640962652965751,
'colsample_bytree': 0.7190396685432388, 'min_child_weight': 4, 'gamma':
5.2900759182755725e-06, 'reg_alpha': 1.664986778379984e-08, 'reg_lambda':
6.32450086549987e-08}. Best is trial 270 with value: -1.3014346401217736.
[I 2026-03-13 19:11:37,183] Trial 271 finished with value:
-1.304556757815279 and parameters: {'n_estimators': 549, 'learning_rate':
0.011458555773192951, 'max_depth': 9, 'subsample': 0.7399374803180123,
'colsample_bytree': 0.7441805881122016, 'min_child_weight': 4, 'gamma':
4.873064036457715e-06, 'reg_alpha': 5.12449017023886e-08, 'reg_lambda':
7.617860830198439e-08}. Best is trial 270 with value: -1.3014346401217736.
[I 2026-03-13 19:11:37,684] Trial 272 finished with value:
-1.310898908586985 and parameters: {'n_estimators': 555, 'learning_rate':
0.011298844435543758, 'max_depth': 9, 'subsample': 0.6660173871714614,
'colsample_bytree': 0.7543526005817794, 'min_child_weight': 4, 'gamma':
3.034351771601315e-06, 'reg_alpha': 1.434983289662896e-07, 'reg_lambda':
1.3824533574876408e-07}. Best is trial 270 with value: -1.3014346401217736.
[I 2026-03-13 19:11:38,180] Trial 273 finished with value:
-1.3007401584614298 and parameters: {'n_estimators': 540, 'learning_rate':
0.012830396591835618, 'max_depth': 9, 'subsample': 0.7319650988655815,
'colsample_bytree': 0.7811285350688918, 'min_child_weight': 4, 'gamma':
5.02761433747073e-06, 'reg_alpha': 4.977063144182273e-08, 'reg_lambda':
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2.8845093421390575e-07}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:38,559] Trial 274 finished with value:
-1.5110366736086673 and parameters: {'n_estimators': 546, 'learning_rate':
0.2984792510084404, 'max_depth': 9, 'subsample': 0.7406941135372296,
'colsample_bytree': 0.8060098385933911, 'min_child_weight': 4, 'gamma':
4.283048965097428e-06, 'reg_alpha': 5.76893818673199e-08, 'reg_lambda':
3.200150662037529e-07}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:39,134] Trial 275 finished with value:
-1.3114436300415095 and parameters: {'n_estimators': 566, 'learning_rate':
0.013097820833716477, 'max_depth': 10, 'subsample': 0.6379648573725056,
'colsample_bytree': 0.7748670531641233, 'min_child_weight': 4, 'gamma':
5.388862915668473e-06, 'reg_alpha': 2.7810886852353777e-08, 'reg_lambda':
1.3969376329464567e-07}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:39,615] Trial 276 finished with value:
-1.3106491256514399 and parameters: {'n_estimators': 519, 'learning_rate':
0.012679271321330038, 'max_depth': 9, 'subsample': 0.738715383979357,
'colsample_bytree': 0.7801089728227052, 'min_child_weight': 4, 'gamma':
1.9159126968185596e-06, 'reg_alpha': 9.161282701885079e-08, 'reg_lambda':
9.204898755089233e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:40,120] Trial 277 finished with value:
-1.306994314379643 and parameters: {'n_estimators': 577, 'learning_rate':
0.015422855558132317, 'max_depth': 9, 'subsample': 0.6772212084104194,
'colsample_bytree': 0.7855444948840375, 'min_child_weight': 4, 'gamma':
5.7629953842897275e-06, 'reg_alpha': 1.3608631815660097e-08, 'reg_lambda':
7.901030076627855e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:40,580] Trial 278 finished with value:
-1.3453703738576355 and parameters: {'n_estimators': 546, 'learning_rate':
0.06186153196698542, 'max_depth': 10, 'subsample': 0.7459255581378371,
'colsample_bytree': 0.7595161188058556, 'min_child_weight': 4, 'gamma':
2.9893635853378167e-06, 'reg_alpha': 5.137514188829846e-08, 'reg_lambda':
2.0620720720930404e-07}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:41,116] Trial 279 finished with value:
-1.3113103645935669 and parameters: {'n_estimators': 598, 'learning_rate':
0.010907421889899868, 'max_depth': 9, 'subsample': 0.7719517319638067,
'colsample_bytree': 0.7427756380973274, 'min_child_weight': 4, 'gamma':
7.152808076461433e-06, 'reg_alpha': 3.7982642133672045e-08, 'reg_lambda':
5.120257830938939e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:41,631] Trial 280 finished with value:
-1.3076931210576488 and parameters: {'n_estimators': 561, 'learning_rate':
0.014230599056457304, 'max_depth': 9, 'subsample': 0.8103422955968028,
'colsample_bytree': 0.740129675108519, 'min_child_weight': 4, 'gamma':
4.291648757149176e-06, 'reg_alpha': 1.244334143761046e-07, 'reg_lambda':
9.529985746248201e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:42,177] Trial 281 finished with value:
-1.305412060118431 and parameters: {'n_estimators': 509, 'learning_rate':
0.010015004641720868, 'max_depth': 10, 'subsample': 0.7250766906799517,
'colsample_bytree': 0.7196032992978476, 'min_child_weight': 4, 'gamma':
7.61552658665456e-06, 'reg_alpha': 8.28250158130558e-08, 'reg_lambda':
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3.451245603525831e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:42,713] Trial 282 finished with value:
-1.308959597128532 and parameters: {'n_estimators': 514, 'learning_rate':
0.011876277015034416, 'max_depth': 10, 'subsample': 0.7237762533105346,
'colsample_bytree': 0.724087540201734, 'min_child_weight': 4, 'gamma':
6.680201415230171e-06, 'reg_alpha': 5.240154289133115e-08, 'reg_lambda':
3.1842322257604515e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:43,259] Trial 283 finished with value:
-1.3092925064279795 and parameters: {'n_estimators': 502, 'learning_rate':
0.010032346453455118, 'max_depth': 10, 'subsample': 0.6821273932119645,
'colsample_bytree': 0.7519220481736244, 'min_child_weight': 4, 'gamma':
9.5078732125459e-06, 'reg_alpha': 9.667514044589368e-08, 'reg_lambda':
6.437256347042856e-05}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:43,897] Trial 284 finished with value:
-1.3082379545905245 and parameters: {'n_estimators': 534, 'learning_rate':
0.012926002279502604, 'max_depth': 11, 'subsample': 0.7324133858416332,
'colsample_bytree': 0.7920268787791537, 'min_child_weight': 4, 'gamma':
2.656696283643086e-06, 'reg_alpha': 7.306342468578657e-08, 'reg_lambda':
2.3403701494934288e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:44,463] Trial 285 finished with value:
-1.3050342796568064 and parameters: {'n_estimators': 537, 'learning_rate':
0.011085181537171617, 'max_depth': 10, 'subsample': 0.734068015490496,
'colsample_bytree': 0.7323943372928935, 'min_child_weight': 4, 'gamma':
5.112522579101094e-06, 'reg_alpha': 1.5027223834048308e-07, 'reg_lambda':
1.6470384262222492e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:44,883] Trial 286 finished with value:
-1.3917559522815892 and parameters: {'n_estimators': 525, 'learning_rate':
0.14263860052034766, 'max_depth': 10, 'subsample': 0.7559446969654203,
'colsample_bytree': 0.7148599797194363, 'min_child_weight': 4, 'gamma':
3.9811759031527374e-06, 'reg_alpha': 1.6681820535566094e-07, 'reg_lambda':
1.05873412046699e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:45,242] Trial 287 finished with value:
-1.3407733549935983 and parameters: {'n_estimators': 236, 'learning_rate':
0.010693264769191888, 'max_depth': 11, 'subsample': 0.7539250157509314,
'colsample_bytree': 0.7343589124348561, 'min_child_weight': 4, 'gamma':
4.706601351754568e-06, 'reg_alpha': 7.576059378321435e-08, 'reg_lambda':
1.4366167251265857e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:45,810] Trial 288 finished with value:
-1.306289621213273 and parameters: {'n_estimators': 541, 'learning_rate':
0.010028198203985914, 'max_depth': 10, 'subsample': 0.718168667391667,
'colsample_bytree': 0.7248154910549139, 'min_child_weight': 4, 'gamma':
6.257640260559635e-06, 'reg_alpha': 4.1330311892489604e-07, 'reg_lambda':
5.085205440938377e-07}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:46,296] Trial 289 finished with value:
-1.307837348078186 and parameters: {'n_estimators': 499, 'learning_rate':
0.011334092389127144, 'max_depth': 10, 'subsample': 0.7017794280714582,
'colsample_bytree': 0.6056680615713324, 'min_child_weight': 4, 'gamma':
3.1184884332174056e-06, 'reg_alpha': 2.3102157865343398e-07, 'reg_lambda':

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1.69384967969644e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:46,766] Trial 290 finished with value:
-1.3066195845583193 and parameters: {'n_estimators': 482, 'learning_rate':
0.010935659157560361, 'max_depth': 9, 'subsample': 0.7367297708434268,
'colsample_bytree': 0.7155518886565599, 'min_child_weight': 4, 'gamma':
7.372761991627148e-06, 'reg_alpha': 3.224718344129258e-08, 'reg_lambda':
2.084903048524236e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:47,497] Trial 291 finished with value:
-1.3057836189206498 and parameters: {'n_estimators': 583, 'learning_rate':
0.011914244759160315, 'max_depth': 10, 'subsample': 0.7320185450631596,
'colsample_bytree': 0.7402810198917328, 'min_child_weight': 4, 'gamma':
5.508482537262892e-06, 'reg_alpha': 1.3889288966876234e-07, 'reg_lambda':
3.764877892108313e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:48,107] Trial 292 finished with value:
-1.3056355291751225 and parameters: {'n_estimators': 511, 'learning_rate':
0.010660605071160999, 'max_depth': 9, 'subsample': 0.7264374085119933,
'colsample_bytree': 0.7275578903995517, 'min_child_weight': 4, 'gamma':
1.2006130900785654e-05, 'reg_alpha': 6.814665840627557e-08, 'reg_lambda':
2.5697199048574354e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:48,628] Trial 293 finished with value:
-1.3055995324745948 and parameters: {'n_estimators': 507, 'learning_rate':
0.010597201364872947, 'max_depth': 9, 'subsample': 0.7175928134511576,
'colsample_bytree': 0.7195093988549858, 'min_child_weight': 4, 'gamma':
9.636276126423957e-06, 'reg_alpha': 6.937667137464787e-08, 'reg_lambda':
2.5654058703151442e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:49,137] Trial 294 finished with value:
-1.3092978575981764 and parameters: {'n_estimators': 456, 'learning_rate':
0.011303242329031241, 'max_depth': 10, 'subsample': 0.7107843742616657,
'colsample_bytree': 0.7052354043566464, 'min_child_weight': 4, 'gamma':
2.283058206411011e-06, 'reg_alpha': 4.158290207200785e-08, 'reg_lambda':
2.0283281460450826e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:49,597] Trial 295 finished with value:
-1.3077964376538926 and parameters: {'n_estimators': 488, 'learning_rate':
0.01045732024099547, 'max_depth': 9, 'subsample': 0.7113350456864199,
'colsample_bytree': 0.7160731663869001, 'min_child_weight': 4, 'gamma':
8.738975451831778e-06, 'reg_alpha': 8.149324379989037e-08, 'reg_lambda':
1.4859629813463056e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:50,228] Trial 296 finished with value:
-1.3075510827984562 and parameters: {'n_estimators': 821, 'learning_rate':
0.01165754181310857, 'max_depth': 9, 'subsample': 0.7489110204539654,
'colsample_bytree': 0.6226577327429974, 'min_child_weight': 4, 'gamma':
4.7443130697267995e-06, 'reg_alpha': 5.2005110389459526e-08, 'reg_lambda':
3.169165810577935e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:50,721] Trial 297 finished with value:
-1.3061392994786092 and parameters: {'n_estimators': 525, 'learning_rate':
0.010008641904559126, 'max_depth': 9, 'subsample': 0.7212371279747727,
'colsample_bytree': 0.7213509026524003, 'min_child_weight': 4, 'gamma':
6.867358381283797e-06, 'reg_alpha': 1.536491441285758e-07, 'reg_lambda':
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3.720472782384539e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:51,111] Trial 298 finished with value:
-1.3100997652671822 and parameters: {'n_estimators': 474, 'learning_rate':
0.010992562947907802, 'max_depth': 8, 'subsample': 0.7389027003125304,
'colsample_bytree': 0.7505762011374644, 'min_child_weight': 4, 'gamma':
3.461499629942625e-06, 'reg_alpha': 1.1640686834693628e-07, 'reg_lambda':
1.2153761942357422e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:51,732] Trial 299 finished with value:
-1.3093970694267538 and parameters: {'n_estimators': 540, 'learning_rate':
0.011752907685015879, 'max_depth': 10, 'subsample': 0.7619702951877037,
'colsample_bytree': 0.8787714783770865, 'min_child_weight': 4, 'gamma':
1.1865844831000671e-05, 'reg_alpha': 2.2143725915315424e-08, 'reg_lambda':
2.6472563881673794e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:51,997] Trial 300 finished with value:
-1.3294952378407228 and parameters: {'n_estimators': 506, 'learning_rate':
0.012503046738456158, 'max_depth': 6, 'subsample': 0.7324702124772651,
'colsample_bytree': 0.7304844463366137, 'min_child_weight': 4, 'gamma':
8.871070580223446e-06, 'reg_alpha': 1.671117000998136e-08, 'reg_lambda':
4.719352630757213e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:52,659] Trial 301 finished with value:
-1.3094894706263414 and parameters: {'n_estimators': 784, 'learning_rate':
0.010981531792753347, 'max_depth': 9, 'subsample': 0.7259437503873373,
'colsample_bytree': 0.7089343463989027, 'min_child_weight': 4, 'gamma':
6.802848647565647e-08, 'reg_alpha': 9.689201117646042e-08, 'reg_lambda':
1.5923051312385836e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:53,101] Trial 302 finished with value:
-1.3049477884988496 and parameters: {'n_estimators': 522, 'learning_rate':
0.01220771705315826, 'max_depth': 9, 'subsample': 0.7458626175912495,
'colsample_bytree': 0.5949817165352606, 'min_child_weight': 4, 'gamma':
1.8286494763797747e-06, 'reg_alpha': 4.212587010559548e-08, 'reg_lambda':
4.194783218663938e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:53,606] Trial 303 finished with value:
-1.305507656987733 and parameters: {'n_estimators': 532, 'learning_rate':
0.012402211505935233, 'max_depth': 10, 'subsample': 0.7415504337403847,
'colsample_bytree': 0.591839850684793, 'min_child_weight': 4, 'gamma':
1.6552589815444984e-06, 'reg_alpha': 3.880494706190466e-08, 'reg_lambda':
4.296835395129343e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:54,382] Trial 304 finished with value:
-1.3173439258538948 and parameters: {'n_estimators': 944, 'learning_rate':
0.012812436922628383, 'max_depth': 10, 'subsample': 0.7522366162617092,
'colsample_bytree': 0.5766404201314452, 'min_child_weight': 4, 'gamma':
1.4765160543342171e-06, 'reg_alpha': 3.334452946785025e-08, 'reg_lambda':
4.3556117677471004e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:54,992] Trial 305 finished with value:
-1.3118781797497927 and parameters: {'n_estimators': 528, 'learning_rate':
0.012431686946867257, 'max_depth': 10, 'subsample': 0.7462987669351637,
'colsample_bytree': 0.7458610295316999, 'min_child_weight': 4, 'gamma':
2.70958351498148e-06, 'reg_alpha': 4.098331413920713e-08, 'reg_lambda':
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2.209241885906539e-06}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:55,509] Trial 306 finished with value:
-1.3093594626943204 and parameters: {'n_estimators': 551, 'learning_rate':
0.013636248227854688, 'max_depth': 10, 'subsample': 0.7573372486410547,
'colsample_bytree': 0.5855449192719848, 'min_child_weight': 4, 'gamma':
1.1601920704930887e-06, 'reg_alpha': 2.673596662772676e-08, 'reg_lambda':
5.540559446346917e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:56,028] Trial 307 finished with value:
-1.3089013337738677 and parameters: {'n_estimators': 540, 'learning_rate':
0.012166091155605647, 'max_depth': 10, 'subsample': 0.76902631479958,
'colsample_bytree': 0.6069628199095574, 'min_child_weight': 4, 'gamma':
5.136994731258478e-07, 'reg_alpha': 4.714515052990737e-08, 'reg_lambda':
5.7145938165396234e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:56,546] Trial 308 finished with value:
-1.3055036192671696 and parameters: {'n_estimators': 517, 'learning_rate':
0.011824844573486069, 'max_depth': 11, 'subsample': 0.7427475511332902,
'colsample_bytree': 0.5957170268661419, 'min_child_weight': 5, 'gamma':
1.932775152657663e-06, 'reg_alpha': 2.2087988032137332e-08, 'reg_lambda':
3.759981996287675e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:56,906] Trial 309 finished with value:
-1.3087951605703985 and parameters: {'n_estimators': 518, 'learning_rate':
0.01305610904235462, 'max_depth': 8, 'subsample': 0.7798199792832983,
'colsample_bytree': 0.5483801712904994, 'min_child_weight': 5, 'gamma':
2.144015774545098e-06, 'reg_alpha': 3.115292592495321e-08, 'reg_lambda':
2.904284351338308e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:57,453] Trial 310 finished with value:
-1.3133958931213767 and parameters: {'n_estimators': 525, 'learning_rate':
0.012207582228650725, 'max_depth': 12, 'subsample': 0.73642459170218,
'colsample_bytree': 0.5711528331383451, 'min_child_weight': 5, 'gamma':
5.2323597500726364e-06, 'reg_alpha': 1.1723611366247567e-08, 'reg_lambda':
5.125885680654621e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:57,993] Trial 311 finished with value:
-1.3122568389783476 and parameters: {'n_estimators': 575, 'learning_rate':
0.013610455953948103, 'max_depth': 11, 'subsample': 0.7443650859030737,
'colsample_bytree': 0.5955637616507922, 'min_child_weight': 5, 'gamma':
9.73438669642879e-07, 'reg_alpha': 4.5399665757259616e-08, 'reg_lambda':
2.0593467868610458e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:58,491] Trial 312 finished with value:
-1.3072801078034741 and parameters: {'n_estimators': 489, 'learning_rate':
0.011728335196889735, 'max_depth': 11, 'subsample': 0.7654198455141282,
'colsample_bytree': 0.6172424681792961, 'min_child_weight': 5, 'gamma':
1.9422803846236005e-06, 'reg_alpha': 2.0454030452324844e-08, 'reg_lambda':
7.556022578739058e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:58,842] Trial 313 finished with value:
-1.315523418831664 and parameters: {'n_estimators': 558, 'learning_rate':
0.01265196312530091, 'max_depth': 8, 'subsample': 0.7506138266760941,
'colsample_bytree': 0.588764259944126, 'min_child_weight': 6, 'gamma':
3.093472710571386e-06, 'reg_alpha': 2.871651098652578e-08, 'reg_lambda':
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3.8351944424912106e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:59,382] Trial 314 finished with value:
-1.3097758980435543 and parameters: {'n_estimators': 596, 'learning_rate':
0.011539878461411671, 'max_depth': 11, 'subsample': 0.7373038921340389,
'colsample_bytree': 0.5528414777285592, 'min_child_weight': 5, 'gamma':
1.5812779999670674e-06, 'reg_alpha': 1.2040246316514194e-08, 'reg_lambda':
0.0016351040336531433}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:11:59,961] Trial 315 finished with value:
-1.3105051717524816 and parameters: {'n_estimators': 505, 'learning_rate':
0.013816004140357246, 'max_depth': 11, 'subsample': 0.7426185367763765,
'colsample_bytree': 0.7424078897629756, 'min_child_weight': 4, 'gamma':
8.43665155587265e-07, 'reg_alpha': 5.807393270172182e-08, 'reg_lambda':
5.8949748548470537e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:00,280] Trial 316 finished with value:
-1.3160960488509916 and parameters: {'n_estimators': 336, 'learning_rate':
0.012203173917794689, 'max_depth': 9, 'subsample': 0.7284495441473364,
'colsample_bytree': 0.5836538769698697, 'min_child_weight': 4, 'gamma':
3.754875021677159e-06, 'reg_alpha': 2.1151459569953296e-08, 'reg_lambda':
2.0832733697755504e-05}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:00,589] Trial 317 finished with value:
-1.3154996323220767 and parameters: {'n_estimators': 537, 'learning_rate':
0.011612651416477385, 'max_depth': 7, 'subsample': 0.7537329509483387,
'colsample_bytree': 0.5657599204398671, 'min_child_weight': 4, 'gamma':
1.9056106140103552e-06, 'reg_alpha': 0.00019583169857404354, 'reg_lambda':
3.072122732923007e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:00,993] Trial 318 finished with value:
-1.330605972716073 and parameters: {'n_estimators': 549, 'learning_rate':
0.03215029115976798, 'max_depth': 9, 'subsample': 0.7431770867227325,
'colsample_bytree': 0.6090299615659862, 'min_child_weight': 4, 'gamma':
6.706782405646999e-07, 'reg_alpha': 1.7367744642415226e-08, 'reg_lambda':
4.4358370105612804e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:01,397] Trial 319 finished with value:
-1.305658443719908 and parameters: {'n_estimators': 523, 'learning_rate':
0.01265266782407637, 'max_depth': 9, 'subsample': 0.7876956215528549,
'colsample_bytree': 0.5999721262141365, 'min_child_weight': 5, 'gamma':
3.1936216272942456e-08, 'reg_alpha': 9.9229548876861e-08, 'reg_lambda':
9.97829526381988e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:01,842] Trial 320 finished with value:
-1.308657408290932 and parameters: {'n_estimators': 573, 'learning_rate':
0.01327207464420029, 'max_depth': 9, 'subsample': 0.6347642665464029,
'colsample_bytree': 0.5971180281831934, 'min_child_weight': 4, 'gamma':
4.067781077228959e-06, 'reg_alpha': 3.167404025011993e-08, 'reg_lambda':
1.74346814926998e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:02,579] Trial 321 finished with value:
-1.3279983651743332 and parameters: {'n_estimators': 500, 'learning_rate':
0.01462956584239391, 'max_depth': 9, 'subsample': 0.7324129053151266,
'colsample_bytree': 0.7588133160824153, 'min_child_weight': 1, 'gamma':
2.4060954452991797e-06, 'reg_alpha': 6.143743508615662e-08, 'reg_lambda':
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1.05502758348215e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:03,025] Trial 322 finished with value:
-1.3061771123183494 and parameters: {'n_estimators': 621, 'learning_rate':
0.011250187391809618, 'max_depth': 8, 'subsample': 0.7606428758173869,
'colsample_bytree': 0.6322491995528857, 'min_child_weight': 4, 'gamma':
5.58267058409487e-06, 'reg_alpha': 2.415725581828582e-07, 'reg_lambda':
6.739895842526107e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:03,481] Trial 323 finished with value:
-1.3105110808905984 and parameters: {'n_estimators': 593, 'learning_rate':
0.012046631736985728, 'max_depth': 9, 'subsample': 0.6461374038767901,
'colsample_bytree': 0.5905663849790661, 'min_child_weight': 4, 'gamma':
1.2831507045950933e-06, 'reg_alpha': 1.5871244874900644e-08, 'reg_lambda':
2.827487310118829e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:03,909] Trial 324 finished with value:
-1.3149933506758036 and parameters: {'n_estimators': 557, 'learning_rate':
0.011113573334492031, 'max_depth': 9, 'subsample': 0.6581247529382661,
'colsample_bytree': 0.7307395440107827, 'min_child_weight': 6, 'gamma':
3.1941984467146716e-07, 'reg_alpha': 2.3938605373361155e-08, 'reg_lambda':
4.6004371247819006e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:04,479] Trial 325 finished with value:
-1.3086959680636578 and parameters: {'n_estimators': 517, 'learning_rate':
0.013242912587294596, 'max_depth': 11, 'subsample': 0.6332011245023189,
'colsample_bytree': 0.7595302522233629, 'min_child_weight': 4, 'gamma':
3.2743150376054955e-06, 'reg_alpha': 4.6504861362075057e-08, 'reg_lambda':
2.0711524164549386e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:04,831] Trial 326 finished with value:
-1.3103927159723674 and parameters: {'n_estimators': 536, 'learning_rate':
0.012350190216640803, 'max_depth': 7, 'subsample': 0.6242063703910167,
'colsample_bytree': 0.7388281855871796, 'min_child_weight': 3, 'gamma':
3.961079315358495e-06, 'reg_alpha': 1.6923675810058192e-07, 'reg_lambda':
1.109526572362548e-07}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:05,307] Trial 327 finished with value:
-1.3047761260502384 and parameters: {'n_estimators': 572, 'learning_rate':
0.011480460299573274, 'max_depth': 9, 'subsample': 0.6455473778336414,
'colsample_bytree': 0.5753725894739777, 'min_child_weight': 3, 'gamma':
6.410352241954169e-06, 'reg_alpha': 7.796488098567326e-08, 'reg_lambda':
6.808023570197301e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:06,169] Trial 328 finished with value:
-1.3096157298053748 and parameters: {'n_estimators': 587, 'learning_rate':
0.011305827041605657, 'max_depth': 12, 'subsample': 0.6428784253449923,
'colsample_bytree': 0.8358995985579246, 'min_child_weight': 3, 'gamma':
7.261207384234033e-06, 'reg_alpha': 1.08635067714385e-07, 'reg_lambda':
2.654068982313838e-07}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:06,341] Trial 329 finished with value:
-1.4367478697008387 and parameters: {'n_estimators': 139, 'learning_rate':
0.010725034887464376, 'max_depth': 9, 'subsample': 0.6530725883847655,
'colsample_bytree': 0.5171383740214882, 'min_child_weight': 3, 'gamma':
5.4927094848879176e-06, 'reg_alpha': 8.950030819213589e-08, 'reg_lambda':
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1.6941512427774302e-07}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:06,881] Trial 330 finished with value:
-1.3127440692917614 and parameters: {'n_estimators': 568, 'learning_rate':
0.011565192385941859, 'max_depth': 9, 'subsample': 0.6374843074824046,
'colsample_bytree': 0.5683336884270853, 'min_child_weight': 2, 'gamma':
1.2638164339939889e-05, 'reg_alpha': 8.722150836703007e-06, 'reg_lambda':
7.655490888811192e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:07,221] Trial 331 finished with value:
-1.3237251304937294 and parameters: {'n_estimators': 604, 'learning_rate':
0.014192060557622337, 'max_depth': 8, 'subsample': 0.6165029446545798,
'colsample_bytree': 0.6667423418323817, 'min_child_weight': 9, 'gamma':
9.710966671809243e-06, 'reg_alpha': 0.0002653125798187846, 'reg_lambda':
7.749410636431372e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:07,645] Trial 332 finished with value:
-1.3550804953207063 and parameters: {'n_estimators': 558, 'learning_rate':
0.08345687871949949, 'max_depth': 9, 'subsample': 0.6630390135706183,
'colsample_bytree': 0.7308094948786442, 'min_child_weight': 3, 'gamma':
6.83198702814242e-06, 'reg_alpha': 0.00040398088971626856, 'reg_lambda':
5.657540473775384e-06}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:08,070] Trial 333 finished with value:
-1.3055292454809038 and parameters: {'n_estimators': 469, 'learning_rate':
0.016055317001705368, 'max_depth': 9, 'subsample': 0.6502665258991494,
'colsample_bytree': 0.684369351039109, 'min_child_weight': 3, 'gamma':
4.943559133937791e-06, 'reg_alpha': 2.9439004991429225e-06, 'reg_lambda':
1.0748560494717197e-07}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:08,598] Trial 334 finished with value:
-1.3056261049096118 and parameters: {'n_estimators': 581, 'learning_rate':
0.011538208185958168, 'max_depth': 9, 'subsample': 0.6284916305040428,
'colsample_bytree': 0.7027530791837869, 'min_child_weight': 3, 'gamma':
1.6415654931217496e-05, 'reg_alpha': 1.427222360095732e-07, 'reg_lambda':
5.694446243767207e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:09,055] Trial 335 finished with value:
-1.3132399366880658 and parameters: {'n_estimators': 609, 'learning_rate':
0.010504463708706294, 'max_depth': 8, 'subsample': 0.8348404514240729,
'colsample_bytree': 0.5576581821114942, 'min_child_weight': 3, 'gamma':
8.464817792327775e-06, 'reg_alpha': 5.836012085854548e-08, 'reg_lambda':
1.4873216878983938e-06}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:09,574] Trial 336 finished with value:
-1.3010993605911503 and parameters: {'n_estimators': 551, 'learning_rate':
0.01309500767725145, 'max_depth': 9, 'subsample': 0.6456818215687876,
'colsample_bytree': 0.7483965861039074, 'min_child_weight': 3, 'gamma':
1.2114652595575533e-05, 'reg_alpha': 3.2527766463005493e-07, 'reg_lambda':
0.00011264184843489514}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:10,135] Trial 337 finished with value:
-1.3088484618883218 and parameters: {'n_estimators': 631, 'learning_rate':
0.013635546729060444, 'max_depth': 9, 'subsample': 0.644361384827996,
'colsample_bytree': 0.7475384259947498, 'min_child_weight': 3, 'gamma':
9.7673743447131e-08, 'reg_alpha': 1.939362043844881e-07, 'reg_lambda':
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0.00010115277514473045}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:10,548] Trial 338 finished with value:
-1.3137973269958845 and parameters: {'n_estimators': 428, 'learning_rate':
0.01527831618263911, 'max_depth': 9, 'subsample': 0.6611297132090654,
'colsample_bytree': 0.7637163667235973, 'min_child_weight': 3, 'gamma':
1.24374872987733e-05, 'reg_alpha': 8.180157652513391e-08, 'reg_lambda':
7.703313599696012e-05}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:11,068] Trial 339 finished with value:
-1.3094272044139412 and parameters: {'n_estimators': 572, 'learning_rate':
0.01288242377287129, 'max_depth': 9, 'subsample': 0.6382015910591141,
'colsample_bytree': 0.7221680829838785, 'min_child_weight': 3, 'gamma':
1.616425817438752e-05, 'reg_alpha': 4.803380190636553e-07, 'reg_lambda':
1.387141633206447e-07}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:11,576] Trial 340 finished with value:
-1.3125007930079875 and parameters: {'n_estimators': 547, 'learning_rate':
0.012705417525459955, 'max_depth': 9, 'subsample': 0.653050086541766,
'colsample_bytree': 0.7459034273077999, 'min_child_weight': 3, 'gamma':
2.0429246523666732e-05, 'reg_alpha': 2.0534649126846497e-07, 'reg_lambda':
0.00013530951733686824}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:12,106] Trial 341 finished with value:
-1.307494989133459 and parameters: {'n_estimators': 554, 'learning_rate':
0.010950042960192056, 'max_depth': 9, 'subsample': 0.6185302282928659,
'colsample_bytree': 0.7392253942257011, 'min_child_weight': 3, 'gamma':
7.754782481966394e-06, 'reg_alpha': 0.1983504831542038, 'reg_lambda':
8.47694058958951e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:12,625] Trial 342 finished with value:
-1.3702892633612178 and parameters: {'n_estimators': 588, 'learning_rate':
0.05309001901340051, 'max_depth': 9, 'subsample': 0.6309596644705413,
'colsample_bytree': 0.7548620967468529, 'min_child_weight': 2, 'gamma':
1.2174391317908767e-05, 'reg_alpha': 0.0005472180160550785, 'reg_lambda':
6.231236136014901e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:13,134] Trial 343 finished with value:
-1.306702223382943 and parameters: {'n_estimators': 569, 'learning_rate':
0.014540806033950419, 'max_depth': 9, 'subsample': 0.6444746052661473,
'colsample_bytree': 0.7293315080606362, 'min_child_weight': 3, 'gamma':
4.172808353732588e-05, 'reg_alpha': 3.842216569931537e-07, 'reg_lambda':
0.00010690278493077088}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:13,443] Trial 344 finished with value:
-1.4052570580694599 and parameters: {'n_estimators': 547, 'learning_rate':
0.10792664737239242, 'max_depth': 8, 'subsample': 0.6263528242855472,
'colsample_bytree': 0.419857956214379, 'min_child_weight': 3, 'gamma':
6.517136187664687e-06, 'reg_alpha': 2.5841268977460857e-07, 'reg_lambda':
2.8038025443538017e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:13,919] Trial 345 finished with value:
-1.3497907600843284 and parameters: {'n_estimators': 611, 'learning_rate':
0.04637279953808528, 'max_depth': 9, 'subsample': 0.6560389165544614,
'colsample_bytree': 0.7097416007556141, 'min_child_weight': 3, 'gamma':
2.0871624020309344e-05, 'reg_alpha': 1.261390288498931e-07, 'reg_lambda':
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0.00019675522844898137}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:14,406] Trial 346 finished with value:
-1.3114290532551507 and parameters: {'n_estimators': 538, 'learning_rate':
0.013586442735929237, 'max_depth': 9, 'subsample': 0.6144998869970619,
'colsample_bytree': 0.719821009568812, 'min_child_weight': 3, 'gamma':
1.3976570247227749e-05, 'reg_alpha': 0.0007614965649736092, 'reg_lambda':
0.00023709386925292055}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:14,809] Trial 347 finished with value:
-1.3333826999238527 and parameters: {'n_estimators': 569, 'learning_rate':
0.03808793170117754, 'max_depth': 8, 'subsample': 0.6395853845372123,
'colsample_bytree': 0.7698614727806434, 'min_child_weight': 3, 'gamma':
2.931758832078348e-05, 'reg_alpha': 3.115479184443161e-07, 'reg_lambda':
2.119485840514538e-08}. Best is trial 273 with value: -1.3007401584614298.
[I 2026-03-13 19:12:15,243] Trial 348 finished with value:
-1.3001254686484316 and parameters: {'n_estimators': 493, 'learning_rate':
0.012060024388503027, 'max_depth': 9, 'subsample': 0.6488591464024076,
'colsample_bytree': 0.7372583915337819, 'min_child_weight': 4, 'gamma':
8.588483486286481e-06, 'reg_alpha': 0.00023766633798349672, 'reg_lambda':
2.1202846002582458e-07}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:15,604] Trial 349 finished with value:
-1.3184986602970572 and parameters: {'n_estimators': 477, 'learning_rate':
0.011238727633357436, 'max_depth': 9, 'subsample': 0.648246323679865,
'colsample_bytree': 0.4790688155052045, 'min_child_weight': 4, 'gamma':
4.808393909641931e-06, 'reg_alpha': 0.00020668266690920256, 'reg_lambda':
3.0783051619687835e-07}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:16,048] Trial 350 finished with value:
-1.3141006884986808 and parameters: {'n_estimators': 447, 'learning_rate':
0.012071398523106703, 'max_depth': 9, 'subsample': 0.6611251699094753,
'colsample_bytree': 0.7494537392438068, 'min_child_weight': 3, 'gamma':
7.3773526825908664e-06, 'reg_alpha': 0.0001269085244144352, 'reg_lambda':
1.524357157463173e-07}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:16,607] Trial 351 finished with value:
-1.319468815482045 and parameters: {'n_estimators': 495, 'learning_rate':
0.01065530925839332, 'max_depth': 9, 'subsample': 0.6653638457048541,
'colsample_bytree': 0.7378355573134033, 'min_child_weight': 2, 'gamma':
1.6565145791588698e-08, 'reg_alpha': 7.389579195597131e-08, 'reg_lambda':
1.9190316495667332e-07}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:17,042] Trial 352 finished with value:
-1.3092206436525873 and parameters: {'n_estimators': 507, 'learning_rate':
0.011915534946083765, 'max_depth': 9, 'subsample': 0.6317170577895012,
'colsample_bytree': 0.7305589105154484, 'min_child_weight': 4, 'gamma':
9.82015529489793e-06, 'reg_alpha': 1.1659849996572912e-07, 'reg_lambda':
3.6166153535332586e-05}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:17,308] Trial 353 finished with value:
-1.3399718968857346 and parameters: {'n_estimators': 484, 'learning_rate':
0.010006824630907542, 'max_depth': 6, 'subsample': 0.6517091207680595,
'colsample_bytree': 0.7519466577108813, 'min_child_weight': 3, 'gamma':
3.824371702713153e-06, 'reg_alpha': 0.00027259333122771, 'reg_lambda':
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1.0738824408189327e-07}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:17,843] Trial 354 finished with value:
-1.3095478604446968 and parameters: {'n_estimators': 524, 'learning_rate':
0.011050319022230682, 'max_depth': 10, 'subsample': 0.6700166209562828,
'colsample_bytree': 0.7637129734430143, 'min_child_weight': 4, 'gamma':
8.345002634728116e-06, 'reg_alpha': 0.00017533740724763494, 'reg_lambda':
2.245187126916785e-07}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:18,266] Trial 355 finished with value:
-1.3151832627534334 and parameters: {'n_estimators': 511, 'learning_rate':
0.01262364728696558, 'max_depth': 9, 'subsample': 0.6228159833538426,
'colsample_bytree': 0.5417782735875333, 'min_child_weight': 3, 'gamma':
5.328012621406009e-06, 'reg_alpha': 0.0003614329080795253, 'reg_lambda':
0.0006456498654777051}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:18,898] Trial 356 finished with value:
-1.311403018233304 and parameters: {'n_estimators': 806, 'learning_rate':
0.011907926158026437, 'max_depth': 9, 'subsample': 0.6874553654180842,
'colsample_bytree': 0.72543662776054, 'min_child_weight': 4, 'gamma':
0.1332101993622352, 'reg_alpha': 0.0008959166025734397, 'reg_lambda':
1.3727079904587076e-08}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:19,396] Trial 357 finished with value:
-1.3108659446047268 and parameters: {'n_estimators': 536, 'learning_rate':
0.011122142085164856, 'max_depth': 10, 'subsample': 0.6418447522610551,
'colsample_bytree': 0.7363187785019976, 'min_child_weight': 3, 'gamma':
0.7195949857790485, 'reg_alpha': 9.774712914538124e-07, 'reg_lambda':
4.088480661157759e-07}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:19,852] Trial 358 finished with value:
-1.3051559966249588 and parameters: {'n_estimators': 497, 'learning_rate':
0.010592379071360876, 'max_depth': 9, 'subsample': 0.7147505742346101,
'colsample_bytree': 0.7167556134899808, 'min_child_weight': 4, 'gamma':
1.1545532966481245e-05, 'reg_alpha': 1.474383327002278e-07, 'reg_lambda':
5.087381066873108e-05}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:20,309] Trial 359 finished with value:
-1.3051660610165488 and parameters: {'n_estimators': 492, 'learning_rate':
0.010566239009072848, 'max_depth': 9, 'subsample': 0.7166317261000646,
'colsample_bytree': 0.7139119925668046, 'min_child_weight': 4, 'gamma':
1.1871685226204346e-05, 'reg_alpha': 0.000510838658876491, 'reg_lambda':
5.487266266767302e-08}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:20,744] Trial 360 finished with value:
-1.30415682252581 and parameters: {'n_estimators': 471, 'learning_rate':
0.010659669523697717, 'max_depth': 9, 'subsample': 0.710013756573298,
'colsample_bytree': 0.7107819113856747, 'min_child_weight': 4, 'gamma':
1.2887232060833012e-05, 'reg_alpha': 0.000624466416569133, 'reg_lambda':
5.933441257428783e-07}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:21,169] Trial 361 finished with value:
-1.3128347896243817 and parameters: {'n_estimators': 457, 'learning_rate':
0.010500098345645066, 'max_depth': 9, 'subsample': 0.7083446988229078,
'colsample_bytree': 0.7079796055779772, 'min_child_weight': 4, 'gamma':
1.2694063719897367e-05, 'reg_alpha': 0.00043560655993024437, 'reg_lambda':
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8.839342164517978e-07}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:21,625] Trial 362 finished with value:
-1.308375859266184 and parameters: {'n_estimators': 503, 'learning_rate':
0.010630661746976388, 'max_depth': 9, 'subsample': 0.7044082375736076,
'colsample_bytree': 0.6952741765855444, 'min_child_weight': 4, 'gamma':
2.2356634691097097e-05, 'reg_alpha': 0.0006218276052470245, 'reg_lambda':
4.625639270282771e-07}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:22,061] Trial 363 finished with value:
-1.30574192908533 and parameters: {'n_estimators': 478, 'learning_rate':
0.011414942751335193, 'max_depth': 9, 'subsample': 0.7151360698097494,
'colsample_bytree': 0.7142812173704379, 'min_child_weight': 4, 'gamma':
1.2173326132323208e-05, 'reg_alpha': 0.00029187159487480124, 'reg_lambda':
2.216976811134617e-07}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:22,360] Trial 364 finished with value:
-1.444378687686811 and parameters: {'n_estimators': 419, 'learning_rate':
0.20620287378306182, 'max_depth': 9, 'subsample': 0.7184976829235753,
'colsample_bytree': 0.7040492115689732, 'min_child_weight': 4, 'gamma':
1.5692509655825594e-05, 'reg_alpha': 0.0005509356985688489, 'reg_lambda':
5.176669797794032e-07}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:22,794] Trial 365 finished with value:
-1.3050579423682185 and parameters: {'n_estimators': 486, 'learning_rate':
0.010659044804983224, 'max_depth': 9, 'subsample': 0.7017357185781921,
'colsample_bytree': 0.6877316696872111, 'min_child_weight': 4, 'gamma':
2.828805934495841e-05, 'reg_alpha': 0.0008688933472423805, 'reg_lambda':
0.004944283161387356}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:23,217] Trial 366 finished with value:
-1.307461797449298 and parameters: {'n_estimators': 482, 'learning_rate':
0.01185855186056347, 'max_depth': 9, 'subsample': 0.7089041361469249,
'colsample_bytree': 0.6830642638346844, 'min_child_weight': 4, 'gamma':
2.0276891628122792e-05, 'reg_alpha': 0.0002065619347368574, 'reg_lambda':
0.043490559622027726}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:23,631] Trial 367 finished with value:
-1.3111578948977336 and parameters: {'n_estimators': 450, 'learning_rate':
0.011131147314174385, 'max_depth': 9, 'subsample': 0.7021785505266673,
'colsample_bytree': 0.6922184340029038, 'min_child_weight': 4, 'gamma':
3.9421745448157596e-05, 'reg_alpha': 0.00039417197246468025, 'reg_lambda':
1.3660828531946765e-05}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:24,036] Trial 368 finished with value:
-1.307337355280324 and parameters: {'n_estimators': 458, 'learning_rate':
0.01231580750979416, 'max_depth': 9, 'subsample': 0.7172391651570326,
'colsample_bytree': 0.6780691852954822, 'min_child_weight': 4, 'gamma':
1.1617235070414717e-05, 'reg_alpha': 0.0005146071481323161, 'reg_lambda':
0.0011459110988924874}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:24,419] Trial 369 finished with value:
-1.3122029226540677 and parameters: {'n_estimators': 482, 'learning_rate':
0.02382392729874204, 'max_depth': 9, 'subsample': 0.6941005179167703,
'colsample_bytree': 0.7132983315380349, 'min_child_weight': 4, 'gamma':
2.8942296380668826e-05, 'reg_alpha': 0.0008301337924772223, 'reg_lambda':
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0.009101289168515127}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:24,843] Trial 370 finished with value:
-1.3115810544376811 and parameters: {'n_estimators': 459, 'learning_rate':
0.010602653821426201, 'max_depth': 9, 'subsample': 0.6986578968238186,
'colsample_bytree': 0.6909582751267482, 'min_child_weight': 4, 'gamma':
1.704374408131571e-05, 'reg_alpha': 0.00028327667601773783, 'reg_lambda':
0.0027527199484721847}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:25,278] Trial 371 finished with value:
-1.3058968415018974 and parameters: {'n_estimators': 489, 'learning_rate':
0.011453447748632975, 'max_depth': 9, 'subsample': 0.7156042473619482,
'colsample_bytree': 0.7027227110153047, 'min_child_weight': 4, 'gamma':
5.001590500752435e-05, 'reg_alpha': 3.350542130723528e-07, 'reg_lambda':
0.014288123786705487}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:25,703] Trial 372 finished with value:
-1.3051358047414026 and parameters: {'n_estimators': 498, 'learning_rate':
0.012656745466920647, 'max_depth': 9, 'subsample': 0.6842847371324269,
'colsample_bytree': 0.6737346320383669, 'min_child_weight': 4, 'gamma':
8.783295759330986e-06, 'reg_alpha': 1.90228585222937e-06, 'reg_lambda':
0.020690478741415583}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:26,098] Trial 373 finished with value:
-1.3068727593869813 and parameters: {'n_estimators': 461, 'learning_rate':
0.013180253544392772, 'max_depth': 9, 'subsample': 0.6772556973920362,
'colsample_bytree': 0.6815807710576528, 'min_child_weight': 4, 'gamma':
8.737293106984389e-06, 'reg_alpha': 5.447436395986812e-06, 'reg_lambda':
0.03536407133182132}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:26,522] Trial 374 finished with value:
-1.3089068261679628 and parameters: {'n_estimators': 469, 'learning_rate':
0.012459803913814283, 'max_depth': 9, 'subsample': 0.6854047110144137,
'colsample_bytree': 0.7268199416493861, 'min_child_weight': 4, 'gamma':
2.7984804473391853e-06, 'reg_alpha': 0.00010521925359867812, 'reg_lambda':
0.022924684098418007}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:26,945] Trial 375 finished with value:
-1.3060209746039613 and parameters: {'n_estimators': 498, 'learning_rate':
0.013064325817480488, 'max_depth': 9, 'subsample': 0.6946786218090832,
'colsample_bytree': 0.6749560899245739, 'min_child_weight': 4, 'gamma':
4.875517177196489e-06, 'reg_alpha': 1.9726633040346526e-06, 'reg_lambda':
0.19104370778758833}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:27,377] Trial 376 finished with value:
-1.3037767377677532 and parameters: {'n_estimators': 516, 'learning_rate':
0.012003615070467434, 'max_depth': 9, 'subsample': 0.6883504211076531,
'colsample_bytree': 0.6569012619392994, 'min_child_weight': 4, 'gamma':
2.468452419631003e-05, 'reg_alpha': 2.0636093808510732e-07, 'reg_lambda':
0.0011572725218253709}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:27,748] Trial 377 finished with value:
-1.3104271571288975 and parameters: {'n_estimators': 438, 'learning_rate':
0.013997978676120423, 'max_depth': 9, 'subsample': 0.698608555086107,
'colsample_bytree': 0.6513962894246997, 'min_child_weight': 4, 'gamma':
3.580309486453609e-05, 'reg_alpha': 2.1620773058247952e-07, 'reg_lambda':
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0.012072575607786622}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:28,192] Trial 378 finished with value:
-1.309389340039008 and parameters: {'n_estimators': 520, 'learning_rate':
0.012004442743906834, 'max_depth': 9, 'subsample': 0.678485186440287,
'colsample_bytree': 0.6785543494475489, 'min_child_weight': 4, 'gamma':
6.6017959428528e-05, 'reg_alpha': 5.024979577629909e-07, 'reg_lambda':
0.0918258439051866}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:28,649] Trial 379 finished with value:
-1.3103982976767365 and parameters: {'n_estimators': 551, 'learning_rate':
0.012844886634858476, 'max_depth': 9, 'subsample': 0.6784630189319569,
'colsample_bytree': 0.6632500662080558, 'min_child_weight': 4, 'gamma':
2.7811286671367165e-05, 'reg_alpha': 2.9998877295336567e-07, 'reg_lambda':
0.0021259081351696013}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:29,082] Trial 380 finished with value:
-1.3161871785870565 and parameters: {'n_estimators': 523, 'learning_rate':
0.012203009988631112, 'max_depth': 9, 'subsample': 0.6964192788319589,
'colsample_bytree': 0.9969002347442534, 'min_child_weight': 7, 'gamma':
9.16426545754446e-05, 'reg_alpha': 2.031643383286319e-07, 'reg_lambda':
0.005636558375854284}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:29,547] Trial 381 finished with value:
-1.306641100571375 and parameters: {'n_estimators': 531, 'learning_rate':
0.013661915121316334, 'max_depth': 9, 'subsample': 0.6896015804890324,
'colsample_bytree': 0.7470360342601309, 'min_child_weight': 4, 'gamma':
2.082160232917674e-05, 'reg_alpha': 8.56511372646309e-07, 'reg_lambda':
0.008100093966858533}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:30,002] Trial 382 finished with value:
-1.3058431924909852 and parameters: {'n_estimators': 552, 'learning_rate':
0.011526151114991, 'max_depth': 9, 'subsample': 0.6733275889249288,
'colsample_bytree': 0.6644923936371772, 'min_child_weight': 4, 'gamma':
4.5315632465091186e-05, 'reg_alpha': 1.1901899441339106e-06, 'reg_lambda':
0.0038785562182638562}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:30,437] Trial 383 finished with value:
-1.3045090333186131 and parameters: {'n_estimators': 519, 'learning_rate':
0.01268225690461086, 'max_depth': 9, 'subsample': 0.6852172438713138,
'colsample_bytree': 0.65910974623208, 'min_child_weight': 4, 'gamma':
1.8544365166477108e-05, 'reg_alpha': 1.5040725205945614e-05, 'reg_lambda':
0.02410654828423166}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:30,924] Trial 384 finished with value:
-1.311790047663283 and parameters: {'n_estimators': 562, 'learning_rate':
0.011753751602360436, 'max_depth': 10, 'subsample': 0.6630088290807339,
'colsample_bytree': 0.653409337294896, 'min_child_weight': 5, 'gamma':
2.2260620112072044e-05, 'reg_alpha': 3.298696356751463e-05, 'reg_lambda':
7.579649629776692e-06}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:31,419] Trial 385 finished with value:
-1.3088912297861444 and parameters: {'n_estimators': 517, 'learning_rate':
0.014730349894997016, 'max_depth': 9, 'subsample': 0.6709434950468085,
'colsample_bytree': 0.7866803175805039, 'min_child_weight': 3, 'gamma':
3.184510344122176e-05, 'reg_alpha': 6.126353735950826e-05, 'reg_lambda':
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3.1559480753056836e-07}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:31,769] Trial 386 finished with value:
-1.494831177372648 and parameters: {'n_estimators': 537, 'learning_rate':
0.26586914362039676, 'max_depth': 9, 'subsample': 0.6693695512337421,
'colsample_bytree': 0.660935885337706, 'min_child_weight': 4, 'gamma':
1.7098762819683293e-05, 'reg_alpha': 1.2275271279280454e-05, 'reg_lambda':
0.00037344793148072914}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:32,359] Trial 387 finished with value:
-1.3120192556295966 and parameters: {'n_estimators': 549, 'learning_rate':
0.01326106748943552, 'max_depth': 10, 'subsample': 0.655949794044735,
'colsample_bytree': 0.7377051777154462, 'min_child_weight': 3, 'gamma':
5.045342366334195e-05, 'reg_alpha': 0.00018197029658546723, 'reg_lambda':
0.005763452494407931}. Best is trial 348 with value: -1.3001254686484316.
[I 2026-03-13 19:12:32,865] Trial 388 finished with value:
-1.2994569632941035 and parameters: {'n_estimators': 575, 'learning_rate':
0.012277855697386078, 'max_depth': 9, 'subsample': 0.680723604868237,
'colsample_bytree': 0.7621233428703186, 'min_child_weight': 4, 'gamma':
2.442387927247573e-05, 'reg_alpha': 0.00015325317573310228, 'reg_lambda':
0.0013163052121671264}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:33,382] Trial 389 finished with value:
-1.3030851309520532 and parameters: {'n_estimators': 591, 'learning_rate':
0.011421761326274267, 'max_depth': 9, 'subsample': 0.6839583802470481,
'colsample_bytree': 0.7675593662205858, 'min_child_weight': 4, 'gamma':
2.84135742074009e-05, 'reg_alpha': 2.1425092461036436e-05, 'reg_lambda':
0.005199418121997138}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:33,897] Trial 390 finished with value:
-1.3103935601280694 and parameters: {'n_estimators': 579, 'learning_rate':
0.01128425108455789, 'max_depth': 9, 'subsample': 0.6913727108105922,
'colsample_bytree': 0.7625519890383313, 'min_child_weight': 4, 'gamma':
2.7383887681864357e-05, 'reg_alpha': 8.16427913686395e-05, 'reg_lambda':
0.05878100179629617}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:34,404] Trial 391 finished with value:
-1.3053715849538783 and parameters: {'n_estimators': 570, 'learning_rate':
0.01215979093233814, 'max_depth': 9, 'subsample': 0.6848688019674004,
'colsample_bytree': 0.7705184080791918, 'min_child_weight': 4, 'gamma':
1.549203364462281e-05, 'reg_alpha': 0.0001438031675738868, 'reg_lambda':
0.000834601501908093}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:34,934] Trial 392 finished with value:
-1.3120975203911778 and parameters: {'n_estimators': 584, 'learning_rate':
0.011243640564082358, 'max_depth': 9, 'subsample': 0.6940456659522292,
'colsample_bytree': 0.7939961303722808, 'min_child_weight': 4, 'gamma':
2.2460634519036477e-05, 'reg_alpha': 1.6084773185816716e-05, 'reg_lambda':
0.003677856340800448}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:35,296] Trial 393 finished with value:
-1.3306000142879018 and parameters: {'n_estimators': 530, 'learning_rate':
0.011842933090564575, 'max_depth': 9, 'subsample': 0.7049596795507118,
'colsample_bytree': 0.7796319563001116, 'min_child_weight': 10, 'gamma':
3.09388815743774e-05, 'reg_alpha': 0.0001155235079119973, 'reg_lambda':
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0.0017201150355485O1}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:35,834] Trial 394 finished with value:
-1.3043920068058095 and parameters: {'n_estimators': 597, 'learning_rate':
0.010900879913533971, 'max_depth': 9, 'subsample': 0.6856250144386655,
'colsample_bytree': 0.7844012530595146, 'min_child_weight': 4, 'gamma':
0.002255244565326694, 'reg_alpha': 0.00023818103084288918, 'reg_lambda':
0.0012178716604997932}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:36,281] Trial 395 finished with value:
-1.3217331258134613 and parameters: {'n_estimators': 616, 'learning_rate':
0.010509949391848085, 'max_depth': 9, 'subsample': 0.6853979950000345,
'colsample_bytree': 0.8063182628954443, 'min_child_weight': 8, 'gamma':
0.0015620205409795278, 'reg_alpha': 0.0001493569580220509, 'reg_lambda':
0.001230618114411676}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:36,819] Trial 396 finished with value:
-1.3084596228557952 and parameters: {'n_estimators': 597, 'learning_rate':
0.010077772359957458, 'max_depth': 9, 'subsample': 0.6822025677201736,
'colsample_bytree': 0.7779795453378806, 'min_child_weight': 4, 'gamma':
2.6739020166718776e-06, 'reg_alpha': 0.00016658897009866867, 'reg_lambda':
0.0016823409051493748}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:37,346] Trial 397 finished with value:
-1.3075015581391356 and parameters: {'n_estimators': 592, 'learning_rate':
0.011221862558271897, 'max_depth': 9, 'subsample': 0.701798248035345,
'colsample_bytree': 0.793373758423983, 'min_child_weight': 4, 'gamma':
0.005077949908613925, 'reg_alpha': 1.9905761337857707e-05, 'reg_lambda':
0.00553026904589532}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:37,831] Trial 398 finished with value:
-1.3117141412435303 and parameters: {'n_estimators': 520, 'learning_rate':
0.011040077427291078, 'max_depth': 9, 'subsample': 0.6898632827585421,
'colsample_bytree': 0.7721841447851141, 'min_child_weight': 4, 'gamma':
0.009305219488851426, 'reg_alpha': 0.00021836999045165073, 'reg_lambda':
0.0008761047802979753}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:38,225] Trial 399 finished with value:
-1.3545995206368062 and parameters: {'n_estimators': 549, 'learning_rate':
0.06808102892241978, 'max_depth': 9, 'subsample': 0.6772529457089319,
'colsample_bytree': 0.7807022151445074, 'min_child_weight': 4, 'gamma':
6.035854778352089e-06, 'reg_alpha': 0.00023893456299316603, 'reg_lambda':
0.00029518225474270263}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:38,698] Trial 400 finished with value:
-1.3075387102016256 and parameters: {'n_estimators': 510, 'learning_rate':
0.010075049298650926, 'max_depth': 9, 'subsample': 0.6695331387717676,
'colsample_bytree': 0.8009089373948969, 'min_child_weight': 4, 'gamma':
3.856133048604939e-06, 'reg_alpha': 4.325822190425412e-08, 'reg_lambda':
0.0012468922078148733}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:39,208] Trial 401 finished with value:
-1.30734386245927 and parameters: {'n_estimators': 560, 'learning_rate':
0.011153764300279874, 'max_depth': 9, 'subsample': 0.7002651392508009,
'colsample_bytree': 0.7633615645502417, 'min_child_weight': 4, 'gamma':
1.5027508280818095e-05, 'reg_alpha': 0.00011084948514540059, 'reg_lambda':
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0.0009959755848803558}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:39,714] Trial 402 finished with value:
-1.306499145098425 and parameters: {'n_estimators': 539, 'learning_rate':
0.011966416920004525, 'max_depth': 9, 'subsample': 0.7074153991582659,
'colsample_bytree': 0.7850295633018763, 'min_child_weight': 4, 'gamma':
6.678323504843996e-06, 'reg_alpha': 7.161981523903197e-06, 'reg_lambda':
0.42330773905517505}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:40,242] Trial 403 finished with value:
-1.310748862596931 and parameters: {'n_estimators': 601, 'learning_rate':
0.010807644801362667, 'max_depth': 9, 'subsample': 0.6803087546808999,
'colsample_bytree': 0.7647060104859295, 'min_child_weight': 4, 'gamma':
3.6214024245305805e-05, 'reg_alpha': 0.00026939214044572354, 'reg_lambda':
0.00368778605084811}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:40,729] Trial 404 finished with value:
-1.3103594356637585 and parameters: {'n_estimators': 577, 'learning_rate':
0.012478144995740676, 'max_depth': 9, 'subsample': 0.6929779266761396,
'colsample_bytree': 0.818914504197757, 'min_child_weight': 5, 'gamma':
3.5025188274803504e-06, 'reg_alpha': 0.0003216999270868042, 'reg_lambda':
0.0006615252450504349}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:41,205] Trial 405 finished with value:
-1.3040741444402735 and parameters: {'n_estimators': 516, 'learning_rate':
0.011685220888343987, 'max_depth': 9, 'subsample': 0.7235894776612414,
'colsample_bytree': 0.761859543271926, 'min_child_weight': 4, 'gamma':
1.0749893701509833e-05, 'reg_alpha': 5.847243680753416e-08, 'reg_lambda':
0.002551786027345797}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:41,680] Trial 406 finished with value:
-1.3037211384532617 and parameters: {'n_estimators': 531, 'learning_rate':
0.013688622272138979, 'max_depth': 9, 'subsample': 0.731308929889393,
'colsample_bytree': 0.7604421941725228, 'min_child_weight': 4, 'gamma':
9.771425282641214e-06, 'reg_alpha': 6.230959935943296e-08, 'reg_lambda':
0.0018865956946395091}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:41,852] Trial 407 finished with value:
-1.368880876800737 and parameters: {'n_estimators': 526, 'learning_rate':
0.015551872271644975, 'max_depth': 4, 'subsample': 0.7272176775916848,
'colsample_bytree': 0.7604866851325464, 'min_child_weight': 4, 'gamma':
1.0979025760946711e-05, 'reg_alpha': 5.963815391245935e-08, 'reg_lambda':
0.002422241898278757}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:42,298] Trial 408 finished with value:
-1.3113503371520412 and parameters: {'n_estimators': 506, 'learning_rate':
0.01452768222482453, 'max_depth': 9, 'subsample': 0.6584690854171943,
'colsample_bytree': 0.7765822008561619, 'min_child_weight': 4, 'gamma':
0.00181661230555532, 'reg_alpha': 3.660103221770916e-08, 'reg_lambda':
0.0016481834737117796}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:42,784] Trial 409 finished with value:
-1.3097749137937909 and parameters: {'n_estimators': 549, 'learning_rate':
0.013944106666299931, 'max_depth': 9, 'subsample': 0.7236220616966826,
'colsample_bytree': 0.7872153552098906, 'min_child_weight': 4, 'gamma':
1.0609171936991245e-05, 'reg_alpha': 7.099506107256288e-08, 'reg_lambda':
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0.002587809600123814}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:43,231] Trial 410 finished with value:
-1.3099035512179005 and parameters: {'n_estimators': 562, 'learning_rate':
0.01734196577739792, 'max_depth': 9, 'subsample': 0.730081221661144,
'colsample_bytree': 0.7703764935581671, 'min_child_weight': 5, 'gamma':
8.344826089530693e-06, 'reg_alpha': 4.2177922921877034e-08, 'reg_lambda':
0.003263707229278897}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:43,897] Trial 411 finished with value:
-1.3141259899444415 and parameters: {'n_estimators': 517, 'learning_rate':
0.01334569756062474, 'max_depth': 9, 'subsample': 0.6641784292959414,
'colsample_bytree': 0.7549310338740416, 'min_child_weight': 1, 'gamma':
0.020268185343463353, 'reg_alpha': 6.527996332125813e-08, 'reg_lambda':
0.001391818709107417}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:44,303] Trial 412 finished with value:
-1.311498900894262 and parameters: {'n_estimators': 538, 'learning_rate':
0.012980221710583193, 'max_depth': 8, 'subsample': 0.65137855904456,
'colsample_bytree': 0.7602568061739661, 'min_child_weight': 4, 'gamma':
1.4295200828068195e-05, 'reg_alpha': 5.071208126167108e-08, 'reg_lambda':
0.0006727128420358265}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:44,780] Trial 413 finished with value:
-1.3134934207259243 and parameters: {'n_estimators': 587, 'learning_rate':
0.012507786190202052, 'max_depth': 9, 'subsample': 0.6700480349732512,
'colsample_bytree': 0.7754515303514797, 'min_child_weight': 5, 'gamma':
8.508898489421172e-06, 'reg_alpha': 4.391103708048378e-05, 'reg_lambda':
0.0005395336820309555}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:45,217] Trial 414 finished with value:
-1.3035080437426403 and parameters: {'n_estimators': 512, 'learning_rate':
0.01407500588787337, 'max_depth': 9, 'subsample': 0.6443324578524161,
'colsample_bytree': 0.7518505566514458, 'min_child_weight': 4, 'gamma':
0.03426252913862957, 'reg_alpha': 9.663498805865118e-08, 'reg_lambda':
0.0010162322202556429}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:45,653] Trial 415 finished with value:
-1.3076322822119224 and parameters: {'n_estimators': 501, 'learning_rate':
0.014750348021478326, 'max_depth': 9, 'subsample': 0.7232817520944782,
'colsample_bytree': 0.7555524674192005, 'min_child_weight': 4, 'gamma':
0.095983611635808, 'reg_alpha': 9.382306979353483e-08, 'reg_lambda':
0.0010213917105760036}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:46,099] Trial 416 finished with value:
-1.3049532796761714 and parameters: {'n_estimators': 523, 'learning_rate':
0.016064384426498605, 'max_depth': 9, 'subsample': 0.7330122069724349,
'colsample_bytree': 0.7517947624304528, 'min_child_weight': 4, 'gamma':
0.028684747813957502, 'reg_alpha': 1.2656620644667084e-07, 'reg_lambda':
0.0005455703376517506}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:46,449] Trial 417 finished with value:
-1.308213959783152 and parameters: {'n_estimators': 390, 'learning_rate':
0.013391086954576395, 'max_depth': 9, 'subsample': 0.6445794036439962,
'colsample_bytree': 0.74628269527974, 'min_child_weight': 4, 'gamma':
0.33113872923269166, 'reg_alpha': 8.516678873988553e-08, 'reg_lambda':
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0.0020159340826247886}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:46,884] Trial 418 finished with value:
-1.3083711904339177 and parameters: {'n_estimators': 481, 'learning_rate':
0.014475772044769694, 'max_depth': 9, 'subsample': 0.7114592202676177,
'colsample_bytree': 0.7672356474349742, 'min_child_weight': 4, 'gamma':
0.03343810387605676, 'reg_alpha': 2.9474602299291922e-08, 'reg_lambda':
0.0016759726712055809}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:47,338] Trial 419 finished with value:
-1.3072625221146221 and parameters: {'n_estimators': 513, 'learning_rate':
0.01391495748568621, 'max_depth': 9, 'subsample': 0.646921483673016,
'colsample_bytree': 0.7888487750677783, 'min_child_weight': 4, 'gamma':
6.017704937358747e-06, 'reg_alpha': 2.2712595838651983e-05, 'reg_lambda':
0.0012822778861058036}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:47,804] Trial 420 finished with value:
-1.3193563824105745 and parameters: {'n_estimators': 546, 'learning_rate':
0.015486044275194786, 'max_depth': 9, 'subsample': 0.7496522038995834,
'colsample_bytree': 0.7494639353809431, 'min_child_weight': 4, 'gamma':
0.003993764960364372, 'reg_alpha': 5.6717359893629205e-08, 'reg_lambda':
0.0019728157424571137}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:48,313] Trial 421 finished with value:
-1.311511986228209 and parameters: {'n_estimators': 569, 'learning_rate':
0.013582236295345204, 'max_depth': 9, 'subsample': 0.6373410596154564,
'colsample_bytree': 0.8011588310162655, 'min_child_weight': 4, 'gamma':
0.04866672083753089, 'reg_alpha': 8.084248602720546e-05, 'reg_lambda':
0.0022447577559557863}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:48,779] Trial 422 finished with value:
-1.3090765836206755 and parameters: {'n_estimators': 529, 'learning_rate':
0.012480059512356048, 'max_depth': 9, 'subsample': 0.6561712650888382,
'colsample_bytree': 0.7615772022075511, 'min_child_weight': 4, 'gamma':
1.4926466864547953e-05, 'reg_alpha': 3.860876997777945e-08, 'reg_lambda':
0.0007869484540070634}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:49,241] Trial 423 finished with value:
-1.3041195056443173 and parameters: {'n_estimators': 501, 'learning_rate':
0.01193215184775328, 'max_depth': 9, 'subsample': 0.6290153795346154,
'colsample_bytree': 0.7749966796029225, 'min_child_weight': 4, 'gamma':
0.011847887439153644, 'reg_alpha': 1.1689215840107128e-07, 'reg_lambda':
0.0010194500769655794}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:49,624] Trial 424 finished with value:
-1.3236443468395962 and parameters: {'n_estimators': 469, 'learning_rate':
0.028421994875953063, 'max_depth': 9, 'subsample': 0.6289592937482322,
'colsample_bytree': 0.7822192631807163, 'min_child_weight': 4, 'gamma':
0.00679937376616289, 'reg_alpha': 1.5620105519804387e-07, 'reg_lambda':
0.0010259222014647573}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:50,050] Trial 425 finished with value:
-1.3133948359869587 and parameters: {'n_estimators': 495, 'learning_rate':
0.01691322986603773, 'max_depth': 9, 'subsample': 0.6344316111572081,
'colsample_bytree': 0.7647524832832211, 'min_child_weight': 4, 'gamma':
0.0009708677063687503, 'reg_alpha': 9.273586419033009e-08, 'reg_lambda':
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0.001231735776958407}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:50,473] Trial 426 finished with value:
-1.306850179450208 and parameters: {'n_estimators': 502, 'learning_rate':
0.01177255787096132, 'max_depth': 9, 'subsample': 0.6182386639225227,
'colsample_bytree': 0.7980944252459258, 'min_child_weight': 5, 'gamma':
0.11037228080737999, 'reg_alpha': 1.0408891122275953e-08, 'reg_lambda':
0.0030033046633639033}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:50,962] Trial 427 finished with value:
-1.307201614148539 and parameters: {'n_estimators': 552, 'learning_rate':
0.01293534291988935, 'max_depth': 9, 'subsample': 0.6436664268080551,
'colsample_bytree': 0.7737794709533707, 'min_child_weight': 4, 'gamma':
0.013018407153520654, 'reg_alpha': 1.3225169759954257e-07, 'reg_lambda':
0.0008255259911434654}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:51,450] Trial 428 finished with value:
-1.3143734595737184 and parameters: {'n_estimators': 608, 'learning_rate':
0.01412918286628877, 'max_depth': 9, 'subsample': 0.579411817815598,
'colsample_bytree': 0.7484981908358639, 'min_child_weight': 4, 'gamma':
8.633219339756473e-06, 'reg_alpha': 2.1817677229595112e-07, 'reg_lambda':
0.0004152802745786079}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:51,821] Trial 429 finished with value:
-1.3109110423043047 and parameters: {'n_estimators': 473, 'learning_rate':
0.011706268256782658, 'max_depth': 9, 'subsample': 0.6256768499510438,
'colsample_bytree': 0.5211599113552418, 'min_child_weight': 4, 'gamma':
0.00418027626910828, 'reg_alpha': 2.994566081129948e-07, 'reg_lambda':
0.0005244483400404635}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:52,312] Trial 430 finished with value:
-1.3128487547612575 and parameters: {'n_estimators': 579, 'learning_rate':
0.012970420478582987, 'max_depth': 9, 'subsample': 0.5921063954125915,
'colsample_bytree': 0.7811927635601353, 'min_child_weight': 4, 'gamma':
0.019510064587578462, 'reg_alpha': 1.0760773681002641e-07, 'reg_lambda':
0.0022040625890274263}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:52,723] Trial 431 finished with value:
-1.3098801426029694 and parameters: {'n_estimators': 508, 'learning_rate':
0.011837484142106, 'max_depth': 9, 'subsample': 0.6390060818704002,
'colsample_bytree': 0.7538775949825103, 'min_child_weight': 5, 'gamma':
0.013198854302694718, 'reg_alpha': 1.7875411835563652e-07, 'reg_lambda':
6.443792131029511e-07}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:53,157] Trial 432 finished with value:
-1.3159391547564605 and parameters: {'n_estimators': 538, 'learning_rate':
0.015012520653309595, 'max_depth': 9, 'subsample': 0.6094091416175088,
'colsample_bytree': 0.7427945193192268, 'min_child_weight': 4, 'gamma':
4.661722540767725e-06, 'reg_alpha': 7.281763299861086e-08, 'reg_lambda':
0.0013695815913494077}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:53,644] Trial 433 finished with value:
-1.3102986773645238 and parameters: {'n_estimators': 560, 'learning_rate':
0.013717113308805137, 'max_depth': 9, 'subsample': 0.6618407315680305,
'colsample_bytree': 0.7658337059449514, 'min_child_weight': 4, 'gamma':
0.043038989410791165, 'reg_alpha': 0.00023321817008062438, 'reg_lambda':
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0.003203888516017146}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:53,996] Trial 434 finished with value:
-1.3202449117359494 and parameters: {'n_estimators': 494, 'learning_rate':
0.018531869394071827, 'max_depth': 9, 'subsample': 0.6516612575557428,
'colsample_bytree': 0.7383561352136996, 'min_child_weight': 7, 'gamma':
0.05240644252647174, 'reg_alpha': 1.0550356281910213e-07, 'reg_lambda':
0.12637939701807632}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:54,410] Trial 435 finished with value:
-1.3144810239245148 and parameters: {'n_estimators': 519, 'learning_rate':
0.012522352597959729, 'max_depth': 9, 'subsample': 0.9844151563594755,
'colsample_bytree': 0.5020248485290427, 'min_child_weight': 4, 'gamma':
6.767653823533347e-06, 'reg_alpha': 0.0001503044575619671, 'reg_lambda':
0.000981660962795574}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:54,904] Trial 436 finished with value:
-1.3119650551413562 and parameters: {'n_estimators': 538, 'learning_rate':
0.011563924931420356, 'max_depth': 9, 'subsample': 0.6795820971860969,
'colsample_bytree': 0.7553346237537797, 'min_child_weight': 4, 'gamma':
1.1799918270306794e-05, 'reg_alpha': 6.385628281731247e-08, 'reg_lambda':
0.0015004164989983163}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:55,403] Trial 437 finished with value:
-1.306713916201441 and parameters: {'n_estimators': 600, 'learning_rate':
0.011295180486024493, 'max_depth': 9, 'subsample': 0.6332247478036442,
'colsample_bytree': 0.7281489501555206, 'min_child_weight': 4, 'gamma':
2.0015000800784678e-05, 'reg_alpha': 9.749671974996891e-05, 'reg_lambda':
0.0007120269359114209}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:55,827] Trial 438 finished with value:
-1.3112081573711238 and parameters: {'n_estimators': 567, 'learning_rate':
0.012435517144841765, 'max_depth': 8, 'subsample': 0.6210333309635052,
'colsample_bytree': 0.7735152201378702, 'min_child_weight': 4, 'gamma':
0.002603218638456578, 'reg_alpha': 2.502647468833392e-07, 'reg_lambda':
0.0068387689228652845}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:56,294] Trial 439 finished with value:
-1.31150416088575 and parameters: {'n_estimators': 584, 'learning_rate':
0.010031796240367975, 'max_depth': 9, 'subsample': 0.6676321706269448,
'colsample_bytree': 0.8079233683293293, 'min_child_weight': 6, 'gamma':
0.06715170256531261, 'reg_alpha': 4.5484444908628847e-07, 'reg_lambda':
0.00406764565270566}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:56,783] Trial 440 finished with value:
-1.3211788308632066 and parameters: {'n_estimators': 489, 'learning_rate':
0.013325764658874218, 'max_depth': 9, 'subsample': 0.931560606491814,
'colsample_bytree': 0.7895611308901519, 'min_child_weight': 4, 'gamma':
0.027608513838015852, 'reg_alpha': 1.3705699873844634e-07, 'reg_lambda':
0.0016203739527743482}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:57,260] Trial 441 finished with value:
-1.3046290414273765 and parameters: {'n_estimators': 549, 'learning_rate':
0.010964920201549294, 'max_depth': 9, 'subsample': 0.643635832079019,
'colsample_bytree': 0.7434440814763531, 'min_child_weight': 4, 'gamma':
0.0001286719837421653, 'reg_alpha': 0.000191473424688773, 'reg_lambda':
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0.0027236716629047315}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:57,738] Trial 442 finished with value:
-1.3068333156511924 and parameters: {'n_estimators': 558, 'learning_rate':
0.0119255533759024, 'max_depth': 9, 'subsample': 0.6453431446525147,
'colsample_bytree': 0.7428091803313169, 'min_child_weight': 4, 'gamma':
0.00011974050347157426, 'reg_alpha': 0.00017479494843015717, 'reg_lambda':
0.00246089512694992 8}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:58,207] Trial 443 finished with value:
-1.3063117095568615 and parameters: {'n_estimators': 547, 'learning_rate':
0.014099514173545045, 'max_depth': 9, 'subsample': 0.6565176967077777,
'colsample_bytree': 0.7592665951406126, 'min_child_weight': 4, 'gamma':
0.014313878222826851, 'reg_alpha': 3.5754617600466276e-06, 'reg_lambda':
0.02153211882620707}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:58,674] Trial 444 finished with value:
-1.3101489299172824 and parameters: {'n_estimators': 575, 'learning_rate':
0.01109154922376204, 'max_depth': 9, 'subsample': 0.6862931809918313,
'colsample_bytree': 0.7472763071689233, 'min_child_weight': 5, 'gamma':
4.716137900711179e-06, 'reg_alpha': 0.00018557353151391 68, 'reg_lambda':
0.004257321481755981}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:59,120] Trial 445 finished with value:
-1.308532608041153 and parameters: {'n_estimators': 623, 'learning_rate':
0.012652224582971435, 'max_depth': 8, 'subsample': 0.6479937130409077,
'colsample_bytree': 0.7320849468539515, 'min_child_weight': 4, 'gamma':
9.058953819435476e-06, 'reg_alpha': 0.00011855104148442813, 'reg_lambda':
0.002896267181177479}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:12:59,641] Trial 446 finished with value:
-1.3092451018202487 and parameters: {'n_estimators': 595, 'learning_rate':
0.011871556342115877, 'max_depth': 9, 'subsample': 0.6617474273972701,
'colsample_bytree': 0.7715702829103651, 'min_child_weight': 4, 'gamma':
0.008791737752481746, 'reg_alpha': 0.00026379786328576066, 'reg_lambda':
0.0024540934438544762}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:00,109] Trial 447 finished with value:
-1.3104862971259577 and parameters: {'n_estimators': 529, 'learning_rate':
0.011111492273748219, 'max_depth': 9, 'subsample': 0.6730718760919024,
'colsample_bytree': 0.7473613985626664, 'min_child_weight': 4, 'gamma':
7.304543119560406e-06, 'reg_alpha': 6.67730287572528e-05, 'reg_lambda':
0.0010553871092167208}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:00,564] Trial 448 finished with value:
-1.303737644624911 and parameters: {'n_estimators': 516, 'learning_rate':
0.013368621334678558, 'max_depth': 9, 'subsample': 0.6392860846121117,
'colsample_bytree': 0.7608429064282832, 'min_child_weight': 4, 'gamma':
0.002881485784202815, 'reg_alpha': 5.60086634990094e-08, 'reg_lambda':
0.0015907374325189276}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:00,990] Trial 449 finished with value:
-1.310025992590564 and parameters: {'n_estimators': 486, 'learning_rate':
0.01531559886176773, 'max_depth': 9, 'subsample': 0.6354278774641902,
'colsample_bytree': 0.7585913512215479, 'min_child_weight': 4, 'gamma':
0.006094134262760003, 'reg_alpha': 2.166927747544081e-08, 'reg_lambda':

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0.001766450480916883}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:01,426] Trial 450 finished with value:
-1.304094517086441 and parameters: {'n_estimators': 518, 'learning_rate':
0.014086949891729744, 'max_depth': 9, 'subsample': 0.706054209441183,
'colsample_bytree': 0.7728852336240224, 'min_child_weight': 5, 'gamma':
0.0003826150413969479, 'reg_alpha': 5.550830555204581e-08, 'reg_lambda':
0.00047265637165118927}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:01,841] Trial 451 finished with value:
-1.3091145255642138 and parameters: {'n_estimators': 504, 'learning_rate':
0.015637780600054225, 'max_depth': 9, 'subsample': 0.7089144428585004,
'colsample_bytree': 0.7808583483875768, 'min_child_weight': 5, 'gamma':
0.0006356193252976359, 'reg_alpha': 5.046920921028519e-08, 'reg_lambda':
0.00019241785479708344}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:01,982] Trial 452 finished with value:
-1.402968235558692 and parameters: {'n_estimators': 512, 'learning_rate':
0.014302334512932094, 'max_depth': 3, 'subsample': 0.7035894484057875,
'colsample_bytree': 0.7894510485589805, 'min_child_weight': 5, 'gamma':
0.0028121901059254707, 'reg_alpha': 3.735003270701529e-08, 'reg_lambda':
0.0003147599108727197}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:02,363] Trial 453 finished with value:
-1.3088308991219095 and parameters: {'n_estimators': 436, 'learning_rate':
0.014617559354414483, 'max_depth': 9, 'subsample': 0.6934342963073136,
'colsample_bytree': 0.769699943985774, 'min_child_weight': 5, 'gamma':
0.004455776123332111, 'reg_alpha': 2.7414315292935844e-08, 'reg_lambda':
0.001463623570839902}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:02,767] Trial 454 finished with value:
-1.3038326017302733 and parameters: {'n_estimators': 474, 'learning_rate':
0.01623215240565547, 'max_depth': 9, 'subsample': 0.7194964541616333,
'colsample_bytree': 0.7826748805624978, 'min_child_weight': 5, 'gamma':
0.0018380107978112237, 'reg_alpha': 8.601519376184171e-08, 'reg_lambda':
0.00048358942042689624}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:03,172] Trial 455 finished with value:
-1.3071622010303745 and parameters: {'n_estimators': 455, 'learning_rate':
0.015326461402466239, 'max_depth': 9, 'subsample': 0.7177946992957559,
'colsample_bytree': 0.8113029189813971, 'min_child_weight': 5, 'gamma':
0.001927457665055554, 'reg_alpha': 1.0047507264619039e-07, 'reg_lambda':
0.0006876805249325221}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:03,534] Trial 456 finished with value:
-1.3098202861905774 and parameters: {'n_estimators': 473, 'learning_rate':
0.01626672507469444, 'max_depth': 8, 'subsample': 0.709413029623355,
'colsample_bytree': 0.7977245329846091, 'min_child_weight': 4, 'gamma':
0.000332123706763906, 'reg_alpha': 6.881088133667945e-08, 'reg_lambda':
0.0007111127159862307}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:03,970] Trial 457 finished with value:
-1.313919376923536 and parameters: {'n_estimators': 463, 'learning_rate':
0.01675450180226129, 'max_depth': 10, 'subsample': 0.6960319036421778,
'colsample_bytree': 0.7822108227331328, 'min_child_weight': 5, 'gamma':
0.0009501371682980525, 'reg_alpha': 1.2400453000691893e-07, 'reg_lambda':
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0.00032589811601556535}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:04,375] Trial 458 finished with value:
-1.3068650367239607 and parameters: {'n_estimators': 476, 'learning_rate':
0.014013537249054705, 'max_depth': 9, 'subsample': 0.7181263262090561,
'colsample_bytree': 0.7741778341301393, 'min_child_weight': 5, 'gamma':
0.007828344698408209, 'reg_alpha': 1.7773837772029823e-07, 'reg_lambda':
0.0008858727521920119}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:04,716] Trial 459 finished with value:
-1.3209446498317463 and parameters: {'n_estimators': 411, 'learning_rate':
0.013529846685961733, 'max_depth': 9, 'subsample': 0.5641895615100726,
'colsample_bytree': 0.7853132449989773, 'min_child_weight': 5, 'gamma':
0.0034376356004748743, 'reg_alpha': 7.290854718180794e-08, 'reg_lambda':
0.00037123601689017453}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:05,130] Trial 460 finished with value:
-1.3061654973900214 and parameters: {'n_estimators': 494, 'learning_rate':
0.014769344737781149, 'max_depth': 9, 'subsample': 0.7080035775757738,
'colsample_bytree': 0.8007379942368433, 'min_child_weight': 5, 'gamma':
0.0013987095051551733, 'reg_alpha': 2.8991710220816424e-07, 'reg_lambda':
0.0005447069658761569}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:05,460] Trial 461 finished with value:
-1.3112776817780059 and parameters: {'n_estimators': 447, 'learning_rate':
0.013678015814934346, 'max_depth': 8, 'subsample': 0.686885515156221,
'colsample_bytree': 0.7668145547486419, 'min_child_weight': 5, 'gamma':
0.0027964401853633105, 'reg_alpha': 9.39938080394569e-08, 'reg_lambda':
0.0001482825669005245}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:05,853] Trial 462 finished with value:
-1.3073591379125282 and parameters: {'n_estimators': 485, 'learning_rate':
0.013253921883200228, 'max_depth': 9, 'subsample': 0.7019473477573072,
'colsample_bytree': 0.7911007925032633, 'min_child_weight': 6, 'gamma':
0.0010146414710027196, 'reg_alpha': 5.8915493714828287e-08, 'reg_lambda':
0.0009238691397962323}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:06,289] Trial 463 finished with value:
-1.3155083950096307 and parameters: {'n_estimators': 509, 'learning_rate':
0.01599277653096773, 'max_depth': 10, 'subsample': 0.7243923983012012,
'colsample_bytree': 0.7640540388071781, 'min_child_weight': 6, 'gamma':
0.0012727155082583015, 'reg_alpha': 1.6605185337966318e-07, 'reg_lambda':
0.000536675554784752}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:06,711] Trial 464 finished with value:
-1.3023355988009355 and parameters: {'n_estimators': 516, 'learning_rate':
0.014224619356304338, 'max_depth': 9, 'subsample': 0.6919759025393725,
'colsample_bytree': 0.7738637811523126, 'min_child_weight': 5, 'gamma':
0.00048341431736922567, 'reg_alpha': 3.036087121316872e-08, 'reg_lambda':
0.000426950072254883}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:07,117] Trial 465 finished with value:
-1.3130954349395632 and parameters: {'n_estimators': 469, 'learning_rate':
0.014642463830793435, 'max_depth': 9, 'subsample': 0.7258367747616797,
'colsample_bytree': 0.8219144068239878, 'min_child_weight': 5, 'gamma':
0.0019545315201919514, 'reg_alpha': 1.7488188379287028e-08, 'reg_lambda':
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0.0003678019353199869}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:07,529] Trial 466 finished with value:
-1.3091251924616156 and parameters: {'n_estimators': 500, 'learning_rate':
0.01534744347380736, 'max_depth': 9, 'subsample': 0.693759661357011,
'colsample_bytree': 0.7786930576151586, 'min_child_weight': 5, 'gamma':
0.0007563293028998639, 'reg_alpha': 2.673904286653905e-08, 'reg_lambda':
0.0002688501286801172}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:07,964] Trial 467 finished with value:
-1.304598198732339 and parameters: {'n_estimators': 518, 'learning_rate':
0.014343980991243819, 'max_depth': 9, 'subsample': 0.717757920920822,
'colsample_bytree': 0.7743744704917065, 'min_child_weight': 5, 'gamma':
0.00023382362958590143, 'reg_alpha': 3.8199663811017344e-08, 'reg_lambda':
0.0005125301471129769}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:08,359] Trial 468 finished with value:
-1.3037815521236908 and parameters: {'n_estimators': 490, 'learning_rate':
0.01717854665458552, 'max_depth': 9, 'subsample': 0.7121432892367928,
'colsample_bytree': 0.7590528070555725, 'min_child_weight': 5, 'gamma':
0.0018816151972003918, 'reg_alpha': 3.336108527243313e-08, 'reg_lambda':
0.0010801438925970I2}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:08,553] Trial 469 finished with value:
-1.3379901780066896 and parameters: {'n_estimators': 478, 'learning_rate':
0.01941258302319444, 'max_depth': 5, 'subsample': 0.7096956857017963,
'colsample_bytree': 0.7591259193566682, 'min_child_weight': 5, 'gamma':
0.0006482093202343053, 'reg_alpha': 1.5322446667627888e-08, 'reg_lambda':
0.0010994186799429925}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:08,976] Trial 470 finished with value:
-1.3061862009654404 and parameters: {'n_estimators': 449, 'learning_rate':
0.01608854626842219, 'max_depth': 10, 'subsample': 0.7114313635320998,
'colsample_bytree': 0.7889069994838928, 'min_child_weight': 5, 'gamma':
0.001455087400040774, 'reg_alpha': 2.7182241249152795e-08, 'reg_lambda':
0.0004499938678461772}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:09,379] Trial 471 finished with value:
-1.3096668593657388 and parameters: {'n_estimators': 487, 'learning_rate':
0.016202152505084905, 'max_depth': 9, 'subsample': 0.6971112883155927,
'colsample_bytree': 0.7738111120626006, 'min_child_weight': 5, 'gamma':
0.0004248716439607792, 'reg_alpha': 2.4011124794141734e-08, 'reg_lambda':
0.0012551600910603428}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:09,781] Trial 472 finished with value:
-1.3001611014358727 and parameters: {'n_estimators': 496, 'learning_rate':
0.01749905733402106, 'max_depth': 9, 'subsample': 0.7237550722475301,
'colsample_bytree': 0.7567815891137858, 'min_child_weight': 5, 'gamma':
0.0019069962906744487, 'reg_alpha': 3.537650930814344e-08, 'reg_lambda':
0.0008146749112135963}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:10,174] Trial 473 finished with value:
-1.306641050161944 and parameters: {'n_estimators': 469, 'learning_rate':
0.015350852425088966, 'max_depth': 9, 'subsample': 0.7252174279523115,
'colsample_bytree': 0.7569885856307925, 'min_child_weight': 5, 'gamma':
0.002218156662327645, 'reg_alpha': 1.3735363336299809e-08, 'reg_lambda':
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0.0007007896393258884}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:10,515] Trial 474 finished with value:
-1.3048434289896536 and parameters: {'n_estimators': 497, 'learning_rate':
0.01878791544288306, 'max_depth': 8, 'subsample': 0.7217691375545948,
'colsample_bytree': 0.7593422554962368, 'min_child_weight': 5, 'gamma':
0.0001689860626102402, 'reg_alpha': 3.6160055503743744e-08, 'reg_lambda':
0.0007891549836726802}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:10,973] Trial 475 finished with value:
-1.3133631248122202 and parameters: {'n_estimators': 494, 'learning_rate':
0.018384707982164343, 'max_depth': 10, 'subsample': 0.7327820870811508,
'colsample_bytree': 0.7685888450758549, 'min_child_weight': 5, 'gamma':
0.0034955609129477803, 'reg_alpha': 1.9356551955306176e-08, 'reg_lambda':
0.00022614017132672595}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:11,335] Trial 476 finished with value:
-1.3003760582421258 and parameters: {'n_estimators': 435, 'learning_rate':
0.016092519376150936, 'max_depth': 9, 'subsample': 0.7134607801056129,
'colsample_bytree': 0.7573356360173973, 'min_child_weight': 5, 'gamma':
0.000488633067787836, 'reg_alpha': 3.3855476240392656e-08, 'reg_lambda':
0.000489282938866371}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:11,698] Trial 477 finished with value:
-1.3033920003019526 and parameters: {'n_estimators': 443, 'learning_rate':
0.01790204138441561, 'max_depth': 9, 'subsample': 0.7220475321207371,
'colsample_bytree': 0.7552981550936938, 'min_child_weight': 5, 'gamma':
0.0007496958369087295, 'reg_alpha': 3.0883859358889466e-08, 'reg_lambda':
0.0003965282883445219}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:12,030] Trial 478 finished with value:
-1.3072953865195842 and parameters: {'n_estimators': 377, 'learning_rate':
0.017882624772103332, 'max_depth': 9, 'subsample': 0.7331219689174852,
'colsample_bytree': 0.756979674729174, 'min_child_weight': 5, 'gamma':
0.00043983988143539855, 'reg_alpha': 2.6539525891111125e-08, 'reg_lambda':
0.00039556955668930015}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:12,383] Trial 479 finished with value:
-1.3075658799969938 and parameters: {'n_estimators': 419, 'learning_rate':
0.020632682271893556, 'max_depth': 9, 'subsample': 0.7250911580085392,
'colsample_bytree': 0.7699718848296527, 'min_child_weight': 5, 'gamma':
0.00047146636899171465, 'reg_alpha': 4.2207465568453315e-08, 'reg_lambda':
0.00047764093195897775}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:12,714] Trial 480 finished with value:
-1.3034219138338952 and parameters: {'n_estimators': 391, 'learning_rate':
0.01725322238272439, 'max_depth': 9, 'subsample': 0.7152572727483348,
'colsample_bytree': 0.7583714959271346, 'min_child_weight': 5, 'gamma':
0.0006660782104671804, 'reg_alpha': 3.1326253933255425e-08, 'reg_lambda':
0.00023927261599045012}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:12,995] Trial 481 finished with value:
-1.3069271966962848 and parameters: {'n_estimators': 375, 'learning_rate':
0.016805334745344465, 'max_depth': 8, 'subsample': 0.716889604577653 5,
'colsample_bytree': 0.7530495357674872, 'min_child_weight': 5, 'gamma':
0.0009000150687498525, 'reg_alpha': 1.866846763791989e-08, 'reg_lambda':
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0.00025499483674549167}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:13,305] Trial 482 finished with value:
-1.3083878236152486 and parameters: {'n_estimators': 354, 'learning_rate':
0.017485517071011765, 'max_depth': 9, 'subsample': 0.7184107781567631,
'colsample_bytree': 0.754349648026418, 'min_child_weight': 5, 'gamma':
0.0005985599393096493, 'reg_alpha': 3.3908033469923684e-08, 'reg_lambda':
0.00011193160645651033}. Best is trial 388 with value: -1.2994569632941035.
[I 2026-03-13 19:13:13,667] Trial 483 finished with value:
-1.298542171258924 and parameters: {'n_estimators': 413, 'learning_rate':
0.01763895823216975, 'max_depth': 9, 'subsample': 0.712091737933571,
'colsample_bytree': 0.752845441622297, 'min_child_weight': 5, 'gamma':
0.0006543449151942095, 'reg_alpha': 3.095474945139113e-08, 'reg_lambda':
0.00045119924266977414}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:14,022] Trial 484 finished with value:
-1.3136149890165199 and parameters: {'n_estimators': 415, 'learning_rate':
0.020118113830810705, 'max_depth': 9, 'subsample': 0.7357416210090332,
'colsample_bytree': 0.7492734678623599, 'min_child_weight': 5, 'gamma':
0.0012056051538579275, 'reg_alpha': 1.4555912652023725e-08, 'reg_lambda':
0.0001329633004716334}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:14,375] Trial 485 finished with value:
-1.3044678562489993 and parameters: {'n_estimators': 430, 'learning_rate':
0.01806114941694199, 'max_depth': 9, 'subsample': 0.7304280328687296,
'colsample_bytree': 0.7438256174404698, 'min_child_weight': 5, 'gamma':
0.0006067764869803034, 'reg_alpha': 2.3858767781931934e-08, 'reg_lambda':
0.00017664864709473766}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:14,696] Trial 486 finished with value:
-1.313170039368445 and parameters: {'n_estimators': 389, 'learning_rate':
0.022117090816458195, 'max_depth': 9, 'subsample': 0.7168641708131709,
'colsample_bytree': 0.7565689766134637, 'min_child_weight': 6, 'gamma':
0.0002821132451316208, 'reg_alpha': 3.0355093115347275e-08, 'reg_lambda':
0.00025233853215843706}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:15,039] Trial 487 finished with value:
-1.29897589109981 and parameters: {'n_estimators': 403, 'learning_rate':
0.01702238733031566, 'max_depth': 9, 'subsample': 0.7257671574119132,
'colsample_bytree': 0.743611628226318, 'min_child_weight': 5, 'gamma':
0.0008595149785481212, 'reg_alpha': 2.1005247147465492e-08, 'reg_lambda':
0.0002803981068697081}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:15,372] Trial 488 finished with value:
-1.3019639003824361 and parameters: {'n_estimators': 405, 'learning_rate':
0.01743229359381003, 'max_depth': 9, 'subsample': 0.7105862406817777,
'colsample_bytree': 0.7367673245617591, 'min_child_weight': 5, 'gamma':
0.0009806590333516093, 'reg_alpha': 1.0177425862766184e-08, 'reg_lambda':
0.00030431773423612163}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:15,703] Trial 489 finished with value:
-1.3017597828142211 and parameters: {'n_estimators': 400, 'learning_rate':
0.018907526743624132, 'max_depth': 9, 'subsample': 0.70688330575286,
'colsample_bytree': 0.7449813415670862, 'min_child_weight': 5, 'gamma':
0.0009025287690878705, 'reg_alpha': 1.1312163427247885e-08, 'reg_lambda':
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0.00021996915892563937}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:16,034] Trial 490 finished with value:
-1.304647685932525 and parameters: {'n_estimators': 394, 'learning_rate':
0.017955675703461335, 'max_depth': 9, 'subsample': 0.7019466351202188,
'colsample_bytree': 0.7384285275416301, 'min_child_weight': 5, 'gamma':
0.0007226893120849685, 'reg_alpha': 1.5427770644855725e-08, 'reg_lambda':
0.00020716965346425842}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:16,333] Trial 491 finished with value:
-1.305553910997795 and parameters: {'n_estimators': 406, 'learning_rate':
0.02076345196190796, 'max_depth': 8, 'subsample': 0.7098871088116105,
'colsample_bytree': 0.7436992276983987, 'min_child_weight': 5, 'gamma':
0.0013054714966878292, 'reg_alpha': 1.0669792966321815e-08, 'reg_lambda':
0.00025840837120291266}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:16,683] Trial 492 finished with value:
-1.3046301502666413 and parameters: {'n_estimators': 402, 'learning_rate':
0.017114034063767398, 'max_depth': 9, 'subsample': 0.7024583933182685,
'colsample_bytree': 0.7397991064996118, 'min_child_weight': 5, 'gamma':
0.001004921955946865, 'reg_alpha': 1.1046769582128009e-08, 'reg_lambda':
0.00032167406657644275}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:16,992] Trial 493 finished with value:
-1.3015831612879456 and parameters: {'n_estimators': 358, 'learning_rate':
0.0192667811776035, 'max_depth': 9, 'subsample': 0.7099012277983057,
'colsample_bytree': 0.7511238932058814, 'min_child_weight': 5, 'gamma':
0.0005619446540622425, 'reg_alpha': 1.0513815841419066e-08, 'reg_lambda':
0.00013086635447424799}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:17,281] Trial 494 finished with value:
-1.306324328726869 and parameters: {'n_estimators': 336, 'learning_rate':
0.019442991872990995, 'max_depth': 9, 'subsample': 0.6974188843962146,
'colsample_bytree': 0.7487083597431184, 'min_child_weight': 5, 'gamma':
0.0006952073097417804, 'reg_alpha': 1.2967045276819788e-08, 'reg_lambda':
0.00010166916141530909}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:17,571] Trial 495 finished with value:
-1.3085325840249968 and parameters: {'n_estimators': 352, 'learning_rate':
0.018473110827799722, 'max_depth': 9, 'subsample': 0.7323390059455256,
'colsample_bytree': 0.7329288483989164, 'min_child_weight': 6, 'gamma':
0.0005824001216779386, 'reg_alpha': 1.0917768058243578e-08, 'reg_lambda':
0.00011902512539661436}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:17,879] Trial 496 finished with value:
-1.3079333726639926 and parameters: {'n_estimators': 362, 'learning_rate':
0.01980073209091755, 'max_depth': 9, 'subsample': 0.7102177146492953,
'colsample_bytree': 0.7478467023541161, 'min_child_weight': 5, 'gamma':
0.0003566446441746107, 'reg_alpha': 1.6066226683853313e-08, 'reg_lambda':
0.00016450387031053046}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:18,118] Trial 497 finished with value:
-1.3105741252949914 and parameters: {'n_estimators': 309, 'learning_rate':
0.019576864631789207, 'max_depth': 8, 'subsample': 0.6974804101962203,
'colsample_bytree': 0.7324815234599394, 'min_child_weight': 5, 'gamma':
0.00047210112838360713, 'reg_alpha': 1.78746585549281e-08, 'reg_lambda':
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7.157660179368989e-05}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:18,448] Trial 498 finished with value:
-1.3049305076458502 and parameters: {'n_estimators': 395, 'learning_rate':
0.021228236505386078, 'max_depth': 9, 'subsample': 0.7270476431877386,
'colsample_bytree': 0.7493857320365961, 'min_child_weight': 5, 'gamma':
0.0011090815303089212, 'reg_alpha': 1.1719047106100289e-08, 'reg_lambda':
0.00017159302117664412}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:18,779] Trial 499 finished with value:
-1.3019432235858501 and parameters: {'n_estimators': 378, 'learning_rate':
0.0175515244482423, 'max_depth': 9, 'subsample': 0.7113099002225065,
'colsample_bytree': 0.7381870831330751, 'min_child_weight': 5, 'gamma':
0.0009390276209379839, 'reg_alpha': 1.0637489265451098e-08, 'reg_lambda':
5.533729710721271e-05}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:19,108] Trial 500 finished with value:
-1.304035109583585 and parameters: {'n_estimators': 379, 'learning_rate':
0.017345692386198618, 'max_depth': 9, 'subsample': 0.7196980066901316,
'colsample_bytree': 0.738607603042929, 'min_child_weight': 5, 'gamma':
0.000753851015420478, 'reg_alpha': 1.666781377797143e-08, 'reg_lambda':
0.00022177358816726693}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:19,395] Trial 501 finished with value:
-1.3012552688066534 and parameters: {'n_estimators': 339, 'learning_rate':
0.018683385766289355, 'max_depth': 9, 'subsample': 0.7080527911370829,
'colsample_bytree': 0.7267715186872105, 'min_child_weight': 5, 'gamma':
0.0009961250332672631, 'reg_alpha': 1.0202359631920896e-08, 'reg_lambda':
3.953180350602217e-05}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:19,695] Trial 502 finished with value:
-1.308040266057896 and parameters: {'n_estimators': 359, 'learning_rate':
0.019105234125425696, 'max_depth': 9, 'subsample': 0.7104923535304414,
'colsample_bytree': 0.7220532135679623, 'min_child_weight': 5, 'gamma':
0.0008521659743044052, 'reg_alpha': 1.0611081405053762e-08, 'reg_lambda':
3.7536920158764594e-05}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:20,045] Trial 503 finished with value:
-1.3109141580991308 and parameters: {'n_estimators': 370, 'learning_rate':
0.017868711056845576, 'max_depth': 9, 'subsample': 0.7028385476230568,
'colsample_bytree': 0.9125756727197964, 'min_child_weight': 5, 'gamma':
0.0005502214280694107, 'reg_alpha': 1.0183382537239248e-08, 'reg_lambda':
2.389803020348508e-05}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:20,302] Trial 504 finished with value:
-1.3092962856432875 and parameters: {'n_estimators': 337, 'learning_rate':
0.02173364594789113, 'max_depth': 8, 'subsample': 0.7129584991411775,
'colsample_bytree': 0.7293633729160159, 'min_child_weight': 5, 'gamma':
0.001015029487455361, 'reg_alpha': 1.6532128763182413e-08, 'reg_lambda':
3.429642776998664e-06}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:20,541] Trial 505 finished with value:
-1.3065384544126468 and parameters: {'n_estimators': 275, 'learning_rate':
0.02462101059141227, 'max_depth': 9, 'subsample': 0.7345125327938399,
'colsample_bytree': 0.734639629538524, 'min_child_weight': 6, 'gamma':
0.00044848942107460427, 'reg_alpha': 1.0100462543418325e-08, 'reg_lambda':
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7.629536482366397e-05}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:20,883] Trial 506 finished with value:
-1.303008302017653 and parameters: {'n_estimators': 415, 'learning_rate':
0.019000347208218506, 'max_depth': 9, 'subsample': 0.7237406656990734,
'colsample_bytree': 0.724676285909298, 'min_child_weight': 5, 'gamma':
0.0003404317040709663, 'reg_alpha': 2.0920122707643935e-08, 'reg_lambda':
7.036931350603873e-05}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:21,216] Trial 507 finished with value:
-1.3067567283711956 and parameters: {'n_estimators': 401, 'learning_rate':
0.01876364344721786, 'max_depth': 9, 'subsample': 0.7181158651068998,
'colsample_bytree': 0.7312440187320576, 'min_child_weight': 5, 'gamma':
0.00030331120662093456, 'reg_alpha': 2.065966422426789e-08, 'reg_lambda':
6.293144520254369e-05}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:21,566] Trial 508 finished with value:
-1.3001435408598188 and parameters: {'n_estimators': 428, 'learning_rate':
0.017588568269015156, 'max_depth': 9, 'subsample': 0.705334287759322,
'colsample_bytree': 0.7234967151139733, 'min_child_weight': 5, 'gamma':
0.000578224581973125, 'reg_alpha': 1.8698698622088784e-08, 'reg_lambda':
0.00014736591974070783}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:21,908] Trial 509 finished with value:
-1.3057196985281294 and parameters: {'n_estimators': 417, 'learning_rate':
0.01973279841362457, 'max_depth': 9, 'subsample': 0.704918135750705,
'colsample_bytree': 0.7305263826798098, 'min_child_weight': 5, 'gamma':
0.00034882187764575146, 'reg_alpha': 1.5903442021513974e-08, 'reg_lambda':
0.00010769858935556535}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:22,259] Trial 510 finished with value:
-1.3080286623733628 and parameters: {'n_estimators': 430, 'learning_rate':
0.018143272042660454, 'max_depth': 9, 'subsample': 0.7020050720078135,
'colsample_bytree': 0.7211881124942539, 'min_child_weight': 5, 'gamma':
0.0005670140952169533, 'reg_alpha': 1.0038263810380867e-08, 'reg_lambda':
0.0001482465410392849}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:22,538] Trial 511 finished with value:
-1.3061527384323093 and parameters: {'n_estimators': 385, 'learning_rate':
0.017460014932167672, 'max_depth': 8, 'subsample': 0.713843769712886,
'colsample_bytree': 0.7201139584104114, 'min_child_weight': 5, 'gamma':
0.0008660680783872419, 'reg_alpha': 2.2379459600868635e-08, 'reg_lambda':
6.684137387249024e-05}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:22,878] Trial 512 finished with value:
-1.3039990136513955 and parameters: {'n_estimators': 407, 'learning_rate':
0.021077879547959966, 'max_depth': 9, 'subsample': 0.7006979958257625,
'colsample_bytree': 0.7393719212307821, 'min_child_weight': 5, 'gamma':
0.0011818418735841582, 'reg_alpha': 1.5283542095874616e-08, 'reg_lambda':
4.2049757315587496e-05}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:23,219] Trial 513 finished with value:
-1.3089577619526709 and parameters: {'n_estimators': 420, 'learning_rate':
0.019650033450016133, 'max_depth': 9, 'subsample': 0.7214785604039331,
'colsample_bytree': 0.7230229288426494, 'min_child_weight': 5, 'gamma':
0.0005196677113293373, 'reg_alpha': 2.0862092310418793e-08, 'reg_lambda':
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8.489474329231962e-05}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:23,539] Trial 514 finished with value:
-1.305585186992106 and parameters: {'n_estimators': 372, 'learning_rate':
0.01692329938766273, 'max_depth': 9, 'subsample': 0.7111705991954569,
'colsample_bytree': 0.7430798607211341, 'min_child_weight': 5, 'gamma':
0.000252037743478821, 'reg_alpha': 1.4907145084319895e-08, 'reg_lambda':
5.5439013724047807e-05}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:23,641] Trial 515 finished with value:
-1.4557455331720464 and parameters: {'n_estimators': 391, 'learning_rate':
0.017746209286196038, 'max_depth': 2, 'subsample': 0.694749385092972,
'colsample_bytree': 0.7228711673011521, 'min_child_weight': 5, 'gamma':
0.0007680760595104047, 'reg_alpha': 2.210989895809783e-08, 'reg_lambda':
0.00015225775570056808}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:23,921] Trial 516 finished with value:
-1.3081334247962968 and parameters: {'n_estimators': 299, 'learning_rate':
0.018700338333478502, 'max_depth': 9, 'subsample': 0.7408294479267984,
'colsample_bytree': 0.7469673112706066, 'min_child_weight': 5, 'gamma':
0.00037800317821801903, 'reg_alpha': 1.4028542620382628e-08, 'reg_lambda':
0.0002478634636882983}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:24,264] Trial 517 finished with value:
-1.3129139449427518 and parameters: {'n_estimators': 435, 'learning_rate':
0.016852241465338938, 'max_depth': 9, 'subsample': 0.7213273490023173,
'colsample_bytree': 0.7362729616620947, 'min_child_weight': 6, 'gamma':
0.0007681566333599954, 'reg_alpha': 2.263629191277317e-08, 'reg_lambda':
0.00029948098860901017}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:24,512] Trial 518 finished with value:
-1.3071069046115429 and parameters: {'n_estimators': 321, 'learning_rate':
0.02031939868751696, 'max_depth': 8, 'subsample': 0.726698387924946,
'colsample_bytree': 0.7177460725334819, 'min_child_weight': 5, 'gamma':
0.0012466654555452256, 'reg_alpha': 1.4200992270054118e-08, 'reg_lambda':
0.00014871043668870568}. Best is trial 483 with value: -1.298542171258924.
[I 2026-03-13 19:13:24,823] Trial 519 finished with value:
-1.2974087478779242 and parameters: {'n_estimators': 347, 'learning_rate':
0.018406245617040438, 'max_depth': 9, 'subsample': 0.7081430240167019,
'colsample_bytree': 0.7492260699417507, 'min_child_weight': 5, 'gamma':
0.0005405407805244604, 'reg_alpha': 1.9961594381733318e-08, 'reg_lambda':
4.6445909700439096e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:25,115] Trial 520 finished with value:
-1.3080005069673954 and parameters: {'n_estimators': 337, 'learning_rate':
0.02290521855739736, 'max_depth': 9, 'subsample': 0.7054666013492604,
'colsample_bytree': 0.732452810308751, 'min_child_weight': 5, 'gamma':
0.0005339940458887145, 'reg_alpha': 1.012712782684827e-08, 'reg_lambda':
2.7057395884227068e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:25,417] Trial 521 finished with value:
-1.3097328352813262 and parameters: {'n_estimators': 349, 'learning_rate':
0.018787335627775823, 'max_depth': 9, 'subsample': 0.6946090137335058,
'colsample_bytree': 0.7461675299081312, 'min_child_weight': 5, 'gamma':
0.0014554968165155476, 'reg_alpha': 2.117030189040233e-08, 'reg_lambda':
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3.412228827311723e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:25,706] Trial 522 finished with value:
-1.3009989514082607 and parameters: {'n_estimators': 328, 'learning_rate':
0.016885533409259385, 'max_depth': 9, 'subsample': 0.7147332001108567,
'colsample_bytree': 0.717728274446904, 'min_child_weight': 5, 'gamma':
0.00027982592594300064, 'reg_alpha': 1.5926544930186494e-08, 'reg_lambda':
5.560393928960627e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:25,955] Trial 523 finished with value:
-1.3125118848880053 and parameters: {'n_estimators': 316, 'learning_rate':
0.018676032016393465, 'max_depth': 8, 'subsample': 0.710174343469521,
'colsample_bytree': 0.7127293078090251, 'min_child_weight': 5, 'gamma':
0.00019332287267286957, 'reg_alpha': 1.5009609684851656e-08, 'reg_lambda':
5.343817955571958e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:26,267] Trial 524 finished with value:
-1.3083246676212934 and parameters: {'n_estimators': 359, 'learning_rate':
0.01667064967714967, 'max_depth': 9, 'subsample': 0.7236662599131567,
'colsample_bytree': 0.7054447254349264, 'min_child_weight': 5, 'gamma':
0.0003010391814314016, 'reg_alpha': 1.0045959898083542e-08, 'reg_lambda':
8.920754892370067e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:26,548] Trial 525 finished with value:
-1.3126768458660247 and parameters: {'n_estimators': 325, 'learning_rate':
0.020723554724520143, 'max_depth': 9, 'subsample': 0.6926053736085895,
'colsample_bytree': 0.7248329332874283, 'min_child_weight': 5, 'gamma':
0.00043655119121438675, 'reg_alpha': 1.932630418363285e-08, 'reg_lambda':
1.9183382015199722e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:26,808] Trial 526 finished with value:
-1.3053597156784775 and parameters: {'n_estimators': 291, 'learning_rate':
0.018059204540018546, 'max_depth': 9, 'subsample': 0.7034910271077843,
'colsample_bytree': 0.7157764434577248, 'min_child_weight': 5, 'gamma':
0.00024173850175735156, 'reg_alpha': 1.4546057598444446e-08, 'reg_lambda':
4.3864262774573604e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:27,097] Trial 527 finished with value:
-1.3018548560907413 and parameters: {'n_estimators': 338, 'learning_rate':
0.019525696649085423, 'max_depth': 9, 'subsample': 0.7136135606395713,
'colsample_bytree': 0.730371881266507, 'min_child_weight': 5, 'gamma':
0.0008901807066939816, 'reg_alpha': 2.1737859379655818e-08, 'reg_lambda':
4.948685210218122e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:27,368] Trial 528 finished with value:
-1.3104842050188925 and parameters: {'n_estimators': 341, 'learning_rate':
0.021481886633474976, 'max_depth': 9, 'subsample': 0.7013903257759766,
'colsample_bytree': 0.7236228478238594, 'min_child_weight': 6, 'gamma':
0.0013902330681512297, 'reg_alpha': 2.0592001203200806e-08, 'reg_lambda':
6.0017209923667315e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:27,689] Trial 529 finished with value:
-1.298680605735996 and parameters: {'n_estimators': 322, 'learning_rate':
0.01919791762942248, 'max_depth': 10, 'subsample': 0.7111808636636572,
'colsample_bytree': 0.735181834490884, 'min_child_weight': 5, 'gamma':
0.0009318897045893878, 'reg_alpha': 1.3824058425268529e-08, 'reg_lambda':
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2.8998880456479553e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:27,978] Trial 530 finished with value:
-1.3043782913164064 and parameters: {'n_estimators': 289, 'learning_rate':
0.020172036944516296, 'max_depth': 10, 'subsample': 0.7139195902336815,
'colsample_bytree': 0.7349104904348855, 'min_child_weight': 5, 'gamma':
0.0008934763060103139, 'reg_alpha': 1.000283744357421e-08, 'reg_lambda':
3.4890178303868794e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:28,329] Trial 531 finished with value:
-1.3050726529597705 and parameters: {'n_estimators': 365, 'learning_rate':
0.019009328401143423, 'max_depth': 10, 'subsample': 0.6938640798182613,
'colsample_bytree': 0.7377825484531085, 'min_child_weight': 5, 'gamma':
0.00039949612722663705, 'reg_alpha': 1.482703366735834e-08, 'reg_lambda':
4.759090099918722e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:28,650] Trial 532 finished with value:
-1.3076602460928657 and parameters: {'n_estimators': 321, 'learning_rate':
0.016550504173292502, 'max_depth': 10, 'subsample': 0.7050304266438612,
'colsample_bytree': 0.7083871012611916, 'min_child_weight': 5, 'gamma':
0.0005226200414380725, 'reg_alpha': 2.2694502528364342e-08, 'reg_lambda':
2.208882749174994e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:28,980] Trial 533 finished with value:
-1.310431686013238 and parameters: {'n_estimators': 346, 'learning_rate':
0.023810377968113132, 'max_depth': 10, 'subsample': 0.8658795402055035,
'colsample_bytree': 0.7269935572480415, 'min_child_weight': 5, 'gamma':
0.0011445089970318464, 'reg_alpha': 1.4792138537617706e-08, 'reg_lambda':
9.832138006362418e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:29,290] Trial 534 finished with value:
-1.3040935072960915 and parameters: {'n_estimators': 308, 'learning_rate':
0.022052050637735087, 'max_depth': 10, 'subsample': 0.728679049248425,
'colsample_bytree': 0.7415734543142526, 'min_child_weight': 5, 'gamma':
0.0009297024847927176, 'reg_alpha': 2.5182566763364037e-08, 'reg_lambda':
2.962864703949703e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:29,644] Trial 535 finished with value:
-1.3039907570832694 and parameters: {'n_estimators': 374, 'learning_rate':
0.019322326014571616, 'max_depth': 10, 'subsample': 0.7108993206425027,
'colsample_bytree': 0.7273255464464823, 'min_child_weight': 5, 'gamma':
0.00031498951264214873, 'reg_alpha': 1.4456636614638579e-08, 'reg_lambda':
7.630694783708098e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:29,956] Trial 536 finished with value:
-1.3075383922032777 and parameters: {'n_estimators': 337, 'learning_rate':
0.016667050736089793, 'max_depth': 10, 'subsample': 0.7160498444800478,
'colsample_bytree': 0.7166341131440797, 'min_child_weight': 6, 'gamma':
0.001596031432747947, 'reg_alpha': 1.0266541560296424e-08, 'reg_lambda':
1.0982602805719577e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:30,269] Trial 537 finished with value:
-1.3086409979437486 and parameters: {'n_estimators': 324, 'learning_rate':
0.019869670571236436, 'max_depth': 10, 'subsample': 0.6907147754121159,
'colsample_bytree': 0.7367437516294904, 'min_child_weight': 5, 'gamma':
0.000526670275118979, 'reg_alpha': 1.9475335863940385e-08, 'reg_lambda':
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4.7812281963256246e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:30,558] Trial 538 finished with value:
-1.3244651085656467 and parameters: {'n_estimators': 365, 'learning_rate':
0.017616911129221476, 'max_depth': 9, 'subsample': 0.5320563144124852,
'colsample_bytree': 0.7483745747902084, 'min_child_weight': 5, 'gamma':
0.0007450730147589154, 'reg_alpha': 2.8534291548810844e-08, 'reg_lambda':
8.027424695380109e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:30,933] Trial 539 finished with value:
-1.3095111875690408 and parameters: {'n_estimators': 402, 'learning_rate':
0.018789801848164506, 'max_depth': 10, 'subsample': 0.7340817312985416,
'colsample_bytree': 0.7123350907617849, 'min_child_weight': 5, 'gamma':
0.0011380790434558136, 'reg_alpha': 1.4619362420965192e-08, 'reg_lambda':
4.4190811550765326e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:31,172] Trial 540 finished with value:
-1.3024696601172072 and parameters: {'n_estimators': 255, 'learning_rate':
0.020428028566378483, 'max_depth': 9, 'subsample': 0.7026841813389516,
'colsample_bytree': 0.7327428795685343, 'min_child_weight': 5, 'gamma':
0.00021570280062786255, 'reg_alpha': 2.3345998846510982e-08, 'reg_lambda':
0.00012560164311385783}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:31,449] Trial 541 finished with value:
-1.3153441942048494 and parameters: {'n_estimators': 383, 'learning_rate':
0.021176471728714497, 'max_depth': 8, 'subsample': 0.7067811998556859,
'colsample_bytree': 0.727332302099519, 'min_child_weight': 5, 'gamma':
0.00017071792869550564, 'reg_alpha': 2.0761057411875857e-08, 'reg_lambda':
0.00012192537286578497}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:31,666] Trial 542 finished with value:
-1.308367647971172 and parameters: {'n_estimators': 223, 'learning_rate':
0.020309637303551422, 'max_depth': 9, 'subsample': 0.7244109343684417,
'colsample_bytree': 0.7014291769013101, 'min_child_weight': 5, 'gamma':
0.00021519873385558822, 'reg_alpha': 2.6882395602387553e-08, 'reg_lambda':
6.557314965744058e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:31,925] Trial 543 finished with value:
-1.3115865908811382 and parameters: {'n_estimators': 270, 'learning_rate':
0.018045993392042867, 'max_depth': 9, 'subsample': 0.712750387691667,
'colsample_bytree': 0.7352336252464179, 'min_child_weight': 5, 'gamma':
0.00042450934995552956, 'reg_alpha': 1.7837398931622916e-08, 'reg_lambda':
0.00016704605745148615}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:32,310] Trial 544 finished with value:
-1.3087903797584597 and parameters: {'n_estimators': 418, 'learning_rate':
0.016510766578503193, 'max_depth': 10, 'subsample': 0.6970898035634413,
'colsample_bytree': 0.7226318367367278, 'min_child_weight': 5, 'gamma':
0.00030185152311913043, 'reg_alpha': 1.0055150429896342e-08, 'reg_lambda':
8.965738863780325e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:32,548] Trial 545 finished with value:
-1.3091926507735534 and parameters: {'n_estimators': 249, 'learning_rate':
0.019177182089456967, 'max_depth': 9, 'subsample': 0.7197473766625644,
'colsample_bytree': 0.7416885254986407, 'min_child_weight': 5, 'gamma':
0.0005891447553106122, 'reg_alpha': 2.8055180265293032e-08, 'reg_lambda':
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2.9212421605683597e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:32,849] Trial 546 finished with value:
-1.3050887901046802 and parameters: {'n_estimators': 353, 'learning_rate':
0.02292314722791973, 'max_depth': 9, 'subsample': 0.7376969238942825,
'colsample_bytree': 0.7079439593112842, 'min_child_weight': 5, 'gamma':
0.000294530536043614, 'reg_alpha': 1.42233415657256e-08, 'reg_lambda':
1.757773200206596e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:33,043] Trial 547 finished with value:
-1.3338229256430318 and parameters: {'n_estimators': 173, 'learning_rate':
0.017521638549290366, 'max_depth': 9, 'subsample': 0.7048920638581088,
'colsample_bytree': 0.7295869556114072, 'min_child_weight': 6, 'gamma':
0.0009195090673184404, 'reg_alpha': 1.8707447991445974e-08, 'reg_lambda':
0.00010842892358994081}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:33,376] Trial 548 finished with value:
-1.3090834033586194 and parameters: {'n_estimators': 410, 'learning_rate':
0.025122350724182528, 'max_depth': 9, 'subsample': 0.7285193229578202,
'colsample_bytree': 0.7457003132723983, 'min_child_weight': 5, 'gamma':
0.00038368477577918614, 'reg_alpha': 2.8855443902667995e-08, 'reg_lambda':
5.179039843110724e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:33,604] Trial 549 finished with value:
-1.3238913104754035 and parameters: {'n_estimators': 211, 'learning_rate':
0.016199444592151568, 'max_depth': 9, 'subsample': 0.7108650447616585,
'colsample_bytree': 0.717389472270424, 'min_child_weight': 5, 'gamma':
0.0017686130802886365, 'reg_alpha': 1.4278835921029936e-08, 'reg_lambda':
0.00012642973506802482}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:33,959] Trial 550 finished with value:
-1.3061106600981576 and parameters: {'n_estimators': 378, 'learning_rate':
0.019979873091825473, 'max_depth': 10, 'subsample': 0.7033151539095699,
'colsample_bytree': 0.7358241592520938, 'min_child_weight': 5, 'gamma':
0.000634397768639455, 'reg_alpha': 1.945934767401162e-08, 'reg_lambda':
6.96218031816558e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:34,314] Trial 551 finished with value:
-1.3088726171224547 and parameters: {'n_estimators': 428, 'learning_rate':
0.01826816356333926, 'max_depth': 9, 'subsample': 0.7162181729169669,
'colsample_bytree': 0.7477000099309805, 'min_child_weight': 5, 'gamma':
0.00018769794557248654, 'reg_alpha': 2.4740063572845095e-08, 'reg_lambda':
4.063861105224478e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:34,563] Trial 552 finished with value:
-1.310158968926508 and parameters: {'n_estimators': 336, 'learning_rate':
0.021170704136844968, 'max_depth': 8, 'subsample': 0.7429192320250414,
'colsample_bytree': 0.7214306374576076, 'min_child_weight': 5, 'gamma':
0.0010648883534775137, 'reg_alpha': 1.3874058547188337e-08, 'reg_lambda':
0.0001837892373083156}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:34,763] Trial 553 finished with value:
-1.3353765146471617 and parameters: {'n_estimators': 183, 'learning_rate':
0.017066797794266293, 'max_depth': 9, 'subsample': 0.6967028950824546,
'colsample_bytree': 0.7501412706066329, 'min_child_weight': 6, 'gamma':
0.0005053157858657932, 'reg_alpha': 3.4746596759673526e-08, 'reg_lambda':
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0.00019192353759346658}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:35,066] Trial 554 finished with value:
-1.304640107687172 and parameters: {'n_estimators': 360, 'learning_rate':
0.0191331383251307, 'max_depth': 9, 'subsample': 0.7287497651896021,
'colsample_bytree': 0.7035699080633809, 'min_child_weight': 5, 'gamma':
0.0006957040786021861, 'reg_alpha': 1.0071694778519064e-08, 'reg_lambda':
9.747633355376809e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:35,337] Trial 555 finished with value:
-1.3092274633808423 and parameters: {'n_estimators': 283, 'learning_rate':
0.015868018425321134, 'max_depth': 9, 'subsample': 0.6882713209475837,
'colsample_bytree': 0.7370430341886132, 'min_child_weight': 5, 'gamma':
0.000249471694190925, 'reg_alpha': 1.9838818508101132e-08, 'reg_lambda':
2.892553516100681e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:35,450] Trial 556 finished with value:
-1.3820098009591908 and parameters: {'n_estimators': 307, 'learning_rate':
0.017605337240412147, 'max_depth': 4, 'subsample': 0.7198042156317481,
'colsample_bytree': 0.7258971475379418, 'min_child_weight': 5, 'gamma':
0.0015141311798800203, 'reg_alpha': 1.3553960385172355e-08, 'reg_lambda':
6.316417531390656e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:35,781] Trial 557 finished with value:
-1.303575150392117 and parameters: {'n_estimators': 398, 'learning_rate':
0.018540490242187378, 'max_depth': 9, 'subsample': 0.7012752408329599,
'colsample_bytree': 0.7534704755477366, 'min_child_weight': 5, 'gamma':
0.00034331740232604275, 'reg_alpha': 3.0176475522763e-08, 'reg_lambda':
0.00012883519772219901}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:36,136] Trial 558 finished with value:
-1.310753243231838 and parameters: {'n_estimators': 378, 'learning_rate':
0.022044857439229067, 'max_depth': 10, 'subsample': 0.7125942085090742,
'colsample_bytree': 0.7128783145877223, 'min_child_weight': 5, 'gamma':
0.0008093634167719805, 'reg_alpha': 2.4269039871444668e-08, 'reg_lambda':
1.1558311821126158e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:36,427] Trial 559 finished with value:
-1.3009197801786254 and parameters: {'n_estimators': 325, 'learning_rate':
0.020353581789054647, 'max_depth': 9, 'subsample': 0.72537159166242,
'colsample_bytree': 0.7314489582697052, 'min_child_weight': 5, 'gamma':
0.0004270284219702253, 'reg_alpha': 1.0035458062009451e-08, 'reg_lambda':
0.00019303739809542012}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:36,717] Trial 560 finished with value:
-1.3105333663790384 and parameters: {'n_estimators': 318, 'learning_rate':
0.02017794733853596, 'max_depth': 9, 'subsample': 0.7403746014032353,
'colsample_bytree': 0.7426602761008791, 'min_child_weight': 5, 'gamma':
0.001117520398772157, 'reg_alpha': 1.3131003411532775e-08, 'reg_lambda':
0.0003180991234952943}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:36,955] Trial 561 finished with value:
-1.318404275692725 and parameters: {'n_estimators': 297, 'learning_rate':
0.0160209430996492, 'max_depth': 8, 'subsample': 0.6910103644345713,
'colsample_bytree': 0.7583935885970055, 'min_child_weight': 5, 'gamma':
0.00048552954726219173, 'reg_alpha': 1.0050250947015301e-08, 'reg_lambda':
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0.00024123580945364633}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:37,248] Trial 562 finished with value:
-1.3045545208321778 and parameters: {'n_estimators': 250, 'learning_rate':
0.02224061488813119, 'max_depth': 11, 'subsample': 0.7093401700116977,
'colsample_bytree': 0.7348480374038914, 'min_child_weight': 5, 'gamma':
0.000673778817438517, 'reg_alpha': 1.0025849960257002e-08, 'reg_lambda':
0.00033304097763747835}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:37,621] Trial 563 finished with value:
-1.307966074915358 and parameters: {'n_estimators': 332, 'learning_rate':
0.01709211515248539, 'max_depth': 12, 'subsample': 0.7259483927415836,
'colsample_bytree': 0.7496110318199166, 'min_child_weight': 6, 'gamma':
0.002047984916570039, 'reg_alpha': 1.722555367868531e-08, 'reg_lambda':
0.00015274098217786574}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:37,922] Trial 564 finished with value:
-1.307154916577794 and parameters: {'n_estimators': 343, 'learning_rate':
0.020065769897288928, 'max_depth': 9, 'subsample': 0.7008208173894929,
'colsample_bytree': 0.7646203166388059, 'min_child_weight': 5, 'gamma':
0.0014072513481669575, 'reg_alpha': 1.504193003515868e-08, 'reg_lambda':
0.00019965936225405304}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:38,235] Trial 565 finished with value:
-1.3008267915185765 and parameters: {'n_estimators': 352, 'learning_rate':
0.017948083663706805, 'max_depth': 9, 'subsample': 0.7182021936071247,
'colsample_bytree': 0.7350627400446127, 'min_child_weight': 5, 'gamma':
0.0008897009162257884, 'reg_alpha': 3.584447353340919e-08, 'reg_lambda':
0.00035769925812391835}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:38,546] Trial 566 finished with value:
-1.3095977794380769 and parameters: {'n_estimators': 349, 'learning_rate':
0.017924834804738988, 'max_depth': 9, 'subsample': 0.7347701438936414,
'colsample_bytree': 0.7412863361028494, 'min_child_weight': 5, 'gamma':
0.0007975012528074158, 'reg_alpha': 3.703916370639853e-08, 'reg_lambda':
0.00039812186657817384}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:38,868] Trial 567 finished with value:
-1.302439132356047 and parameters: {'n_estimators': 319, 'learning_rate':
0.016049970296682845, 'max_depth': 10, 'subsample': 0.717073644294327,
'colsample_bytree': 0.75296071527808, 'min_child_weight': 5, 'gamma':
0.0010190823277954856, 'reg_alpha': 3.073237931192529e-08, 'reg_lambda':
0.0002722418832344503}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:39,190] Trial 568 finished with value:
-1.3053754948499667 and parameters: {'n_estimators': 316, 'learning_rate':
0.01604591889396188, 'max_depth': 10, 'subsample': 0.7169267343747284,
'colsample_bytree': 0.7588599316936994, 'min_child_weight': 5, 'gamma':
0.0010747314920171711, 'reg_alpha': 3.497955588628128e-08, 'reg_lambda':
0.0002904107770423281}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:39,556] Trial 569 finished with value:
-1.3021784231592093 and parameters: {'n_estimators': 360, 'learning_rate':
0.017084741213641546, 'max_depth': 10, 'subsample': 0.7170335280485461,
'colsample_bytree': 0.7650751210345226, 'min_child_weight': 5, 'gamma':
0.001616481026842832, 'reg_alpha': 2.094389465369614e-08, 'reg_lambda':
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0.0001893229707902892}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:39,910] Trial 570 finished with value:
-1.3146236497890138 and parameters: {'n_estimators': 353, 'learning_rate':
0.01699220363832057, 'max_depth': 10, 'subsample': 0.7470942000680675,
'colsample_bytree': 0.7639539005915363, 'min_child_weight': 5, 'gamma':
0.002361619617266543, 'reg_alpha': 3.8895508509108413e-08, 'reg_lambda':
0.00021085472158057896}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:40,254] Trial 571 finished with value:
-1.3020066995627229 and parameters: {'n_estimators': 326, 'learning_rate':
0.015966054416379386, 'max_depth': 10, 'subsample': 0.7327033086415965,
'colsample_bytree': 0.7704992568997929, 'min_child_weight': 5, 'gamma':
0.0016732669531772514, 'reg_alpha': 1.8421154220756092e-08, 'reg_lambda':
0.0004795064741563636}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:40,629] Trial 572 finished with value:
-1.3159302857216246 and parameters: {'n_estimators': 368, 'learning_rate':
0.017692868592921365, 'max_depth': 10, 'subsample': 0.7515693122884878,
'colsample_bytree': 0.7702413750014604, 'min_child_weight': 5, 'gamma':
0.002321311427427318, 'reg_alpha': 1.4320755820216259e-08, 'reg_lambda':
0.0005382442129560791}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:40,973] Trial 573 finished with value:
-1.3071115383808403 and parameters: {'n_estimators': 331, 'learning_rate':
0.016763436345744433, 'max_depth': 10, 'subsample': 0.7365496319686053,
'colsample_bytree': 0.7711958767397091, 'min_child_weight': 5, 'gamma':
0.0016312543916196637, 'reg_alpha': 1.8892281111310222e-08, 'reg_lambda':
0.0004405154226942148}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:41,286] Trial 574 finished with value:
-1.3007964570873856 and parameters: {'n_estimators': 302, 'learning_rate':
0.018245697631240374, 'max_depth': 10, 'subsample': 0.727469754016753,
'colsample_bytree': 0.7677805673372101, 'min_child_weight': 5, 'gamma':
0.0013055347750826787, 'reg_alpha': 1.4202287313428984e-08, 'reg_lambda':
0.0003559500790184634}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:41,608] Trial 575 finished with value:
-1.3064332146702027 and parameters: {'n_estimators': 304, 'learning_rate':
0.018027928399988395, 'max_depth': 10, 'subsample': 0.7362137695237502,
'colsample_bytree': 0.7787497246217705, 'min_child_weight': 5, 'gamma':
0.0016678531908198278, 'reg_alpha': 1.3914177982756433e-08, 'reg_lambda':
0.00029972366359305307}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:41,913] Trial 576 finished with value:
-1.3116549771805197 and parameters: {'n_estimators': 292, 'learning_rate':
0.01887472945344667, 'max_depth': 10, 'subsample': 0.7266652687537399,
'colsample_bytree': 0.7666071640468192, 'min_child_weight': 6, 'gamma':
0.0014549807290376387, 'reg_alpha': 1.0322670118347033e-08, 'reg_lambda':
0.00036486209486366616}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:42,257] Trial 577 finished with value:
-1.3074065530207972 and parameters: {'n_estimators': 332, 'learning_rate':
0.01653167301733027, 'max_depth': 10, 'subsample': 0.7296208023772099,
'colsample_bytree': 0.7669451300934867, 'min_child_weight': 5, 'gamma':
0.002563798405198598, 'reg_alpha': 1.9924073163595097e-08, 'reg_lambda':
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0.0005599991865613161}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:42,664] Trial 578 finished with value:
-1.3076999906956457 and parameters: {'n_estimators': 352, 'learning_rate':
0.01580944443800251, 'max_depth': 11, 'subsample': 0.74841839436448,
'colsample_bytree': 0.7836899830961664, 'min_child_weight': 5, 'gamma':
0.001171337167897083, 'reg_alpha': 2.426598553310576e-08, 'reg_lambda':
0.0001943732206455034}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:42,985] Trial 579 finished with value:
-1.3004904132831765 and parameters: {'n_estimators': 305, 'learning_rate':
0.019097556914801073, 'max_depth': 10, 'subsample': 0.7260986399026101,
'colsample_bytree': 0.7957252899878028, 'min_child_weight': 5, 'gamma':
0.0007608709388572422, 'reg_alpha': 1.3676648853312716e-08, 'reg_lambda':
0.00044708066703958387}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:43,358] Trial 580 finished with value:
-1.3281341655827739 and parameters: {'n_estimators': 304, 'learning_rate':
0.01915359329780431, 'max_depth': 10, 'subsample': 0.9099373339450886,
'colsample_bytree': 0.9504135381526367, 'min_child_weight': 5, 'gamma':
0.001782611378246046, 'reg_alpha': 1.0131699832006349e-08, 'reg_lambda':
0.0006236542510379648}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:43,679] Trial 581 finished with value:
-1.3063611323204292 and parameters: {'n_estimators': 283, 'learning_rate':
0.01822715822555297, 'max_depth': 10, 'subsample': 0.7418602752872525,
'colsample_bytree': 0.8031396761997881, 'min_child_weight': 5, 'gamma':
0.0008607820396208688, 'reg_alpha': 1.521395784738744e-08, 'reg_lambda':
0.000311207939224218}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:44,043] Trial 582 finished with value:
-1.3100517541682344 and parameters: {'n_estimators': 323, 'learning_rate':
0.020327662179387033, 'max_depth': 11, 'subsample': 0.7265608361785408,
'colsample_bytree': 0.7964546203797813, 'min_child_weight': 5, 'gamma':
0.0010736803322221723, 'reg_alpha': 1.469471928425065e-08, 'reg_lambda':
0.00020791657095893567}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:44,422] Trial 583 finished with value:
-1.308020017225788 and parameters: {'n_estimators': 362, 'learning_rate':
0.018995771580399338, 'max_depth': 10, 'subsample': 0.739462773647323,
'colsample_bytree': 0.8259106854983116, 'min_child_weight': 5, 'gamma':
0.0006846497605192831, 'reg_alpha': 1.926401103471399e-08, 'reg_lambda':
0.00015944928821790615}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:44,743] Trial 584 finished with value:
-1.3094620867053772 and parameters: {'n_estimators': 306, 'learning_rate':
0.017716695143941336, 'max_depth': 10, 'subsample': 0.7313935789853836,
'colsample_bytree': 0.7576536747972787, 'min_child_weight': 5, 'gamma':
0.0014978911521282743, 'reg_alpha': 1.3764444766143302e-08, 'reg_lambda':
1.5277670820952338e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:45,087] Trial 585 finished with value:
-1.3144336905855702 and parameters: {'n_estimators': 337, 'learning_rate':
0.021495979797643443, 'max_depth': 10, 'subsample': 0.7546789835775755,
'colsample_bytree': 0.7924842823893342, 'min_child_weight': 5, 'gamma':
0.0008414525467043064, 'reg_alpha': 2.0919529700270536e-08, 'reg_lambda':
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0.00043911584305031246}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:45,452] Trial 586 finished with value:
-1.3060959731917265 and parameters: {'n_estimators': 382, 'learning_rate':
0.01937266120429129, 'max_depth': 10, 'subsample': 0.7222389934755145,
'colsample_bytree': 0.7524028123244557, 'min_child_weight': 5, 'gamma':
0.0028807234277073864, 'reg_alpha': 1.0117076320482734e-08, 'reg_lambda':
0.0002576788571283001}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:45,732] Trial 587 finished with value:
-1.3174670199893728 and parameters: {'n_estimators': 269, 'learning_rate':
0.016956385886729014, 'max_depth': 10, 'subsample': 0.716594958315983,
'colsample_bytree': 0.8131303854386773, 'min_child_weight': 6, 'gamma':
0.0012748174731290346, 'reg_alpha': 1.7746505603885983e-08, 'reg_lambda':
0.00010806533734609216}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:46,077] Trial 588 finished with value:
-1.3087034772278097 and parameters: {'n_estimators': 347, 'learning_rate':
0.018071498958109798, 'max_depth': 10, 'subsample': 0.7352167382443825,
'colsample_bytree': 0.7662865443091157, 'min_child_weight': 5, 'gamma':
0.001959136330656226, 'reg_alpha': 1.4112086444292433e-08, 'reg_lambda':
0.0004128518554968229}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:46,389] Trial 589 finished with value:
-1.3025005721412433 and parameters: {'n_estimators': 317, 'learning_rate':
0.02322005910666431, 'max_depth': 10, 'subsample': 0.7273953069626663,
'colsample_bytree': 0.7809917702303425, 'min_child_weight': 5, 'gamma':
0.0006778443266066428, 'reg_alpha': 2.6311835290120022e-08, 'reg_lambda':
0.0006605988454759697}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:46,734] Trial 590 finished with value:
-1.3059894557216614 and parameters: {'n_estimators': 369, 'learning_rate':
0.02668197622513918, 'max_depth': 10, 'subsample': 0.7153136229491797,
'colsample_bytree': 0.7507822474348498, 'min_child_weight': 5, 'gamma':
0.0005414364704890335, 'reg_alpha': 1.012762374846669e-08, 'reg_lambda':
4.626413236266797e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:47,037] Trial 591 finished with value:
-1.3135249411089136 and parameters: {'n_estimators': 326, 'learning_rate':
0.04136991617626825, 'max_depth': 10, 'subsample': 0.7437719860016961,
'colsample_bytree': 0.7590092492936574, 'min_child_weight': 5, 'gamma':
0.0009482995554569106, 'reg_alpha': 1.983191386327203e-08, 'reg_lambda':
0.00016013542700173394}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:47,338] Trial 592 finished with value:
-1.3072974474525432 and parameters: {'n_estimators': 290, 'learning_rate':
0.021087930899664868, 'max_depth': 10, 'subsample': 0.7200096937590995,
'colsample_bytree': 0.7431875682383855, 'min_child_weight': 5, 'gamma':
0.0013233237778014504, 'reg_alpha': 1.4131286553162596e-08, 'reg_lambda':
8.885098654744776e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:47,713] Trial 593 finished with value:
-1.3194518019034127 and parameters: {'n_estimators': 387, 'learning_rate':
0.019428214402308117, 'max_depth': 11, 'subsample': 0.7110885609194565,
'colsample_bytree': 0.7756325743904151, 'min_child_weight': 6, 'gamma':
0.0021894164769366525, 'reg_alpha': 2.5596702119318486e-08, 'reg_lambda':
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0.0003377880667218279}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:48,067] Trial 594 finished with value:
-1.3111953423089708 and parameters: {'n_estimators': 351, 'learning_rate':
0.015845719184429573, 'max_depth': 10, 'subsample': 0.7319713220826711,
'colsample_bytree': 0.7534998374663633, 'min_child_weight': 5, 'gamma':
0.0005924671773142376, 'reg_alpha': 4.118916504940913e-08, 'reg_lambda':
3.247092708379278e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:48,400] Trial 595 finished with value:
-1.304480942607956 and parameters: {'n_estimators': 304, 'learning_rate':
0.017484662115581216, 'max_depth': 10, 'subsample': 0.7239893269114381,
'colsample_bytree': 0.7914675360534954, 'min_child_weight': 5, 'gamma':
0.0007935671245263196, 'reg_alpha': 1.819884137463049e-08, 'reg_lambda':
0.0002488772409886524}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:48,786] Trial 596 finished with value:
-1.3016344237774975 and parameters: {'n_estimators': 403, 'learning_rate':
0.016829855360946784, 'max_depth': 10, 'subsample': 0.7095250845931722,
'colsample_bytree': 0.7441306748907045, 'min_child_weight': 5, 'gamma':
0.0004078015355521147, 'reg_alpha': 1.3267666763320466e-08, 'reg_lambda':
0.0005717562548938865}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:49,213] Trial 597 finished with value:
-1.3089689564265543 and parameters: {'n_estimators': 402, 'learning_rate':
0.018475089612727497, 'max_depth': 11, 'subsample': 0.7590154618900637,
'colsample_bytree': 0.7439569394095747, 'min_child_weight': 5, 'gamma':
0.00035759195083366916, 'reg_alpha': 1.0256714482621434e-08, 'reg_lambda':
0.0005158087114483582}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:49,598] Trial 598 finished with value:
-1.3098223318755933 and parameters: {'n_estimators': 429, 'learning_rate':
0.019865345107590582, 'max_depth': 10, 'subsample': 0.7043687408610658,
'colsample_bytree': 0.7377964235221429, 'min_child_weight': 5, 'gamma':
0.0004975735721922124, 'reg_alpha': 1.328899302269352e-08, 'reg_lambda':
0.000647318095173735}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:49,980] Trial 599 finished with value:
-1.302820823838361 and parameters: {'n_estimators': 396, 'learning_rate':
0.015504440702380676, 'max_depth': 10, 'subsample': 0.7127727454859174,
'colsample_bytree': 0.7264732687146551, 'min_child_weight': 5, 'gamma':
0.0006498380735532802, 'reg_alpha': 1.3638154831694761e-08, 'reg_lambda':
0.0004537599611007055}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:50,365] Trial 600 finished with value:
-1.3102450525333624 and parameters: {'n_estimators': 384, 'learning_rate':
0.016747611689385773, 'max_depth': 10, 'subsample': 0.7429093773855047,
'colsample_bytree': 0.7486140238815202, 'min_child_weight': 5, 'gamma':
0.00041181818977731706, 'reg_alpha': 2.7705971836588926e-08, 'reg_lambda':
0.0006994826122253879}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:50,666] Trial 601 finished with value:
-1.3124141479234481 and parameters: {'n_estimators': 329, 'learning_rate':
0.0187336741861909, 'max_depth': 10, 'subsample': 0.7286141555869533,
'colsample_bytree': 0.7379522959391699, 'min_child_weight': 6, 'gamma':
0.0009650610512231213, 'reg_alpha': 1.5260930989734792e-08, 'reg_lambda':

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2.4396713884046612e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:51,031] Trial 602 finished with value:
-1.3132195130248918 and parameters: {'n_estimators': 416, 'learning_rate':
0.03504349777071205, 'max_depth': 10, 'subsample': 0.7084824050153111,
'colsample_bytree': 0.722445029050739, 'min_child_weight': 5, 'gamma':
0.00043235070682819544, 'reg_alpha': 1.8776253426367448e-08, 'reg_lambda':
0.0003337313116278776}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:51,375] Trial 603 finished with value:
-1.3107971051437382 and parameters: {'n_estimators': 371, 'learning_rate':
0.020313692738914323, 'max_depth': 10, 'subsample': 0.7193107335243892,
'colsample_bytree': 0.7134058175521857, 'min_child_weight': 5, 'gamma':
0.0006185584118589774, 'reg_alpha': 3.5024274384439394e-08, 'reg_lambda':
5.0434911022179054e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:51,722] Trial 604 finished with value:
-1.2980337360022045 and parameters: {'n_estimators': 344, 'learning_rate':
0.017934620067628824, 'max_depth': 10, 'subsample': 0.7064634487019088,
'colsample_bytree': 0.752532078921236, 'min_child_weight': 5, 'gamma':
0.0009614752134394202, 'reg_alpha': 1.3459211474129577e-08, 'reg_lambda':
0.0005846959114844589}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:52,075] Trial 605 finished with value:
-1.3170774271120151 and parameters: {'n_estimators': 346, 'learning_rate':
0.021866892704051358, 'max_depth': 10, 'subsample': 0.7055542362722254,
'colsample_bytree': 0.747222454394534, 'min_child_weight': 5, 'gamma':
0.0009291974738082809, 'reg_alpha': 0.9983297435423247, 'reg_lambda':
1.8287130365891807e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:52,502] Trial 606 finished with value:
-1.3063139785395999 and parameters: {'n_estimators': 409, 'learning_rate':
0.01775963971027372, 'max_depth': 11, 'subsample': 0.7005525091355964,
'colsample_bytree': 0.7371122429841896, 'min_child_weight': 5, 'gamma':
0.0005026579111617923, 'reg_alpha': 1.0033327412614205e-08, 'reg_lambda':
0.0001253052081201643}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:52,896] Trial 607 finished with value:
-1.3092677612125136 and parameters: {'n_estimators': 439, 'learning_rate':
0.018538332414469665, 'max_depth': 10, 'subsample': 0.7103984536421766,
'colsample_bytree': 0.7292996920568378, 'min_child_weight': 5, 'gamma':
0.00029870845421110423, 'reg_alpha': 1.0058715961923477e-08, 'reg_lambda':
0.0007176429337512673}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:53,287] Trial 608 finished with value:
-1.3190177032313255 and parameters: {'n_estimators': 388, 'learning_rate':
0.02035845481509046, 'max_depth': 10, 'subsample': 0.6960381170020343,
'colsample_bytree': 0.8533361284180669, 'min_child_weight': 5, 'gamma':
0.0007886046396117256, 'reg_alpha': 1.3301624936325066e-08, 'reg_lambda':
0.0003353198103044366}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:53,660] Trial 609 finished with value:
-1.3013977082786676 and parameters: {'n_estimators': 375, 'learning_rate':
0.017474912585527146, 'max_depth': 10, 'subsample': 0.7075702143407745,
'colsample_bytree': 0.7537419307458764, 'min_child_weight': 5, 'gamma':
0.0011129210963756236, 'reg_alpha': 1.0012038691760022e-08, 'reg_lambda':
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3.402724032710517e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:54,012] Trial 610 finished with value:
-1.3059663127692358 and parameters: {'n_estimators': 366, 'learning_rate':
0.018722248514270665, 'max_depth': 10, 'subsample': 0.72113991149367,
'colsample_bytree': 0.7562704752312872, 'min_child_weight': 5, 'gamma':
0.0004082121859441847, 'reg_alpha': 2.6406241152226805e-08, 'reg_lambda':
1.8717735370250188e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:54,358] Trial 611 finished with value:
-1.3076833686275668 and parameters: {'n_estimators': 349, 'learning_rate':
0.01954070355811061, 'max_depth': 10, 'subsample': 0.6864258923338206,
'colsample_bytree': 0.7562321120875962, 'min_child_weight': 5, 'gamma':
0.0011782492475835844, 'reg_alpha': 1.5401280999717717e-08, 'reg_lambda':
4.9303730674928906e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:54,680] Trial 612 finished with value:
-1.3017820947732217 and parameters: {'n_estimators': 315, 'learning_rate':
0.016890013680402872, 'max_depth': 10, 'subsample': 0.6982788950214661,
'colsample_bytree': 0.747079839096955, 'min_child_weight': 5, 'gamma':
0.0006521320851628311, 'reg_alpha': 2.4942687990445035e-08, 'reg_lambda':
1.3111919932406924e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:54,994] Trial 613 finished with value:
-1.304890661154363 and parameters: {'n_estimators': 301, 'learning_rate':
0.016579962694475908, 'max_depth': 10, 'subsample': 0.6962118319247426,
'colsample_bytree': 0.7672126065804128, 'min_child_weight': 5, 'gamma':
0.0005881728880827567, 'reg_alpha': 4.220808355850023e-08, 'reg_lambda':
1.3111265774030905e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:55,297] Trial 614 finished with value:
-1.3114018667368057 and parameters: {'n_estimators': 313, 'learning_rate':
0.01535570298237934, 'max_depth': 10, 'subsample': 0.700320326884716,
'colsample_bytree': 0.7508624613376162, 'min_child_weight': 6, 'gamma':
0.000378854989515221, 'reg_alpha': 2.836262402845174e-08, 'reg_lambda':
8.625564415770033e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:55,609] Trial 615 finished with value:
-1.3029338271734048 and parameters: {'n_estimators': 282, 'learning_rate':
0.02088448475257783, 'max_depth': 11, 'subsample': 0.6917029192232506,
'colsample_bytree': 0.7633498127292411, 'min_child_weight': 5, 'gamma':
0.0006002374535004465, 'reg_alpha': 2.4113297755871504e-08, 'reg_lambda':
2.4249972603385814e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:55,933] Trial 616 finished with value:
-1.3044135310448493 and parameters: {'n_estimators': 328, 'learning_rate':
0.016546975505681356, 'max_depth': 10, 'subsample': 0.6823986179264633,
'colsample_bytree': 0.7447402493516332, 'min_child_weight': 5, 'gamma':
0.0002648152103146464, 'reg_alpha': 3.4738252007299655e-08, 'reg_lambda':
3.269249380734796e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:56,247] Trial 617 finished with value:
-1.2980252840714563 and parameters: {'n_estimators': 309, 'learning_rate':
0.018197879449025147, 'max_depth': 10, 'subsample': 0.7048908167198763,
'colsample_bytree': 0.7546549102297396, 'min_child_weight': 5, 'gamma':
0.0007143731981522746, 'reg_alpha': 1.9785926053986504e-08, 'reg_lambda':
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3.4948270725808706e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:56,551] Trial 618 finished with value:
-1.3100050470279627 and parameters: {'n_estimators': 273, 'learning_rate':
0.01797876283402607, 'max_depth': 10, 'subsample': 0.6966426537474103,
'colsample_bytree': 0.7786247034642528, 'min_child_weight': 5, 'gamma':
0.0004852568015388372, 'reg_alpha': 4.0742335137614306e-08, 'reg_lambda':
1.181364270606647e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:56,875] Trial 619 finished with value:
-1.3018311868907335 and parameters: {'n_estimators': 299, 'learning_rate':
0.015599164514696067, 'max_depth': 10, 'subsample': 0.7059700211325592,
'colsample_bytree': 0.7566149767846914, 'min_child_weight': 5, 'gamma':
0.00016282379551789088, 'reg_alpha': 1.7824341454509106e-08, 'reg_lambda':
2.0276907289918712e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:57,177] Trial 620 finished with value:
-1.3083770952920462 and parameters: {'n_estimators': 293, 'learning_rate':
0.0172561859477998, 'max_depth': 10, 'subsample': 0.6838529935131799,
'colsample_bytree': 0.7597478743627086, 'min_child_weight': 5, 'gamma':
0.0006204999995368026, 'reg_alpha': 2.5674399067546696e-08, 'reg_lambda':
3.903248350609695e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:57,501] Trial 621 finished with value:
-1.3009012633712387 and parameters: {'n_estimators': 319, 'learning_rate':
0.018255441349101287, 'max_depth': 10, 'subsample': 0.7029181266564394,
'colsample_bytree': 0.7714350928929772, 'min_child_weight': 5, 'gamma':
0.0003130250028990696, 'reg_alpha': 1.4011004491118337e-08, 'reg_lambda':
2.959156953028116e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:57,824] Trial 622 finished with value:
-1.30148171862154 and parameters: {'n_estimators': 310, 'learning_rate':
0.01857916792127007, 'max_depth': 10, 'subsample': 0.7195895621796118,
'colsample_bytree': 0.7840750198784837, 'min_child_weight': 5, 'gamma':
0.00028523910610630637, 'reg_alpha': 1.4329560469090878e-08, 'reg_lambda':
2.8195244393939683e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:58,106] Trial 623 finished with value:
-1.3107198420639996 and parameters: {'n_estimators': 279, 'learning_rate':
0.018295785893572187, 'max_depth': 10, 'subsample': 0.7218357389115486,
'colsample_bytree': 0.7917599365954423, 'min_child_weight': 6, 'gamma':
0.00019143181234307634, 'reg_alpha': 1.550990552887423e-08, 'reg_lambda':
3.664909397835295e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:58,272] Trial 624 finished with value:
-1.359594789035982 and parameters: {'n_estimators': 100, 'learning_rate':
0.021660848836307103, 'max_depth': 11, 'subsample': 0.7217268813278815,
'colsample_bytree': 0.7827171545066013, 'min_child_weight': 5, 'gamma':
0.0002092176935169447, 'reg_alpha': 1.687575746209519e-08, 'reg_lambda':
3.270039858615197e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:58,605] Trial 625 finished with value:
-1.3051614319848603 and parameters: {'n_estimators': 314, 'learning_rate':
0.01938149187634183, 'max_depth': 10, 'subsample': 0.7311296368091618,
'colsample_bytree': 0.8015262855508491, 'min_child_weight': 5, 'gamma':
0.0002669629457554169, 'reg_alpha': 1.4523319582595529e-08, 'reg_lambda':
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1.8180859805774655e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:58,897] Trial 626 finished with value:
-1.3219307915919174 and parameters: {'n_estimators': 303, 'learning_rate':
0.04628887065894976, 'max_depth': 10, 'subsample': 0.7155393037976476,
'colsample_bytree': 0.8017657113391897, 'min_child_weight': 5, 'gamma':
0.00012915392989170185, 'reg_alpha': 1.894790150562348e-08, 'reg_lambda':
2.8385373524120674e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:59,211] Trial 627 finished with value:
-1.3179739262798065 and parameters: {'n_estimators': 322, 'learning_rate':
0.01757651140512261, 'max_depth': 10, 'subsample': 0.7387097975837277,
'colsample_bytree': 0.7839396597495706, 'min_child_weight': 6, 'gamma':
0.000291465331289124, 'reg_alpha': 3.177121325359396e-08, 'reg_lambda':
2.5231610275169896e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:59,512] Trial 628 finished with value:
-1.3103322329946074 and parameters: {'n_estimators': 267, 'learning_rate':
0.016228941967288416, 'max_depth': 10, 'subsample': 0.7267314979132908,
'colsample_bytree': 0.7718749094405418, 'min_child_weight': 5, 'gamma':
0.0003582842786032427, 'reg_alpha': 1.4611398206435959e-08, 'reg_lambda':
1.8119632396055306e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:13:59,856] Trial 629 finished with value:
-1.3026133747090867 and parameters: {'n_estimators': 340, 'learning_rate':
0.02026874993906606, 'max_depth': 10, 'subsample': 0.7081130509972163,
'colsample_bytree': 0.7796700688192589, 'min_child_weight': 5, 'gamma':
0.0001406000833115068, 'reg_alpha': 4.354651963198653e-08, 'reg_lambda':
7.252910581350517e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:00,201] Trial 630 finished with value:
-1.3018427589104342 and parameters: {'n_estimators': 338, 'learning_rate':
0.018490785207102414, 'max_depth': 10, 'subsample': 0.720419102863062,
'colsample_bytree': 0.7983818887731395, 'min_child_weight': 5, 'gamma':
0.00029445207440026243, 'reg_alpha': 1.8867892315647184e-08, 'reg_lambda':
2.6337389048623516e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:00,460] Trial 631 finished with value:
-1.4177120804536887 and parameters: {'n_estimators': 295, 'learning_rate':
0.15279022234866682, 'max_depth': 10, 'subsample': 0.7040055268551866,
'colsample_bytree': 0.812017175381425, 'min_child_weight': 5, 'gamma':
0.00040186576780899895, 'reg_alpha': 2.3958252674297293e-08, 'reg_lambda':
0.000662476459594543}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:00,798] Trial 632 finished with value:
-1.308989675470653 and parameters: {'n_estimators': 312, 'learning_rate':
0.015375737471754287, 'max_depth': 10, 'subsample': 0.7130461373051509,
'colsample_bytree': 0.770945412151277, 'min_child_weight': 5, 'gamma':
0.0002334601557492156, 'reg_alpha': 1.4716859131628905e-08, 'reg_lambda':
3.220285129596993e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:01,413] Trial 633 finished with value:
-1.3329903510968106 and parameters: {'n_estimators': 343, 'learning_rate':
0.022544789770872358, 'max_depth': 10, 'subsample': 0.7221238045054242,
'colsample_bytree': 0.7824673577367034, 'min_child_weight': 1, 'gamma':
0.00042894689832204683, 'reg_alpha': 1.0334735067376208e-08, 'reg_lambda':
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8.07084613436707e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:01,830] Trial 634 finished with value:
-1.31053621222305 and parameters: {'n_estimators': 357, 'learning_rate':
0.016922029278420413, 'max_depth': 11, 'subsample': 0.7481675719778185,
'colsample_bytree': 0.8261487965992298, 'min_child_weight': 5, 'gamma':
0.00041915583921631216, 'reg_alpha': 3.0208839819436516e-08, 'reg_lambda':
4.183685136092591e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:02,193] Trial 635 finished with value:
-1.3117620319426344 and parameters: {'n_estimators': 320, 'learning_rate':
0.01818968657764286, 'max_depth': 11, 'subsample': 0.7317879942673228,
'colsample_bytree': 0.7671724170779091, 'min_child_weight': 5, 'gamma':
0.00022217388512184148, 'reg_alpha': 2.0522520511408097e-08, 'reg_lambda':
3.5406449219362615e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:02,494] Trial 636 finished with value:
-1.2990397010765224 and parameters: {'n_estimators': 282, 'learning_rate':
0.020834939952624594, 'max_depth': 10, 'subsample': 0.6908918567073408,
'colsample_bytree': 0.7654776018078069, 'min_child_weight': 5, 'gamma':
0.000515118463898492, 'reg_alpha': 1.3367676297478092e-08, 'reg_lambda':
6.756914377405992e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:02,776] Trial 637 finished with value:
-1.315797692586677 and parameters: {'n_estimators': 279, 'learning_rate':
0.03002022841665993, 'max_depth': 10, 'subsample': 0.6776855639354468,
'colsample_bytree': 0.7906425662138729, 'min_child_weight': 5, 'gamma':
0.0012070781133382456, 'reg_alpha': 4.367186973310125e-08, 'reg_lambda':
2.5904376900631293e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:03,058] Trial 638 finished with value:
-1.3102330302831071 and parameters: {'n_estimators': 260, 'learning_rate':
0.02082954816414577, 'max_depth': 10, 'subsample': 0.6890473772003606,
'colsample_bytree': 0.7644792856742227, 'min_child_weight': 5, 'gamma':
0.0007233929187063118, 'reg_alpha': 2.1672218656846003e-08, 'reg_lambda':
1.1942526701652414e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:03,329] Trial 639 finished with value:
-1.3065699674822204 and parameters: {'n_estimators': 243, 'learning_rate':
0.02171038299648747, 'max_depth': 10, 'subsample': 0.6810023618265817,
'colsample_bytree': 0.7758518674996299, 'min_child_weight': 5, 'gamma':
0.0005475932126229765, 'reg_alpha': 1.3249079561998564e-08, 'reg_lambda':
2.015512753825845e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:03,619] Trial 640 finished with value:
-1.3152684783333815 and parameters: {'n_estimators': 291, 'learning_rate':
0.023112614499195397, 'max_depth': 10, 'subsample': 0.6938003906102378,
'colsample_bytree': 0.8104321798459039, 'min_child_weight': 6, 'gamma':
0.0007178371772943448, 'reg_alpha': 1.0174795720077394e-08, 'reg_lambda':
5.006374076795547e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:03,933] Trial 641 finished with value:
-1.313050818279019 and parameters: {'n_estimators': 307, 'learning_rate':
0.01980861720350503, 'max_depth': 10, 'subsample': 0.6945901109107427,
'colsample_bytree': 0.779074952896888, 'min_child_weight': 5, 'gamma':
0.0013198558867490184, 'reg_alpha': 2.7001928108170425e-08, 'reg_lambda':
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1.540747599638451e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:04,215] Trial 642 finished with value:
-1.3079808196962597 and parameters: {'n_estimators': 269, 'learning_rate':
0.024444760113908594, 'max_depth': 10, 'subsample': 0.7009577282203495,
'colsample_bytree': 0.789644134717734, 'min_child_weight': 5, 'gamma':
0.0002894878511043816, 'reg_alpha': 1.781197199717662e-08, 'reg_lambda':
6.251504890480139e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:04,514] Trial 643 finished with value:
-1.304354666463795 and parameters: {'n_estimators': 292, 'learning_rate':
0.019058746683919714, 'max_depth': 10, 'subsample': 0.6845912430685599,
'colsample_bytree': 0.7607378648548869, 'min_child_weight': 5, 'gamma':
0.0005422652414345801, 'reg_alpha': 3.700761369322821e-08, 'reg_lambda':
9.75544848089355e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:04,838] Trial 644 finished with value:
-1.2987473412734274 and parameters: {'n_estimators': 318, 'learning_rate':
0.020325094050031436, 'max_depth': 10, 'subsample': 0.7064988724180049,
'colsample_bytree': 0.7620375013581975, 'min_child_weight': 5, 'gamma':
0.003211787125439423, 'reg_alpha': 1.4597313764085415e-08, 'reg_lambda':
7.48695030230837e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:05,170] Trial 645 finished with value:
-1.3102263834688845 and parameters: {'n_estimators': 330, 'learning_rate':
0.021842307464183756, 'max_depth': 10, 'subsample': 0.7024215046339563,
'colsample_bytree': 0.771881869293976, 'min_child_weight': 5, 'gamma':
0.0034178234441456443, 'reg_alpha': 3.0159987906118855e-08, 'reg_lambda':
3.310992179126175e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:05,496] Trial 646 finished with value:
-1.3044507300629131 and parameters: {'n_estimators': 305, 'learning_rate':
0.020770251627323447, 'max_depth': 10, 'subsample': 0.7388100713992778,
'colsample_bytree': 0.7766665710582159, 'min_child_weight': 5, 'gamma':
0.0016082504327210917, 'reg_alpha': 1.94014202018367e-08, 'reg_lambda':
1.5545556775575007e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:05,778] Trial 647 finished with value:
-1.3122546096675938 and parameters: {'n_estimators': 284, 'learning_rate':
0.024209058199136838, 'max_depth': 10, 'subsample': 0.7180117770444844,
'colsample_bytree': 0.7877128738911577, 'min_child_weight': 6, 'gamma':
0.0020271521897388175, 'reg_alpha': 4.582712298215931e-08, 'reg_lambda':
6.7066324904970866e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:06,100] Trial 648 finished with value:
-1.3098556867038005 and parameters: {'n_estimators': 318, 'learning_rate':
0.020457618194293353, 'max_depth': 10, 'subsample': 0.6925810905539697,
'colsample_bytree': 0.7982120986481417, 'min_child_weight': 5, 'gamma':
0.003827938816462029, 'reg_alpha': 2.4652792503733692e-08, 'reg_lambda':
8.349724805782829e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:06,434] Trial 649 finished with value:
-1.3135112205499966 and parameters: {'n_estimators': 330, 'learning_rate':
0.01908517677494935, 'max_depth': 10, 'subsample': 0.7318416002130527,
'colsample_bytree': 0.7665871128117903, 'min_child_weight': 5, 'gamma':
0.005465180154140785, 'reg_alpha': 1.5592907845168816e-08, 'reg_lambda':
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2.567628201764838e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:06,683] Trial 650 finished with value:
-1.4062248494751683 and parameters: {'n_estimators': 259, 'learning_rate':
0.12379637447147217, 'max_depth': 11, 'subsample': 0.7071611248555064,
'colsample_bytree': 0.7613304566766275, 'min_child_weight': 5, 'gamma':
0.002790632527616882, 'reg_alpha': 3.3318882281143454e-08, 'reg_lambda':
4.251887950790339e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:06,995] Trial 651 finished with value:
-1.3170583218762182 and parameters: {'n_estimators': 301, 'learning_rate':
0.01810578799690546, 'max_depth': 10, 'subsample': 0.6789416185901526,
'colsample_bytree': 0.7822553118506327, 'min_child_weight': 5, 'gamma':
0.0023780658309521976, 'reg_alpha': 1.9632255758930567e-08, 'reg_lambda':
1.947714391108543e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:07,328] Trial 652 finished with value:
-1.3000382045994947 and parameters: {'n_estimators': 323, 'learning_rate':
0.017834444007941184, 'max_depth': 10, 'subsample': 0.7234352067484733,
'colsample_bytree': 0.7655083818518359, 'min_child_weight': 5, 'gamma':
0.0011170364517520668, 'reg_alpha': 1.3956147700941858e-08, 'reg_lambda':
4.2664348672906394e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:07,713] Trial 653 finished with value:
-1.2992191034646092 and parameters: {'n_estimators': 343, 'learning_rate':
0.016079612642958305, 'max_depth': 11, 'subsample': 0.7253286621543283,
'colsample_bytree': 0.7584067340579698, 'min_child_weight': 5, 'gamma':
0.001228965537850491, 'reg_alpha': 2.427196436247319e-08, 'reg_lambda':
5.984567933848161e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:08,140] Trial 654 finished with value:
-1.3046346047541857 and parameters: {'n_estimators': 343, 'learning_rate':
0.015557712925621735, 'max_depth': 12, 'subsample': 0.738742232124017,
'colsample_bytree': 0.7620287196769673, 'min_child_weight': 5, 'gamma':
0.0013210060059429725, 'reg_alpha': 4.1970040143946484e-08, 'reg_lambda':
4.0633601717225513e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:08,525] Trial 655 finished with value:
-1.3085505832111775 and parameters: {'n_estimators': 339, 'learning_rate':
0.016253127677415397, 'max_depth': 11, 'subsample': 0.7560490167272814,
'colsample_bytree': 0.7652649814971635, 'min_child_weight': 5, 'gamma':
0.0031936120519450583, 'reg_alpha': 2.7247640785737955e-08, 'reg_lambda':
6.327296227279721e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:08,868] Trial 656 finished with value:
-1.312513051484345 and parameters: {'n_estimators': 330, 'learning_rate':
0.01725401181604535, 'max_depth': 11, 'subsample': 0.731575806683306,
'colsample_bytree': 0.7548425881685146, 'min_child_weight': 6, 'gamma':
0.0021599823882222473, 'reg_alpha': 4.553802191446841e-08, 'reg_lambda':
4.3632789143151804e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:09,296] Trial 657 finished with value:
-1.3009816078018035 and parameters: {'n_estimators': 349, 'learning_rate':
0.015359526113771787, 'max_depth': 12, 'subsample': 0.724253421704841,
'colsample_bytree': 0.7701370567553593, 'min_child_weight': 5, 'gamma':
0.0011849851354310754, 'reg_alpha': 2.2944200481851197e-08, 'reg_lambda':
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8.492946698352848e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:09,608] Trial 658 finished with value:
-1.3078760533360563 and parameters: {'n_estimators': 283, 'learning_rate':
0.015250075391492755, 'max_depth': 10, 'subsample': 0.7409318302532037,
'colsample_bytree': 0.7711956016637957, 'min_child_weight': 5, 'gamma':
0.0016059379788529039, 'reg_alpha': 3.1801975551840944e-08, 'reg_lambda':
0.00010403717650640257}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:10,019] Trial 659 finished with value:
-1.3051054117424254 and parameters: {'n_estimators': 316, 'learning_rate':
0.015110230546114076, 'max_depth': 12, 'subsample': 0.7511831424095444,
'colsample_bytree': 0.7670586954434667, 'min_child_weight': 5, 'gamma':
0.001036046026271077, 'reg_alpha': 2.4430276989710932e-08, 'reg_lambda':
6.56926164002046e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:10,434] Trial 660 finished with value:
-1.2986598493607344 and parameters: {'n_estimators': 329, 'learning_rate':
0.015807384909876678, 'max_depth': 12, 'subsample': 0.7275029021218141,
'colsample_bytree': 0.776871458342162, 'min_child_weight': 5, 'gamma':
0.0013567995697878477, 'reg_alpha': 4.338843424888318e-08, 'reg_lambda':
9.641768362037988e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:10,843] Trial 661 finished with value:
-1.307933496020301 and parameters: {'n_estimators': 321, 'learning_rate':
0.015331408175813162, 'max_depth': 12, 'subsample': 0.7455557163405787,
'colsample_bytree': 0.7912490241938249, 'min_child_weight': 5, 'gamma':
0.0022233004207701085, 'reg_alpha': 4.8351938883760774e-08, 'reg_lambda':
5.697249364557349e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:11,230] Trial 662 finished with value:
-1.308231382515163 and parameters: {'n_estimators': 295, 'learning_rate':
0.016125337501906197, 'max_depth': 12, 'subsample': 0.7646224295954803,
'colsample_bytree': 0.7787367706863345, 'min_child_weight': 5, 'gamma':
0.004240397080791541, 'reg_alpha': 4.063005039302465e-08, 'reg_lambda':
0.00011636157722491959}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:11,595] Trial 663 finished with value:
-1.3502308315115386 and parameters: {'n_estimators': 353, 'learning_rate':
0.05976423750851033, 'max_depth': 12, 'subsample': 0.7330797105063899,
'colsample_bytree': 0.7757969732525893, 'min_child_weight': 5, 'gamma':
0.0013805124967215076, 'reg_alpha': 3.6418540772373255e-08, 'reg_lambda':
0.0001776798864052588}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:12,012] Trial 664 finished with value:
-1.304173980866073 and parameters: {'n_estimators': 326, 'learning_rate':
0.015954343377588706, 'max_depth': 12, 'subsample': 0.7253426201237234,
'colsample_bytree': 0.7993483164084311, 'min_child_weight': 5, 'gamma':
0.0031180075535808404, 'reg_alpha': 3.0468221984743405e-08, 'reg_lambda':
7.691874990658954e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:12,379] Trial 665 finished with value:
-1.3027368335538612 and parameters: {'n_estimators': 298, 'learning_rate':
0.014829156077686468, 'max_depth': 12, 'subsample': 0.7239365929065585,
'colsample_bytree': 0.772812716937508, 'min_child_weight': 6, 'gamma':
0.001975923689793338, 'reg_alpha': 5.4106703589345445e-08, 'reg_lambda':
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7.293039858749212e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:12,745] Trial 666 finished with value:
-1.3095974522864917 and parameters: {'n_estimators': 274, 'learning_rate':
0.017183059516662814, 'max_depth': 12, 'subsample': 0.7468653557651374,
'colsample_bytree': 0.7915416754279457, 'min_child_weight': 5, 'gamma':
0.0008738187780088502, 'reg_alpha': 5.3984403695759766e-08, 'reg_lambda':
9.260861685096269e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:13,057] Trial 667 finished with value:
-1.36306088249737 and parameters: {'n_estimators': 312, 'learning_rate':
0.09350924712543722, 'max_depth': 12, 'subsample': 0.7289032545747391,
'colsample_bytree': 0.7623466294302876, 'min_child_weight': 5, 'gamma':
0.0014668086571902514, 'reg_alpha': 2.5088904235881867e-08, 'reg_lambda':
0.00011120621705843254}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:13,485] Trial 668 finished with value:
-1.3078279922906855 and parameters: {'n_estimators': 355, 'learning_rate':
0.01622852415037332, 'max_depth': 12, 'subsample': 0.7373275116914831,
'colsample_bytree': 0.7755578609854065, 'min_child_weight': 5, 'gamma':
0.0008116613011880426, 'reg_alpha': 3.105271871590695e-08, 'reg_lambda':
8.972688908210256e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:13,892] Trial 669 finished with value:
-1.3045205184744768 and parameters: {'n_estimators': 334, 'learning_rate':
0.015148075547988001, 'max_depth': 12, 'subsample': 0.7227015585811245,
'colsample_bytree': 0.7581847271194081, 'min_child_weight': 5, 'gamma':
0.0013408543498242482, 'reg_alpha': 2.158475516945257e-08, 'reg_lambda':
3.6805905451521466e-07}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:14,185] Trial 670 finished with value:
-1.3169160146498686 and parameters: {'n_estimators': 311, 'learning_rate':
0.017083556555466006, 'max_depth': 12, 'subsample': 0.7177704423020818,
'colsample_bytree': 0.7866696057127928, 'min_child_weight': 9, 'gamma':
0.0017015885824350357, 'reg_alpha': 4.306484057945413e-08, 'reg_lambda':
5.127377367322648e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:14,573] Trial 671 finished with value:
-1.3049982636050643 and parameters: {'n_estimators': 287, 'learning_rate':
0.016246512844909036, 'max_depth': 12, 'subsample': 0.7311796163625445,
'colsample_bytree': 0.8094796088982362, 'min_child_weight': 5, 'gamma':
0.0008603193722004882, 'reg_alpha': 2.0414827918818224e-08, 'reg_lambda':
0.00016429379445750782}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:14,958] Trial 672 finished with value:
-1.3083026895000052 and parameters: {'n_estimators': 349, 'learning_rate':
0.01795713757856357, 'max_depth': 11, 'subsample': 0.7407956163336992,
'colsample_bytree': 0.7694028506551858, 'min_child_weight': 5, 'gamma':
0.0010898092393224442, 'reg_alpha': 3.286853220157105e-08, 'reg_lambda':
0.00013512041273322837}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:15,333] Trial 673 finished with value:
-1.3029642071701149 and parameters: {'n_estimators': 332, 'learning_rate':
0.01545857712360155, 'max_depth': 11, 'subsample': 0.7168409247108266,
'colsample_bytree': 0.7588641270676095, 'min_child_weight': 5, 'gamma':
0.0024341099840503554, 'reg_alpha': 5.259434694716741e-08, 'reg_lambda':
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6.26517805496005e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:15,636] Trial 674 finished with value:
-1.431856497224927 and parameters: {'n_estimators': 320, 'learning_rate':
0.1809416110184088, 'max_depth': 12, 'subsample': 0.7289750970889146,
'colsample_bytree': 0.7791898591624183, 'min_child_weight': 5, 'gamma':
0.0007239765496343016, 'reg_alpha': 2.1097865010332897e-08, 'reg_lambda':
9.703740139426757e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:15,907] Trial 675 finished with value:
-1.318540849430563 and parameters: {'n_estimators': 263, 'learning_rate':
0.01668296546323124, 'max_depth': 11, 'subsample': 0.5171954539913153,
'colsample_bytree': 0.7527887055113023, 'min_child_weight': 5, 'gamma':
0.0011827724848810543, 'reg_alpha': 2.5269484282203967e-08, 'reg_lambda':
0.00024233042622232395}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:16,305] Trial 676 finished with value:
-1.3101673210093052 and parameters: {'n_estimators': 362, 'learning_rate':
0.019704041286811972, 'max_depth': 11, 'subsample': 0.7525832487914803,
'colsample_bytree': 0.7627484940535025, 'min_child_weight': 5, 'gamma':
0.0005332585577540864, 'reg_alpha': 1.7409994264230015e-08, 'reg_lambda':
0.000764845544825301}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:16,662] Trial 677 finished with value:
-1.3105270253111232 and parameters: {'n_estimators': 300, 'learning_rate':
0.018213657539763756, 'max_depth': 12, 'subsample': 0.7165860993297413,
'colsample_bytree': 0.7937907149602264, 'min_child_weight': 6, 'gamma':
0.000731093182569433, 'reg_alpha': 3.469524388436908e-08, 'reg_lambda':
8.320415284759873e-07}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:17,069] Trial 678 finished with value:
-1.3032518459878721 and parameters: {'n_estimators': 332, 'learning_rate':
0.01489554326547645, 'max_depth': 12, 'subsample': 0.7264024305868145,
'colsample_bytree': 0.7544498194887582, 'min_child_weight': 5, 'gamma':
0.0026417470075511804, 'reg_alpha': 1.7339262356702235e-08, 'reg_lambda':
2.3652862697718165e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:17,311] Trial 679 finished with value:
-1.3522917390887557 and parameters: {'n_estimators': 226, 'learning_rate':
0.07720478433001807, 'max_depth': 11, 'subsample': 0.7355470277956089,
'colsample_bytree': 0.7724123009073618, 'min_child_weight': 5, 'gamma':
0.001675312786822972, 'reg_alpha': 5.488566894007653e-08, 'reg_lambda':
0.0003918355858476022}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:17,739] Trial 680 finished with value:
-1.3051445646790025 and parameters: {'n_estimators': 350, 'learning_rate':
0.017442558334402634, 'max_depth': 12, 'subsample': 0.7178520965376797,
'colsample_bytree': 0.7834478524086161, 'min_child_weight': 5, 'gamma':
0.0045519777421865444, 'reg_alpha': 2.5686975204869525e-08, 'reg_lambda':
5.143741559480969e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:18,094] Trial 681 finished with value:
-1.3120928976988158 and parameters: {'n_estimators': 303, 'learning_rate':
0.01647573387106502, 'max_depth': 11, 'subsample': 0.741791916526269,
'colsample_bytree': 0.7690152930830789, 'min_child_weight': 5, 'gamma':
0.0007688638427024982, 'reg_alpha': 1.5659336177994193e-08, 'reg_lambda':
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7.5808781830812e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:18,438] Trial 682 finished with value:
-1.3106544315012114 and parameters: {'n_estimators': 322, 'learning_rate':
0.02044382930427521, 'max_depth': 12, 'subsample': 0.710495515953315,
'colsample_bytree': 0.7597939361253134, 'min_child_weight': 6, 'gamma':
0.001080399985546679, 'reg_alpha': 3.297797698209234e-08, 'reg_lambda':
0.00014388393739077454}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:18,761] Trial 683 finished with value:
-1.3032434146632295 and parameters: {'n_estimators': 283, 'learning_rate':
0.02262024872741121, 'max_depth': 11, 'subsample': 0.7267201586230843,
'colsample_bytree': 0.7495788038632838, 'min_child_weight': 5, 'gamma':
0.0004956579692292364, 'reg_alpha': 2.0536373302662414e-08, 'reg_lambda':
0.00047928232618934564}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:19,148] Trial 684 finished with value:
-1.3038448912777414 and parameters: {'n_estimators': 358, 'learning_rate':
0.017733749106843076, 'max_depth': 11, 'subsample': 0.6989433612925731,
'colsample_bytree': 0.7455975306556862, 'min_child_weight': 5, 'gamma':
0.0018814171806016424, 'reg_alpha': 1.427842381293157e-08, 'reg_lambda':
0.00023000812971188067}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:19,671] Trial 685 finished with value:
-1.3121201814956516 and parameters: {'n_estimators': 434, 'learning_rate':
0.014958321526162284, 'max_depth': 12, 'subsample': 0.7156564806731558,
'colsample_bytree': 0.8025590988095321, 'min_child_weight': 5, 'gamma':
0.001219280892014928, 'reg_alpha': 4.234886010444693e-08, 'reg_lambda':
9.400912341182517e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:20,522] Trial 686 finished with value:
-1.3259478981217128 and parameters: {'n_estimators': 977, 'learning_rate':
0.01953694328968307, 'max_depth': 11, 'subsample': 0.7246541710583133,
'colsample_bytree': 0.7806473271293337, 'min_child_weight': 5, 'gamma':
0.0005918215230260105, 'reg_alpha': 2.3644916502772603e-08, 'reg_lambda':
5.1703986550861806e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:20,939] Trial 687 finished with value:
-1.3045028383141775 and parameters: {'n_estimators': 337, 'learning_rate':
0.016207153241886933, 'max_depth': 12, 'subsample': 0.7006244662450134,
'colsample_bytree': 0.7662037831258063, 'min_child_weight': 5, 'gamma':
0.000392612681316067, 'reg_alpha': 1.4737908087193394e-08, 'reg_lambda':
0.0008315696725938039}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:21,295] Trial 688 finished with value:
-1.3026718638589978 and parameters: {'n_estimators': 310, 'learning_rate':
0.018277502233810538, 'max_depth': 11, 'subsample': 0.7109998469885832,
'colsample_bytree': 0.7535261539751199, 'min_child_weight': 5, 'gamma':
0.0009331275203178074, 'reg_alpha': 2.6363300181908087e-08, 'reg_lambda':
0.0002980934476296974}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:21,744] Trial 689 finished with value:
-1.3149883359478558 and parameters: {'n_estimators': 369, 'learning_rate':
0.020646281617792306, 'max_depth': 12, 'subsample': 0.7324260142576173,
'colsample_bytree': 0.770855654981562, 'min_child_weight': 5, 'gamma':
0.0031113675262071506, 'reg_alpha': 6.084944419307301e-08, 'reg_lambda':
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0.0004695879510713099}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:22,037] Trial 690 finished with value:
-1.331690115802638 and parameters: {'n_estimators': 280, 'learning_rate':
0.016964495458880395, 'max_depth': 11, 'subsample': 0.6878419194862035,
'colsample_bytree': 0.7891244956162236, 'min_child_weight': 10, 'gamma':
0.0007977300899007552, 'reg_alpha': 1.834372141093992e-08, 'reg_lambda':
0.00012178097775651276}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:22,544] Trial 691 finished with value:
-1.3120766864838886 and parameters: {'n_estimators': 319, 'learning_rate':
0.019132138413446197, 'max_depth': 11, 'subsample': 0.7421134558283398,
'colsample_bytree': 0.7470814956778641, 'min_child_weight': 5, 'gamma':
0.0015183831038903567, 'reg_alpha': 0.054502040862427194, 'reg_lambda':
0.00016856620562820637}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:23,048] Trial 692 finished with value:
-1.3025762493678672 and parameters: {'n_estimators': 296, 'learning_rate':
0.015778593761496405, 'max_depth': 12, 'subsample': 0.722505102765806,
'colsample_bytree': 0.7595649933941202, 'min_child_weight': 5, 'gamma':
0.0005410553769911759, 'reg_alpha': 3.630641628554278e-08, 'reg_lambda':
3.9623592938516994e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:23,545] Trial 693 finished with value:
-1.3072706336351325 and parameters: {'n_estimators': 340, 'learning_rate':
0.01749082870038762, 'max_depth': 10, 'subsample': 0.7037125114786901,
'colsample_bytree': 0.7799189415519449, 'min_child_weight': 5, 'gamma':
0.000387206432154403, 'reg_alpha': 1.3834128870428298e-08, 'reg_lambda':
0.0005320017441961542}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:24,032] Trial 694 finished with value:
-1.3142595139503632 and parameters: {'n_estimators': 445, 'learning_rate':
0.021811182621895138, 'max_depth': 10, 'subsample': 0.7147529173279361,
'colsample_bytree': 0.7458109345448027, 'min_child_weight': 6, 'gamma':
0.0011966260717528653, 'reg_alpha': 2.437851983279449e-08, 'reg_lambda':
8.100252242332763e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:24,425] Trial 695 finished with value:
-1.3139709717654753 and parameters: {'n_estimators': 345, 'learning_rate':
0.019393274659047725, 'max_depth': 10, 'subsample': 0.7495544990159206,
'colsample_bytree': 0.43091138740487867, 'min_child_weight': 5, 'gamma':
0.001918484442549351, 'reg_alpha': 1.4237183742475259e-08, 'reg_lambda':
0.0003544987713911569}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:24,983] Trial 696 finished with value:
-1.3103761303647061 and parameters: {'n_estimators': 367, 'learning_rate':
0.01803454840022937, 'max_depth': 11, 'subsample': 0.6948845369982534,
'colsample_bytree': 0.7681556375109098, 'min_child_weight': 5, 'gamma':
0.0006949088367718017, 'reg_alpha': 3.3424586414443305e-08, 'reg_lambda':
0.0002135113332944592}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:25,410] Trial 697 finished with value:
-1.3132641174692774 and parameters: {'n_estimators': 257, 'learning_rate':
0.015024143161684378, 'max_depth': 10, 'subsample': 0.7331739700340036,
'colsample_bytree': 0.7528838124841429, 'min_child_weight': 5, 'gamma':
0.0009606054664167863, 'reg_alpha': 4.8639553259431234e-08, 'reg_lambda':
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1.1874624137926977e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:25,790] Trial 698 finished with value:
-1.3034487508361718 and parameters: {'n_estimators': 315, 'learning_rate':
0.016682468485667856, 'max_depth': 10, 'subsample': 0.712214215999468,
'colsample_bytree': 0.7421524143984053, 'min_child_weight': 5, 'gamma':
0.0004045801933266704, 'reg_alpha': 1.978445478797827e-08, 'reg_lambda':
0.0008258066412837945}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:26,205] Trial 699 finished with value:
-1.3090409750520386 and parameters: {'n_estimators': 329, 'learning_rate':
0.020824805239425037, 'max_depth': 10, 'subsample': 0.7243792922914091,
'colsample_bytree': 0.8110687801311982, 'min_child_weight': 7, 'gamma':
0.005651324447308952, 'reg_alpha': 1.376490545478731e-08, 'reg_lambda':
6.117461266601862e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:26,651] Trial 700 finished with value:
-1.3037302563322937 and parameters: {'n_estimators': 294, 'learning_rate':
0.01859708866830051, 'max_depth': 10, 'subsample': 0.7047556947571162,
'colsample_bytree': 0.7923917252083011, 'min_child_weight': 5, 'gamma':
0.0006131542625656568, 'reg_alpha': 2.782201269102383e-08, 'reg_lambda':
0.00016078387942609383}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:26,884] Trial 701 finished with value:
-1.3529079571442042 and parameters: {'n_estimators': 349, 'learning_rate':
0.015867555963108595, 'max_depth': 5, 'subsample': 0.6894018848449762,
'colsample_bytree': 0.7650940702077756, 'min_child_weight': 5, 'gamma':
0.0015693833261240487, 'reg_alpha': 1.913421339333537e-08, 'reg_lambda':
4.106276585366079e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:27,444] Trial 702 finished with value:
-1.303070722308417 and parameters: {'n_estimators': 309, 'learning_rate':
0.01721370723012127, 'max_depth': 12, 'subsample': 0.7217646402262593,
'colsample_bytree': 0.7784279654338947, 'min_child_weight': 5, 'gamma':
0.002204696609397434, 'reg_alpha': 4.160400756814625e-08, 'reg_lambda':
0.0003086560948952052}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:27,919] Trial 703 finished with value:
-1.3066785961890715 and parameters: {'n_estimators': 326, 'learning_rate':
0.019549172879338123, 'max_depth': 11, 'subsample': 0.6758057848875191,
'colsample_bytree': 0.754380051499232, 'min_child_weight': 6, 'gamma':
0.00016939493466640456, 'reg_alpha': 1.3601605631936775e-08, 'reg_lambda':
2.3991890144844314e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:28,267] Trial 704 finished with value:
-1.3064038343915776 and parameters: {'n_estimators': 276, 'learning_rate':
0.023516571661029474, 'max_depth': 10, 'subsample': 0.7586188105049452,
'colsample_bytree': 0.7388102601756822, 'min_child_weight': 5, 'gamma':
0.0011659260335972448, 'reg_alpha': 0.5278456685907925, 'reg_lambda':
9.84528022930456e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:28,733] Trial 705 finished with value:
-1.313484527205419 and parameters: {'n_estimators': 372, 'learning_rate':
0.015035276879072529, 'max_depth': 10, 'subsample': 0.9474800070611029,
'colsample_bytree': 0.7703994282597134, 'min_child_weight': 8, 'gamma':
0.000764395733174311, 'reg_alpha': 2.473766911498931e-08, 'reg_lambda':
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1.4854483205369825e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:29,230] Trial 706 finished with value:
-1.3075514651410731 and parameters: {'n_estimators': 341, 'learning_rate':
0.01810478338387917, 'max_depth': 10, 'subsample': 0.7351732200182884,
'colsample_bytree': 0.7559861892003831, 'min_child_weight': 5, 'gamma':
0.00028327873904137544, 'reg_alpha': 1.3629242261220661e-08, 'reg_lambda':
0.0005274442603663015}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:29,771] Trial 707 finished with value:
-1.301408620450733 and parameters: {'n_estimators': 296, 'learning_rate':
0.01629904516178042, 'max_depth': 12, 'subsample': 0.7107610810122195,
'colsample_bytree': 0.7356305583071252, 'min_child_weight': 5, 'gamma':
0.0005207553031101932, 'reg_alpha': 2.0308275430981607e-08, 'reg_lambda':
0.00023173314673231678}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:30,237] Trial 708 finished with value:
-1.3499720065635357 and parameters: {'n_estimators': 356, 'learning_rate':
0.05389872566593263, 'max_depth': 10, 'subsample': 0.7728948838135606,
'colsample_bytree': 0.7972496849043244, 'min_child_weight': 5, 'gamma':
0.003453243720569186 7, 'reg_alpha': 6.587812343643226e-08, 'reg_lambda':
5.2689633788036365e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:30,808] Trial 709 finished with value:
-1.3201571042927613 and parameters: {'n_estimators': 451, 'learning_rate':
0.019639960768635233, 'max_depth': 11, 'subsample': 0.6964294760088475,
'colsample_bytree': 0.7793130886654355, 'min_child_weight': 5, 'gamma':
0.0008405427021891364, 'reg_alpha': 3.486936928447676e-08, 'reg_lambda':
0.0001365439399578754}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:31,452] Trial 710 finished with value:
-1.327089384900719 and parameters: {'n_estimators': 431, 'learning_rate':
0.021403798471997942, 'max_depth': 10, 'subsample': 0.5513677010934668,
'colsample_bytree': 0.745961527798535, 'min_child_weight': 2, 'gamma':
0.00036007927473696166, 'reg_alpha': 1.0125970866481358e-08, 'reg_lambda':
5.402544716159241e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:31,954] Trial 711 finished with value:
-1.3035008721406809 and parameters: {'n_estimators': 316, 'learning_rate':
0.014665325764787287, 'max_depth': 10, 'subsample': 0.7250275379635825,
'colsample_bytree': 0.7627155063700842, 'min_child_weight': 5, 'gamma':
0.0012403990511774016, 'reg_alpha': 1.8285134501213513e-08, 'reg_lambda':
0.0008109600536131548}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:32,420] Trial 712 finished with value:
-1.3090240107935691 and parameters: {'n_estimators': 335, 'learning_rate':
0.017802954088242288, 'max_depth': 10, 'subsample': 0.7418093087557478,
'colsample_bytree': 0.7386576914881093, 'min_child_weight': 5, 'gamma':
0.0005782961275714281, 'reg_alpha': 3.120909828501697e-08, 'reg_lambda':
7.144101996457072e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:32,818] Trial 713 finished with value:
-1.3163292315621888 and parameters: {'n_estimators': 282, 'learning_rate':
0.01685605823408062, 'max_depth': 11, 'subsample': 0.7035990834884379,
'colsample_bytree': 0.46354323599832203, 'min_child_weight': 5, 'gamma':
0.0023829683363473316, 'reg_alpha': 4.794814074773371e-08, 'reg_lambda':
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0.00038467842688111423}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:33,294] Trial 714 finished with value:
-1.3096647556631191 and parameters: {'n_estimators': 237, 'learning_rate':
0.015858856097583973, 'max_depth': 12, 'subsample': 0.7188653223252555,
'colsample_bytree': 0.8220724260838841, 'min_child_weight': 5, 'gamma':
0.0014030780823148538, 'reg_alpha': 1.4272894604147135e-08, 'reg_lambda':
4.025785190292894e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:33,560] Trial 715 finished with value:
-1.3252723347617381 and parameters: {'n_estimators': 378, 'learning_rate':
0.019074852277551062, 'max_depth': 6, 'subsample': 0.7120913742454614,
'colsample_bytree': 0.7548110090651944, 'min_child_weight': 6, 'gamma':
0.00076436747499123, 'reg_alpha': 2.3009957719148093e-08, 'reg_lambda':
7.821924189710174e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:33,710] Trial 716 finished with value:
-1.421071200708133 and parameters: {'n_estimators': 323, 'learning_rate':
0.017314461758677055, 'max_depth': 3, 'subsample': 0.7310726962249053,
'colsample_bytree': 0.7862280723491674, 'min_child_weight': 5, 'gamma':
0.0003994518530480377, 'reg_alpha': 1.022697223768828e-08, 'reg_lambda':
0.0006126381699437394}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:34,157] Trial 717 finished with value:
-1.308453927030999 and parameters: {'n_estimators': 305, 'learning_rate':
0.014820996697807688, 'max_depth': 10, 'subsample': 0.6848749309314963,
'colsample_bytree': 0.7673064169270036, 'min_child_weight': 5, 'gamma':
0.0009455480142217374, 'reg_alpha': 1.8026213473561626e-08, 'reg_lambda':
9.004968206029255e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:34,666] Trial 718 finished with value:
-1.3096679960190445 and parameters: {'n_estimators': 358, 'learning_rate':
0.018880124627719593, 'max_depth': 10, 'subsample': 0.7025712490139572,
'colsample_bytree': 0.7482723053291442, 'min_child_weight': 5, 'gamma':
9.773022992635496e-05, 'reg_alpha': 2.7813550664361053e-08, 'reg_lambda':
2.8287352287012864e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:34,938] Trial 719 finished with value:
-1.4355280234179162 and parameters: {'n_estimators': 266, 'learning_rate':
0.21900940155754725, 'max_depth': 8, 'subsample': 0.7504100659992804,
'colsample_bytree': 0.7744379949581799, 'min_child_weight': 5, 'gamma':
0.0002311954371051382, 'reg_alpha': 4.8986250403295355e-08, 'reg_lambda':
0.00024284213540842937}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:35,393] Trial 720 finished with value:
-1.3125339899327555 and parameters: {'n_estimators': 346, 'learning_rate':
0.020243482058591812, 'max_depth': 10, 'subsample': 0.7162054884508846,
'colsample_bytree': 0.7343225998905428, 'min_child_weight': 5, 'gamma':
0.001805684907855489, 'reg_alpha': 1.3911743936848031e-08, 'reg_lambda':
0.0001383934893285636}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:35,886] Trial 721 finished with value:
-1.3093163988361682 and parameters: {'n_estimators': 304, 'learning_rate':
0.01632898372743385, 'max_depth': 11, 'subsample': 0.7378980414520335,
'colsample_bytree': 0.7602533707088639, 'min_child_weight': 5, 'gamma':
0.0005069884330833936, 'reg_alpha': 3.301481479780906e-08, 'reg_lambda':
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0.0003910133232516269}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:36,341] Trial 722 finished with value:
-1.300406639815941 and parameters: {'n_estimators': 320, 'learning_rate':
0.018098151374780626, 'max_depth': 10, 'subsample': 0.723621033868647,
'colsample_bytree': 0.745861926110717, 'min_child_weight': 5, 'gamma':
0.0010970782739644954, 'reg_alpha': 0.025102888208162175, 'reg_lambda':
0.00018067492098381322}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:36,841] Trial 723 finished with value:
-1.306373840143772 and parameters: {'n_estimators': 330, 'learning_rate':
0.017788758164516007, 'max_depth': 10, 'subsample': 0.727929989289438,
'colsample_bytree': 0.7877746325703129, 'min_child_weight': 5, 'gamma':
0.0010608469979484178, 'reg_alpha': 0.35307847586515717, 'reg_lambda':
0.0002693474668209363}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:37,250] Trial 724 finished with value:
-1.3026982194765135 and parameters: {'n_estimators': 289, 'learning_rate':
0.021162077887890523, 'max_depth': 10, 'subsample': 0.7203748384622358,
'colsample_bytree': 0.7315830259157362, 'min_child_weight': 5, 'gamma':
0.0016277681111723078, 'reg_alpha': 0.02571521744457064, 'reg_lambda':
0.0005717380567386534}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:37,732] Trial 725 finished with value:
-1.309926196111153 and parameters: {'n_estimators': 323, 'learning_rate':
0.018457529915861366, 'max_depth': 10, 'subsample': 0.7446221366515452,
'colsample_bytree': 0.7464506011016983, 'min_child_weight': 5, 'gamma':
0.002617654386716232, 'reg_alpha': 3.833045302497472e-05, 'reg_lambda':
0.00020171220115518137}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:38,162] Trial 726 finished with value:
-1.304892976766262 and parameters: {'n_estimators': 312, 'learning_rate':
0.019859255497467792, 'max_depth': 10, 'subsample': 0.7102365645609215,
'colsample_bytree': 0.80086434583473, 'min_child_weight': 5, 'gamma':
0.0006968222307658328, 'reg_alpha': 0.01196820808004448, 'reg_lambda':
2.839263218716986e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:38,627] Trial 727 finished with value:
-1.3081518236371084 and parameters: {'n_estimators': 348, 'learning_rate':
0.02252844648291573, 'max_depth': 10, 'subsample': 0.7307774230662234,
'colsample_bytree': 0.7721213035970752, 'min_child_weight': 5, 'gamma':
0.0010160685874081208, 'reg_alpha': 2.0090552429382697e-08, 'reg_lambda':
0.0009049533995426444}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:39,148] Trial 728 finished with value:
-1.3058827804787407 and parameters: {'n_estimators': 389, 'learning_rate':
0.01701078547805745, 'max_depth': 10, 'subsample': 0.698253027322854,
'colsample_bytree': 0.758392907695477, 'min_child_weight': 5, 'gamma':
0.0013618104202351182, 'reg_alpha': 1.3221312452563639e-08, 'reg_lambda':
2.9687452065048644e-07}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:39,590] Trial 729 finished with value:
-1.3538454073407724 and parameters: {'n_estimators': 367, 'learning_rate':
0.06567082563952797, 'max_depth': 11, 'subsample': 0.7138128548190721,
'colsample_bytree': 0.7411799619827821, 'min_child_weight': 6, 'gamma':
0.0006547597352955794, 'reg_alpha': 6.30736078544194e-06, 'reg_lambda':
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0.0003419723084281337}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:40,044] Trial 730 finished with value:
-1.297577134744733 and parameters: {'n_estimators': 296, 'learning_rate':
0.018448917793766513, 'max_depth': 10, 'subsample': 0.7257295594248939,
'colsample_bytree': 0.7649916489925863, 'min_child_weight': 5, 'gamma':
0.0019707578219981714, 'reg_alpha': 0.018570053002780775, 'reg_lambda':
1.56393752291065e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:40,437] Trial 731 finished with value:
-1.3063738606706894 and parameters: {'n_estimators': 246, 'learning_rate':
0.02045611032718273, 'max_depth': 10, 'subsample': 0.7348376684153074,
'colsample_bytree': 0.7783929672360006, 'min_child_weight': 5, 'gamma':
0.004560908080404672, 'reg_alpha': 2.5055469521317746e-08, 'reg_lambda':
1.768468540133802e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:40,859] Trial 732 finished with value:
-1.3047468859011726 and parameters: {'n_estimators': 301, 'learning_rate':
0.018602427223845946, 'max_depth': 10, 'subsample': 0.7385720869769126,
'colsample_bytree': 0.7665179391263828, 'min_child_weight': 5, 'gamma':
0.003298777080141394, 'reg_alpha': 0.0366975134373881, 'reg_lambda':
2.4027597188293383e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:41,275] Trial 733 finished with value:
-1.3043892113093563 and parameters: {'n_estimators': 268, 'learning_rate':
0.019032853957406892, 'max_depth': 10, 'subsample': 0.7232477137644101,
'colsample_bytree': 0.7867709118020093, 'min_child_weight': 5, 'gamma':
0.0018891565676280627, 'reg_alpha': 3.793482319740488e-08, 'reg_lambda':
9.518920324760163e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:41,685] Trial 734 finished with value:
-1.3008285016491614 and parameters: {'n_estimators': 279, 'learning_rate':
0.02554327879788497, 'max_depth': 10, 'subsample': 0.7596628003452877,
'colsample_bytree': 0.7605693524703415, 'min_child_weight': 5, 'gamma':
0.002322458222295009, 'reg_alpha': 6.356104557834536e-08, 'reg_lambda':
1.1477708278827966e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:42,035] Trial 735 finished with value:
-1.322608848703705 and parameters: {'n_estimators': 268, 'learning_rate':
0.0325152276040246, 'max_depth': 10, 'subsample': 0.7850605099524647,
'colsample_bytree': 0.7547832801442949, 'min_child_weight': 5, 'gamma':
0.0034786325478222985, 'reg_alpha': 0.014766378332812647, 'reg_lambda':
1.009626012240354e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:42,374] Trial 736 finished with value:
-1.3090731914617182 and parameters: {'n_estimators': 252, 'learning_rate':
0.02205633602531123, 'max_depth': 10, 'subsample': 0.74771726515294,
'colsample_bytree': 0.7580411583221318, 'min_child_weight': 5, 'gamma':
0.005705241677619333, 'reg_alpha': 6.04824742917991e-08, 'reg_lambda':
6.449711153254022e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:42,759] Trial 737 finished with value:
-1.3063917710654136 and parameters: {'n_estimators': 279, 'learning_rate':
0.024693083618122797, 'max_depth': 10, 'subsample': 0.7637344865904576,
'colsample_bytree': 0.763579184017937, 'min_child_weight': 6, 'gamma':
0.002564128788499646, 'reg_alpha': 0.02734843567717184, 'reg_lambda':
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6.452809739418649e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:43,186] Trial 738 finished with value:
-1.3095825499842741 and parameters: {'n_estimators': 288, 'learning_rate':
0.021375619542952955, 'max_depth': 10, 'subsample': 0.7596488905099833,
'colsample_bytree': 0.7502798109759217, 'min_child_weight': 5, 'gamma':
0.002427861404086009, 'reg_alpha': 0.022653755662072887, 'reg_lambda':
1.7507379490544004e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:43,709] Trial 739 finished with value:
-1.3269259426246995 and parameters: {'n_estimators': 288, 'learning_rate':
0.020433489488689702, 'max_depth': 10, 'subsample': 0.9986066249852572,
'colsample_bytree': 0.7767535205732226, 'min_child_weight': 5, 'gamma':
0.00458664665366841, 'reg_alpha': 0.022964539861452256, 'reg_lambda':
1.0714596535901521e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:44,124] Trial 740 finished with value:
-1.3118801125165458 and parameters: {'n_estimators': 270, 'learning_rate':
0.01920472662991484, 'max_depth': 10, 'subsample': 0.7491552864699867,
'colsample_bytree': 0.7478267199101483, 'min_child_weight': 5, 'gamma':
0.0020013916849911098, 'reg_alpha': 0.012380326509667772, 'reg_lambda':
1.3683623634188079e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:44,486] Trial 741 finished with value:
-1.309446068000281 and parameters: {'n_estimators': 240, 'learning_rate':
0.022504744659947895, 'max_depth': 10, 'subsample': 0.7468747543858947,
'colsample_bytree': 0.764688473677572, 'min_child_weight': 5, 'gamma':
0.0031209273826991873, 'reg_alpha': 0.05179596533214309, 'reg_lambda':
1.5163723860928235e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:44,903] Trial 742 finished with value:
-1.3017718973875938 and parameters: {'n_estimators': 300, 'learning_rate':
0.02415687509589046, 'max_depth': 10, 'subsample': 0.7595157829366872,
'colsample_bytree': 0.7380339538023492, 'min_child_weight': 5, 'gamma':
0.00753001001476221, 'reg_alpha': 6.893464610994148e-08, 'reg_lambda':
4.214521239979414e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:45,337] Trial 743 finished with value:
-1.3129569145766065 and parameters: {'n_estimators': 283, 'learning_rate':
0.020217874769769863, 'max_depth': 10, 'subsample': 0.7680819584154258,
'colsample_bytree': 0.7904995375971474, 'min_child_weight': 5, 'gamma':
0.0016413003534171348, 'reg_alpha': 0.1689945329247875, 'reg_lambda':
1.427138347624196e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:45,731] Trial 744 finished with value:
-1.315700157426744 and parameters: {'n_estimators': 301, 'learning_rate':
0.03740918615462347, 'max_depth': 10, 'subsample': 0.7379216470935381,
'colsample_bytree': 0.7558295075089522, 'min_child_weight': 5, 'gamma':
0.0019663104005398246, 'reg_alpha': 4.7419102923714756e-08, 'reg_lambda':
8.967782578085738e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:46,294] Trial 745 finished with value:
-1.3185809258377865 and parameters: {'n_estimators': 421, 'learning_rate':
0.02600252734567505, 'max_depth': 10, 'subsample': 0.7554232217196662,
'colsample_bytree': 0.7720755805738728, 'min_child_weight': 5, 'gamma':
0.001235500211968627 2, 'reg_alpha': 0.06279039871659385, 'reg_lambda':
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7.576102571165125e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:46,842] Trial 746 finished with value:
-1.3143453982956494 and parameters: {'n_estimators': 308, 'learning_rate':
0.01824694269549807, 'max_depth': 10, 'subsample': 0.7701063049144714,
'colsample_bytree': 0.9729655463184976, 'min_child_weight': 5, 'gamma':
0.0034697183051209824, 'reg_alpha': 0.11627752569714277, 'reg_lambda':
1.2736046536092792e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:47,223] Trial 747 finished with value:
-1.3144553226310565 and parameters: {'n_estimators': 273, 'learning_rate':
0.02944554156168134, 'max_depth': 10, 'subsample': 0.6910738510003714,
'colsample_bytree': 0.8074500611690163, 'min_child_weight': 5, 'gamma':
0.0014844119527946092, 'reg_alpha': 0.07559641167664774, 'reg_lambda':
2.3849678311869467e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:47,546] Trial 748 finished with value:
-1.3078821028543823 and parameters: {'n_estimators': 258, 'learning_rate':
0.023059759913727936, 'max_depth': 10, 'subsample': 0.7042954322404592,
'colsample_bytree': 0.7467565395964091, 'min_child_weight': 6, 'gamma':
0.0009232490920075373, 'reg_alpha': 0.018766733437791096, 'reg_lambda':
1.867975574563178e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:48,188] Trial 749 finished with value:
-1.3312275470367985 and parameters: {'n_estimators': 453, 'learning_rate':
0.021108886663407798, 'max_depth': 10, 'subsample': 0.7934870192171631,
'colsample_bytree': 0.8705838552567008, 'min_child_weight': 5, 'gamma':
0.0013650392849422794, 'reg_alpha': 0.017093034774340043, 'reg_lambda':
1.2517787964057295e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:48,622] Trial 750 finished with value:
-1.3076526992021427 and parameters: {'n_estimators': 322, 'learning_rate':
0.028312234752811475, 'max_depth': 10, 'subsample': 0.7276793998868357,
'colsample_bytree': 0.7799183796760133, 'min_child_weight': 5, 'gamma':
0.0008716187991643149, 'reg_alpha': 3.715649127810468e-08, 'reg_lambda':
6.6419055034253436e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:49,059] Trial 751 finished with value:
-1.3030649795554954 and parameters: {'n_estimators': 289, 'learning_rate':
0.0179584582741824, 'max_depth': 11, 'subsample': 0.7169256764035713,
'colsample_bytree': 0.7627714526784201, 'min_child_weight': 5, 'gamma':
0.002136444172883648, 'reg_alpha': 7.235290825863698e-05, 'reg_lambda':
2.3694922964359732e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:49,481] Trial 752 finished with value:
-1.3076887741744248 and parameters: {'n_estimators': 304, 'learning_rate':
0.025633606270393683, 'max_depth': 10, 'subsample': 0.7758288932509281,
'colsample_bytree': 0.7396094930117045, 'min_child_weight': 5, 'gamma':
0.0006256180378677212, 'reg_alpha': 0.009437285325082676, 'reg_lambda':
4.876987997970502e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:49,924] Trial 753 finished with value:
-1.3191678191547445 and parameters: {'n_estimators': 313, 'learning_rate':
0.019485462044951513, 'max_depth': 10, 'subsample': 0.8902489417189212,
'colsample_bytree': 0.7566121537155771, 'min_child_weight': 5, 'gamma':
0.0009915504356070891, 'reg_alpha': 7.586672917305203e-08, 'reg_lambda':
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0.00043248506378994395}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:50,420] Trial 754 finished with value: -1.3064720138171955 and parameters: {'n_estimators': 333, 'learning_rate': 0.017839773996465, 'max_depth': 10, 'subsample': 0.7359435732685286, 'colsample_bytree': 0.7728138523246467, 'min_child_weight': 5, 'gamma': 0.0023051112175040307, 'reg_alpha': 1.0315864424198316e-08, 'reg_lambda': 1.2097860287887473e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:50,843] Trial 755 finished with value: -1.3048053805661102 and parameters: {'n_estimators': 281, 'learning_rate': 0.01951960713395489, 'max_depth': 11, 'subsample': 0.701860013570614, 'colsample_bytree': 0.7305662970172982, 'min_child_weight': 5, 'gamma': 0.0012734065233403142, 'reg_alpha': 0.005789218827799502, 'reg_lambda': 0.0008595389969316696}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:51,353] Trial 756 finished with value: -1.3168230735358857 and parameters: {'n_estimators': 427, 'learning_rate': 0.02113262518151053, 'max_depth': 10, 'subsample': 0.7264142276218616, 'colsample_bytree': 0.7489047330072278, 'min_child_weight': 6, 'gamma': 0.0007327391512102279, 'reg_alpha': 0.26276348831843643, 'reg_lambda': 2.0279370181510787e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:51,808] Trial 757 finished with value: -1.308931879877034 and parameters: {'n_estimators': 318, 'learning_rate': 0.018550510693771987, 'max_depth': 10, 'subsample': 0.6913412923149702, 'colsample_bytree': 0.7937278102758665, 'min_child_weight': 5, 'gamma': 0.00046635857801159177, 'reg_alpha': 5.710253426317569e-08, 'reg_lambda': 8.849069044155421e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:52,209] Trial 758 finished with value: -1.3082132585660353 and parameters: {'n_estimators': 257, 'learning_rate': 0.017305597290361446, 'max_depth': 10, 'subsample': 0.7083389044767946, 'colsample_bytree': 0.8331909083032889, 'min_child_weight': 5, 'gamma': 0.002882045219110933, 'reg_alpha': 0.024702248838431955, 'reg_lambda': 0.0005985170980575173}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:52,665] Trial 759 finished with value: -1.3090199174187644 and parameters: {'n_estimators': 293, 'learning_rate': 0.019207677941535974, 'max_depth': 10, 'subsample': 0.7453238375525649, 'colsample_bytree': 0.7659594564993566, 'min_child_weight': 5, 'gamma': 0.0018808827033051018, 'reg_alpha': 2.9710628902332407e-08, 'reg_lambda': 0.00033479106495403203}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:53,182] Trial 760 finished with value: -1.3144089163081165 and parameters: {'n_estimators': 335, 'learning_rate': 0.017403386488014302, 'max_depth': 10, 'subsample': 0.6797741237729158, 'colsample_bytree': 0.7786804921604324, 'min_child_weight': 5, 'gamma': 0.003940406583974936, 'reg_alpha': 0.033956892082802864, 'reg_lambda': 2.8960951373101935e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:53,469] Trial 761 finished with value: -1.3279908039750075 and parameters: {'n_estimators': 140, 'learning_rate': 0.020625004843888305, 'max_depth': 11, 'subsample': 0.7200529409674451, 'colsample_bytree': 0.7446230937881995, 'min_child_weight': 5, 'gamma': 0.0009298037431922764, 'reg_alpha': 1.63701730830996e-08, 'reg_lambda':

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0.00046305999672565485}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:54,034] Trial 762 finished with value:
-1.30475521422932 and parameters: {'n_estimators': 445, 'learning_rate':
0.01660149819690962, 'max_depth': 10, 'subsample': 0.8150586088953367,
'colsample_bytree': 0.7571889095081202, 'min_child_weight': 6, 'gamma':
0.0016209542808249223, 'reg_alpha': 3.9590585797320766e-08, 'reg_lambda':
3.1569497685993846e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:54,465] Trial 763 finished with value:
-1.3056987894002852 and parameters: {'n_estimators': 320, 'learning_rate':
0.022977202302828327, 'max_depth': 10, 'subsample': 0.7365888967533052,
'colsample_bytree': 0.7322023957774797, 'min_child_weight': 5, 'gamma':
0.00048027690308692357, 'reg_alpha': 0.04975505882717662, 'reg_lambda':
0.0010954246222178755}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:54,865] Trial 764 finished with value:
-1.3044989096388842 and parameters: {'n_estimators': 301, 'learning_rate':
0.01841893299415947, 'max_depth': 10, 'subsample': 0.7081731107556591,
'colsample_bytree': 0.7870114432050294, 'min_child_weight': 5, 'gamma':
0.0007124810227847933, 'reg_alpha': 2.9481864877791075e-06, 'reg_lambda':
0.00030069378895833724}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:55,362] Trial 765 finished with value:
-1.3100097660324748 and parameters: {'n_estimators': 339, 'learning_rate':
0.019310188141270664, 'max_depth': 10, 'subsample': 0.6962194553560767,
'colsample_bytree': 0.765346081705857, 'min_child_weight': 5, 'gamma':
0.0011207488252878942, 'reg_alpha': 2.056828589449918e-08, 'reg_lambda':
0.0007398618027094721}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:55,865] Trial 766 finished with value:
-1.311409144833646 and parameters: {'n_estimators': 390, 'learning_rate':
0.017616500137810794, 'max_depth': 10, 'subsample': 0.7283559057767311,
'colsample_bytree': 0.7416980766892801, 'min_child_weight': 5, 'gamma':
0.0005774454411264904, 'reg_alpha': 1.0019615340068779e-08, 'reg_lambda':
0.00021067477807354195}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:56,294] Trial 767 finished with value:
-1.3226113507043233 and parameters: {'n_estimators': 272, 'learning_rate':
0.020193920247652923, 'max_depth': 11, 'subsample': 0.9672087428472389,
'colsample_bytree': 0.753256328223791, 'min_child_weight': 5, 'gamma':
0.0014605732752147134, 'reg_alpha': 0.03455764817941927, 'reg_lambda':
2.033906255534408e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:56,779] Trial 768 finished with value:
-1.3058204542301846 and parameters: {'n_estimators': 311, 'learning_rate':
0.016184190716353825, 'max_depth': 10, 'subsample': 0.7170953834498839,
'colsample_bytree': 0.7784721127257551, 'min_child_weight': 5, 'gamma':
0.000355173545062277, 'reg_alpha': 8.894599579569858e-06, 'reg_lambda':
5.355894855565249e-07}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:57,307] Trial 769 finished with value:
-1.3019444854410682 and parameters: {'n_estimators': 366, 'learning_rate':
0.018554897621307638, 'max_depth': 10, 'subsample': 0.7064569869736581,
'colsample_bytree': 0.7972296554538827, 'min_child_weight': 5, 'gamma':
0.0024750435690879535, 'reg_alpha': 0.0412281346045638, 'reg_lambda':
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0.00044020041540346664}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:57,612] Trial 770 finished with value:
-1.310871868628903 and parameters: {'n_estimators': 229, 'learning_rate':
0.022115406768038665, 'max_depth': 11, 'subsample': 0.7288299228668639,
'colsample_bytree': 0.7305540707518059, 'min_child_weight': 5, 'gamma':
0.0008962481307416998, 'reg_alpha': 2.476927024946732e-08, 'reg_lambda':
3.348801339519887e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:58,019] Trial 771 finished with value:
-1.3131357075830519 and parameters: {'n_estimators': 296, 'learning_rate':
0.017050211255306494, 'max_depth': 10, 'subsample': 0.7545028028904073,
'colsample_bytree': 0.7586308112171533, 'min_child_weight': 6, 'gamma':
0.0013244375228961573, 'reg_alpha': 0.016253028107996108, 'reg_lambda':
1.536954937277449e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:58,561] Trial 772 finished with value:
-1.3079469176080867 and parameters: {'n_estimators': 412, 'learning_rate':
0.019788543369791148, 'max_depth': 10, 'subsample': 0.7184959022074369,
'colsample_bytree': 0.7684814867251175, 'min_child_weight': 5, 'gamma':
0.0006409395351165802, 'reg_alpha': 1.4268757046976367e-08, 'reg_lambda':
2.3623926857869378e-07}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:14:59,612] Trial 773 finished with value:
-1.323622665165601 and parameters: {'n_estimators': 938, 'learning_rate':
0.01811036364707669, 'max_depth': 10, 'subsample': 0.6863552261233745,
'colsample_bytree': 0.741848681233457, 'min_child_weight': 5, 'gamma':
0.005074207940914392, 'reg_alpha': 5.3435719432098045e-08, 'reg_lambda':
0.0002708653564232664}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:15:00,166] Trial 774 finished with value:
-1.3151492549984278 and parameters: {'n_estimators': 326, 'learning_rate':
0.01612900332842242, 'max_depth': 11, 'subsample': 0.7369435650276619,
'colsample_bytree': 0.7500983633419569, 'min_child_weight': 5, 'gamma':
0.00044471986792472536, 'reg_alpha': 1.491112453710028e-06, 'reg_lambda':
0.0001768896568842284}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:15:00,671] Trial 775 finished with value:
-1.3114390050856826 and parameters: {'n_estimators': 356, 'learning_rate':
0.021507157175068226, 'max_depth': 10, 'subsample': 0.7017914310089968,
'colsample_bytree': 0.7814671507657515, 'min_child_weight': 5, 'gamma':
0.0020128117586636378, 'reg_alpha': 3.0455473440263836e-08, 'reg_lambda':
0.0006600932724377289}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:15:01,167] Trial 776 finished with value:
-1.310241172408853 and parameters: {'n_estimators': 339, 'learning_rate':
0.01743032982706061, 'max_depth': 10, 'subsample': 0.713185838166702,
'colsample_bytree': 0.8110141638876871, 'min_child_weight': 5, 'gamma':
0.0009732682612084766, 'reg_alpha': 1.9028037675687055e-08, 'reg_lambda':
0.0003958178908548112}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:15:01,435] Trial 777 finished with value:
-1.3211184955958575 and parameters: {'n_estimators': 287, 'learning_rate':
0.019749868910653084, 'max_depth': 7, 'subsample': 0.7449002951444132,
'colsample_bytree': 0.7293231722769106, 'min_child_weight': 5, 'gamma':
0.0006352200261861002, 'reg_alpha': 1.4529950745680291e-08, 'reg_lambda':
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4.742020508185265e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:15:02,051] Trial 778 finished with value:
-1.3159214197382345 and parameters: {'n_estimators': 437, 'learning_rate':
0.016249601850779498, 'max_depth': 10, 'subsample': 0.7244864712299217,
'colsample_bytree': 0.8898327228368532, 'min_child_weight': 5, 'gamma':
0.0027567659818988365, 'reg_alpha': 1.000874282632732e-08, 'reg_lambda':
0.0011675557211101032}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:15:02,475] Trial 779 finished with value:
-1.2999470253489558 and parameters: {'n_estimators': 309, 'learning_rate':
0.017749181132299705, 'max_depth': 10, 'subsample': 0.6991444243079614,
'colsample_bytree': 0.7640191817387754, 'min_child_weight': 5, 'gamma':
0.0015820128070622842, 'reg_alpha': 0.008613526878380294, 'reg_lambda':
3.951166499679808e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:15:02,815] Trial 780 finished with value:
-1.3667948205406004 and parameters: {'n_estimators': 271, 'learning_rate':
0.11386817868794101, 'max_depth': 10, 'subsample': 0.6778758160827137,
'colsample_bytree': 0.7710172868393848, 'min_child_weight': 5, 'gamma':
0.0016144278983317045, 'reg_alpha': 0.021141291111214232, 'reg_lambda':
4.109045234574779e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:15:03,208] Trial 781 finished with value:
-1.3189728056954433 and parameters: {'n_estimators': 259, 'learning_rate':
0.015568753751533685, 'max_depth': 10, 'subsample': 0.6895776239900315,
'colsample_bytree': 0.7851009131930493, 'min_child_weight': 6, 'gamma':
0.003390073699526616, 'reg_alpha': 0.006025992666307816, 'reg_lambda':
4.0164422028093804e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:15:03,645] Trial 782 finished with value:
-1.3055124040828028 and parameters: {'n_estimators': 307, 'learning_rate':
0.016644023277086046, 'max_depth': 10, 'subsample': 0.6943058130801053,
'colsample_bytree': 0.7654466464149408, 'min_child_weight': 5, 'gamma':
0.0019843159804154484, 'reg_alpha': 0.009217087587134506, 'reg_lambda':
2.7067257503114694e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:15:04,049] Trial 783 finished with value:
-1.3041967374277537 and parameters: {'n_estimators': 286, 'learning_rate':
0.01788965592692049, 'max_depth': 10, 'subsample': 0.6992080230863723,
'colsample_bytree': 0.7738758937066957, 'min_child_weight': 5, 'gamma':
0.0013501654213741315, 'reg_alpha': 0.018577815958519477, 'reg_lambda':
1.011810530471294e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:15:04,511] Trial 784 finished with value:
-1.313534544217065 and parameters: {'n_estimators': 311, 'learning_rate':
0.017100273609286027, 'max_depth': 10, 'subsample': 0.6825647936504221,
'colsample_bytree': 0.7944510835277478, 'min_child_weight': 5, 'gamma':
0.0025371163062703353, 'reg_alpha': 4.530181335655825e-08, 'reg_lambda':
4.253180731624158e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:15:04,841] Trial 785 finished with value:
-1.3149540994100632 and parameters: {'n_estimators': 205, 'learning_rate':
0.01838276249360974, 'max_depth': 10, 'subsample': 0.6962803542493959,
'colsample_bytree': 0.7598517861207952, 'min_child_weight': 5, 'gamma':
0.0010410722323660401, 'reg_alpha': 0.02127050512706799, 'reg_lambda':
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2.5475531245945142e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:15:05,437] Trial 786 finished with value:
-1.3056529067075913 and parameters: {'n_estimators': 381, 'learning_rate':
0.01560119745995189, 'max_depth': 11, 'subsample': 0.7083319444416961,
'colsample_bytree': 0.7766895756501849, 'min_child_weight': 5, 'gamma':
0.0013469091573253336, 'reg_alpha': 2.6438911391113047e-05, 'reg_lambda':
2.3134121956358683e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:15:05,839] Trial 787 finished with value:
-1.3035655881384665 and parameters: {'n_estimators': 285, 'learning_rate':
0.016806139307815274, 'max_depth': 10, 'subsample': 0.702330159633901,
'colsample_bytree': 0.755780318254587, 'min_child_weight': 5, 'gamma':
0.003846913540354942, 'reg_alpha': 0.02986721763251037, 'reg_lambda':
1.6313165815620228e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:15:06,209] Trial 788 finished with value:
-1.3152864113720006 and parameters: {'n_estimators': 244, 'learning_rate':
0.018486505890321345, 'max_depth': 10, 'subsample': 0.6766045943557176,
'colsample_bytree': 0.8030752818452994, 'min_child_weight': 5, 'gamma':
0.0016692614935169112, 'reg_alpha': 0.039259809698469116, 'reg_lambda':
3.0700697158613744e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:15:06,714] Trial 789 finished with value:
-1.3062229550665005 and parameters: {'n_estimators': 316, 'learning_rate':
0.017625076063017633, 'max_depth': 11, 'subsample': 0.6871115389256577,
'colsample_bytree': 0.7660108175505645, 'min_child_weight': 5, 'gamma':
0.0008892731090531673, 'reg_alpha': 0.011663036830161039, 'reg_lambda':
1.8549411528677595e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:15:07,221] Trial 790 finished with value:
-1.3582293265693341 and parameters: {'n_estimators': 457, 'learning_rate':
0.04889657425444977, 'max_depth': 10, 'subsample': 0.7141375433562166,
'colsample_bytree': 0.78951462668547, 'min_child_weight': 5, 'gamma':
0.002408341542510339, 'reg_alpha': 3.650278172112199e-08, 'reg_lambda':
5.46343372458695e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:15:07,667] Trial 791 finished with value:
-1.3070057459218156 and parameters: {'n_estimators': 342, 'learning_rate':
0.015145242488381451, 'max_depth': 10, 'subsample': 0.7054921152346106,
'colsample_bytree': 0.7522760240346237, 'min_child_weight': 6, 'gamma':
0.007383306078244069, 'reg_alpha': 6.813278534658572e-08, 'reg_lambda':
4.016258899819354e-05}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:15:08,204] Trial 792 finished with value:
-1.312795561567532 and parameters: {'n_estimators': 422, 'learning_rate':
0.01886761541823577, 'max_depth': 10, 'subsample': 0.6910529146473428,
'colsample_bytree': 0.7765322060468127, 'min_child_weight': 5, 'gamma':
0.0011523466169078944, 'reg_alpha': 2.8774987984876452e-08, 'reg_lambda':
9.952979314242812e-06}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:15:08,677] Trial 793 finished with value:
-1.3102478055617697 and parameters: {'n_estimators': 300, 'learning_rate':
0.01626446206627939, 'max_depth': 10, 'subsample': 0.7804254579709111,
'colsample_bytree': 0.7613245481599181, 'min_child_weight': 5, 'gamma':
0.0017372505329575443, 'reg_alpha': 0.008571453799310706, 'reg_lambda':
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0.0006058615375645021}. Best is trial 519 with value: -1.2974087478779242.
[I 2026-03-13 19:15:09,172] Trial 794 finished with value:
-1.297373640685575 and parameters: {'n_estimators': 328, 'learning_rate':
0.017436092070699592, 'max_depth': 10, 'subsample': 0.7098597504900896,
'colsample_bytree': 0.748950604387795, 'min_child_weight': 5, 'gamma':
0.0009244441393580168, 'reg_alpha': 0.0153085974034161, 'reg_lambda':
0.0009952136009229449}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:10,150] Trial 795 finished with value:
-1.3185149154052882 and parameters: {'n_estimators': 869, 'learning_rate':
0.014838449949563667, 'max_depth': 10, 'subsample': 0.716225599356777,
'colsample_bytree': 0.7415626231877815, 'min_child_weight': 5, 'gamma':
0.0019930195497527762, 'reg_alpha': 0.016748143257026785, 'reg_lambda':
0.0014407640572762761}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:10,774] Trial 796 finished with value:
-1.3062407059012877 and parameters: {'n_estimators': 397, 'learning_rate':
0.016728066398098925, 'max_depth': 11, 'subsample': 0.7158216169356914,
'colsample_bytree': 0.7420129550080152, 'min_child_weight': 5, 'gamma':
0.004827941567800575, 'reg_alpha': 0.017202187503585566, 'reg_lambda':
0.0008869548572036947}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:11,276] Trial 797 finished with value:
-1.3108339789764478 and parameters: {'n_estimators': 364, 'learning_rate':
0.015993009598588306, 'max_depth': 10, 'subsample': 0.7297218544587306,
'colsample_bytree': 0.7510426073429586, 'min_child_weight': 6, 'gamma':
0.21827695556597543, 'reg_alpha': 0.04510550845537647, 'reg_lambda':
0.0008209782539894574}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:11,733] Trial 798 finished with value:
-1.2988513844459946 and parameters: {'n_estimators': 341, 'learning_rate':
0.01759662884693415, 'max_depth': 10, 'subsample': 0.708901259637711,
'colsample_bytree': 0.7487951221355792, 'min_child_weight': 5, 'gamma':
0.0012041342531665445, 'reg_alpha': 0.014929150468527084, 'reg_lambda':
0.0014673133391317852}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:12,194] Trial 799 finished with value:
-1.2999206697203236 and parameters: {'n_estimators': 354, 'learning_rate':
0.017356736469586097, 'max_depth': 10, 'subsample': 0.704489564631847,
'colsample_bytree': 0.7402797788529457, 'min_child_weight': 5, 'gamma':
0.0008842060262077407, 'reg_alpha': 0.011042319354772992, 'reg_lambda':
0.0014684017799375768}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:12,692] Trial 800 finished with value:
-1.304036843291943 and parameters: {'n_estimators': 379, 'learning_rate':
0.015618425795739953, 'max_depth': 10, 'subsample': 0.696828332297943,
'colsample_bytree': 0.7461416188866359, 'min_child_weight': 5, 'gamma':
0.0011359779140786815, 'reg_alpha': 0.012454429543304288, 'reg_lambda':
0.0012542649794100344}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:13,201] Trial 801 finished with value:
-1.301172622282589 and parameters: {'n_estimators': 353, 'learning_rate':
0.017376081297954104, 'max_depth': 10, 'subsample': 0.7062259853890417,
'colsample_bytree': 0.742343148760836, 'min_child_weight': 5, 'gamma':
0.0007514465096820578, 'reg_alpha': 0.008049790599653862, 'reg_lambda':
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0.0017648425501233034}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:13,667] Trial 802 finished with value:
-1.3090119538390517 and parameters: {'n_estimators': 343, 'learning_rate':
0.016393443531667225, 'max_depth': 10, 'subsample': 0.6945836139069679,
'colsample_bytree': 0.7230807410833868, 'min_child_weight': 5, 'gamma':
0.0013809716144721734, 'reg_alpha': 0.011581187378172672, 'reg_lambda':
0.0015244562455840075}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:14,166] Trial 803 finished with value:
-1.3081918045082184 and parameters: {'n_estimators': 332, 'learning_rate':
0.014835930689226473, 'max_depth': 10, 'subsample': 0.6847554848985337,
'colsample_bytree': 0.7522259491797845, 'min_child_weight': 5, 'gamma':
0.0008919166282013895, 'reg_alpha': 0.010164724591732536, 'reg_lambda':
0.0015586762872718551}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:14,657] Trial 804 finished with value:
-1.3029454506323699 and parameters: {'n_estimators': 364, 'learning_rate':
0.01762138987942021, 'max_depth': 10, 'subsample': 0.7054077564683107,
'colsample_bytree': 0.7385675537695247, 'min_child_weight': 5, 'gamma':
0.0013451119980541586, 'reg_alpha': 0.010367457556789335, 'reg_lambda':
0.0011636432484534375}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:15,167] Trial 805 finished with value:
-1.31274702304105 and parameters: {'n_estimators': 376, 'learning_rate':
0.01889243495802899, 'max_depth': 11, 'subsample': 0.6719692291751547,
'colsample_bytree': 0.7596806057249864, 'min_child_weight': 5, 'gamma':
0.0007322278086694463, 'reg_alpha': 0.0066173171184787044, 'reg_lambda':
0.0016565052558548683}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:15,656] Trial 806 finished with value:
-1.3011126736710994 and parameters: {'n_estimators': 332, 'learning_rate':
0.016683585607295968, 'max_depth': 10, 'subsample': 0.710560940420993,
'colsample_bytree': 0.7523785868553877, 'min_child_weight': 5, 'gamma':
0.0011158856458396854, 'reg_alpha': 0.022231657093560373, 'reg_lambda':
0.002073405073069963}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:16,166] Trial 807 finished with value:
-1.30831413800183 and parameters: {'n_estimators': 398, 'learning_rate':
0.014529531026680835, 'max_depth': 10, 'subsample': 0.6976696600911201,
'colsample_bytree': 0.7300230487186566, 'min_child_weight': 6, 'gamma':
0.0016718742732166143, 'reg_alpha': 0.013819124875139062, 'reg_lambda':
0.0011939228536764906}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:16,688] Trial 808 finished with value:
-1.2989316769417938 and parameters: {'n_estimators': 352, 'learning_rate':
0.017432112942424485, 'max_depth': 10, 'subsample': 0.7097989268037521,
'colsample_bytree': 0.7465689315000776, 'min_child_weight': 5, 'gamma':
0.0008177140529769602, 'reg_alpha': 0.008204923719062741, 'reg_lambda':
0.002311651830111999}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:17,198] Trial 809 finished with value:
-1.3067670633912913 and parameters: {'n_estimators': 374, 'learning_rate':
0.015787246287703255, 'max_depth': 10, 'subsample': 0.6938040089051865,
'colsample_bytree': 0.7221821573465104, 'min_child_weight': 5, 'gamma':
0.0006673867085092402, 'reg_alpha': 0.004225515915523821, 'reg_lambda':
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0.002695128527957248}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:17,720] Trial 810 finished with value: -1.3042048910086506 and parameters: {'n_estimators': 346, 'learning_rate': 0.017048380341724765, 'max_depth': 11, 'subsample': 0.7073485449888278, 'colsample_bytree': 0.7382012474972071, 'min_child_weight': 5, 'gamma': 0.0008049407555558606, 'reg_alpha': 0.01500734162477615, 'reg_lambda': 0.002281292918535011}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:18,124] Trial 811 finished with value: -1.3255719064827207 and parameters: {'n_estimators': 363, 'learning_rate': 0.015540015913715973, 'max_depth': 10, 'subsample': 0.6825238661773712, 'colsample_bytree': 0.7446984999236576, 'min_child_weight': 7, 'gamma': 0.0005322570608311425, 'reg_alpha': 0.006675163859152032, 'reg_lambda': 0.0016716320486260413}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:18,662] Trial 812 finished with value: -1.3084958939816997 and parameters: {'n_estimators': 391, 'learning_rate': 0.017205838881547496, 'max_depth': 10, 'subsample': 0.6994557023841393, 'colsample_bytree': 0.7275643433198729, 'min_child_weight': 5, 'gamma': 0.0009880838660289045, 'reg_alpha': 0.005560943564519965, 'reg_lambda': 0.002936080735060333}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:19,087] Trial 813 finished with value: -1.3014935581511644 and parameters: {'n_estimators': 357, 'learning_rate': 0.019255608914700004, 'max_depth': 10, 'subsample': 0.7077329956089026, 'colsample_bytree': 0.7516648403173555, 'min_child_weight': 5, 'gamma': 0.0006564126085505044, 'reg_alpha': 0.00995043220167055, 'reg_lambda': 0.003985568495362804}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:19,672] Trial 814 finished with value: -1.3095033361775228 and parameters: {'n_estimators': 417, 'learning_rate': 0.015806517940402522, 'max_depth': 10, 'subsample': 0.689235409045524, 'colsample_bytree': 0.7356203107651115, 'min_child_weight': 5, 'gamma': 0.0009414298294234287, 'reg_alpha': 0.007635046524853054, 'reg_lambda': 0.0010235512030374548}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:20,182] Trial 815 finished with value: -1.302396651380865 and parameters: {'n_estimators': 340, 'learning_rate': 0.014727446531614348, 'max_depth': 11, 'subsample': 0.7159290618715807, 'colsample_bytree': 0.7171257998739872, 'min_child_weight': 5, 'gamma': 0.0006175069834531072, 'reg_alpha': 0.015575286728582726, 'reg_lambda': 0.0020671188906482506}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:20,721] Trial 816 finished with value: -1.3123982581248683 and parameters: {'n_estimators': 365, 'learning_rate': 0.01755213584390919, 'max_depth': 10, 'subsample': 0.8418746293928013, 'colsample_bytree': 0.747275488319514, 'min_child_weight': 5, 'gamma': 0.0012913263388043496, 'reg_alpha': 0.024330479835217182, 'reg_lambda': 0.0013372334662691005}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:21,123] Trial 817 finished with value: -1.307422127719551 and parameters: {'n_estimators': 334, 'learning_rate': 0.019412360450536126, 'max_depth': 10, 'subsample': 0.699305025332554, 'colsample_bytree': 0.7371280785965043, 'min_child_weight': 6, 'gamma': 0.0008216002148139926, 'reg_alpha': 0.012466800702370036, 'reg_lambda':

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0.0020038182119635835}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:21,610] Trial 818 finished with value:
-1.300702449694243 and parameters: {'n_estimators': 382, 'learning_rate':
0.0166021259567074, 'max_depth': 10, 'subsample': 0.7117315110640368,
'colsample_bytree': 0.7549007386342139, 'min_child_weight': 5, 'gamma':
0.0017180441450484556, 'reg_alpha': 0.004460353479185403, 'reg_lambda':
0.0009231681936593378}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:22,138] Trial 819 finished with value:
-1.309347101512573 and parameters: {'n_estimators': 387, 'learning_rate':
0.01616198809387747, 'max_depth': 10, 'subsample': 0.6890299736087174,
'colsample_bytree': 0.7494762235407069, 'min_child_weight': 5, 'gamma':
0.0020248997333051043, 'reg_alpha': 0.003508407700788591, 'reg_lambda':
0.001068984468796736}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:22,679] Trial 820 finished with value:
-1.3098457482804304 and parameters: {'n_estimators': 398, 'learning_rate':
0.016910752420352223, 'max_depth': 10, 'subsample': 0.674652575934439,
'colsample_bytree': 0.7278351020913507, 'min_child_weight': 5, 'gamma':
0.0033056650279904925, 'reg_alpha': 0.005302883656599953, 'reg_lambda':
0.001025104504205336}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:23,142] Trial 821 finished with value:
-1.3950263806653151 and parameters: {'n_estimators': 413, 'learning_rate':
0.13194357900078704, 'max_depth': 10, 'subsample': 0.7062391872871394,
'colsample_bytree': 0.755880221198716, 'min_child_weight': 5, 'gamma':
0.001776475044850551, 'reg_alpha': 0.007601849814266754, 'reg_lambda':
0.003501253304428247}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:23,596] Trial 822 finished with value:
-1.3747960459161899 and parameters: {'n_estimators': 385, 'learning_rate':
0.10301034097106415, 'max_depth': 10, 'subsample': 0.6989802892202989,
'colsample_bytree': 0.736444686493671, 'min_child_weight': 5, 'gamma':
0.0012649252558357474, 'reg_alpha': 0.01801132680444819, 'reg_lambda':
0.0008294121451361341}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:24,101] Trial 823 finished with value:
-1.3068163683641452 and parameters: {'n_estimators': 377, 'learning_rate':
0.018335725337918992, 'max_depth': 10, 'subsample': 0.712681789019682,
'colsample_bytree': 0.7466698490683578, 'min_child_weight': 5, 'gamma':
0.00257043176763652, 'reg_alpha': 0.004960807993837629, 'reg_lambda':
0.001338869641080307}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:24,599] Trial 824 finished with value:
-1.3092288676565789 and parameters: {'n_estimators': 367, 'learning_rate':
0.01661653791841727, 'max_depth': 10, 'subsample': 0.6926602164955618,
'colsample_bytree': 0.7611004918159219, 'min_child_weight': 5, 'gamma':
0.0014138307635216752, 'reg_alpha': 0.00926909491157349, 'reg_lambda':
0.0018729859227666675}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:25,202] Trial 825 finished with value:
-1.3009021470662796 and parameters: {'n_estimators': 401, 'learning_rate':
0.018009044032565383, 'max_depth': 11, 'subsample': 0.711339946937452,
'colsample_bytree': 0.7187751241258242, 'min_child_weight': 5, 'gamma':
0.000985114546333928, 'reg_alpha': 0.028236179448961014, 'reg_lambda':
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0.0015074230144876415}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:25,690] Trial 826 finished with value:
-1.3070949128363631 and parameters: {'n_estimators': 355, 'learning_rate':
0.020086367694442386, 'max_depth': 10, 'subsample': 0.6821837867015528,
'colsample_bytree': 0.7384593191145883, 'min_child_weight': 5, 'gamma':
0.000520095616920419, 'reg_alpha': 0.012806745384700492, 'reg_lambda':
0.0006859400156847609}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:25,925] Trial 827 finished with value:
-1.3514956708569794 and parameters: {'n_estimators': 382, 'learning_rate':
0.01573519012119444, 'max_depth': 5, 'subsample': 0.701801731905979,
'colsample_bytree': 0.7526175088091203, 'min_child_weight': 6, 'gamma':
0.0016639638174917178, 'reg_alpha': 0.006985514251592931, 'reg_lambda':
0.0009901104518264842}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:26,452] Trial 828 finished with value:
-1.3053882271126527 and parameters: {'n_estimators': 412, 'learning_rate':
0.0174177311511801, 'max_depth': 10, 'subsample': 0.7164577177300222,
'colsample_bytree': 0.7609767170957792, 'min_child_weight': 5, 'gamma':
0.0008578874821277706, 'reg_alpha': 0.013922691031692917, 'reg_lambda':
0.0012713504356987493}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:27,039] Trial 829 finished with value:
-1.3094836686072928 and parameters: {'n_estimators': 433, 'learning_rate':
0.018630482444459993, 'max_depth': 10, 'subsample': 0.7058420872799095,
'colsample_bytree': 0.7305224038512368, 'min_child_weight': 5, 'gamma':
0.002872961898107966, 'reg_alpha': 0.021098479600706575, 'reg_lambda':
0.0029612669686701204}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:27,512] Trial 830 finished with value:
-1.3011722264201298 and parameters: {'n_estimators': 349, 'learning_rate':
0.016716766188424593, 'max_depth': 10, 'subsample': 0.7196171351442193,
'colsample_bytree': 0.7441060999187983, 'min_child_weight': 5, 'gamma':
0.0011075354110687762, 'reg_alpha': 0.004071289339766842, 'reg_lambda':
0.0007184023989266646}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:28,029] Trial 831 finished with value:
-1.309036491309134 and parameters: {'n_estimators': 356, 'learning_rate':
0.01931836073198186, 'max_depth': 10, 'subsample': 0.689300683882477,
'colsample_bytree': 0.7632606971663443, 'min_child_weight': 5, 'gamma':
0.0005025350997722033, 'reg_alpha': 0.00911342484155115, 'reg_lambda':
0.0018781852505501916}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:28,609] Trial 832 finished with value:
-1.304436625804432 and parameters: {'n_estimators': 372, 'learning_rate':
0.015235810689109001, 'max_depth': 11, 'subsample': 0.6995668574740429,
'colsample_bytree': 0.7502388802660483, 'min_child_weight': 5, 'gamma':
0.0016601888415676866, 'reg_alpha': 0.015430560335762837, 'reg_lambda':
0.0011503325779480859}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:29,087] Trial 833 finished with value:
-1.3031353172797153 and parameters: {'n_estimators': 328, 'learning_rate':
0.01793193545149292, 'max_depth': 10, 'subsample': 0.7100711712798615,
'colsample_bytree': 0.7240379993387284, 'min_child_weight': 5, 'gamma':
0.0007099243389519067, 'reg_alpha': 0.009107518473992035, 'reg_lambda':
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0.0008010055696465043}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:29,597] Trial 834 finished with value:
-1.3057395257313165 and parameters: {'n_estimators': 394, 'learning_rate':
0.01684339712398001, 'max_depth': 10, 'subsample': 0.7210390122831173,
'colsample_bytree': 0.7388921106224802, 'min_child_weight': 5, 'gamma':
0.0023990158293367304, 'reg_alpha': 0.030052605324802386, 'reg_lambda':
0.0022295674117842204}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:30,090] Trial 835 finished with value:
-1.3045975017718725 and parameters: {'n_estimators': 344, 'learning_rate':
0.019762476010259863, 'max_depth': 10, 'subsample': 0.6962514226196652,
'colsample_bytree': 0.7658504563160125, 'min_child_weight': 5, 'gamma':
0.0011109599102640951, 'reg_alpha': 0.0072262164112403635, 'reg_lambda':
0.0006346974013929219}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:30,627] Trial 836 finished with value:
-1.3015305943086306 and parameters: {'n_estimators': 331, 'learning_rate':
0.01564741063842112, 'max_depth': 11, 'subsample': 0.7092638824015935,
'colsample_bytree': 0.7532507037396332, 'min_child_weight': 5, 'gamma':
0.0007941352172842564, 'reg_alpha': 0.01316373824553081, 'reg_lambda':
0.0008548812495194513}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:31,149] Trial 837 finished with value:
-1.3114146886098765 and parameters: {'n_estimators': 365, 'learning_rate':
0.018474394680377818, 'max_depth': 10, 'subsample': 0.6757300816063199,
'colsample_bytree': 0.7421755280684627, 'min_child_weight': 5, 'gamma':
0.0017639423819507906, 'reg_alpha': 0.018700279764206077, 'reg_lambda':
0.0013600630868179345}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:31,596] Trial 838 finished with value:
-1.310882050496826 and parameters: {'n_estimators': 346, 'learning_rate':
0.020751641265152602, 'max_depth': 10, 'subsample': 0.7192399777788345,
'colsample_bytree': 0.7678695802855973, 'min_child_weight': 6, 'gamma':
0.0004507359340566507, 'reg_alpha': 0.0042311440580176385, 'reg_lambda':
0.003057724738078358}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:32,185] Trial 839 finished with value:
-1.3054050865973355 and parameters: {'n_estimators': 427, 'learning_rate':
0.014778253796149913, 'max_depth': 10, 'subsample': 0.686231483805952,
'colsample_bytree': 0.7153719385619455, 'min_child_weight': 5, 'gamma':
0.001178335289208409, 'reg_alpha': 0.03410428554045026, 'reg_lambda':
0.0005789390772530089}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:32,581] Trial 840 finished with value:
-1.3125424203086518 and parameters: {'n_estimators': 322, 'learning_rate':
0.0170710806412138, 'max_depth': 10, 'subsample': 0.5022952321766351,
'colsample_bytree': 0.7317811152767322, 'min_child_weight': 5, 'gamma':
0.003463254728819698, 'reg_alpha': 0.011294089315787325, 'reg_lambda':
0.0009416896639052411}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:33,153] Trial 841 finished with value:
-1.308432333081616 and parameters: {'n_estimators': 402, 'learning_rate':
0.01848900233946867, 'max_depth': 11, 'subsample': 0.7018726682840888,
'colsample_bytree': 0.7559951072710158, 'min_child_weight': 5, 'gamma':
0.0006199957434954429, 'reg_alpha': 0.005541500996891525, 'reg_lambda':
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0.0013513414312831067}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:33,558] Trial 842 finished with value:
-1.3048539384232256 and parameters: {'n_estimators': 320, 'learning_rate':
0.01662258389035766, 'max_depth': 10, 'subsample': 0.7217402108388813,
'colsample_bytree': 0.7458962963262475, 'min_child_weight': 5, 'gamma':
0.0020146777359177615, 'reg_alpha': 0.029490311763789405, 'reg_lambda':
0.002124794980594805}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:34,076] Trial 843 finished with value:
-1.306576823122806 and parameters: {'n_estimators': 376, 'learning_rate':
0.01984893817266733, 'max_depth': 10, 'subsample': 0.7111658313569089,
'colsample_bytree': 0.7706934519195262, 'min_child_weight': 5, 'gamma':
0.0009096312765829533, 'reg_alpha': 0.019003427911375114, 'reg_lambda':
0.0006071007822153661}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:34,640] Trial 844 finished with value:
-1.315106127203632 and parameters: {'n_estimators': 447, 'learning_rate':
0.015866779442124997, 'max_depth': 10, 'subsample': 0.6923078932241244,
'colsample_bytree': 0.7581943937751895, 'min_child_weight': 5, 'gamma':
0.0014098128402076715, 'reg_alpha': 0.007356567482030029, 'reg_lambda':
0.0009865921026400066}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:35,100] Trial 845 finished with value:
-1.3016006952170143 and parameters: {'n_estimators': 350, 'learning_rate':
0.01781928000827134, 'max_depth': 10, 'subsample': 0.7103095385490982,
'colsample_bytree': 0.7290052524052515, 'min_child_weight': 5, 'gamma':
0.0004805960514458048, 'reg_alpha': 0.010537261399296083, 'reg_lambda':
0.005802904137025236}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:35,662] Trial 846 finished with value:
-1.3030020507584739 and parameters: {'n_estimators': 331, 'learning_rate':
0.01459512414578631, 'max_depth': 11, 'subsample': 0.721808320395668,
'colsample_bytree': 0.7394976453781165, 'min_child_weight': 5, 'gamma':
0.0006732967014485994, 'reg_alpha': 0.07426550785843573, 'reg_lambda':
0.0016288135331762683}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:36,121] Trial 847 finished with value:
-1.3165248286105984 and parameters: {'n_estimators': 384, 'learning_rate':
0.02103728332955652, 'max_depth': 10, 'subsample': 0.6972096968416184,
'colsample_bytree': 0.7715061048843556, 'min_child_weight': 6, 'gamma':
0.0011855954704339778, 'reg_alpha': 0.014042121811391018, 'reg_lambda':
0.0005317424473879753}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:36,575] Trial 848 finished with value:
-1.3051538385269947 and parameters: {'n_estimators': 365, 'learning_rate':
0.017449236416887354, 'max_depth': 10, 'subsample': 0.7061529224364443,
'colsample_bytree': 0.7527202461692916, 'min_child_weight': 5, 'gamma':
0.00037784775892298967, 'reg_alpha': 0.027173116047166348, 'reg_lambda':
0.0007917244875591329}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:37,006] Trial 849 finished with value:
-1.305931716352448 and parameters: {'n_estimators': 314, 'learning_rate':
0.01875069528664047, 'max_depth': 10, 'subsample': 0.6730721474989723,
'colsample_bytree': 0.7195076891786059, 'min_child_weight': 5, 'gamma':
0.006039144669136001, 'reg_alpha': 0.008973371092945777, 'reg_lambda':
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6.462063712367949e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:37,451] Trial 850 finished with value: -1.3179972170763017 and parameters: {'n_estimators': 407, 'learning_rate': 0.01603651601006386, 'max_depth': 10, 'subsample': 0.7253993984801033, 'colsample_bytree': 0.40522209613956484, 'min_child_weight': 5, 'gamma': 0.0024825921491339843, 'reg_alpha': 0.019786988441963575, 'reg_lambda': 0.0010215977448372663}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:37,930] Trial 851 finished with value: -1.3061249850995846 and parameters: {'n_estimators': 344, 'learning_rate': 0.019204931302205328, 'max_depth': 10, 'subsample': 0.7148627287804765, 'colsample_bytree': 0.7591221713684173, 'min_child_weight': 5, 'gamma': 0.0008443369830789491, 'reg_alpha': 0.003242074207592063, 'reg_lambda': 0.004476453333222367}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:38,405] Trial 852 finished with value: -1.3081224386477113 and parameters: {'n_estimators': 317, 'learning_rate': 0.01726210931124109, 'max_depth': 11, 'subsample': 0.6829350079031113, 'colsample_bytree': 0.7429731345409379, 'min_child_weight': 5, 'gamma': 0.001438511461246266, 'reg_alpha': 0.013542598444940246, 'reg_lambda': 0.002087555715066719}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:38,939] Trial 853 finished with value: -1.3163395179751782 and parameters: {'n_estimators': 428, 'learning_rate': 0.020353227057866085, 'max_depth': 10, 'subsample': 0.7003380979965231, 'colsample_bytree': 0.7054491272968699, 'min_child_weight': 6, 'gamma': 0.003987397960953902, 'reg_alpha': 0.004599453158893844, 'reg_lambda': 0.0004738838257988521}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:39,497] Trial 854 finished with value: -1.3050958224439775 and parameters: {'n_estimators': 357, 'learning_rate': 0.016121526207159266, 'max_depth': 10, 'subsample': 0.7260349268621041, 'colsample_bytree': 0.775324154259322, 'min_child_weight': 5, 'gamma': 0.0005564368524891509, 'reg_alpha': 0.006979771966326955, 'reg_lambda': 5.3819033329085e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:39,649] Trial 855 finished with value: -1.4173387565800502 and parameters: {'n_estimators': 331, 'learning_rate': 0.01799391591810399, 'max_depth': 3, 'subsample': 0.7138249292162875, 'colsample_bytree': 0.7325832275427754, 'min_child_weight': 5, 'gamma': 0.0018512127135816737, 'reg_alpha': 0.04027152818951883, 'reg_lambda': 0.001331657960904956}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:39,782] Trial 856 finished with value: -1.4222549377160283 and parameters: {'n_estimators': 383, 'learning_rate': 0.042542141304389755, 'max_depth': 2, 'subsample': 0.6911185810014008, 'colsample_bytree': 0.7602072119998734, 'min_child_weight': 5, 'gamma': 0.0009914194467257284, 'reg_alpha': 0.021963500244512615, 'reg_lambda': 0.000712863368070202}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:40,238] Trial 857 finished with value: -1.3076555241432641 and parameters: {'n_estimators': 309, 'learning_rate': 0.015003181252323566, 'max_depth': 10, 'subsample': 0.706688978770524, 'colsample_bytree': 0.7447399280018853, 'min_child_weight': 5, 'gamma': 0.0007148101573654724, 'reg_alpha': 0.010370677860288238, 'reg_lambda':

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0.000500496600790537}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:40,514] Trial 858 finished with value:
-1.3275884200381916 and parameters: {'n_estimators': 343, 'learning_rate':
0.016714711311268258, 'max_depth': 6, 'subsample': 0.7202533741960329,
'colsample_bytree': 0.7666332932445393, 'min_child_weight': 5, 'gamma':
0.00290716657691411, 'reg_alpha': 4.181485689090042e-06, 'reg_lambda':
0.0008895781356071113}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:40,719] Trial 859 finished with value:
-1.3799567989807613 and parameters: {'n_estimators': 363, 'learning_rate':
0.018946150007926557, 'max_depth': 4, 'subsample': 0.7296053754159426,
'colsample_bytree': 0.7534303023933542, 'min_child_weight': 5, 'gamma':
0.0015407288879019862, 'reg_alpha': 0.006607480571419987, 'reg_lambda':
6.645329461307991e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:41,146] Trial 860 finished with value:
-1.3183213794844952 and parameters: {'n_estimators': 329, 'learning_rate':
0.021460461436089203, 'max_depth': 11, 'subsample': 0.6943466706071244,
'colsample_bytree': 0.4814531179531133, 'min_child_weight': 5, 'gamma':
0.00032646396255385805, 'reg_alpha': 0.01209984741073463, 'reg_lambda':
0.003174506377922928}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:41,761] Trial 861 finished with value:
-1.3056051598743543 and parameters: {'n_estimators': 412, 'learning_rate':
0.014296186411145143, 'max_depth': 10, 'subsample': 0.7053820305664853,
'colsample_bytree': 0.7782198201880999, 'min_child_weight': 5, 'gamma':
0.001062031113840472, 'reg_alpha': 9.033991953180004e-05, 'reg_lambda':
0.001621806002047501}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:42,200] Trial 862 finished with value:
-1.3028446240985352 and parameters: {'n_estimators': 309, 'learning_rate':
0.017817872135000693, 'max_depth': 10, 'subsample': 0.7153425969424695,
'colsample_bytree': 0.7258738250000659, 'min_child_weight': 5, 'gamma':
0.000514991462776432, 'reg_alpha': 0.016965191134122648, 'reg_lambda':
3.671483722664494e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:42,478] Trial 863 finished with value:
-1.3315986208962356 and parameters: {'n_estimators': 297, 'learning_rate':
0.01922914350261972, 'max_depth': 7, 'subsample': 0.6840156548767715,
'colsample_bytree': 0.7463687478757871, 'min_child_weight': 6, 'gamma':
0.0007767142496154101, 'reg_alpha': 0.031097602233580404, 'reg_lambda':
0.5883106661812205}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:42,845] Trial 864 finished with value:
-1.3260832450626718 and parameters: {'n_estimators': 346, 'learning_rate':
0.015748482244333298, 'max_depth': 10, 'subsample': 0.7321826731515608,
'colsample_bytree': 0.7636764019541752, 'min_child_weight': 10, 'gamma':
0.0017031900249616985, 'reg_alpha': 0.023759527430410033, 'reg_lambda':
0.0005328703694062691}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:43,593] Trial 865 finished with value:
-1.3181924719069795 and parameters: {'n_estimators': 841, 'learning_rate':
0.017640912126510686, 'max_depth': 10, 'subsample': 0.7034154277989221,
'colsample_bytree': 0.7322296756863347, 'min_child_weight': 5, 'gamma':
0.001158179155357028, 'reg_alpha': 0.013898209241775083, 'reg_lambda':
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0.002283665977053278}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:44,148] Trial 866 finished with value:
-1.3142723322331438 and parameters: {'n_estimators': 442, 'learning_rate':
0.01991400737336563, 'max_depth': 10, 'subsample': 0.7214948710904382,
'colsample_bytree': 0.7817733630049033, 'min_child_weight': 5, 'gamma':
0.00038644044691415926, 'reg_alpha': 0.008673888609180948, 'reg_lambda':
5.366074665193244e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:44,709] Trial 867 finished with value:
-1.3042734789016819 and parameters: {'n_estimators': 377, 'learning_rate':
0.01650918651847112, 'max_depth': 11, 'subsample': 0.7132303390364869,
'colsample_bytree': 0.7525431644003384, 'min_child_weight': 5, 'gamma':
0.0006708027563335809, 'reg_alpha': 0.04594128823889397, 'reg_lambda':
9.278400859984578e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:45,179] Trial 868 finished with value:
-1.3115213230456935 and parameters: {'n_estimators': 320, 'learning_rate':
0.018517694472835227, 'max_depth': 10, 'subsample': 0.696868862328579,
'colsample_bytree': 0.7118404749600774, 'min_child_weight': 5, 'gamma':
0.0009497285950406973, 'reg_alpha': 0.026890211448461498, 'reg_lambda':
0.0011405144333330849}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:45,746] Trial 869 finished with value:
-1.3056856954952731 and parameters: {'n_estimators': 394, 'learning_rate':
0.017164947972838643, 'max_depth': 10, 'subsample': 0.7315808999144712,
'colsample_bytree': 0.7389827074143517, 'min_child_weight': 5, 'gamma':
0.0026591829456635, 'reg_alpha': 0.0035938005257907775, 'reg_lambda':
0.0007150947996845134}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:46,222] Trial 870 finished with value:
-1.305341405697593 and parameters: {'n_estimators': 335, 'learning_rate':
0.015320034124557284, 'max_depth': 10, 'subsample': 0.7060087273481453,
'colsample_bytree': 0.7675244840258969, 'min_child_weight': 5, 'gamma':
0.002062457089686202, 'reg_alpha': 0.0180862148515278, 'reg_lambda':
4.440989186441913e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:46,698] Trial 871 finished with value:
-1.3109167203753105 and parameters: {'n_estimators': 363, 'learning_rate':
0.020975217183105098, 'max_depth': 10, 'subsample': 0.7208912251787991,
'colsample_bytree': 0.7486735242914317, 'min_child_weight': 5, 'gamma':
0.00133442676877007, 'reg_alpha': 0.007104367603118474, 'reg_lambda':
0.0017391892209713104}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:47,030] Trial 872 finished with value:
-1.3197396337101732 and parameters: {'n_estimators': 300, 'learning_rate':
0.019073464939893762, 'max_depth': 8, 'subsample': 0.6884497128246824,
'colsample_bytree': 0.720781918983578, 'min_child_weight': 6, 'gamma':
0.0005137428470974825, 'reg_alpha': 0.09558053925909818, 'reg_lambda':
0.00013402491356427774}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:47,446] Trial 873 finished with value:
-1.3622074455417927 and parameters: {'n_estimators': 351, 'learning_rate':
0.07888827836727885, 'max_depth': 11, 'subsample': 0.671656220631059,
'colsample_bytree': 0.7627008538539486, 'min_child_weight': 5, 'gamma':
0.0008062823241755457, 'reg_alpha': 0.009933881855223037, 'reg_lambda':
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0.0009809247804217673}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:47,933] Trial 874 finished with value:
-1.2990270160734252 and parameters: {'n_estimators': 325, 'learning_rate':
0.01629030255881041, 'max_depth': 10, 'subsample': 0.7109971887477816,
'colsample_bytree': 0.7767808450440045, 'min_child_weight': 5, 'gamma':
0.004502419974658405, 'reg_alpha': 0.019085035220295254, 'reg_lambda':
1.9108281679662687e-06}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:48,431] Trial 875 finished with value:
-1.302944956139447 and parameters: {'n_estimators': 315, 'learning_rate':
0.014124085128399841, 'max_depth': 10, 'subsample': 0.7006498180800184,
'colsample_bytree': 0.7879914466007417, 'min_child_weight': 5, 'gamma':
0.005382395532051593, 'reg_alpha': 0.031285463999324914, 'reg_lambda':
3.4579471571701964e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:48,896] Trial 876 finished with value:
-1.3049696540689872 and parameters: {'n_estimators': 297, 'learning_rate':
0.01785991941392478, 'max_depth': 10, 'subsample': 0.7264988589402501,
'colsample_bytree': 0.7907796235205558, 'min_child_weight': 5, 'gamma':
0.006514721397322135, 'reg_alpha': 5.9481418755626856e-05, 'reg_lambda':
5.547828018107424e-06}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:49,395] Trial 877 finished with value:
-1.3052039963048128 and parameters: {'n_estimators': 319, 'learning_rate':
0.015411453330318054, 'max_depth': 10, 'subsample': 0.7140540843789189,
'colsample_bytree': 0.7806612640675902, 'min_child_weight': 5, 'gamma':
0.004869350081905106, 'reg_alpha': 0.05258750466854441, 'reg_lambda':
2.674256661201403e-06}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:49,834] Trial 878 finished with value:
-1.3110773893871437 and parameters: {'n_estimators': 291, 'learning_rate':
0.019856745622061318, 'max_depth': 10, 'subsample': 0.7390663687014822,
'colsample_bytree': 0.8208750677788135, 'min_child_weight': 5, 'gamma':
0.009270029087758652, 'reg_alpha': 0.021409433949016277, 'reg_lambda':
1.1946453556377532e-06}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:50,298] Trial 879 finished with value:
-1.3090601838645124 and parameters: {'n_estimators': 329, 'learning_rate':
0.01713655357987133, 'max_depth': 10, 'subsample': 0.6940970616259958,
'colsample_bytree': 0.8028331135734349, 'min_child_weight': 5, 'gamma':
0.009844481445052868, 'reg_alpha': 0.01635775552025061, 'reg_lambda':
2.567298498444031e-06}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:50,690] Trial 880 finished with value:
-1.3242526536790482 and parameters: {'n_estimators': 311, 'learning_rate':
0.01638276755657574, 'max_depth': 10, 'subsample': 0.6789721704814741,
'colsample_bytree': 0.775336293624272, 'min_child_weight': 9, 'gamma':
0.0031965498800603062, 'reg_alpha': 0.04593228825439804, 'reg_lambda':
8.98074827888387e-07}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:51,042] Trial 881 finished with value:
-1.3120161535227286 and parameters: {'n_estimators': 326, 'learning_rate':
0.02228802061513007, 'max_depth': 8, 'subsample': 0.7218972084550114,
'colsample_bytree': 0.7863645765279331, 'min_child_weight': 6, 'gamma':
0.004089762084291106, 'reg_alpha': 0.03339875134430611, 'reg_lambda':
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6.998303723033938e-06}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:51,524] Trial 882 finished with value:
-1.3033714402420427 and parameters: {'n_estimators': 299, 'learning_rate':
0.018093728653001493, 'max_depth': 11, 'subsample': 0.7039219146836937,
'colsample_bytree': 0.7709363742518331, 'min_child_weight': 5, 'gamma':
0.00034164076010461444, 'reg_alpha': 0.02101523403095842, 'reg_lambda':
1.623324948196182e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:51,970] Trial 883 finished with value:
-1.3146080697370677 and parameters: {'n_estimators': 281, 'learning_rate':
0.014636861743011622, 'max_depth': 10, 'subsample': 0.7142181053787267,
'colsample_bytree': 0.7741047374114653, 'min_child_weight': 5, 'gamma':
0.003870928204550614, 'reg_alpha': 0.018655015631885628, 'reg_lambda':
1.5961217332399003e-06}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:52,355] Trial 884 finished with value:
-1.4840566047460346 and parameters: {'n_estimators': 335, 'learning_rate':
0.24393938274783467, 'max_depth': 10, 'subsample': 0.728909098445708,
'colsample_bytree': 0.7624781833919531, 'min_child_weight': 5, 'gamma':
0.00042339629163971795, 'reg_alpha': 4.4339767094198536e-05, 'reg_lambda':
1.8830237882296047e-06}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:52,793] Trial 885 finished with value:
-1.3162865777389168 and parameters: {'n_estimators': 310, 'learning_rate':
0.020515114378046092, 'max_depth': 11, 'subsample': 0.6899104888569888,
'colsample_bytree': 0.7840758003538133, 'min_child_weight': 5, 'gamma':
0.0006778039414728824, 'reg_alpha': 0.14436719235344445, 'reg_lambda':
2.2922784927400605e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:53,177] Trial 886 finished with value:
-1.3056807331176685 and parameters: {'n_estimators': 344, 'learning_rate':
0.01877602304355782, 'max_depth': 10, 'subsample': 0.7022336937136919,
'colsample_bytree': 0.7377600798669144, 'min_child_weight': 5, 'gamma':
0.005816126134345257, 'reg_alpha': 0.01350105253012338, 'reg_lambda':
3.7767009618513954e-06}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:53,662] Trial 887 finished with value:
-1.3012246115243638 and parameters: {'n_estimators': 319, 'learning_rate':
0.01601899176225335, 'max_depth': 10, 'subsample': 0.711269105407952,
'colsample_bytree': 0.7531616482598836, 'min_child_weight': 5, 'gamma':
0.001038906916696609, 'reg_alpha': 0.07646399165920495, 'reg_lambda':
3.6333944218706328e-06}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:54,372] Trial 888 finished with value:
-1.3078186916849985 and parameters: {'n_estimators': 453, 'learning_rate':
0.017474652705388293, 'max_depth': 11, 'subsample': 0.7356027074588263,
'colsample_bytree': 0.8016330986644988, 'min_child_weight': 5, 'gamma':
0.0006269321391962623, 'reg_alpha': 0.015381822351287718, 'reg_lambda':
5.657425811375235e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:54,791] Trial 889 finished with value:
-1.3114548089957698 and parameters: {'n_estimators': 288, 'learning_rate':
0.01549551012963575, 'max_depth': 10, 'subsample': 0.7219236742784719,
'colsample_bytree': 0.76761797555602, 'min_child_weight': 6, 'gamma':
0.002464616965389173 8, 'reg_alpha': 0.03011867585305139, 'reg_lambda':
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8.96831519636455e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:55,281] Trial 890 finished with value:
-1.3039214170377873 and parameters: {'n_estimators': 339, 'learning_rate':
0.019626629381099464, 'max_depth': 10, 'subsample': 0.6989179597071148,
'colsample_bytree': 0.7412865007002785, 'min_child_weight': 5, 'gamma':
0.0005034066407738297, 'reg_alpha': 0.00011968458458181963, 'reg_lambda':
7.003293754243418e-07}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:55,683] Trial 891 finished with value:
-1.3098697377267172 and parameters: {'n_estimators': 305, 'learning_rate':
0.018404209210016818, 'max_depth': 10, 'subsample': 0.683845852951171,
'colsample_bytree': 0.7268369459227437, 'min_child_weight': 5, 'gamma':
0.0002648806654643566, 'reg_alpha': 0.010960348913936204, 'reg_lambda':
2.775966665796641e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:56,047] Trial 892 finished with value:
-1.3056506045652496 and parameters: {'n_estimators': 268, 'learning_rate':
0.016733909034960102, 'max_depth': 9, 'subsample': 0.7110383717554487,
'colsample_bytree': 0.7591710036199549, 'min_child_weight': 5, 'gamma':
0.0010627350071467827, 'reg_alpha': 0.025255570224622874, 'reg_lambda':
4.991072054472175e-06}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:56,453] Trial 893 finished with value:
-1.3141816094164156 and parameters: {'n_estimators': 325, 'learning_rate':
0.01465159055382369, 'max_depth': 8, 'subsample': 0.7233455351318254,
'colsample_bytree': 0.7785380273657795, 'min_child_weight': 5, 'gamma':
0.0075009898916748256, 'reg_alpha': 0.04147425431001396, 'reg_lambda':
3.780591949048734e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:57,079] Trial 894 finished with value:
-1.3166199737305715 and parameters: {'n_estimators': 464, 'learning_rate':
0.022044961531093997, 'max_depth': 10, 'subsample': 0.7401250071946507,
'colsample_bytree': 0.7479574850460635, 'min_child_weight': 5, 'gamma':
0.0013893437481760787, 'reg_alpha': 0.015154903958609052, 'reg_lambda':
7.250798494391782e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:57,503] Trial 895 finished with value:
-1.3104060346839024 and parameters: {'n_estimators': 345, 'learning_rate':
0.017803797888427533, 'max_depth': 9, 'subsample': 0.6972817229099287,
'colsample_bytree': 0.7697270735244768, 'min_child_weight': 5, 'gamma':
0.0007753790848967536, 'reg_alpha': 1.430650614007264e-05, 'reg_lambda':
0.0002703768862450242}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:57,824] Trial 896 finished with value:
-1.3259275133823822 and parameters: {'n_estimators': 293, 'learning_rate':
0.020164249633040063, 'max_depth': 7, 'subsample': 0.7158123453313642,
'colsample_bytree': 0.9218717992888118, 'min_child_weight': 5, 'gamma':
0.0021333095981133088, 'reg_alpha': 2.097318810582579e-08, 'reg_lambda':
0.0001229620007423591}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:58,314] Trial 897 finished with value:
-1.3019571541081338 and parameters: {'n_estimators': 312, 'learning_rate':
0.016175204848144276, 'max_depth': 10, 'subsample': 0.7059197805684194,
'colsample_bytree': 0.7956333342188701, 'min_child_weight': 5, 'gamma':
0.0004911908282444246, 'reg_alpha': 0.010530278136047302, 'reg_lambda':
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1.9700582810123158e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:58,797] Trial 898 finished with value:
-1.300505878128812 and parameters: {'n_estimators': 332, 'learning_rate':
0.01886974799251382, 'max_depth': 10, 'subsample': 0.7310732831399007,
'colsample_bytree': 0.7319680237276303, 'min_child_weight': 5, 'gamma':
0.002922256167009853, 'reg_alpha': 0.021208301421472238, 'reg_lambda':
8.115565955556746e-06}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:59,232] Trial 899 finished with value:
-1.313877944804579 and parameters: {'n_estimators': 355, 'learning_rate':
0.017352828605026967, 'max_depth': 9, 'subsample': 0.6901514050944861,
'colsample_bytree': 0.7577903319554217, 'min_child_weight': 6, 'gamma':
0.004064109774707964, 'reg_alpha': 1.8572618276171368e-08, 'reg_lambda':
4.6889421550388886e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:15:59,761] Trial 900 finished with value:
-1.3071872875025894 and parameters: {'n_estimators': 325, 'learning_rate':
0.015523590569973631, 'max_depth': 11, 'subsample': 0.7201326770330627,
'colsample_bytree': 0.7092289893005038, 'min_child_weight': 5, 'gamma':
0.001249468915102163, 'reg_alpha': 0.0230210681115906, 'reg_lambda':
0.0001526715101368425}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:00,197] Trial 901 finished with value:
-1.310204728845762 and parameters: {'n_estimators': 307, 'learning_rate':
0.02125167682042408, 'max_depth': 10, 'subsample': 0.6737339487543353,
'colsample_bytree': 0.7450628885206858, 'min_child_weight': 5, 'gamma':
0.0007272974688522709, 'reg_alpha': 2.197830989541577e-08, 'reg_lambda':
0.0004198986482830441}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:00,807] Trial 902 finished with value:
-1.306106344665283 and parameters: {'n_estimators': 433, 'learning_rate':
0.01861232159582151, 'max_depth': 10, 'subsample': 0.7081547031344078,
'colsample_bytree': 0.7837086002580016, 'min_child_weight': 5, 'gamma':
0.0003593643361972629, 'reg_alpha': 0.014087703451626134, 'reg_lambda':
6.68738653654386e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:01,182] Trial 903 finished with value:
-1.3535310826130738 and parameters: {'n_estimators': 280, 'learning_rate':
0.08539861051435406, 'max_depth': 10, 'subsample': 0.7357225013838998,
'colsample_bytree': 0.7669032565334101, 'min_child_weight': 5, 'gamma':
1.0584272160240503e-08, 'reg_alpha': 0.23949911634488666, 'reg_lambda':
2.8592401607156964e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:01,625] Trial 904 finished with value:
-1.311120282471079 and parameters: {'n_estimators': 355, 'learning_rate':
0.01690719479210412, 'max_depth': 9, 'subsample': 0.6996968499461319,
'colsample_bytree': 0.7203350317022134, 'min_child_weight': 5, 'gamma':
0.0009237800510228814, 'reg_alpha': 2.3625246099339373e-08, 'reg_lambda':
1.423903880118726e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:02,108] Trial 905 finished with value:
-1.3032601762881528 and parameters: {'n_estimators': 339, 'learning_rate':
0.014250788023576693, 'max_depth': 10, 'subsample': 0.7228609076819256,
'colsample_bytree': 0.7402273399380045, 'min_child_weight': 5, 'gamma':
0.0017546252506036354, 'reg_alpha': 1.7072179800987968e-08, 'reg_lambda':
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0.00368987194067063}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:02,482] Trial 906 finished with value:
-1.306967774117709 and parameters: {'n_estimators': 296, 'learning_rate':
0.01979742499384155, 'max_depth': 9, 'subsample': 0.7118833294973247,
'colsample_bytree': 0.7504049094137297, 'min_child_weight': 5, 'gamma':
0.0006173295750513339, 'reg_alpha': 0.0080994305989674, 'reg_lambda':
3.3830392588643937e-06}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:02,953] Trial 907 finished with value:
-1.308063495714408 and parameters: {'n_estimators': 320, 'learning_rate':
0.01633191636556897, 'max_depth': 10, 'subsample': 0.6833595455658217,
'colsample_bytree': 0.6978396064946121, 'min_child_weight': 5, 'gamma':
0.001261674081727981, 'reg_alpha': 0.00012542981712857784, 'reg_lambda':
9.427528104805091e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:03,337] Trial 908 finished with value:
-1.3137448504664473 and parameters: {'n_estimators': 268, 'learning_rate':
0.017890763145464155, 'max_depth': 9, 'subsample': 0.6945029794045741,
'colsample_bytree': 0.7745531287479913, 'min_child_weight': 5, 'gamma':
0.00021706463625936466, 'reg_alpha': 0.029075501045189186, 'reg_lambda':
0.0001953895056462436}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:03,832] Trial 909 finished with value:
-1.3146674003644199 and parameters: {'n_estimators': 365, 'learning_rate':
0.015170264280281166, 'max_depth': 10, 'subsample': 0.7295574873895613,
'colsample_bytree': 0.761635779862437, 'min_child_weight': 6, 'gamma':
0.002129060524874433, 'reg_alpha': 2.5956508570195426e-08, 'reg_lambda':
0.00036096726742813434}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:04,311] Trial 910 finished with value:
-1.30269165846264 and parameters: {'n_estimators': 339, 'learning_rate':
0.0188277348091445, 'max_depth': 11, 'subsample': 0.7161405562326075,
'colsample_bytree': 0.7379555261559433, 'min_child_weight': 5, 'gamma':
0.0009163514378058349, 'reg_alpha': 0.012010841034835781, 'reg_lambda':
3.7285501193519324e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:04,824] Trial 911 finished with value:
-1.307463033335196 and parameters: {'n_estimators': 454, 'learning_rate':
0.017068777368108558, 'max_depth': 10, 'subsample': 0.7022667908413288,
'colsample_bytree': 0.7275670323689425, 'min_child_weight': 5, 'gamma':
0.0004972972137544132, 'reg_alpha': 1.5211201437821356e-08, 'reg_lambda':
1.839858068339477e-06}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:05,239] Trial 912 finished with value:
-1.3049214931807667 and parameters: {'n_estimators': 305, 'learning_rate':
0.020378950630136186, 'max_depth': 9, 'subsample': 0.7082775953209071,
'colsample_bytree': 0.8173658437547786, 'min_child_weight': 5, 'gamma':
0.0014852068094588086, 'reg_alpha': 1.7330466431216536e-08, 'reg_lambda':
0.0005605704090685945}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:05,730] Trial 913 finished with value:
-1.3022663855002048 and parameters: {'n_estimators': 324, 'learning_rate':
0.017919084895067744, 'max_depth': 10, 'subsample': 0.7255001527253784,
'colsample_bytree': 0.7893493597385158, 'min_child_weight': 5, 'gamma':
0.00032600346634352234, 'reg_alpha': 0.038599111609018244, 'reg_lambda':
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0.002574504692196306}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:06,114] Trial 914 finished with value:
-1.315638850639793 and parameters: {'n_estimators': 351, 'learning_rate':
0.023375339190398726, 'max_depth': 9, 'subsample': 0.7432368857958197,
'colsample_bytree': 0.8465139805054154, 'min_child_weight': 6, 'gamma':
0.0031298449531563005, 'reg_alpha': 0.018442901794555683, 'reg_lambda':
2.18561589205663e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:06,559] Trial 915 finished with value:
-1.3024213401306703 and parameters: {'n_estimators': 297, 'learning_rate':
0.016411813094835574, 'max_depth': 11, 'subsample': 0.7174191885064088,
'colsample_bytree': 0.7544772242093684, 'min_child_weight': 5, 'gamma':
0.000828676560071462, 'reg_alpha': 2.826562776145213e-08, 'reg_lambda':
0.0002643587678687263}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:07,038] Trial 916 finished with value:
-1.30331227727624 and parameters: {'n_estimators': 328, 'learning_rate':
0.019409028831801466, 'max_depth': 10, 'subsample': 0.6926236470782853,
'colsample_bytree': 0.7736878206778809, 'min_child_weight': 5, 'gamma':
0.00012485430609920456, 'reg_alpha': 1.9735847034687475e-08, 'reg_lambda':
4.172649020940461e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:07,537] Trial 917 finished with value:
-1.305629268215542 and parameters: {'n_estimators': 364, 'learning_rate':
0.014280479699991511, 'max_depth': 10, 'subsample': 0.6683836748965085,
'colsample_bytree': 0.7607180153144544, 'min_child_weight': 5, 'gamma':
0.0005931399992594318, 'reg_alpha': 0.05896662594381521, 'reg_lambda':
6.360105534610862e-06}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:08,079] Trial 918 finished with value:
-1.3081177486128859 and parameters: {'n_estimators': 423, 'learning_rate':
0.015599209610225803, 'max_depth': 9, 'subsample': 0.7331204540346663,
'colsample_bytree': 0.746890300635984, 'min_child_weight': 5, 'gamma':
0.002008974508705954, 'reg_alpha': 0.005390793072489735, 'reg_lambda':
7.377895362177916e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:08,525] Trial 919 finished with value:
-1.3012167091077205 and parameters: {'n_estimators': 315, 'learning_rate':
0.021550278117243048, 'max_depth': 10, 'subsample': 0.7051478619712999,
'colsample_bytree': 0.7297844619595989, 'min_child_weight': 5, 'gamma':
0.0009666759056454014, 'reg_alpha': 0.0027072244621222093, 'reg_lambda':
0.00011225355410321105}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:08,891] Trial 920 finished with value:
-1.3185636515130807 and parameters: {'n_estimators': 279, 'learning_rate':
0.017562981518879296, 'max_depth': 9, 'subsample': 0.6810314877732304,
'colsample_bytree': 0.7828031091697919, 'min_child_weight': 5, 'gamma':
0.0013778961763180103, 'reg_alpha': 1.2824769914183307e-08, 'reg_lambda':
0.0014644241292266534}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:09,223] Trial 921 finished with value:
-1.3190255731411058 and parameters: {'n_estimators': 250, 'learning_rate':
0.018822784096265654, 'max_depth': 10, 'subsample': 0.7164991790660159,
'colsample_bytree': 0.7650945834284377, 'min_child_weight': 8, 'gamma':
0.0005024928622356373, 'reg_alpha': 2.719311474500033e-08, 'reg_lambda':
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5.269135057770625e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:09,617] Trial 922 finished with value:
-1.3093397366461614 and parameters: {'n_estimators': 351, 'learning_rate':
0.016378388741710423, 'max_depth': 8, 'subsample': 0.6967701135396941,
'colsample_bytree': 0.7495226127214601, 'min_child_weight': 5, 'gamma':
0.00111075439512578, 'reg_alpha': 0.01453249587357513, 'reg_lambda':
0.0006196424099770459}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:10,103] Trial 923 finished with value:
-1.3066589063459622 and parameters: {'n_estimators': 333, 'learning_rate':
0.020210960867981655, 'max_depth': 10, 'subsample': 0.7091031895547808,
'colsample_bytree': 0.716133339295749, 'min_child_weight': 5, 'gamma':
0.004374914544787512, 'reg_alpha': 1.4019538068245353e-08, 'reg_lambda':
0.0004145748913099011}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:10,629] Trial 924 finished with value:
-1.3027395116637217 and parameters: {'n_estimators': 292, 'learning_rate':
0.017355743715374346, 'max_depth': 11, 'subsample': 0.7252123087543775,
'colsample_bytree': 0.7974951455507602, 'min_child_weight': 5, 'gamma':
0.0007588447365456369, 'reg_alpha': 0.0085796294916977, 'reg_lambda':
1.215966954301664e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:11,239] Trial 925 finished with value:
-1.3047875948415852 and parameters: {'n_estimators': 444, 'learning_rate':
0.015230326660075113, 'max_depth': 10, 'subsample': 0.7389282789040474,
'colsample_bytree': 0.7377505498502923, 'min_child_weight': 5, 'gamma':
0.0018272663953651685, 'reg_alpha': 3.341242616584301e-08, 'reg_lambda':
2.648210191661348e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:11,635] Trial 926 finished with value:
-1.3084760872721644 and parameters: {'n_estimators': 312, 'learning_rate':
0.018438206290263333, 'max_depth': 9, 'subsample': 0.6892540703244335,
'colsample_bytree': 0.7724226616564723, 'min_child_weight': 6, 'gamma':
0.002873148726544392, 'reg_alpha': 1.91592502108663e-08, 'reg_lambda':
0.00019603504534267837}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:12,152] Trial 927 finished with value:
-1.3057721495730708 and parameters: {'n_estimators': 338, 'learning_rate':
0.016873820801571446, 'max_depth': 10, 'subsample': 0.7180691666596102,
'colsample_bytree': 0.7566193134128488, 'min_child_weight': 5, 'gamma':
0.0003478654179792204, 'reg_alpha': 9.634120307549302e-05, 'reg_lambda':
0.002158713559781056}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:12,579] Trial 928 finished with value:
-1.3422969307144914 and parameters: {'n_estimators': 371, 'learning_rate':
0.07042028298721974, 'max_depth': 9, 'subsample': 0.703040312836004,
'colsample_bytree': 0.7388121691229257, 'min_child_weight': 5, 'gamma':
0.0006602011916832084, 'reg_alpha': 0.5636338846397795, 'reg_lambda':
0.0007119180253653396}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:13,030] Trial 929 finished with value:
-1.3083133481760703 and parameters: {'n_estimators': 307, 'learning_rate':
0.019752695469847526, 'max_depth': 10, 'subsample': 0.7299768368781462,
'colsample_bytree': 0.7524783333973429, 'min_child_weight': 5, 'gamma':
0.0012415419978419084, 'reg_alpha': 0.02331450244149663, 'reg_lambda':
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0.00013888069212014518}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:13,510] Trial 930 finished with value:
-1.3065016095183595 and parameters: {'n_estimators': 403, 'learning_rate':
0.018228634445101935, 'max_depth': 10, 'subsample': 0.7104468060392687,
'colsample_bytree': 0.7793463877524072, 'min_child_weight': 5, 'gamma':
0.0009602229479444406, 'reg_alpha': 1.9088524717997768e-08, 'reg_lambda':
0.00032004217725592476}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:13,842] Trial 931 finished with value:
-1.3241801047379989 and parameters: {'n_estimators': 265, 'learning_rate':
0.01410377618348966, 'max_depth': 8, 'subsample': 0.6953280168278277,
'colsample_bytree': 0.7622230096863153, 'min_child_weight': 5, 'gamma':
0.0004265409997673262, 'reg_alpha': 1.4369932371491524e-08, 'reg_lambda':
6.293062532699774e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:14,314] Trial 932 finished with value:
-1.3107073077564282 and parameters: {'n_estimators': 352, 'learning_rate':
0.015934689558879684, 'max_depth': 9, 'subsample': 0.7240618595010644,
'colsample_bytree': 0.7231807548998765, 'min_child_weight': 5, 'gamma':
0.006297283224733272, 'reg_alpha': 1.0057222168480283e-08, 'reg_lambda':
0.26275103247250914}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:14,714] Trial 933 finished with value:
-1.3092910743245856 and parameters: {'n_estimators': 288, 'learning_rate':
0.021673992506132802, 'max_depth': 10, 'subsample': 0.6804118282666963,
'colsample_bytree': 0.7434240876572573, 'min_child_weight': 5, 'gamma':
0.011051442477602603, 'reg_alpha': 2.108829474008328e-06, 'reg_lambda':
0.001327488667675315}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:15,056] Trial 934 finished with value:
-1.4400580938210137 and parameters: {'n_estimators': 325, 'learning_rate':
0.19304917650039313, 'max_depth': 9, 'subsample': 0.7139590447115765,
'colsample_bytree': 0.769499843119324, 'min_child_weight': 5, 'gamma':
0.0022581786685668464, 'reg_alpha': 2.716332395723155e-08, 'reg_lambda':
1.7797453865341874e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:15,577] Trial 935 finished with value:
-1.3154115485163151 and parameters: {'n_estimators': 341, 'learning_rate':
0.017753135892576388, 'max_depth': 11, 'subsample': 0.7360302524865322,
'colsample_bytree': 0.7879116977310812, 'min_child_weight': 6, 'gamma':
0.0015726053919673183, 'reg_alpha': 0.011570163901509542, 'reg_lambda':
3.244905699932747e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:16,131] Trial 936 finished with value:
-1.3046337873152705 and parameters: {'n_estimators': 381, 'learning_rate':
0.01633967524654827, 'max_depth': 10, 'subsample': 0.6987238177545453,
'colsample_bytree': 0.8090521308409977, 'min_child_weight': 5, 'gamma':
0.0006869633228415937, 'reg_alpha': 0.01731860185556684, 'reg_lambda':
7.741436463836906e-06}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:16,673] Trial 937 finished with value:
-1.3040430059385557 and parameters: {'n_estimators': 420, 'learning_rate':
0.02052802126170873, 'max_depth': 10, 'subsample': 0.7212089232681639,
'colsample_bytree': 0.7298449517026936, 'min_child_weight': 5, 'gamma':
0.0010101537662459239, 'reg_alpha': 0.035061727734210485, 'reg_lambda':
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0.0005806963925133494}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:17,034] Trial 938 finished with value:
-1.3330592527611023 and parameters: {'n_estimators': 313, 'learning_rate':
0.05667479303267643, 'max_depth': 9, 'subsample': 0.7462493329990156,
'colsample_bytree': 0.7526951679816308, 'min_child_weight': 5, 'gamma':
0.0006122990485281641, 'reg_alpha': 3.5369713401067384e-08, 'reg_lambda':
0.0009486410964871304}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:17,587] Trial 939 finished with value:
-1.3063262444472816 and parameters: {'n_estimators': 367, 'learning_rate':
0.019115273042974657, 'max_depth': 11, 'subsample': 0.7068436984960506,
'colsample_bytree': 0.7634990221356085, 'min_child_weight': 5, 'gamma':
0.00026232469358535053, 'reg_alpha': 1.3368255302220233e-08, 'reg_lambda':
4.6689495939815493e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:17,971] Trial 940 finished with value:
-1.316706136286409 and parameters: {'n_estimators': 463, 'learning_rate':
0.023445289381753564, 'max_depth': 7, 'subsample': 0.689132591414812,
'colsample_bytree': 0.7430894513351897, 'min_child_weight': 5, 'gamma':
0.0015599934253196694, 'reg_alpha': 2.2587786855481758e-08, 'reg_lambda':
8.443207630216857e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:18,347] Trial 941 finished with value:
-1.3079830192603792 and parameters: {'n_estimators': 296, 'learning_rate':
0.01518095896645376, 'max_depth': 10, 'subsample': 0.7299351511517312,
'colsample_bytree': 0.7117687555726141, 'min_child_weight': 5, 'gamma':
0.0032712911746284232, 'reg_alpha': 2.717888415416895e-08, 'reg_lambda':
0.0002480118086649343}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:18,794] Trial 942 finished with value:
-1.3232825141718891 and parameters: {'n_estimators': 326, 'learning_rate':
0.03864145416433717, 'max_depth': 10, 'subsample': 0.7123733671193627,
'colsample_bytree': 0.7764942747844171, 'min_child_weight': 5, 'gamma':
0.000178400714795686, 'reg_alpha': 2.0117871175207287e-08, 'reg_lambda':
0.00045061687330588966}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:19,189] Trial 943 finished with value:
-1.3103185920878448 and parameters: {'n_estimators': 350, 'learning_rate':
0.017373764759504227, 'max_depth': 9, 'subsample': 0.7016261297700759,
'colsample_bytree': 0.7561928200274142, 'min_child_weight': 6, 'gamma':
0.00046126951665630784, 'reg_alpha': 0.005991732973776074, 'reg_lambda':
0.0025553872909566885}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:19,584] Trial 944 finished with value:
-1.2982870221585288 and parameters: {'n_estimators': 283, 'learning_rate':
0.018990866032976856, 'max_depth': 10, 'subsample': 0.7206857123156732,
'colsample_bytree': 0.7307330148885012, 'min_child_weight': 5, 'gamma':
0.0008896884290202636, 'reg_alpha': 1.280466205373433e-08, 'reg_lambda':
3.2726956810267096e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:19,989] Trial 945 finished with value:
-1.3059199615149306 and parameters: {'n_estimators': 272, 'learning_rate':
0.016613539696112987, 'max_depth': 10, 'subsample': 0.7034015757349861,
'colsample_bytree': 0.7172096888369459, 'min_child_weight': 5, 'gamma':
0.001280840671070558, 'reg_alpha': 0.009709044606602867, 'reg_lambda':
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2.4064255002542996e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:20,522] Trial 946 finished with value:
-1.3088408700238439 and parameters: {'n_estimators': 436, 'learning_rate':
0.018344136730901274, 'max_depth': 9, 'subsample': 0.7174511047410026,
'colsample_bytree': 0.7099739711119992, 'min_child_weight': 5, 'gamma':
0.0019423809926259101, 'reg_alpha': 0.02187186059108823, 'reg_lambda':
3.265512274656982e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:20,887] Trial 947 finished with value:
-1.3182579810040793 and parameters: {'n_estimators': 228, 'learning_rate':
0.014425467732801128, 'max_depth': 10, 'subsample': 0.6881212749552081,
'colsample_bytree': 0.7014814144182371, 'min_child_weight': 5, 'gamma':
0.0009270605331391231, 'reg_alpha': 1.0305222468265611e-08, 'reg_lambda':
1.6850528671113278e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:21,308] Trial 948 finished with value:
-1.3071400572090732 and parameters: {'n_estimators': 271, 'learning_rate':
0.015458077091175639, 'max_depth': 10, 'subsample': 0.7102217271532405,
'colsample_bytree': 0.7289345713447126, 'min_child_weight': 5, 'gamma':
0.0022659968789987034, 'reg_alpha': 1.7184681073486777e-08, 'reg_lambda':
3.289254495234801e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:21,800] Trial 949 finished with value:
-1.312743932891328 and parameters: {'n_estimators': 399, 'learning_rate':
0.017154981837795753, 'max_depth': 9, 'subsample': 0.6714526788124148,
'colsample_bytree': 0.7239226796883642, 'min_child_weight': 5, 'gamma':
0.004556158208556429, 'reg_alpha': 3.863966852173156e-08, 'reg_lambda':
2.260023342019936e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:22,226] Trial 950 finished with value:
-1.3140338397264977 and parameters: {'n_estimators': 249, 'learning_rate':
0.019639465459427335, 'max_depth': 11, 'subsample': 0.856863433039884,
'colsample_bytree': 0.7338475203402541, 'min_child_weight': 5, 'gamma':
0.0013440050553635491, 'reg_alpha': 5.346808537111782e-05, 'reg_lambda':
5.575672405227863e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:22,401] Trial 951 finished with value:
-1.3473683787392083 and parameters: {'n_estimators': 283, 'learning_rate':
0.021404211186786908, 'max_depth': 5, 'subsample': 0.6948138475016877,
'colsample_bytree': 0.7363744611879786, 'min_child_weight': 6, 'gamma':
0.0009281308889703798, 'reg_alpha': 0.014614344560026034, 'reg_lambda':
4.560197147648861e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:22,785] Trial 952 finished with value:
-1.3075336961300283 and parameters: {'n_estimators': 260, 'learning_rate':
0.018099461383546497, 'max_depth': 10, 'subsample': 0.719457389827171,
'colsample_bytree': 0.7185392715899843, 'min_child_weight': 5, 'gamma':
0.0005402375460039425, 'reg_alpha': 1.3268512652429829e-08, 'reg_lambda':
1.184801770175921e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:23,166] Trial 953 finished with value:
-1.309159556036619 and parameters: {'n_estimators': 366, 'learning_rate':
0.01604626431081164, 'max_depth': 8, 'subsample': 0.701004768090315,
'colsample_bytree': 0.7442520846343202, 'min_child_weight': 5, 'gamma':
0.0016744859461791089, 'reg_alpha': 2.4046403779036562e-08, 'reg_lambda':
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3.605227921444662e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:23,549] Trial 954 finished with value: -1.3133657485849464 and parameters: {'n_estimators': 293, 'learning_rate': 0.01911343104311368, 'max_depth': 9, 'subsample': 0.6817309546721374, 'colsample_bytree': 0.6955362435917175, 'min_child_weight': 5, 'gamma': 0.0028026840661661096, 'reg_alpha': 0.031148634056798182, 'reg_lambda': 1.5706931106336735e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:23,903] Trial 955 finished with value: -1.4739456685507488 and parameters: {'n_estimators': 335, 'learning_rate': 0.2739931135423994, 'max_depth': 10, 'subsample': 0.7102693606779019, 'colsample_bytree': 0.7330514194066475, 'min_child_weight': 5, 'gamma': 0.0012212082699436446, 'reg_alpha': 1.8922781773402242e-08, 'reg_lambda': 7.507214102797587e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:24,415] Trial 956 finished with value: -1.308341880240631 and parameters: {'n_estimators': 387, 'learning_rate': 0.01722320513067178, 'max_depth': 10, 'subsample': 0.7342112652153651, 'colsample_bytree': 0.7489915288974368, 'min_child_weight': 5, 'gamma': 0.0005962251198140895, 'reg_alpha': 3.786297625495609e-08, 'reg_lambda': 2.588596085454011e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:24,831] Trial 957 finished with value: -1.427793362547992 and parameters: {'n_estimators': 416, 'learning_rate': 0.1693187399562397, 'max_depth': 9, 'subsample': 0.7233654729122645, 'colsample_bytree': 0.7390821356545699, 'min_child_weight': 5, 'gamma': 0.0008338142266074272, 'reg_alpha': 1.0132676665746165e-08, 'reg_lambda': 4.087067983203667e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:25,300] Trial 958 finished with value: -1.2995279726620885 and parameters: {'n_estimators': 315, 'learning_rate': 0.020243023363445935, 'max_depth': 10, 'subsample': 0.7070232712992776, 'colsample_bytree': 0.7494865763317771, 'min_child_weight': 4, 'gamma': 0.0003627544859709625, 'reg_alpha': 2.7520537266881606e-08, 'reg_lambda': 5.819640111159996e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:25,748] Trial 959 finished with value: -1.310112416998247 and parameters: {'n_estimators': 287, 'learning_rate': 0.022214693096185827, 'max_depth': 9, 'subsample': 0.9209327287215973, 'colsample_bytree': 0.7588086627238929, 'min_child_weight': 4, 'gamma': 0.00012947923653379963, 'reg_alpha': 4.444894576399639e-08, 'reg_lambda': 6.402950764064871e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:26,878] Trial 960 finished with value: -1.333009767705448 and parameters: {'n_estimators': 915, 'learning_rate': 0.023714817647598852, 'max_depth': 11, 'subsample': 0.6924782756068586, 'colsample_bytree': 0.7273487617801626, 'min_child_weight': 4, 'gamma': 0.00024307391451608011, 'reg_alpha': 2.579392895553215e-08, 'reg_lambda': 5.097016543312196e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:27,296] Trial 961 finished with value: -1.3022202269754561 and parameters: {'n_estimators': 309, 'learning_rate': 0.021055872014013165, 'max_depth': 10, 'subsample': 0.6996257235539309, 'colsample_bytree': 0.7517569085752825, 'min_child_weight': 4, 'gamma': 0.00027954370009980483, 'reg_alpha': 4.112334623093938e-08, 'reg_lambda':

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4.8748588932098186e-05}. Best is trial 794 with value: -1.297373640685575.
[I 2026-03-13 19:16:27,701] Trial 962 finished with value:
-1.296562810511752 and parameters: {'n_estimators': 256, 'learning_rate':
0.020012253811940252, 'max_depth': 10, 'subsample': 0.6665846829614864,
'colsample_bytree': 0.7638386854926835, 'min_child_weight': 4, 'gamma':
0.00033931717208469817, 'reg_alpha': 3.021289454045435e-08, 'reg_lambda':
3.26613626701184e-05}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:28,127] Trial 963 finished with value:
-1.3028559754389508 and parameters: {'n_estimators': 258, 'learning_rate':
0.023935324918800738, 'max_depth': 10, 'subsample': 0.6632425989858008,
'colsample_bytree': 0.7668903774084245, 'min_child_weight': 4, 'gamma':
0.00017622356865849509, 'reg_alpha': 1.850379987392429e-08, 'reg_lambda':
2.899582979703346e-05}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:28,502] Trial 964 finished with value:
-1.3050205333149933 and parameters: {'n_estimators': 252, 'learning_rate':
0.02713282193242486, 'max_depth': 10, 'subsample': 0.6864382228153595,
'colsample_bytree': 0.7672395310909954, 'min_child_weight': 4, 'gamma':
0.000328391118158834, 'reg_alpha': 3.024778474528004e-08, 'reg_lambda':
3.2954705540122124e-05}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:28,882] Trial 965 finished with value:
-1.2967164013481955 and parameters: {'n_estimators': 244, 'learning_rate':
0.02179435722187038, 'max_depth': 10, 'subsample': 0.6660086896484012,
'colsample_bytree': 0.7466792423088894, 'min_child_weight': 4, 'gamma':
0.00033900824683431326, 'reg_alpha': 2.7575509097260125e-08, 'reg_lambda':
2.313987366226712e-05}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:29,225] Trial 966 finished with value:
-1.2984199130927356 and parameters: {'n_estimators': 213, 'learning_rate':
0.023800375350880996, 'max_depth': 10, 'subsample': 0.6814692215517391,
'colsample_bytree': 0.7180839857441531, 'min_child_weight': 4, 'gamma':
0.00021219354998472517, 'reg_alpha': 1.5572482160870747e-08, 'reg_lambda':
1.6050396894911126e-05}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:29,590] Trial 967 finished with value:
-1.3065551449268007 and parameters: {'n_estimators': 225, 'learning_rate':
0.024332753035906922, 'max_depth': 10, 'subsample': 0.6695347696849161,
'colsample_bytree': 0.7006613550656173, 'min_child_weight': 4, 'gamma':
0.00015574963596899798, 'reg_alpha': 1.4088578846813447e-08, 'reg_lambda':
1.8298608354518318e-05}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:29,934] Trial 968 finished with value:
-1.3008332971125809 and parameters: {'n_estimators': 206, 'learning_rate':
0.023030066158048833, 'max_depth': 10, 'subsample': 0.6680888554346204,
'colsample_bytree': 0.7082979794363822, 'min_child_weight': 4, 'gamma':
0.0001692871119699674, 'reg_alpha': 2.1358648166835434e-08, 'reg_lambda':
1.3122283084293544e-05}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:30,281] Trial 969 finished with value:
-1.2998293902040923 and parameters: {'n_estimators': 198, 'learning_rate':
0.026105592299625162, 'max_depth': 10, 'subsample': 0.6635568310584693,
'colsample_bytree': 0.7214492327334356, 'min_child_weight': 4, 'gamma':
7.965679888865286e-05, 'reg_alpha': 1.3301445263418089e-08, 'reg_lambda':
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1.8991749209954413e-05}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:30,540] Trial 970 finished with value:
-1.3099784048971614 and parameters: {'n_estimators': 153, 'learning_rate':
0.024089037665647337, 'max_depth': 10, 'subsample': 0.6677805132479441,
'colsample_bytree': 0.6943896977756109, 'min_child_weight': 4, 'gamma':
0.00010357426120976147, 'reg_alpha': 1.4136925680091216e-08, 'reg_lambda':
1.008813012437331e-05}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:30,852] Trial 971 finished with value:
-1.3038414417605695 and parameters: {'n_estimators': 236, 'learning_rate':
0.02517760305686178, 'max_depth': 10, 'subsample': 0.6575240836853332,
'colsample_bytree': 0.7262529131592773, 'min_child_weight': 4, 'gamma':
0.0001254573003726545, 'reg_alpha': 1.0178407372340802e-08, 'reg_lambda':
1.4498911889555646e-05}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:31,187] Trial 972 finished with value:
-1.29873820051902 and parameters: {'n_estimators': 218, 'learning_rate':
0.02377961066056701, 'max_depth': 10, 'subsample': 0.6773962747992184,
'colsample_bytree': 0.7106716364300409, 'min_child_weight': 4, 'gamma':
8.825403891838357e-05, 'reg_alpha': 1.3250903243264914e-08, 'reg_lambda':
1.643267958448071e-05}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:31,514] Trial 973 finished with value:
-1.3049880267027358 and parameters: {'n_estimators': 193, 'learning_rate':
0.023611104220144973, 'max_depth': 10, 'subsample': 0.6657785746754586,
'colsample_bytree': 0.6914566610269317, 'min_child_weight': 4, 'gamma':
0.00011243704637772349, 'reg_alpha': 1.360589007770969e-08, 'reg_lambda':
2.0235985041894703e-05}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:31,814] Trial 974 finished with value:
-1.3066577229976617 and parameters: {'n_estimators': 204, 'learning_rate':
0.026950936618317267, 'max_depth': 10, 'subsample': 0.6693345444570113,
'colsample_bytree': 0.6980453147049036, 'min_child_weight': 4, 'gamma':
6.673559427133286e-05, 'reg_alpha': 1.8188594305797857e-08, 'reg_lambda':
1.5254399266584187e-05}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:32,135] Trial 975 finished with value:
-1.3089435730529493 and parameters: {'n_estimators': 202, 'learning_rate':
0.025911413707252605, 'max_depth': 10, 'subsample': 0.6608793192328317,
'colsample_bytree': 0.7195981920008876, 'min_child_weight': 4, 'gamma':
6.025303005228034e-05, 'reg_alpha': 1.4674751555178927e-08, 'reg_lambda':
9.280331957599619e-06}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:32,447] Trial 976 finished with value:
-1.3018434991888876 and parameters: {'n_estimators': 212, 'learning_rate':
0.028543811478281743, 'max_depth': 10, 'subsample': 0.6620680459468344,
'colsample_bytree': 0.711314026803311, 'min_child_weight': 4, 'gamma':
9.238122839571363e-05, 'reg_alpha': 2.1888316280708888e-08, 'reg_lambda':
1.0953100294339026e-05}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:32,769] Trial 977 finished with value:
-1.2988608167511344 and parameters: {'n_estimators': 180, 'learning_rate':
0.025206473216426887, 'max_depth': 10, 'subsample': 0.6684527592669807,
'colsample_bytree': 0.6994397957064131, 'min_child_weight': 4, 'gamma':
0.0001520435799179849, 'reg_alpha': 2.545809351819555e-08, 'reg_lambda':
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2.0276081377788843e-05}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:33,077] Trial 978 finished with value:
-1.3035898038951734 and parameters: {'n_estimators': 180, 'learning_rate':
0.025771239484385325, 'max_depth': 10, 'subsample': 0.6613630016088196,
'colsample_bytree': 0.7000895549465307, 'min_child_weight': 4, 'gamma':
0.0002023715181282643, 'reg_alpha': 2.688770987956318e-08, 'reg_lambda':
1.7200524131070307e-05}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:33,325] Trial 979 finished with value:
-1.3053303401158445 and parameters: {'n_estimators': 190, 'learning_rate':
0.025513936909763942, 'max_depth': 10, 'subsample': 0.6719338837842077,
'colsample_bytree': 0.685978224386488, 'min_child_weight': 4, 'gamma':
8.017831241953798e-05, 'reg_alpha': 2.6990901882134973e-08, 'reg_lambda':
1.0083441341623876e-05}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:33,572] Trial 980 finished with value:
-1.323560844638757 and parameters: {'n_estimators': 125, 'learning_rate':
0.027569699661811907, 'max_depth': 10, 'subsample': 0.6716130497154539,
'colsample_bytree': 0.7058119427581572, 'min_child_weight': 4, 'gamma':
0.00015036697737704012, 'reg_alpha': 1.8003173016228104e-08, 'reg_lambda':
2.052437152671942e-05}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:33,891] Trial 981 finished with value:
-1.304797501623953 and parameters: {'n_estimators': 218, 'learning_rate':
0.029708276260968117, 'max_depth': 10, 'subsample': 0.6593036318431428,
'colsample_bytree': 0.6780455713477901, 'min_child_weight': 4, 'gamma':
0.00014017320488549415, 'reg_alpha': 1.3490681336696888e-08, 'reg_lambda':
2.2089792068179777e-05}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:34,171] Trial 982 finished with value:
-1.3123060804118525 and parameters: {'n_estimators': 163, 'learning_rate':
0.026449270143228124, 'max_depth': 10, 'subsample': 0.6643015627229232,
'colsample_bytree': 0.6969855084817721, 'min_child_weight': 4, 'gamma':
7.858053117425387e-05, 'reg_alpha': 2.9161006976835414e-08, 'reg_lambda':
8.525920254823503e-06}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:34,474] Trial 983 finished with value:
-1.3028455688818157 and parameters: {'n_estimators': 173, 'learning_rate':
0.026782749543169372, 'max_depth': 11, 'subsample': 0.6741819377205056,
'colsample_bytree': 0.7063142907231004, 'min_child_weight': 4, 'gamma':
0.00012043541680118216, 'reg_alpha': 1.013176427239449e-08, 'reg_lambda':
1.2494372391998639e-05}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:34,773] Trial 984 finished with value:
-1.3095542931314068 and parameters: {'n_estimators': 170, 'learning_rate':
0.024020146209006756, 'max_depth': 10, 'subsample': 0.6726813731634478,
'colsample_bytree': 0.6800156483152103, 'min_child_weight': 4, 'gamma':
0.00019843587829099357, 'reg_alpha': 2.117561338828668e-08, 'reg_lambda':
1.545822426912162e-05}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:35,084] Trial 985 finished with value:
-1.302019368546018 and parameters: {'n_estimators': 187, 'learning_rate':
0.023300529314852374, 'max_depth': 10, 'subsample': 0.6571848072829758,
'colsample_bytree': 0.6884777052675564, 'min_child_weight': 4, 'gamma':
0.00019490971388666257, 'reg_alpha': 4.612841248312709e-08, 'reg_lambda':
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2.1611932826827013e-05}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:35,410] Trial 986 finished with value:
-1.3044540272484688 and parameters: {'n_estimators': 194, 'learning_rate':
0.028242654893487363, 'max_depth': 10, 'subsample': 0.6771117439694826,
'colsample_bytree': 0.7109755558055739, 'min_child_weight': 4, 'gamma':
6.598027066948517e-05, 'reg_alpha': 1.7736841157600738e-08, 'reg_lambda':
1.4922318037868827e-05}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:35,752] Trial 987 finished with value:
-1.3007087583493024 and parameters: {'n_estimators': 222, 'learning_rate':
0.02281604265777739, 'max_depth': 10, 'subsample': 0.6767601427913382,
'colsample_bytree': 0.7046286449284346, 'min_child_weight': 4, 'gamma':
0.00015229300423465268, 'reg_alpha': 3.262887269947272e-08, 'reg_lambda':
5.2194232757133095e-06}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:36,028] Trial 988 finished with value:
-1.3017300397239389 and parameters: {'n_estimators': 217, 'learning_rate':
0.025112932966961078, 'max_depth': 10, 'subsample': 0.666900964759516,
'colsample_bytree': 0.7124122506268736, 'min_child_weight': 4, 'gamma':
0.00010337140553413986, 'reg_alpha': 1.337140880056377e-08, 'reg_lambda':
2.2653172216327107e-05}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:36,349] Trial 989 finished with value:
-1.3065088973376289 and parameters: {'n_estimators': 199, 'learning_rate':
0.03157820565751664, 'max_depth': 11, 'subsample': 0.6767428524150088,
'colsample_bytree': 0.7120228616483122, 'min_child_weight': 4, 'gamma':
0.00022473809237375306, 'reg_alpha': 2.5853094293068007e-08, 'reg_lambda':
7.771346456426015e-06}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:36,680] Trial 990 finished with value:
-1.308912802524462 and parameters: {'n_estimators': 211, 'learning_rate':
0.021820588935083082, 'max_depth': 10, 'subsample': 0.6597229969211263,
'colsample_bytree': 0.6934124575396049, 'min_child_weight': 4, 'gamma':
9.893020142372529e-05, 'reg_alpha': 1.020114397214067e-08, 'reg_lambda':
1.4114593061970619e-05}. Best is trial 962 with value: -1.296562810511752.
[I 2026-03-13 19:16:37,036] Trial 991 finished with value:
-1.2959463136220624 and parameters: {'n_estimators': 238, 'learning_rate':
0.024893528510341163, 'max_depth': 10, 'subsample': 0.6648252398755385,
'colsample_bytree': 0.7170908662074246, 'min_child_weight': 4, 'gamma':
0.000237574238044691, 'reg_alpha': 1.846656718857583e-08, 'reg_lambda':
1.0553507893726066e-05}. Best is trial 991 with value: -1.2959463136220624.
[I 2026-03-13 19:16:37,411] Trial 992 finished with value:
-1.3119585786908292 and parameters: {'n_estimators': 233, 'learning_rate':
0.024525962179493765, 'max_depth': 10, 'subsample': 0.6799837934391205,
'colsample_bytree': 0.6855826227193064, 'min_child_weight': 4, 'gamma':
8.791934865415865e-05, 'reg_alpha': 1.8572839918517196e-08, 'reg_lambda':
1.0170927307353227e-05}. Best is trial 991 with value: -1.2959463136220624.
[I 2026-03-13 19:16:37,710] Trial 993 finished with value:
-1.3103676308084966 and parameters: {'n_estimators': 154, 'learning_rate':
0.0253464303907562, 'max_depth': 10, 'subsample': 0.6616298679500476,
'colsample_bytree': 0.695793376975512, 'min_child_weight': 4, 'gamma':
0.00018628933038597345, 'reg_alpha': 2.753899786267102e-08, 'reg_lambda':
```

```
5.874486479363064e-06}. Best is trial 991 with value: -1.2959463136220624.
[I 2026-03-13 19:16:38,074] Trial 994 finished with value:
-1.3073186768464389 and parameters: {'n_estimators': 228, 'learning_rate':
0.026852213015965975, 'max_depth': 10, 'subsample': 0.6684654867684399,
'colsample_bytree': 0.709744647538295, 'min_child_weight': 4, 'gamma':
0.00023095811043267390, 'reg_alpha': 1.87294383112711e-08, 'reg_lambda':
1.057008028310048e-05}. Best is trial 991 with value: -1.2959463136220624.
[I 2026-03-13 19:16:38,399] Trial 995 finished with value:
-1.3068910623281382 and parameters: {'n_estimators': 192, 'learning_rate':
0.02405069524637866, 'max_depth': 10, 'subsample': 0.6562580485285254,
'colsample_bytree': 0.7047787489899829, 'min_child_weight': 4, 'gamma':
0.00023999567181163473, 'reg_alpha': 3.5710631340733184e-08, 'reg_lambda':
7.2912689091940186e-06}. Best is trial 991 with value: -1.2959463136220624.
[I 2026-03-13 19:16:38,786] Trial 996 finished with value:
-1.3038546012094183 and parameters: {'n_estimators': 231, 'learning_rate':
0.02281440320678455, 'max_depth': 11, 'subsample': 0.6548908055659481,
'colsample_bytree': 0.6912916756334117, 'min_child_weight': 4, 'gamma':
0.00015069584232265980, 'reg_alpha': 1.4563556882017327e-08, 'reg_lambda':
1.4534864137432251e-05}. Best is trial 991 with value: -1.2959463136220624.
[I 2026-03-13 19:16:39,084] Trial 997 finished with value:
-1.3009447907402354 and parameters: {'n_estimators': 176, 'learning_rate':
0.02513420209879101, 'max_depth': 10, 'subsample': 0.6774535771761658,
'colsample_bytree': 0.7157920771254711, 'min_child_weight': 4, 'gamma':
0.00013341259280126970, 'reg_alpha': 2.2183933234644935e-08, 'reg_lambda':
1.9557589363223554e-05}. Best is trial 991 with value: -1.2959463136220624.
[I 2026-03-13 19:16:39,407] Trial 998 finished with value:
-1.3081746327889163 and parameters: {'n_estimators': 214, 'learning_rate':
0.03476059572523435, 'max_depth': 10, 'subsample': 0.664075855774584,
'colsample_bytree': 0.7180344240136862, 'min_child_weight': 4, 'gamma':
7.968709660543028e-05, 'reg_alpha': 4.995125909746383e-08, 'reg_lambda':
2.3619796225426285e-05}. Best is trial 991 with value: -1.2959463136220624.
[I 2026-03-13 19:16:39,782] Trial 999 finished with value:
-1.3039955970297583 and parameters: {'n_estimators': 230, 'learning_rate':
0.022352713703134207, 'max_depth': 10, 'subsample': 0.6789120695856109,
'colsample_bytree': 0.6985020750168021, 'min_child_weight': 4, 'gamma':
0.00023623811620267313, 'reg_alpha': 1.6150095904431542e-08, 'reg_lambda':
1.2164360424456178e-05}. Best is trial 991 with value: -1.2959463136220624.

Best Optuna parameters: {'n_estimators': 238, 'learning_rate':
0.024893528510341163, 'max_depth': 10, 'subsample': 0.6648252398755385,
'colsample_bytree': 0.7170908662074246, 'min_child_weight': 4, 'gamma':
0.000237574238044691, 'reg_alpha': 1.846656718857583e-08, 'reg_lambda':
1.0553507893726066e-05}


Tuning completed in 435.21 seconds
Model:                   XGBoost (Optuna tuned)
MAE (log):               1.3669
```

```
RMSE (log):                 1.9002
MAE (minutes):              1731.78 minutes
RMSE (minutes):             3663.05 minutes
R^2:                        0.5310
Training Time:              0.17 seconds
```

[54]:
```
class OutageHurdleModel(BaseEstimator, RegressorMixin):
    def __init__(self, classifier, regressor):
        self.classifier = classifier
        self.regressor = regressor

    def _get_augmented_X(self, X, probs):
        # Adds the probability column as a new feature to the input matrix
        return np.column_stack([X, probs])

    def fit(self, X, y):
        # Stage 1: Train the gatekeeper
        y_binary = (y > 0).astype(int)
        print("Fitting Stage 1: Hurdle Classifier...")
        self.classifier.fit(X, y_binary)

        # Get probabilities for the training set to use as a feature for Stage 2
        probs = self.classifier.predict_proba(X)[:, 1]
        X_augmented = self._get_augmented_X(X, probs)

        # Stage 2: Train the estimator only on rows where duration > 0
        print("Fitting Stage 2: Duration Regressor (with Stage 1 context)...")
        mask = y_binary == 1
        self.regressor.fit(X_augmented[mask], y[mask])
        return self

    def predict(self, X):
        # 1. Get Stage 1 Probability
        probs = self.classifier.predict_proba(X)[:, 1]

        # 2. Augment X with that probability
        X_augmented = self._get_augmented_X(X, probs)

        # 3. Predict Raw Duration
        raw_preds = self.regressor.predict(X_augmented)

        # 4. Soft Hurdle: Probability * Predicted Duration
        return probs * raw_preds
```

[55]:
```
# 1. Manually transform the features to numeric matrices
# This ensures Stage 1 and Stage 2 are looking at the same numeric data
X_train_proc = preprocessor_dense.fit_transform(X_train)
```

```python
X_test_proc = preprocessor_dense.transform(X_test)

# 2. Initialize the sub-models with your best-found parameters
h_clf = RandomForestClassifier(n_estimators=200, max_depth=10, random_state=42)
h_reg = XGBRegressor(**optuna_xgb.named_steps['regressor'].get_params()) # Use␣
↪the best XGBoost params found with Optuna

# 3. Initialize and fit the Hurdle Model
reaching_model = OutageHurdleModel(classifier=h_clf, regressor=h_reg)
reaching_model.fit(X_train_proc, y_train.values)

# 4. Generate Predictions
hurdle_preds = reaching_model.predict(X_test_proc)
```

```
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
```

[56]:

```python
y_test_minutes = np.expm1(y_test)
hurdle_preds_minutes = np.expm1(hurdle_preds)

# Calculate Final Metrics
OHM_metrics = {
    'Model': "Advanced Hurdle (RF + XGB Stacked)",
    'MAE (log)': mean_absolute_error(y_test, hurdle_preds),
    'RMSE (log)': root_mean_squared_error(y_test, hurdle_preds),
    'MAE (minutes)': mean_absolute_error(y_test_minutes, hurdle_preds_minutes),
    'RMSE (minutes)': root_mean_squared_error(y_test_minutes,␣
↪hurdle_preds_minutes),
    'R^2': r2_score(y_test, hurdle_preds)
}

print("\n--- REAching MODEL PERFORMANCE ---")
for k, v in OHM_metrics.items():
    if isinstance(v, float):
        print(f"{k:15}: {v:.4f}")
    else:
        print(f"{k:15}: {v}")

results_log.append(OHM_metrics)
```

```
--- REAching MODEL PERFORMANCE ---
Model          : Advanced Hurdle (RF + XGB Stacked)
MAE (log)      : 1.3898
RMSE (log)     : 1.8705
MAE (minutes)  : 1786.1129
RMSE (minutes) : 3849.9978
R^2            : 0.5456
```

[57]:
```
# Residuals for the OHM model
residual_df_ohm = pd.DataFrame({
    'Predicted Log Duration': hurdle_preds,
    'Residuals (log)': y_test - hurdle_preds,
    'Predicted Duration (min)': hurdle_preds_minutes,
    'Residuals (min)': y_test_minutes - hurdle_preds_minutes
})
ohm_fig = plot_log_and_reg_residuals(residual_df_ohm, "Advanced Hurdle Model")

# Export plot for website
ohm_fig.write_html('docs/assets/plots/advanced_hurdle_model_residuals.html',␣
 ↪include_plotlyjs='cdn')
```

[58]:
```
def objective_hurdle(trial):
    # Random Forest Classifier params
    clf_params = {
        'n_estimators': trial.suggest_int('clf_n_estimators', 100, 800),
        'max_depth': trial.suggest_int('clf_max_depth', 3, 25),
        'min_samples_leaf': trial.suggest_int('clf_min_samples_leaf', 2, 20),
        'min_samples_split': trial.suggest_int('clf_min_samples_split', 2, 20),
        'max_features': trial.suggest_categorical('clf_max_features', ['sqrt',␣
 ↪'log2', 0.5]),
        'class_weight': trial.suggest_categorical('clf_class_weight',␣
 ↪[None,'balanced']),
        'random_state': 42,
        'n_jobs': -1,
    }

    # XGBoost Regressor params
    reg_params = {
        'n_estimators': trial.suggest_int('n_estimators', 100, 1000),
        'learning_rate': trial.suggest_float('learning_rate', 0.005, 0.2,␣
 ↪log=True),
        'max_depth': trial.suggest_int('max_depth', 2, 12),
        'subsample': trial.suggest_float('subsample', 0.5, 1.0),
        'colsample_bytree': trial.suggest_float('colsample_bytree', 0.4, 1.0),
        'min_child_weight': trial.suggest_int('min_child_weight', 1, 10),
        'gamma': trial.suggest_float('gamma', 1e-8, 1.0, log=True),
        'reg_alpha': trial.suggest_float('reg_alpha', 1e-8, 1.0, log=True),
        'reg_lambda': trial.suggest_float('reg_lambda', 1e-8, 1.0, log=True),
        'tree_method': 'hist',
        'random_state': 42,
        'n_jobs': -1,
    }

    # Manual cross-validation
    from sklearn.model_selection import KFold
```

```python
    kf = KFold(n_splits=3, shuffle=True, random_state=42)
    fold_mae = []

    for train_idx, val_idx in kf.split(X_train_proc):
        X_tr, X_val = X_train_proc[train_idx], X_train_proc[val_idx]
        y_tr, y_val = y_train.values[train_idx], y_train.values[val_idx]

        model = OutageHurdleModel(
            classifier=RandomForestClassifier(**clf_params),
            regressor=XGBRegressor(**reg_params),
        )
        model.fit(X_tr, y_tr)
        preds = model.predict(X_val)
        fold_mae.append(mean_absolute_error(y_val, preds))

        # Prune unpromising trials after first fold
        trial.report(np.mean(fold_mae), step=len(fold_mae))
        if trial.should_prune():
            raise optuna.exceptions.TrialPruned()

    return np.mean(fold_mae)


study = optuna.create_study(
    direction='minimize',
    sampler=TPESampler(seed=42, n_startup_trials=20),
    pruner=MedianPruner(n_startup_trials=20, n_warmup_steps=1),
)
study.optimize(objective_hurdle, n_trials=1000, show_progress_bar=True)

print("Best MAE:", study.best_value)
print("Best params:", study.best_params)
```

```
[I 2026-03-13 19:16:40,486] A new study created in memory with name:
no-name-e7fef154-0e40-4e3d-97b3-80755422c7b6

  0%|          | 0/1000 [00:00<?, ?it/s]

Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:16:44,545] Trial 0 finished with value:
1.392185716115278 and parameters: {'clf_n_estimators': 362, 'clf_max_depth': 24,
'clf_min_samples_leaf': 15, 'clf_min_samples_split': 13, 'clf_max_features':
'sqrt', 'clf_class_weight': None, 'n_estimators': 737, 'learning_rate':
0.005394455304087533, 'max_depth': 12, 'subsample': 0.9162213204002109,
```

```
'colsample_bytree': 0.5274034664069657, 'min_child_weight': 2, 'gamma':
2.9324868872723725e-07, 'reg_alpha': 2.716051144654844e-06, 'reg_lambda':
0.00015777981883364995}. Best is trial 0 with value: 1.392185716115278.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:16:46,239] Trial 1 finished with value:
1.4411671286898466 and parameters: {'clf_n_estimators': 402, 'clf_max_depth': 9,
'clf_min_samples_leaf': 13, 'clf_min_samples_split': 4, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 563, 'learning_rate':
0.04446862319918233, 'max_depth': 2, 'subsample': 0.8037724259507192,
'colsample_bytree': 0.502314474212375, 'min_child_weight': 1, 'gamma':
0.39001768308022033, 'reg_alpha': 0.530953226900921, 'reg_lambda':
0.02932100047183291}. Best is trial 0 with value: 1.392185716115278.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:16:48,230] Trial 2 finished with value:
1.4427723796611367 and parameters: {'clf_n_estimators': 313, 'clf_max_depth': 5,
'clf_min_samples_leaf': 15, 'clf_min_samples_split': 10, 'clf_max_features':
'log2', 'clf_class_weight': None, 'n_estimators': 696, 'learning_rate':
0.015788826347222458, 'max_depth': 7, 'subsample': 0.7733551396716398,
'colsample_bytree': 0.5109126733153162, 'min_child_weight': 10, 'gamma':
0.01588775693167255, 'reg_alpha': 0.32808889626606236, 'reg_lambda':
0.14408501080722544}. Best is trial 0 with value: 1.392185716115278.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:16:50,589] Trial 3 finished with value:
1.414399156911184 and parameters: {'clf_n_estimators': 519, 'clf_max_depth': 24,
'clf_min_samples_leaf': 3, 'clf_min_samples_split': 5, 'clf_max_features': 0.5,
'clf_class_weight': 'balanced', 'n_estimators': 421, 'learning_rate':
0.014094313993387368, 'max_depth': 7, 'subsample': 0.5704621124873813,
'colsample_bytree': 0.8813181884524238, 'min_child_weight': 1, 'gamma':
0.7854083114461319, 'reg_alpha': 0.015064619068942013, 'reg_lambda':
3.8879928024075543e-07}. Best is trial 0 with value: 1.392185716115278.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
```

```
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:16:51,407] Trial 4 finished with value:
1.7606164670023539 and parameters: {'clf_n_estimators': 103, 'clf_max_depth':
21, 'clf_min_samples_leaf': 15, 'clf_min_samples_split': 15, 'clf_max_features':
'sqrt', 'clf_class_weight': 'balanced', 'n_estimators': 661, 'learning_rate':
0.01694683071232564, 'max_depth': 2, 'subsample': 0.6554911608578311,
'colsample_bytree': 0.5951099932160482, 'min_child_weight': 8, 'gamma':
0.0012602588933700108, 'reg_alpha': 0.12522814303053625, 'reg_lambda':
5.994036749692399e-05}. Best is trial 0 with value: 1.392185716115278.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:16:52,458] Trial 5 finished with value:
1.5569648411931258 and parameters: {'clf_n_estimators': 183, 'clf_max_depth':
19, 'clf_min_samples_leaf': 16, 'clf_min_samples_split': 12, 'clf_max_features':
'sqrt', 'clf_class_weight': None, 'n_estimators': 197, 'learning_rate':
0.005614633922822615, 'max_depth': 9, 'subsample': 0.6571779905381634,
'colsample_bytree': 0.7051424146988217, 'min_child_weight': 10, 'gamma':
9.8704700073667e-07, 'reg_alpha': 1.918948786414487e-05, 'reg_lambda':
0.011076667222301087}. Best is trial 0 with value: 1.392185716115278.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:16:53,755] Trial 6 finished with value:
1.4486436771096258 and parameters: {'clf_n_estimators': 260, 'clf_max_depth': 4,
'clf_min_samples_leaf': 7, 'clf_min_samples_split': 5, 'clf_max_features':
'sqrt', 'clf_class_weight': None, 'n_estimators': 268, 'learning_rate':
0.13455599923054787, 'max_depth': 7, 'subsample': 0.9037200775820313,
'colsample_bytree': 0.937654779954096, 'min_child_weight': 4, 'gamma':
7.593034903208851e-08, 'reg_alpha': 6.660108438734075e-07, 'reg_lambda':
2.6113333123958975e-05}. Best is trial 0 with value: 1.392185716115278.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:16:56,483] Trial 7 finished with value:
1.4680254553624141 and parameters: {'clf_n_estimators': 673, 'clf_max_depth':
22, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 11, 'clf_max_features':
```

```
'sqrt', 'clf_class_weight': 'balanced', 'n_estimators': 391, 'learning_rate':
0.0338925076935574, 'max_depth': 9, 'subsample': 0.681814801189647,
'colsample_bytree': 0.9830692496325764, 'min_child_weight': 10, 'gamma':
1.033375989766645e-06, 'reg_alpha': 9.505786310154948e-05, 'reg_lambda':
2.5528569432507475e-06}. Best is trial 0 with value: 1.392185716115278.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:16:58,014] Trial 8 finished with value:
1.4989231569595072 and parameters: {'clf_n_estimators': 299, 'clf_max_depth': 3,
'clf_min_samples_leaf': 13, 'clf_min_samples_split': 11, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 540, 'learning_rate':
0.18968856989494792, 'max_depth': 4, 'subsample': 0.8360677737029393,
'colsample_bytree': 0.8569717691972305, 'min_child_weight': 3, 'gamma':
0.0066946948903108105, 'reg_alpha': 8.755179646617154e-06, 'reg_lambda':
0.001144054318382189}. Best is trial 0 with value: 1.392185716115278.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:16:59,947] Trial 9 finished with value:
1.5305651029798746 and parameters: {'clf_n_estimators': 544, 'clf_max_depth':
15, 'clf_min_samples_leaf': 3, 'clf_min_samples_split': 17, 'clf_max_features':
'sqrt', 'clf_class_weight': 'balanced', 'n_estimators': 114, 'learning_rate':
0.033065401830212754, 'max_depth': 4, 'subsample': 0.822586395204725,
'colsample_bytree': 0.5046198574029949, 'min_child_weight': 7, 'gamma':
1.2413171922260375e-05, 'reg_alpha': 0.31177401074836186, 'reg_lambda':
1.2594112057549154e-07}. Best is trial 0 with value: 1.392185716115278.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:17:01,945] Trial 10 finished with value:
1.4064045151822064 and parameters: {'clf_n_estimators': 339, 'clf_max_depth': 5,
'clf_min_samples_leaf': 19, 'clf_min_samples_split': 18, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 317, 'learning_rate':
0.007048979113298945, 'max_depth': 11, 'subsample': 0.9502090285816652,
'colsample_bytree': 0.7798608743639608, 'min_child_weight': 4, 'gamma':
6.218370535731491e-06, 'reg_alpha': 0.006421632248858025, 'reg_lambda':
0.15027338858202638}. Best is trial 0 with value: 1.392185716115278.
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:17:04,613] Trial 11 finished with value:
1.9697560728188985 and parameters: {'clf_n_estimators': 721, 'clf_max_depth':
20, 'clf_min_samples_leaf': 14, 'clf_min_samples_split': 3, 'clf_max_features':
'log2', 'clf_class_weight': 'balanced', 'n_estimators': 697, 'learning_rate':
0.005094234882279176, 'max_depth': 3, 'subsample': 0.7743668946832931,
'colsample_bytree': 0.815137118615616, 'min_child_weight': 7, 'gamma':
6.225216726492816e-07, 'reg_alpha': 0.004982344702102993, 'reg_lambda':
7.906653376497142e-07}. Best is trial 0 with value: 1.392185716115278.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:17:06,026] Trial 12 finished with value:
1.4749388594492567 and parameters: {'clf_n_estimators': 328, 'clf_max_depth':
20, 'clf_min_samples_leaf': 14, 'clf_min_samples_split': 18, 'clf_max_features':
'sqrt', 'clf_class_weight': None, 'n_estimators': 319, 'learning_rate':
0.18104698016273907, 'max_depth': 6, 'subsample': 0.9460232775885566,
'colsample_bytree': 0.7786831755983578, 'min_child_weight': 8, 'gamma':
0.0001049776254156589, 'reg_alpha': 0.00041231684401416913, 'reg_lambda':
8.712475112920381e-05}. Best is trial 0 with value: 1.392185716115278.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:17:07,663] Trial 13 finished with value:
1.4233506441296162 and parameters: {'clf_n_estimators': 236, 'clf_max_depth':
19, 'clf_min_samples_leaf': 7, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 433, 'learning_rate':
0.005293372188801157, 'max_depth': 12, 'subsample': 0.7140920741586572,
'colsample_bytree': 0.9799928914262017, 'min_child_weight': 10, 'gamma':
0.06669229211616488, 'reg_alpha': 2.2677289093956233e-06, 'reg_lambda':
1.2044307332953856e-05}. Best is trial 0 with value: 1.392185716115278.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:17:11,030] Trial 14 finished with value:
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1.5213587957336816 and parameters: {'clf_n_estimators': 696, 'clf_max_depth':
10, 'clf_min_samples_leaf': 5, 'clf_min_samples_split': 12, 'clf_max_features':
'sqrt', 'clf_class_weight': 'balanced', 'n_estimators': 992, 'learning_rate':
0.008382875198464706, 'max_depth': 7, 'subsample': 0.9386865359639778,
'colsample_bytree': 0.8444611706525227, 'min_child_weight': 7, 'gamma':
0.0041674718203826185, 'reg_alpha': 7.515003890531393e-06, 'reg_lambda':
2.2322084823690722e-06}. Best is trial 0 with value: 1.392185716115278.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:17:13,779] Trial 15 finished with value:
1.6281810655696438 and parameters: {'clf_n_estimators': 667, 'clf_max_depth':
21, 'clf_min_samples_leaf': 18, 'clf_min_samples_split': 19, 'clf_max_features':
0.5, 'clf_class_weight': 'balanced', 'n_estimators': 817, 'learning_rate':
0.13329441547774673, 'max_depth': 5, 'subsample': 0.687791476319972,
'colsample_bytree': 0.45638916390452144, 'min_child_weight': 6, 'gamma':
1.9388231219417792e-08, 'reg_alpha': 5.3062064728788974e-05, 'reg_lambda':
0.0002193598182745354}. Best is trial 0 with value: 1.392185716115278.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:17:14,966] Trial 16 finished with value:
1.473802681022715 and parameters: {'clf_n_estimators': 300, 'clf_max_depth': 16,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 2, 'clf_max_features':
'sqrt', 'clf_class_weight': 'balanced', 'n_estimators': 294, 'learning_rate':
0.049759718479373743, 'max_depth': 2, 'subsample': 0.5258408605843039,
'colsample_bytree': 0.7188127789408888, 'min_child_weight': 6, 'gamma':
0.001257300858183282, 'reg_alpha': 0.006437699040892684, 'reg_lambda':
0.6409389074134212}. Best is trial 0 with value: 1.392185716115278.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:17:16,849] Trial 17 finished with value:
1.4353800831982406 and parameters: {'clf_n_estimators': 461, 'clf_max_depth':
10, 'clf_min_samples_leaf': 17, 'clf_min_samples_split': 7, 'clf_max_features':
'sqrt', 'clf_class_weight': None, 'n_estimators': 727, 'learning_rate':
0.022601494372818134, 'max_depth': 3, 'subsample': 0.5782185213355431,
'colsample_bytree': 0.5501457388987572, 'min_child_weight': 6, 'gamma':
0.005209156216369749, 'reg_alpha': 0.001912401227325431, 'reg_lambda':
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1.7356860603349111e-06}. Best is trial 0 with value: 1.392185716115278.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:17:19,380] Trial 18 finished with value:
1.6337752986943916 and parameters: {'clf_n_estimators': 769, 'clf_max_depth':
19, 'clf_min_samples_leaf': 12, 'clf_min_samples_split': 13, 'clf_max_features':
'sqrt', 'clf_class_weight': None, 'n_estimators': 204, 'learning_rate':
0.005924735907317127, 'max_depth': 2, 'subsample': 0.9277302920055036,
'colsample_bytree': 0.8221947156280143, 'min_child_weight': 5, 'gamma':
6.06280060403243e-08, 'reg_alpha': 8.568938007937746e-05, 'reg_lambda':
6.134429310056972e-05}. Best is trial 0 with value: 1.392185716115278.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:17:20,687] Trial 19 finished with value:
1.48647975239183 and parameters: {'clf_n_estimators': 221, 'clf_max_depth': 12,
'clf_min_samples_leaf': 9, 'clf_min_samples_split': 13, 'clf_max_features':
'sqrt', 'clf_class_weight': None, 'n_estimators': 871, 'learning_rate':
0.05678540376125823, 'max_depth': 3, 'subsample': 0.5352843737002149,
'colsample_bytree': 0.7854515669237894, 'min_child_weight': 1, 'gamma':
0.00048551723386735715, 'reg_alpha': 0.33253849842431976, 'reg_lambda':
0.00040159974003412744}. Best is trial 0 with value: 1.392185716115278.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:17:23,695] Trial 20 finished with value:
1.4243582212638781 and parameters: {'clf_n_estimators': 414, 'clf_max_depth': 7,
'clf_min_samples_leaf': 20, 'clf_min_samples_split': 15, 'clf_max_features':
'log2', 'clf_class_weight': None, 'n_estimators': 583, 'learning_rate':
0.009266610539943303, 'max_depth': 12, 'subsample': 0.9984416307355438,
'colsample_bytree': 0.6208351661552375, 'min_child_weight': 3, 'gamma':
1.0040417609079498e-05, 'reg_alpha': 5.8141765792516835e-08, 'reg_lambda':
3.4878848782898526e-08}. Best is trial 0 with value: 1.392185716115278.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:17:24,743] Trial 21 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:17:27,322] Trial 22 finished with value:
1.3874239027268807 and parameters: {'clf_n_estimators': 483, 'clf_max_depth':
25, 'clf_min_samples_leaf': 18, 'clf_min_samples_split': 20, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 375, 'learning_rate':
0.012745830104519391, 'max_depth': 11, 'subsample': 0.8798528463875022,
'colsample_bytree': 0.6480484598156278, 'min_child_weight': 2, 'gamma':
7.629245137807497e-05, 'reg_alpha': 0.0208797602807589, 'reg_lambda':
1.8134313304390152e-07}. Best is trial 22 with value: 1.3874239027268807.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:17:29,448] Trial 23 finished with value:
1.4016861657723474 and parameters: {'clf_n_estimators': 383, 'clf_max_depth':
23, 'clf_min_samples_leaf': 18, 'clf_min_samples_split': 16, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 356, 'learning_rate':
0.007945073002086534, 'max_depth': 11, 'subsample': 0.8894677195361091,
'colsample_bytree': 0.4064487177080809, 'min_child_weight': 3, 'gamma':
7.763533399402654e-05, 'reg_alpha': 0.000636035940167732, 'reg_lambda':
0.01018539528271691}. Best is trial 22 with value: 1.3874239027268807.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:17:32,370] Trial 24 finished with value:
1.3885713774131385 and parameters: {'clf_n_estimators': 605, 'clf_max_depth':
24, 'clf_min_samples_leaf': 17, 'clf_min_samples_split': 20, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 492, 'learning_rate':
0.01227676371804577, 'max_depth': 10, 'subsample': 0.8740866131350146,
'colsample_bytree': 0.4217828879392712, 'min_child_weight': 2, 'gamma':
0.00015278417568556754, 'reg_alpha': 0.0009203707477521335, 'reg_lambda':
0.004682547040429204}. Best is trial 22 with value: 1.3874239027268807.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:17:35,795] Trial 25 finished with value:
1.3852975066029778 and parameters: {'clf_n_estimators': 620, 'clf_max_depth':
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25, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 20, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 799, 'learning_rate':
0.012389818804043967, 'max_depth': 9, 'subsample': 0.8699353394462971,
'colsample_bytree': 0.6461322535535858, 'min_child_weight': 2, 'gamma':
0.0001232257750560202, 'reg_alpha': 8.295894203810815e-08, 'reg_lambda':
2.2225767522138785e-08}. Best is trial 25 with value: 1.3852975066029778.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:17:38,740] Trial 26 finished with value:
1.3832518520504353 and parameters: {'clf_n_estimators': 608, 'clf_max_depth':
25, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 20, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 507, 'learning_rate':
0.012347536689235845, 'max_depth': 9, 'subsample': 0.8735397073693515,
'colsample_bytree': 0.6290955078759742, 'min_child_weight': 2, 'gamma':
8.239545763093374e-05, 'reg_alpha': 1.551056990171232e-08, 'reg_lambda':
3.934261072425912e-08}. Best is trial 26 with value: 1.3832518520504353.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:17:41,666] Trial 27 finished with value:
1.3896905425418182 and parameters: {'clf_n_estimators': 611, 'clf_max_depth':
25, 'clf_min_samples_leaf': 10, 'clf_min_samples_split': 20, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 614, 'learning_rate':
0.021064579452035343, 'max_depth': 9, 'subsample': 0.853982049615971,
'colsample_bytree': 0.6165021222144075, 'min_child_weight': 4, 'gamma':
0.0003624464121508663, 'reg_alpha': 2.1013007259699696e-08, 'reg_lambda':
1.2789761055302295e-08}. Best is trial 26 with value: 1.3832518520504353.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:17:44,574] Trial 28 finished with value:
1.3969827057587276 and parameters: {'clf_n_estimators': 473, 'clf_max_depth':
17, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 20, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 978, 'learning_rate':
0.02302505868518139, 'max_depth': 8, 'subsample': 0.7424759841452724,
'colsample_bytree': 0.6608773393343692, 'min_child_weight': 2, 'gamma':
5.140342619058579e-05, 'reg_alpha': 2.069685504616195e-07, 'reg_lambda':
8.201033159731042e-08}. Best is trial 26 with value: 1.3832518520504353.
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:17:48,279] Trial 29 finished with value:
1.3840762244407758 and parameters: {'clf_n_estimators': 615, 'clf_max_depth':
23, 'clf_min_samples_leaf': 9, 'clf_min_samples_split': 18, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 790, 'learning_rate':
0.010823203811826033, 'max_depth': 11, 'subsample': 0.8597085023720237,
'colsample_bytree': 0.6593372142756072, 'min_child_weight': 3, 'gamma':
4.0148938627186095e-06, 'reg_alpha': 1.1636589634877293e-07, 'reg_lambda':
2.0448494030669717e-08}. Best is trial 26 with value: 1.3832518520504353.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:17:51,675] Trial 30 finished with value:
1.3864014364603134 and parameters: {'clf_n_estimators': 594, 'clf_max_depth':
23, 'clf_min_samples_leaf': 8, 'clf_min_samples_split': 18, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 803, 'learning_rate':
0.010502274110158834, 'max_depth': 10, 'subsample': 0.8047726952605683,
'colsample_bytree': 0.5512601674227222, 'min_child_weight': 3, 'gamma':
2.794286683298008e-06, 'reg_alpha': 1.0479230759588874e-08, 'reg_lambda':
1.0130812496176829e-08}. Best is trial 26 with value: 1.3832518520504353.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:17:55,108] Trial 31 finished with value:
1.3912284718271863 and parameters: {'clf_n_estimators': 615, 'clf_max_depth':
23, 'clf_min_samples_leaf': 8, 'clf_min_samples_split': 18, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 802, 'learning_rate':
0.010580204748461138, 'max_depth': 10, 'subsample': 0.7943345592508133,
'colsample_bytree': 0.5635969763239952, 'min_child_weight': 3, 'gamma':
2.5063573341095373e-06, 'reg_alpha': 1.2100059663303262e-08, 'reg_lambda':
1.0596175546844685e-08}. Best is trial 26 with value: 1.3832518520504353.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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[I 2026-03-13 19:17:58,424] Trial 32 finished with value:
1.3950239450231 and parameters: {'clf_n_estimators': 580, 'clf_max_depth': 23,
'clf_min_samples_leaf': 10, 'clf_min_samples_split': 16, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 915, 'learning_rate':
0.019577697636226046, 'max_depth': 8, 'subsample': 0.8589278520980982,
'colsample_bytree': 0.6707572652597491, 'min_child_weight': 1, 'gamma':
3.064940341837637e-05, 'reg_alpha': 1.0017526287617406e-07, 'reg_lambda':
4.170739425673799e-08}. Best is trial 26 with value: 1.3832518520504353.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:18:02,108] Trial 33 finished with value:
1.3894525476996369 and parameters: {'clf_n_estimators': 799, 'clf_max_depth':
22, 'clf_min_samples_leaf': 6, 'clf_min_samples_split': 17, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 803, 'learning_rate':
0.006996575448847217, 'max_depth': 8, 'subsample': 0.8156367309350179,
'colsample_bytree': 0.5759108329841005, 'min_child_weight': 4, 'gamma':
2.8907816973127876e-06, 'reg_alpha': 3.890783912008925e-07, 'reg_lambda':
2.9322044294030382e-08}. Best is trial 26 with value: 1.3832518520504353.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:18:05,872] Trial 34 finished with value:
1.425739218423302 and parameters: {'clf_n_estimators': 650, 'clf_max_depth': 17,
'clf_min_samples_leaf': 11, 'clf_min_samples_split': 19, 'clf_max_features':
'log2', 'clf_class_weight': None, 'n_estimators': 764, 'learning_rate':
0.01625233016361774, 'max_depth': 10, 'subsample': 0.7459754712040433,
'colsample_bytree': 0.731688528201038, 'min_child_weight': 1, 'gamma':
3.4515120531876944e-07, 'reg_alpha': 3.5945195384585444e-08, 'reg_lambda':
3.4135673714197527e-07}. Best is trial 26 with value: 1.3832518520504353.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:18:09,561] Trial 35 finished with value:
1.3969054996209909 and parameters: {'clf_n_estimators': 734, 'clf_max_depth':
25, 'clf_min_samples_leaf': 9, 'clf_min_samples_split': 15, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 883, 'learning_rate':
0.012373782235390986, 'max_depth': 9, 'subsample': 0.7880855274798678,
'colsample_bytree': 0.47119119623484157, 'min_child_weight': 3, 'gamma':
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2.5878895220005766e-05, 'reg_alpha': 1.2348867951127395e-06, 'reg_lambda':
6.346477100577844e-08}. Best is trial 26 with value: 1.3832518520504353.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:18:10,486] Trial 36 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:18:13,231] Trial 37 finished with value:
1.3962070927107153 and parameters: {'clf_n_estimators': 517, 'clf_max_depth':
13, 'clf_min_samples_leaf': 9, 'clf_min_samples_split': 17, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 621, 'learning_rate':
0.026977465070806332, 'max_depth': 9, 'subsample': 0.9113617901397565,
'colsample_bytree': 0.614928324932083, 'min_child_weight': 2, 'gamma':
0.0003552561451019697, 'reg_alpha': 1.1332663461699557e-08, 'reg_lambda':
3.0215019603230225e-07}. Best is trial 26 with value: 1.3832518520504353.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:18:17,914] Trial 38 finished with value:
1.386336575697425 and parameters: {'clf_n_estimators': 660, 'clf_max_depth': 23,
'clf_min_samples_leaf': 5, 'clf_min_samples_split': 14, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 665, 'learning_rate':
0.015284735544374425, 'max_depth': 12, 'subsample': 0.9652742867089056,
'colsample_bytree': 0.6840681783956121, 'min_child_weight': 1, 'gamma':
1.3539528150849e-07, 'reg_alpha': 3.6332393651362844e-08, 'reg_lambda':
6.157010045288949e-06}. Best is trial 26 with value: 1.3832518520504353.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:18:22,599] Trial 39 finished with value:
1.4044143874730317 and parameters: {'clf_n_estimators': 658, 'clf_max_depth':
21, 'clf_min_samples_leaf': 5, 'clf_min_samples_split': 16, 'clf_max_features':
'log2', 'clf_class_weight': None, 'n_estimators': 659, 'learning_rate':
0.014888833026020988, 'max_depth': 12, 'subsample': 0.9784982569914974,
'colsample_bytree': 0.6922284738546912, 'min_child_weight': 1, 'gamma':
1.409628090842609e-07, 'reg_alpha': 3.579538804107093e-07, 'reg_lambda':
1.5063354516312998e-05}. Best is trial 26 with value: 1.3832518520504353.
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:18:26,754] Trial 40 finished with value:
1.3900699756265464 and parameters: {'clf_n_estimators': 718, 'clf_max_depth':
24, 'clf_min_samples_leaf': 5, 'clf_min_samples_split': 14, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 657, 'learning_rate':
0.01900906764721793, 'max_depth': 12, 'subsample': 0.9661320679997403,
'colsample_bytree': 0.7240741692661061, 'min_child_weight': 2, 'gamma':
1.7608143743123135e-08, 'reg_alpha': 5.3429498570207275e-08, 'reg_lambda':
9.126225274750818e-07}. Best is trial 26 with value: 1.3832518520504353.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:18:31,076] Trial 41 finished with value:
1.388246619727078 and parameters: {'clf_n_estimators': 640, 'clf_max_depth': 23,
'clf_min_samples_leaf': 7, 'clf_min_samples_split': 18, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 749, 'learning_rate':
0.011079129031295928, 'max_depth': 11, 'subsample': 0.8332114628822781,
'colsample_bytree': 0.6435624607713131, 'min_child_weight': 1, 'gamma':
1.2551498582214496e-06, 'reg_alpha': 3.27629926451034e-08, 'reg_lambda':
2.7099310289397716e-08}. Best is trial 26 with value: 1.3832518520504353.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:18:34,579] Trial 42 finished with value:
1.3911458664855063 and parameters: {'clf_n_estimators': 595, 'clf_max_depth':
24, 'clf_min_samples_leaf': 8, 'clf_min_samples_split': 19, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 835, 'learning_rate':
0.014721579050001923, 'max_depth': 10, 'subsample': 0.9012767575195575,
'colsample_bytree': 0.7481571791332023, 'min_child_weight': 3, 'gamma':
1.8415361627645494e-07, 'reg_alpha': 1.0544764893586974e-07, 'reg_lambda':
1.0840423551471574e-07}. Best is trial 26 with value: 1.3832518520504353.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:18:39,251] Trial 43 finished with value:
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1.3828579192540527 and parameters: {'clf_n_estimators': 689, 'clf_max_depth':
21, 'clf_min_samples_leaf': 10, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 702, 'learning_rate':
0.00923760135647521, 'max_depth': 11, 'subsample': 0.918116510457643,
'colsample_bytree': 0.5904573119822194, 'min_child_weight': 1, 'gamma':
5.089215420700353e-07, 'reg_alpha': 1.2387593454187301e-08, 'reg_lambda':
8.134427926298699e-06}. Best is trial 43 with value: 1.3828579192540527.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:18:44,761] Trial 44 finished with value:
1.380708717645845 and parameters: {'clf_n_estimators': 754, 'clf_max_depth': 21,
'clf_min_samples_leaf': 12, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 703, 'learning_rate':
0.006808603403720034, 'max_depth': 12, 'subsample': 0.9247648738325163,
'colsample_bytree': 0.5899988469194871, 'min_child_weight': 1, 'gamma':
7.210099699200402e-08, 'reg_alpha': 8.467917115230667e-07, 'reg_lambda':
6.121901548554442e-06}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:18:49,155] Trial 45 finished with value:
1.3860548838650433 and parameters: {'clf_n_estimators': 760, 'clf_max_depth':
20, 'clf_min_samples_leaf': 13, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 717, 'learning_rate':
0.006896490311726811, 'max_depth': 11, 'subsample': 0.9187770211000945,
'colsample_bytree': 0.5847079624137402, 'min_child_weight': 2, 'gamma':
3.389318530344901e-08, 'reg_alpha': 9.116563472473815e-07, 'reg_lambda':
4.958892962964776e-06}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:18:53,039] Trial 46 finished with value:
1.3961685191825917 and parameters: {'clf_n_estimators': 700, 'clf_max_depth':
18, 'clf_min_samples_leaf': 12, 'clf_min_samples_split': 6, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 945, 'learning_rate':
0.008666896225976996, 'max_depth': 8, 'subsample': 0.8740070505466628,
'colsample_bytree': 0.6005126727079204, 'min_child_weight': 1, 'gamma':
4.583123810283896e-07, 'reg_alpha': 6.099767538677484e-06, 'reg_lambda':
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9.027250624974719e-07}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:18:56,547] Trial 47 finished with value:
1.399504148780996 and parameters: {'clf_n_estimators': 753, 'clf_max_depth': 21,
'clf_min_samples_leaf': 14, 'clf_min_samples_split': 10, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 756, 'learning_rate':
0.007628742448200208, 'max_depth': 6, 'subsample': 0.9008504297992375,
'colsample_bytree': 0.642506617251808, 'min_child_weight': 1, 'gamma':
8.152462455254754e-07, 'reg_alpha': 2.5661652557569857e-07, 'reg_lambda':
2.0725322640349741e-07}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:18:57,891] Trial 48 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:18:59,632] Trial 49 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:03,595] Trial 50 finished with value:
1.3815324778835583 and parameters: {'clf_n_estimators': 639, 'clf_max_depth':
24, 'clf_min_samples_leaf': 13, 'clf_min_samples_split': 5, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 696, 'learning_rate':
0.006142829232947632, 'max_depth': 11, 'subsample': 0.9249841619933722,
'colsample_bytree': 0.6314355802530203, 'min_child_weight': 4, 'gamma':
4.908886841159966e-08, 'reg_alpha': 2.1905527351402795e-05, 'reg_lambda':
4.944796937473789e-07}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:07,648] Trial 51 finished with value:
1.384160104723234 and parameters: {'clf_n_estimators': 628, 'clf_max_depth': 24,
'clf_min_samples_leaf': 13, 'clf_min_samples_split': 5, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 683, 'learning_rate':
0.00605107219859534, 'max_depth': 11, 'subsample': 0.9160015615247954,
'colsample_bytree': 0.6226261100392151, 'min_child_weight': 4, 'gamma':
5.0484520990691175e-08, 'reg_alpha': 1.8531994847888676e-05, 'reg_lambda':
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6.016483661447171e-07}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:10,973] Trial 52 finished with value:
1.3814554447089595 and parameters: {'clf_n_estimators': 521, 'clf_max_depth':
24, 'clf_min_samples_leaf': 15, 'clf_min_samples_split': 4, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 618, 'learning_rate':
0.005806088349528734, 'max_depth': 11, 'subsample': 0.9205238957113256,
'colsample_bytree': 0.6078836951004122, 'min_child_weight': 4, 'gamma':
1.1540930400274776e-08, 'reg_alpha': 1.7883249171758162e-05, 'reg_lambda':
5.691415073371387e-07}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:14,269] Trial 53 finished with value:
1.3813299109811592 and parameters: {'clf_n_estimators': 526, 'clf_max_depth':
22, 'clf_min_samples_leaf': 15, 'clf_min_samples_split': 4, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 615, 'learning_rate':
0.005079847129439592, 'max_depth': 12, 'subsample': 0.9305769572622085,
'colsample_bytree': 0.587303017616827, 'min_child_weight': 5, 'gamma':
1.1417350399881526e-08, 'reg_alpha': 0.00022129347419376567, 'reg_lambda':
1.993442958062834e-06}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:16,188] Trial 54 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:19,033] Trial 55 finished with value:
1.386232983997785 and parameters: {'clf_n_estimators': 419, 'clf_max_depth': 21,
'clf_min_samples_leaf': 16, 'clf_min_samples_split': 4, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 623, 'learning_rate':
0.005671862482484605, 'max_depth': 12, 'subsample': 0.9645508970230999,
'colsample_bytree': 0.5283866721178251, 'min_child_weight': 5, 'gamma':
2.823713556531901e-08, 'reg_alpha': 3.480320174483228e-05, 'reg_lambda':
2.477447674995697e-06}. Best is trial 44 with value: 1.380708717645845.
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:20,161] Trial 56 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:20,763] Trial 57 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:21,655] Trial 58 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:24,438] Trial 59 finished with value:
1.386546580962306 and parameters: {'clf_n_estimators': 526, 'clf_max_depth': 21,
'clf_min_samples_leaf': 13, 'clf_min_samples_split': 6, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 485, 'learning_rate':
0.005643737150731823, 'max_depth': 11, 'subsample': 0.9592874980672663,
'colsample_bytree': 0.6301079684994753, 'min_child_weight': 5, 'gamma':
2.8978479745616465e-08, 'reg_alpha': 2.287574910139359e-06, 'reg_lambda':
0.00014104098357508658}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:25,417] Trial 60 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:26,644] Trial 61 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:29,834] Trial 62 finished with value:
1.3861748287756075 and parameters: {'clf_n_estimators': 454, 'clf_max_depth':
20, 'clf_min_samples_leaf': 13, 'clf_min_samples_split': 5, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 773, 'learning_rate':
0.006394973758396934, 'max_depth': 11, 'subsample': 0.9167633741317817,
'colsample_bytree': 0.7066873419024803, 'min_child_weight': 4, 'gamma':
1.0013532261993077e-08, 'reg_alpha': 4.194450881392716e-06, 'reg_lambda':
3.555184543515252e-06}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:33,439] Trial 63 finished with value:
1.3843951314857161 and parameters: {'clf_n_estimators': 572, 'clf_max_depth':
24, 'clf_min_samples_leaf': 10, 'clf_min_samples_split': 10, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 721, 'learning_rate':
0.008400760371064565, 'max_depth': 10, 'subsample': 0.9850915790585494,
'colsample_bytree': 0.6807371816697187, 'min_child_weight': 3, 'gamma':
3.847214108095147e-08, 'reg_alpha': 1.471216101496699e-05, 'reg_lambda':
7.545027011409255e-08}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:36,588] Trial 64 finished with value:
1.3881156996759854 and parameters: {'clf_n_estimators': 530, 'clf_max_depth':
22, 'clf_min_samples_leaf': 15, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 675, 'learning_rate':
0.009305004639134131, 'max_depth': 11, 'subsample': 0.6443270699828453,
'colsample_bytree': 0.5516943208456393, 'min_child_weight': 4, 'gamma':
3.1087363639075905e-07, 'reg_alpha': 0.0018538793149055527, 'reg_lambda':
1.178840100407929e-06}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:37,698] Trial 65 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:41,785] Trial 66 finished with value:
1.388927520902481 and parameters: {'clf_n_estimators': 734, 'clf_max_depth': 25,
'clf_min_samples_leaf': 17, 'clf_min_samples_split': 12, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 693, 'learning_rate':
0.007139778138108414, 'max_depth': 10, 'subsample': 0.8819455692528998,
'colsample_bytree': 0.5734394899716315, 'min_child_weight': 3, 'gamma':
2.4481261552764225e-08, 'reg_alpha': 6.701301690163722e-07, 'reg_lambda':
3.47655618644214e-07}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:43,157] Trial 67 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:46,043] Trial 68 finished with value:
1.3830350179374264 and parameters: {'clf_n_estimators': 582, 'clf_max_depth':
23, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 11, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 409, 'learning_rate':
0.011368449754472147, 'max_depth': 12, 'subsample': 0.843586551767787,
'colsample_bytree': 0.6780408524944445, 'min_child_weight': 9, 'gamma':
5.717091949281564e-07, 'reg_alpha': 5.852765462088879e-05, 'reg_lambda':
2.431709136185508e-06}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:48,806] Trial 69 finished with value:
1.3813838597932122 and parameters: {'clf_n_estimators': 544, 'clf_max_depth':
19, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 11, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 410, 'learning_rate':
0.00782158767341439, 'max_depth': 12, 'subsample': 0.9033147528713034,
'colsample_bytree': 0.6681521916743922, 'min_child_weight': 7, 'gamma':
4.914108259248619e-07, 'reg_alpha': 7.490945499971261e-05, 'reg_lambda':
3.6027776567148774e-05}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:49,717] Trial 70 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:52,393] Trial 71 finished with value:
1.387715690704727 and parameters: {'clf_n_estimators': 581, 'clf_max_depth': 21,
'clf_min_samples_leaf': 11, 'clf_min_samples_split': 11, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 367, 'learning_rate':
0.009927829145657362, 'max_depth': 12, 'subsample': 0.8999884970043324,
'colsample_bytree': 0.7037440083467705, 'min_child_weight': 9, 'gamma':
5.691278315991525e-07, 'reg_alpha': 0.00014070078240047273, 'reg_lambda':
4.3233697744729254e-05}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:55,113] Trial 72 finished with value:
1.3861671979770487 and parameters: {'clf_n_estimators': 553, 'clf_max_depth':
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19, 'clf_min_samples_leaf': 10, 'clf_min_samples_split': 11, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 402, 'learning_rate':
0.008088019171969746, 'max_depth': 12, 'subsample': 0.8402506058443113,
'colsample_bytree': 0.6669984401871191, 'min_child_weight': 7, 'gamma':
2.375485766769883e-07, 'reg_alpha': 0.0003823013209798141, 'reg_lambda':
2.587942538684751e-06}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:55,942] Trial 73 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:56,923] Trial 74 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:19:57,806] Trial 75 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:20:00,436] Trial 76 finished with value:
1.3857858057446937 and parameters: {'clf_n_estimators': 566, 'clf_max_depth':
25, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 5, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 463, 'learning_rate':
0.011360308772067828, 'max_depth': 10, 'subsample': 0.9332048818695929,
'colsample_bytree': 0.5411880110063824, 'min_child_weight': 8, 'gamma':
0.6234023141819203, 'reg_alpha': 1.0977521346982092e-05, 'reg_lambda':
3.0412788761491752e-06}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:20:01,375] Trial 77 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:20:04,925] Trial 78 finished with value:
1.384515793254723 and parameters: {'clf_n_estimators': 675, 'clf_max_depth': 21,
'clf_min_samples_leaf': 9, 'clf_min_samples_split': 13, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 650, 'learning_rate':
0.005552986753493665, 'max_depth': 12, 'subsample': 0.866051454842048,
'colsample_bytree': 0.6099017109618239, 'min_child_weight': 7, 'gamma':
6.735812944846747e-07, 'reg_alpha': 7.36050807033109e-05, 'reg_lambda':
5.356079055333866e-07}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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[I 2026-03-13 19:20:05,519] Trial 79 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:20:06,568] Trial 80 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:20:07,730] Trial 81 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:20:10,683] Trial 82 finished with value:
1.3835254171422846 and parameters: {'clf_n_estimators': 625, 'clf_max_depth':
23, 'clf_min_samples_leaf': 8, 'clf_min_samples_split': 4, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 341, 'learning_rate':
0.01102020279716818, 'max_depth': 11, 'subsample': 0.8782405163981959,
'colsample_bytree': 0.6659971365095408, 'min_child_weight': 3, 'gamma':
5.317131020591092e-06, 'reg_alpha': 5.3725192100767494e-08, 'reg_lambda':
2.3087354016690957e-07}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:20:13,636] Trial 83 finished with value:
1.3823489544665033 and parameters: {'clf_n_estimators': 634, 'clf_max_depth':
25, 'clf_min_samples_leaf': 15, 'clf_min_samples_split': 4, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 345, 'learning_rate':
0.011503925266492212, 'max_depth': 12, 'subsample': 0.8776908520348569,
'colsample_bytree': 0.623958583354904, 'min_child_weight': 4, 'gamma':
5.778175735942123e-06, 'reg_alpha': 5.577687507758684e-08, 'reg_lambda':
0.000751948011956789}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:20:14,610] Trial 84 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:20:15,515] Trial 85 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:20:16,494] Trial 86 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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[I 2026-03-13 19:20:18,909] Trial 87 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:20:20,002] Trial 88 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:20:21,225] Trial 89 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:20:21,971] Trial 90 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:20:25,070] Trial 91 finished with value:
1.3834608106757174 and parameters: {'clf_n_estimators': 629, 'clf_max_depth':
23, 'clf_min_samples_leaf': 8, 'clf_min_samples_split': 4, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 353, 'learning_rate':
0.00894784465050072, 'max_depth': 11, 'subsample': 0.8790732169047081,
'colsample_bytree': 0.6728486073873531, 'min_child_weight': 4, 'gamma':
4.592306468978554e-06, 'reg_alpha': 5.8776602682664447e-08, 'reg_lambda':
2.2004959441830425e-07}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:20:28,385] Trial 92 finished with value:
1.386820728607394 and parameters: {'clf_n_estimators': 669, 'clf_max_depth': 23,
'clf_min_samples_leaf': 11, 'clf_min_samples_split': 5, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 346, 'learning_rate':
0.008652525692416821, 'max_depth': 11, 'subsample': 0.930720470023554,
'colsample_bytree': 0.7141051638870973, 'min_child_weight': 4, 'gamma':
1.8312432254354596e-06, 'reg_alpha': 4.368108755801621e-08, 'reg_lambda':
1.7347551295787696e-06}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:20:32,186] Trial 93 finished with value:
1.3816851396548009 and parameters: {'clf_n_estimators': 631, 'clf_max_depth':
12, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 3, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 636, 'learning_rate':
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0.00526888997423869, 'max_depth': 11, 'subsample': 0.9717826094205201,
'colsample_bytree': 0.6896079655798683, 'min_child_weight': 4, 'gamma':
1.3773108570443712e-07, 'reg_alpha': 7.1236065436815e-08, 'reg_lambda':
6.572550810246932e-07}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:20:35,800] Trial 94 finished with value:
1.382296323342837 and parameters: {'clf_n_estimators': 579, 'clf_max_depth': 11,
'clf_min_samples_leaf': 11, 'clf_min_samples_split': 3, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 634, 'learning_rate':
0.005038619919541043, 'max_depth': 12, 'subsample': 0.9725539749290755,
'colsample_bytree': 0.7438834747047652, 'min_child_weight': 5, 'gamma':
3.7099511623962376e-07, 'reg_alpha': 2.6306376778789113e-07, 'reg_lambda':
7.425813005526466e-07}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:20:39,534] Trial 95 finished with value:
1.389207447382205 and parameters: {'clf_n_estimators': 581, 'clf_max_depth': 11,
'clf_min_samples_leaf': 13, 'clf_min_samples_split': 2, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 631, 'learning_rate':
0.005004746384918655, 'max_depth': 12, 'subsample': 0.9824532475571414,
'colsample_bytree': 0.7358703223042888, 'min_child_weight': 5, 'gamma':
1.5949292183116682e-07, 'reg_alpha': 1.744601830600392e-07, 'reg_lambda':
8.05746057197255e-07}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:20:42,834] Trial 96 finished with value:
1.3812770007237702 and parameters: {'clf_n_estimators': 556, 'clf_max_depth':
12, 'clf_min_samples_leaf': 12, 'clf_min_samples_split': 3, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 586, 'learning_rate':
0.005458718192471301, 'max_depth': 12, 'subsample': 0.9761863036669172,
'colsample_bytree': 0.6880310333492803, 'min_child_weight': 5, 'gamma':
2.8011141900441733e-07, 'reg_alpha': 6.621468589396338e-07, 'reg_lambda':
5.147698018021616e-07}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:20:46,329] Trial 97 finished with value:
1.3858571134648752 and parameters: {'clf_n_estimators': 564, 'clf_max_depth':
13, 'clf_min_samples_leaf': 12, 'clf_min_samples_split': 3, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 601, 'learning_rate':
0.005413927696009384, 'max_depth': 12, 'subsample': 0.9994637322931257,
'colsample_bytree': 0.7899422328077351, 'min_child_weight': 5, 'gamma':
3.15487660033729e-07, 'reg_alpha': 6.258511524642807e-07, 'reg_lambda':
4.609488366768946e-06}. Best is trial 44 with value: 1.380708717645845.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:20:47,199] Trial 98 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:20:50,032] Trial 99 finished with value:
1.3796198346927098 and parameters: {'clf_n_estimators': 436, 'clf_max_depth':
12, 'clf_min_samples_leaf': 10, 'clf_min_samples_split': 4, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 568, 'learning_rate':
0.006008417890247713, 'max_depth': 11, 'subsample': 0.9707649554438408,
'colsample_bytree': 0.7648296120558737, 'min_child_weight': 5, 'gamma':
6.661741966733032e-08, 'reg_alpha': 1.6723069533614929e-06, 'reg_lambda':
3.923193807469724e-07}. Best is trial 99 with value: 1.3796198346927098.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:20:52,702] Trial 100 finished with value:
1.3828916365312722 and parameters: {'clf_n_estimators': 435, 'clf_max_depth':
12, 'clf_min_samples_leaf': 13, 'clf_min_samples_split': 4, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 563, 'learning_rate':
0.005969600569072745, 'max_depth': 11, 'subsample': 0.9768518360931845,
'colsample_bytree': 0.7645369460482152, 'min_child_weight': 6, 'gamma':
3.6258451231305244e-08, 'reg_alpha': 1.674530653032553e-06, 'reg_lambda':
4.533137858889353e-07}. Best is trial 99 with value: 1.3796198346927098.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:20:55,756] Trial 101 finished with value:
1.3841053619054684 and parameters: {'clf_n_estimators': 471, 'clf_max_depth':
10, 'clf_min_samples_leaf': 10, 'clf_min_samples_split': 3, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 588, 'learning_rate':
0.005263748171867506, 'max_depth': 11, 'subsample': 0.9744424662100094,
'colsample_bytree': 0.8008427831401616, 'min_child_weight': 5, 'gamma':
6.211274004720318e-08, 'reg_alpha': 5.60566736850035e-07, 'reg_lambda':
1.2296942506210951e-06}. Best is trial 99 with value: 1.3796198346927098.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:20:59,151] Trial 102 finished with value:
1.379028042764923 and parameters: {'clf_n_estimators': 492, 'clf_max_depth': 14,
'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 664, 'learning_rate':
0.006370651585162557, 'max_depth': 12, 'subsample': 0.9921897857798312,
'colsample_bytree': 0.8290893746270718, 'min_child_weight': 5, 'gamma':
8.304010566064056e-08, 'reg_alpha': 9.276210252135867e-07, 'reg_lambda':
2.712271627247829e-07}. Best is trial 102 with value: 1.379028042764923.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:21:02,532] Trial 103 finished with value:
1.3840238330782677 and parameters: {'clf_n_estimators': 494, 'clf_max_depth':
14, 'clf_min_samples_leaf': 12, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 669, 'learning_rate':
0.006115002849523296, 'max_depth': 12, 'subsample': 0.988841201329127,
'colsample_bytree': 0.7348887896677496, 'min_child_weight': 5, 'gamma':
7.632000926563707e-08, 'reg_alpha': 1.2369449535440326e-06, 'reg_lambda':
9.722535644161646e-08}. Best is trial 102 with value: 1.379028042764923.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:21:05,501] Trial 104 finished with value:
1.3895103622580474 and parameters: {'clf_n_estimators': 393, 'clf_max_depth':
13, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 534, 'learning_rate':
0.005410038201426876, 'max_depth': 12, 'subsample': 0.9660288721130617,
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'colsample_bytree': 0.8393164269215516, 'min_child_weight': 4, 'gamma':
2.1200310996879996e-07, 'reg_alpha': 3.116448014390557e-06, 'reg_lambda':
3.590166172717561e-07}. Best is trial 102 with value: 1.379028042764923.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:21:06,452] Trial 105 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:21:09,538] Trial 106 finished with value:
1.3778175997557447 and parameters: {'clf_n_estimators': 446, 'clf_max_depth':
15, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 4, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 682, 'learning_rate':
0.006554128280746907, 'max_depth': 12, 'subsample': 0.9731476503760688,
'colsample_bytree': 0.8271793414364144, 'min_child_weight': 6, 'gamma':
1.1800700526200399e-07, 'reg_alpha': 4.265066494771375e-07, 'reg_lambda':
5.603751722504028e-07}. Best is trial 106 with value: 1.3778175997557447.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:21:12,721] Trial 107 finished with value:
1.3747161589700159 and parameters: {'clf_n_estimators': 442, 'clf_max_depth':
11, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 5, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 688, 'learning_rate':
0.005744228639121590, 'max_depth': 11, 'subsample': 0.9690452863445002,
'colsample_bytree': 0.81613388382731, 'min_child_weight': 6, 'gamma':
1.1193116303415551e-07, 'reg_alpha': 2.6074800955226583e-07, 'reg_lambda':
5.834812666737395e-07}. Best is trial 107 with value: 1.3747161589700159.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:21:13,733] Trial 108 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:21:14,818] Trial 109 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:21:15,813] Trial 110 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:21:18,833] Trial 111 finished with value:
1.3783912140123078 and parameters: {'clf_n_estimators': 448, 'clf_max_depth':
11, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 4, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 590, 'learning_rate':
0.005271544538005566, 'max_depth': 12, 'subsample': 0.9700344942763987,
'colsample_bytree': 0.7710559562996916, 'min_child_weight': 5, 'gamma':
1.2988415581487395e-07, 'reg_alpha': 2.0684196322347955e-07, 'reg_lambda':
6.694509741365115e-07}. Best is trial 107 with value: 1.3747161589700159.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:21:21,698] Trial 112 finished with value:
1.3741298962869888 and parameters: {'clf_n_estimators': 487, 'clf_max_depth':
13, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 4, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 578, 'learning_rate':
0.007465389289240077, 'max_depth': 12, 'subsample': 0.967688198760221,
'colsample_bytree': 0.7764964885000805, 'min_child_weight': 7, 'gamma':
1.3255364190289443e-07, 'reg_alpha': 2.3430346276448603e-07, 'reg_lambda':
2.5576204743644264e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:21:24,547] Trial 113 finished with value:
1.379411514037849 and parameters: {'clf_n_estimators': 484, 'clf_max_depth': 13,
'clf_min_samples_leaf': 9, 'clf_min_samples_split': 4, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 576, 'learning_rate':
0.007633504570919192, 'max_depth': 12, 'subsample': 0.9573707645230864,
'colsample_bytree': 0.7765087424584752, 'min_child_weight': 7, 'gamma':
5.993253330818722e-08, 'reg_alpha': 4.518996906294865e-07, 'reg_lambda':
2.74603149760112e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:21:27,255] Trial 114 finished with value:
1.3792962681711127 and parameters: {'clf_n_estimators': 420, 'clf_max_depth':
13, 'clf_min_samples_leaf': 9, 'clf_min_samples_split': 4, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 582, 'learning_rate':
0.007753738377591531, 'max_depth': 12, 'subsample': 0.96445053999578,
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'colsample_bytree': 0.8295361410563078, 'min_child_weight': 7, 'gamma':
2.411660097353005e-08, 'reg_alpha': 3.371182877505037e-07, 'reg_lambda':
3.286204153455458e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:21:30,063] Trial 115 finished with value:
1.3826324953908007 and parameters: {'clf_n_estimators': 418, 'clf_max_depth':
14, 'clf_min_samples_leaf': 9, 'clf_min_samples_split': 4, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 572, 'learning_rate':
0.007921218871375526, 'max_depth': 12, 'subsample': 0.9606081360030763,
'colsample_bytree': 0.8286670124119009, 'min_child_weight': 7, 'gamma':
6.788365938036346e-08, 'reg_alpha': 3.7779167638264744e-07, 'reg_lambda':
2.730165524485807e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:21:32,902] Trial 116 finished with value:
1.3808091088513652 and parameters: {'clf_n_estimators': 450, 'clf_max_depth':
13, 'clf_min_samples_leaf': 9, 'clf_min_samples_split': 4, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 551, 'learning_rate':
0.007386712292692088, 'max_depth': 12, 'subsample': 0.9901566187070674,
'colsample_bytree': 0.7733252404689743, 'min_child_weight': 7, 'gamma':
1.1270671075086151e-07, 'reg_alpha': 2.0577274324233376e-07, 'reg_lambda':
1.1717037262240228e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:21:33,834] Trial 117 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:21:36,728] Trial 118 finished with value:
1.3863278740453755 and parameters: {'clf_n_estimators': 465, 'clf_max_depth':
15, 'clf_min_samples_leaf': 9, 'clf_min_samples_split': 5, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 493, 'learning_rate':
0.007358450744696671, 'max_depth': 12, 'subsample': 0.9935095894441,
'colsample_bytree': 0.8072116644556364, 'min_child_weight': 7, 'gamma':
2.47154568140358e-07, 'reg_alpha': 1.542356509207878e-07, 'reg_lambda':
7.990627498077405e-08}. Best is trial 112 with value: 1.3741298962869888.
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:21:39,793] Trial 119 finished with value:
1.3773665495616332 and parameters: {'clf_n_estimators': 488, 'clf_max_depth':
11, 'clf_min_samples_leaf': 8, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 553, 'learning_rate':
0.006408706898405737, 'max_depth': 12, 'subsample': 0.9512617495670252,
'colsample_bytree': 0.7732049841356826, 'min_child_weight': 7, 'gamma':
1.0886884301658052e-07, 'reg_alpha': 1.1361134081068987e-07, 'reg_lambda':
5.0122601042462695e-08}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:21:42,787] Trial 120 finished with value:
1.3833393492284018 and parameters: {'clf_n_estimators': 482, 'clf_max_depth':
10, 'clf_min_samples_leaf': 7, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 549, 'learning_rate':
0.006579721286576952, 'max_depth': 12, 'subsample': 0.9517213948669849,
'colsample_bytree': 0.7697883247572073, 'min_child_weight': 7, 'gamma':
1.3466375493585616e-07, 'reg_alpha': 1.17007277794826e-07, 'reg_lambda':
6.269660688430603e-08}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:21:45,639] Trial 121 finished with value:
1.3903249515688338 and parameters: {'clf_n_estimators': 428, 'clf_max_depth':
12, 'clf_min_samples_leaf': 6, 'clf_min_samples_split': 4, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 582, 'learning_rate':
0.005776452614419727, 'max_depth': 12, 'subsample': 0.9658864901078545,
'colsample_bytree': 0.8662045195025471, 'min_child_weight': 8, 'gamma':
4.1471204706298946e-08, 'reg_alpha': 4.5126512799765917e-07, 'reg_lambda':
4.754496814975505e-08}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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[I 2026-03-13 19:21:48,398] Trial 122 finished with value:
1.382786088309988 and parameters: {'clf_n_estimators': 407, 'clf_max_depth': 11,
'clf_min_samples_leaf': 8, 'clf_min_samples_split': 3, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 591, 'learning_rate':
0.007282723597196525, 'max_depth': 12, 'subsample': 0.9761611540973075,
'colsample_bytree': 0.7897972767334388, 'min_child_weight': 7, 'gamma':
1.131445026981941e-07, 'reg_alpha': 9.261030677732806e-07, 'reg_lambda':
1.797597833197002e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:21:51,134] Trial 123 finished with value:
1.3856070956376183 and parameters: {'clf_n_estimators': 455, 'clf_max_depth':
15, 'clf_min_samples_leaf': 7, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 546, 'learning_rate':
0.006334214163351525, 'max_depth': 12, 'subsample': 0.9547284060330657,
'colsample_bytree': 0.8930879155072122, 'min_child_weight': 7, 'gamma':
2.780176380816084e-08, 'reg_alpha': 2.990993375108574e-07, 'reg_lambda':
1.0987689458492707e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:21:54,404] Trial 124 finished with value:
1.3856749715265912 and parameters: {'clf_n_estimators': 491, 'clf_max_depth':
13, 'clf_min_samples_leaf': 10, 'clf_min_samples_split': 4, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 610, 'learning_rate':
0.00547569653721092, 'max_depth': 12, 'subsample': 0.9886795993186136,
'colsample_bytree': 0.8389887941829722, 'min_child_weight': 6, 'gamma':
1.8071037566257794e-07, 'reg_alpha': 8.433104015758115e-08, 'reg_lambda':
3.2536199683188327e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:21:57,190] Trial 125 finished with value:
1.3807824641721709 and parameters: {'clf_n_estimators': 385, 'clf_max_depth':
12, 'clf_min_samples_leaf': 9, 'clf_min_samples_split': 3, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 560, 'learning_rate':
0.007869902608139854, 'max_depth': 12, 'subsample': 0.963189407133793,
'colsample_bytree': 0.7766753638804117, 'min_child_weight': 8, 'gamma':
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5.810246265419322e-08, 'reg_alpha': 1.8152862642347955e-07, 'reg_lambda':
2.1505002567261893e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:21:59,809] Trial 126 finished with value:
1.3853371969821913 and parameters: {'clf_n_estimators': 364, 'clf_max_depth':
13, 'clf_min_samples_leaf': 9, 'clf_min_samples_split': 3, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 524, 'learning_rate':
0.008199600695405124, 'max_depth': 12, 'subsample': 0.966770044194214,
'colsample_bytree': 0.8111414289362754, 'min_child_weight': 8, 'gamma':
5.978505718397383e-08, 'reg_alpha': 1.995098267840522e-07, 'reg_lambda':
2.0384266194099805e-08}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:00,795] Trial 127 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:01,934] Trial 128 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:05,004] Trial 129 finished with value:
1.3807562886231188 and parameters: {'clf_n_estimators': 445, 'clf_max_depth':
11, 'clf_min_samples_leaf': 9, 'clf_min_samples_split': 6, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 569, 'learning_rate':
0.007632531293127822, 'max_depth': 12, 'subsample': 0.9596210250904037,
'colsample_bytree': 0.822338748036486, 'min_child_weight': 7, 'gamma':
2.3114357684686007e-07, 'reg_alpha': 4.7016418023075294e-07, 'reg_lambda':
8.704810224188288e-08}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:05,912] Trial 130 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:08,615] Trial 131 finished with value:
1.3760703767883038 and parameters: {'clf_n_estimators': 469, 'clf_max_depth':
12, 'clf_min_samples_leaf': 9, 'clf_min_samples_split': 4, 'clf_max_features':
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0.5, 'clf_class_weight': None, 'n_estimators': 582, 'learning_rate':
0.006994104214612547, 'max_depth': 12, 'subsample': 0.9499374787485504,
'colsample_bytree': 0.7928477596991393, 'min_child_weight': 7, 'gamma':
3.2178566650605457e-07, 'reg_alpha': 6.855222515822439e-07, 'reg_lambda':
3.402136430223274e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:10,252] Trial 132 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:13,344] Trial 133 finished with value:
1.38179074343181 and parameters: {'clf_n_estimators': 479, 'clf_max_depth': 12,
'clf_min_samples_leaf': 9, 'clf_min_samples_split': 4, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 574, 'learning_rate':
0.007416788891987626, 'max_depth': 12, 'subsample': 0.7241056082588,
'colsample_bytree': 0.7744563133806182, 'min_child_weight': 7, 'gamma':
4.938340365648734e-08, 'reg_alpha': 9.303066844026104e-07, 'reg_lambda':
1.571272077610966e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:14,166] Trial 134 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:17,031] Trial 135 finished with value:
1.3799989450465808 and parameters: {'clf_n_estimators': 443, 'clf_max_depth':
11, 'clf_min_samples_leaf': 9, 'clf_min_samples_split': 6, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 551, 'learning_rate':
0.00826385401228295, 'max_depth': 12, 'subsample': 0.9901114822499772,
'colsample_bytree': 0.7581828817466076, 'min_child_weight': 6, 'gamma':
7.343951448276138e-08, 'reg_alpha': 1.1228880537847766e-07, 'reg_lambda':
3.8754800590341677e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:18,046] Trial 136 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:20,904] Trial 137 finished with value:
1.37999242548083 and parameters: {'clf_n_estimators': 506, 'clf_max_depth': 11,
'clf_min_samples_leaf': 8, 'clf_min_samples_split': 6, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 477, 'learning_rate':
0.008882530163404598, 'max_depth': 11, 'subsample': 0.9678443992188028,
'colsample_bytree': 0.7954877237794435, 'min_child_weight': 6, 'gamma':
7.096343340176917e-08, 'reg_alpha': 1.0881444050517787e-07, 'reg_lambda':
2.1831269149841922e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:23,921] Trial 138 finished with value:
1.3816821118600011 and parameters: {'clf_n_estimators': 512, 'clf_max_depth':
11, 'clf_min_samples_leaf': 8, 'clf_min_samples_split': 6, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 503, 'learning_rate':
0.008520103620243451, 'max_depth': 11, 'subsample': 0.9544847358803815,
'colsample_bytree': 0.8573538856270594, 'min_child_weight': 6, 'gamma':
3.4821007772867585e-07, 'reg_alpha': 8.521792715577775e-08, 'reg_lambda':
9.175867642427908e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:25,922] Trial 139 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:28,694] Trial 140 finished with value:
1.3828053456047584 and parameters: {'clf_n_estimators': 472, 'clf_max_depth': 8,
'clf_min_samples_leaf': 8, 'clf_min_samples_split': 6, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 475, 'learning_rate':
0.008968306822423007, 'max_depth': 11, 'subsample': 0.9696185457356468,
'colsample_bytree': 0.7967701296263026, 'min_child_weight': 6, 'gamma':
7.893276428878146e-08, 'reg_alpha': 3.3720974940910424e-07, 'reg_lambda':
3.739131288231899e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:32,044] Trial 141 finished with value:
1.3797529145377203 and parameters: {'clf_n_estimators': 437, 'clf_max_depth':
12, 'clf_min_samples_leaf': 9, 'clf_min_samples_split': 5, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 573, 'learning_rate':
0.006216513772769247, 'max_depth': 12, 'subsample': 0.9651221233800092,
'colsample_bytree': 0.8096137815346052, 'min_child_weight': 6, 'gamma':
4.808858719224738e-08, 'reg_alpha': 4.5746524525749043e-07, 'reg_lambda':
2.231437142467069e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:35,076] Trial 142 finished with value:
1.3845821822181914 and parameters: {'clf_n_estimators': 437, 'clf_max_depth':
10, 'clf_min_samples_leaf': 10, 'clf_min_samples_split': 6, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 525, 'learning_rate':
0.005990261715516125, 'max_depth': 12, 'subsample': 0.9828279775053176,
'colsample_bytree': 0.8155028997510028, 'min_child_weight': 6, 'gamma':
3.9858850458696126e-08, 'reg_alpha': 1.2206121190398114e-06, 'reg_lambda':
2.67028718619952e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:38,104] Trial 143 finished with value:
1.380238805467651 and parameters: {'clf_n_estimators': 458, 'clf_max_depth': 11,
'clf_min_samples_leaf': 9, 'clf_min_samples_split': 5, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 606, 'learning_rate':
0.006402090576519132, 'max_depth': 12, 'subsample': 0.9371309482840736,
'colsample_bytree': 0.8456413987290724, 'min_child_weight': 7, 'gamma':
7.570755959526121e-08, 'reg_alpha': 4.814734597039674e-07, 'reg_lambda':
4.7372721996760593e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:41,104] Trial 144 finished with value:
1.3790537368844704 and parameters: {'clf_n_estimators': 417, 'clf_max_depth':
12, 'clf_min_samples_leaf': 8, 'clf_min_samples_split': 5, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 654, 'learning_rate':
0.006349318635316839, 'max_depth': 11, 'subsample': 0.9408642295076186,
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'colsample_bytree': 0.8479642844348412, 'min_child_weight': 6, 'gamma':
2.0424343181340762e-08, 'reg_alpha': 1.2534469684783162e-07, 'reg_lambda':
6.389678848983367e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:44,154] Trial 145 finished with value:
1.3794939175199763 and parameters: {'clf_n_estimators': 413, 'clf_max_depth':
12, 'clf_min_samples_leaf': 8, 'clf_min_samples_split': 5, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 660, 'learning_rate':
0.006328488397337997, 'max_depth': 11, 'subsample': 0.9386979755624933,
'colsample_bytree': 0.8458096188879836, 'min_child_weight': 6, 'gamma':
2.0880004936300486e-08, 'reg_alpha': 3.033320718727747e-07, 'reg_lambda':
5.913267158357456e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:45,149] Trial 146 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:48,276] Trial 147 finished with value:
1.3873384966103 and parameters: {'clf_n_estimators': 423, 'clf_max_depth': 13,
'clf_min_samples_leaf': 7, 'clf_min_samples_split': 5, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 644, 'learning_rate':
0.005733113529508098, 'max_depth': 11, 'subsample': 0.9683236949786242,
'colsample_bytree': 0.8580989286087328, 'min_child_weight': 6, 'gamma':
2.803353968154944e-08, 'reg_alpha': 2.8647680755415566e-07, 'reg_lambda':
6.624041494151583e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:49,416] Trial 148 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:52,778] Trial 149 finished with value:
1.3823689104316184 and parameters: {'clf_n_estimators': 469, 'clf_max_depth':
12, 'clf_min_samples_leaf': 6, 'clf_min_samples_split': 4, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 631, 'learning_rate':
0.006352333847258148, 'max_depth': 10, 'subsample': 0.9484040616446022,
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'colsample_bytree': 0.8907969540172732, 'min_child_weight': 6, 'gamma':
1.5957112518780735e-08, 'reg_alpha': 7.67982297487226e-08, 'reg_lambda':
4.1068435233419143e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:53,817] Trial 150 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:56,757] Trial 151 finished with value:
1.3775299824824188 and parameters: {'clf_n_estimators': 456, 'clf_max_depth':
11, 'clf_min_samples_leaf': 8, 'clf_min_samples_split': 4, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 617, 'learning_rate':
0.006384586588130877, 'max_depth': 11, 'subsample': 0.9391783846330318,
'colsample_bytree': 0.8449830547526967, 'min_child_weight': 7, 'gamma':
4.1986517899137536e-08, 'reg_alpha': 4.573619866436637e-07, 'reg_lambda':
4.4741434523078074e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:22:59,435] Trial 152 finished with value:
1.3795928588999307 and parameters: {'clf_n_estimators': 409, 'clf_max_depth':
12, 'clf_min_samples_leaf': 8, 'clf_min_samples_split': 4, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 616, 'learning_rate':
0.0058439893110222695, 'max_depth': 11, 'subsample': 0.9344231615661773,
'colsample_bytree': 0.8312761151556853, 'min_child_weight': 6, 'gamma':
4.2103843271081503e-08, 'reg_alpha': 3.9803906341319837e-07, 'reg_lambda':
1.2863486689915508e-06}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:23:02,125] Trial 153 finished with value:
1.3791591921266928 and parameters: {'clf_n_estimators': 401, 'clf_max_depth':
12, 'clf_min_samples_leaf': 8, 'clf_min_samples_split': 4, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 669, 'learning_rate':
0.005868762624938245, 'max_depth': 10, 'subsample': 0.9400554666751405,
'colsample_bytree': 0.8393843033748243, 'min_child_weight': 6, 'gamma':
4.383674960760233e-08, 'reg_alpha': 3.562396803532592e-07, 'reg_lambda':
1.3080942137834888e-06}. Best is trial 112 with value: 1.3741298962869888.
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:23:03,006] Trial 154 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:23:05,507] Trial 155 finished with value:
1.382401130438293 and parameters: {'clf_n_estimators': 373, 'clf_max_depth': 13,
'clf_min_samples_leaf': 7, 'clf_min_samples_split': 4, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 616, 'learning_rate':
0.005839639881815457, 'max_depth': 10, 'subsample': 0.9301459107350024,
'colsample_bytree': 0.8597114136285425, 'min_child_weight': 6, 'gamma':
2.961254141249154e-08, 'reg_alpha': 7.099008784475572e-07, 'reg_lambda':
2.1116617377231405e-06}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:23:08,491] Trial 156 finished with value:
1.3809108754621253 and parameters: {'clf_n_estimators': 411, 'clf_max_depth':
12, 'clf_min_samples_leaf': 8, 'clf_min_samples_split': 4, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 675, 'learning_rate':
0.005000428757450467, 'max_depth': 11, 'subsample': 0.9409152555638282,
'colsample_bytree': 0.8420753483869756, 'min_child_weight': 7, 'gamma':
1.4668649977388315e-08, 'reg_alpha': 2.6115139223256947e-07, 'reg_lambda':
1.8015014421867726e-06}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:23:11,674] Trial 157 finished with value:
1.380763979953069 and parameters: {'clf_n_estimators': 481, 'clf_max_depth': 15,
'clf_min_samples_leaf': 6, 'clf_min_samples_split': 4, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 619, 'learning_rate':
0.006300410862961874, 'max_depth': 11, 'subsample': 0.9524802435364169,
'colsample_bytree': 0.880203137000207, 'min_child_weight': 5, 'gamma':
5.1112455216771664e-08, 'reg_alpha': 1.2556329363166058e-06, 'reg_lambda':
1.4429656169489753e-06}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:23:12,327] Trial 158 pruned.
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:23:15,073] Trial 159 finished with value:
1.3782366192489277 and parameters: {'clf_n_estimators': 463, 'clf_max_depth':
13, 'clf_min_samples_leaf': 10, 'clf_min_samples_split': 5, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 582, 'learning_rate':
0.006147897872657034, 'max_depth': 11, 'subsample': 0.9135522710874386,
'colsample_bytree': 0.8323251478216452, 'min_child_weight': 6, 'gamma':
2.0242924349432415e-08, 'reg_alpha': 4.2481286087245733e-07, 'reg_lambda':
6.375900762640091e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:23:18,394] Trial 160 finished with value:
1.3770040308041882 and parameters: {'clf_n_estimators': 464, 'clf_max_depth':
14, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 3, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 702, 'learning_rate':
0.006874661442940132, 'max_depth': 11, 'subsample': 0.9193391939973609,
'colsample_bytree': 0.8323864248366548, 'min_child_weight': 5, 'gamma':
1.94384259097971e-08, 'reg_alpha': 0.08016547022342357, 'reg_lambda':
6.540061052816361e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:23:21,510] Trial 161 finished with value:
1.378049315417722 and parameters: {'clf_n_estimators': 460, 'clf_max_depth': 14,
'clf_min_samples_leaf': 11, 'clf_min_samples_split': 3, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 696, 'learning_rate':
0.0068773374250586106, 'max_depth': 11, 'subsample': 0.9172830519856239,
'colsample_bytree': 0.8349386010210484, 'min_child_weight': 5, 'gamma':
2.07443644208663e-08, 'reg_alpha': 1.6424988570286668e-07, 'reg_lambda':
5.780631139724428e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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[I 2026-03-13 19:23:24,595] Trial 162 finished with value:
1.3780687045342426 and parameters: {'clf_n_estimators': 464, 'clf_max_depth':
14, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 3, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 699, 'learning_rate':
0.006975703642348683, 'max_depth': 11, 'subsample': 0.9135417422576162,
'colsample_bytree': 0.913578920222857, 'min_child_weight': 5, 'gamma':
1.916541969478192e-08, 'reg_alpha': 1.6653745717783692e-07, 'reg_lambda':
6.74314341328586e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:23:27,881] Trial 163 finished with value:
1.3818949793259085 and parameters: {'clf_n_estimators': 463, 'clf_max_depth':
14, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 717, 'learning_rate':
0.006950930568635242, 'max_depth': 11, 'subsample': 0.9232762620839778,
'colsample_bytree': 0.941494058610877, 'min_child_weight': 5, 'gamma':
1.0392084684017283e-08, 'reg_alpha': 0.006700408241463331, 'reg_lambda':
5.87524193629827e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:23:30,937] Trial 164 finished with value:
1.3790197321085094 and parameters: {'clf_n_estimators': 471, 'clf_max_depth':
14, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 3, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 691, 'learning_rate':
0.0071832941371507645, 'max_depth': 11, 'subsample': 0.9095982571106381,
'colsample_bytree': 0.8483257928013317, 'min_child_weight': 5, 'gamma':
2.0678393287078366e-08, 'reg_alpha': 1.6310767590240147e-07, 'reg_lambda':
8.01404854634498e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:23:34,012] Trial 165 finished with value:
1.3818045562671835 and parameters: {'clf_n_estimators': 490, 'clf_max_depth':
15, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 3, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 685, 'learning_rate':
0.007400439275086657, 'max_depth': 10, 'subsample': 0.920353901086007,
'colsample_bytree': 0.9258526935631528, 'min_child_weight': 5, 'gamma':
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1.6556435030393434e-08, 'reg_alpha': 1.7498395006914468e-07, 'reg_lambda':
8.217704385479275e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:23:34,967] Trial 166 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:23:37,851] Trial 167 finished with value:
1.3802776555411775 and parameters: {'clf_n_estimators': 459, 'clf_max_depth':
16, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 702, 'learning_rate':
0.006980768304029519, 'max_depth': 10, 'subsample': 0.9079664704467927,
'colsample_bytree': 0.8669459937526076, 'min_child_weight': 5, 'gamma':
1.1936953932955643e-08, 'reg_alpha': 5.796026566452204e-08, 'reg_lambda':
2.920073533568165e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:23:40,929] Trial 168 finished with value:
1.3795009026284035 and parameters: {'clf_n_estimators': 467, 'clf_max_depth':
15, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 3, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 730, 'learning_rate':
0.006672424497480334, 'max_depth': 11, 'subsample': 0.8954018809955057,
'colsample_bytree': 0.9564398853681694, 'min_child_weight': 5, 'gamma':
2.6994832653647597e-08, 'reg_alpha': 1.4635261680832822e-07, 'reg_lambda':
4.833147907346665e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:23:43,548] Trial 169 finished with value:
1.3861863771749316 and parameters: {'clf_n_estimators': 485, 'clf_max_depth':
13, 'clf_min_samples_leaf': 10, 'clf_min_samples_split': 3, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 700, 'learning_rate':
0.0077198926378905805, 'max_depth': 8, 'subsample': 0.9188507613908918,
'colsample_bytree': 0.8189195024856623, 'min_child_weight': 5, 'gamma':
1.530864199490131e-08, 'reg_alpha': 2.280382227399884e-07, 'reg_lambda':
1.4188928976887016e-06}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:23:46,662] Trial 170 finished with value:
1.3747010720555164 and parameters: {'clf_n_estimators': 513, 'clf_max_depth':
14, 'clf_min_samples_leaf': 10, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 673, 'learning_rate':
0.005272065120879482, 'max_depth': 11, 'subsample': 0.909800291983154,
'colsample_bytree': 0.8347109802465756, 'min_child_weight': 5, 'gamma':
2.983464822290137e-08, 'reg_alpha': 0.019421910008057253, 'reg_lambda':
7.453285736638095e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:23:49,803] Trial 171 finished with value:
1.375384317582178 and parameters: {'clf_n_estimators': 515, 'clf_max_depth': 14,
'clf_min_samples_leaf': 10, 'clf_min_samples_split': 2, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 682, 'learning_rate':
0.005524543309533737, 'max_depth': 11, 'subsample': 0.9108189458702842,
'colsample_bytree': 0.8341227229203906, 'min_child_weight': 5, 'gamma':
2.9959897129386746e-08, 'reg_alpha': 0.026159477428354363, 'reg_lambda':
6.991371743532199e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:23:52,936] Trial 172 finished with value:
1.3767386980452845 and parameters: {'clf_n_estimators': 524, 'clf_max_depth':
14, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 675, 'learning_rate':
0.005469852710534025, 'max_depth': 11, 'subsample': 0.9010598056887845,
'colsample_bytree': 0.8353997967855922, 'min_child_weight': 5, 'gamma':
3.081045680393435e-08, 'reg_alpha': 0.07817550454194937, 'reg_lambda':
7.557494478836657e-07}. Best is trial 112 with value: 1.3741298962869888.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:23:56,099] Trial 173 finished with value:
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1.374048201093393 and parameters: {'clf_n_estimators': 524, 'clf_max_depth': 14,
'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 676, 'learning_rate':
0.005370273218896229, 'max_depth': 11, 'subsample': 0.9078645405455606,
'colsample_bytree': 0.8443908098826469, 'min_child_weight': 5, 'gamma':
3.472092131387786e-08, 'reg_alpha': 0.06488695728903746, 'reg_lambda':
9.107682261318007e-07}. Best is trial 173 with value: 1.374048201093393.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:23:59,297] Trial 174 finished with value:
1.3742257343324116 and parameters: {'clf_n_estimators': 526, 'clf_max_depth':
14, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 692, 'learning_rate':
0.005484352072937818, 'max_depth': 11, 'subsample': 0.9079278030708211,
'colsample_bytree': 0.8542086445313873, 'min_child_weight': 5, 'gamma':
2.982023819657321e-08, 'reg_alpha': 0.05983183352091487, 'reg_lambda':
7.998591310403043e-07}. Best is trial 173 with value: 1.374048201093393.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:24:02,479] Trial 175 finished with value:
1.375489782412611 and parameters: {'clf_n_estimators': 527, 'clf_max_depth': 14,
'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 687, 'learning_rate':
0.005233306150018524, 'max_depth': 11, 'subsample': 0.9064623281902654,
'colsample_bytree': 0.8641628546734108, 'min_child_weight': 5, 'gamma':
3.6019433104860616e-08, 'reg_alpha': 0.040649697566515444, 'reg_lambda':
2.2212036359905302e-06}. Best is trial 173 with value: 1.374048201093393.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:24:05,917] Trial 176 finished with value:
1.376115468249708 and parameters: {'clf_n_estimators': 536, 'clf_max_depth': 14,
'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 749, 'learning_rate':
0.005280332776798223, 'max_depth': 11, 'subsample': 0.9011287488051465,
'colsample_bytree': 0.8853565440111082, 'min_child_weight': 5, 'gamma':
3.231082235408298e-08, 'reg_alpha': 0.08811237883334154, 'reg_lambda':
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2.7150610228614053e-06}. Best is trial 173 with value: 1.374048201093393.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:24:07,038] Trial 177 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:24:10,325] Trial 178 finished with value:
1.375584985454099 and parameters: {'clf_n_estimators': 534, 'clf_max_depth': 15,
'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 719, 'learning_rate':
0.00532000856410942, 'max_depth': 11, 'subsample': 0.9009050536298888,
'colsample_bytree': 0.8843074933163229, 'min_child_weight': 5, 'gamma':
0.012255151581725448, 'reg_alpha': 0.03763486683942308, 'reg_lambda':
3.501954822270207e-06}. Best is trial 173 with value: 1.374048201093393.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:24:13,692] Trial 179 finished with value:
1.3805763121973953 and parameters: {'clf_n_estimators': 531, 'clf_max_depth':
16, 'clf_min_samples_leaf': 10, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 757, 'learning_rate':
0.005002258874364939, 'max_depth': 11, 'subsample': 0.9008245425901483,
'colsample_bytree': 0.8839047737799076, 'min_child_weight': 5, 'gamma':
0.0021231717893952433, 'reg_alpha': 0.026802310699315546, 'reg_lambda':
2.2288887380928975e-06}. Best is trial 173 with value: 1.374048201093393.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:24:16,980] Trial 180 finished with value:
1.3766605858703078 and parameters: {'clf_n_estimators': 537, 'clf_max_depth':
15, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 724, 'learning_rate':
0.005446829651581548, 'max_depth': 11, 'subsample': 0.891527860850721,
'colsample_bytree': 0.8732602275607466, 'min_child_weight': 5, 'gamma':
0.014817441757255575, 'reg_alpha': 0.08191324106768691, 'reg_lambda':
3.5760966240299913e-06}. Best is trial 173 with value: 1.374048201093393.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:24:20,567] Trial 181 finished with value:
1.3760007786129431 and parameters: {'clf_n_estimators': 538, 'clf_max_depth':
15, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 721, 'learning_rate':
0.00549356222184123, 'max_depth': 11, 'subsample': 0.9092351888204532,
'colsample_bytree': 0.9014484931520602, 'min_child_weight': 5, 'gamma':
0.012901909528277387, 'reg_alpha': 0.15963049959361264, 'reg_lambda':
4.3671601111511066e-06}. Best is trial 173 with value: 1.374048201093393.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:24:24,036] Trial 182 finished with value:
1.3774158340104563 and parameters: {'clf_n_estimators': 539, 'clf_max_depth':
15, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 743, 'learning_rate':
0.005474582144640177, 'max_depth': 11, 'subsample': 0.8905712044648164,
'colsample_bytree': 0.904153346305324, 'min_child_weight': 5, 'gamma':
0.015384693959370263, 'reg_alpha': 0.19043419818853285, 'reg_lambda':
3.244612980926312e-06}. Best is trial 173 with value: 1.374048201093393.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:24:27,405] Trial 183 finished with value:
1.381595911213968 and parameters: {'clf_n_estimators': 535, 'clf_max_depth': 15,
'clf_min_samples_leaf': 12, 'clf_min_samples_split': 2, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 749, 'learning_rate':
0.005432853115146465, 'max_depth': 11, 'subsample': 0.8877087071985126,
'colsample_bytree': 0.8845612701268833, 'min_child_weight': 5, 'gamma':
0.013084757592173462, 'reg_alpha': 0.17369964139354963, 'reg_lambda':
3.878966260622448e-06}. Best is trial 173 with value: 1.374048201093393.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:24:30,636] Trial 184 finished with value:
1.3764867999493038 and parameters: {'clf_n_estimators': 517, 'clf_max_depth':
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15, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 738, 'learning_rate':
0.005444393395408464, 'max_depth': 11, 'subsample': 0.9002430126643663,
'colsample_bytree': 0.8716540403702182, 'min_child_weight': 5, 'gamma':
0.012762764423178944, 'reg_alpha': 0.06147651743038985, 'reg_lambda':
6.01442970991167e-06}. Best is trial 173 with value: 1.374048201093393.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:24:33,889] Trial 185 finished with value:
1.3801517445492744 and parameters: {'clf_n_estimators': 518, 'clf_max_depth':
16, 'clf_min_samples_leaf': 12, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 731, 'learning_rate':
0.005483608146083307, 'max_depth': 11, 'subsample': 0.8962303136477427,
'colsample_bytree': 0.8773876018390888, 'min_child_weight': 5, 'gamma':
0.014187499734825023, 'reg_alpha': 0.048802745317111026, 'reg_lambda':
3.2645697776994307e-06}. Best is trial 173 with value: 1.374048201093393.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:24:37,186] Trial 186 finished with value:
1.3748482956729404 and parameters: {'clf_n_estimators': 541, 'clf_max_depth':
15, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 715, 'learning_rate':
0.005385359989930946, 'max_depth': 11, 'subsample': 0.905165201164344,
'colsample_bytree': 0.8986432522416254, 'min_child_weight': 5, 'gamma':
0.006921966823803718, 'reg_alpha': 0.14795234906282362, 'reg_lambda':
5.200413244174139e-06}. Best is trial 173 with value: 1.374048201093393.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:24:40,527] Trial 187 finished with value:
1.3758644989638444 and parameters: {'clf_n_estimators': 545, 'clf_max_depth':
17, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 724, 'learning_rate':
0.005311797754899166, 'max_depth': 11, 'subsample': 0.9049750702351621,
'colsample_bytree': 0.8985329762399337, 'min_child_weight': 5, 'gamma':
0.008205163840923884, 'reg_alpha': 0.16650293375948635, 'reg_lambda':
5.967060839066985e-06}. Best is trial 173 with value: 1.374048201093393.
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:24:43,877] Trial 188 finished with value:
1.375433814239918 and parameters: {'clf_n_estimators': 542, 'clf_max_depth': 17,
'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 718, 'learning_rate':
0.005412318483066974, 'max_depth': 11, 'subsample': 0.9022655687800313,
'colsample_bytree': 0.8935996613516937, 'min_child_weight': 5, 'gamma':
0.007960799554897539, 'reg_alpha': 0.17274699631666876, 'reg_lambda':
5.982051717675171e-06}. Best is trial 173 with value: 1.374048201093393.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:24:44,974] Trial 189 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:24:48,289] Trial 190 finished with value:
1.3744734371751555 and parameters: {'clf_n_estimators': 509, 'clf_max_depth':
17, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 794, 'learning_rate':
0.005360033420891113, 'max_depth': 11, 'subsample': 0.8712524320908963,
'colsample_bytree': 0.9140114307583571, 'min_child_weight': 5, 'gamma':
0.02750349428268114, 'reg_alpha': 0.028320342272845273, 'reg_lambda':
8.769745600443921e-06}. Best is trial 173 with value: 1.374048201093393.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:24:51,510] Trial 191 finished with value:
1.3744742076752603 and parameters: {'clf_n_estimators': 510, 'clf_max_depth':
17, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 717, 'learning_rate':
0.005331240737779661, 'max_depth': 11, 'subsample': 0.902706858726543,
'colsample_bytree': 0.9114112740330054, 'min_child_weight': 5, 'gamma':
0.003750623278514656, 'reg_alpha': 0.029277880542798795, 'reg_lambda':
1.2700416207161215e-05}. Best is trial 173 with value: 1.374048201093393.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:24:54,926] Trial 192 finished with value:
1.3757733495694335 and parameters: {'clf_n_estimators': 511, 'clf_max_depth':
18, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 825, 'learning_rate':
0.005352751666753888, 'max_depth': 11, 'subsample': 0.8651584716741032,
'colsample_bytree': 0.9154425015462982, 'min_child_weight': 5, 'gamma':
0.02164011151166553, 'reg_alpha': 0.0270846642741387, 'reg_lambda':
1.107434843536291e-05}. Best is trial 173 with value: 1.374048201093393.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:24:58,264] Trial 193 finished with value:
1.3760912167038548 and parameters: {'clf_n_estimators': 514, 'clf_max_depth':
17, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 786, 'learning_rate':
0.005383089323422584, 'max_depth': 11, 'subsample': 0.8650769881767346,
'colsample_bytree': 0.9279493885085952, 'min_child_weight': 5, 'gamma':
0.0274111856048773, 'reg_alpha': 0.022853653683787317, 'reg_lambda':
1.0842018405414058e-05}. Best is trial 173 with value: 1.374048201093393.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:25:01,714] Trial 194 finished with value:
1.3756831843665243 and parameters: {'clf_n_estimators': 509, 'clf_max_depth':
17, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 852, 'learning_rate':
0.0052751554743139546, 'max_depth': 11, 'subsample': 0.8665352837608837,
'colsample_bytree': 0.9217054167127493, 'min_child_weight': 5, 'gamma':
0.029444985030835154, 'reg_alpha': 0.023569377496931266, 'reg_lambda':
1.1628118679236038e-05}. Best is trial 173 with value: 1.374048201093393.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:25:05,117] Trial 195 finished with value:
1.3784865144091114 and parameters: {'clf_n_estimators': 508, 'clf_max_depth':
17, 'clf_min_samples_leaf': 10, 'clf_min_samples_split': 2, 'clf_max_features':
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0.5, 'clf_class_weight': None, 'n_estimators': 818, 'learning_rate':
0.005009302052608029, 'max_depth': 11, 'subsample': 0.8565095131432287,
'colsample_bytree': 0.9272622882787789, 'min_child_weight': 5, 'gamma':
0.03342725457004945, 'reg_alpha': 0.029027121610122577, 'reg_lambda':
1.2753854380348848e-05}. Best is trial 173 with value: 1.374048201093393.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:25:06,334] Trial 196 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:25:09,852] Trial 197 finished with value:
1.3792216108899409 and parameters: {'clf_n_estimators': 512, 'clf_max_depth':
18, 'clf_min_samples_leaf': 10, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 868, 'learning_rate':
0.005007711110955994, 'max_depth': 11, 'subsample': 0.8655092365792032,
'colsample_bytree': 0.9409754627113711, 'min_child_weight': 5, 'gamma':
0.006939382300680122, 'reg_alpha': 0.013819968594664918, 'reg_lambda':
2.26437808595664e-05}. Best is trial 173 with value: 1.374048201093393.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:25:13,112] Trial 198 finished with value:
1.3780319786234896 and parameters: {'clf_n_estimators': 525, 'clf_max_depth':
17, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 782, 'learning_rate':
0.005641822939686477, 'max_depth': 10, 'subsample': 0.8802293843400332,
'colsample_bytree': 0.9519630495686116, 'min_child_weight': 5, 'gamma':
0.004035537865784354, 'reg_alpha': 0.04233279254459902, 'reg_lambda':
5.591062208538761e-06}. Best is trial 173 with value: 1.374048201093393.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:25:14,264] Trial 199 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:25:15,411] Trial 200 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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[I 2026-03-13 19:25:18,952] Trial 201 finished with value:
1.3777762855050575 and parameters: {'clf_n_estimators': 548, 'clf_max_depth':
17, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 770, 'learning_rate':
0.005311479486532774, 'max_depth': 11, 'subsample': 0.8922652020579624,
'colsample_bytree': 0.9270413369190188, 'min_child_weight': 5, 'gamma':
0.010267599110068639, 'reg_alpha': 0.15260133351215516, 'reg_lambda':
4.823557594538477 5e-06}. Best is trial 173 with value: 1.374048201093393.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:25:22,365] Trial 202 finished with value:
1.3744430545077322 and parameters: {'clf_n_estimators': 534, 'clf_max_depth':
18, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 806, 'learning_rate':
0.005605259232866854, 'max_depth': 11, 'subsample': 0.8699800920308429,
'colsample_bytree': 0.8884274414422507, 'min_child_weight': 5, 'gamma':
0.004168558395119644, 'reg_alpha': 0.020677153915731467, 'reg_lambda':
5.081328070450845e-06}. Best is trial 173 with value: 1.374048201093393.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:25:25,774] Trial 203 finished with value:
1.3730843257119691 and parameters: {'clf_n_estimators': 529, 'clf_max_depth':
18, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 809, 'learning_rate':
0.005322129622015431, 'max_depth': 11, 'subsample': 0.8673433515754215,
'colsample_bytree': 0.892793528347274, 'min_child_weight': 5, 'gamma':
0.004805753175460478, 'reg_alpha': 0.0097931176444647, 'reg_lambda':
7.3331051436349866e-06}. Best is trial 203 with value: 1.3730843257119691.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:25:29,297] Trial 204 finished with value:
1.3831400983482969 and parameters: {'clf_n_estimators': 566, 'clf_max_depth':
18, 'clf_min_samples_leaf': 10, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 804, 'learning_rate':
0.0058471586391068585, 'max_depth': 11, 'subsample': 0.8645475620042328,
'colsample_bytree': 0.9057088761121928, 'min_child_weight': 5, 'gamma':
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0.004573981748533599, 'reg_alpha': 0.020025971535262838, 'reg_lambda':
1.170741731372602e-05}. Best is trial 203 with value: 1.3730843257119691.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:25:32,575] Trial 205 finished with value:
1.3785908066216723 and parameters: {'clf_n_estimators': 533, 'clf_max_depth':
18, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 808, 'learning_rate':
0.005048363106543042, 'max_depth': 10, 'subsample': 0.8488212111397508,
'colsample_bytree': 0.8890769553757385, 'min_child_weight': 5, 'gamma':
0.003534246830399618, 'reg_alpha': 0.030477744986973455, 'reg_lambda':
7.886301625021626e-06}. Best is trial 203 with value: 1.3730843257119691.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:25:33,718] Trial 206 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:25:37,165] Trial 207 finished with value:
1.3791154040368188 and parameters: {'clf_n_estimators': 526, 'clf_max_depth':
17, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 761, 'learning_rate':
0.005339794692271515, 'max_depth': 11, 'subsample': 0.8683105652553265,
'colsample_bytree': 0.8944130532745488, 'min_child_weight': 4, 'gamma':
0.04619568851444052, 'reg_alpha': 0.10993297156394714, 'reg_lambda':
4.489397447880034e-06}. Best is trial 203 with value: 1.3730843257119691.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:25:38,348] Trial 208 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:25:41,495] Trial 209 finished with value:
1.3805204497040264 and parameters: {'clf_n_estimators': 503, 'clf_max_depth':
19, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 715, 'learning_rate':
0.005306422927134957, 'max_depth': 11, 'subsample': 0.8349472489253934,
'colsample_bytree': 0.9373268875437305, 'min_child_weight': 5, 'gamma':
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0.005888295361899234, 'reg_alpha': 0.011738258409096267, 'reg_lambda':
5.9892264397271365e-05}. Best is trial 203 with value: 1.3730843257119691.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:25:44,965] Trial 210 finished with value:
1.3770671998478514 and parameters: {'clf_n_estimators': 534, 'clf_max_depth':
17, 'clf_min_samples_leaf': 10, 'clf_min_samples_split': 3, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 783, 'learning_rate':
0.005004940295505447, 'max_depth': 11, 'subsample': 0.9073063256402807,
'colsample_bytree': 0.901433831578974, 'min_child_weight': 5, 'gamma':
0.007458557109020258, 'reg_alpha': 0.2327188310736931, 'reg_lambda':
1.1208101459184751e-05}. Best is trial 203 with value: 1.3730843257119691.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:25:48,354] Trial 211 finished with value:
1.3764094094986579 and parameters: {'clf_n_estimators': 514, 'clf_max_depth':
16, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 763, 'learning_rate':
0.005531633998824399, 'max_depth': 11, 'subsample': 0.9007182390358921,
'colsample_bytree': 0.8639044067147504, 'min_child_weight': 5, 'gamma':
0.010999269440813765, 'reg_alpha': 0.05646170721381094, 'reg_lambda':
6.072156738439097e-06}. Best is trial 203 with value: 1.3730843257119691.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:25:51,706] Trial 212 finished with value:
1.3737973903709968 and parameters: {'clf_n_estimators': 521, 'clf_max_depth':
17, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 764, 'learning_rate':
0.005788387612014433, 'max_depth': 11, 'subsample': 0.9095174479734108,
'colsample_bytree': 0.8833591159557622, 'min_child_weight': 5, 'gamma':
0.01807954409943281, 'reg_alpha': 0.1196303363152475, 'reg_lambda':
5.035580805943101e-06}. Best is trial 203 with value: 1.3730843257119691.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:25:55,156] Trial 213 finished with value:
1.3819316726327315 and parameters: {'clf_n_estimators': 541, 'clf_max_depth':
17, 'clf_min_samples_leaf': 12, 'clf_min_samples_split': 3, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 792, 'learning_rate':
0.005945341340044811, 'max_depth': 11, 'subsample': 0.9071787203060951,
'colsample_bytree': 0.8852336362381161, 'min_child_weight': 5, 'gamma':
0.021467232474394728, 'reg_alpha': 0.1220695698948795, 'reg_lambda':
2.4484505121589915e-06}. Best is trial 203 with value: 1.3730843257119691.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:25:58,687] Trial 214 finished with value:
1.3794295656546183 and parameters: {'clf_n_estimators': 570, 'clf_max_depth':
18, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 808, 'learning_rate':
0.00570541960855318, 'max_depth': 11, 'subsample': 0.8609988537766263,
'colsample_bytree': 0.9077326807187026, 'min_child_weight': 5, 'gamma':
0.04459686364694605, 'reg_alpha': 0.03720042167315306, 'reg_lambda':
1.8901884213717636e-05}. Best is trial 203 with value: 1.3730843257119691.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:26:02,235] Trial 215 finished with value:
1.3775222752126906 and parameters: {'clf_n_estimators': 522, 'clf_max_depth':
16, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 868, 'learning_rate':
0.005273699333527137, 'max_depth': 11, 'subsample': 0.887987617414783,
'colsample_bytree': 0.9327224802313016, 'min_child_weight': 5, 'gamma':
0.02139783052810067, 'reg_alpha': 0.02072367245862192, 'reg_lambda':
8.645516677698168e-06}. Best is trial 203 with value: 1.3730843257119691.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:26:03,136] Trial 216 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:26:07,084] Trial 217 finished with value:
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1.3809995362124015 and parameters: {'clf_n_estimators': 551, 'clf_max_depth':
17, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 902, 'learning_rate':
0.005000439691099546, 'max_depth': 11, 'subsample': 0.9105001559513664,
'colsample_bytree': 0.9202096786712043, 'min_child_weight': 4, 'gamma':
0.0029757789011413338, 'reg_alpha': 0.39158574479264674, 'reg_lambda':
3.1600199301081834e-06}. Best is trial 203 with value: 1.3730843257119691.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:26:10,527] Trial 218 finished with value:
1.3800195594072433 and parameters: {'clf_n_estimators': 529, 'clf_max_depth':
18, 'clf_min_samples_leaf': 10, 'clf_min_samples_split': 3, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 854, 'learning_rate':
0.0056227672123683745, 'max_depth': 11, 'subsample': 0.8986516381759265,
'colsample_bytree': 0.8749747965336128, 'min_child_weight': 5, 'gamma':
0.009840287574622403, 'reg_alpha': 0.02783923276936944, 'reg_lambda':
1.2813478908537548e-05}. Best is trial 203 with value: 1.3730843257119691.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:26:13,930] Trial 219 finished with value:
1.3782490211194462 and parameters: {'clf_n_estimators': 498, 'clf_max_depth':
17, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 821, 'learning_rate':
0.005994791913107687, 'max_depth': 11, 'subsample': 0.9246593897942038,
'colsample_bytree': 0.9031828854079393, 'min_child_weight': 5, 'gamma':
0.006026386897101776, 'reg_alpha': 0.11311692382320764, 'reg_lambda':
8.684671267768935e-06}. Best is trial 203 with value: 1.3730843257119691.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:26:17,014] Trial 220 finished with value:
1.3800050587355532 and parameters: {'clf_n_estimators': 515, 'clf_max_depth':
16, 'clf_min_samples_leaf': 10, 'clf_min_samples_split': 3, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 719, 'learning_rate':
0.005329438348851586, 'max_depth': 10, 'subsample': 0.8932655058649919,
'colsample_bytree': 0.8621231661219774, 'min_child_weight': 5, 'gamma':
0.001110564411894482, 'reg_alpha': 0.008910521708233778, 'reg_lambda':
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4.455777792771483e-06}. Best is trial 203 with value: 1.3730843257119691.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:26:20,339] Trial 221 finished with value:
1.3771132749557904 and parameters: {'clf_n_estimators': 515, 'clf_max_depth':
16, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 768, 'learning_rate':
0.005585089803897287, 'max_depth': 11, 'subsample': 0.9030526014743544,
'colsample_bytree': 0.8843150064220282, 'min_child_weight': 5, 'gamma':
0.034354898745461826, 'reg_alpha': 0.05631277402440432, 'reg_lambda':
6.208000985868932e-06}. Best is trial 203 with value: 1.3730843257119691.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:26:23,732] Trial 222 finished with value:
1.3708078024605301 and parameters: {'clf_n_estimators': 540, 'clf_max_depth':
17, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 767, 'learning_rate':
0.0053532497193304815, 'max_depth': 11, 'subsample': 0.91012964461159,
'colsample_bytree': 0.8637114182265963, 'min_child_weight': 5, 'gamma':
0.019732802459394735, 'reg_alpha': 0.03701026030802366, 'reg_lambda':
6.6929726279484965e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:26:27,213] Trial 223 finished with value:
1.378096172105187 and parameters: {'clf_n_estimators': 540, 'clf_max_depth': 17,
'clf_min_samples_leaf': 12, 'clf_min_samples_split': 2, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 788, 'learning_rate':
0.0050083690567419275, 'max_depth': 11, 'subsample': 0.9125881111951574,
'colsample_bytree': 0.894172797205625, 'min_child_weight': 5, 'gamma':
0.01725918494798505, 'reg_alpha': 0.017895379782868343, 'reg_lambda':
3.420781155247788e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:26:30,548] Trial 224 finished with value:
1.3770443772826615 and parameters: {'clf_n_estimators': 553, 'clf_max_depth':
18, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 3, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 752, 'learning_rate':
0.0060194087056719804, 'max_depth': 11, 'subsample': 0.8712457300682392,
'colsample_bytree': 0.8810529775902743, 'min_child_weight': 5, 'gamma':
0.019765690009527066, 'reg_alpha': 0.02752795720568645, 'reg_lambda':
2.578153478445797e-05}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:26:33,837] Trial 225 finished with value:
1.3780242196092976 and parameters: {'clf_n_estimators': 532, 'clf_max_depth':
18, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 733, 'learning_rate':
0.005356805426525425, 'max_depth': 11, 'subsample': 0.8481514380179535,
'colsample_bytree': 0.9136221916362626, 'min_child_weight': 5, 'gamma':
0.008466061313233337, 'reg_alpha': 0.036213912201113765, 'reg_lambda':
2.620607290175909e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:26:37,538] Trial 226 finished with value:
1.3802806222623818 and parameters: {'clf_n_estimators': 563, 'clf_max_depth':
17, 'clf_min_samples_leaf': 10, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 709, 'learning_rate':
0.005686891017561781, 'max_depth': 11, 'subsample': 0.9248250587725725,
'colsample_bytree': 0.8590114911670178, 'min_child_weight': 5, 'gamma':
0.04081921143004388, 'reg_alpha': 0.15259732929082423, 'reg_lambda':
7.696079931156194e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:26:38,822] Trial 227 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:26:39,859] Trial 228 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:26:41,138] Trial 229 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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[I 2026-03-13 19:26:42,512] Trial 230 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:26:46,286] Trial 231 finished with value:
1.3758418842690106 and parameters: {'clf_n_estimators': 515, 'clf_max_depth':
16, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 759, 'learning_rate':
0.005352007784117606, 'max_depth': 11, 'subsample': 0.9021464415573422,
'colsample_bytree': 0.8653523636240807, 'min_child_weight': 5, 'gamma':
0.010011078370358579, 'reg_alpha': 0.051409151589255804, 'reg_lambda':
6.236597692328486e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:26:50,206] Trial 232 finished with value:
1.3756742530752393 and parameters: {'clf_n_estimators': 526, 'clf_max_depth':
17, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 799, 'learning_rate':
0.005270604098889861, 'max_depth': 11, 'subsample': 0.9046030681474212,
'colsample_bytree': 0.8791750258903863, 'min_child_weight': 5, 'gamma':
0.009401338117736057, 'reg_alpha': 0.10643108723063262, 'reg_lambda':
2.716488855382552e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:26:53,953] Trial 233 finished with value:
1.3759619114193729 and parameters: {'clf_n_estimators': 497, 'clf_max_depth':
17, 'clf_min_samples_leaf': 10, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 797, 'learning_rate':
0.005034410384636493, 'max_depth': 11, 'subsample': 0.91284327264284,
'colsample_bytree': 0.8596968193539848, 'min_child_weight': 5, 'gamma':
0.009425714146094475, 'reg_alpha': 0.11033901593854258, 'reg_lambda':
7.306675883683324e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:26:58,593] Trial 234 finished with value:
1.376739928796755 and parameters: {'clf_n_estimators': 491, 'clf_max_depth': 16,
'clf_min_samples_leaf': 10, 'clf_min_samples_split': 2, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 814, 'learning_rate':
0.005766649501824803, 'max_depth': 11, 'subsample': 0.916048343609484,
'colsample_bytree': 0.8642779374865475, 'min_child_weight': 5, 'gamma':
0.009860200581660946, 'reg_alpha': 0.003996714260628697, 'reg_lambda':
6.853039284712941e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:27:02,296] Trial 235 finished with value:
1.3788379367503918 and parameters: {'clf_n_estimators': 526, 'clf_max_depth':
17, 'clf_min_samples_leaf': 10, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 795, 'learning_rate':
0.00500878184522993, 'max_depth': 11, 'subsample': 0.9263420211627481,
'colsample_bytree': 0.8532622213146333, 'min_child_weight': 5, 'gamma':
0.005030169005319543, 'reg_alpha': 0.127265450110013, 'reg_lambda':
4.622795604668242e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:27:05,981] Trial 236 finished with value:
1.3784204344870743 and parameters: {'clf_n_estimators': 502, 'clf_max_depth':
16, 'clf_min_samples_leaf': 10, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 732, 'learning_rate':
0.0060274339125568795, 'max_depth': 11, 'subsample': 0.9087502910174574,
'colsample_bytree': 0.8763386583578306, 'min_child_weight': 5, 'gamma':
0.00789618791545471, 'reg_alpha': 0.2129084608180925, 'reg_lambda':
1.800127117545289e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:27:07,223] Trial 237 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:27:08,409] Trial 238 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:27:12,153] Trial 239 finished with value:
1.377019656907245 and parameters: {'clf_n_estimators': 505, 'clf_max_depth': 17,
'clf_min_samples_leaf': 10, 'clf_min_samples_split': 2, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 768, 'learning_rate':
0.005330973005994701, 'max_depth': 11, 'subsample': 0.9138536397035304,
'colsample_bytree': 0.8535701374019293, 'min_child_weight': 5, 'gamma':
0.010389806249870527, 'reg_alpha': 0.0478150210039725, 'reg_lambda':
0.38617329072220247}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:27:12,936] Trial 240 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:27:16,555] Trial 241 finished with value:
1.3774425319109718 and parameters: {'clf_n_estimators': 513, 'clf_max_depth':
17, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 795, 'learning_rate':
0.0053412901752234905, 'max_depth': 11, 'subsample': 0.8756099259105194,
'colsample_bytree': 0.916949279761702, 'min_child_weight': 5, 'gamma':
0.014598311951233832, 'reg_alpha': 0.029822974948571623, 'reg_lambda':
9.695878712659433e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:27:20,081] Trial 242 finished with value:
1.3763855326938905 and parameters: {'clf_n_estimators': 519, 'clf_max_depth':
17, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 789, 'learning_rate':
0.0056594723028620345, 'max_depth': 11, 'subsample': 0.8953068671055474,
'colsample_bytree': 0.9289886350520695, 'min_child_weight': 5, 'gamma':
0.02362755358289918, 'reg_alpha': 0.03542684616036163, 'reg_lambda':
1.6568826211875874e-05}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:27:23,734] Trial 243 finished with value:
1.3780531522233035 and parameters: {'clf_n_estimators': 533, 'clf_max_depth':
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18, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 830, 'learning_rate':
0.0053378892884868205, 'max_depth': 11, 'subsample': 0.9221231835795165,
'colsample_bytree': 0.9083095559467512, 'min_child_weight': 5, 'gamma':
0.00979662434477048, 'reg_alpha': 0.017601852033064247, 'reg_lambda':
5.93420051830777e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:27:24,874] Trial 244 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:27:28,233] Trial 245 finished with value:
1.373780226926458 and parameters: {'clf_n_estimators': 541, 'clf_max_depth': 16,
'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 678, 'learning_rate':
0.005705190854487684, 'max_depth': 11, 'subsample': 0.9074871057019098,
'colsample_bytree': 0.8789565592039499, 'min_child_weight': 5, 'gamma':
0.007499457549953126, 'reg_alpha': 0.04449356983629024, 'reg_lambda':
4.624082541952214e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:27:31,506] Trial 246 finished with value:
1.3768859161066882 and parameters: {'clf_n_estimators': 542, 'clf_max_depth':
16, 'clf_min_samples_leaf': 10, 'clf_min_samples_split': 3, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 672, 'learning_rate':
0.00586160395817737, 'max_depth': 11, 'subsample': 0.908528979577477,
'colsample_bytree': 0.8726818333885018, 'min_child_weight': 5, 'gamma':
0.007248701269604422, 'reg_alpha': 0.09742356304288638, 'reg_lambda':
4.0993897715529214e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:27:32,619] Trial 247 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:27:33,779] Trial 248 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:27:37,114] Trial 249 finished with value:
1.3778884308085007 and parameters: {'clf_n_estimators': 521, 'clf_max_depth':
14, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 3, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 683, 'learning_rate':
0.005025941604078347, 'max_depth': 11, 'subsample': 0.8996892303220884,
'colsample_bytree': 0.884543843064589, 'min_child_weight': 5, 'gamma':
0.005367199135887312, 'reg_alpha': 0.27931510965069745, 'reg_lambda':
3.200917742869895e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:27:40,728] Trial 250 finished with value:
1.3803527806609373 and parameters: {'clf_n_estimators': 554, 'clf_max_depth':
16, 'clf_min_samples_leaf': 10, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 733, 'learning_rate':
0.005689085766730984, 'max_depth': 11, 'subsample': 0.9158231933825819,
'colsample_bytree': 0.8996231788575587, 'min_child_weight': 5, 'gamma':
0.01744189886157624, 'reg_alpha': 0.06607583155890948, 'reg_lambda':
6.849819320183206e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:27:41,538] Trial 251 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:27:42,745] Trial 252 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:27:43,954] Trial 253 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:27:45,211] Trial 254 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:27:48,646] Trial 255 finished with value:
1.3806126627652076 and parameters: {'clf_n_estimators': 525, 'clf_max_depth':
14, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 740, 'learning_rate':
0.006476625506554387, 'max_depth': 10, 'subsample': 0.9069744716430207,
'colsample_bytree': 0.8902675611275026, 'min_child_weight': 5, 'gamma':
0.018791544217602485, 'reg_alpha': 0.03880995995927142, 'reg_lambda':
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4.690731693918858e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:27:52,589] Trial 256 finished with value:
1.3757972201127562 and parameters: {'clf_n_estimators': 502, 'clf_max_depth':
16, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 3, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 766, 'learning_rate':
0.005396380846482918, 'max_depth': 11, 'subsample': 0.9291024139722537,
'colsample_bytree': 0.8750943480292411, 'min_child_weight': 5, 'gamma':
0.010667346640831266, 'reg_alpha': 0.024267531910076856, 'reg_lambda':
1.519416603241617e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:27:54,017] Trial 257 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:27:58,476] Trial 258 finished with value:
1.3788264661874752 and parameters: {'clf_n_estimators': 530, 'clf_max_depth':
16, 'clf_min_samples_leaf': 10, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 804, 'learning_rate':
0.005011504598995264, 'max_depth': 11, 'subsample': 0.8981627122300396,
'colsample_bytree': 0.8570282038386737, 'min_child_weight': 5, 'gamma':
0.01715235296298511, 'reg_alpha': 0.05488170987735091, 'reg_lambda':
1.9593153369423015e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:27:59,314] Trial 259 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:28:01,905] Trial 260 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:28:03,121] Trial 261 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:28:07,137] Trial 262 finished with value:
1.374686550117176 and parameters: {'clf_n_estimators': 513, 'clf_max_depth': 16,
'clf_min_samples_leaf': 11, 'clf_min_samples_split': 3, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 742, 'learning_rate':
0.005413966414347992, 'max_depth': 11, 'subsample': 0.9222749815212796,
'colsample_bytree': 0.8463940225433138, 'min_child_weight': 5, 'gamma':
0.00424854351901628, 'reg_alpha': 0.06333346074792053, 'reg_lambda':
5.811904819271958e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:28:11,308] Trial 263 finished with value:
1.377169281034116 and parameters: {'clf_n_estimators': 510, 'clf_max_depth': 16,
'clf_min_samples_leaf': 10, 'clf_min_samples_split': 3, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 815, 'learning_rate':
0.00529741693257524, 'max_depth': 11, 'subsample': 0.9211043249175371,
'colsample_bytree': 0.844082156535533, 'min_child_weight': 5, 'gamma':
0.0029970731757469355, 'reg_alpha': 0.0689318141353242, 'reg_lambda':
7.51774118515709e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:28:15,046] Trial 264 finished with value:
1.3754862044009328 and parameters: {'clf_n_estimators': 516, 'clf_max_depth':
17, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 3, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 745, 'learning_rate':
0.005974980019889992, 'max_depth': 11, 'subsample': 0.930230371976755,
'colsample_bytree': 0.861370040232198, 'min_child_weight': 5, 'gamma':
0.004945272147812039, 'reg_alpha': 0.0438367250600442, 'reg_lambda':
1.8780914584771544e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:28:16,582] Trial 265 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:28:20,495] Trial 266 finished with value:
1.3771348002695227 and parameters: {'clf_n_estimators': 522, 'clf_max_depth':
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17, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 3, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 739, 'learning_rate':
0.005973556827398003, 'max_depth': 11, 'subsample': 0.9313735396614455,
'colsample_bytree': 0.8733240305577833, 'min_child_weight': 5, 'gamma':
0.0040362140914281894, 'reg_alpha': 0.023462565638133037, 'reg_lambda':
1.1075680497714133e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:28:21,329] Trial 267 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:28:22,577] Trial 268 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:28:23,683] Trial 269 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:28:24,826] Trial 270 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:28:28,386] Trial 271 finished with value:
1.377105166257664 and parameters: {'clf_n_estimators': 490, 'clf_max_depth': 15,
'clf_min_samples_leaf': 11, 'clf_min_samples_split': 3, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 773, 'learning_rate':
0.005972740678200011, 'max_depth': 11, 'subsample': 0.8853889275307981,
'colsample_bytree': 0.8686919355261441, 'min_child_weight': 5, 'gamma':
0.003067050478473216, 'reg_alpha': 0.020016165728175793, 'reg_lambda':
1.7205002525911332e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:28:32,248] Trial 272 finished with value:
1.3806247022602751 and parameters: {'clf_n_estimators': 562, 'clf_max_depth':
18, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 725, 'learning_rate':
0.0053528508890755356, 'max_depth': 11, 'subsample': 0.903222459590515,
'colsample_bytree': 0.8211443571486917, 'min_child_weight': 4, 'gamma':
0.004827222473244102, 'reg_alpha': 0.045731736072633956, 'reg_lambda':
5.410774428828157e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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[I 2026-03-13 19:28:33,431] Trial 273 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:28:36,972] Trial 274 finished with value:
1.3759683366798496 and parameters: {'clf_n_estimators': 531, 'clf_max_depth':
19, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 700, 'learning_rate':
0.0061289132653208015, 'max_depth': 11, 'subsample': 0.8924172060967289,
'colsample_bytree': 0.841749186207827, 'min_child_weight': 5, 'gamma':
0.0016188720192015856, 'reg_alpha': 0.08154282644738226, 'reg_lambda':
3.759600595758829e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:28:38,275] Trial 275 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:28:39,580] Trial 276 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:28:41,813] Trial 277 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:28:45,430] Trial 278 finished with value:
1.3787970442602362 and parameters: {'clf_n_estimators': 505, 'clf_max_depth':
16, 'clf_min_samples_leaf': 11, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 784, 'learning_rate':
0.00629579859467052, 'max_depth': 11, 'subsample': 0.8734561703480228,
'colsample_bytree': 0.8960360417871824, 'min_child_weight': 5, 'gamma':
0.003640023251735484, 'reg_alpha': 0.03285827827787282, 'reg_lambda':
9.1591293732002e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:28:49,193] Trial 279 finished with value:
1.3807318537981381 and parameters: {'clf_n_estimators': 530, 'clf_max_depth':
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18, 'clf_min_samples_leaf': 12, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 723, 'learning_rate':
0.00564433733400295, 'max_depth': 11, 'subsample': 0.9320670199590051,
'colsample_bytree': 0.8728517224197957, 'min_child_weight': 4, 'gamma':
0.035229570877375126, 'reg_alpha': 0.04898618842301212, 'reg_lambda':
2.5727984959021683e-06}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:28:50,104] Trial 280 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:28:53,670] Trial 281 finished with value:
1.3708984741668149 and parameters: {'clf_n_estimators': 496, 'clf_max_depth':
14, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 672, 'learning_rate':
0.005242651509466289, 'max_depth': 11, 'subsample': 0.8981840190379574,
'colsample_bytree': 0.9099288183236031, 'min_child_weight': 5, 'gamma':
0.008438757703353469, 'reg_alpha': 0.060904681379088894, 'reg_lambda':
1.9411177401270853e-05}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:28:54,802] Trial 282 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:28:58,274] Trial 283 finished with value:
1.3736409887294279 and parameters: {'clf_n_estimators': 513, 'clf_max_depth':
14, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 674, 'learning_rate':
0.006556934410226789, 'max_depth': 10, 'subsample': 0.8953781157035707,
'colsample_bytree': 0.994748036774838, 'min_child_weight': 5, 'gamma':
0.05840037874529677, 'reg_alpha': 0.03477392545764943, 'reg_lambda':
1.406416450471556e-05}. Best is trial 222 with value: 1.3708078024605301.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:00,385] Trial 284 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:01,137] Trial 285 pruned.
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:02,239] Trial 286 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:03,395] Trial 287 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:04,740] Trial 288 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:08,294] Trial 289 finished with value:
1.3701498117273536 and parameters: {'clf_n_estimators': 499, 'clf_max_depth':
15, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 3, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 651, 'learning_rate':
0.0056275162712253225, 'max_depth': 11, 'subsample': 0.9185753652388833,
'colsample_bytree': 0.8882045819060106, 'min_child_weight': 5, 'gamma':
0.22303611014285765, 'reg_alpha': 0.04343120629667995, 'reg_lambda':
1.8328691694630553e-05}. Best is trial 289 with value: 1.3701498117273536.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:10,639] Trial 290 finished with value:
1.3685052188525793 and parameters: {'clf_n_estimators': 144, 'clf_max_depth':
15, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 2, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 642, 'learning_rate':
0.006578244936529025, 'max_depth': 10, 'subsample': 0.8665278153590641,
'colsample_bytree': 0.9983424708839623, 'min_child_weight': 5, 'gamma':
0.08943481757470267, 'reg_alpha': 0.06412306910134431, 'reg_lambda':
2.5728298694608342e-05}. Best is trial 290 with value: 1.3685052188525793.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:14,198] Trial 291 finished with value:
1.376103140320397 and parameters: {'clf_n_estimators': 577, 'clf_max_depth': 15,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 2, 'clf_max_features':
'sqrt', 'clf_class_weight': None, 'n_estimators': 645, 'learning_rate':
0.006609039759612925, 'max_depth': 9, 'subsample': 0.915200740367111,
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'colsample_bytree': 0.8877434300127401, 'min_child_weight': 5, 'gamma':
0.2490115588556754, 'reg_alpha': 0.06991083848219047, 'reg_lambda':
4.682602985911352e-05}. Best is trial 290 with value: 1.3685052188525793.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:15,064] Trial 292 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:16,559] Trial 293 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:20,154] Trial 294 finished with value:
1.38551306903537 and parameters: {'clf_n_estimators': 538, 'clf_max_depth': 15,
'clf_min_samples_leaf': 3, 'clf_min_samples_split': 2, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 687, 'learning_rate':
0.006683492103727873, 'max_depth': 10, 'subsample': 0.9174871132696033,
'colsample_bytree': 0.9870096247818231, 'min_child_weight': 5, 'gamma':
0.07958190844211534, 'reg_alpha': 0.06471965029807913, 'reg_lambda':
1.6435535872155932e-05}. Best is trial 290 with value: 1.3685052188525793.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:23,238] Trial 295 finished with value:
1.372414613690455 and parameters: {'clf_n_estimators': 278, 'clf_max_depth': 14,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 15, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 674, 'learning_rate':
0.0059436829271172845, 'max_depth': 11, 'subsample': 0.8557080405096217,
'colsample_bytree': 0.956378409082965, 'min_child_weight': 5, 'gamma':
0.2710897792958334, 'reg_alpha': 0.10477072290277681, 'reg_lambda':
2.8051634183917118e-05}. Best is trial 290 with value: 1.3685052188525793.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:24,177] Trial 296 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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[I 2026-03-13 19:29:26,489] Trial 297 finished with value:
1.3692362928624913 and parameters: {'clf_n_estimators': 189, 'clf_max_depth':
13, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 13, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 649, 'learning_rate':
0.005810231090082255, 'max_depth': 11, 'subsample': 0.8538743032439318,
'colsample_bytree': 0.971403840134293, 'min_child_weight': 5, 'gamma':
0.7155447701352508, 'reg_alpha': 0.11805521107988598, 'reg_lambda':
1.942724262885626e-05}. Best is trial 290 with value: 1.3685052188525793.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:28,686] Trial 298 finished with value:
1.3736968694178564 and parameters: {'clf_n_estimators': 158, 'clf_max_depth':
13, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 13, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 644, 'learning_rate':
0.0066826222998654945, 'max_depth': 11, 'subsample': 0.8457767964764888,
'colsample_bytree': 0.9911686535800475, 'min_child_weight': 5, 'gamma':
0.6340846902851869, 'reg_alpha': 0.15742884323855552, 'reg_lambda':
2.3048764240481734e-05}. Best is trial 290 with value: 1.3685052188525793.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:29,375] Trial 299 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:29,777] Trial 300 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:31,809] Trial 301 finished with value:
1.376758770850186 and parameters: {'clf_n_estimators': 133, 'clf_max_depth': 13,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 13, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 669, 'learning_rate':
0.006521469448983959, 'max_depth': 10, 'subsample': 0.8558846937462133,
'colsample_bytree': 0.9545168808092482, 'min_child_weight': 4, 'gamma':
0.44465003288597704, 'reg_alpha': 0.13409928732266194, 'reg_lambda':
1.9450213693236787e-05}. Best is trial 290 with value: 1.3685052188525793.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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[I 2026-03-13 19:29:33,852] Trial 302 finished with value:
1.3726438578850015 and parameters: {'clf_n_estimators': 122, 'clf_max_depth':
14, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 12, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 642, 'learning_rate':
0.00634360748236767, 'max_depth': 11, 'subsample': 0.8373959602072022,
'colsample_bytree': 0.9808196386331987, 'min_child_weight': 5, 'gamma':
0.0947197345299144, 'reg_alpha': 0.1891559220773717, 'reg_lambda':
3.235943539245165e-05}. Best is trial 290 with value: 1.3685052188525793.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:34,601] Trial 303 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:35,942] Trial 304 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:38,054] Trial 305 finished with value:
1.373607719531625 and parameters: {'clf_n_estimators': 122, 'clf_max_depth': 14,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 12, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 649, 'learning_rate':
0.006152174301989423, 'max_depth': 11, 'subsample': 0.8374886392940666,
'colsample_bytree': 0.9803094855127578, 'min_child_weight': 5, 'gamma':
0.8985612255342696, 'reg_alpha': 0.11433999623100223, 'reg_lambda':
2.6919216061153636e-05}. Best is trial 290 with value: 1.3685052188525793.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:39,241] Trial 306 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:40,035] Trial 307 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:41,457] Trial 308 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:43,760] Trial 309 finished with value:
1.3726625508172938 and parameters: {'clf_n_estimators': 148, 'clf_max_depth':
14, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 13, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 668, 'learning_rate':
0.005862286624783977, 'max_depth': 11, 'subsample': 0.86440123417923,
'colsample_bytree': 0.9641784534651752, 'min_child_weight': 5, 'gamma':
0.5546396847198278, 'reg_alpha': 0.14638339146628196, 'reg_lambda':
2.0348493979291925e-05}. Best is trial 290 with value: 1.3685052188525793.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:44,585] Trial 310 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:45,457] Trial 311 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:46,732] Trial 312 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:49,324] Trial 313 finished with value:
1.3750075505370198 and parameters: {'clf_n_estimators': 295, 'clf_max_depth':
14, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 13, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 644, 'learning_rate':
0.0071074454086151925, 'max_depth': 11, 'subsample': 0.8487831884792111,
'colsample_bytree': 0.9769816493946645, 'min_child_weight': 5, 'gamma':
0.7038873669644498, 'reg_alpha': 0.0020814069510573492, 'reg_lambda':
4.091165626634303e-05}. Best is trial 290 with value: 1.3685052188525793.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:52,001] Trial 314 finished with value:
1.3736260226446522 and parameters: {'clf_n_estimators': 292, 'clf_max_depth':
14, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 13, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 619, 'learning_rate':
0.007111631785048098, 'max_depth': 11, 'subsample': 0.8524320240374955,
'colsample_bytree': 0.9784813277866328, 'min_child_weight': 5, 'gamma':
0.6015799356888213, 'reg_alpha': 0.0013726373712049616, 'reg_lambda':
6.177073662403503e-05}. Best is trial 290 with value: 1.3685052188525793.
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:54,630] Trial 315 finished with value:
1.376656010022557 3 and parameters: {'clf_n_estimators': 242, 'clf_max_depth':
15, 'clf_min_samples_leaf': 3, 'clf_min_samples_split': 13, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 609, 'learning_rate':
0.006840149060613513, 'max_depth': 11, 'subsample': 0.860229206480665,
'colsample_bytree': 0.962385427198929, 'min_child_weight': 5, 'gamma':
0.572133540435853, 'reg_alpha': 0.07458575364998597, 'reg_lambda':
0.00019977808582415424}. Best is trial 290 with value: 1.3685052188525793.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:55,313] Trial 316 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:56,197] Trial 317 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:56,937] Trial 318 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:29:58,844] Trial 319 finished with value:
1.378837092088398 and parameters: {'clf_n_estimators': 115, 'clf_max_depth': 13,
'clf_min_samples_leaf': 3, 'clf_min_samples_split': 13, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 658, 'learning_rate':
0.007685388713367892, 'max_depth': 11, 'subsample': 0.85377642620494,
'colsample_bytree': 0.9690561103865822, 'min_child_weight': 5, 'gamma':
0.6463017971739434, 'reg_alpha': 0.000419606058839029, 'reg_lambda':
3.496316679418537e-05}. Best is trial 290 with value: 1.3685052188525793.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:00,153] Trial 320 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:00,746] Trial 321 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:01,442] Trial 322 pruned.
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:01,874] Trial 323 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:04,436] Trial 324 finished with value:
1.3837373425763815 and parameters: {'clf_n_estimators': 233, 'clf_max_depth':
13, 'clf_min_samples_leaf': 3, 'clf_min_samples_split': 13, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 675, 'learning_rate':
0.005809757089807047, 'max_depth': 11, 'subsample': 0.8742080174607433,
'colsample_bytree': 0.9695145655624762, 'min_child_weight': 4, 'gamma':
0.31471159725548425, 'reg_alpha': 0.22023972971918918, 'reg_lambda':
1.6048689675116278e-05}. Best is trial 290 with value: 1.3685052188525793.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:05,889] Trial 325 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:06,618] Trial 326 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:08,069] Trial 327 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:09,362] Trial 328 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:10,066] Trial 329 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:12,132] Trial 330 finished with value:
1.3756400288252202 and parameters: {'clf_n_estimators': 125, 'clf_max_depth':
14, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 13, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 620, 'learning_rate':
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0.00573996294022751, 'max_depth': 11, 'subsample': 0.8755472783839817,
'colsample_bytree': 0.9869811722961171, 'min_child_weight': 5, 'gamma':
0.05311653312133655, 'reg_alpha': 0.0882864175670923, 'reg_lambda':
6.118472199173456e-05}. Best is trial 290 with value: 1.3685052188525793.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:13,077] Trial 331 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:15,101] Trial 332 finished with value:
1.370493460376813 and parameters: {'clf_n_estimators': 136, 'clf_max_depth': 13,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 11, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 657, 'learning_rate':
0.00683919542557729, 'max_depth': 12, 'subsample': 0.8292688562751295,
'colsample_bytree': 0.9428164502040803, 'min_child_weight': 5, 'gamma':
0.635516613890179, 'reg_alpha': 0.20415202346436967, 'reg_lambda':
1.7486821260776566e-05}. Best is trial 290 with value: 1.3685052188525793.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:17,231] Trial 333 finished with value:
1.3699802395547183 and parameters: {'clf_n_estimators': 130, 'clf_max_depth':
13, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 11, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 657, 'learning_rate':
0.007730034702963886, 'max_depth': 12, 'subsample': 0.8297248757440934,
'colsample_bytree': 0.9440590108145448, 'min_child_weight': 5, 'gamma':
0.6103006743896496, 'reg_alpha': 0.11728585760028166, 'reg_lambda':
3.681012276261703e-05}. Best is trial 290 with value: 1.3685052188525793.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:19,176] Trial 334 finished with value:
1.372805319198731 and parameters: {'clf_n_estimators': 131, 'clf_max_depth': 13,
'clf_min_samples_leaf': 3, 'clf_min_samples_split': 11, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 636, 'learning_rate':
0.008222678118966353, 'max_depth': 12, 'subsample': 0.8344386143560534,
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'colsample_bytree': 0.945385043335037, 'min_child_weight': 5, 'gamma':
0.709216737321479, 'reg_alpha': 0.2730769234241924, 'reg_lambda':
3.9897234282361246e-05}. Best is trial 290 with value: 1.3685052188525793.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:19,859] Trial 335 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:21,781] Trial 336 finished with value:
1.371675133645213 and parameters: {'clf_n_estimators': 117, 'clf_max_depth': 13,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 11, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 640, 'learning_rate':
0.007564011823809577, 'max_depth': 12, 'subsample': 0.8285004379734124,
'colsample_bytree': 0.9466518726018023, 'min_child_weight': 4, 'gamma':
0.7125372426359751, 'reg_alpha': 0.2146728870868672, 'reg_lambda':
3.83987857621566e-05}. Best is trial 290 with value: 1.3685052188525793.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:22,509] Trial 337 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:23,099] Trial 338 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:23,839] Trial 339 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:26,151] Trial 340 finished with value:
1.372524203218133 and parameters: {'clf_n_estimators': 139, 'clf_max_depth': 13,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 10, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 653, 'learning_rate':
0.008190699483293064, 'max_depth': 12, 'subsample': 0.8138119199243036,
'colsample_bytree': 0.9402614873939874, 'min_child_weight': 4, 'gamma':
0.4217079174396852, 'reg_alpha': 0.21499120667595156, 'reg_lambda':
2.736665938168743e-05}. Best is trial 290 with value: 1.3685052188525793.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:28,305] Trial 341 finished with value:
1.3769368724339441 and parameters: {'clf_n_estimators': 141, 'clf_max_depth':
12, 'clf_min_samples_leaf': 4, 'clf_min_samples_split': 11, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 599, 'learning_rate':
0.008041154120128098, 'max_depth': 12, 'subsample': 0.8178077176855186,
'colsample_bytree': 0.956966250950772, 'min_child_weight': 4, 'gamma':
0.362181979024282, 'reg_alpha': 0.2000568687004574, 'reg_lambda':
2.391565845910991e-05}. Best is trial 290 with value: 1.3685052188525793.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:30,325] Trial 342 finished with value:
1.375141323875219 and parameters: {'clf_n_estimators': 152, 'clf_max_depth': 13,
'clf_min_samples_leaf': 3, 'clf_min_samples_split': 10, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 658, 'learning_rate':
0.009115929956780353, 'max_depth': 12, 'subsample': 0.8098778068034538,
'colsample_bytree': 0.9378598970511679, 'min_child_weight': 4, 'gamma':
0.6773778103849802, 'reg_alpha': 0.28856039470331013, 'reg_lambda':
3.910074397703358e-05}. Best is trial 290 with value: 1.3685052188525793.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:32,453] Trial 343 finished with value:
1.3729165205366485 and parameters: {'clf_n_estimators': 161, 'clf_max_depth':
13, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 10, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 948, 'learning_rate':
0.009825228729186827, 'max_depth': 12, 'subsample': 0.8280111561740685,
'colsample_bytree': 0.9594532800544197, 'min_child_weight': 4, 'gamma':
0.554013227082865, 'reg_alpha': 0.7577779056828792, 'reg_lambda':
3.022476106400345e-05}. Best is trial 290 with value: 1.3685052188525793.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:34,931] Trial 344 finished with value:
1.3749488359837787 and parameters: {'clf_n_estimators': 129, 'clf_max_depth':
13, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 10, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 939, 'learning_rate':
0.009605372792715549, 'max_depth': 12, 'subsample': 0.8277980865753499,
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'colsample_bytree': 0.9588188130240672, 'min_child_weight': 4, 'gamma':
0.23965673918688823, 'reg_alpha': 0.7454997939098755, 'reg_lambda':
2.9348781727024978e-05}. Best is trial 290 with value: 1.3685052188525793.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:35,738] Trial 345 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:37,845] Trial 346 finished with value:
1.3668000915767153 and parameters: {'clf_n_estimators': 169, 'clf_max_depth':
13, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 635, 'learning_rate':
0.008791816647903449, 'max_depth': 12, 'subsample': 0.8227062060958198,
'colsample_bytree': 0.9448095365749982, 'min_child_weight': 4, 'gamma':
0.5459145040238707, 'reg_alpha': 0.27490537019050554, 'reg_lambda':
7.357205084178703e-05}. Best is trial 346 with value: 1.3668000915767153.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:39,246] Trial 347 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:40,114] Trial 348 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:41,510] Trial 349 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:43,367] Trial 350 finished with value:
1.3715076056940312 and parameters: {'clf_n_estimators': 139, 'clf_max_depth':
13, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 647, 'learning_rate':
0.008147465982396294, 'max_depth': 12, 'subsample': 0.8276899948186205,
'colsample_bytree': 0.972227742170927, 'min_child_weight': 4, 'gamma':
0.9856005432718224, 'reg_alpha': 0.0011992999709170133, 'reg_lambda':
4.636164670536063e-05}. Best is trial 346 with value: 1.3668000915767153.
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:44,616] Trial 351 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:46,390] Trial 352 finished with value:
1.377920707240409 and parameters: {'clf_n_estimators': 101, 'clf_max_depth': 12,
'clf_min_samples_leaf': 3, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 628, 'learning_rate':
0.007990183208724836, 'max_depth': 12, 'subsample': 0.825007460782345,
'colsample_bytree': 0.9756229942352612, 'min_child_weight': 4, 'gamma':
0.6923119053824004, 'reg_alpha': 0.28479430412679313, 'reg_lambda':
7.436679172689953e-05}. Best is trial 346 with value: 1.3668000915767153.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:48,394] Trial 353 finished with value:
1.369182939105939 and parameters: {'clf_n_estimators': 137, 'clf_max_depth': 13,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 600, 'learning_rate':
0.007572212319288237, 'max_depth': 12, 'subsample': 0.8062692275271673,
'colsample_bytree': 0.9551362921231065, 'min_child_weight': 4, 'gamma':
0.4619287849493686, 'reg_alpha': 0.0016123575166383309, 'reg_lambda':
2.984551211970414e-05}. Best is trial 346 with value: 1.3668000915767153.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:50,080] Trial 354 finished with value:
1.3699228176798712 and parameters: {'clf_n_estimators': 139, 'clf_max_depth':
13, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 612, 'learning_rate':
0.009407294593149677, 'max_depth': 12, 'subsample': 0.8067463291908126,
'colsample_bytree': 0.9600889177496326, 'min_child_weight': 4, 'gamma':
0.9906510310480524, 'reg_alpha': 0.0016062575154955152, 'reg_lambda':
3.128525448080672e-05}. Best is trial 346 with value: 1.3668000915767153.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:51,693] Trial 355 finished with value:
1.3699906411850666 and parameters: {'clf_n_estimators': 134, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 600, 'learning_rate':
0.009963418364306236, 'max_depth': 12, 'subsample': 0.8003105916954651,
'colsample_bytree': 0.9567557368082982, 'min_child_weight': 4, 'gamma':
0.9810920484666188, 'reg_alpha': 0.0004678501671792555, 'reg_lambda':
3.210653058441896e-05}. Best is trial 346 with value: 1.3668000915767153.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:53,350] Trial 356 finished with value:
1.3701156813626962 and parameters: {'clf_n_estimators': 135, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 595, 'learning_rate':
0.009990139505503964, 'max_depth': 12, 'subsample': 0.7840419687701511,
'colsample_bytree': 0.9543686463129825, 'min_child_weight': 4, 'gamma':
0.985263386723616, 'reg_alpha': 0.0014188203366156857, 'reg_lambda':
3.0245076639092444e-05}. Best is trial 346 with value: 1.3668000915767153.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:54,985] Trial 357 finished with value:
1.3681924240057661 and parameters: {'clf_n_estimators': 137, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 596, 'learning_rate':
0.0097915621687116, 'max_depth': 12, 'subsample': 0.7935389636949164,
'colsample_bytree': 0.9581969214948869, 'min_child_weight': 4, 'gamma':
0.9998604096236677, 'reg_alpha': 0.0014276545420467425, 'reg_lambda':
3.238407813766309e-05}. Best is trial 346 with value: 1.3668000915767153.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:56,642] Trial 358 finished with value:
1.3699897509889754 and parameters: {'clf_n_estimators': 135, 'clf_max_depth':
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12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 593, 'learning_rate':
0.009949992284329502, 'max_depth': 12, 'subsample': 0.7903011891806222,
'colsample_bytree': 0.9564360830106248, 'min_child_weight': 4, 'gamma':
0.8465514874458442, 'reg_alpha': 0.0008597265316874267, 'reg_lambda':
5.11262335160901e-05}. Best is trial 346 with value: 1.3668000915767153.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:58,287] Trial 359 finished with value:
1.3670565426015309 and parameters: {'clf_n_estimators': 137, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 591, 'learning_rate':
0.010044819473174376, 'max_depth': 12, 'subsample': 0.7672158624603729,
'colsample_bytree': 0.9553603040124818, 'min_child_weight': 4, 'gamma':
0.998425751685386, 'reg_alpha': 0.0011730674712008683, 'reg_lambda':
4.874977081842411 5e-05}. Best is trial 346 with value: 1.3668000915767153.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:30:59,933] Trial 360 finished with value:
1.3709135211434162 and parameters: {'clf_n_estimators': 137, 'clf_max_depth':
12, 'clf_min_samples_leaf': 3, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 592, 'learning_rate':
0.010677887297558699, 'max_depth': 12, 'subsample': 0.7757098704822853,
'colsample_bytree': 0.9559355365664481, 'min_child_weight': 4, 'gamma':
0.98278003731424, 'reg_alpha': 0.0012593538686021045, 'reg_lambda':
3.115825932775228e-05}. Best is trial 346 with value: 1.3668000915767153.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:00,523] Trial 361 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:01,158] Trial 362 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:03,246] Trial 363 finished with value:
1.3659202654348466 and parameters: {'clf_n_estimators': 116, 'clf_max_depth':
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12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 590, 'learning_rate':
0.011989888172483584, 'max_depth': 12, 'subsample': 0.7597598484958172,
'colsample_bytree': 0.9405244400555998, 'min_child_weight': 4, 'gamma':
0.4486737021215353, 'reg_alpha': 0.0004741459229429197, 'reg_lambda':
0.00011568708137128439}. Best is trial 363 with value: 1.3659202654348466.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:05,461] Trial 364 finished with value:
1.3679838376142925 and parameters: {'clf_n_estimators': 122, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 586, 'learning_rate':
0.00996876210084541, 'max_depth': 12, 'subsample': 0.7415356632934527,
'colsample_bytree': 0.9426363213056479, 'min_child_weight': 4, 'gamma':
0.9947223794446493, 'reg_alpha': 0.0005513095848336597, 'reg_lambda':
0.0001701953316385689}. Best is trial 363 with value: 1.3659202654348466.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:06,156] Trial 365 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:06,726] Trial 366 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:08,309] Trial 367 finished with value:
1.3712493471741196 and parameters: {'clf_n_estimators': 132, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 595, 'learning_rate':
0.012757653788177638, 'max_depth': 12, 'subsample': 0.788433382140303,
'colsample_bytree': 0.955460449354863, 'min_child_weight': 4, 'gamma':
0.9766653695813484, 'reg_alpha': 0.0029594499648323497, 'reg_lambda':
0.00016258556244335147}. Best is trial 363 with value: 1.3659202654348466.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:10,286] Trial 368 finished with value:
1.373426480000515 and parameters: {'clf_n_estimators': 142, 'clf_max_depth': 11,
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'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 596, 'learning_rate':
0.011481436433954908, 'max_depth': 12, 'subsample': 0.7866305632473298,
'colsample_bytree': 0.9590134378823203, 'min_child_weight': 4, 'gamma':
0.35874293651355366, 'reg_alpha': 0.00024940249080050215, 'reg_lambda':
0.00017621420960185492}. Best is trial 363 with value: 1.3659202654348466.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:11,860] Trial 369 finished with value:
1.3691073644300271 and parameters: {'clf_n_estimators': 115, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 566, 'learning_rate':
0.01012054531785374, 'max_depth': 12, 'subsample': 0.7280069124360936,
'colsample_bytree': 0.9342451294816294, 'min_child_weight': 4, 'gamma':
0.9543946937934573, 'reg_alpha': 0.002823001870066483, 'reg_lambda':
8.676461869341734e-05}. Best is trial 363 with value: 1.3659202654348466.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:13,256] Trial 370 finished with value:
1.3695009984141973 and parameters: {'clf_n_estimators': 112, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 560, 'learning_rate':
0.013803277299973812, 'max_depth': 12, 'subsample': 0.7157452627869523,
'colsample_bytree': 0.9325440782111833, 'min_child_weight': 4, 'gamma':
0.9917891295885453, 'reg_alpha': 0.0028080541304105456, 'reg_lambda':
0.00013904584221239213}. Best is trial 363 with value: 1.3659202654348466.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:14,665] Trial 371 finished with value:
1.3705127218700535 and parameters: {'clf_n_estimators': 114, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 549, 'learning_rate':
0.013457862232207555, 'max_depth': 12, 'subsample': 0.7149223017190905,
'colsample_bytree': 0.93264721922569, 'min_child_weight': 4, 'gamma':
0.954058186944482, 'reg_alpha': 0.002339141478390868, 'reg_lambda':
0.0001445981167776363}. Best is trial 363 with value: 1.3659202654348466.
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:16,027] Trial 372 finished with value:
1.3697194519028335 and parameters: {'clf_n_estimators': 102, 'clf_max_depth':
10, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 556, 'learning_rate':
0.012913246073982384, 'max_depth': 12, 'subsample': 0.7315746468465298,
'colsample_bytree': 0.9278176331974302, 'min_child_weight': 4, 'gamma':
0.99299829015785, 'reg_alpha': 0.002819085789544435, 'reg_lambda':
0.00025512967453260966}. Best is trial 363 with value: 1.3659202654348466.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:16,653] Trial 373 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:17,153] Trial 374 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:18,616] Trial 375 finished with value:
1.369000658395575 and parameters: {'clf_n_estimators': 114, 'clf_max_depth': 11,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 530, 'learning_rate':
0.012376885800922968, 'max_depth': 12, 'subsample': 0.7321933918429551,
'colsample_bytree': 0.9329175818233825, 'min_child_weight': 3, 'gamma':
0.9760414869587127, 'reg_alpha': 0.0014547303697247053, 'reg_lambda':
0.00027721107486777076}. Best is trial 363 with value: 1.3659202654348466.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:20,090] Trial 376 finished with value:
1.370440025085978 and parameters: {'clf_n_estimators': 106, 'clf_max_depth': 11,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 526, 'learning_rate':
0.014958592439934701, 'max_depth': 12, 'subsample': 0.7327848697206046,
'colsample_bytree': 0.9322235652614721, 'min_child_weight': 3, 'gamma':
0.9955255531430816, 'reg_alpha': 0.0013668501439063508, 'reg_lambda':
0.0004962718517118497}. Best is trial 363 with value: 1.3659202654348466.
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:20,565] Trial 377 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:22,106] Trial 378 finished with value:
1.3710866142103206 and parameters: {'clf_n_estimators': 114, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 557, 'learning_rate':
0.012471242740235294, 'max_depth': 12, 'subsample': 0.7334560908672347,
'colsample_bytree': 0.9270872339381099, 'min_child_weight': 3, 'gamma':
0.9636616682954222, 'reg_alpha': 0.0005389100031979429, 'reg_lambda':
0.000379088075745604}. Best is trial 363 with value: 1.3659202654348466.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:23,538] Trial 379 finished with value:
1.3703040820618122 and parameters: {'clf_n_estimators': 111, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 552, 'learning_rate':
0.014350416364867958, 'max_depth': 12, 'subsample': 0.7058991825165485,
'colsample_bytree': 0.9270717257670165, 'min_child_weight': 3, 'gamma':
0.995907410892153, 'reg_alpha': 0.0005266422544184206, 'reg_lambda':
0.00039342978487936017}. Best is trial 363 with value: 1.3659202654348466.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:24,046] Trial 380 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:24,717] Trial 381 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:26,556] Trial 382 finished with value:
1.3661310022264825 and parameters: {'clf_n_estimators': 117, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 535, 'learning_rate':
0.010806954768487103, 'max_depth': 12, 'subsample': 0.7040291813753449,
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'colsample_bytree': 0.9405972029785491, 'min_child_weight': 3, 'gamma':
0.4904622181416657, 'reg_alpha': 0.0006581541557025127, 'reg_lambda':
0.000295153626380109}. Best is trial 363 with value: 1.3659202654348466.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:28,137] Trial 383 finished with value:
1.3713476123583586 and parameters: {'clf_n_estimators': 105, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 526, 'learning_rate':
0.013834732059853916, 'max_depth': 12, 'subsample': 0.7065799283938161,
'colsample_bytree': 0.9342378924464497, 'min_child_weight': 3, 'gamma':
0.4771449581732221, 'reg_alpha': 0.0003539781045516523, 'reg_lambda':
0.00032932743428334526}. Best is trial 363 with value: 1.3659202654348466.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:29,340] Trial 384 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:30,907] Trial 385 finished with value:
1.3706562221333012 and parameters: {'clf_n_estimators': 114, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 505, 'learning_rate':
0.016520401171067928, 'max_depth': 12, 'subsample': 0.6775136401050006,
'colsample_bytree': 0.9444057146310615, 'min_child_weight': 2, 'gamma':
0.43500999667970713, 'reg_alpha': 0.0007162729214409774, 'reg_lambda':
0.0003076521977722511}. Best is trial 363 with value: 1.3659202654348466.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:32,638] Trial 386 finished with value:
1.3674141709978134 and parameters: {'clf_n_estimators': 115, 'clf_max_depth': 9,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 516, 'learning_rate':
0.01464517112623865, 'max_depth': 12, 'subsample': 0.757085830459782,
'colsample_bytree': 0.9388641592544947, 'min_child_weight': 3, 'gamma':
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0.4302676883310672, 'reg_alpha': 0.0010662902544322436, 'reg_lambda':
0.0013747070888564337}. Best is trial 363 with value: 1.3659202654348466.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:33,307] Trial 387 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:35,050] Trial 388 finished with value:
1.3681157625521176 and parameters: {'clf_n_estimators': 125, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 517, 'learning_rate':
0.013398960388291819, 'max_depth': 12, 'subsample': 0.759600520031318,
'colsample_bytree': 0.9496271600379825, 'min_child_weight': 2, 'gamma':
0.5002734264253575, 'reg_alpha': 0.0001441281266913227, 'reg_lambda':
0.0007770003996732629}. Best is trial 363 with value: 1.3659202654348466.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:35,680] Trial 389 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:37,282] Trial 390 finished with value:
1.3725766933896226 and parameters: {'clf_n_estimators': 100, 'clf_max_depth':
10, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 543, 'learning_rate':
0.014096815117166784, 'max_depth': 12, 'subsample': 0.7394604924603208,
'colsample_bytree': 0.9484769119139075, 'min_child_weight': 3, 'gamma':
0.9988237647126579, 'reg_alpha': 0.00045912404347797447, 'reg_lambda':
0.00044505011829683164}. Best is trial 363 with value: 1.3659202654348466.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:38,911] Trial 391 finished with value:
1.3715151721584193 and parameters: {'clf_n_estimators': 129, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 7, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 524, 'learning_rate':
0.015082542156234825, 'max_depth': 12, 'subsample': 0.760166901956033,
'colsample_bytree': 0.9256401855340456, 'min_child_weight': 3, 'gamma':
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0.5087859146525417, 'reg_alpha': 0.0019404010727101535, 'reg_lambda':
0.000995669062939063}. Best is trial 363 with value: 1.3659202654348466.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:40,207] Trial 392 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:41,368] Trial 393 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:43,235] Trial 394 finished with value:
1.3674295665421312 and parameters: {'clf_n_estimators': 149, 'clf_max_depth': 9,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 483, 'learning_rate':
0.011280813125953153, 'max_depth': 12, 'subsample': 0.7373234126999206,
'colsample_bytree': 0.9353727973764596, 'min_child_weight': 2, 'gamma':
0.47206730753468507, 'reg_alpha': 0.00016463252959959048, 'reg_lambda':
0.0017916486381769402}. Best is trial 363 with value: 1.3659202654348466.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:44,481] Trial 395 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:45,122] Trial 396 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:45,865] Trial 397 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:46,541] Trial 398 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:48,260] Trial 399 finished with value:
1.3653981179322419 and parameters: {'clf_n_estimators': 132, 'clf_max_depth': 9,
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'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 519, 'learning_rate':
0.011966390324487966, 'max_depth': 12, 'subsample': 0.7273132825459552,
'colsample_bytree': 0.9399232588392998, 'min_child_weight': 2, 'gamma':
0.471876094591401, 'reg_alpha': 0.003767461498186566, 'reg_lambda':
0.004138337495702043}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:48,920] Trial 400 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:49,535] Trial 401 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:51,127] Trial 402 finished with value:
1.3700661511828471 and parameters: {'clf_n_estimators': 100, 'clf_max_depth': 9,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 503, 'learning_rate':
0.011086004863466586, 'max_depth': 12, 'subsample': 0.7630350080375887,
'colsample_bytree': 0.9392181046724489, 'min_child_weight': 2, 'gamma':
0.4709853765374755, 'reg_alpha': 0.0017079000596253982, 'reg_lambda':
0.0007798086382701973}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:51,752] Trial 403 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:53,933] Trial 404 finished with value:
1.3708679362441103 and parameters: {'clf_n_estimators': 175, 'clf_max_depth': 9,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 505, 'learning_rate':
0.01129782109873806, 'max_depth': 12, 'subsample': 0.7590229528221194,
'colsample_bytree': 0.9409290371396577, 'min_child_weight': 2, 'gamma':
0.3327757744634613, 'reg_alpha': 0.0022993402190384805, 'reg_lambda':
0.0017277814530512262}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:54,618] Trial 405 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:56,539] Trial 406 finished with value:
1.3711554946880549 and parameters: {'clf_n_estimators': 100, 'clf_max_depth': 9,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 580, 'learning_rate':
0.010210758512177011, 'max_depth': 12, 'subsample': 0.7802962781597171,
'colsample_bytree': 0.9227430528758911, 'min_child_weight': 2, 'gamma':
0.2391205425226317, 'reg_alpha': 0.0006892767994821505, 'reg_lambda':
0.0008113441804700954}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:57,228] Trial 407 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:57,977] Trial 408 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:31:59,688] Trial 409 finished with value:
1.373263137683488 and parameters: {'clf_n_estimators': 116, 'clf_max_depth': 12,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 575, 'learning_rate':
0.01299686954807708, 'max_depth': 12, 'subsample': 0.699722207924216,
'colsample_bytree': 0.9543120596900762, 'min_child_weight': 2, 'gamma':
0.668765814176933, 'reg_alpha': 0.005715078517670722, 'reg_lambda':
0.003961377663942401}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:00,977] Trial 410 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:03,011] Trial 411 finished with value:
1.3772195289344564 and parameters: {'clf_n_estimators': 126, 'clf_max_depth': 9,
'clf_min_samples_leaf': 3, 'clf_min_samples_split': 7, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 597, 'learning_rate':
0.010252432633126902, 'max_depth': 12, 'subsample': 0.8032516398109599,
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'colsample_bytree': 0.9673469819218621, 'min_child_weight': 2, 'gamma':
0.17172878117307477, 'reg_alpha': 0.0008636214691123962, 'reg_lambda':
9.589835007674954e-05}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:04,840] Trial 412 finished with value:
1.3663468041539009 and parameters: {'clf_n_estimators': 113, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 512, 'learning_rate':
0.008971209706633902, 'max_depth': 12, 'subsample': 0.7743068652334794,
'colsample_bytree': 0.9383593132039463, 'min_child_weight': 4, 'gamma':
0.4541643448807806, 'reg_alpha': 0.0004062689327558654, 'reg_lambda':
0.0007235308212693155}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:05,521] Trial 413 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:07,456] Trial 414 finished with value:
1.3741756945723906 and parameters: {'clf_n_estimators': 130, 'clf_max_depth': 8,
'clf_min_samples_leaf': 3, 'clf_min_samples_split': 9, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 500, 'learning_rate':
0.00919783516391559, 'max_depth': 12, 'subsample': 0.7700543420585761,
'colsample_bytree': 0.9555476166210212, 'min_child_weight': 4, 'gamma':
0.26431356611885937, 'reg_alpha': 0.00033168382891445616, 'reg_lambda':
0.000841253736176881}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:08,753] Trial 415 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:10,461] Trial 416 finished with value:
1.369428931025827 and parameters: {'clf_n_estimators': 100, 'clf_max_depth': 12,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features': 0.5,
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'clf_class_weight': None, 'n_estimators': 602, 'learning_rate':
0.010131534043255043, 'max_depth': 12, 'subsample': 0.7439829839046944,
'colsample_bytree': 0.9509053320577141, 'min_child_weight': 4, 'gamma':
0.6734828058080969, 'reg_alpha': 0.001452841146705292, 'reg_lambda':
0.00235013454396003}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:11,757] Trial 417 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:12,364] Trial 418 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:14,238] Trial 419 finished with value:
1.3674273996733097 and parameters: {'clf_n_estimators': 100, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 580, 'learning_rate':
0.008968732570769761, 'max_depth': 12, 'subsample': 0.790773444832632,
'colsample_bytree': 0.9622235961308727, 'min_child_weight': 4, 'gamma':
0.38580645784623313, 'reg_alpha': 0.002113484251639981, 'reg_lambda':
0.0017744409652399964}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:14,929] Trial 420 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:15,494] Trial 421 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:17,320] Trial 422 finished with value:
1.374722653311031 and parameters: {'clf_n_estimators': 115, 'clf_max_depth': 10,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 7, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 483, 'learning_rate':
0.009074638802968495, 'max_depth': 12, 'subsample': 0.757321543561868,
'colsample_bytree': 0.9342348837194373, 'min_child_weight': 4, 'gamma':
0.33593810670225305, 'reg_alpha': 0.0006637210044776759, 'reg_lambda':
0.003035502883827299}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:19,010] Trial 423 finished with value:
1.3684033207533155 and parameters: {'clf_n_estimators': 101, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 536, 'learning_rate':
0.010694102966298023, 'max_depth': 12, 'subsample': 0.7385928483860991,
'colsample_bytree': 0.9461941687217924, 'min_child_weight': 4, 'gamma':
0.6689817362300428, 'reg_alpha': 0.003982650997586932, 'reg_lambda':
0.001629160131298211}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:19,594] Trial 424 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:21,354] Trial 425 finished with value:
1.367264634140082 and parameters: {'clf_n_estimators': 146, 'clf_max_depth': 12,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 566, 'learning_rate':
0.010542909085881465, 'max_depth': 12, 'subsample': 0.739885515859226,
'colsample_bytree': 0.9651915388633832, 'min_child_weight': 4, 'gamma':
0.6834149732486557, 'reg_alpha': 0.0026221756013234273, 'reg_lambda':
0.002004841061946117}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:23,201] Trial 426 finished with value:
1.371056006133615 and parameters: {'clf_n_estimators': 150, 'clf_max_depth': 11,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 558, 'learning_rate':
0.010641523522072847, 'max_depth': 12, 'subsample': 0.7414043348749186,
'colsample_bytree': 0.9717200319026547, 'min_child_weight': 4, 'gamma':
0.6641598200463381, 'reg_alpha': 0.00433273943588886, 'reg_lambda':
0.0012196662824303889}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:25,120] Trial 427 finished with value:
1.3661950927377724 and parameters: {'clf_n_estimators': 115, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 528, 'learning_rate':
0.008853037532543543, 'max_depth': 12, 'subsample': 0.7365516141646585,
'colsample_bytree': 0.9671044067792556, 'min_child_weight': 4, 'gamma':
0.30675653352293636, 'reg_alpha': 0.002490848578147573, 'reg_lambda':
0.006863330149730784}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:27,314] Trial 428 finished with value:
1.3762066010814775 and parameters: {'clf_n_estimators': 114, 'clf_max_depth':
11, 'clf_min_samples_leaf': 3, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 533, 'learning_rate':
0.009148446564367425, 'max_depth': 12, 'subsample': 0.7171741778462057,
'colsample_bytree': 0.9721105335615866, 'min_child_weight': 4, 'gamma':
0.18188533244529193, 'reg_alpha': 0.003188404374464101, 'reg_lambda':
0.0020590595862664254}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:29,404] Trial 429 finished with value:
1.3670394044920517 and parameters: {'clf_n_estimators': 116, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 523, 'learning_rate':
0.008778372533871225, 'max_depth': 12, 'subsample': 0.7367732472925854,
'colsample_bytree': 0.9215027796272107, 'min_child_weight': 4, 'gamma':
0.28910960834797966, 'reg_alpha': 0.002606323728829129, 'reg_lambda':
0.007130628845566871}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:30,021] Trial 430 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:31,818] Trial 431 finished with value:
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1.366492012723526 and parameters: {'clf_n_estimators': 100, 'clf_max_depth': 12,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 535, 'learning_rate':
0.008886826752191367, 'max_depth': 12, 'subsample': 0.7244417280707409,
'colsample_bytree': 0.9281790179948712, 'min_child_weight': 4, 'gamma':
0.35632150189024236, 'reg_alpha': 0.002877006330159745, 'reg_lambda':
0.005870024580733869}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:33,970] Trial 432 finished with value:
1.3771943948598473 and parameters: {'clf_n_estimators': 103, 'clf_max_depth':
12, 'clf_min_samples_leaf': 3, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 531, 'learning_rate':
0.00876754614109204, 'max_depth': 12, 'subsample': 0.7270394642239429,
'colsample_bytree': 0.9214890099263151, 'min_child_weight': 4, 'gamma':
0.1589397719991858, 'reg_alpha': 0.0029968091475618915, 'reg_lambda':
0.006962138599805687}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:34,703] Trial 433 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:35,327] Trial 434 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:36,029] Trial 435 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:37,579] Trial 436 finished with value:
1.3751762225468394 and parameters: {'clf_n_estimators': 112, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 571, 'learning_rate':
0.018615513545957068, 'max_depth': 12, 'subsample': 0.7255747280111429,
'colsample_bytree': 0.9432055327698751, 'min_child_weight': 4, 'gamma':
0.4181962174051677, 'reg_alpha': 0.0037158609186899554, 'reg_lambda':
0.026669294276328706}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:39,443] Trial 437 finished with value:
1.3692345039810638 and parameters: {'clf_n_estimators': 123, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 485, 'learning_rate':
0.008626242139457015, 'max_depth': 12, 'subsample': 0.7375696199782178,
'colsample_bytree': 0.9323087491497998, 'min_child_weight': 4, 'gamma':
0.21008516984768105, 'reg_alpha': 0.0021428110736614993, 'reg_lambda':
0.0024025346281623685}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:40,167] Trial 438 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:42,283] Trial 439 finished with value:
1.3659471402456422 and parameters: {'clf_n_estimators': 173, 'clf_max_depth': 7,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 448, 'learning_rate':
0.009476300147829192, 'max_depth': 12, 'subsample': 0.7570446051746585,
'colsample_bytree': 0.9691858977208968, 'min_child_weight': 4, 'gamma':
0.19799417048659654, 'reg_alpha': 0.0012583790759366528, 'reg_lambda':
0.004698379696695061}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:44,492] Trial 440 finished with value:
1.3772744593600876 and parameters: {'clf_n_estimators': 177, 'clf_max_depth':
12, 'clf_min_samples_leaf': 4, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 493, 'learning_rate':
0.008847807980463302, 'max_depth': 12, 'subsample': 0.754911790452291,
'colsample_bytree': 0.9743800482053222, 'min_child_weight': 4, 'gamma':
0.18598005109842614, 'reg_alpha': 0.0012549523414077549, 'reg_lambda':
0.01702653046480602}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:45,085] Trial 441 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:47,081] Trial 442 finished with value:
1.3760633289421984 and parameters: {'clf_n_estimators': 153, 'clf_max_depth': 5,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 453, 'learning_rate':
0.00956852715859485, 'max_depth': 12, 'subsample': 0.7513002642380694,
'colsample_bytree': 0.9668972492417225, 'min_child_weight': 4, 'gamma':
0.20898654004087597, 'reg_alpha': 0.001623301887571916, 'reg_lambda':
0.016163630993982016}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:47,758] Trial 443 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:48,659] Trial 444 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:50,733] Trial 445 finished with value:
1.3757412471252888 and parameters: {'clf_n_estimators': 125, 'clf_max_depth': 6,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 495, 'learning_rate':
0.01033163033565464, 'max_depth': 12, 'subsample': 0.728488860385759,
'colsample_bytree': 0.9533779837713123, 'min_child_weight': 4, 'gamma':
0.2327563127307837, 'reg_alpha': 0.004319709160997006, 'reg_lambda':
0.0016467112633030158}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:52,803] Trial 446 finished with value:
1.3683931825270343 and parameters: {'clf_n_estimators': 144, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 517, 'learning_rate':
0.008723680089887213, 'max_depth': 12, 'subsample': 0.7584912334569617,
'colsample_bytree': 0.968783877389108, 'min_child_weight': 4, 'gamma':
0.4794231615851678, 'reg_alpha': 0.0002702202314974178, 'reg_lambda':
0.002881162467281437}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:53,505] Trial 447 pruned.
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:55,580] Trial 448 finished with value:
1.3694442288575635 and parameters: {'clf_n_estimators': 143, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 490, 'learning_rate':
0.008745924584584995, 'max_depth': 12, 'subsample': 0.7778649839468705,
'colsample_bytree': 0.9685602214364929, 'min_child_weight': 3, 'gamma':
0.2185841940733619, 'reg_alpha': 0.0002738609509270654, 'reg_lambda':
0.005210086053625613}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:56,098] Trial 449 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:56,890] Trial 450 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:57,672] Trial 451 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:32:59,908] Trial 452 finished with value:
1.3708456722095061 and parameters: {'clf_n_estimators': 127, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 534, 'learning_rate':
0.009585509696857647, 'max_depth': 12, 'subsample': 0.7676834394224462,
'colsample_bytree': 0.9754833781033245, 'min_child_weight': 4, 'gamma':
0.2973142680055079, 'reg_alpha': 0.0003679232852966801, 'reg_lambda':
0.002896074841204817}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:00,592] Trial 453 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:01,149] Trial 454 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:01,526] Trial 455 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:03,580] Trial 456 finished with value:
1.3730101012472236 and parameters: {'clf_n_estimators': 126, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 514, 'learning_rate':
0.009582933543345329, 'max_depth': 12, 'subsample': 0.7796034777984876,
'colsample_bytree': 0.9160871770191679, 'min_child_weight': 4, 'gamma':
0.24366088033696118, 'reg_alpha': 0.0036361435995258175, 'reg_lambda':
0.0043285788941251}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:05,704] Trial 457 finished with value:
1.3776938777413321 and parameters: {'clf_n_estimators': 187, 'clf_max_depth':
12, 'clf_min_samples_leaf': 3, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 541, 'learning_rate':
0.00896085235833828, 'max_depth': 12, 'subsample': 0.7495798514119643,
'colsample_bytree': 0.9901850565425488, 'min_child_weight': 3, 'gamma':
0.5059662749426491, 'reg_alpha': 0.0004343835068755388, 'reg_lambda':
0.0010617164756781278}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:07,742] Trial 458 finished with value:
1.3716837418209007 and parameters: {'clf_n_estimators': 142, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 7, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 512, 'learning_rate':
0.010449379872976436, 'max_depth': 12, 'subsample': 0.7547395690981143,
'colsample_bytree': 0.9343742028855597, 'min_child_weight': 4, 'gamma':
0.31051224145822875, 'reg_alpha': 0.00011596585841730435, 'reg_lambda':
0.0038459320549304875}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:09,902] Trial 459 finished with value:
1.3676100581833595 and parameters: {'clf_n_estimators': 161, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
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0.5, 'clf_class_weight': None, 'n_estimators': 541, 'learning_rate':
0.008204971806202338, 'max_depth': 12, 'subsample': 0.6804501800920909,
'colsample_bytree': 0.9773253853233683, 'min_child_weight': 2, 'gamma':
0.467675260008211, 'reg_alpha': 0.0022529675139695577, 'reg_lambda':
0.008319467150994487}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:10,725] Trial 460 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:11,376] Trial 461 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:12,122] Trial 462 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:13,511] Trial 463 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:14,161] Trial 464 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:14,733] Trial 465 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:16,711] Trial 466 finished with value:
1.369023291142953 and parameters: {'clf_n_estimators': 118, 'clf_max_depth': 11,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 576, 'learning_rate':
0.009860984518591695, 'max_depth': 12, 'subsample': 0.7421458326164454,
'colsample_bytree': 0.9635483493326856, 'min_child_weight': 3, 'gamma':
0.37373768579490635, 'reg_alpha': 0.0011283772206614918, 'reg_lambda':
0.004589937430305331}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:17,319] Trial 467 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:18,033] Trial 468 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:20,163] Trial 469 finished with value:
1.3679983354314142 and parameters: {'clf_n_estimators': 140, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 544, 'learning_rate':
0.009385715586821438, 'max_depth': 12, 'subsample': 0.6464252257782441,
'colsample_bytree': 0.9608209666830668, 'min_child_weight': 3, 'gamma':
0.5915939414722796, 'reg_alpha': 8.352036476371783e-05, 'reg_lambda':
0.012275073774417599}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:20,844] Trial 470 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:21,462] Trial 471 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:22,108] Trial 472 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:22,790] Trial 473 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:23,456] Trial 474 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:24,611] Trial 475 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:26,111] Trial 476 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:26,751] Trial 477 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:28,381] Trial 478 finished with value:
1.3704594878951308 and parameters: {'clf_n_estimators': 100, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
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0.5, 'clf_class_weight': None, 'n_estimators': 510, 'learning_rate':
0.010994368708418953, 'max_depth': 12, 'subsample': 0.775332100543915,
'colsample_bytree': 0.9564001895067193, 'min_child_weight': 2, 'gamma':
0.6641445873518822, 'reg_alpha': 0.0002226007311737252, 'reg_lambda':
0.01150737047028202}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:29,065] Trial 479 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:29,758] Trial 480 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:30,338] Trial 481 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:30,784] Trial 482 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:31,431] Trial 483 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:33,364] Trial 484 finished with value:
1.3657790514393362 and parameters: {'clf_n_estimators': 111, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 573, 'learning_rate':
0.008609104785813828, 'max_depth': 12, 'subsample': 0.7119860842820366,
'colsample_bytree': 0.9258075819282783, 'min_child_weight': 2, 'gamma':
0.5205574092166614, 'reg_alpha': 0.0025739746007772584, 'reg_lambda':
0.0016984359154463046}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:34,123] Trial 485 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:34,811] Trial 486 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:35,509] Trial 487 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:36,145] Trial 488 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:38,091] Trial 489 finished with value:
1.3744529489145532 and parameters: {'clf_n_estimators': 146, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 19, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 446, 'learning_rate':
0.009021396572708328, 'max_depth': 12, 'subsample': 0.7728616159373755,
'colsample_bytree': 0.91276839528693, 'min_child_weight': 2, 'gamma':
0.3439343655256595, 'reg_alpha': 0.0031373917199469443, 'reg_lambda':
0.00265793446164142}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:39,489] Trial 490 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:40,147] Trial 491 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:42,052] Trial 492 finished with value:
1.3676556349026472 and parameters: {'clf_n_estimators': 111, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 553, 'learning_rate':
0.008985107365205754, 'max_depth': 12, 'subsample': 0.7185605766090072,
'colsample_bytree': 0.9974273588093654, 'min_child_weight': 3, 'gamma':
0.4296497432629059, 'reg_alpha': 0.001483311066495688, 'reg_lambda':
0.0008972986447281411}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:42,821] Trial 493 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:44,517] Trial 494 finished with value:
1.3667546338625367 and parameters: {'clf_n_estimators': 110, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 465, 'learning_rate':
0.008607924505492637, 'max_depth': 12, 'subsample': 0.6943330812316261,
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'colsample_bytree': 0.9979382848809637, 'min_child_weight': 3, 'gamma':
0.4979834097872027, 'reg_alpha': 0.0015855912431466217, 'reg_lambda':
0.0004955991142137345}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:45,249] Trial 495 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:47,060] Trial 496 finished with value:
1.368042950200956 and parameters: {'clf_n_estimators': 150, 'clf_max_depth': 6,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 426, 'learning_rate':
0.007806169022291316, 'max_depth': 12, 'subsample': 0.7006066375364667,
'colsample_bytree': 0.9958789155494995, 'min_child_weight': 4, 'gamma':
0.3188009653265916, 'reg_alpha': 0.0038464822218218547, 'reg_lambda':
0.000980657580382407}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:48,850] Trial 497 finished with value:
1.372794653000625 and parameters: {'clf_n_estimators': 156, 'clf_max_depth': 6,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 421, 'learning_rate':
0.00785323118067445, 'max_depth': 12, 'subsample': 0.7002415003382947,
'colsample_bytree': 0.9987640929870301, 'min_child_weight': 4, 'gamma':
0.4354664524177953, 'reg_alpha': 0.005116143871873144, 'reg_lambda':
0.0004140241728986345}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:49,375] Trial 498 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:51,090] Trial 499 finished with value:
1.374631441858816 and parameters: {'clf_n_estimators': 128, 'clf_max_depth': 6,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 429, 'learning_rate':
0.007953264358555963, 'max_depth': 12, 'subsample': 0.683752115015008,
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'colsample_bytree': 0.9987964168617727, 'min_child_weight': 4, 'gamma':
0.4830435835897515, 'reg_alpha': 0.0020467396854877327, 'reg_lambda':
0.00080019497405794}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:51,767] Trial 500 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:52,373] Trial 501 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:53,017] Trial 502 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:53,717] Trial 503 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:54,339] Trial 504 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:54,946] Trial 505 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:56,712] Trial 506 finished with value:
1.3702465067792888 and parameters: {'clf_n_estimators': 130, 'clf_max_depth': 6,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 469, 'learning_rate':
0.009730544084466662, 'max_depth': 12, 'subsample': 0.7010428918355904,
'colsample_bytree': 0.9977892888777564, 'min_child_weight': 3, 'gamma':
0.5131065735964311, 'reg_alpha': 0.0003077707669822834, 'reg_lambda':
0.0009193376295075221}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:58,894] Trial 507 finished with value:
1.3738326908671086 and parameters: {'clf_n_estimators': 167, 'clf_max_depth': 8,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 513, 'learning_rate':
0.01065257593348266, 'max_depth': 12, 'subsample': 0.7194455629897075,
'colsample_bytree': 0.9576239779338006, 'min_child_weight': 4, 'gamma':
0.26479391341099234, 'reg_alpha': 0.00174738942160508, 'reg_lambda':
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0.0019978555311710106}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:33:59,660] Trial 508 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:00,359] Trial 509 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:02,605] Trial 510 finished with value:
1.3737872524009458 and parameters: {'clf_n_estimators': 127, 'clf_max_depth': 9,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 566, 'learning_rate':
0.00968921941224121, 'max_depth': 12, 'subsample': 0.7790341893138817,
'colsample_bytree': 0.965524463946581, 'min_child_weight': 4, 'gamma':
1.671315753625119e-05, 'reg_alpha': 0.0013914107262295976, 'reg_lambda':
0.0006112681721398736}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:04,047] Trial 511 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:04,689] Trial 512 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:05,074] Trial 513 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:05,446] Trial 514 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:06,075] Trial 515 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:06,909] Trial 516 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:07,655] Trial 517 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:09,630] Trial 518 finished with value:
1.3671020929556785 and parameters: {'clf_n_estimators': 141, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 564, 'learning_rate':
0.00915868287665161, 'max_depth': 12, 'subsample': 0.7487489977867541,
'colsample_bytree': 0.9130746195143491, 'min_child_weight': 4, 'gamma':
0.5068804845200511, 'reg_alpha': 0.0038297431131602395, 'reg_lambda':
0.003332141194493501 4}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:10,373] Trial 519 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:11,067] Trial 520 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:13,323] Trial 521 finished with value:
1.3669383983164272 and parameters: {'clf_n_estimators': 141, 'clf_max_depth':
10, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 559, 'learning_rate':
0.008337708321670708, 'max_depth': 12, 'subsample': 0.7496330151068054,
'colsample_bytree': 0.9136609266903813, 'min_child_weight': 3, 'gamma':
0.22138715473550205, 'reg_alpha': 0.0008029658437968319, 'reg_lambda':
0.004280964266957874}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:14,810] Trial 522 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:15,357] Trial 523 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:16,172] Trial 524 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:17,619] Trial 525 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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[I 2026-03-13 19:34:18,439] Trial 526 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:19,941] Trial 527 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:20,703] Trial 528 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:21,981] Trial 529 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:23,283] Trial 530 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:27,261] Trial 531 finished with value:
1.3719747104632003 and parameters: {'clf_n_estimators': 783, 'clf_max_depth':
11, 'clf_min_samples_leaf': 3, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 421, 'learning_rate':
0.009883135402552227, 'max_depth': 12, 'subsample': 0.6954842876753956,
'colsample_bytree': 0.9365253010259347, 'min_child_weight': 3, 'gamma':
0.2445199738107684, 'reg_alpha': 0.0065552264569979325, 'reg_lambda':
0.0019705974273722957}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:28,831] Trial 532 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:29,944] Trial 533 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:30,389] Trial 534 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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[I 2026-03-13 19:34:31,167] Trial 535 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:33,127] Trial 536 finished with value:
1.3697240250878835 and parameters: {'clf_n_estimators': 130, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 609, 'learning_rate':
0.009495076995198512, 'max_depth': 12, 'subsample': 0.7212214873243687,
'colsample_bytree': 0.9545989916820752, 'min_child_weight': 2, 'gamma':
0.31374021366835225, 'reg_alpha': 7.304578044200112e-05, 'reg_lambda':
0.00017271994130694875}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:33,714] Trial 537 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:34,369] Trial 538 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:34,940] Trial 539 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:35,356] Trial 540 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:35,697] Trial 541 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:37,576] Trial 542 finished with value:
1.3750879902661115 and parameters: {'clf_n_estimators': 125, 'clf_max_depth': 9,
'clf_min_samples_leaf': 3, 'clf_min_samples_split': 7, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 556, 'learning_rate':
0.008980015177213169, 'max_depth': 12, 'subsample': 0.7404999059660183,
'colsample_bytree': 0.9043689797517722, 'min_child_weight': 3, 'gamma':
0.47478862018450474, 'reg_alpha': 0.0015888425844319003, 'reg_lambda':
0.00635133947173303}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:39,581] Trial 543 finished with value:
1.3748780890326016 and parameters: {'clf_n_estimators': 164, 'clf_max_depth': 8,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 581, 'learning_rate':
0.013137201814885036, 'max_depth': 12, 'subsample': 0.7127478645706748,
'colsample_bytree': 0.9466845743420161, 'min_child_weight': 3, 'gamma':
0.17570070011245087, 'reg_alpha': 0.0012283115898330114, 'reg_lambda':
0.0027664371311142107}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:40,243] Trial 544 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:40,781] Trial 545 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:41,950] Trial 546 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:42,515] Trial 547 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:43,277] Trial 548 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:45,736] Trial 549 finished with value:
1.37520008361651 and parameters: {'clf_n_estimators': 187, 'clf_max_depth': 12,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 496, 'learning_rate':
0.009195903376902565, 'max_depth': 12, 'subsample': 0.780784009233255,
'colsample_bytree': 0.9685868076717353, 'min_child_weight': 2, 'gamma':
0.0004954136702757776, 'reg_alpha': 0.0012101360099720563, 'reg_lambda':
0.0002347263558278311}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:47,284] Trial 550 finished with value:
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1.3738207736417098 and parameters: {'clf_n_estimators': 123, 'clf_max_depth': 7,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 7, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 402, 'learning_rate':
0.012145466999739521, 'max_depth': 12, 'subsample': 0.7624534938612196,
'colsample_bytree': 0.986256099242131, 'min_child_weight': 4, 'gamma':
0.1150561076010967, 'reg_alpha': 0.0004471702573616578, 'reg_lambda':
0.0006975027650876279}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:47,917] Trial 551 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:49,920] Trial 552 finished with value:
1.3670090058900388 and parameters: {'clf_n_estimators': 147, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 589, 'learning_rate':
0.008702487891239924, 'max_depth': 12, 'subsample': 0.7470110919052858,
'colsample_bytree': 0.930369597937232, 'min_child_weight': 3, 'gamma':
0.3741663123891519, 'reg_alpha': 0.003475895723631841, 'reg_lambda':
0.022392678969152757}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:52,121] Trial 553 finished with value:
1.367446415319128 and parameters: {'clf_n_estimators': 154, 'clf_max_depth': 13,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 565, 'learning_rate':
0.00753839545153384, 'max_depth': 12, 'subsample': 0.7508173731945302,
'colsample_bytree': 0.9241984216727309, 'min_child_weight': 3, 'gamma':
0.2232549892608822, 'reg_alpha': 0.0037459923937094783, 'reg_lambda':
0.03598570495246592}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:53,611] Trial 554 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:54,412] Trial 555 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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[I 2026-03-13 19:34:54,860] Trial 556 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:56,613] Trial 557 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:57,341] Trial 558 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:34:59,424] Trial 559 finished with value:
1.3738670775715773 and parameters: {'clf_n_estimators': 101, 'clf_max_depth':
13, 'clf_min_samples_leaf': 3, 'clf_min_samples_split': 7, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 550, 'learning_rate':
0.008672684836600804, 'max_depth': 12, 'subsample': 0.7492873737010107,
'colsample_bytree': 0.932625405691268, 'min_child_weight': 3, 'gamma':
0.34158106852659376, 'reg_alpha': 0.008294585031779239, 'reg_lambda':
0.02139646291624267}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:01,618] Trial 560 finished with value:
1.3697908797262135 and parameters: {'clf_n_estimators': 156, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 534, 'learning_rate':
0.008054418249805143, 'max_depth': 12, 'subsample': 0.734715584255653,
'colsample_bytree': 0.9174806589713643, 'min_child_weight': 3, 'gamma':
0.2482512069137186, 'reg_alpha': 0.002693963534763744, 'reg_lambda':
0.034414242132845156}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:02,441] Trial 561 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:03,122] Trial 562 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:04,774] Trial 563 pruned.
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:07,002] Trial 564 finished with value:
1.3704961840939713 and parameters: {'clf_n_estimators': 144, 'clf_max_depth':
13, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 539, 'learning_rate':
0.008667675783882639, 'max_depth': 12, 'subsample': 0.7390656257334084,
'colsample_bytree': 0.9163713228785229, 'min_child_weight': 3, 'gamma':
0.15976963640176958, 'reg_alpha': 0.0018354207376260186, 'reg_lambda':
0.004650351998259982}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:07,749] Trial 565 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:08,408] Trial 566 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:09,167] Trial 567 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:11,084] Trial 568 finished with value:
1.3679912750629744 and parameters: {'clf_n_estimators': 122, 'clf_max_depth':
13, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 546, 'learning_rate':
0.00957361750447408, 'max_depth': 12, 'subsample': 0.7497595761297002,
'colsample_bytree': 0.9404538466812227, 'min_child_weight': 3, 'gamma':
0.4233989004989167, 'reg_alpha': 0.004885382046917691, 'reg_lambda':
0.011900076732204256}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:11,785] Trial 569 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:13,464] Trial 570 finished with value:
1.3670259713543451 and parameters: {'clf_n_estimators': 121, 'clf_max_depth':
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13, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 6, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 556, 'learning_rate':
0.01036205335514911, 'max_depth': 12, 'subsample': 0.7602988644561562,
'colsample_bytree': 0.9234881348829684, 'min_child_weight': 3, 'gamma':
0.4482115463435169, 'reg_alpha': 0.005356532070237992, 'reg_lambda':
0.0095971838696848}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:14,011] Trial 571 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:14,663] Trial 572 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:15,233] Trial 573 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:15,620] Trial 574 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:16,308] Trial 575 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:18,078] Trial 576 finished with value:
1.3662350840570652 and parameters: {'clf_n_estimators': 112, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 555, 'learning_rate':
0.009489376799283715, 'max_depth': 12, 'subsample': 0.7713046960958511,
'colsample_bytree': 0.9120783207741138, 'min_child_weight': 3, 'gamma':
0.3833352698690944, 'reg_alpha': 0.0016379249630093969, 'reg_lambda':
0.0031212848653603678}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:18,732] Trial 577 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:20,735] Trial 578 finished with value:
1.3722122668470427 and parameters: {'clf_n_estimators': 100, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 10, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 562, 'learning_rate':
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0.008761041397896074, 'max_depth': 12, 'subsample': 0.7759671186362315,
'colsample_bytree': 0.9142716135658355, 'min_child_weight': 3, 'gamma':
0.392985760857136, 'reg_alpha': 0.0008431322999485397, 'reg_lambda':
0.0026165170919328017}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:21,439] Trial 579 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:23,278] Trial 580 finished with value:
1.3686186846363313 and parameters: {'clf_n_estimators': 134, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 6, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 500, 'learning_rate':
0.009774757729294517, 'max_depth': 12, 'subsample': 0.7709894040682141,
'colsample_bytree': 0.914995325845132, 'min_child_weight': 3, 'gamma':
0.5947421458109581, 'reg_alpha': 0.0011664815153010254, 'reg_lambda':
0.0032576264072183155}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:25,228] Trial 581 finished with value:
1.3678046287157652 and parameters: {'clf_n_estimators': 129, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 595, 'learning_rate':
0.008600949886490813, 'max_depth': 12, 'subsample': 0.7791680812630085,
'colsample_bytree': 0.9237996490442305, 'min_child_weight': 4, 'gamma':
0.3139950567867297, 'reg_alpha': 0.003062461973957132, 'reg_lambda':
0.05920872471764608}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:27,163] Trial 582 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:27,868] Trial 583 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:29,977] Trial 584 finished with value:
1.3757514304954768 and parameters: {'clf_n_estimators': 110, 'clf_max_depth':
12, 'clf_min_samples_leaf': 3, 'clf_min_samples_split': 7, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 595, 'learning_rate':
0.008125091373862914, 'max_depth': 12, 'subsample': 0.7825643052497413,
'colsample_bytree': 0.9009209892792328, 'min_child_weight': 3, 'gamma':
0.19824824704887853, 'reg_alpha': 0.003238779038380887, 'reg_lambda':
0.04885876526594588}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:31,770] Trial 585 finished with value:
1.373375772943847 and parameters: {'clf_n_estimators': 127, 'clf_max_depth': 8,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 10, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 530, 'learning_rate':
0.00897018900370099, 'max_depth': 12, 'subsample': 0.7737574214597975,
'colsample_bytree': 0.927155840256618, 'min_child_weight': 4, 'gamma':
0.3024237253758813, 'reg_alpha': 0.0019311889212283073, 'reg_lambda':
0.07122570454122369}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:33,588] Trial 586 finished with value:
1.3696027425624335 and parameters: {'clf_n_estimators': 101, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 572, 'learning_rate':
0.007676931009044368, 'max_depth': 12, 'subsample': 0.5237314264574642,
'colsample_bytree': 0.9164108525609425, 'min_child_weight': 3, 'gamma':
0.3462934895079728, 'reg_alpha': 0.003072718204062062, 'reg_lambda':
0.002186142142596787}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:34,278] Trial 587 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:36,180] Trial 588 finished with value:
1.3671187103598905 and parameters: {'clf_n_estimators': 119, 'clf_max_depth':
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11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 548, 'learning_rate':
0.009204415027920156, 'max_depth': 12, 'subsample': 0.7718684391711426,
'colsample_bytree': 0.9170507914999615, 'min_child_weight': 3, 'gamma':
0.19469040254337072, 'reg_alpha': 0.002259708804470377, 'reg_lambda':
0.004156414638035476}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:38,058] Trial 589 finished with value:
1.3722763685542407 and parameters: {'clf_n_estimators': 119, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 523, 'learning_rate':
0.00942958263867218, 'max_depth': 12, 'subsample': 0.7690684588585647,
'colsample_bytree': 0.8998660116878436, 'min_child_weight': 3, 'gamma':
0.13333292761625828, 'reg_alpha': 0.0020078511302356957, 'reg_lambda':
0.004528323021416652}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:38,512] Trial 590 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:39,245] Trial 591 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:41,045] Trial 592 finished with value:
1.3731654375004332 and parameters: {'clf_n_estimators': 116, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 504, 'learning_rate':
0.011779804662332822, 'max_depth': 12, 'subsample': 0.759890852819001,
'colsample_bytree': 0.9345220043664122, 'min_child_weight': 3, 'gamma':
0.21206541514557195, 'reg_alpha': 0.0014832160736581894, 'reg_lambda':
0.0060040461475669386}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:41,692] Trial 593 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:42,263] Trial 594 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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[I 2026-03-13 19:35:42,976] Trial 595 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:43,352] Trial 596 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:44,596] Trial 597 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:45,132] Trial 598 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:46,788] Trial 599 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:47,487] Trial 600 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:49,212] Trial 601 finished with value:
1.3680545482150757 and parameters: {'clf_n_estimators': 127, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 474, 'learning_rate':
0.010844612936498427, 'max_depth': 12, 'subsample': 0.7380897323300111,
'colsample_bytree': 0.9472812752216613, 'min_child_weight': 3, 'gamma':
0.6513382481517772, 'reg_alpha': 0.0009950051649831218, 'reg_lambda':
0.005518882874021829}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:49,892] Trial 602 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:52,032] Trial 603 finished with value:
1.371089316061907 and parameters: {'clf_n_estimators': 137, 'clf_max_depth': 12,
'clf_min_samples_leaf': 3, 'clf_min_samples_split': 10, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 550, 'learning_rate':
0.009353610091131276, 'max_depth': 12, 'subsample': 0.7561127334393318,
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'colsample_bytree': 0.8963112140284383, 'min_child_weight': 3, 'gamma':
0.26711191249285904, 'reg_alpha': 0.0006451093708009883, 'reg_lambda':
0.004239806359236129}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:53,978] Trial 604 finished with value:
1.3707208630857457 and parameters: {'clf_n_estimators': 115, 'clf_max_depth': 8,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 531, 'learning_rate':
0.008312851379978206, 'max_depth': 12, 'subsample': 0.7254130535555036,
'colsample_bytree': 0.9143804789665183, 'min_child_weight': 2, 'gamma':
0.42942120870530187, 'reg_alpha': 0.002966643810861747, 'reg_lambda':
0.008815722238731397}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:55,818] Trial 605 finished with value:
1.3718959402014341 and parameters: {'clf_n_estimators': 153, 'clf_max_depth':
13, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 565, 'learning_rate':
0.010285012511232905, 'max_depth': 12, 'subsample': 0.7813006254918082,
'colsample_bytree': 0.9542892236897621, 'min_child_weight': 2, 'gamma':
0.5682581153783232, 'reg_alpha': 0.0015321107310072315, 'reg_lambda':
0.001807245699991691}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:57,262] Trial 606 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:57,878] Trial 607 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:35:59,833] Trial 608 finished with value:
1.3686928628190014 and parameters: {'clf_n_estimators': 113, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
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0.5, 'clf_class_weight': None, 'n_estimators': 540, 'learning_rate':
0.00961342110689553, 'max_depth': 12, 'subsample': 0.7316780221941332,
'colsample_bytree': 0.5740314589390577, 'min_child_weight': 4, 'gamma':
0.35785241666153306, 'reg_alpha': 0.0023654516474230377, 'reg_lambda':
0.0021863397434271612}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:00,502] Trial 609 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:01,925] Trial 610 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:02,533] Trial 611 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:02,846] Trial 612 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:03,371] Trial 613 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:04,018] Trial 614 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:04,737] Trial 615 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:05,313] Trial 616 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:06,052] Trial 617 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:07,990] Trial 618 finished with value:
1.374429137023301 and parameters: {'clf_n_estimators': 122, 'clf_max_depth': 9,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 7, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 535, 'learning_rate':
0.008140185343278516, 'max_depth': 12, 'subsample': 0.7464170962634944,
'colsample_bytree': 0.9012620315743547, 'min_child_weight': 3, 'gamma':
0.3499343587088592, 'reg_alpha': 0.0015205124353736554, 'reg_lambda':
0.003590764970952049}. Best is trial 399 with value: 1.3653981179322419.
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:08,696] Trial 619 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:11,170] Trial 620 finished with value:
1.37582778965025 and parameters: {'clf_n_estimators': 111, 'clf_max_depth': 13,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 596, 'learning_rate':
0.007255166440992021, 'max_depth': 12, 'subsample': 0.7810084801349763,
'colsample_bytree': 0.9431513069002236, 'min_child_weight': 2, 'gamma':
0.11141392792655204, 'reg_alpha': 0.0021054191251781785, 'reg_lambda':
0.0018041328573032527}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:11,774] Trial 621 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:13,599] Trial 622 finished with value:
1.3656553675614405 and parameters: {'clf_n_estimators': 122, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 573, 'learning_rate':
0.010571617577702624, 'max_depth': 12, 'subsample': 0.7376253161241408,
'colsample_bytree': 0.916607981919063, 'min_child_weight': 2, 'gamma':
0.28883029860956666, 'reg_alpha': 0.005923123173221856, 'reg_lambda':
0.016700454276252368}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:14,362] Trial 623 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:15,655] Trial 624 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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[I 2026-03-13 19:36:19,709] Trial 625 finished with value:
1.3755472612540693 and parameters: {'clf_n_estimators': 704, 'clf_max_depth':
10, 'clf_min_samples_leaf': 3, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 597, 'learning_rate':
0.01016979583510332, 'max_depth': 12, 'subsample': 0.7543007770361656,
'colsample_bytree': 0.893294091572571, 'min_child_weight': 2, 'gamma':
0.19608504193108903, 'reg_alpha': 0.006699434254705345, 'reg_lambda':
0.019153802092603648}. Best is trial 399 with value: 1.3653981179322419.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:21,521] Trial 626 finished with value:
1.3648000719976008 and parameters: {'clf_n_estimators': 132, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 568, 'learning_rate':
0.013103856001398089, 'max_depth': 12, 'subsample': 0.741276441798004,
'colsample_bytree': 0.9134266778170689, 'min_child_weight': 2, 'gamma':
0.30810402510660184, 'reg_alpha': 0.006776358701828121, 'reg_lambda':
0.014101331476886999}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:22,262] Trial 627 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:24,064] Trial 628 finished with value:
1.3667757568851238 and parameters: {'clf_n_estimators': 123, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 590, 'learning_rate':
0.012955624714342175, 'max_depth': 12, 'subsample': 0.7332714230882131,
'colsample_bytree': 0.8968991585544166, 'min_child_weight': 2, 'gamma':
0.26975849607037855, 'reg_alpha': 0.007723677078071416, 'reg_lambda':
0.025385906270307627}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:24,686] Trial 629 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:25,763] Trial 630 pruned.
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:27,080] Trial 631 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:27,716] Trial 632 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:28,517] Trial 633 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:30,157] Trial 634 finished with value:
1.3689721073881485 and parameters: {'clf_n_estimators': 136, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 573, 'learning_rate':
0.01320770427626862, 'max_depth': 12, 'subsample': 0.7272388119993899,
'colsample_bytree': 0.9145251100491542, 'min_child_weight': 2, 'gamma':
0.5112500044158198, 'reg_alpha': 0.0051578653071519395, 'reg_lambda':
0.030459038099128546}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:30,498] Trial 635 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:30,942] Trial 636 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:31,495] Trial 637 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:32,133] Trial 638 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:32,831] Trial 639 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:33,378] Trial 640 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:33,978] Trial 641 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:34,565] Trial 642 pruned.
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:35,123] Trial 643 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:37,356] Trial 644 finished with value:
1.367806501490515 and parameters: {'clf_n_estimators': 143, 'clf_max_depth': 10,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 534, 'learning_rate':
0.010601832453877634, 'max_depth': 12, 'subsample': 0.7160270314333924,
'colsample_bytree': 0.9427236848649418, 'min_child_weight': 2, 'gamma':
0.0007723082100455948, 'reg_alpha': 0.0006122546895187653, 'reg_lambda':
0.005508622334038242}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:37,906] Trial 645 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:38,508] Trial 646 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:39,073] Trial 647 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:40,432] Trial 648 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:41,055] Trial 649 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:42,053] Trial 650 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:43,602] Trial 651 finished with value:
1.3689148463435075 and parameters: {'clf_n_estimators': 113, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
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0.5, 'clf_class_weight': None, 'n_estimators': 471, 'learning_rate':
0.01005492149430561, 'max_depth': 12, 'subsample': 0.7473097820605239,
'colsample_bytree': 0.9343942764310663, 'min_child_weight': 4, 'gamma':
0.4454027351912363, 'reg_alpha': 0.000703084707421004, 'reg_lambda':
0.003596545496233167}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:44,186] Trial 652 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:46,281] Trial 653 finished with value:
1.3700334470299254 and parameters: {'clf_n_estimators': 145, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 504, 'learning_rate':
0.010717717144902508, 'max_depth': 12, 'subsample': 0.7345548911604123,
'colsample_bytree': 0.9268361641925018, 'min_child_weight': 2, 'gamma':
0.11580516886697695, 'reg_alpha': 0.0011709072361847262, 'reg_lambda':
0.004995540900470091}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:46,955] Trial 654 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:48,177] Trial 655 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:48,717] Trial 656 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:49,059] Trial 657 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:49,748] Trial 658 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:50,338] Trial 659 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:50,924] Trial 660 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:52,173] Trial 661 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:53,797] Trial 662 finished with value:
1.3678404590221358 and parameters: {'clf_n_estimators': 114, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 536, 'learning_rate':
0.010047123757091695, 'max_depth': 12, 'subsample': 0.7538560966205348,
'colsample_bytree': 0.9446664989904047, 'min_child_weight': 4, 'gamma':
0.9871279510644841, 'reg_alpha': 0.0018248346256891492, 'reg_lambda':
0.006035507332508164}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:54,516] Trial 663 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:55,121] Trial 664 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:55,831] Trial 665 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:56,291] Trial 666 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:56,694] Trial 667 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:57,441] Trial 668 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:59,263] Trial 669 finished with value:
1.3767737646711316 and parameters: {'clf_n_estimators': 100, 'clf_max_depth':
11, 'clf_min_samples_leaf': 4, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 622, 'learning_rate':
0.013187352571703307, 'max_depth': 12, 'subsample': 0.7846401601136186,
'colsample_bytree': 0.8913125675529497, 'min_child_weight': 4, 'gamma':
0.283947890489467, 'reg_alpha': 0.0021535431728009007, 'reg_lambda':
0.008226705804121168}. Best is trial 626 with value: 1.3648000719976008.
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:36:59,931] Trial 670 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:01,100] Trial 671 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:02,368] Trial 672 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:03,075] Trial 673 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:03,816] Trial 674 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:04,380] Trial 675 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:05,086] Trial 676 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:05,722] Trial 677 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:06,408] Trial 678 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:07,765] Trial 679 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:08,321] Trial 680 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:08,811] Trial 681 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:10,089] Trial 682 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:11,319] Trial 683 pruned.
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:11,995] Trial 684 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:12,603] Trial 685 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:13,297] Trial 686 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:14,035] Trial 687 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:15,198] Trial 688 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:15,797] Trial 689 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:16,360] Trial 690 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:16,790] Trial 691 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:17,478] Trial 692 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:18,190] Trial 693 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:18,936] Trial 694 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:19,615] Trial 695 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:20,287] Trial 696 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:21,730] Trial 697 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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[I 2026-03-13 19:37:22,514] Trial 698 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:22,880] Trial 699 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:23,493] Trial 700 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:24,071] Trial 701 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:25,228] Trial 702 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:26,566] Trial 703 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:27,117] Trial 704 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:27,529] Trial 705 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:27,952] Trial 706 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:28,896] Trial 707 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:29,635] Trial 708 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:31,604] Trial 709 finished with value:
1.367840262992961 and parameters: {'clf_n_estimators': 172, 'clf_max_depth': 12,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 513, 'learning_rate':
0.009753007982114571, 'max_depth': 12, 'subsample': 0.7747519411184269,
'colsample_bytree': 0.9311817399581035, 'min_child_weight': 2, 'gamma':
0.5697510216021089, 'reg_alpha': 0.007162685058446812, 'reg_lambda':
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0.002160068414793442}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:32,015] Trial 710 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:33,050] Trial 711 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:33,631] Trial 712 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:34,296] Trial 713 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:34,953] Trial 714 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:36,824] Trial 715 finished with value:
1.3717959208912796 and parameters: {'clf_n_estimators': 109, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 556, 'learning_rate':
0.011945845767093568, 'max_depth': 12, 'subsample': 0.7675204944475871,
'colsample_bytree': 0.9407612506507756, 'min_child_weight': 4, 'gamma':
0.00018524372536911308, 'reg_alpha': 0.0020115951912996854, 'reg_lambda':
0.0006764965313503749}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:38,533] Trial 716 finished with value:
1.3661198693348098 and parameters: {'clf_n_estimators': 100, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 536, 'learning_rate':
0.009077517370962294, 'max_depth': 12, 'subsample': 0.7487837514909654,
'colsample_bytree': 0.9110188508999055, 'min_child_weight': 3, 'gamma':
0.3719519531307627, 'reg_alpha': 0.0009155815680570432, 'reg_lambda':
6.05323174263115e-05}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:40,545] Trial 717 finished with value:
1.3721004047301344 and parameters: {'clf_n_estimators': 100, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 7, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 546, 'learning_rate':
0.007748862700694364, 'max_depth': 12, 'subsample': 0.7834015630376264,
'colsample_bytree': 0.8939681247739713, 'min_child_weight': 3, 'gamma':
0.1893636283792486, 'reg_alpha': 0.0010592323539962243, 'reg_lambda':
0.00011918083637616755}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:42,607] Trial 718 finished with value:
1.3699016531194452 and parameters: {'clf_n_estimators': 100, 'clf_max_depth':
13, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 532, 'learning_rate':
0.008973379252845817, 'max_depth': 12, 'subsample': 0.7578822843646273,
'colsample_bytree': 0.9105889603738575, 'min_child_weight': 3, 'gamma':
0.26839544007193494, 'reg_alpha': 0.0012534040923788657, 'reg_lambda':
6.965252339062911e-05}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:43,129] Trial 719 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:43,910] Trial 720 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:45,018] Trial 721 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:46,627] Trial 722 finished with value:
1.3706658582824283 and parameters: {'clf_n_estimators': 125, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 541, 'learning_rate':
0.026975668528272014, 'max_depth': 12, 'subsample': 0.7445765050656532,
'colsample_bytree': 0.9117830282280386, 'min_child_weight': 3, 'gamma':
0.20472109871856667, 'reg_alpha': 0.0025896003172274644, 'reg_lambda':
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6.07657929451512e-05}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:47,399] Trial 723 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:48,105] Trial 724 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:49,862] Trial 725 finished with value:
1.3702128002750291 and parameters: {'clf_n_estimators': 100, 'clf_max_depth':
13, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 525, 'learning_rate':
0.008576700692679424, 'max_depth': 12, 'subsample': 0.7681316712067316,
'colsample_bytree': 0.9231514658975702, 'min_child_weight': 3, 'gamma':
0.6420876268712938, 'reg_alpha': 0.01480387435137013, 'reg_lambda':
5.4492299257576496e-05}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:51,126] Trial 726 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:51,805] Trial 727 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:52,387] Trial 728 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:54,524] Trial 729 finished with value:
1.3720462828127011 and parameters: {'clf_n_estimators': 112, 'clf_max_depth':
20, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 622, 'learning_rate':
0.010071053523324887, 'max_depth': 12, 'subsample': 0.7311463013088539,
'colsample_bytree': 0.9249422384348294, 'min_child_weight': 4, 'gamma':
0.31829849408784, 'reg_alpha': 0.0018635489874956933, 'reg_lambda':
0.007339830398286357}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:55,348] Trial 730 pruned.
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:57,320] Trial 731 finished with value:
1.3674948268560911 and parameters: {'clf_n_estimators': 100, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 607, 'learning_rate':
0.007294280365646187, 'max_depth': 12, 'subsample': 0.755126995629732,
'colsample_bytree': 0.912115070276944, 'min_child_weight': 4, 'gamma':
0.7520115318392224, 'reg_alpha': 0.006281244423488257, 'reg_lambda':
0.01864897194850128}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:37:59,695] Trial 732 finished with value:
1.3700211105187323 and parameters: {'clf_n_estimators': 122, 'clf_max_depth':
13, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 536, 'learning_rate':
0.008949819392011464, 'max_depth': 12, 'subsample': 0.7378637769962114,
'colsample_bytree': 0.9711977980085058, 'min_child_weight': 3, 'gamma':
0.15082325856356843, 'reg_alpha': 0.0010334337591022635, 'reg_lambda':
0.009025054486678718}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:01,736] Trial 733 finished with value:
1.3771384413481105 and parameters: {'clf_n_estimators': 136, 'clf_max_depth':
11, 'clf_min_samples_leaf': 3, 'clf_min_samples_split': 7, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 567, 'learning_rate':
0.009483127374757172, 'max_depth': 12, 'subsample': 0.7879331122405192,
'colsample_bytree': 0.9517084433274058, 'min_child_weight': 4, 'gamma':
0.378269245710557, 'reg_alpha': 0.0004951093627967107, 'reg_lambda':
0.0003438245675973395}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:02,665] Trial 734 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:03,280] Trial 735 pruned.
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:03,907] Trial 736 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:04,492] Trial 737 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:05,277] Trial 738 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:06,043] Trial 739 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:06,718] Trial 740 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:07,982] Trial 741 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:08,417] Trial 742 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:09,146] Trial 743 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:09,875] Trial 744 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:10,578] Trial 745 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:11,701] Trial 746 finished with value:
1.3660347212610737 and parameters: {'clf_n_estimators': 136, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 519, 'learning_rate':
0.05131920972746744, 'max_depth': 12, 'subsample': 0.7318514593575592,
'colsample_bytree': 0.9472766012744624, 'min_child_weight': 4, 'gamma':
0.9865312014260177, 'reg_alpha': 0.0006112944673136657, 'reg_lambda':
0.002065378129604661}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:12,157] Trial 747 pruned.
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:12,639] Trial 748 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:13,035] Trial 749 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:13,508] Trial 750 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:14,064] Trial 751 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:14,464] Trial 752 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:14,815] Trial 753 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:15,918] Trial 754 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:16,335] Trial 755 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:17,052] Trial 756 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:17,633] Trial 757 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:18,190] Trial 758 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:19,038] Trial 759 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:21,044] Trial 760 finished with value:
1.3722724448770274 and parameters: {'clf_n_estimators': 168, 'clf_max_depth':
12, 'clf_min_samples_leaf': 3, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 494, 'learning_rate':
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0.014525234917950606, 'max_depth': 12, 'subsample': 0.7471009800286187,
'colsample_bytree': 0.8985603008256001, 'min_child_weight': 2, 'gamma':
0.2940187777776004, 'reg_alpha': 0.004973259377569315, 'reg_lambda':
0.00015153249309092937}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:21,688] Trial 761 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:22,410] Trial 762 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:23,081] Trial 763 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:23,466] Trial 764 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:23,882] Trial 765 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:24,574] Trial 766 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:25,921] Trial 767 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:26,523] Trial 768 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:27,155] Trial 769 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:27,602] Trial 770 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:28,176] Trial 771 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:30,067] Trial 772 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:30,846] Trial 773 pruned.
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:33,301] Trial 774 finished with value:
1.368600522863603 and parameters: {'clf_n_estimators': 112, 'clf_max_depth': 13,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 555, 'learning_rate':
0.009116855655576997, 'max_depth': 12, 'subsample': 0.7520609022626608,
'colsample_bytree': 0.9355716504297469, 'min_child_weight': 3, 'gamma':
1.9124630406512835e-05, 'reg_alpha': 0.007769949205360521, 'reg_lambda':
0.004756491287026852}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:33,790] Trial 775 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:34,312] Trial 776 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:35,712] Trial 777 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:36,889] Trial 778 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:37,804] Trial 779 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:40,428] Trial 780 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:41,254] Trial 781 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:41,995] Trial 782 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:42,538] Trial 783 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:44,719] Trial 784 finished with value:
1.3656078448282 and parameters: {'clf_n_estimators': 131, 'clf_max_depth': 13,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 491, 'learning_rate':
0.00813920501412115, 'max_depth': 12, 'subsample': 0.7400890191089176,
'colsample_bytree': 0.929476601662843, 'min_child_weight': 4, 'gamma':
0.2764958712904037, 'reg_alpha': 0.0006820462842260191, 'reg_lambda':
0.0023396518875712408}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:45,500] Trial 785 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:46,205] Trial 786 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:47,758] Trial 787 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:49,332] Trial 788 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:49,928] Trial 789 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:50,624] Trial 790 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:51,966] Trial 791 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:52,805] Trial 792 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:53,594] Trial 793 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:55,533] Trial 794 finished with value:
1.3700973485330248 and parameters: {'clf_n_estimators': 124, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 515, 'learning_rate':
0.008455960121856778, 'max_depth': 12, 'subsample': 0.7510917874204401,
'colsample_bytree': 0.6187068465278482, 'min_child_weight': 4, 'gamma':
0.229015022389901, 'reg_alpha': 0.0029626431857665628, 'reg_lambda':
0.00029148471381293525}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:55,974] Trial 795 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:56,783] Trial 796 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:57,584] Trial 797 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:38:58,408] Trial 798 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:00,190] Trial 799 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:01,906] Trial 800 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:02,850] Trial 801 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:03,702] Trial 802 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:04,373] Trial 803 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:05,313] Trial 804 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:06,247] Trial 805 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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[I 2026-03-13 19:39:06,906] Trial 806 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:08,183] Trial 807 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:09,242] Trial 808 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:11,716] Trial 809 finished with value:
1.3743082467968637 and parameters: {'clf_n_estimators': 109, 'clf_max_depth':
14, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 524, 'learning_rate':
0.009905550407495144, 'max_depth': 12, 'subsample': 0.5092917876340266,
'colsample_bytree': 0.9178584018965101, 'min_child_weight': 2, 'gamma':
0.05541895286935451, 'reg_alpha': 0.0039086371042641515, 'reg_lambda':
0.00037512159742909883}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:12,756] Trial 810 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:13,481] Trial 811 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:14,321] Trial 812 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:15,153] Trial 813 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:16,566] Trial 814 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:18,899] Trial 815 finished with value:
1.3712297410375083 and parameters: {'clf_n_estimators': 124, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 10, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 537, 'learning_rate':
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0.008592594674168912, 'max_depth': 12, 'subsample': 0.7346439119179498,
'colsample_bytree': 0.9268525240531739, 'min_child_weight': 3, 'gamma':
0.21808198377580754, 'reg_alpha': 0.009379100814943739, 'reg_lambda':
0.00020940425716534344}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:19,726] Trial 816 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:20,555] Trial 817 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:21,342] Trial 818 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:22,154] Trial 819 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:22,988] Trial 820 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:25,541] Trial 821 finished with value:
1.3742856005594541 and parameters: {'clf_n_estimators': 169, 'clf_max_depth':
13, 'clf_min_samples_leaf': 3, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 549, 'learning_rate':
0.009325113253188013, 'max_depth': 12, 'subsample': 0.6713622773044388,
'colsample_bytree': 0.9197128105240983, 'min_child_weight': 2, 'gamma':
0.28760004724046334, 'reg_alpha': 0.005187422238340897, 'reg_lambda':
0.003724547927543662}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:26,257] Trial 822 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:26,726] Trial 823 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:28,270] Trial 824 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:28,985] Trial 825 pruned.
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:29,782] Trial 826 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:30,230] Trial 827 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:31,028] Trial 828 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:31,879] Trial 829 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:32,523] Trial 830 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:34,772] Trial 831 finished with value:
1.3664204761099727 and parameters: {'clf_n_estimators': 111, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 543, 'learning_rate':
0.007882635543614635, 'max_depth': 11, 'subsample': 0.7192696579125375,
'colsample_bytree': 0.9743060509974375, 'min_child_weight': 4, 'gamma':
0.3745793302439652, 'reg_alpha': 0.002600133411970888, 'reg_lambda':
0.013678180207120355}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:36,990] Trial 832 finished with value:
1.375353767522353 and parameters: {'clf_n_estimators': 100, 'clf_max_depth': 10,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 7, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 517, 'learning_rate':
0.007201938888072303, 'max_depth': 11, 'subsample': 0.6936854166403512,
'colsample_bytree': 0.9835034996438754, 'min_child_weight': 2, 'gamma':
0.1828871869691347, 'reg_alpha': 0.0010779966624295085, 'reg_lambda':
0.028853372865986222}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:37,789] Trial 833 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:38,534] Trial 834 pruned.
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:39,240] Trial 835 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:40,529] Trial 836 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:41,113] Trial 837 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:41,737] Trial 838 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:42,658] Trial 839 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:43,550] Trial 840 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:44,217] Trial 841 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:46,605] Trial 842 finished with value:
1.371138413746339 and parameters: {'clf_n_estimators': 113, 'clf_max_depth': 11,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 550, 'learning_rate':
0.008015383434576678, 'max_depth': 12, 'subsample': 0.7781956771521152,
'colsample_bytree': 0.9256527492604203, 'min_child_weight': 3, 'gamma':
0.1648840364613127, 'reg_alpha': 2.3309247448309558e-08, 'reg_lambda':
0.00902985460641083}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:48,171] Trial 843 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:48,734] Trial 844 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:49,173] Trial 845 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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[I 2026-03-13 19:39:49,908] Trial 846 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:51,406] Trial 847 finished with value:
1.3659484237062693 and parameters: {'clf_n_estimators': 136, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 501, 'learning_rate':
0.0217484882844147, 'max_depth': 12, 'subsample': 0.7454763592047331,
'colsample_bytree': 0.9896045400051219, 'min_child_weight': 3, 'gamma':
0.5548626474195129, 'reg_alpha': 0.0009618985380299058, 'reg_lambda':
5.429502327557098e-05}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:51,976] Trial 848 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:52,469] Trial 849 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:53,021] Trial 850 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:53,516] Trial 851 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:54,198] Trial 852 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:54,873] Trial 853 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:57,120] Trial 854 finished with value:
1.375409241354913 and parameters: {'clf_n_estimators': 112, 'clf_max_depth': 13,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 5, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 505, 'learning_rate':
0.006984303827352806, 'max_depth': 12, 'subsample': 0.749358494537228,
'colsample_bytree': 0.9869619202786079, 'min_child_weight': 2, 'gamma':
0.3826643846022148, 'reg_alpha': 0.001187653732066234, 'reg_lambda':
6.984517738220393e-05}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:57,980] Trial 855 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:39:59,655] Trial 856 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:00,103] Trial 857 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:00,724] Trial 858 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:01,261] Trial 859 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:01,874] Trial 860 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:03,765] Trial 861 finished with value:
1.3680422124138651 and parameters: {'clf_n_estimators': 112, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 456, 'learning_rate':
0.009037309783359185, 'max_depth': 12, 'subsample': 0.7172047820183106,
'colsample_bytree': 0.965712449549844, 'min_child_weight': 4, 'gamma':
0.381347816301646, 'reg_alpha': 0.0008562364239226266, 'reg_lambda':
6.241510258767974e-05}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:04,500] Trial 862 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:05,274] Trial 863 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:05,774] Trial 864 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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[I 2026-03-13 19:40:07,765] Trial 865 finished with value:
1.3691443027481782 and parameters: {'clf_n_estimators': 111, 'clf_max_depth':
13, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 494, 'learning_rate':
0.008202039456419822, 'max_depth': 12, 'subsample': 0.7008319670166457,
'colsample_bytree': 0.9707432129121473, 'min_child_weight': 3, 'gamma':
0.49935343456407394, 'reg_alpha': 0.0018213689989012758, 'reg_lambda':
0.012811914033385404}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:08,492] Trial 866 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:09,205] Trial 867 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:10,871] Trial 868 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:11,552] Trial 869 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:12,136] Trial 870 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:12,738] Trial 871 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:13,361] Trial 872 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:14,089] Trial 873 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:14,821] Trial 874 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:15,757] Trial 875 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:16,377] Trial 876 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:17,235] Trial 877 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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[I 2026-03-13 19:40:17,705] Trial 878 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:18,284] Trial 879 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:20,559] Trial 880 finished with value:
1.3741477484492022 and parameters: {'clf_n_estimators': 189, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 547, 'learning_rate':
0.009341857230127931, 'max_depth': 12, 'subsample': 0.7618696093473196,
'colsample_bytree': 0.9987523218921867, 'min_child_weight': 3, 'gamma':
0.7058808702314727, 'reg_alpha': 0.0027105832703713255, 'reg_lambda':
5.571939053391742e-05}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:21,100] Trial 881 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:21,687] Trial 882 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:21,994] Trial 883 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:22,720] Trial 884 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:23,386] Trial 885 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:25,186] Trial 886 finished with value:
1.3694118928596384 and parameters: {'clf_n_estimators': 112, 'clf_max_depth':
13, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 594, 'learning_rate':
0.00979316386839558, 'max_depth': 12, 'subsample': 0.7176120410825805,
'colsample_bytree': 0.978281366266238, 'min_child_weight': 2, 'gamma':
0.6592928114147365, 'reg_alpha': 0.0005724210893545788, 'reg_lambda':
0.004816396721960267}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:25,760] Trial 887 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:26,335] Trial 888 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:27,096] Trial 889 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:27,823] Trial 890 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:28,615] Trial 891 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:29,037] Trial 892 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:29,707] Trial 893 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:30,425] Trial 894 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:32,110] Trial 895 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:32,675] Trial 896 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:33,429] Trial 897 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:34,151] Trial 898 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:34,840] Trial 899 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:35,488] Trial 900 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:35,962] Trial 901 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:38,187] Trial 902 finished with value:
1.3664110192357395 and parameters: {'clf_n_estimators': 137, 'clf_max_depth':
13, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 10, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 548, 'learning_rate':
0.010200216761690592, 'max_depth': 12, 'subsample': 0.7257972117065457,
'colsample_bytree': 0.9832560915733246, 'min_child_weight': 3, 'gamma':
0.30617368907072734, 'reg_alpha': 0.005810888556256481, 'reg_lambda':
0.0004344240391911239}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:39,475] Trial 903 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:41,564] Trial 904 finished with value:
1.3713264052579708 and parameters: {'clf_n_estimators': 124, 'clf_max_depth':
13, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 10, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 464, 'learning_rate':
0.01101566737032119, 'max_depth': 12, 'subsample': 0.7201240053683154,
'colsample_bytree': 0.9826521872134877, 'min_child_weight': 3, 'gamma':
0.1835947793383897, 'reg_alpha': 0.006347047290832854, 'reg_lambda':
0.0006008971804043978}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:42,331] Trial 905 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:43,790] Trial 906 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:45,771] Trial 907 finished with value:
1.3720361309338112 and parameters: {'clf_n_estimators': 129, 'clf_max_depth':
13, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 10, 'clf_max_features':
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0.5, 'clf_class_weight': None, 'n_estimators': 524, 'learning_rate':
0.011519899470463564, 'max_depth': 12, 'subsample': 0.7294166191527993,
'colsample_bytree': 0.9746893806696141, 'min_child_weight': 3, 'gamma':
0.2904750555423896, 'reg_alpha': 0.0007890702129079459, 'reg_lambda':
0.00019876813454829844}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:46,512] Trial 908 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:48,779] Trial 909 finished with value:
1.3724203970484565 and parameters: {'clf_n_estimators': 202, 'clf_max_depth':
14, 'clf_min_samples_leaf': 3, 'clf_min_samples_split': 10, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 515, 'learning_rate':
0.009849232555214856, 'max_depth': 12, 'subsample': 0.704768486380145,
'colsample_bytree': 0.9700165432901616, 'min_child_weight': 3, 'gamma':
0.33476409377158495, 'reg_alpha': 0.00032066155875287857, 'reg_lambda':
0.0004717980767008089}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:49,502] Trial 910 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:50,261] Trial 911 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:50,806] Trial 912 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:51,478] Trial 913 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:52,998] Trial 914 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:53,617] Trial 915 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:54,457] Trial 916 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:54,932] Trial 917 pruned.
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:55,672] Trial 918 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:56,468] Trial 919 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:57,275] Trial 920 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:58,031] Trial 921 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:40:58,894] Trial 922 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:01,134] Trial 923 finished with value:
1.3672178807571402 and parameters: {'clf_n_estimators': 122, 'clf_max_depth':
12, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 541, 'learning_rate':
0.008038650581190488, 'max_depth': 12, 'subsample': 0.7424269740859727,
'colsample_bytree': 0.9415123571300522, 'min_child_weight': 3, 'gamma':
0.6979468323692116, 'reg_alpha': 0.0003683588032356984, 'reg_lambda':
7.148110169226053e-05}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:02,027] Trial 924 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:04,181] Trial 925 finished with value:
1.3677771307262236 and parameters: {'clf_n_estimators': 100, 'clf_max_depth':
13, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 9, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 557, 'learning_rate':
0.00904062690292758, 'max_depth': 12, 'subsample': 0.7315888634139047,
'colsample_bytree': 0.9619372559728865, 'min_child_weight': 3, 'gamma':
0.3900791867695727, 'reg_alpha': 0.00117723220057066, 'reg_lambda':
0.00019192377098771872}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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[I 2026-03-13 19:41:05,032] Trial 926 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:05,779] Trial 927 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:06,604] Trial 928 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:07,427] Trial 929 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:08,190] Trial 930 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:09,030] Trial 931 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:11,067] Trial 932 finished with value:
1.3708544915818346 and parameters: {'clf_n_estimators': 112, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 557, 'learning_rate':
0.011281919625928161, 'max_depth': 12, 'subsample': 0.7438163744237413,
'colsample_bytree': 0.9463650580748855, 'min_child_weight': 3, 'gamma':
0.27894789944584114, 'reg_alpha': 0.0013953914937741192, 'reg_lambda':
0.0007565910347300484}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:11,543] Trial 933 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:12,354] Trial 934 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:12,977] Trial 935 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:13,824] Trial 936 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:14,810] Trial 937 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:15,631] Trial 938 pruned.
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:17,117] Trial 939 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:19,275] Trial 940 finished with value:
1.3656564819186903 and parameters: {'clf_n_estimators': 113, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 586, 'learning_rate':
0.007772667365749624, 'max_depth': 12, 'subsample': 0.7388181362770108,
'colsample_bytree': 0.9211030190876376, 'min_child_weight': 4, 'gamma':
0.6801379772848094, 'reg_alpha': 0.0047576345782820995, 'reg_lambda':
5.592366604186021e-05}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:19,973] Trial 941 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:21,275] Trial 942 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:22,167] Trial 943 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:24,364] Trial 944 finished with value:
1.3676775021175247 and parameters: {'clf_n_estimators': 121, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 496, 'learning_rate':
0.007885538999681875, 'max_depth': 12, 'subsample': 0.7426868565220639,
'colsample_bytree': 0.9195915555391853, 'min_child_weight': 3, 'gamma':
0.4190671451322295, 'reg_alpha': 0.01073863254429225, 'reg_lambda':
0.0001728721495581471}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:25,169] Trial 945 pruned.
Fitting Stage 1: Hurdle Classifier…
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Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:25,987] Trial 946 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:27,576] Trial 947 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:28,466] Trial 948 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:29,148] Trial 949 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:30,720] Trial 950 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:31,428] Trial 951 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:32,168] Trial 952 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:33,799] Trial 953 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:34,762] Trial 954 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:36,317] Trial 955 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:39,181] Trial 956 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:39,808] Trial 957 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:40,701] Trial 958 pruned.
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:43,083] Trial 959 finished with value:
1.3701816412402954 and parameters: {'clf_n_estimators': 121, 'clf_max_depth':
10, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 602, 'learning_rate':
0.00904021791017995, 'max_depth': 12, 'subsample': 0.7071400598344858,
'colsample_bytree': 0.921904625938906, 'min_child_weight': 3, 'gamma':
0.00013087240621350365, 'reg_alpha': 0.004814082482224466, 'reg_lambda':
8.10495680825389e-05}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:43,832] Trial 960 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:44,600] Trial 961 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:46,020] Trial 962 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:46,851] Trial 963 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:47,559] Trial 964 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:49,500] Trial 965 finished with value:
1.367519687327647 and parameters: {'clf_n_estimators': 136, 'clf_max_depth': 9,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 464, 'learning_rate':
0.009507084233709905, 'max_depth': 12, 'subsample': 0.7248262853132216,
'colsample_bytree': 0.9494071270899561, 'min_child_weight': 4, 'gamma':
0.3281771601230489, 'reg_alpha': 0.009283035636225474, 'reg_lambda':
0.02421543202888889}. Best is trial 626 with value: 1.3648000719976008.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:50,116] Trial 966 pruned.
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:50,626] Trial 967 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:51,222] Trial 968 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:51,983] Trial 969 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:53,800] Trial 970 finished with value:
1.36334649565753 and parameters: {'clf_n_estimators': 124, 'clf_max_depth': 11,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 440, 'learning_rate':
0.010649894593413789, 'max_depth': 11, 'subsample': 0.7358207531734963,
'colsample_bytree': 0.9329209758123674, 'min_child_weight': 4, 'gamma':
0.2612050507806572, 'reg_alpha': 1.0038537204297696e-08, 'reg_lambda':
0.06377470739305285}. Best is trial 970 with value: 1.36334649565753.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:54,553] Trial 971 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:55,222] Trial 972 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:55,863] Trial 973 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:57,536] Trial 974 finished with value:
1.365504839033857 and parameters: {'clf_n_estimators': 129, 'clf_max_depth': 11,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 411, 'learning_rate':
0.010440209027963946, 'max_depth': 10, 'subsample': 0.7593288446267137,
'colsample_bytree': 0.9451591778207241, 'min_child_weight': 4, 'gamma':
0.3580129794842825, 'reg_alpha': 6.3639873468972205e-06, 'reg_lambda':
0.060761833716251615}. Best is trial 970 with value: 1.36334649565753.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:58,805] Trial 975 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:41:59,377] Trial 976 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:42:01,082] Trial 977 finished with value:
1.3686707951751398 and parameters: {'clf_n_estimators': 156, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 391, 'learning_rate':
0.010641154731868028, 'max_depth': 10, 'subsample': 0.7143484431969948,
'colsample_bytree': 0.9465011669275692, 'min_child_weight': 4, 'gamma':
0.3055023504419256, 'reg_alpha': 3.1055309536707e-08, 'reg_lambda':
0.0769085820988843}. Best is trial 970 with value: 1.36334649565753.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:42:02,683] Trial 978 finished with value:
1.3650031469272026 and parameters: {'clf_n_estimators': 133, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 385, 'learning_rate':
0.01224343512145819, 'max_depth': 10, 'subsample': 0.758815830506808,
'colsample_bytree': 0.9584103399813604, 'min_child_weight': 4, 'gamma':
0.33293786198381997, 'reg_alpha': 3.3614816976461264e-06, 'reg_lambda':
0.09615548388411502}. Best is trial 970 with value: 1.36334649565753.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:42:03,447] Trial 979 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:42:04,009] Trial 980 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:42:05,543] Trial 981 finished with value:
1.3649720165326371 and parameters: {'clf_n_estimators': 126, 'clf_max_depth':
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21, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 349, 'learning_rate':
0.013073398246091394, 'max_depth': 10, 'subsample': 0.7736981973272357,
'colsample_bytree': 0.9740454182038464, 'min_child_weight': 4, 'gamma':
0.20122253183358674, 'reg_alpha': 3.2724995468154354e-08, 'reg_lambda':
0.04363627211704475}. Best is trial 970 with value: 1.36334649565753.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:42:06,591] Trial 982 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:42:07,987] Trial 983 finished with value:
1.372386182812985 and parameters: {'clf_n_estimators': 100, 'clf_max_depth': 11,
'clf_min_samples_leaf': 3, 'clf_min_samples_split': 7, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 344, 'learning_rate':
0.014290163705804567, 'max_depth': 10, 'subsample': 0.7787100651850526,
'colsample_bytree': 0.9733354183263775, 'min_child_weight': 4, 'gamma':
0.20308808788901841, 'reg_alpha': 3.112568174884295e-08, 'reg_lambda':
0.09977555908916307}. Best is trial 970 with value: 1.36334649565753.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:42:09,474] Trial 984 finished with value:
1.3657915509279135 and parameters: {'clf_n_estimators': 116, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 19, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 359, 'learning_rate':
0.013161740440032068, 'max_depth': 10, 'subsample': 0.7956977578685644,
'colsample_bytree': 0.9832942474803408, 'min_child_weight': 4, 'gamma':
5.902635808186677e-07, 'reg_alpha': 1.3601503607046108e-06, 'reg_lambda':
0.0005593006442808883}. Best is trial 970 with value: 1.36334649565753.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:42:09,980] Trial 985 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
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[I 2026-03-13 19:42:11,584] Trial 986 finished with value:
1.3690471440278749 and parameters: {'clf_n_estimators': 123, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 20, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 378, 'learning_rate':
0.011878833887238859, 'max_depth': 10, 'subsample': 0.8009095252110277,
'colsample_bytree': 0.987946125700554, 'min_child_weight': 4, 'gamma':
0.14114770240117663, 'reg_alpha': 3.7709332985384424e-06, 'reg_lambda':
0.047754316978369495}. Best is trial 970 with value: 1.36334649565753.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:42:15,191] Trial 987 finished with value:
1.3681699066232407 and parameters: {'clf_n_estimators': 743, 'clf_max_depth':
25, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 17, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 294, 'learning_rate':
0.014207073765877272, 'max_depth': 10, 'subsample': 0.7958158479320029,
'colsample_bytree': 0.9884365716569342, 'min_child_weight': 4, 'gamma':
6.972033278458045e-07, 'reg_alpha': 4.700069280490009e-06, 'reg_lambda':
0.06952714621844096}. Best is trial 970 with value: 1.36334649565753.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:42:16,646] Trial 988 finished with value:
1.3717372565812145 and parameters: {'clf_n_estimators': 112, 'clf_max_depth':
11, 'clf_min_samples_leaf': 3, 'clf_min_samples_split': 7, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 344, 'learning_rate':
0.01543401448654701, 'max_depth': 10, 'subsample': 0.772821827645373,
'colsample_bytree': 0.9800375005435341, 'min_child_weight': 4, 'gamma':
0.2488634091413096, 'reg_alpha': 1.2847176325146126e-08, 'reg_lambda':
0.05084793816339574}. Best is trial 970 with value: 1.36334649565753.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:42:18,227] Trial 989 finished with value:
1.3764000475908482 and parameters: {'clf_n_estimators': 123, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 20, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 374, 'learning_rate':
0.013747500404989687, 'max_depth': 10, 'subsample': 0.7949058441500894,
'colsample_bytree': 0.9987557199310221, 'min_child_weight': 4, 'gamma':
```

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3.518702208360118e-07, 'reg_alpha': 2.867902274801424e-06, 'reg_lambda':
0.11058129437991202}. Best is trial 970 with value: 1.36334649565753.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:42:18,763] Trial 990 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:42:20,049] Trial 991 finished with value:
1.3753908042254075 and parameters: {'clf_n_estimators': 100, 'clf_max_depth':
11, 'clf_min_samples_leaf': 3, 'clf_min_samples_split': 19, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 323, 'learning_rate':
0.013747788534976171, 'max_depth': 9, 'subsample': 0.7892067979894816,
'colsample_bytree': 0.9876675273192005, 'min_child_weight': 4, 'gamma':
1.3648615601101491e-05, 'reg_alpha': 2.3665295209690092e-08, 'reg_lambda':
0.06359078230665227}. Best is trial 970 with value: 1.36334649565753.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:42:21,114] Trial 992 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:42:22,731] Trial 993 finished with value:
1.3730010029850659 and parameters: {'clf_n_estimators': 129, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 7, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 355, 'learning_rate':
0.012637448799191391, 'max_depth': 10, 'subsample': 0.807675389802581,
'colsample_bytree': 0.9797126653814842, 'min_child_weight': 4, 'gamma':
0.17839813939461446, 'reg_alpha': 1.0794661816866274e-06, 'reg_lambda':
0.18544362386029695}. Best is trial 970 with value: 1.36334649565753.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:42:23,298] Trial 994 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:42:23,870] Trial 995 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
```

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Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:42:25,405] Trial 996 finished with value:
1.3650549801533243 and parameters: {'clf_n_estimators': 133, 'clf_max_depth':
11, 'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features':
0.5, 'clf_class_weight': None, 'n_estimators': 389, 'learning_rate':
0.014883771642011287, 'max_depth': 9, 'subsample': 0.778471849197188,
'colsample_bytree': 0.9658018703705012, 'min_child_weight': 4, 'gamma':
2.280910008313271e-06, 'reg_alpha': 5.354643601993525e-06, 'reg_lambda':
0.13879848054629781}. Best is trial 970 with value: 1.36334649565753.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:42:25,951] Trial 997 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:42:26,927] Trial 998 pruned.
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
[I 2026-03-13 19:42:28,415] Trial 999 finished with value:
1.375250503404177 and parameters: {'clf_n_estimators': 100, 'clf_max_depth': 11,
'clf_min_samples_leaf': 3, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 381, 'learning_rate':
0.012934036100178186, 'max_depth': 10, 'subsample': 0.7922238437393438,
'colsample_bytree': 0.9999052618342055, 'min_child_weight': 4, 'gamma':
3.630512463427837e-06, 'reg_alpha': 2.7761906982098174e-06, 'reg_lambda':
0.19038609655297356}. Best is trial 970 with value: 1.36334649565753.
Best MAE: 1.36334649565753
Best params: {'clf_n_estimators': 124, 'clf_max_depth': 11,
'clf_min_samples_leaf': 2, 'clf_min_samples_split': 8, 'clf_max_features': 0.5,
'clf_class_weight': None, 'n_estimators': 440, 'learning_rate':
0.010649894593413789, 'max_depth': 11, 'subsample': 0.7358207531734963,
'colsample_bytree': 0.9329209758123674, 'min_child_weight': 4, 'gamma':
0.2612050507806572, 'reg_alpha': 1.0038537204297696e-08, 'reg_lambda':
0.06377470739305285}
```

[59]:
```python
best_OHM = study.best_params

best_clf = RandomForestClassifier(
    n_estimators=best_OHM['clf_n_estimators'],
    max_depth=best_OHM['clf_max_depth'],
    min_samples_leaf=best_OHM['clf_min_samples_leaf'],
```

```
    max_features=best_OHM['clf_max_features'],
    random_state=42,
)
best_reg = XGBRegressor(
    n_estimators=best_OHM['n_estimators'],
    learning_rate=best_OHM['learning_rate'],
    max_depth=best_OHM['max_depth'],
    subsample=best_OHM['subsample'],
    colsample_bytree=best_OHM['colsample_bytree'],
    min_child_weight=best_OHM['min_child_weight'],
    gamma=best_OHM['gamma'],
    reg_alpha=best_OHM['reg_alpha'],
    reg_lambda=best_OHM['reg_lambda'],
    random_state=42,
    objective='reg:squarederror',
)

reaching_model_tuned = OutageHurdleModel(classifier=best_clf,␣
 ↪regressor=best_reg)
reaching_model_tuned.fit(X_train_proc, y_train.values)
tuned_hurdle_preds = reaching_model_tuned.predict(X_test_proc)

tuned_hurdle_preds_minutes = np.expm1(tuned_hurdle_preds)

# Calculate Final Metrics
tuned_OHM_metrics = {
    'Model': "Advanced Hurdle (Optuna tuned)",
    'MAE (log)': mean_absolute_error(y_test, tuned_hurdle_preds),
    'RMSE (log)': root_mean_squared_error(y_test, tuned_hurdle_preds),
    'MAE (minutes)': mean_absolute_error(y_test_minutes,␣
 ↪tuned_hurdle_preds_minutes),
    'RMSE (minutes)': root_mean_squared_error(y_test_minutes,␣
 ↪tuned_hurdle_preds_minutes),
    'R^2': r2_score(y_test, tuned_hurdle_preds)
}

print("\n--- REAching MODEL PERFORMANCE ---")
for k, v in tuned_OHM_metrics.items():
    if isinstance(v, float):
        print(f"{k:15}: {v:.4f}")
    else:
        print(f"{k:15}: {v}")

results_log.append(tuned_OHM_metrics)
```

```
Fitting Stage 1: Hurdle Classifier…
Fitting Stage 2: Duration Regressor (with Stage 1 context)…
```

```
--- REAching MODEL PERFORMANCE ---
Model          : Advanced Hurdle (Optuna tuned)
MAE (log)      : 1.3791
RMSE (log)     : 1.8828
MAE (minutes)  : 1794.9446
RMSE (minutes) : 3793.8704
R^2            : 0.5396
```

[60]:
```
# Residuals for tuned OHM model
residual_df_tuned_ohm = pd.DataFrame({
    'Predicted Log Duration': tuned_hurdle_preds,
    'Residuals (log)': y_test - tuned_hurdle_preds,
    'Predicted Duration (min)': tuned_hurdle_preds_minutes,
    'Residuals (min)': y_test_minutes - tuned_hurdle_preds_minutes
})
tuned_ohm_fig = plot_log_and_reg_residuals(residual_df_tuned_ohm, "Advanced␣
↪Hurdle Model (Optuna Tuned)")
tuned_ohm_fig.write_html('docs/assets/plots/
↪advanced_hurdle_model_residuals_tuned.html', include_plotlyjs='cdn')
```

[70]:
```
results_df = pd.DataFrame(results_log)
results_df['Model'] = results_df['Model'].replace({
    'HistGradientBoostingRegressor (tuned)': 'HGB Regressor (tuned)'
})
results_df
```

[70]:
```
                             Model  MAE (log)  RMSE (log)  MAE (minutes)  \
0        Linear Regression (Baseline)       1.46        1.93        1812.06
1            Ridge Regression (tuned)       1.48        1.91        1802.69
2      Random Forest Regressor (tuned)      1.34        1.83        1739.02
3            XGBoost Regressor (tuned)      1.37        1.90        1755.67
4                HGB Regressor (tuned)      1.34        1.86        1691.86
5               XGBoost (Optuna tuned)      1.37        1.90        1731.78
6  Advanced Hurdle (RF + XGB Stacked)       1.39        1.87        1786.11
7      Advanced Hurdle (Optuna tuned)       1.38        1.88        1794.94

   RMSE (minutes)   R^2
0         4378.00  0.52
1         3925.18  0.52
2         3719.70  0.57
3         3729.29  0.53
4         3557.68  0.55
5         3663.05  0.53
6         3850.00  0.55
7         3793.87  0.54
```

[126]:
```
# To markdown for website
print(results_df.to_markdown(index=False))
```

```
| Model                              |   MAE (log) |   RMSE (log) |   MAE
(minutes) |   RMSE (minutes) |      R^2 |
|:-----------------------------------|------------:|-------------:|-------------
---:|-----------------:|---------:|
| Linear Regression (Baseline)       |     1.46448 |      1.92543 |
1812.06 |          4378    | 0.518461 |
| Ridge Regression (tuned)           |     1.47886 |      1.91423 |
1802.69 |          3925.18 | 0.524049 |
| Random Forest Regressor (tuned)    |     1.33684 |      1.82861 |
1739.02 |          3719.7  | 0.565669 |
| XGBoost Regressor (tuned)          |     1.37451 |      1.89939 |
1755.67 |          3729.29 | 0.531397 |
| HGB Regressor (tuned)              |     1.34207 |      1.86236 |
1691.86 |          3557.68 | 0.549491 |
| XGBoost (Optuna tuned)             |     1.36692 |      1.90019 |
1731.78 |          3663.05 | 0.531003 |
| Advanced Hurdle (RF + XGB Stacked) |     1.38978 |      1.87048 |
1786.11 |          3850    | 0.545554 |
| Advanced Hurdle (Optuna tuned)     |     1.37911 |      1.88275 |
1794.94 |          3793.87 | 0.539571 |
```

[139]:
```
baseline = results_df[results_df['Model'] == 'Linear Regression (Baseline)'].
↪iloc[0]
best = results_df[results_df['Model'] == 'Random Forest Regressor (tuned)'].
↪iloc[0]

metrics = ['MAE (log)', 'RMSE (log)', 'MAE (minutes)', 'RMSE (minutes)', 'R^2']

comparison_df = pd.DataFrame({
    'Metric': metrics,
    'Linear Regression (Baseline)': [baseline[m] for m in metrics],
    'Random Forest Regressor (tuned)': [best[m] for m in metrics],
})

# For R^2 higher is better, for everything else lower is better
def pct_improvement(metric, base_val, best_val):
    if metric == 'R^2':
        return (best_val - base_val) / abs(base_val) * 100
    else:
        return (base_val - best_val) / abs(base_val) * 100

comparison_df['Absolute Improvement'] = comparison_df.apply(
    lambda row: np.abs(row['Linear Regression (Baseline)'] - row['Random Forest␣
↪Regressor (tuned)']),
```

```
    axis=1
)
comparison_df['% Improvement'] = comparison_df.apply(
    lambda row: pct_improvement(row['Metric'], row['Linear Regression␣
↪(Baseline)'], row['Random Forest Regressor (tuned)']),
    axis=1
)

comparison_df['% Improvement'] = comparison_df['% Improvement'].map('{:+.1f}%'.
↪format)

comparison_df
```

[139]:
```
           Metric  Linear Regression (Baseline)  \
0       MAE (log)                          1.46
1      RMSE (log)                          1.93
2   MAE (minutes)                       1812.06
3  RMSE (minutes)                       4378.00
4             R^2                          0.52

   Random Forest Regressor (tuned)  Absolute Improvement % Improvement
0                             1.34                  0.13         +8.7%
1                             1.83                  0.10         +5.0%
2                          1739.02                 73.03         +4.0%
3                          3719.70                658.29        +15.0%
4                             0.57                  0.05         +9.1%
```

[140]:
```
print(comparison_df.to_markdown(index=False))
```

```
| Metric         |   Linear Regression (Baseline) |   Random Forest Regressor
(tuned) |   Absolute Improvement | % Improvement   |
|:---------------|-------------------------------:|-------------------------------
-----:|-----------------------:|:----------------|
| MAE (log)      |                        1.46448 |
1.33684  |              0.127637  | +8.7%           |
| RMSE (log)     |                        1.92543 |
1.82861  |              0.0968146 | +5.0%           |
| MAE (minutes)  |                     1812.06     |
1739.02     |             73.0343    | +4.0%           |
| RMSE (minutes) |                     4378        |                         3719.7
|             658.295      | +15.0%          |
| R^2            |                        0.518461 |
0.565669 |              0.0472081 | +9.1%           |
```

[63]:
```
metrics = ['MAE (log)', 'RMSE (log)', 'MAE (minutes)', 'RMSE (minutes)', 'R^2']
ascending = {'MAE (log)': False, 'RMSE (log)': False, 'MAE (minutes)': False,␣
↪'RMSE (minutes)': False, 'R^2': True}
```

```python
fig = go.Figure()

for i, metric in enumerate(metrics):
    ordered = results_df.sort_values(metric, ascending=ascending[metric])
    fig.add_trace(go.Bar(
        y=ordered['Model'],
        x=ordered[metric],
        name=metric,
        visible=(i == 0),
        text=ordered[metric].apply(lambda x: f'{x:.4f}'),
        textposition='outside',
        marker_color=ordered[metric],
        marker_colorscale='Teal',
        orientation='h'
    ))

fig.update_layout(
    title=f'<b>Model Performance Comparison - {metrics[0]}</b>',
    showlegend=False,
    template='plotly_white',
    updatemenus=[dict(
        buttons=[
            dict(
                label=metric,
                method='update',
                args=[
                    {'visible': [j == i for j in range(len(metrics))]},
                    {'title': f'<b>Model Performance Comparison - {metric}</
 ↪b>'},
                ]
            )
            for i, metric in enumerate(metrics)
        ],
        direction='down',
        x=1.0,
        xanchor='right',
        y=1.15,
        yanchor='top',
        showactive=True,
    )]
)

fig.show()

# Export plot for website
```

```python
fig.write_html('docs/assets/plots/model_performance_comparison.html',␣
↪include_plotlyjs='cdn')
```

3.8 Step 8: Fairness Analysis

Our approach for the model's fairness analysis will be comparing its performance in urban vs rural states. We will quantify "urban states" as those where the POPPCT_URBAN (Percentage of the total population of the U.S. state represented by the urban population (in %)) is above the median, with "rural states" being below.

Null Hypothesis: Our model is fair with respect to urban states and rural states, and any observed difference (in RMSE) is due to random chance. **Alternative Hypothesis:** Our model is unfair with respect to urban states and rural states. The RMSE differs significantly between urban and rural states.

We will be using the OutageHurdleModel as our selected best model.

[142]:
```python
# Selecting OutageHurdleModel as best model, but can use any to apply␣
↪methodology
selected_model = reaching_model_tuned

# Preprocess X_test before passing to hurdle model
X_test_proc  = preprocessor_dense.transform(X_test)
y_pred_log = selected_model.predict(X_test_proc)

y_test_minutes = np.expm1(y_test)
y_pred_minutes = np.expm1(y_pred_log)
```

[65]:
```python
# Pull POPPCT_URBAN from df_engineered using original index
poppct_urban_test = df_engineered.loc[X_test.index, 'POPPCT_URBAN'].to_numpy()
urban_threshold = np.median(poppct_urban_test)
is_urban = (poppct_urban_test >= urban_threshold)
is_rural = ~is_urban

print(f"Median POPPCT_URBAN in test set: {urban_threshold:.2f}%")
print(f"Urban group size: {is_urban.sum()}")
print(f"Rural group size: {is_rural.sum()}")
```

```
Median POPPCT_URBAN in test set: 84.05%
Urban group size: 162
Rural group size: 128
```

[66]:
```python
# Minutes scale
rmse_urban = root_mean_squared_error(y_test_minutes[is_urban],␣
↪y_pred_minutes[is_urban])
rmse_rural = root_mean_squared_error(y_test_minutes[is_rural],␣
↪y_pred_minutes[is_rural])

observed_diff = np.abs(rmse_rural - rmse_urban)
```

```
print(f"RMSE (urban): {rmse_urban:.0f} minutes")
print(f"RMSE (rural): {rmse_rural:.0f} minutes")
print(f"RMSE Absolute Observed Difference: {observed_diff:.0f} minutes␣
↪({observed_diff/60:.1f} hours)")
```

```
RMSE (urban): 4024 minutes
RMSE (rural): 3481 minutes
RMSE Absolute Observed Difference: 543 minutes (9.1 hours)
```

[67]:
```
mae_urban = mean_absolute_error(y_test_minutes[is_urban],␣
↪y_pred_minutes[is_urban])
mae_rural = mean_absolute_error(y_test_minutes[is_rural],␣
↪y_pred_minutes[is_rural])

mae_diff = np.abs(mae_rural - mae_urban)

print(f"MAE (Urban): {mae_urban:.0f} minutes")
print(f"MAE (Rural): {mae_rural:.0f} minutes")
print(f"MAE Absolute Observed Difference: {mae_diff:.0f} minutes")
```

```
MAE (Urban): 1811 minutes
MAE (Rural): 1774 minutes
MAE Absolute Observed Difference: 37 minutes
```

[68]:
```
np.random.seed(42)
fairness_diffs = []
for _ in range(1000):
    shuffled_urban = np.random.permutation(is_urban)
    shuffled_rural = ~shuffled_urban # For readability

    rmse_urban_shuffled =␣
↪root_mean_squared_error(y_test_minutes[shuffled_urban],␣
↪y_pred_minutes[shuffled_urban])
    rmse_rural_shuffled =␣
↪root_mean_squared_error(y_test_minutes[shuffled_rural],␣
↪y_pred_minutes[shuffled_rural])
    fairness_diffs.append(np.abs(rmse_rural_shuffled - rmse_urban_shuffled))

fairness_diffs = np.array(fairness_diffs)
p_value = np.mean(fairness_diffs >= observed_diff)

print(f"P-value: {p_value:.4f}")
print(f"Conclusion: {'Reject H0: evidence of unfairness' if p_value < 0.05 else␣
↪'Fail to reject H0: no significant evidence of unfairness'}")
```

```
P-value: 0.5080
Conclusion: Fail to reject H0: no significant evidence of unfairness
```

[69]:

```
fig = px.histogram(
    x=fairness_diffs, nbins=50,
    title='<b>Fairness Test: Permutation Test of Absolute RMSE Differences␣
↪(Urban - Rural)</b>',
    labels={'x': 'Absolute RMSE Differences (minutes)'},
)

fig.add_vline(
    x=observed_diff,
    line_dash='dash',
    line_color='red',
    line_width=2,
    annotation_text=f'Observed Diff: {observed_diff:.0f} min',
    annotation_position='top right',
    annotation=dict(font_size=12, font_color='red')
)

fig.update_layout(template='plotly_white', bargap=0.03)
fig.show()

# Export plot for website
fig.write_html('docs/assets/plots/fairness_permutation_test.html',␣
↪include_plotlyjs='cdn')
```

Our observed test statistic was the difference in RMSE between rural and urban states, where we performed a permutation test (n=1000) that yielded a p-value of 0.5080. As such, we do not have sufficient evidence to conclude that the model is less fair for rural states.

Because the model was trained on log-transformed outage duration but evaluated here on the original minutes scale, this fairness result reflects real-world prediction error in outage duration.